PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

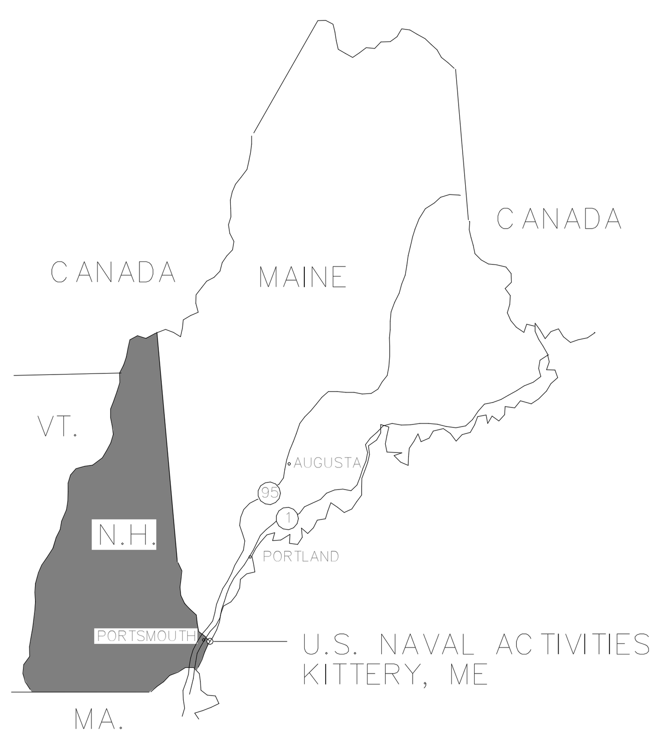
B60 REPAIRS
(DBB)
VOLUME 2

HAVING PARTICIPATED IN THE DESIGN OF EPROJECT NO. 1740579, PORTSMOUTH NAVAL SHIPYARD, KITTERY MAINE, AND HAVING THOROUGHLY REVIEWED THE COMPLETED PROJECT DOCUMENTS, I DECLARE THAT THE FACILITY DESIGN INCLUDED HEREIN COMPLIES WITH ALL APPLICABLE UNITED STATES CODES AND LAWS.

10/01/2025

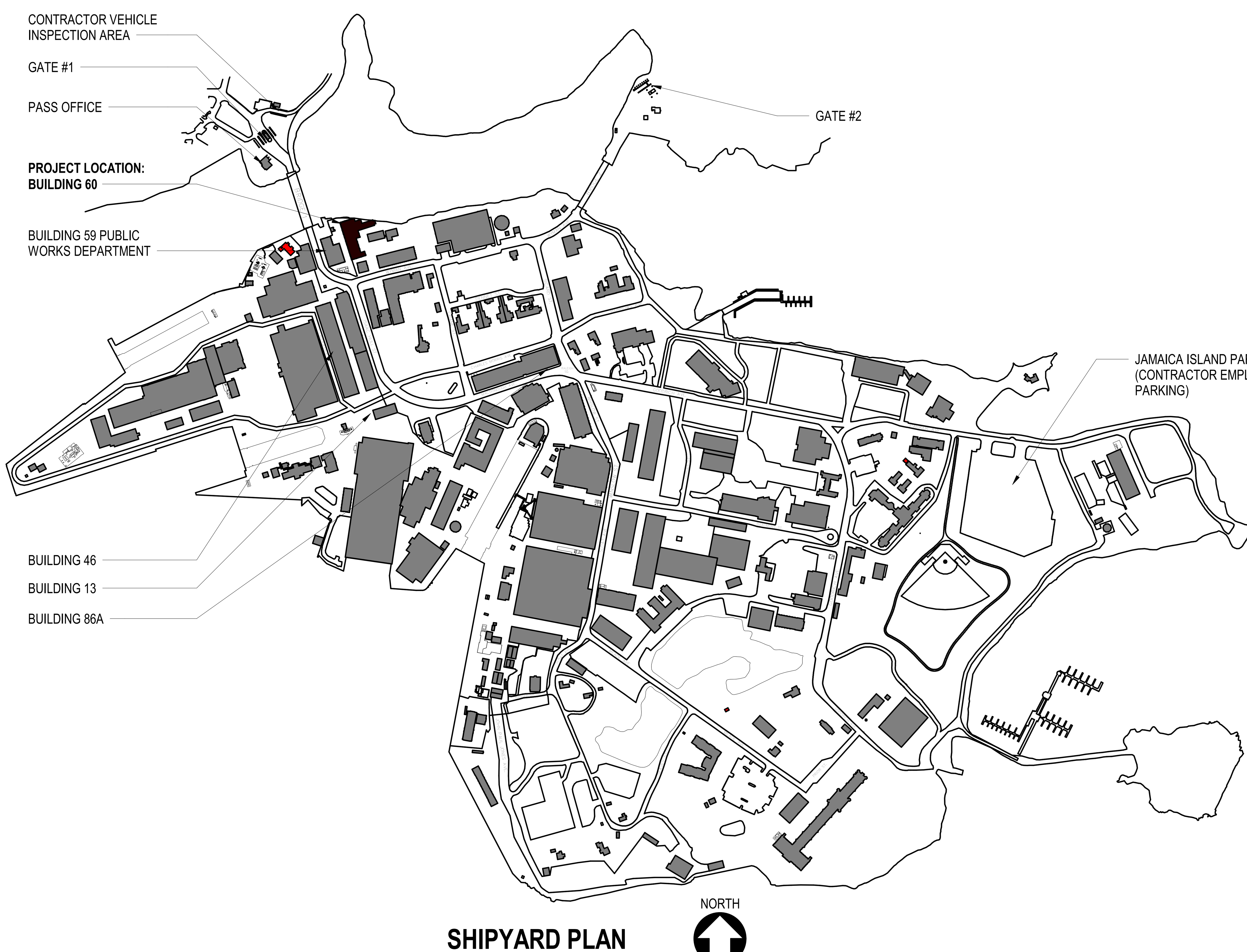
SIGNATURE DATE PROFESSIONAL SEAL

LICENSED ARCHITECT JOSHUA M. RILEY No. 04278 THE STATE OF NEW HAMPSHIRE



AREA MAP
NOT TO SCALE

VICINITY MAP
NOT TO SCALE



SHIPYARD PLAN
NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 04278 THE STATE OF NEW HAMPSHIRE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED
STODDARD.BRIAN.ROBERT.1612130970 2025.12.03 11:58:30-05'00'
FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER	JMR
DRAWN BY	MJD
CHECKED BY	RSH
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME

PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

COVER SHEET - VOLUME 2

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 149 OF 282

G-001B

DRAWFORM REVISION: 14 SEP 2023

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LIST OF DRAWINGS 1 - VOLUME 2

SHEET NO. NO.	SHEET	NAVFAC DWG. NO.	PWD DWG. NO.	SHEET TITLE
GENERAL				
1	G-001A			COVER SHEET - VOLUME 1
2	G-002A			LIST OF DRAWINGS 1 - VOLUME 1
3	G-003A			LIST OF DRAWINGS 2 - VOLUME 1
4	G-004A			ABBREVIATIONS, LEGEND, AND GENERAL PROJECT NOTES - VOLUME 1
5	G-005			ARCHAEOLOGICAL CONSTRAINTS AND REQUIREMENTS PLAN
6	G-006			SITE CONSTRAINTS AND REQUIREMENTS PLAN
7	G-007			CONSTRUCTION SEQUENCE STRATEGY
8	G-008			OUTAGE SCHEDULE
9	G-009			OUTAGE SCHEDULE CONTINUED
10	G-010			CONSTRUCTION BARRIER DETAILS
11	G-100			CODE SUMMARY
12	G-101			FIRST FLOOR PART PLAN - EAST - LIFE SAFETY
13	G-102			FIRST FLOOR PART PLAN - WEST - LIFE SAFETY
14	G-103			FIRST FLOOR MEZZANINE - LIFE SAFETY
15	G-104			SECOND FLOOR - LIFE SAFETY
16	G-105			ATTIC - LIFE SAFETY
HAZARDOUS MATERIALS				
17	HM001			HAZARDOUS MATERIALS NOTES
18	HM101			FIRST FLOOR HAZARDOUS MATERIALS PART PLAN - EAST
19	HM102			FIRST FLOOR HAZARDOUS MATERIALS PART PLAN - WEST
20	HM103			FIRST FLOOR MEZZANINE HAZARDOUS MATERIALS PLAN
21	HM104			SECOND FLOOR HAZARDOUS MATERIALS PLAN
22	HM105			SECOND FLOOR MEZZANINE HAZARDOUS MATERIALS PLAN
23	HM106			SECTION HAZARDOUS MATERIALS
24	HM601			FIRST FLOOR HAZARDOUS MATERIAL - SAMPLING KEYPLAN
25	HM602			SECOND FLOOR HAZARDOUS MATERIAL - SAMPLING KEYPLAN
26	HM603			HAZARDOUS MATERIALS SCHEDULES 1
27	HM604			HAZARDOUS MATERIALS SCHEDULES 2
CIVIL				
28	C-001			CIVIL LEGEND, NOTES AND ABBREVIATIONS
29	CX101			SITE EXISTING CONDITIONS PLAN
30	CR101			SITE REMOVALS PLAN
31	CS101			SITE AND UTILITIES PLAN
32	CG101			GRADING, DRAINAGE AND EROSION CONTROL PLAN
33	C-201			UTILITY PROFILES 1
34	C-501			EROSION AND SEDIMENT CONTROL NOTES
35	C-502			SITE DETAILS 1
36	C-503			SITE DETAILS 2
37	C-504			SITE DETAILS 3
38	B-001			BORING LOGS 1
39	B-002			BORING LOGS 2
40	B-003			BORING LOGS 3
41	B-004			HISTORICAL RECORD BORING LOGS AND TEST PITS
42	B-005			ENVIRONMENTAL TEST RESULTS
STRUCTURAL				
43	S-001			STRUCTURAL NOTES
44	S-002			STRUCTURAL DESIGN LOADS, GENERAL NOTES, AND ABBREVIATIONS
45	S-003			WALL AND ROOF COMPONENT AND CLADDING WIND PRESSURE DIAGRAMS
46	SD101			FOUNDATION REMOVALS PART PLAN - EAST
47	SD102			FOUNDATION REMOVALS PART PLAN - WEST
48	SD103			FIRST FLOOR MEZZANINE FRAMING REMOVALS PART PLAN - EAST
49	SD104			FIRST FLOOR MEZZANINE REMOVALS PART PLAN - WEST
50	SD105			ROOF FRAMING REMOVALS PART PLAN - EAST
51	SD106			SECOND FLOOR FRAMING REMOVALS PLAN
52	SD107			ROOFTOP EQUIPMENT PLATFORM FRAMING REMOVALS PART PLAN - EAST
53	SD108			SECOND FLOOR MEZZANINE FRAMING REMOVALS PLAN
54	SD109			CEILING FRAMING REMOVALS PART PLAN - WEST
55	SB101			FOUNDATION PART PLAN - EAST
56	SB102			FOUNDATION PART PLAN - WEST
57	SB401			NORTH MEZZANINE PILE CAP PLAN AND DETAILS
58	SB501			FOUNDATION DETAILS 1
59	SB502			FOUNDATION DETAILS 2
60	SB503			FOUNDATION DETAILS 3
61	SF101			FIRST FLOOR MEZZANINE FRAMING PART PLAN - EAST
62	SF102			FIRST FLOOR MEZZANINE FRAMING PART PLAN - WEST
63	SF103			SECOND FLOOR FRAMING PLAN
64	SF104			ROOFTOP EQUIPMENT PLATFORM FRAMING PART PLAN - EAST
65	SF105			SECOND FLOOR CEILING FRAMING PLAN
66	SF201			ELEVATOR MASONRY WALL ELEVATIONS
67	SF501			STRUCTURAL DETAILS 1
68	SF502			STRUCTURAL DETAILS 2
69	SF503			STRUCTURAL DETAILS 3
70	SF504			STRUCTURAL DETAILS 4
71	SF505			STRUCTURAL DETAILS 5
72	SF506			STRUCTURAL DETAILS 6
73	SF601			COLUMN AND CONNECTION SCHEDULES AND BASE PLATE DETAILS
74	SP101			STRUCTURAL PROTECTION PART PLAN AND DETAILS

SHEET NO. NO.	SHEET	NAVFAC DWG. NO.	PWD DWG. NO.	SHEET TITLE
ARCHITECTURAL				
75	AD101			FIRST FLOOR REMOVALS PART PLAN - EAST
76	AD102			FIRST FLOOR REMOVALS PART PLAN - WEST
77	AD103			FIRST FLOOR MEZZANINE REMOVALS PLAN
78	AD104			SECOND FLOOR REMOVALS PLAN
79	AD105			SECOND FLOOR MEZZANINE REMOVALS PLAN
80	AD120			ROOF REMOVALS PART PLAN - EAST
81	AD121			ROOF REMOVALS PART PLAN - WEST
82	AD201			EXTERIOR ELEVATION REMOVALS 1
83	AD202			EXTERIOR ELEVATION REMOVALS 2
84	AD203			EXTERIOR ELEVATION REMOVALS 3
85	AD221			BUILDING SECTION REMOVALS 1
86	AD222			BUILDING SECTION REMOVALS 2
87	AD401			ENLARGED REMOVAL PART PLANS
88	AD521			ROOF REMOVALS DETAILS 1
89	AD522			ROOF REMOVALS DETAILS 2
90	AE001			WALL & CEILING TYPES AND DETAILS
91	AE002			FIRESTOPPING DETAILS AND STANDARD MOUNTING HEIGHTS
92	AE101			FIRST FLOOR PART PLAN - EAST
93	AE102			FIRST FLOOR PART PLAN - WEST
94	AE103			FIRST FLOOR MEZZANINE PART PLAN - EAST
95	AE104			FIRST FLOOR MEZZANINE PART PLAN - WEST
96	AE105			SECOND FLOOR PLAN
97	AE106			SECOND FLOOR MEZZANINE PLAN
98	AE107			ATTIC FLOOR PLAN
99	AE108			B59 FIRST FLOOR PLAN
100	AE120			ROOF PART PLAN - EAST
101	AE121			ROOF PART PLAN - WEST
102	AE131			TYPICAL SIGNAGE DETAILS
103	AE132			SIGNAGE SCHEDULE AND DETAILS
104	AE201			BUILDING AXONS
105	AE202			EXTERIOR ELEVATIONS 1
106	AE203			EXTERIOR ELEVATIONS 2
107	AE204			EXTERIOR ELEVATIONS 3
108	AE205			MASONRY REPAIR DETAILS
109	AE221			BUILDING SECTIONS 1
110	AE222			BUILDING SECTIONS 2
111	AE223			BUILDING SECTIONS 3
112	AE301			WALL SECTIONS
113	AE302			SECTION DETAILS
114	AE401			ENLARGED FIRST FLOOR TOILET ROOM AND INTERIOR ELEVATION
115	AE402			ENLARGED SECOND FLOOR TOILET ROOM PLANS AND ELEVATIONS
116	AE411			ENLARGED ELEVATOR PLANS AND SECTIONS
117	AE412			ELEVATOR DETAILS
118	AE421			ENLARGED ELECTRICAL ROOM PLAN AND SECTION
119	AE440			ENLARGED STAIR PLANS 1
120	AE441			ENLARGED STAIR PLANS 2
121	AE442			STAIR SECTIONS
122	AE443			LADDER SECTIONS
123	AE444			STAIR DETAILS
124	AE445			RAILING DETAILS
125	AE501			PLAN DETAILS
126	AE511			METAL PANEL DETAILS
127	AE521			ROOF DETAILS 1
128	AE522			ROOF DETAILS 2
129	AE530			EXPANSION JOINT DETAILS 1
130	AE531			EXPANSION JOINT DETAILS 2
131	AE540			CASEWORK DETAILS
132	AE601			DOOR SCHEDULE AND DETAILS
133	AE602			DOOR DETAILS
134	AE621			WINDOW TYPES
135	AE622			WINDOW AND DOOR REPAIR AND RESTORATION NOTES AND SCHEDULE
136	AE623			WINDOW AND DOOR REPAIR AND RESTORATION TYPES
137	AE626			WINDOW DETAILS
138	AE640			ROOM FINISH SCHEDULE
139	AE701			FIRST FLOOR REFLECTED CEILING PART PLAN - EAST
140	AE702			FIRST FLOOR REFLECTED CEILING PART PLAN - WEST
141	AE703			FIRST FLOOR MEZZANINE REFLECTED CEILING PLAN
142	AE704			SECOND FLOOR REFLECTED CEILING PLAN
143	AE720			CEILING AND SOFFIT DETAILS
144	AE741			INTERIOR ELEVATIONS
145	AE801			FIRST FLOOR FF&E PART PLAN - EAST (OPTION 1)
146	AE802			FIRST FLOOR FF&E PART PLAN - WEST (OPTION 1)
147	AE803			FIRST FLOOR MEZZANINE FF&E PART PLAN - WEST (OPTION 1)
148	AE804			SECOND FLOOR FF&E PLAN (OPTION 1)

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 4275 THE STATE OF NEW HAMPSHIRE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

LIST OF DRAWINGS 1 - VOLUME 2

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 150 OF 282

G-002B

DRAWFORM REVISION: 14 SEP 2023

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LIST OF DRAWINGS 2 - VOLUME 2

SHEET NO. NO.	SHEET NO. SHEET	NAVFAC DWG. NO.	PWD DWG. NO.	SHEET TITLE
GENERAL				
149	G-001B			COVER SHEET - VOLUME 2
150	G-002B			LIST OF DRAWINGS 1 - VOLUME 2
151	G-003B			LIST OF DRAWINGS 2 - VOLUME 2
152	G-004B			ABBREVIATIONS, LEGEND, AND GENERAL PROJECT NOTES - VOLUME 2
EQUIPMENT				
153	QD101			FIRST FLOOR EQUIPMENT REMOVALS PART PLAN - EAST (CLIN 002)
154	QD102			FIRST FLOOR EQUIPMENT REMOVALS PART PLAN - WEST (CLIN 002)
155	QD103			FIRST FLOOR MEZZANINE EQUIPMENT REMOVALS PLAN (CLIN 002)
156	QD601			EQUIPMENT REMOVALS SCHEDULE 1 (CLIN 002)
157	QD602			EQUIPMENT REMOVALS SCHEDULE 2 (CLIN 002)
158	Q-101			FIRST FLOOR EQUIPMENT PART PLAN - EAST (OPTION 2)
159	Q-102			FIRST FLOOR EQUIPMENT PART PLAN - WEST (OPTION 2)
160	Q-103			SECOND FLOOR EQUIPMENT PLAN (OPTION 2)
161	Q-601			EQUIPMENT SCHEDULE (OPTION 2)
PLUMBING				
162	P-001			PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES
163	PD101			UNDERSLAB AND FIRST FLOOR TRENCH PLUMBING REMOVALS PLAN - WEST
164	PD102			FIRST FLOOR PLUMBING REMOVALS PLAN - EAST
165	PD103			FIRST FLOOR PLUMBING REMOVALS PLAN - WEST
166	PD104			FIRST FLOOR MEZZANINE PLUMBING REMOVALS PLAN - WEST
167	PD105			SECOND FLOOR PLUMBING REMOVALS PLAN
168	PD106			SECOND FLOOR MEZZANINE PLUMBING REMOVALS PLAN
169	P-101			UNDERSLAB PLUMBING PLAN - EAST
170	P-102			UNDERSLAB PLUMBING PLAN - WEST
171	P-103			FIRST FLOOR PLUMBING PLAN - EAST
172	P-104			FIRST FLOOR PLUMBING PLAN - WEST
173	P-105			FIRST FLOOR MEZZANINE PLUMBING PLAN - EAST
174	P-106			FIRST FLOOR MEZZANINE PLUMBING PLAN - WEST
175	P-107			SECOND FLOOR PLUMBING PLAN
176	P-108			SECOND FLOOR MEZZANINE PLUMBING PLAN
177	P-109			ATTIC PLUMBING PLAN
178	P-401			ENLARGED PLUMBING PART PLANS 1
179	P-402			ENLARGED PLUMBING PART PLANS 2
180	P-403			ENLARGED PLUMBING PART PLANS 3
181	P-501			PLUMBING DETAILS
182	P-601			PLUMBING SCHEDULES AND DETAIL
183	P-701			PLUMBING RISER DIAGRAM
MECHANICAL				
184	M-001			MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES
185	MD101			FIRST FLOOR TRENCH MECHANICAL PIPING REMOVALS PLAN - WEST
186	MD102			FIRST FLOOR MECHANICAL DUCTWORK REMOVALS PLAN - EAST
187	MD103			FIRST FLOOR MECHANICAL DUCTWORK REMOVALS PLAN - WEST
188	MD104			FIRST FLOOR MECHANICAL PIPING REMOVALS PLAN - EAST
189	MD105			FIRST FLOOR MECHANICAL PIPING REMOVALS PLAN - WEST
190	MD106			FIRST FLOOR MEZZANINE MECHANICAL REMOVALS PLAN - WEST
191	MD107			SECOND FLOOR MECHANICAL REMOVALS PLAN
192	MD108			SECOND FLOOR MEZZANINE MECHANICAL REMOVALS PLAN
193	MD109			ATTIC MECHANICAL REMOVALS PLAN
194	MD120			ROOF MECHANICAL REMOVALS PLAN - EAST
195	MD201			MECHANICAL EXTERIOR ELEVATION REMOVALS
196	MD221			MECHANICAL BUILDING SECTION REMOVALS 1
197	MD222			MECHANICAL BUILDING SECTION REMOVALS 2
198	MD223			MECHANICAL BUILDING SECTION REMOVALS 3
199	MH101			FIRST FLOOR MECHANICAL DUCTWORK PLAN - WEST
200	MH102			FIRST FLOOR MEZZANINE MECHANICAL DUCTWORK PLAN - WEST
201	MH103			SECOND FLOOR MECHANICAL DUCTWORK PLAN
202	MH104			SECOND FLOOR MEZZANINE MECHANICAL DUCTWORK PLAN
203	MH105			ATTIC MECHANICAL DUCTWORK PLAN
204	MP101			FIRST FLOOR MECHANICAL PIPING PLAN - EAST
205	MP102			FIRST FLOOR MECHANICAL PIPING PLAN - WEST
206	MP103			FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - EAST
207	MP104			FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - WEST
208	MP105			SECOND FLOOR MECHANICAL PIPING PLAN
209	MP106			SECOND FLOOR MEZZANINE MECHANICAL PLAN
210	MP107			ATTIC MECHANICAL PIPING PLAN
211	M-401			ENLARGED MECHANICAL PART PLANS
212	M-501			MECHANICAL DETAILS 1
213	M-502			MECHANICAL DETAILS 2
214	M-503			MECHANICAL DETAILS 3

SHEET NO. NO.	SHEET NO. SHEET	NAVFAC DWG. NO.	PWD DWG. NO.	SHEET TITLE
215	M-504			MECHANICAL DETAILS 4
216	M-505			MECHANICAL DETAILS 5
217	M-601			MECHANICAL SCHEDULES 1
218	M-602			MECHANICAL SCHEDULES 2
219	M-603			MECHANICAL SCHEDULES 3
220	M-701			MECHANICAL CONTROL DIAGRAMS 1
221	M-702			MECHANICAL CONTROL DIAGRAMS 2
222	M-703			MECHANICAL CONTROL DIAGRAMS 3
223	M-704			MECHANICAL CONTROL DIAGRAMS 4
224	M-705			MECHANICAL CONTROL DIAGRAMS 5
FIRE PROTECTION				
225	FA001			FIRE ALARM NOTES
226	FA101			FIRST FLOOR PART PLAN - EAST - FIRE ALARM
227	FA102			FIRST FLOOR PART PLAN - WEST - FIRE ALARM
228	FA103			FIRST FLOOR MEZZANINE - FIRE ALARM
229	FA104			SECOND FLOOR - FIRE ALARM
230	FA105			ATTIC - FIRE ALARM
231	FA501			DETAILS - FIRE ALARM
232	FX001			FIRE PROTECTION NOTES
233	FX101			FIRST FLOOR PART PLAN - EAST - FIRE PROTECTION
234	FX102			FIRST FLOOR PART PLAN - WEST - FIRE PROTECTION
235	FX103			FIRST FLOOR MEZZANINE - FIRE PROTECTION
236	FX104			SECOND FLOOR - FIRE PROTECTION
237	FX105			ATTIC - FIRE PROTECTION
238	FX400			ENLARGED FIRE PUMP ROOM
239	FX501			DETAILS - FIRE PROTECTION
ELECTRICAL				
240	E-001			ELECTRICAL LEGEND, ABBREVIATIONS, AND NOTES
241	ED101			FIRST FLOOR ELECTRICAL REMOVALS PART PLAN - EAST
242	ED102			FIRST FLOOR ELECTRICAL REMOVALS PART PLAN - WEST
243	ED103			FIRST FLOOR MEZZANINE ELECTRICAL REMOVALS PART PLAN
244	ED104			SECOND FLOOR ELECTRICAL REMOVALS PLAN
245	ED105			SECOND FLOOR MEZZANINE ELECTRICAL REMOVALS PLAN
246	ES101			ELECTRICAL SITE PLAN
247	ES102			ELECTIRCAL SITE PLAN 2
248	EP101			FIRST FLOOR POWER PART PLAN - EAST
249	EP102			FIRST FLOOR POWER PART PLAN - WEST
250	EP103			FIRST FLOOR MEZZANINE POWER PART PLAN - EAST
251	EP104			FIRST FLOOR MEZZANINE POWER PART PLAN - WEST
252	EP105			SECOND FLOOR POWER PLAN
253	EP106			SECOND FLOOR MEZZANINE POWER PLAN
254	EP107			ATTIC POWER PART PLAN - WEST
255	EP501			ONE-LINE DIAGRAM AND SCHEDULES
256	EP502			ELECTRICAL POWER DETAILS
257	EP503			MECHANICAL, PLUMBING, AND MISCELLANEOUS EQUIPMENT WIRING SCHEDULE
258	EP504			MECHANICAL, PLUMBING, AND MISCELLANEOUS EQUIPMENT SCHEDULE CONTINUATION
259	EP505			SYSTEMS FURNITURE POWER AND POKE THROUGH DETAILS
260	EP601			PANELBOARD SCHEDULES - 1
261	EP602			PANELBOARD SCHEDULES - 2
262	EP603			PANELBOARD SCHEDULES - 3
263	EP604			PANELBOARD SCHEDULES - 4
264	EP701			ELECTRICAL EQUIPMENT SCHEDULES
265	EL101			FIRST FLOOR LIGHTING PART PLAN - EAST
266	EL102			FIRST FLOOR LIGHTING PART PLAN - WEST
267	EL103			FIRST FLOOR MEZZANINE LIGHTING PART PLAN - EAST
268	EL104			FIRST FLOOR MEZZANINE LIGHTING PART PLAN - WEST
269	EL105			SECOND FLOOR LIGHTING PLAN
270	EL106			SECOND FLOOR MEZZANINE LIGHTING PLAN
271	EL107			ATTIC LIGHTING PART PLAN - WEST
272	EL701			LIGHTING FIXTURE SCHEDULE AND DETAILS
273	EL702			LIGHTING FIXTURE DETAILS 1
274	ET101			FIRST FLOOR TECHNOLOGY AND SECURITY PART PLAN - EAST
275	ET102			FIRST FLOOR TECHNOLOGY AND SECURITY PART PLAN - WEST
276	ET103			FIRST FLOOR MEZZANINE TECHNOLOGY AND SECURITY PART PLAN - EAST
277	ET104			FIRST FLOOR MEZZANINE TECHNOLOGY AND SECURITY PART PLAN - WEST
278	ET105			SECOND FLOOR TECHNOLOGY AND SECURITY PLAN
279	ET106			B59 FIRST FLOOR TECHNOLOGY PLAN
280	ET501			TELECOMMUNICATIONS RISER DIAGRAMS
281	ET502			TELECOMMUNICATIONS DETAILS
282	EY501			SECURITY SYSTEM RISER AND DETAILS

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 4275 THE STATE OF NEW HAMPSHIRE 10-01-25

SEAL

OAK POINT ASSOCIATES

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ACTIVITY

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DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME

PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

LIST OF DRAWINGS 2 - VOLUME 2

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 151 OF 282

G-003B

DRAWFORM REVISION: 14 SEP 2023

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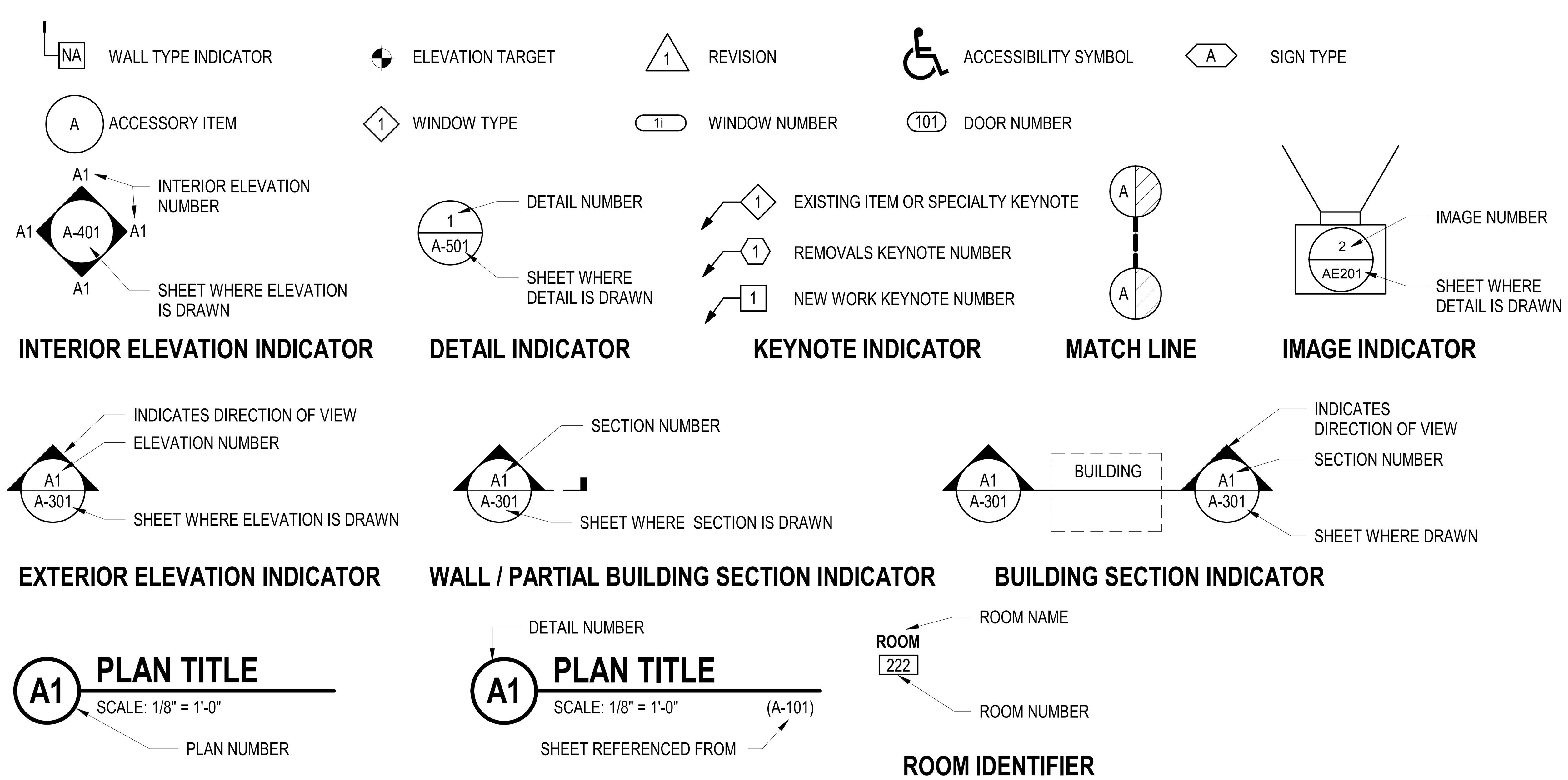
ABBREVIATIONS

"	INCH, INCHES	INT	INTERIOR
'	FOOT, FEET	IR	INSTALLATION RESTORATION
±	PLUS OR MINUS	LBP	LEAD-BASED PAINT
ACM	ASBESTOS CONTAINING MATERIAL	MAX	MAXIMUM
ACT	ACOUSTICAL CEILING TILE	MECH	MECHANICAL
AFF	ABOVE FINISH FLOOR	MFR	MANUFACTURER
AHJ	AUTHORITY HAVING JURISDICTION	MIN	MINIMUM
ALUM	ALUMINUM	MIR	MIRROR
BF	BARRIER FREE	MTL	METAL
BLDG	BUILDING	N	NORTH
CJ	CONTROL JOINT	NAT	NATURAL FINISH
℄	CENTERLINE	NIC	NOT IN CONTRACT
CLG	CEILING	NMCI	NAVY/MARINE CORPS INTRANET
CLR	CLEAR	NO.#	NUMBER
CMU	CONCRETE MASONRY UNIT	OC	ON CENTER
COL	COLUMN	PCB'S	POLYCHLORINATED BIPHENYLS
CONC	CONCRETE	PETG	POLYETHYLENE TEREPHTHALATE GLYCOL
CON	CONNECTION	PMI	PLANNED MODIFICATION ITEM
CONT	CONTINUOUS	PNT	PAINT, PAINTED
CPT	CARPET	PLAM	PLASTIC LAMINATE
CT	CERAMIC TILE	PLY	PLYWOOD
CWT	CERAMIC WALL TILE	PQP	PRIVATE QUALIFIED PERSON
DIA	DIAMETER	PT	PRESSURE TREATED
DN	DOWN	PWD	PUBLIC WORKS DEPARTMENT
DWG	DRAWING	R	RADIUS
E	EAST	RCB	RUBBER COVE BASE
EA	EACH	RCP	REFLECTED CEILING PLAN
EF	EXHAUST FAN	REINF	REINFORCED
EJ	EXPANSION JOINT	RO	ROUGH OPENING
ELEC	ELECTRIC/ELECTRICAL	S	SOUTH
ELEV	ELEVATION	SIM	SIMILAR
EMR	ELEVATOR MACHINE ROOM	SF	SQUARE FEET
EQ	EQUAL	STL	STEEL
EXIST	EXISTING	STRUCT	STRUCTURAL
EXT	EXTERIOR	TYP	TYPICAL
FCO	FLOOR CLEANOUT	UL	UNDERWRITERS LABORATORIES
FD	FLOOR DRAIN	UON	UNLESS OTHERWISE NOTED
FND	FOUNDATION	VERT	VERTICAL
FE	FIRE EXTINGUISHER	VIF	VERIFY IN FIELD
FEC	FIRE EXTINGUISHER CABINET	W	WEST, WIDE
FF	FINISH FLOOR	W/	WITH
FIN	FINISH	WB	WOOD BASE
FL,FLR	FLOOR	WD	WOOD
FT	FOOT, FEET	WP	WALL PROTECTION
GA	GAUGE		
GALV	GALVANIZED		
GYP BD	GYPSUM BOARD		
H,HT,HGT	HEIGHT		
HM	HOLLOW METAL		
HORIZ	HORIZONTAL		
HP	HIGH POINT		
IN	INCHES, INCH		
INSUL	INSULATION, INSULATED		

GENERAL SYMBOLS LEGEND



GENERAL HAZARDOUS MATERIALS NOTES

1. ITEMS IDENTIFIED ON THE HAZARDOUS MATERIALS DRAWINGS ARE EXISTING. SEE SHEETS HM001-HM604 FOR ADDITIONAL HAZARDOUS MATERIALS INFORMATION. COORDINATE HAZARDOUS MATERIALS WORK WITH BUILDING REMOVALS AND CONSTRUCTION SEQUENCE PLAN.
2. OBTAIN PERMITS AND LICENSES REQUIRED FOR THE REMOVAL, TRANSPORT, AND DISPOSAL OF HAZARDOUS MATERIALS AND DEBRIS AT NO ADDITIONAL COST TO THE GOVERNMENT.
3. CONTRACTOR MUST PERFORM NECESSARY MANAGEMENT OF IDENTIFIED HAZARDOUS MATERIALS, IN ACCORDANCE WITH EM 385-1-1, APPLICABLE PROJECT SPECIFICATIONS, AND FEDERAL, STATE, AND LOCAL REGULATIONS.
4. CONTRACTOR MUST SUBMIT WORK PLANS AND HIRE PQP TO COORDINATE IMPLEMENTATION OF THE WORK PLAN, PERFORM INDEPENDENT INSPECTIONS, TESTING, AND VERIFICATION OF THE HAZARDOUS COMPONENTS WORK INCLUDING, BUT NOT LIMITED TO: ASBESTOS, LEAD-, CADMIUM, AND CHROMIUM-CONTAINING PAINT, AND PCBs.
5. EXISTING PAINT IS ASSUMED TO CONTAIN LEAD, CADMIUM, AND CHROMIUM (VI) (HEAVY METALS). MANAGE HEAVY METALS PAINT IN AREAS INDICATED FOR REMOVAL SEPARATE FROM THE SUBSTRATE AND TREAT THE REMOVED PAINT AS HAZARDOUS MATERIAL IN ACCORDANCE WITH THE SPECIFICATIONS. HEAVY METALS MUST BE MANAGED, REMOVED, AND DISPOSED OF IN ACCORDANCE WITH FEDERAL, STATE, AND LOCAL REGULATIONS.
6. EXISTING LIGHT FIXTURE BALLASTS ARE ASSUMED TO CONTAIN PCB'S AND EXISTING LAMPS ARE ASSUMED TO CONTAIN MERCURY. SEE ELECTRICAL DRAWINGS FOR EXTENT OF REMOVALS. REMOVE IN ACCORDANCE WITH SPECIFICATIONS.
7. EXISTING INDUSTRIAL EQUIPMENT/PROCESSES HAVE PRODUCED HEAVY METALS RESIDUE. REFER TO HAZARDOUS MATERIAL SHEETS FOR FURTHER INFORMATION.
8. THE FOLLOWING SPECIFICATIONS APPLY:
   - 02 82 00.00 22 ASBESTOS REMEDIATION (PNSY PROJECTS)
   - 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, REMOVAL, AND ABATEMENT (PNSY PROJECTS)
   - 02 84 16.00 22 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCB'S AND MERCURY (PNSY PROJECTS)
   - 02 84 33.00 22 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBS) (PNSY PROJECTS)

CONTRACT LINE ITEM NUMBERS (CLIN)

001 BASE BID
002 INDUSTRIAL EQUIPMENT REMOVAL AND PROCESS HAZARDOUS MATERIALS MANAGEMENT/ABATEMENT
003 HAZARDOUS MATERIALS MANAGEMENT/ABATEMENT
004 EXISTING TRAILER UTILITIES

OPTION ITEMS

1. FURNITURE, FIXTURES, AND EQUIPMENT (FF&E)
2. INDUSTRIAL EQUIPMENT
3. RADON FANS
4. SECURITY DEVICES
5. ADDITIONAL WINDOW RESTORATIONS
6. NOT APPLICABLE
7. ADDITIONAL ANNEX REPAIRS
8. ADDITIONAL FINISHES
9. ELECTRICAL SERVICE INFRASTRUCTURE
10. RESINOUS FLOORING
11. COMMUNICATION SERVICE ROUTING

GENERAL REMOVALS NOTES

1. COORDINATE REMOVAL OF DEBRIS WITH THE CONTRACTING OFFICER TO MINIMIZE DISRUPTION OF BASE ACTIVITIES. COORDINATE WITH WASTE MANAGEMENT PLAN.
2. ITEMS DEPICTED FOR REMOVAL OR DEMOLITION ARE SHOWN FOR INFORMATIONAL PURPOSES. THESE DRAWINGS DO NOT INDICATE EACH AND EVERY ITEM TO BE REMOVED OR DEMOLISHED. (EX: VALVE, HANGER, FASTENER, ETC.) VERIFY IN FIELD COMPLETE WORK ASSOCIATED WITH REMOVALS OR DEMOLITION.
3. WHERE MATERIALS THAT ARE SUSPECTED TO CONTAIN HAZARDOUS MATERIALS ARE ENCOUNTERED, IMMEDIATELY STOP THAT WORK AND REPORT THE CONDITION TO THE CONTRACTING OFFICER. SEE HAZARDOUS MATERIALS NOTES AND HAZARDOUS MATERIALS REMOVALS PLANS.
4. REMOVALS KEYNOTES INDICATED ON "AD" SHEETS ARE REPRESENTATIVE OF GENERAL ITEMS TO BE REMOVED AND ARE NOT INTENDED TO IDENTIFY EACH ITEM ON EVERY WALL, FLOOR, CEILING, OR ELEVATION. VERIFY IN FIELD COMPLETE WORK ASSOCIATED WITH REMOVALS.
5. BUILDING 60 IS A HISTORIC FACILITY AND IS A CONTRIBUTING RESOURCE TO THE PORTSMOUTH NAVAL SHIPYARD HISTORIC DISTRICT. REMOVE ITEMS IN SUCH A MANNER THAT THE HISTORIC STRUCTURE (e.g. MORTAR JOINTS, GRANITE, BRICK, WOOD WINDOW AND DOOR TRIM, ETC.) TO REMAIN IS NOT DAMAGED DURING DEMOLITION AND CONSTRUCTION ACTIVITIES. ANY ITEMS DAMAGED BY THE CONTRACTOR MUST BE REPAIRED TO ITS ORIGINAL CONDITION AT NO ADDITIONAL COST TO THE GOVERNMENT. SUBMIT A WORK PLAN TO ADDRESS REPAIRS TO DAMAGED ITEMS. PWD MAINE CULTURAL RESOURCES MANAGER MUST REVIEW AND APPROVE THE CONTRACTOR'S QUALITY CONTROL PLAN PRIOR TO COMMENCING REPAIR WORK.
6. HISTORIC ITEMS INDICATED ARE TO BE SALVAGED FOR REINSTALLATION OR ARE TO REMAIN IN PLACE (e.g. WOOD TRIM, METAL DOOR, PAINTED SIGNAGE, WINDOWS, AND STRUCTURAL ELEMENTS, ETC). PROTECT HISTORIC ITEMS DURING DEMOLITION AND CONSTRUCTION. ITEMS TO BE STORED FOR REINSTALLATION OR TURNED OVER TO GOVERNMENT MUST BE STORED ON SITE IN SAFE AND SECURE LOCATION.
7. COORDINATE GENERAL BUILDING REMOVAL CONSTRUCTION SEQUENCE WITH HAZARDOUS MATERIALS REMOVALS.
8. PATCH ANCHORAGE LOCATIONS AT REMOVED ITEMS. CUT FLUSH TO FLOOR/WALLS ANY EQUIPMENT ATTACHMENT/ANCHORAGE AS TO PROVIDE THE REQUIRED FLOOR/WALL FINISH.

GENERAL MASONRY CLEANING/REPAIR/REPOINTING NOTES

1. BUILDING 60 AND ITS MASONRY ARE HISTORIC. COMPLY WITH SPECIFICATION SECTION 04 01 00.91 22 RESTORATION AND CLEANING OF MASONRY IN HISTORIC STRUCTURES (PWD ME).
2. PROTECT MASONRY AND STRUCTURE THROUGHOUT PROJECT. USE THE GENTLEST MEANS TO PERFORM THE WORK.
3. ENGAGE THE SERVICES OF A HISTORIC MASONRY CONSULTANT TO DETERMINE THE CONDITION OF THE EXISTING MASONRY AND MATERIALS AND TO RECOMMEND CLEANING AND RESTORATION PROCEDURES. REFER TO SPECIFICATION SECTION 04 01 00.91 22 RESTORATION AND CLEANING OF MASONRY IN HISTORIC STRUCTURES (PWD ME).
4. PREPARE A MOCK-UP FOR MASONRY REPAIR/RESTORATION WORK. REVIEW WITH CONTRACTING OFFICER AND PWD-ME CULTURAL RESOURCES MANAGER.

CODE SUMMARY

BUILDING CODE:

UNIFIED FACILITIES CRITERIA (UFC) 1-200-01 (26 FEB. 2024)

UNIFIED FACILITIES CRITERIA (UFC) 1-200-02 (01 JUNE 2022)

UNIFIED FACILITIES CRITERIA (UFC) 3-101-01 (08 JAN. 2024)

UNIFIED FACILITIES CRITERIA (UFC) 3-110-03 (01 MAY 2012)

ACCESSIBILITY CODE:

UNIFORM ACCESSIBILITY STANDARDS (UFAS)

ARCHITECTURAL BARRIERS ACT (ABA)

AMERICANS WITH DISABILITIES ACT (ADA)

ACCESSIBILITY GUIDELINES FOR BUILDINGS AND FACILITIES (ADAAG)

MECHANICAL CODE:

INTERNATIONAL MECHANICAL CODE (IMC) 2021

ELECTRICAL CODE:

NFPA 70, NATIONAL ELECTRIC CODE

PLUMBING CODE:

INTERNATIONAL PLUMBING CODE (IPC) 2021

FIRE PREVENTION & LIFE SAFETY CODE:

SEE SHEET G-100.

GENERAL CONSTRUCTION NOTES

1. COMPLY WITH STATE, FEDERAL, NATIONAL, AND OTHER CODES AND ORDINANCES THAT APPLY TO THIS PROJECT.
2. COMPLETE WORK IN ACCORDANCE WITH INDUSTRY STANDARDS AND PERFORM WORK IN A WORKMANLIKE AND PROFESSIONAL MANNER.
3. WHERE CONFLICTS EXIST WITHIN THE DOCUMENTS OR BETWEEN THE DOCUMENTS AND APPLICABLE CODES, ORDINANCES, OR INDUSTRY STANDARDS, THE MORE STRINGENT, AS DETERMINED BY THE CONTRACTING OFFICER, MUST BE PROVIDED AT NO ADDITIONAL COST TO THE GOVERNMENT.
4. MAINTAIN BUILDING EGRESS PATHS DURING THE ENTIRE CONTRACT PERIOD. PROVIDE TEMPORARY PATHS TO MAINTAIN BUILDING EGRESS. COMPLY WITH NFPA 241, STANDARD FOR SAFEGUARDING CONSTRUCTION, ALTERATIONS, AND DEMOLITION OPERATIONS.
5. MAINTAIN THE CONSTRUCTION SITE IN A SECURE, NEAT, WATER TIGHT, CLEAN, AND SAFE CONDITION DURING THE ENTIRE CONTRACT PERIOD. EXTERIOR ENVELOPE WORK MUST BE PERFORMED IN A MANNER TO MAINTAIN WEATHER TIGHTNESS.
6. PROVIDE ITEMS AND COMPONENTS SHOWN ON THE DRAWINGS UNLESS NOTED AS "EXISTING" (EXIST).
7. WORK FROM GIVEN DIMENSIONS AND LARGE SCALE DETAILS. DO NOT SCALE DRAWINGS.
8. DIMENSIONING SYSTEM: UNLESS OTHERWISE NOTED, PLAN DIMENSIONS ARE TO/FROM FINISHED FACE OF EXISTING WALLS, TO/FROM FACE OF FRAMING FOR STUD FRAMED WALLS, AND TO/FROM THE FACE OF MASONRY FOR MASONRY WALLS. DIMENSIONS INDICATED AS "CLEAR" (CLR) MUST BE MAINTAINED AND ARE MEASURED FROM THE FACE OF FINISH. CENTERLINE DIMENSIONS ARE INDICATED BY CENTERLINE SYMBOL (C).
9. LOCATE DOOR OPENINGS 6 INCHES FROM ADJACENT WALL (FACE OF FRAMING TO ROUGH OPENING), UNLESS OTHERWISE DIMENSIONED. L
10. VERIFY EXISTING CONDITIONS AND DIMENSIONS AND REPORT DISCREPANCIES TO THE CONTRACTING OFFICER. PROCEED WITH THE WORK AFTER THE DISCREPANCIES HAVE BEEN RESOLVED BY THE GOVERNMENT.
11. DO NOT PENETRATE JOISTS, BEAMS, OR OTHER STRUCTURAL MEMBERS. OBTAIN WRITTEN APPROVAL FROM THE ENGINEER OF RECORD FOR UNAVOIDABLE PENETRATIONS OF STRUCTURAL MEMBERS.
12. PROVIDE SOLID BLOCKING BEHIND SURFACE APPLIED FIXTURES, TRIM, GRAB BARS, SHELVES, WOOD WALL BASE, BASE AND WALL CABINETS, MARKER BOARDS, TELEVISIONS, AND OTHER ACCESSORIES AND EQUIPMENT INCLUDING GOVERNMENT SUPPLIED ITEMS WHEN MOUNTED TO STUD WALLS.
13. DISPOSE OF AND/OR RECYCLE CONSTRUCTION DEBRIS FROM THE PROJECT SITE AS REQUIRED BY THE STATE OF MAINE AND AS REQUIRED BY THE CONTRACT. OBTAIN REQUIRED DISPOSAL PERMITS.
14. PROTECT EXISTING SYSTEMS AND SURFACES TO REMAIN. REPAIR OR REPLACE DAMAGE RESULTING FROM PROJECT OPERATIONS, AS APPROVED BY THE CONTRACTING OFFICER, AT NO ADDITIONAL COST TO THE GOVERNMENT.
15. OBTAIN APPROVAL FROM THE CONTRACTING OFFICER FOR THE SEQUENCE AND TIMING OF OPERATIONS PRIOR TO COMMENCING WORK AND PER OUTAGE REQUIREMENTS DETAILED IN THE SPECIFICATIONS.
16. FIRESTOP CONDUIT/PIPE/DUCT PENETRATIONS THROUGH FIRE-RATED WALLS, CEILINGS, AND FLOORS. PROVIDE SMOKE BARRIER SEALANT AT FLOOR PENETRATIONS AND PENETRATIONS THROUGH SMOKE PARTITIONS.
17. DO NOT INTERRUPT UTILITY SERVICES WITHOUT AN APPROVED OUTAGE REQUEST. DO NOT RESTORE UTILITIES WITHOUT AN APPROVED 'ONAGE' REQUEST. REFER TO SPECIFICATION SECTION 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) AND SHEETS G-008 AND G-009.
18. BUILDING 60 IS A HISTORIC FACILITY. PERFORM WORK IN COMPLIANCE WITH THE US DEPARTMENT OF THE INTERIOR "SECRETARY OF THE INTERIOR'S STANDARDS FOR THE TREATMENT OF HISTORIC PROPERTIES." HISTORIC ITEMS INDICATED OR SPECIFIED TO BE SALVAGED FOR REINSTALLATION OR TO REMAIN IN PLACE MUST BE PROTECTED DURING CONSTRUCTION.
19. WHEN ATTACHING FASTENERS, BLOCKING, OR ACCESSORIES TO EXISTING HISTORIC BRICK OR STONE, PLACE FASTENERS IN MORTAR JOINTS. EXCEPTION: LIGHTNING PROTECTION.
20. SYSTEMS AND THEIR ASSOCIATED CONDUIT MUST NOT PENETRATE HISTORIC FEATURES OF THE BUILDING (WINDOW AND DOOR SURROUNDS, STAIRWELLS, WOOD BASE, CRANE RAILS, ETC.). MAXIMIZE THE PLACEMENT OF CONDUIT IN SECONDARY SPACES WHERE PRACTICABLE.
21. PRIOR TO INSTALLING FIRE PROTECTION PIPING, MECHANICAL DUCTWORK, PIPING, ELECTRICAL CONDUIT, AND OTHER ASSOCIATED SYSTEM ELEMENTS, SUBMIT COORDINATION DRAWINGS AND OBTAIN APPROVAL BY THE CONTRACTING OFFICER AND THE PWD ME CULTURAL RESOURCES MANAGER. REFER TO SPECIFICATION SECTION 01 30 10.00 22 COORDINATION PROCEDURES AND COORDINATION DRAWINGS (PWD ME).
22. PROVIDE LIGHTNING PROTECTION SYSTEM TYPICAL. SEE ELECTRICAL SHEETS FOR INFORMATION.
23. ACCESS TO THE INTERIOR OF BUILDING 60 MUST BE VIA EXISTING OPENINGS IN EXTERIOR FACADE. UNDER NO CIRCUMSTANCES WILL NEW TEMPORARY OPENINGS BE ENTERTAINED AND MUST NOT BE INCLUDED AS BASIS OF BID.
24. CONTRACTOR MUST PROVIDE PAVEMENT AND ASSOCIATED SELECT SUB BASE FOR AREAS BEYOND THOSE IDENTIFIED ON THE CONTRACT DOCUMENTS THAT ARE IMPACTED AS A RESULT OF THE CONTRACTORS MEANS AND METHODS AS REQUIRED TO PERFORM THE CONTRACT SCOPE OF WORK.

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

LICENSED ARCHITECT JOSHUA M. RILEY No. 04278 THE STATE OF NEW HAMPSHIRE 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED	FOR COMMANDER NAVFAC
ACTIVITY	
SATISFACTORY TO	DATE
DESIGNER	JMR
DRAWN BY	MJD
CHECKED BY	RSH
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
ABBREVIATIONS, LEGEND, AND GENERAL PROJECT NOTES - VOLUME 2

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 152 OF 282
G-004B
DRAWFORM REVISION: 14 SEP 2023

GENERAL NOTES (SHEETS QD101-QD103)

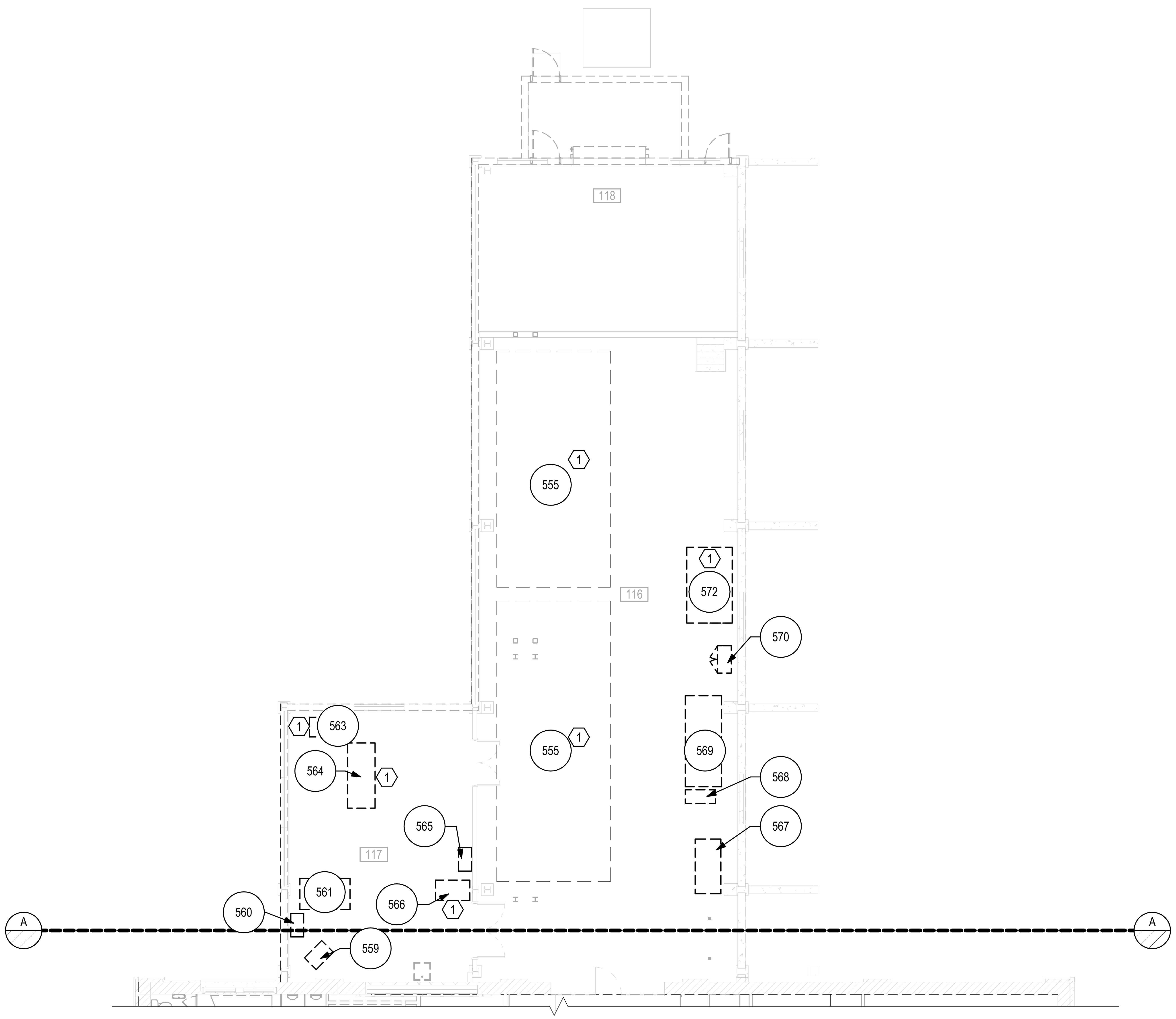
1. SEE QD601 - QD602 FOR EQUIPMENT REMOVALS SCHEDULES.
2. CONTRACTOR MUST VERIFY EXISTING CONDITIONS PRIOR TO PERFORMING WORK.
3. COORDINATE WORK WITH CIVIL, STRUCTURAL, MECHANICAL, PLUMBING AND ELECTRICAL SHEETS.
4. SUBMIT THE SEQUENCING OF REMOVALS AND DISCONNECTION OF UTILITIES. WORK PLAN AND OUTAGES MUST BE APPROVED BY THE CONTRACTING OFFICER PRIOR TO REMOVALS WORK.
5. EQUIPMENT SHOWN ON "QD" SHEETS ARE REPRESENTATIVE OF EQUIPMENT AND ITS CURRENT LOCATION AT THE TIME OF CONTRACT DOCUMENT PREPARATION FOR BIDDING PURPOSES.
6. PROVIDE HAZARDOUS MATERIALS SURVEY AT INDUSTRIAL EQUIPMENT AND SURROUNDING PROCESS AREAS, SEE HAZARDOUS MATERIALS (HM) SHEETS.
7. PATCH ANCHORAGE LOCATIONS AND EQUIPMENT FLOOR DEPRESSIONS WITH NON-SHRINK GROUT ONCE THE EQUIPMENT IS REMOVED. CUT FLUSH TO FLOOR ANY EQUIPMENT ATTACHMENT/ ANCHORAGE AS TO PROVIDE THE REQUIRED FINISHED FLOORING.
8. PROVIDE FOR REMOVAL OF (3)-20 CY DUMPSTERS WORTH OF MISCELLANEOUS MATERIALS/ EQUIPMENT LEFT IN SPACE BEYOND THE EQUIPMENT INDICATED.

REMOVALS KEYNOTES

1 REMOVE UTILITY CONNECTIONS AT INDUSTRIAL EQUIPMENT. SEE ELECTRICAL, MECHANICAL, AND PLUMBING REMOVALS SHEETS.

LEGEND (SHEETS QD101-QD103)

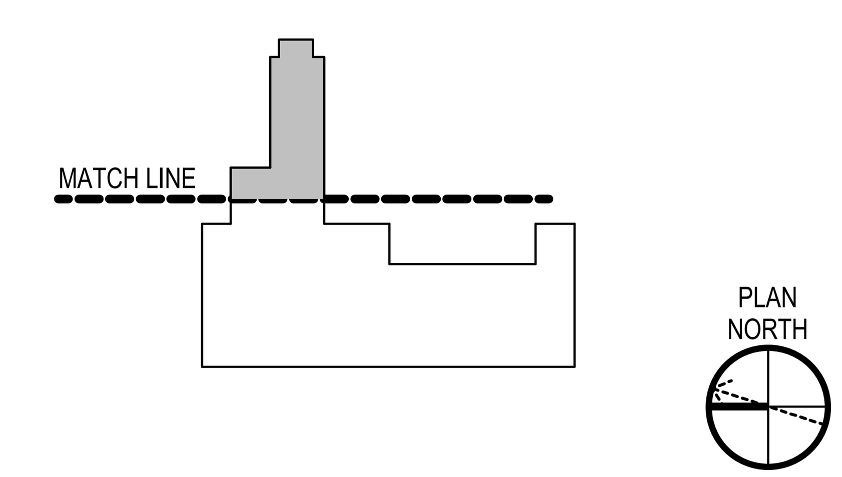
100 EQUIPMENT TAG. SEE EQUIPMENT SCHEDULE ON SHEETS QD601 AND QD602



A1 FIRST FLOOR EQUIPMENT REMOVALS PART PLAN - EAST

SCALE: 1/8" = 1'-0"

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 04215 THE STATE OF NEW HAMPSHIRE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR EQUIPMENT REMOVALS PART PLAN - EAST (CLIN 002)

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 153 OF 282

QD101

DRAWFORM REVISION: 14 SEP 2023

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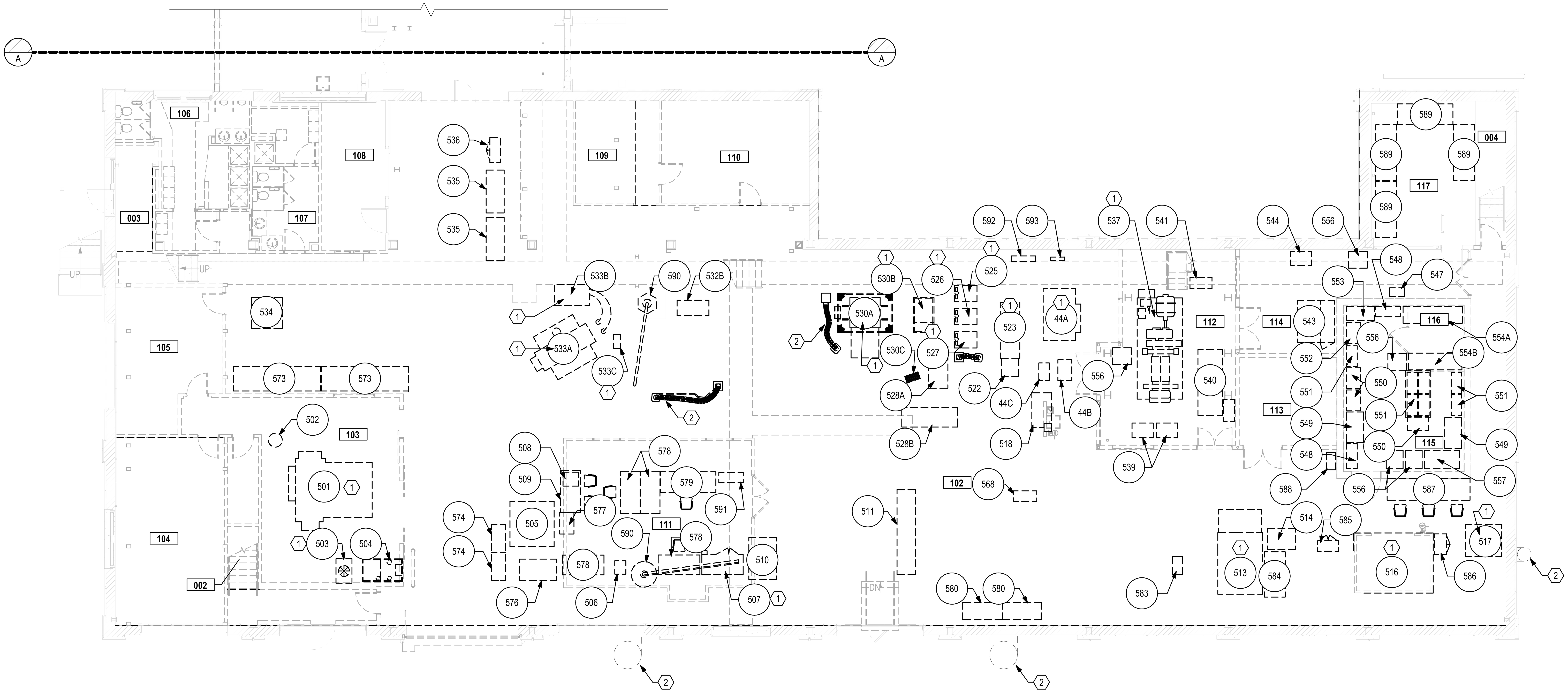


B1 FIRST FLOOR EQUIPMENT REMOVALS PART PLAN - WEST

SCALE: 1/8" = 1'-0"

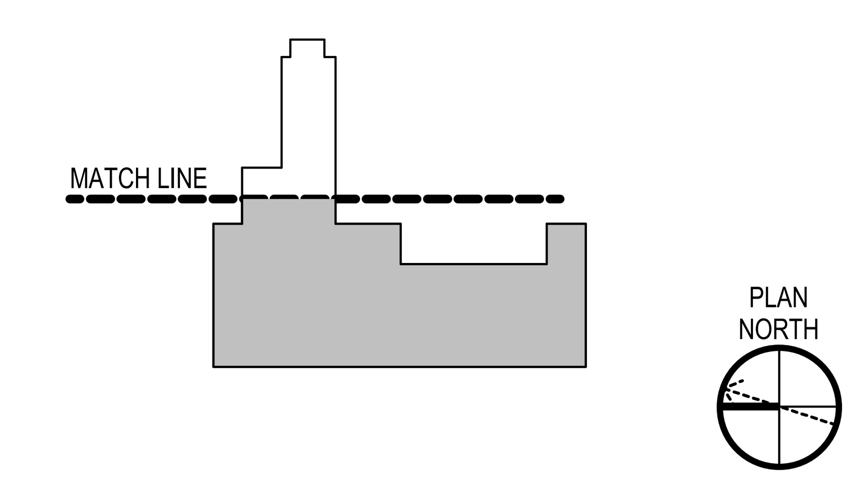
GENERAL NOTE

1. SEE SHEET QD101 FOR GENERAL NOTES AND LEGEND.

REMOVALS KEYNOTES

1. REMOVE UTILITY CONNECTIONS AT INDUSTRIAL EQUIPMENT. SEE ELECTRICAL, MECHANICAL, AND PLUMBING REMOVALS SHEETS.
2. REMOVE DUCTWORK ASSOCIATED WITH INDUSTRIAL EQUIPMENT.

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 04278 THE STATE OF NEW HAMPSHIRE

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME

PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR EQUIPMENT REMOVALS PART PLAN - WEST (CLIN 002)

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 154 OF 282

QD102

DRAWFORM REVISION: 14 SEP 2023

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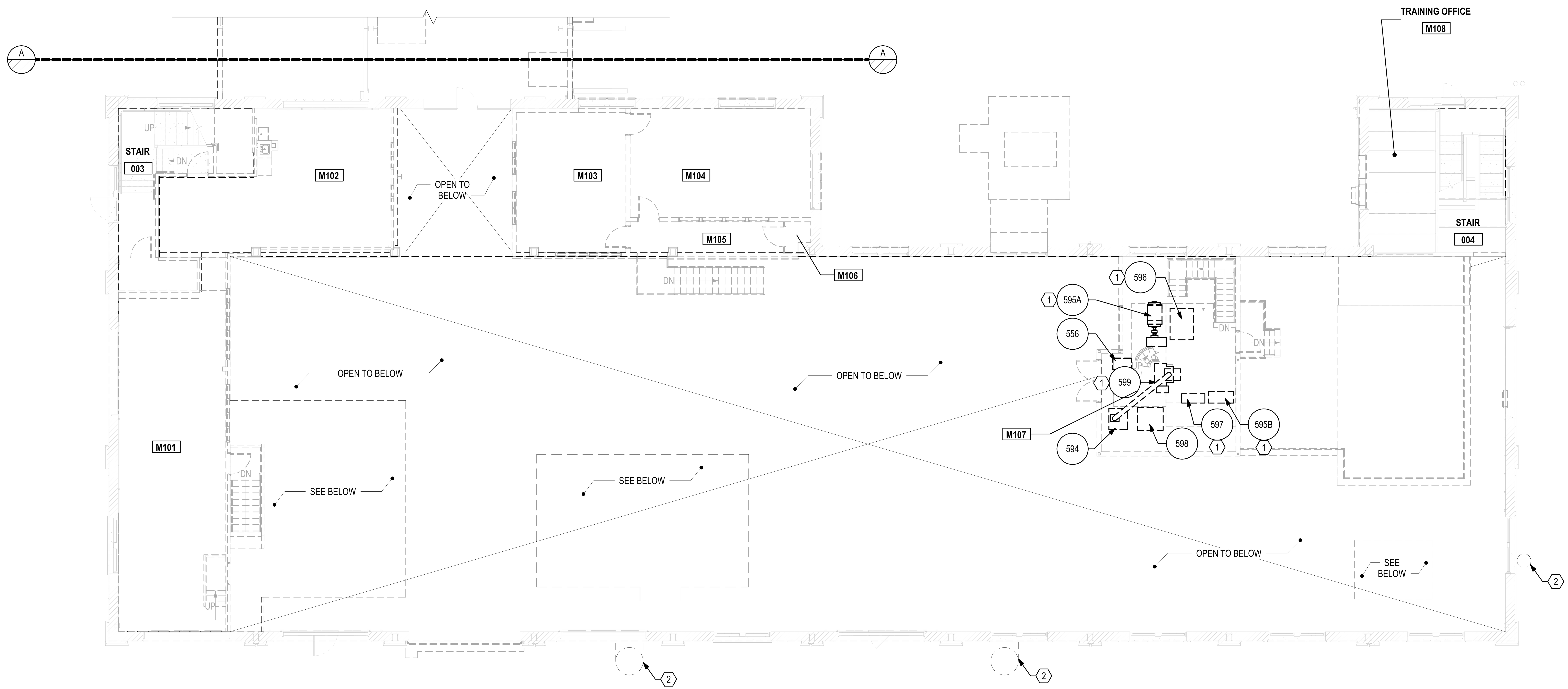
QD102



B1 FIRST FLOOR MEZZANINE EQUIPMENT REMOVALS PLAN

SCALE: 1/8" = 1'-0"

GENERAL NOTE

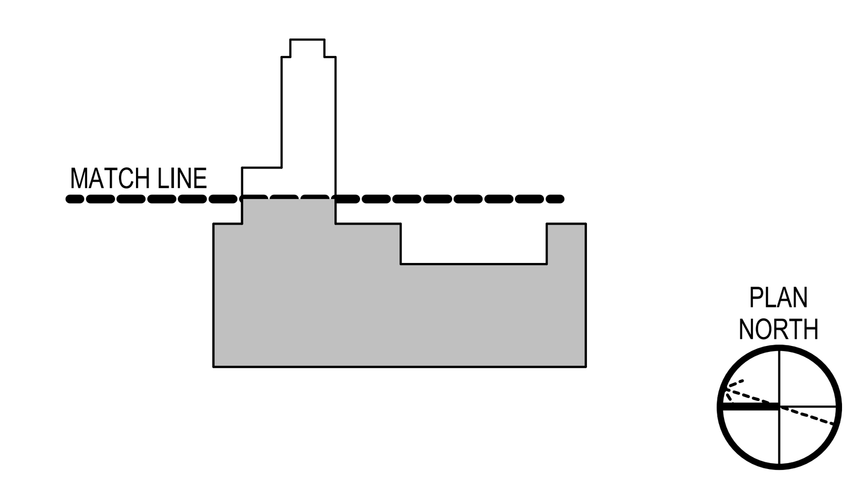
1. SEE SHEET QD101 FOR GENERAL NOTES AND LEGEND.

REMOVALS KEYNOTES

1 REMOVE UTILITY CONNECTIONS AT INDUSTRIAL EQUIPMENT. SEE ELECTRICAL, MECHANICAL, AND PLUMBING REMOVALS SHEETS.

2 REMOVE DUCTWORK ASSOCIATED WITH INDUSTRIAL EQUIPMENT.

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

JOSHUA M. RILEY No. 4275 LICENSED ARCHITECT THE STATE OF NEW HAMPSHIRE SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER JMR
DRAWN BY MJD
CHECKED BY RSH
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD

B60 REPAIRS

FIRST FLOOR MEZZANINE EQUIPMENT REMOVALS PLAN (CLIN 002)

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 155 OF 282
QD103

DRAWFORM REVISION: 14 SEP 2023

UNCLASSIFIED

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BUILDING 60 EQUIPMENT REMOVALS SCHEDULE

EQUIP TAG NO.	ALPHA NO. (SHIPYARD TAG)	QTY	DESCRIPTION	EXISTING LOCATION B60	UTILITIES: VOLTAGE	PHASES	AMPS	HP	FREQ	REMOTE DISCONNECT	SY AIR PSI	GAS	EXHAUST	EXHAUST TYPE	CFM	DUST COLLECTION	WATER	FIRE PROTECTION	DRAIN	STEAM	NRTL	FND	STATUS	DIMENSIONS (INCHES)	WEIGHT (LBS)	CONNECTION/ FASTENER	REMARKS
44A	FRV	1	40" MILL (MAIN UNIT)	102	480	3	80	NA	NA	600	N	N	N	NA	N	NA	N	NA	NA	NA	NA	N	DEMO	66 x 90 x 72	15,000	-	A, B
44B	-	1	40" MILL (CHILLER)	102	460	3	15	NA	NA	30	N	N	N	NA	N	NA	N	NA	NA	NA	NA	N	DEMO	24 x 36 x 36	430	-	A, B
44C	-	1	40" MILL (TEMP CONTROLLER)	102	480	3	40	NA	NA	60	N	N	N	NA	N	NA	N	NA	NA	NA	NA	N	DEMO	18 x 30 x 48	100	-	A, B
501	EYM	1	WATERJET	103	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	144 x 135.6	-	-	A, B
502	-	1	ABRASIVE TANK	103	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	48 x 48	-	-	A, B
503	-	1	COOLING	103	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	42 x 27.48	-	-	A, B
504	-	1	WATER POLISHING SYSTEM	103	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	66.48 x 36	-	-	A, B
505	-	1	FAN PLATFORM	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	78 x 75.96	-	-	A, B
506	-	1	DRILL PRESS	111	120	1	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	23.04 x 18.96 x 66.96	-	-	A, B
507	-	1	OVEN	111	480	3	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	71.04 x 35.28 x 81.48	-	-	A, B
508	-	1	DOWNDRAFT TABLE	111	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	27 x 24 x 56.04	-	-	A, B
509	-	1	BENCH FOR DOWNDRAFT TABLE	111	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	72 x 33.96 x 33.96	-	-	A, B
510	-	1	MOLD RACK	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	72 x 48 x 72	-	-	A, B
511	-	1	WOODEN MOLD RACK	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	146.04 x 32.04 x 68.04	-	-	A, B
513	CWY	1	OVEN, PRECISION QUINCY WITH HOOD	102	460	3	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	146.04 x 77.04	-	-	A, B
514	-	1	HOOD	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	48 x 36	-	-	A, B
516	MXU	1	PLASTISOL BOOTH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	137.04 x 103.56 x 170.04	-	-	A, B
517	ERS	1	RUBBER SHEAR & HYDRAUL. UNIT	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	62.52 x 54.84 x 96	-	-	A, B
518	-	1	BENCH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	60 x 33.96 x 33.96	-	-	A, B
522	-	1	TABLE ON WHEEL WITH VISE	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	32.04 x 30 x 36	-	-	A, B
523	-	1	BENCH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	96 x 33.72 x 38.04	-	-	A, B
525	ATP	1	ELECTRIC PRESS 75 TON	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	39 x 27.96	-	-	A, B
526	ATS	1	ELECTRIC PRESS 75 TON	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	39 x 27.96	-	-	A, B
527	ATN	1	ELECTRIC PRESS 75 TON	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	39 x 27.96	-	-	A, B
528A	-	1	BENCH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	72 x 33.96 x 33.96	-	-	A, B
528B	-	1	BENCH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	96 x 33.72	-	-	A, B
530A	FTB	1	400 TON PRESS (3 COMPONENTS)	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	99.48 x 111.72	-	-	A, B
530B	FSP	1	400 TON PRESS HEATER	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	60 X 36	-	-	A, B
530C	-	1	400 TON PRESS CONTROL PEDESTAL	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	24 X 12	-	-	A, B
532B	-	1	HEATER FOR 500 TON PRESS	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	54.96 x 26.52	-	-	A, B
533A	FSY	1	MILL	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	114 x 84.96	-	-	A, B
533B	-	1	HEATER FOR MILL	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	60.96 x 36	-	-	A, B
533C	-	1	CONTROL PEDESTAL FOR MILL	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	24 x 12	-	-	A, B
534	-	1	CONTAINMENT PALLET	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	53.04 x 53.04 x 12	-	-	A, B
535	-	2	TOOL BOX	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	74.04 x 30.96 x 50.04	-	-	A, B
536	-	1	STORAGE CABINET	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	42.96 x 18 x 65.04	-	-	A, B
537	EKG	1	MILLING MACHINE	112	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	194.52 x 78	-	-	A, B
539	-	2	STORAGE CABINET	112	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36.48 x 25.56	-	-	A, B
540	-	1	BENCH	112	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	114 x 42	-	-	A, B
541	-	1	METAL RACK	112	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	38.04 x 18.96	-	-	A, B
543	-	1	MIXING VENT TABLE	114	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	72 x 63.96	-	-	A, B

EQUIPMENT REMOVALS SCHEDULE GENERAL NOTES (SHEETS QD601-QD602)

1. SEE SHEET G-007 FOR CONSTRUCTION SEQUENCING INFORMATION.
2. SEE SHEETS QD101 THRU QD103 FOR EXISTING EQUIPMENT LOCATIONS. COORDINATE WITH EQUIPMENT SCHEDULE, FOR EXISTING INDUSTRIAL EQUIPMENT TO BE DISCONNECTED AND REMOVED.
3. PATCH ANCHORAGE HOLES IN THE SLAB WITH NON-SHRINK GROUT ONCE THE EQUIPMENT IS REMOVED.
4. SEE CIVIL, STRUCTURAL, ARCHITECTURAL, MECHANICAL, PLUMBING, ELECTRICAL, AND SPRINKLER SHEETS FOR REMOVALS AND EQUIPMENT UTILITY CONNECTIONS.
5. SCHEDULE INFORMATION IS BASED ON AVAILABLE INFORMATION AT THE TIME OF CONTRACT DOCUMENT PREPARATION. COORDINATE WITH CONTRACTING OFFICER FOR ADDITIONAL INFORMATION.
6. PROVIDE HAZARDOUS MATERIALS SURVEY AT INDUSTRIAL EQUIPMENT AND SURROUNDING PROCESS AREAS, SEE HAZARDOUS MATERIALS (HM) SHEETS.

EQUIP REMOVALS SCHEDULE REMARKS NOTES (SHEETS QD601-QD602)

A. REMOVE AND DISPOSE OF INDUSTRIAL EQUIPMENT.

B. REMOVE INDUSTRIAL EQUIPMENT UTILITIES INDICATED. COORDINATE WITH MECHANICAL, PLUMBING, FIRE PROTECTION, AND ELECTRICAL SHEETS.

SCHEDULE KEY (SHEETS QD601-QD602)

B60 BUILDING 60

DEMO CONTRACTOR TO REMOVE AND DISPOSE OF EQUIPMENT

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 04215 THE STATE OF NEW HAMPSHIRE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME

PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

EQUIPMENT REMOVALS SCHEDULE 1 (CLIN 002)

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 156 OF 282

QD601

DRAWFORM REVISION: 14 SEP 2023

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UNCLASSIFIED

3 UNCLASSIFIED

BUILDING 60 EQUIPMENT REMOVALS SCHEDULE (CONTINUED)

EQUIP TAG NO.	ALPHA NO. (SHIPYARD TAG)	QTY	DESCRIPTION	EXISTING LOCATION B60	UTILITIES: VOLTAGE	PHASES	AMPS	HP	FREQ	REMOTE DISCONNECT	SY AIR PSI	GAS	EXHAUST	EXHAUST TYPE	CFM	DUST COLLECTION	WATER	FIRE PROTECTION	DRAIN	STEAM	NRTL	FND	STATUS	DIMENSIONS (INCHES)	WEIGHT (LBS)	CONNECTION/ FASTENER	REMARKS
544	-	1	STORAGE CABINET	114	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36 x 48	-	-	A, B
547	-	1	FILE CABINET	114	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	24.96 x 15 x 72	-	-	A, B
548	-	2	METAL CABINET	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	42.96 x 18	-	-	A, B
549	-	2	STORAGE BOX	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	52.56 x 29.04	-	-	A, B
550	-	3	METAL RACK SHELVES	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36 x 48	-	-	A, B
551	-	7	METAL RACK SHELVES	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36 x 18	-	-	A, B
552	-	1	WIRE RACK SHELVES	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36 x 24	-	-	A, B
553	-	1	WIRE RACK SHELVES	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	48 x 24	-	-	A, B
554A	-	1	WOODEN SHELVES	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	102 x 29.04	-	-	A, B
554B	-	1	WOODEN SHELVES	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	92.04 x 29.04	-	-	A, B
555	AWU, AWG	2	PAINT/VARNISH BOOTH	116	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	153 x 330.96	-	-	A, B
556	-	6	COMPOSITE PALLET	115/112/114/M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	32.04 x 29.04	-	-	A, B
557	-	1	COMPOSITE PALLET	115	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	60 x 32.04	-	-	A, B
559	NLT	1	DRILL PRESS	117	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	32.04 x 21.96 x 80.04	-	-	A, B
560	PWG	1	BENCH	117	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	30.96 x 17.04 x 36	-	-	A, B
561	-	1	BAND SAW	117	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	66 x 41.04 x 110.04	-	-	A, B
563	CMA	1	DUST COLLECTOR	117	460	3	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	26.04 x 23.04 x 99.96	-	-	A, B
564	MYA	1	CABINET SAW	117	460	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	86.04 x 36 x 36.96	-	-	A, B
565	MCS	1	BENCH GRINDER	117	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	30.96 x 17.04 x 51	-	-	A, B
566	FMW	1	PAINT SHAKER	117	115/230	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	45 x 27 x 42	-	-	A, B
567	-	1	BENCH	116	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	72 x 33.96 x 33.96	-	-	A, B
568	-	2	TOOL BOX	106/102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	39.96 x 18 x 60.96	-	-	A, B
569	-	1	BENCH	116	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	120 x 48 x 35.04	-	-	A, B
570	-	1	STORAGE CABINET	116	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36 x 18 x 78	-	-	A, B
572	FNE	1	PAINT BOOTH	116	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	99.96 x 60 x 87	-	-	A, B
573	-	2	SHELVES 6680 CAPACITY	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	150 x 42 x 140.04	-	-	A, B
574	-	2	SHELVES	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	48 x 24 x 84.96	-	-	A, B
576	-	1	SHELVES	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	63.96 x 36.48 x 72	-	-	A, B
577	-	1	SHELVES	111	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	37.56 x 24.48 x 84	-	-	A, B
578	-	4	BENCH	111	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	72 x 33.96 x 33.96	-	-	A, B
579	-	1	BENCH	111	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	96 x 36 x 33.96	-	-	A, B
580	-	2	BENCH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	66 x 30 x 35.04	-	-	A, B
583	-	1	SHELVES	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	30.96 x 17.04 x 36	-	-	A, B
584	-	1	BENCH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	78 x 36 x 36	-	-	A, B
585	-	1	STORAGE CABINET	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36 x 18 x 78	-	-	A, B
586	-	1	STORAGE CABINET	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	36 x 21 x 78	-	-	A, B
587	-	1	WOODEN WORK BENCH	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	143.04 x 36.96 x 39.96	-	-	A, B
588	-	1	STEEL RACK	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	30 x 15.96	-	-	A, B
589	-	4	METAL RACK SHELVES	117	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	96 x 36	-	-	A, B
590	-	2	FREESTANDING JIB CRANE	102/111	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	-	-	-	A, B
591	-	1	TOOL CHEST	111	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	42 x 18 x 57.96	-	-	A, B
592	-	1	CONTROL PANEL	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	42 x 9.96	-	-	A, B
593	-	1	CONTROL PANEL	102	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	24 x 6	-	-	A, B
594	-	1	VIDMAR DUST COLLECTOR	M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	32.04 x 36 x 80.04	-	-	A, B
595A	-	1	RUBBER MIXER	M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	-	-	-	A, B
595B	-	1	RUBBER MIXER CONTROL PANEL	M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	43.8 x 20.04 x 90	-	-	A, B
596	CFE	1	CHILLER DISCONNECT	M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	39.96 x 54 x 63.96	-	-	A, B
597	CFE	1	BANBURY CONTROLS	M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	39.96 x 15.96 x 72	-	-	A, B
598	-	1	DUST COLLECTOR	M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	39.96 x 44.04 x 57.96	-	-	A, B
599	-	1	BANBURY MILL	M107	-	-	-	NA	NA	-	-	-	-	NA	-	NA	-	NA	NA	NA	NA	-	DEMO	20.04 x 27.96	-	-	A, B

GENERAL NOTES

1. SEE SHEET QD601 FOR EQUIPMENT REMOVALS SCHEDULE, GENERAL NOTES, REMARKS NOTES, AND SCHEDULE KEY.

FINAL SUBMISSION | 10/01/2025 | RSH

SYM | DESCRIPTION | DATE | APPR

NAVFAC

LICENSED ARCHITECT
JOSHUA M. RILEY
No. 4215
THE STATE OF NEW HAMPSHIRE
10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER JMR
DRAWN BY MJD
CHECKED BY RSH
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

EQUIPMENT REMOVALS SCHEDULE 2 (CLIN 002)

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 157 OF 282
QD602

DRAWFORM REVISION: 14 SEP 2023

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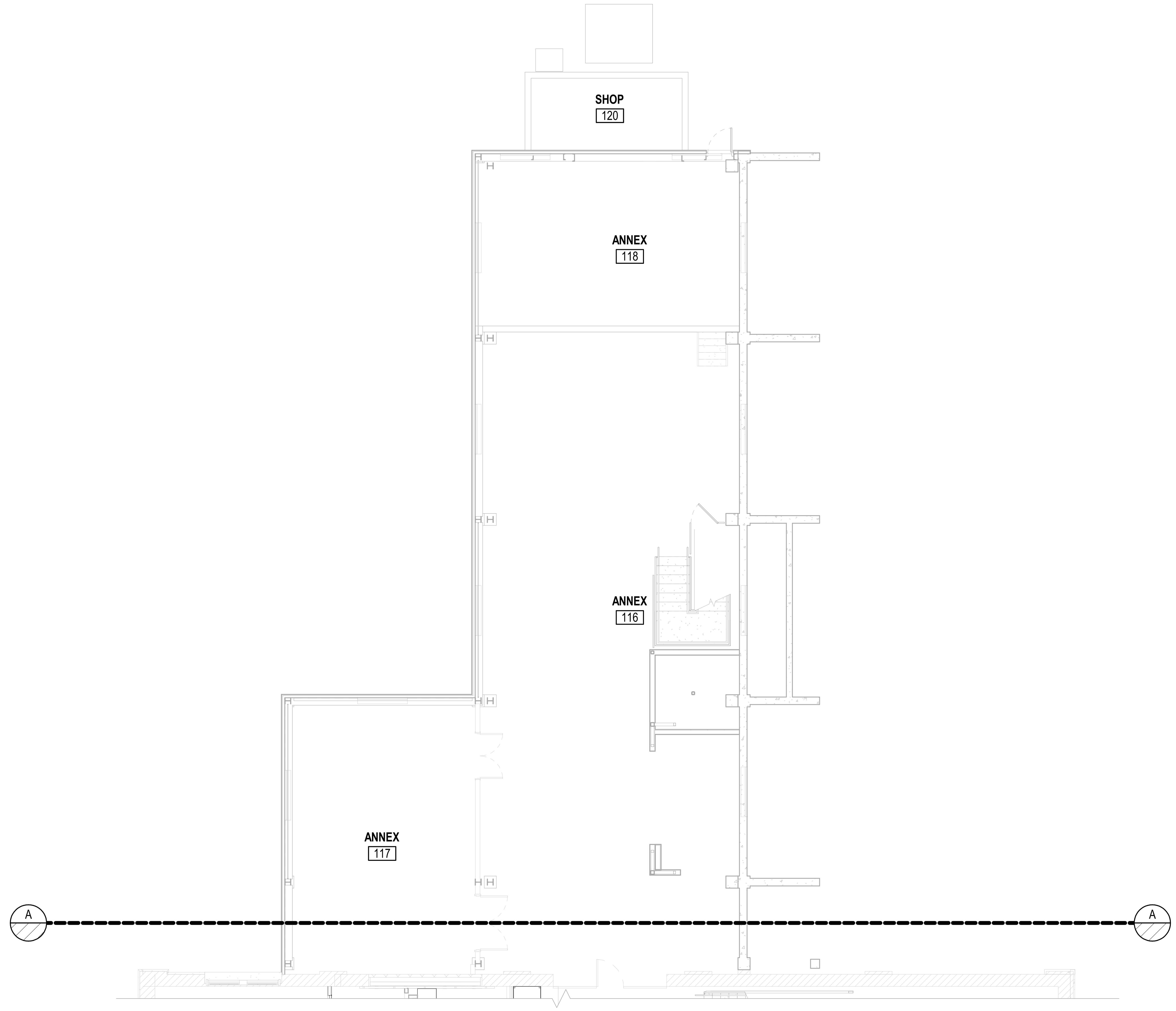


A1 FIRST FLOOR EQUIPMENT PART PLAN - EAST

SCALE: 1/8" = 1'-0"

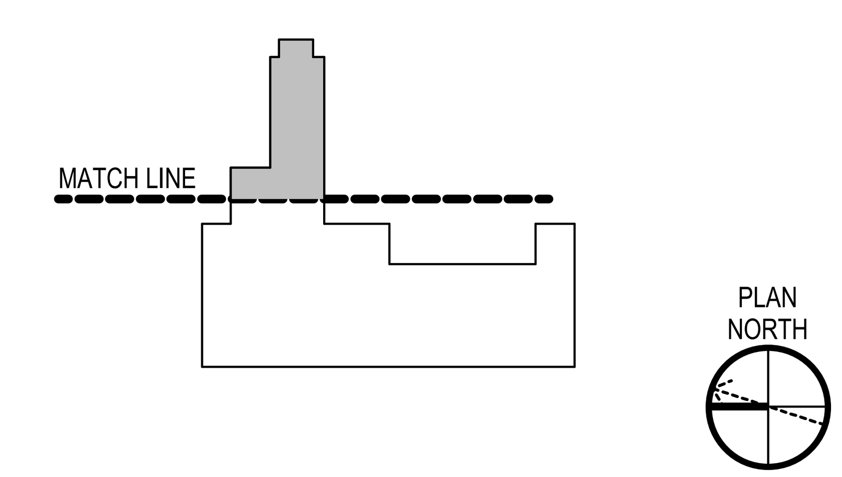
GENERAL NOTES (SHEETS Q-101 - Q-103)

1. PROVIDE UTILITIES FOR AND INSTALLATION OF INDUSTRIAL EQUIPMENT. SEE EQUIPMENT SCHEDULE ON SHEET Q-601.
2. COORDINATE CONSTRUCTION SCHEDULE AND SEQUENCE WITH THE CONTRACTING OFFICER FOR SCHEDULING INDUSTRIAL EQUIPMENT RELOCATIONS. DURATION OF INDUSTRIAL EQUIPMENT MOVES ARE AS FOLLOWS:
   OFF SITE EQUIPMENT TO BUILDING 60: 60 DAYS
3. SEE SHEET G-007 FOR CONSTRUCTION SEQUENCING INFORMATION.
4. COORDINATE LIFTING, HANDLING, AND MOVING INDUSTRIAL EQUIPMENT WITH PHYSICAL CONSTRAINTS, HISTORIC NATURE OF EXISTING BUILDINGS, AND STRUCTURAL LIMITATIONS IDENTIFIED
5. PREPARE AND SUBMIT RELOCATION PLAN. COORDINATE AND VERIFY OPERATIONAL CONDITION OF EXISTING INDUSTRIAL EQUIPMENT WITH CONTRACTING OFFICER OR DESIGNATED GOVERNMENT REPRESENTATIVE PRIOR TO DISCONNECTION AND RELOCATION.
6. PATCH ANCHORAGE HOLES IN THE SLAB WITH NON-SHRINK GROUT ONCE THE EQUIPMENT IS REMOVED FROM ORIGINAL LOCATION.
7. SEE SHEETS AE801-AE804 FOR FF&E PLANS.

LEGEND (SHEETS Q-101 - Q-103)

100 EQUIPMENT TAG. SEE EQUIPMENT SCHEDULE ON SHEET Q-601.

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1'- 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 4275 THE STATE OF NEW HAMPSHIRE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR EQUIPMENT PART PLAN - EAST (OPTION 2)

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 158 OF 282

Q-101

DRAWFORM REVISION: 14 SEP 2023

UNCLASSIFIED

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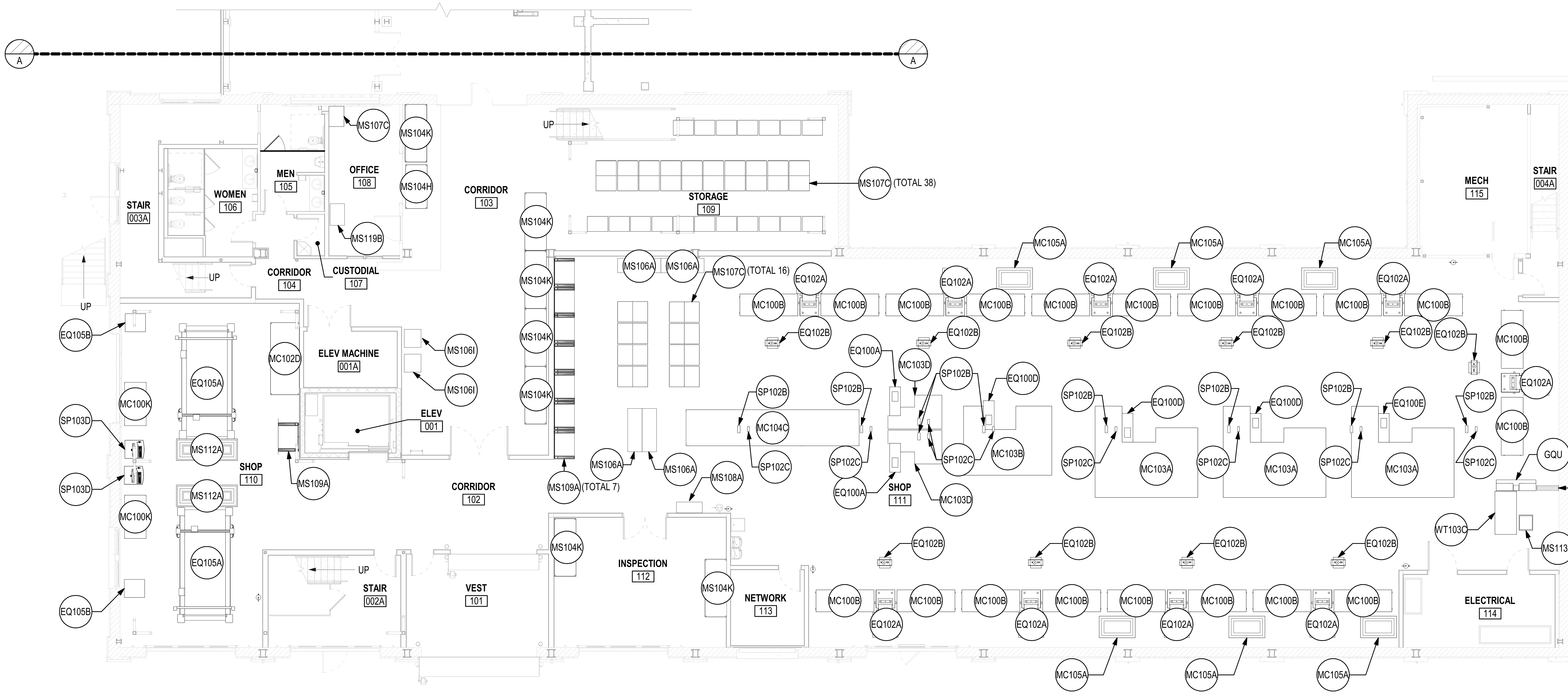
3 UNCLASSIFIED



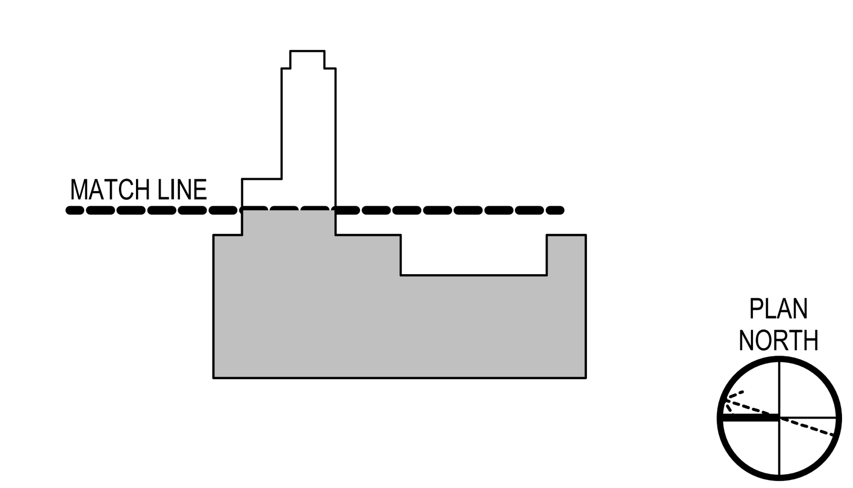
B1 FIRST FLOOR EQUIPMENT PART PLAN - WEST

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE SHEET Q-101 FOR GENERAL NOTES AND LEGEND.

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

JOSHUA M. RILEY No. 4278 LICENSED ARCHITECT THE STATE OF NEW HAMPSHIRE

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER JMR
DRAWN BY MJD
CHECKED BY RSH
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
FIRST FLOOR EQUIPMENT PART PLAN - WEST (OPTION 2)

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 159 OF 282
Q-102
DRAWFORM REVISION: 14 SEP 2023

UNCLASSIFIED

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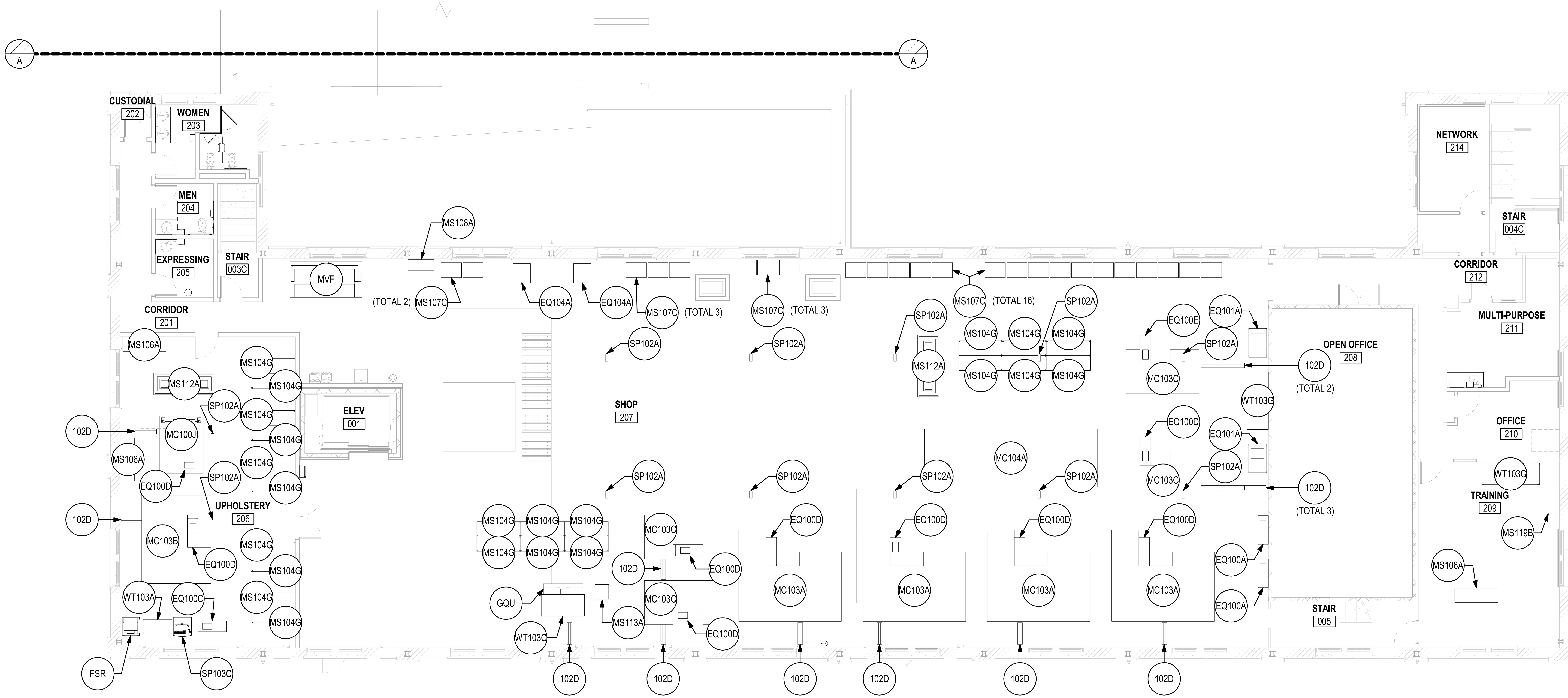
3 UNCLASSIFIED



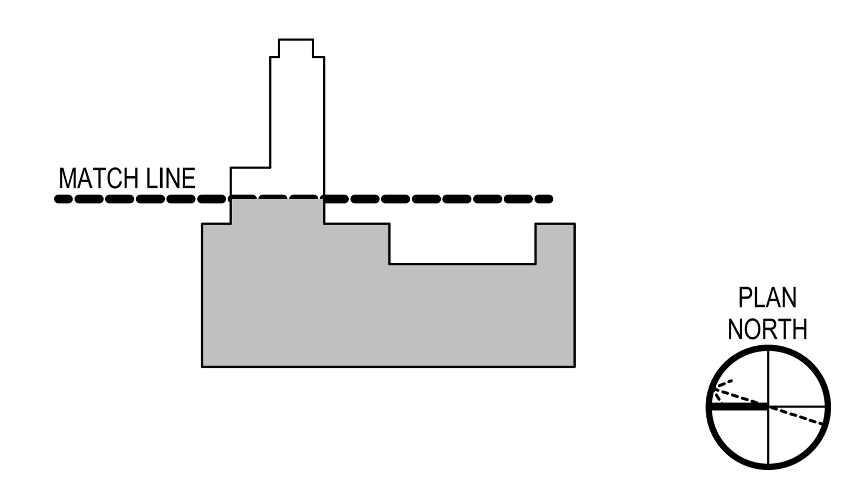
B1 SECOND FLOOR EQUIPMENT PLAN

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE SHEET Q-101 FOR GENERAL NOTES AND LEGEND.

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

LICENSED ARCHITECT JOSHUA M. RILEY No. 04278 THE STATE OF NEW HAMPSHIRE

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

SECOND FLOOR EQUIPMENT PLAN (OPTION 2)

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 160 OF 282

Q-103

DRAWFORM REVISION: 14 SEP 2023

UNCLASSIFIED

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BUILDING 60 EQUIPMENT SCHEDULE

EQUIPMENT TAG NO.	ALPHA NO. (SHIPYARD TAG)	QTY	DESCRIPTION	EXISTING LOCATION	UTILITIES: ELECTRICAL (SEE ELECTRICAL SHEETS)	SY AIR PSI	GAS	EXHAUST	EXHAUST TYPE	CFM	DUST COLLECTION	WATER	FIRE PROTECTION	DRAIN	STEAM	NRTL	FND	STATUS	MANUFACTURER	MODEL NUMBER	REMARKS
102D	-	15	CABLE PROTECTOR RAMP	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	ATLAS	CP9896	F
EQ100A	-	4	MEDIUM WEIGHT SEWING MACHINE	N/A	YES	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	JUKI	LU-1510N-7	A,B,H
EQ100C	-	1	LIGHT WEIGHT SEWING MACHINE	N/A	YES	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	JUKI	DDL 5550N	A,B,H
EQ100D	NDD, ENR, EWY, EXA, EXB, EYH, EYP, EYQ, EYS, FAJ, OR FAS	8	LONG ARM SEWING MACHINE	OFF SITE	YES	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	JUKI	LU-2216N-7	A,B,C
EQ100D	-	4	LONG ARM SEWING MACHINE	N/A	YES	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	JUKI	LU-2216N-7	A,B,H
EQ100E	NDD, ENR, EWY, EXA, EXB, EYH, EYP, EYQ, EYS, FAJ, OR FAS	2	LONG ARM SEWING MACHINE - LG158	OFF SITE	YES	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	JUKI	LG158	A,B,C
EQ101A	FPT, FPS, OR EXC	1	BOX STITCHER	OFF SITE	YES	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	JUKI	AMS-210EN	A,B,C
EQ101A	-	1	BOX STITCHER	N/A	YES	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	JUKI	AMS-210EN	A,B,H
EQ102A	FFF	2	HEAT SEALER	OFF SITE	YES	100	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CONC/PVMT	EXIST	THERMEX THERMATRON	F15-30	A,B,C
EQ102A	-	8	HEAT SEALER	N/A	YES	100	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CONC/PVMT	NEW	THERMEX THERMATRON	F15-30	A,B,G
EQ102B	-	2	RF HEAT SEALER PEDESTAL	OFF SITE	N/A	N/A	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	THERMEX THERMATRON	M23933	A,B,C
EQ102B	-	8	RF HEAT SEALER PEDESTAL	N/A	N/A	N/A	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	THERMEX THERMATRON	M23933	A,B,G
EQ104A	MVY, FHN, EPH, OR MXD	1	GROMMET MACHINE	OFF SITE	YES	80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	LEVEL FLOOR	EXIST	EDWARD SEGAL	92GW	A,B,C
EQ104A	-	1	GROMMET MACHINE	N/A	YES	80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	LEVEL FLOOR	NEW	EDWARD SEGAL	92GW	A,B
EQ105A	FPX	1	CNC FABRIC CUTTER	OFF SITE	YES	100	N/A	N/A	N/A	1500	N/A	N/A	N/A	N/A	N/A	N/A	LEVEL FLOOR	EXIST	GERBER INDUSTRIAL	Z1	A,B,C
EQ105A	-	1	CNC FABRIC CUTTER	N/A	YES	100	N/A	N/A	N/A	1500	N/A	N/A	N/A	N/A	N/A	N/A	LEVEL FLOOR	NEW	GERBER INDUSTRIAL	Z1	A,B,G
EQ105B	-	1	CNC BLOWER	OFF SITE	YES	N/A	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	GERBER INDUSTRIAL	Z1-15HP	A,B,C,E
EQ105B	-	1	CNC BLOWER	N/A	YES	N/A	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GERBER INDUSTRIAL	Z1-15HP	A,B,E,G
FSR	-	1	EMBROIDERY MACHINE	OFF SITE	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	MELCO	EMT16	A,B,C
GQU	ASJ	1	POLY BAG SEALER	OFF SITE	YES	80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	PAC MACHINERY	48PS-KN	A,B,C
GQU	-	1	POLY BAG SEALER	N/A	YES	80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	PAC MACHINERY	48PS-KN	A,B
MC100B	-	20	SHOP BUILT TABLE - 8x3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D
MC100J	-	1	SHOP BUILT TABLE - 8X6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D
MC100K	-	2	SHOP BUILT TABLE - GERBER	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D
MC102D	-	1	SHOP BUILT STORAGE - 10X4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D
MC103A	-	7	SEWING TABLE LARGE	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D,B
MC103B	-	2	SEWING TABLE MEDIUM	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D,B
MC103C	-	4	SEWING TABLE - MEDIUM LONG	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D,B
MC103D	-	2	SEWING TABLE - SMALL	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D,B
MC104A	-	1	LAYOUT TABLE 24'x8'	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D
MC104C	-	1	LAYOUT TABLE - 24'X5'	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D
MC105A	-	6	SHOP BUILT DIE RACK - 5X3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	CUSTOM	CUSTOM	D
MS104G	-	22	HEAVY DUTY RACK - 6'X24"	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GRAINGER	35UR59	A
MS104H	-	1	HEAVY DUTY RACK - 6'X36"	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GRAINGER	35UR60	A
MS104K	-	7	HEAVY DUTY RACK - 8'X36"	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GRAINGER	35UT09	A
MS106A	-	7	CABINET - SHELVES - 6'X24"X78"	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GLOBAL INDUSTRIES	WBB495262	A
MS106I	-	2	CABINET - DRAWERS 30"X28"X59"	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GLOBAL INDUSTRIES	WBB527756	A
MS107C	-	69	TOOL BOX - ROLLING BASE - 34"	OFF SITE	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	GRAINGER	48VA11	A,C
MS107C	-	10	TOOL BOX - ROLLING BASE - 34"	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GRAINGER	48VA11	A
MS108A	-	2	HAZARDOUS MATERIAL CABINET - SMALL	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GRAINGER	55EA93	A
MS109A	-	6	CANTELEVER RACK	OFF SITE	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	GLOBAL INDUSTRIES	WB248383L	A,C
MS109A	-	2	CANTELEVER RACK	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GLOBAL INDUSTRIES	WB248383L	A
MS112A	-	4	CLOTH UNWINDING TRUCK - 102 INCHES	OFF SITE	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	Southwest Sewing Machine LLC	125G	A,C
MS113A	-	2	PLASTIC WASTE CONTAINER	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	MCMASTER CARR	4052T11	A
MS119B	-	2	SHELVING - CLOSED - 36"x24"	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	MCMASTER CARR	4829T325	A
MVF	-	1	ROLL SLITTER	OFF SITE	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	EXIST	COLLINS CRAFT	76-24AR	A,B,C
SP102A	-	12	EXTENSION CORD REEL (25' LENGTH)	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	KH INDUSTRIES	RTBB3LB-1DD520-J12F	A,B
SP102B	-	9	EXTENSION CORD REEL (50' LENGTH)	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	KH INDUSTRIES	RTBB3LB-1GB520-J12K	A,B
SP102C	-	9	COMPRESSED AIR REEL (50' LENGTH)	N/A	N/A	65-75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	REELCRAFT	RS7650 OPL	A,B
SP103C	-	1	COMPUTER - STANDALONE	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	SHIPYARD	N/A	N/A	B,D
SP103D	-	2	CNC - CONTROLLER	N/A	YES	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GERBER INDUSTRIAL	N/A	A,B
WT103A	-	1	WORKBENCH - HARDWOOD TOP - 4'	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GRAINGER	35UX04	A
WT103C	-	2	WORKBENCH - HARDWOOD TOP - 6'	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GRAINGER	35UX07	A
WT103G	-	2	WORKBENCH - HARDWOOD TOP - 8'	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	NEW	GLOBAL INDUSTRIAL	WB183170	A

EQUIPMENT SCHEDULE GENERAL NOTES

1. SEE SHEET G-007 FOR CONSTRUCTION SEQUENCING INFORMATION.
2. SEE SHEETS QD101 THRU QD602 FOR EXISTING EQUIPMENT TO BE DISCONNECTED AND REMOVED.
3. PATCH ANCHORAGE HOLES IN THE SLAB WITH NON-SHRINK GROUT ONCE THE EQUIPMENT IS REMOVED.
4. COORDINATE EQUIPMENT LAYOUT WITH MECHANICAL, PLUMBING, FIRE PROTECTION, AND ELECTRICAL PLANS AND EQUIPMENT SCHEDULE.
5. EQUIPMENT LOCATIONS ARE FOR PLANNING AND COORDINATION PURPOSES. THE CONTRACTOR IS RESPONSIBLE FOR INSTALLING THE INDUSTRIAL EQUIPMENT AS INDICATED AND IS RESPONSIBLE FOR COORDINATING THE RELOCATION OF EXISTING EQUIPMENT BY THE GOVERNMENT. COORDINATE SEQUENCING AND CONNECTION OF UTILITIES TO THE EQUIPMENT WITH THE CONTRACTING OFFICER.
6. COORDINATE TESTING OF EXISTING EQUIPMENT TO BE RELOCATED TO CONFIRM WORKING CONDITION BEFORE AND AFTER RELOCATION. TESTS MUST BE WITNESSED BY SHOP REPRESENTATIVE AND COORDINATED WITH THE CONTRACTING OFFICER.
6. SEE CIVIL, STRUCTURAL, ARCHITECTURAL, MECHANICAL, PLUMBING, ELECTRICAL, AND SPRINKLER SHEETS FOR EQUIPMENT UTILITY CONNECTIONS.
7. PROCURE AND INSTALL EQUIPMENT STATUS MARKED "NEW" (EQUIPMENT TO BE PURCHASED BY THE CONTRACTOR).
8. INSTALL EQUIPMENT STATUS MARKED "EXIST" (EQUIPMENT TO BE RELOCATED BY THE GOVERNMENT).
9. COORDINATE AND PERFORM NEW EQUIPMENT MANUFACTURER'S STANDARD TESTING PROCEDURES.
10. SCHEDULE INFORMATION IS BASED ON AVAILABLE INFORMATION AT THE TIME OF CONTRACT DOCUMENT PREPARATION. COORDINATE WITH CONTRACTING OFFICER FOR ADDITIONAL INFORMATION.

EQUIP SCHEDULE REMARKS NOTES

A. INSTALL EQUIPMENT AT FINAL LOCATION IN BUILDING 60.

B. PROVIDE INDUSTRIAL EQUIPMENT UTILITIES INDICATED. COORDINATE WITH MECHANICAL, PLUMBING, FIRE PROTECTION, AND ELECTRICAL SHEETS.

C. COORDINATE THE EQUIPMENT DISCONNECTION AND RELOCATION BY THE GOVERNMENT.

D. COORDINATE EQUIPMENT INSTALLATION BY GOVERNMENT.

E. EXHAUST TO SPACE. PROVIDE MANUFACTURER'S DUCT SILENCER AND COORDINATE WITH MECHANICAL SHEETS.

F. TURN EQUIPMENT OVER TO THE GOVERNMENT FOR STORAGE. COORDINATE STORAGE LOCATION WITHIN BUILDING 60 WITH THE CONTRACTING OFFICER.

G. THIS ITEM REQUIRES SOLE SOURCING AND IS NOT SUBJECT TO FULL AND OPEN COMPETITION. THIS ITEM IS NOT COMMERCIALLY AVAILABLE "OFF THE SHELF", AND REQUIRES CUSTOMIZED MODIFICATION. SEE ATTACHED SPECIFICATION AND/OR MANUFACTURER'S QUOTE INDICATING EQUIPMENT OPTIONS AT THE END OF THIS PACKAGE.

H. THIS ITEM REQUIRES SOLE SOURCING AND IS NOT SUBJECT TO FULL AND OPEN COMPETITION. PROVIDE SPECIFIC MAKE, MODEL, AND CONFIGURATION INDICATED,

SCHEDULE KEY

B60	BUILDING 60
NEW	CONTRACTOR PURCHASED AND INSTALLED EQUIPMENT
EXIST	GOVERNMENT RELOCATED EQUIPMENT, CONTRACTOR INSTALLED
SHIPYARD	SHIPYARD PROVIDED AND INSTALLED EQUIPMENT

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

JOSHUA M. RILEY No. 04275 LICENSED ARCHITECT THE STATE OF NEW HAMPSHIRE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER JMR

DRAWN BY MJD

CHECKED BY RSH

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

EQUIPMENT SCHEDULE (OPTION 2)

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 161 OF 282

Q-601

DRAWFORM REVISION: 14 SEP 2023

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PLUMBING SYMBOLS

ANNOTATIONS

- P — SYMBOL PER ABBREVIATION LIST
- 2 — EQUIPMENT SEQUENCE NUMBER
- 1 — KEYNOTE
- 1 — REMOVALS KEYNOTE
- CONNECT TO EXISTING
- POINT OF ELEVATION

FIXTURES

- WALL HUNG URINAL WITH FLUSH VALVE
- COUNTERTOP LAVATORY
- WALL HUNG LAVATORY
- FLOOR MOUNTED WATER CLOSET WITH FLUSH VALVE
- MOP RECEPTOR
- WALL MOUNTED, DUAL HEIGHT ELECTRIC WATER COOLER

PIPING AND VALVES

- PIPE ELBOW DOWN
- PIPE ELBOW UP, PIPE UP AND DOWN
- PIPE TEE DOWN
- PIPE TEE UP, PIPE UP AND DOWN
- PIPE CAP
- DIRECTION OF FLOW
- STRAINER
- BALL VALVE
- BUTTERFLY VALVE
- UNION
- GATE VALVE
- CHECK VALVE
- CALIBRATED BALANCING VALVE WITH POSITIVE SHUTOFF
- AUTOMATIC FLOW CONTROL VALVE
- P — PIPE PITCH DOWN
- PRESSURE RELIEF VALVE
- PRESSURE REDUCING VALVE
- PIPE REDUCER/INCREASER
- CONCENTRIC REDUCER
- ECCENTRIC REDUCER
- PRESSURE GAUGE AND COCK
- CO — CLEANOUT
- WCO — WALL CLEANOUT
- FCO — FLOOR CLEANOUT
- P-TRAP

EQUIPMENT AND SPECIALTIES

- TS — TEMPERATURE SENSOR
- PS — PRESSURE SWITCH
- T — INSERTION TEMPERATURE SENSOR (AQUASTAT)
- FLOOR DRAIN
- THERMOMETER
- WATER HAMMER ARRESTOR
- WATER OR GAS METER
- REDUCED PRESSURE ZONE BACKFLOW PREVENTER
- AIR VENT, AUTOMATIC
- AIR VENT, MANUAL
- VACUUM RELIEF VALVE
- AIR GAP FITTING
- HOSE BIBB OR HYDRANT
- IN-LINE PUMP
- PUMP
- TRAP PRIMER ASSEMBLY
- THERMOSTATIC MIXING VALVE

PLUMBING ABBREVIATIONS

ADA	AMERICANS WITH DISABILITIES ACT
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
BFP	BACKFLOW PREVENTER
BP	PACKAGED BOOSTER PUMP
CA	COMPRESSED AIR
CD	CONDENSATE DRAIN
℄	CENTERLINE
CLG	CEILING
CO	CLEANOUT
CONN	CONNECTION
CONC	CONCRETE
COND	CONDENSATE
CW	COLD WATER
ø, DIA	DIAMETER
DN	DOWN
DP	DROP
DWG	DRAWING
DWH	DUAL WALL HYDRANT
ELEC	ELECTRIC/ELECTRICAL
ELEV	ELEVATION, ELEVATOR
EW	EYEWASH
EWC	ELECTRIC WATER COOLER
EWH	ELECTRIC WATER HEATER
EXIST	EXISTING
EXT	EXPANSION TANK
°F	DEGREES FAHRENHEIT
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
FF	FINISHED FLOOR
FLR	FLOOR
FT	FOOT, FEET
G	GAS
GAL	GALLONS
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
H2O	WATER
HB	HOSE BIBB
HP	HORSEPOWER
HVAC	HEATING, VENTILATION AND AIR CONDITIONING
HW	HOT WATER
ID	INSIDE DIAMETER
IN	INCH, INCHES
INV	INVERT
IW	INDIRECT WASTE
JAN	JANITOR (SINK)
KW	KILOWATT, KITCHEN WASTE
L	LAVATORY
LF	LINEAR FEET
LP, LPG	LIQUIFIED PETROLEUM GAS
M	MOTOR
MAX	MAXIMUM
MECH	MECHANICAL
MIN	MINIMUM
MR	MOP RECEPTOR
MTG	MOUNTING
MTL	METAL
N/A	NOT APPLICABLE
NC	NORMALLY CLOSED
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NFWH	NON-FREEZE WALL HYDRANT
NO	NUMBER, NORMALLY OPEN
NPT	NATIONAL PIPE THREAD
NTS	NOT TO SCALE
OC	ON CENTER
OD	OUTSIDE DIAMETER
OE	OPEN ENDED
P	PUMP, PITCH, PRESSURE
PC	PUMPED CONDENSATE
PDI	PLUMBING DRAINAGE INSTITUTE
PEX	CROSS-LINKED POLYETHYLENE
PRESS	PRESSURE
PRV	PRESSURE REDUCING VALVE
PSI	POUNDS PER SQUARE INCH
PSIG	POUNDS PER SQUARE INCH GAUGE
QTY	QUANTITY
R	RADIUS
RD	ROOF DRAIN
REQ'D	REQUIRED
RM	ROOM
RPM	REVOLUTIONS PER MINUTE
RPZ	REDUCED PRESSURE ZONE
S	SINK, SANITARY
SAN	SANITARY
SF	SQUARE FOOT/FEET
SIM	SIMILAR
SP	SUMP PUMP, SPRINKLER
SQ	SQUARE
SS	STAINLESS STEEL
ST	STORM DRAIN
STL	STEEL
TEMP	TEMPERATURE
TM	DOMESTIC HOT WATER TEMPERATURE MAINTENANCE ZONE
TMV	TEMPERATURE MIXING VALVE
TP	TRAP PRIMER ASSEMBLY
TYP	TYPICAL
U	URINAL
UL	UNDERWRITERS LABORATORY
V	VENT
VTR	VENT TO ROOF
W	WASTE
W&T	WASTE AND TRAP
W/	WITH
WC	WATER COLUMN
WCO	WALL CLEANOUT
WH	WALL HYDRANT, WATER HEATER
WHA	WATER HAMMER ARRESTOR
WPD	WATER PRESSURE DROP

GENERAL PLUMBING SYSTEM NOTES

1. PLUMBING SHALL BE DONE IN ACCORDANCE WITH THE 2021 INTERNATIONAL PLUMBING CODE(IPE) AND THE INTERNATIONAL ENERGY AND CONSERVATION CODE (IECC) - CURRENT EDITION.
2. ROOF DRAIN AND RAIN WATER CONDUCTOR SIZES INDICATED ON PLANS ARE BASED UPON A MAXIMUM RAINFALL OF 2.4 INCHES PER HOUR AND A 2% PITCH DOWN IN THE DIRECTION OF FLOW UNLESS OTHERWISE INDICATED ON PLANS.
3. INSTALL SANITARY DRAINAGE WITH A PITCH OF 1/4 INCH PER FOOT FOR BUILDING SANITARY PIPING 3 INCHES AND SMALLER AND A PITCH OF 1/8 INCH PER FOOT FOR BUILDING SANITARY PIPING 4 INCHES AND LARGER.
4. FOR PIPE SIZES NOT SHOWN ON PLANS, SEE APPROPRIATE PART PLANS AND RISER DIAGRAMS.
5. PIPING IS SHOWN DIAGRAMMATICALLY, EXACT LOCATION SHALL BE DETERMINED IN THE FIELD.
6. PIPING SHALL BE INSTALLED CONCEALED ABOVE CEILINGS, IN WALLS AND IN CHASES, UNLESS OTHERWISE NOTED. PIPING SHALL BE INSTALLED PARALLEL TO BUILDING LINES.
7. PROVIDE DRAINS AT DOMESTIC WATER PIPING LOW POINTS AND PITCH PIPING TO LOW POINTS DRAINS.
8. PIPING SHALL BE SUPPORTED FROM BUILDING STRUCTURE. PIPING SHALL BE SUPPORTED FROM TOP CHORD OF JOISTS. NO STRUCTURAL MEMBERS SHALL BE CUT.
9. INSTALL WATER PIPE ON THE WARM SIDE OF BUILDING INSULATION IN EXTERIOR WALLS. SEE ARCHITECTURAL WALL SECTIONS.
10. PROVIDE WATER HAMMER ARRESTORS (WHA) AS SHOWN ON PLANS TO AVOID WATER HAMMER. SIZES INDICATED ON PLANS CONFORM TO THE PLUMBING AND DRAINAGE INSTITUTE (PDI) STANDARDS.
11. PROVIDE ACCESSIBLE CLEANOUTS AT THE BASE OF STACKS, AT HORIZONTAL CHANGES OF DIRECTION GREATER THAN 45° AND WHERE SHOWN ON DRAWINGS.
12. PIPING DROPS TO FIXTURES SHALL BE ANCHORED SOLID TO WALL WITH STEEL SUPPORT BRACKET WITH ADJUSTABLE CLIP.
13. PROVIDE STEEL PIPE SLEEVES AT SLAB PENETRATIONS BOTH ON GRADE AND ABOVE GRADE, AND AT MASONRY WALL AND PARTITION PENETRATIONS. OPENINGS SHALL BE SEALED SMOKE AND WATER TIGHT. SEE PIPE PENETRATIONS DETAILS ON PLUMBING DETAIL SHEETS.
14. HOSE END VALVES SHALL BE PROVIDED WITH HOSE END VACUUM BREAKERS.
15. COORDINATE LOCATION OF PIPING WITH WORK OF OTHER TRADES. PERFORM CUTTING WORK ASSOCIATED WITH PLUMBING SYSTEMS.
16. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

PLUMBING LINE TYPE LEGEND

Line label	Description
	REMOVE ITEMS
	EXISTING ITEMS TO REMAIN
	PROVIDE ITEMS
W	WASTE PIPE
SAN	SANITARY SEWER ABOVE FLOOR
SAN	SANITARY SEWER BELOW FLOOR OR GRADE
CD	CONDENSATE DRAIN ABOVE FLOOR OR GRADE
	VENT
TP	TRAP PRIMER LINE
	DOMESTIC COLD WATER
	DOMESTIC HOT WATER (110°F)
	DOMESTIC HOT WATER RETURN
PW	PUMPED WASTE
	CONTROL WIRING
D	DRAIN LINE
CA	COMPRESSED AIR

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25 SEAL

OAK POINT ASSOCIATES

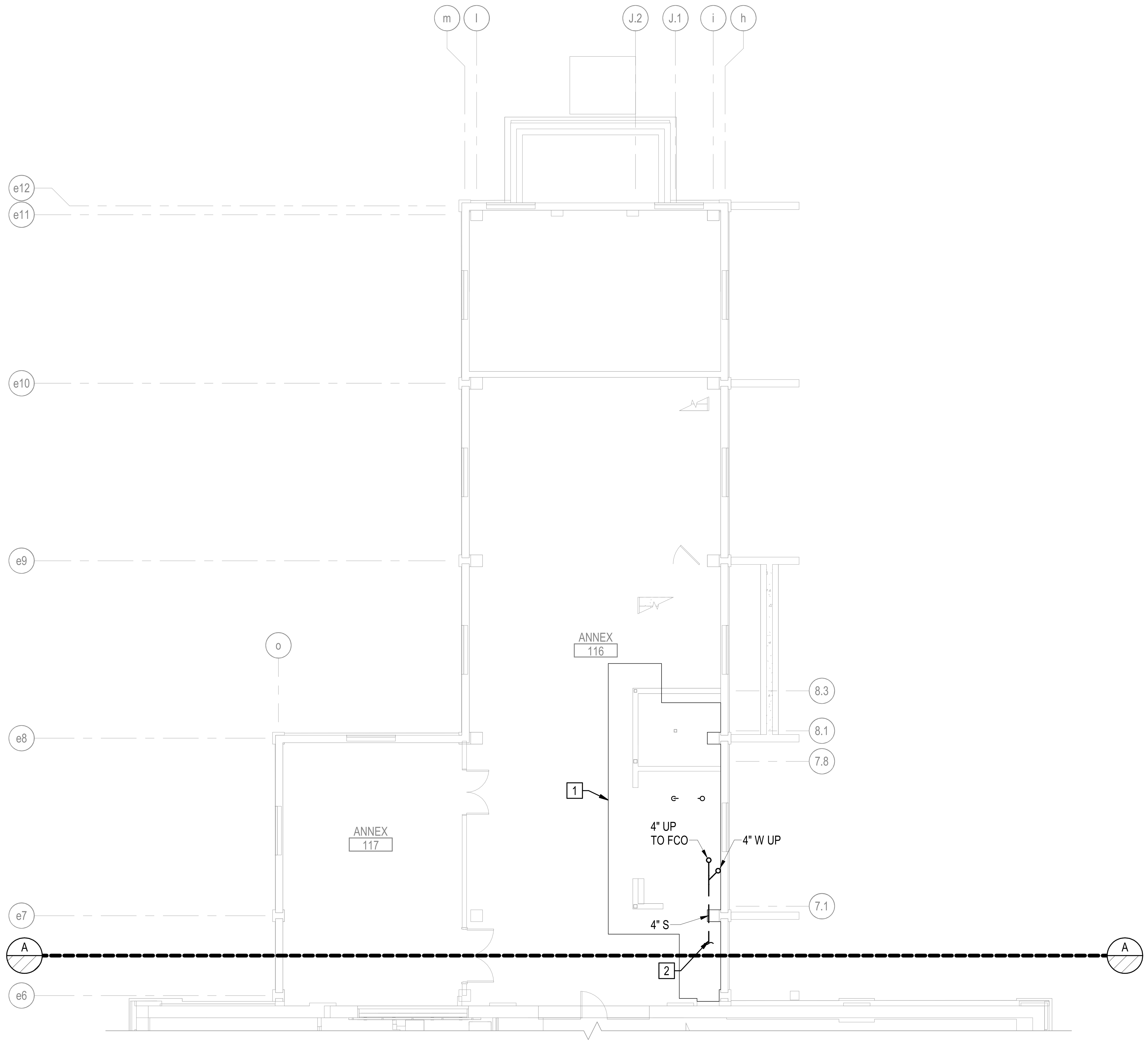
APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CMA
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 162 OF 282
P-001



A1 UNDERSLAB PLUMBING PLAN - EAST
SCALE: 1/8" = 1'-0"

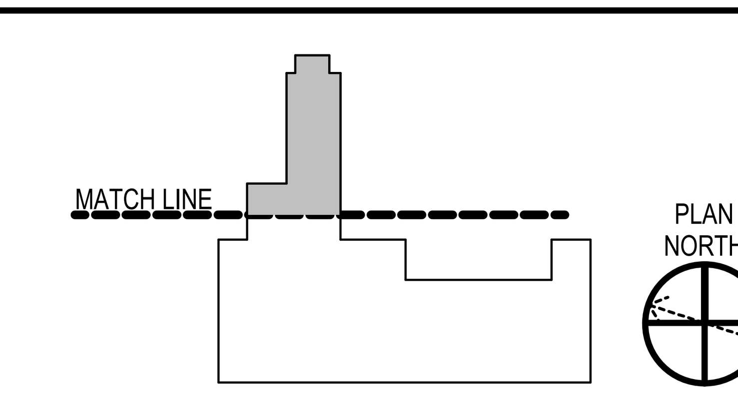
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1. FLOOR SAW CUT INFILL. SEE STRUCTURAL SHEET SB101 FOR INFILL DETAILS.
2. REFER TO SHEET P-102 FOR CONTINUATION.

KEY PLAN



GRAPHIC SCALE

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CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE
RICHARD CRESWELL
No. 13178
LICENSED
PROFESSIONAL ENGINEER
10-01-25
SEAL

OAK POINT ASSOCIATES

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DEPARTMENT OF THE NAVY
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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
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PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
UNDERSLAB PLUMBING PLAN - EAST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 169 OF 282
P-101

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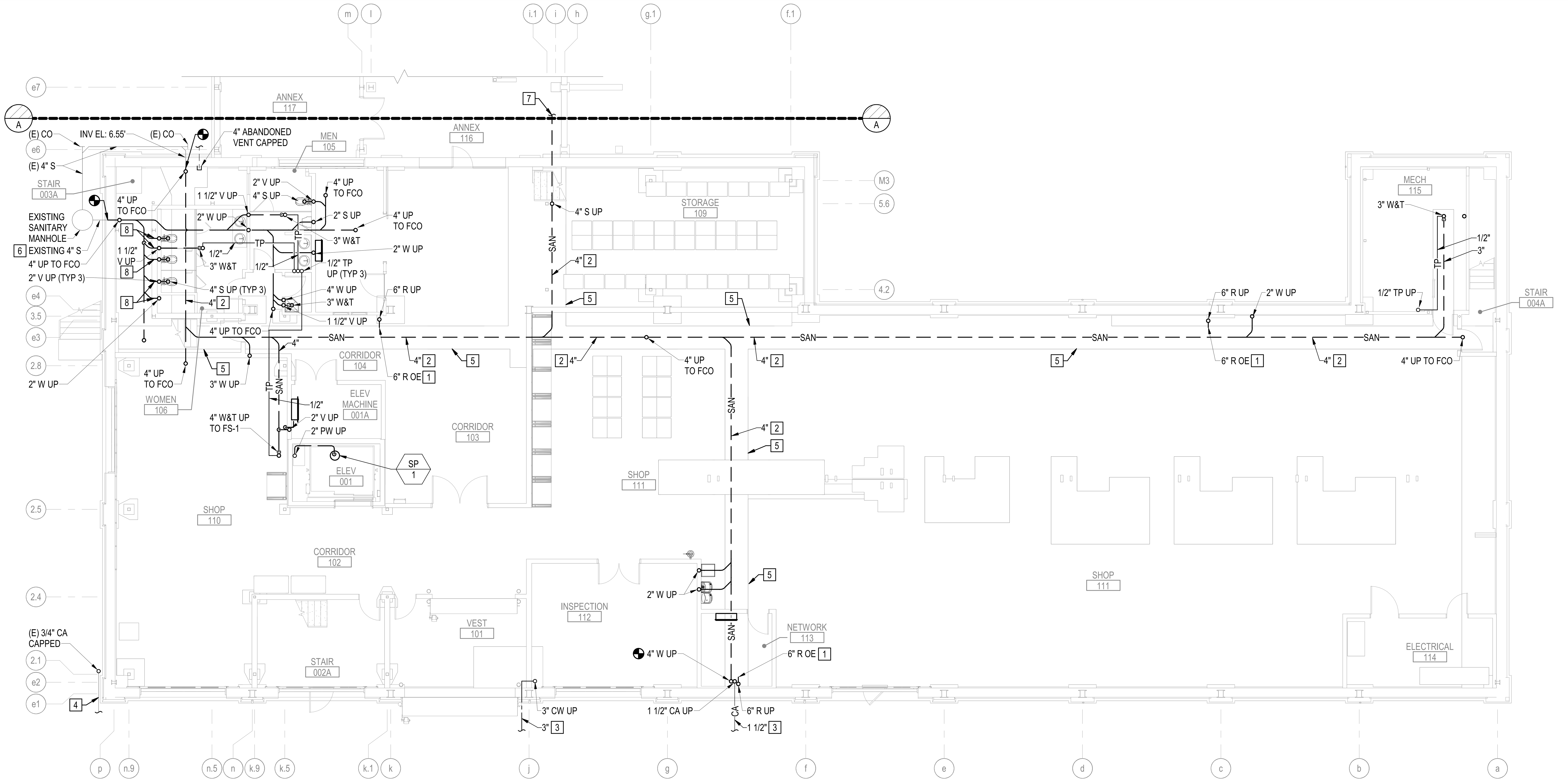


B1 UNDERSLAB PLUMBING PLAN - WEST

SCALE: 1/8" = 1'-0"

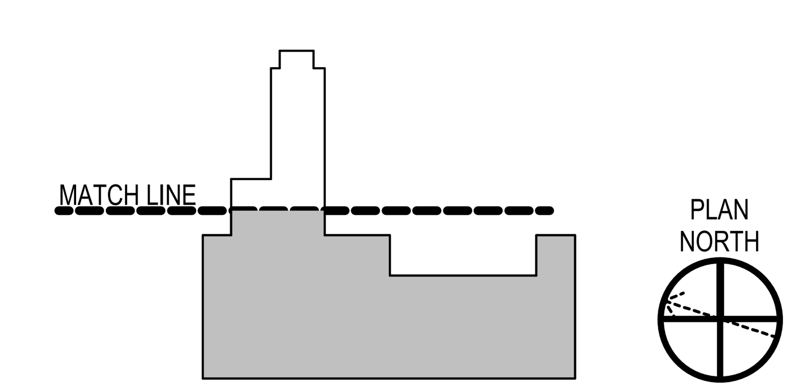
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1. PROVIDE 6" RADON VENT PIPE WITH OPEN END (OE) PIPE FROM ELBOW INTO CRUSHED STONE LAYER OF FILLED FLOOR TRENCH.
2. PITCH PIPE AT 1/8" PER FOOT.
3. SEE CIVIL SHEET CX101 FOR CONTINUATION OF EXISTING UNDER GROUND COMPRESSED AIR PIPING AND DOMESTIC WATER PIPING.
4. EXISTING 3/4" COMPRESSED AIR FROM BUILDING 60 TO BUILDING 292, CUT, CAPPED, AND ABANDONED IN PLACE.
5. EXISTING FLOOR TRENCH TO BE FILLED AND SLAB PATCHED.
6. SEE CIVIL SHEET CX101 FOR CONTINUATION OF EXISTING SANITARY PIPING.
7. SEE SHEET P-101 FOR CONTINUATION.
8. COORDINATE PIPING THROUGH WALL WITH EXISTING STRUCTURAL PIERS. PROVIDE OFFSET AND FITTINGS AS REQUIRED.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
UNDERSLAB PLUMBING PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 170 OF 282
P-102

UNCLASSIFIED

DRAWFORM REVISION: 14 SEP 2023

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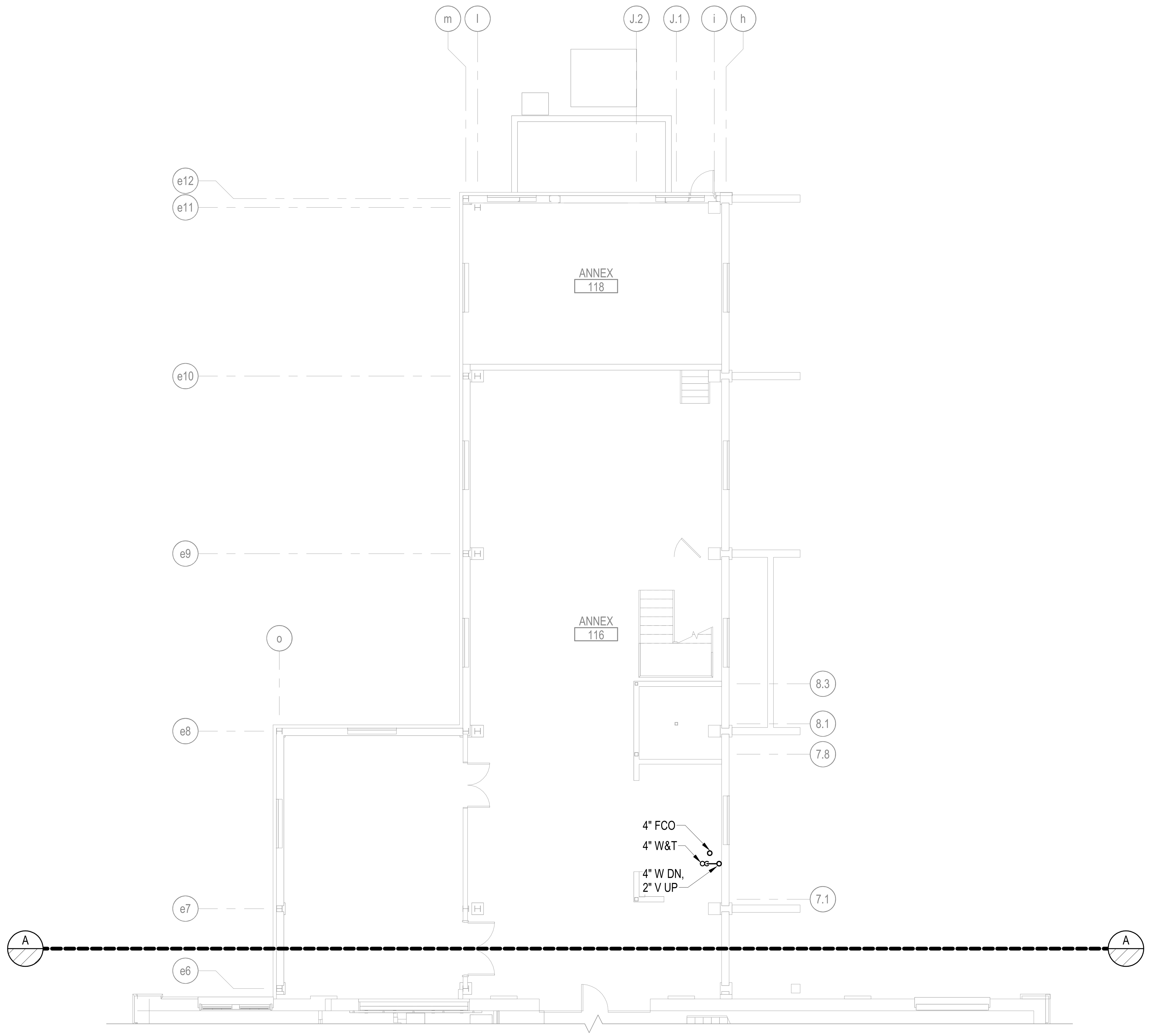
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

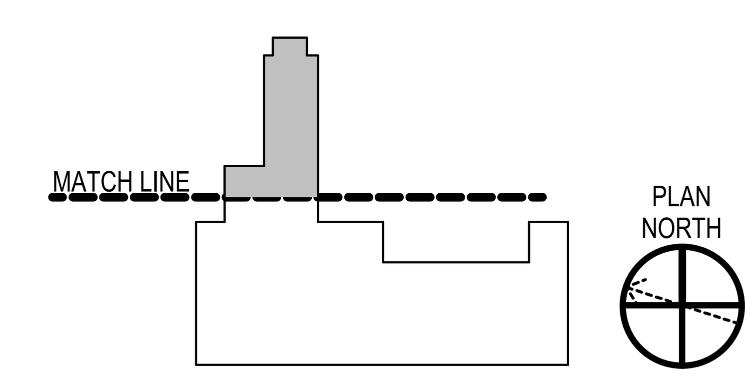


A1 **FIRST FLOOR PLUMBING PLAN - EAST**
SCALE: 1/8" = 1'-0"

KEYNOTES (THIS SHEET ONLY)

1 -

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

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STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25
SEAL

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APPROVED
FOR COMMANDER NAVFAC
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DESIGNER RNC
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FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
FIRST FLOOR PLUMBING PLAN - EAST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 171 OF 282
P-103

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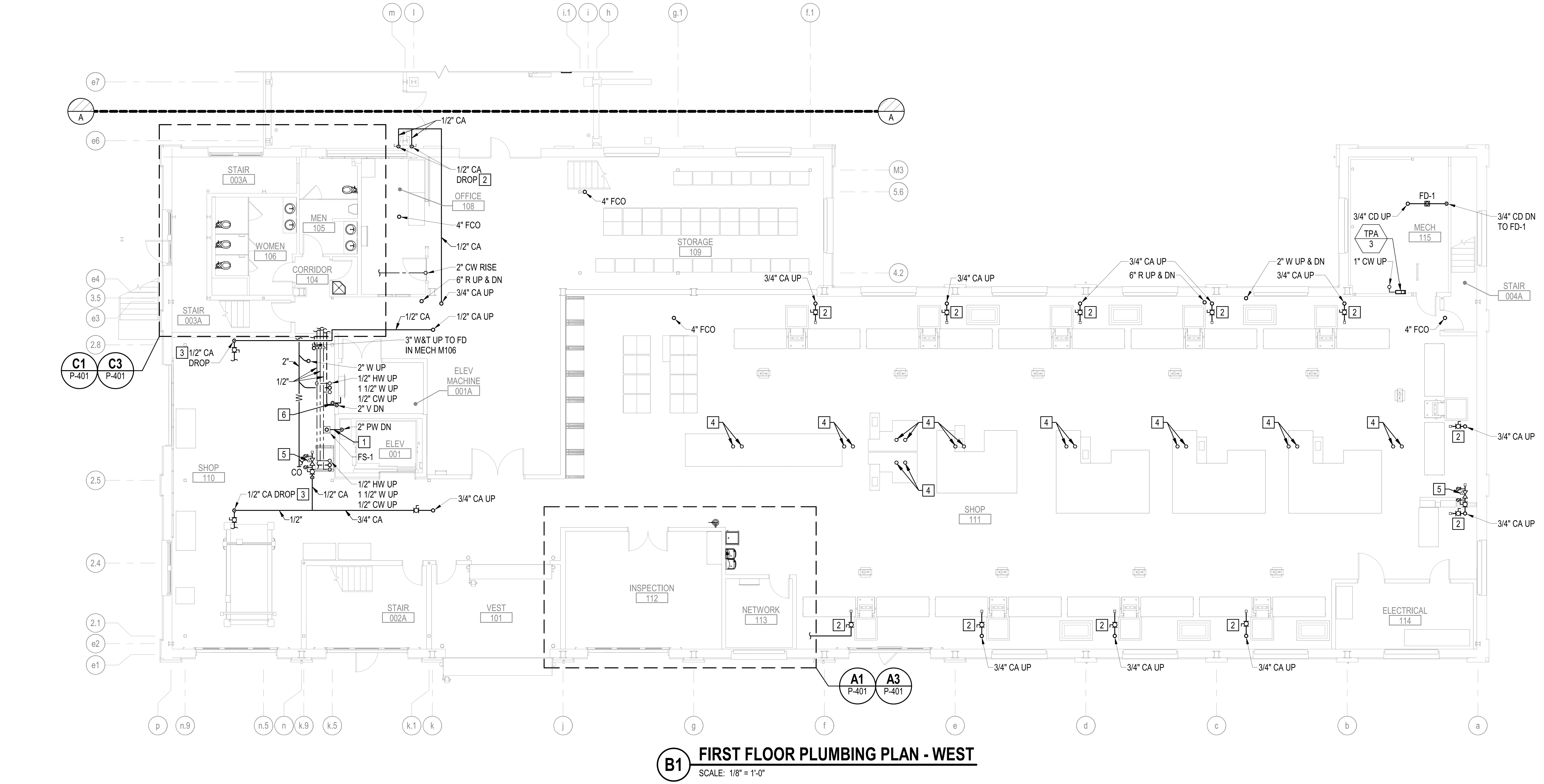
3 UNCLASSIFIED



B1 FIRST FLOOR PLUMBING PLAN - WEST

SCALE: 1/8" = 1'-0"

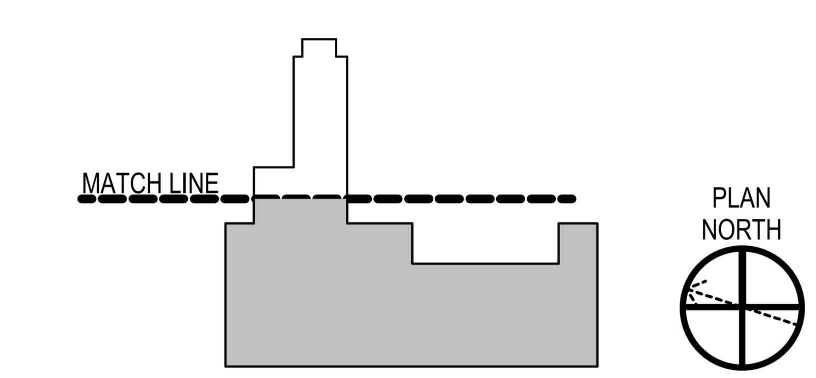
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE WITH CONTRACTING OFFICER THE LOCATIONS FOR COMPRESSED AIR BRANCH SERVICE PIPING OUTLETS, CONSISTENT WITH THE FINAL SHOP EQUIPMENT LAYOUT.
3. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.
4. INDUSTRIAL EQUIPMENT SHOWN FOR REFERENCE.

KEYNOTES (THIS SHEET ONLY)

1. 2" PUMPED WASTE WATER FROM ELEVATOR SUMP PUMP (SP-1). PIPE WILL RUN EXPOSED ALONG FLOOR AND DISCHARGE INTO FS-1.
2. PROVIDE 1/2" CA WITH ISOLATION BALL VALVE AND QUICK CONNECT FEMALE HOSE CONNECTION FITTING 80" AFF. COORDINATE QUICK CONNECT FITTING SELECTION WITH CONTRACTING OFFICER.
3. COMPRESSED AIR SERVICE FOR CNC DOWNDRAFT TABLE OPERATIONS. COORDINATE CA OUTLET LOCATION WITH CONTRACT OFFICER TO ALIGN WITH DOWNDRAFT TABLE INSTALLATION REQUIREMENTS.
4. 3/8" COMPRESSED AIR HOSE AND ELECTRICAL POWER EXTENSION CHORD FROM RETRACTABLE REELS OVERHEAD. SET AIR HOSE AND ELECTRICAL CHORD STOPS TO POSITION THEIR ENDS 7'-0" AFF(ADJUSTABLE). SEE SHEET P-106 FOR OVERHEAD REEL INSTALLATION REQUIREMENTS.
5. PROVIDE CA OUTLET 60" AFF WITH ISOLATION BALL VALVE, CA PRESSURE REGULATOR, PRESSURE GAUGES UPSTREAM AND DOWNSTREAM OF REGULATOR AND QUICK CONNECT HOSE FITTING. COORDINATE QUICK CONNECT FITTING SELECTION WITH CONTRACTING OFFICER.
6. RUN 3/4" IU-1 CONDENSATE DRAIN EXPOSED ON WALL TO NEARBY FLOOR SINK. PROVIDE PVC JACKET ON EXPOSED PIPE INSULATION.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0"   0   5'   10'   15'   30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER, STATE OF MAINE, 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC

DRAWN BY SRW

CHECKED BY MSA

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

FIRST FLOOR PLUMBING PLAN - WEST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 172 OF 282

P-104

DRAWFORM REVISION: 14 SEP 2023

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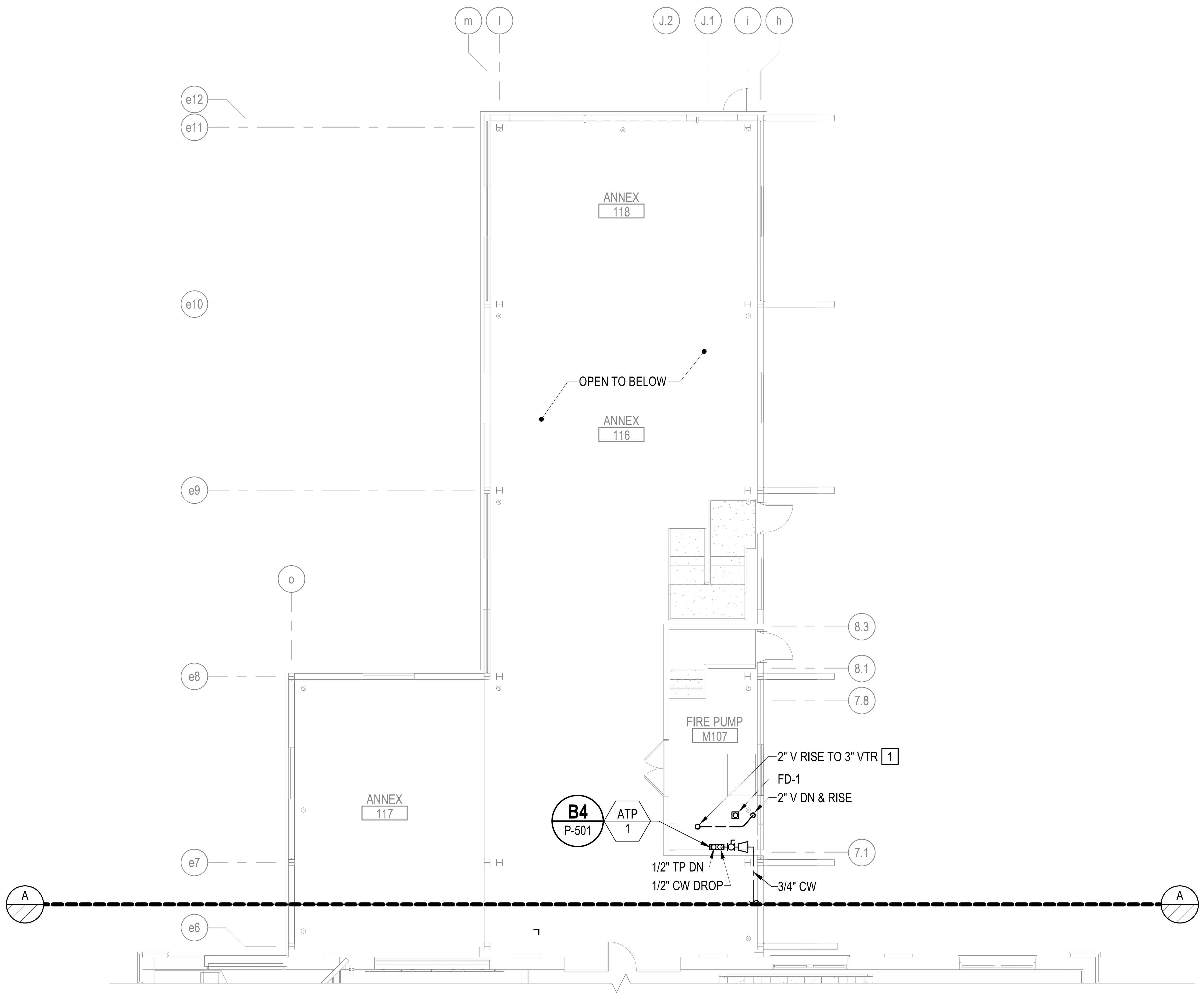
3 UNCLASSIFIED



A1 FIRST FLOOR MEZZANINE PLUMBING PLAN - EAST

SCALE: 1/8" = 1'-0"

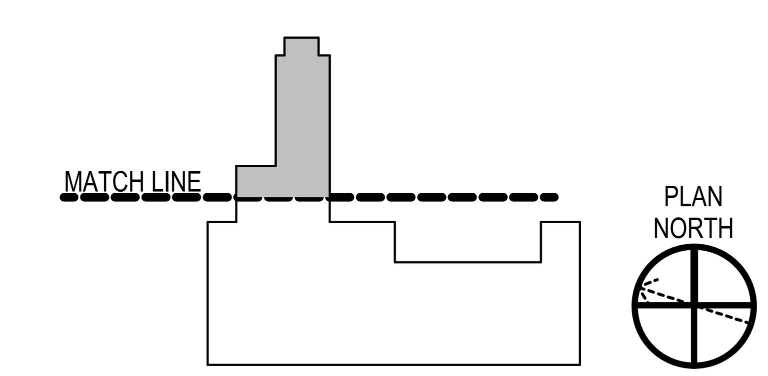
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1 PROVIDE 2" V UP WITH 3" VTR. VTR MUST BE A MINIMUM OF 10'-0" FROM EDGE OF THE WALL.

KEY PLAN



GRAPHIC SCALE

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NAVFAC

RICHARD CRESWELL No. 13178 STATE OF MAINE LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	SRW
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MEZZANINE PLUMBING PLAN - EAST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.

SHEET 173 OF 282

P-105

DRAWFORM REVISION: 14 SEP 2023

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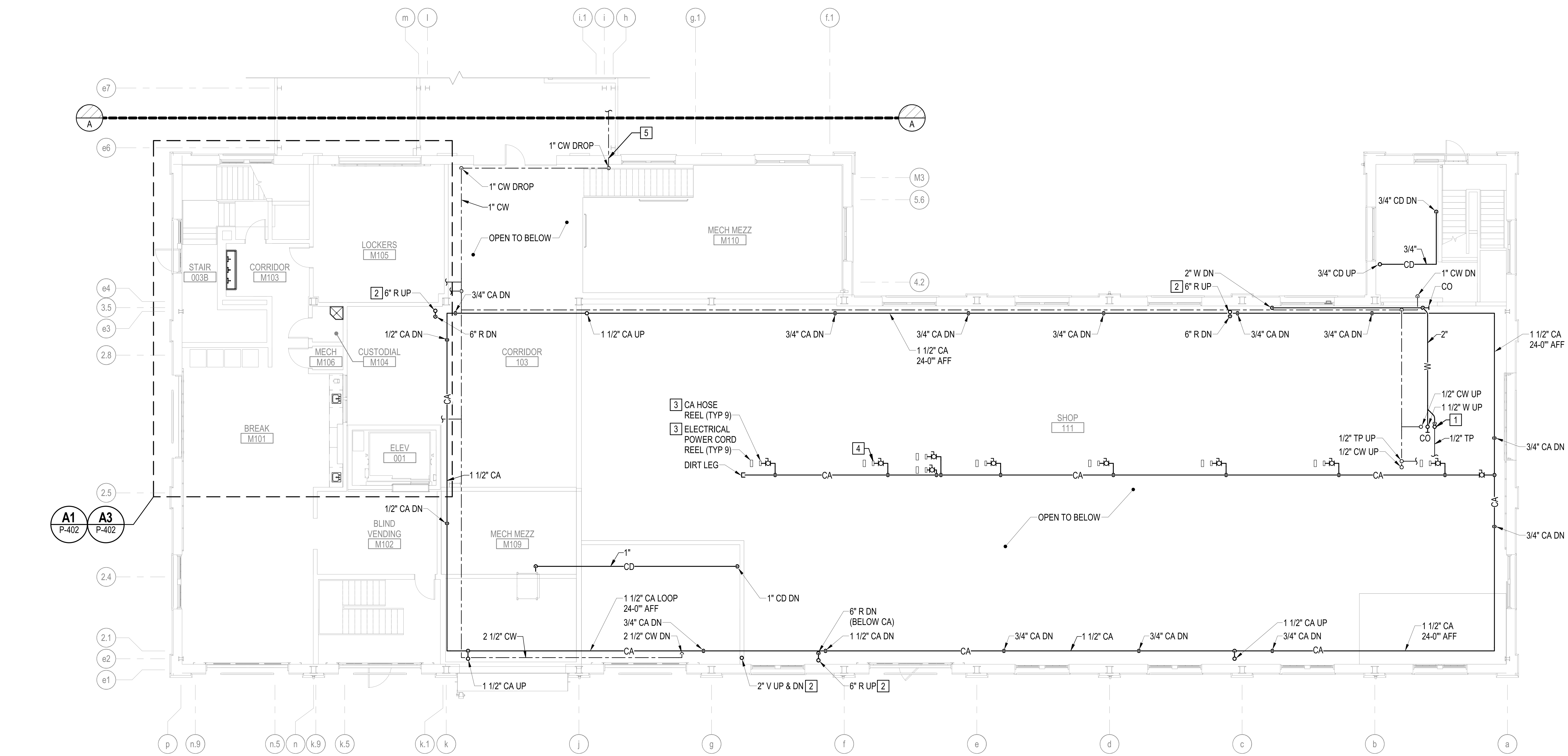
3 UNCLASSIFIED



B1 **FIRST FLOOR MEZZANINE PLUMBING PLAN - WEST**

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. CORRDINATE WITH CONTRACTING OFFICER THE LOCATIONS FOR COMPRESSED AIR BRANCH SERVICE PIPING OUTLETS, CONSISTENT WITH THE FINAL SHOP EQUIPMENT LAYOUT.
3. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1 1-1/2" INDIRECT WASTE DRAIN TRAP. PROVIDE AIR GAP FITTING. 3/4" DRAIN UP FROM WATER HEATER AUXILIARY DRAIN PAN.

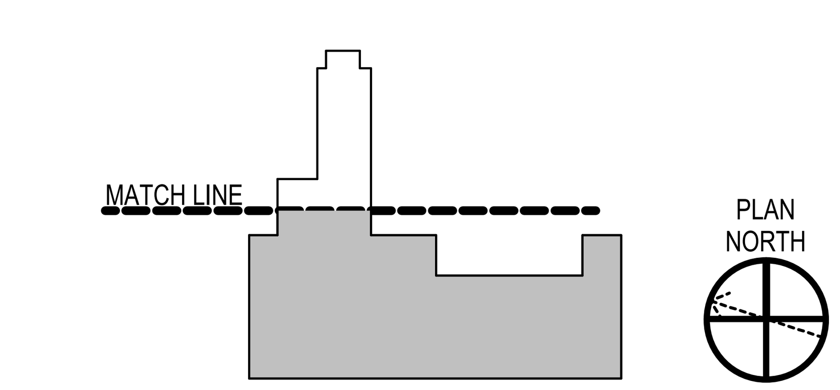
2 PROVIDE OFFSET IN RISER TO AVOID EXISTING CRANE RAILS ON THE FIRST AND SECOND FLOORS.

3 PROVIDE 1/2" COMPRESSED AIR BRANCH SERVICE WITH ISOLATION VALVE AND HARD PIPE HOSE REEL CONNECTION FITTINGS TO HOSE REEL. THE HOSE REELS ARE PROVIDED AS PART OF EQUIPMENT PACKAGE OPTION 2, AND SUSPENDED FROM STRUCTURAL RAILS LOCATED APPROXIMATELY 24'-0" AFF. RUN COMPRESSED AIR MAIN ABOVE THE STRUCTURAL RAIL ON STRUCTURAL STEEL CHANNEL ACROSS THE RAILS WITH PIPE CONNECTION FITTINGS. THE COMPRESSED AIR HOSE REELS WILL BE PAIRED WITH ELECTRICAL CORD REELS. COORDINATE REEL PAIRING ARRANGEMENTS AND LOCATIONS WITH THE CONTRACTING OFFICER TO ALIGN WITH THE SHOP'S FINAL EQUIPMENT LAYOUT. ALSO, COORDINATE PLACEMENT OF COMPRESSED AIR MAIN WITH ELECTRICAL SERVICE CONDUIT SERVING THE ELECTRIC CORD REELS. REFER TO XX### FOR STRUCTURAL RAIL DETAILS.

4 FOR THE CA SERVICE LINE TO THIS CA REEL, PROVIDE ISOLATION BALL VALVE, PRESURE REGULATOR AND PRESURE GAUGE UPSTREAM AND DOWNSTREAM OF REGULATOR.

5 COORDINATE WALL PENETRATION WITH HISTORIC SLIDING DOOR. THE DOOR MUST REMAIN FUNCTIONAL.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0"   0   5'   10'   15'   30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

RICHARD CRESWELL No 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

NAVFAC

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	SRW
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MEZZANINE PLUMBING PLAN - WEST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 174 OF 282

P-106

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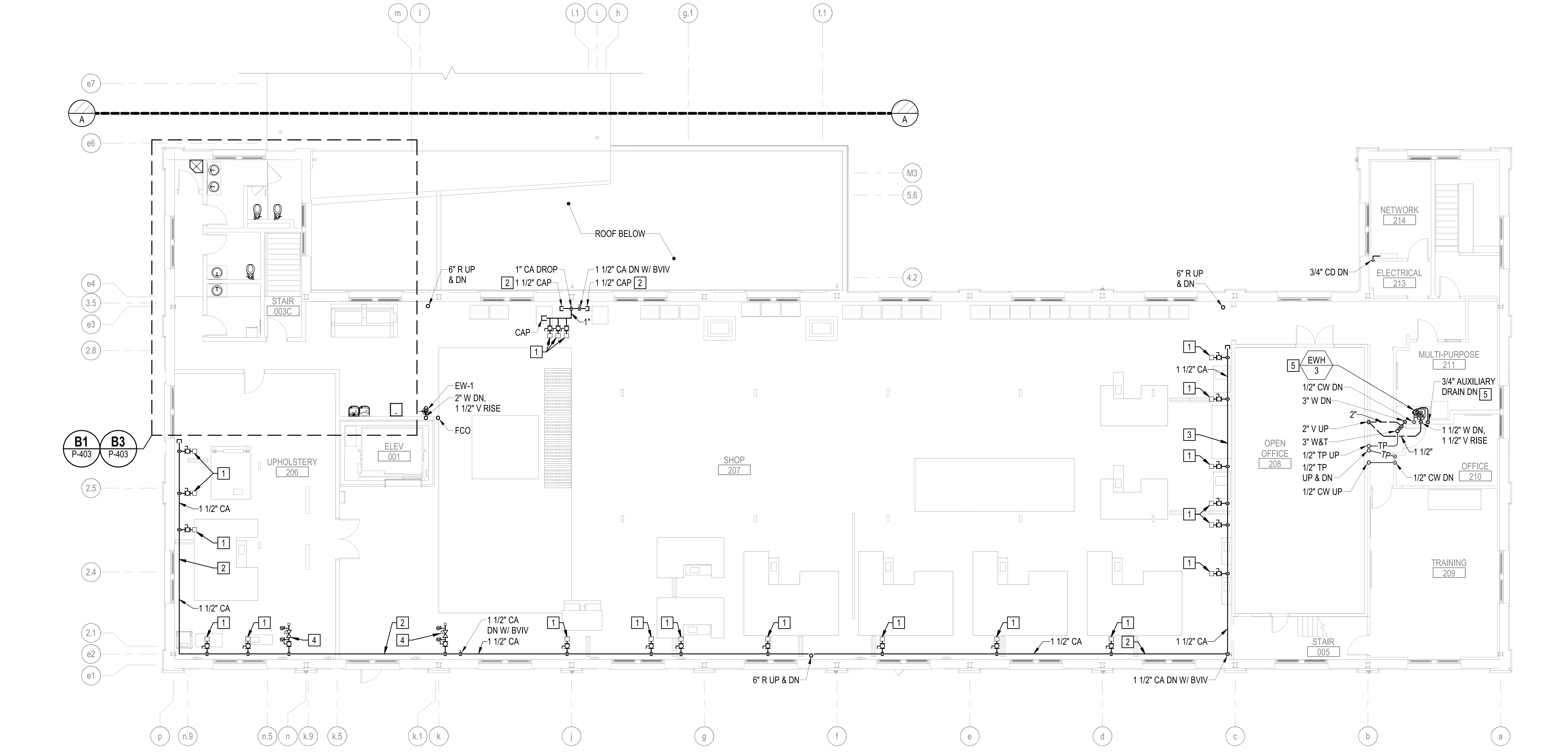
3 UNCLASSIFIED



B1 SECOND FLOOR PLUMBING PLAN - WEST
SCALE: 1/8" = 1'-0"

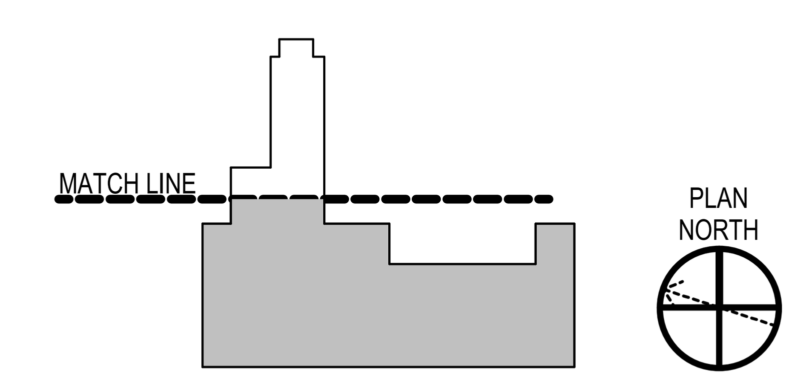
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. CORRDINATE WITH CONTRACTING OFFICER THE LOCATIONS FOR COMPRESSED AIR BRANCH SERVICE PIPING OUTLETS, CONSISTENT WITH THE FINAL SHOP EQUIPMENT LAYOUT.

KEYNOTES (THIS SHEET ONLY)

1. PROVIDE CA OUTLET WITH DIRT LEG AND 1/2" PIPED OUTLET WITH ISOLATION BALL VALVE AND QUICK CONNECT FEMALE FITTING TO RECEIVE A 3/8" CA HOSE W/ CONNECTING MALE COMPONENT OF QUICK CONNECT FITTING. COORDINATE WITH CONTRACTING OFFICER TO ESTABLISH QUICK CONNECT FITTING SELECTIONS. PLACE OUTLETS 48" AFF.
2. RUN HORIZONTAL CA PIPE AT 84" AFF.
3. RUN HORIZONTAL CA PIPE AT 108" AFF.
4. PROVIDE CA OUTLET 48" AFF WITH ISOLATION BALL VALVE, CA PRESSURE REGULATOR, PRESSURE GAUGES UPSTREAM AND DOWNSTREAM OF REGULATOR AND QUICK CONNECT HOSE FITTING. COORDINATE QUICK CONNECT FITTING SELECTION AND OUTLET LOCATIONS WITH CONTRACTING OFFICER TO ALIGN WITH FINAL SHOP EQUIPMENT LAYOUT.
5. PROVIDE AUXILIARY DRAIN PAN FOR EWH-3. PIPE 3/4" DRAIN TO AIR GAP FITTING BELOW (SEE SHEET P-106).

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

FINAL SUBMISSION 10/01/2025 RSH
SYM DESCRIPTION DATE APPR

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RICHARD CRESWELL No. 13178 STATE OF MAINE LICENSED PROFESSIONAL ENGINEER 10-01-25
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
SECOND FLOOR PLUMBING PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 175 OF 282
P-107
DRAWFORM REVISION: 14 SEP 2023

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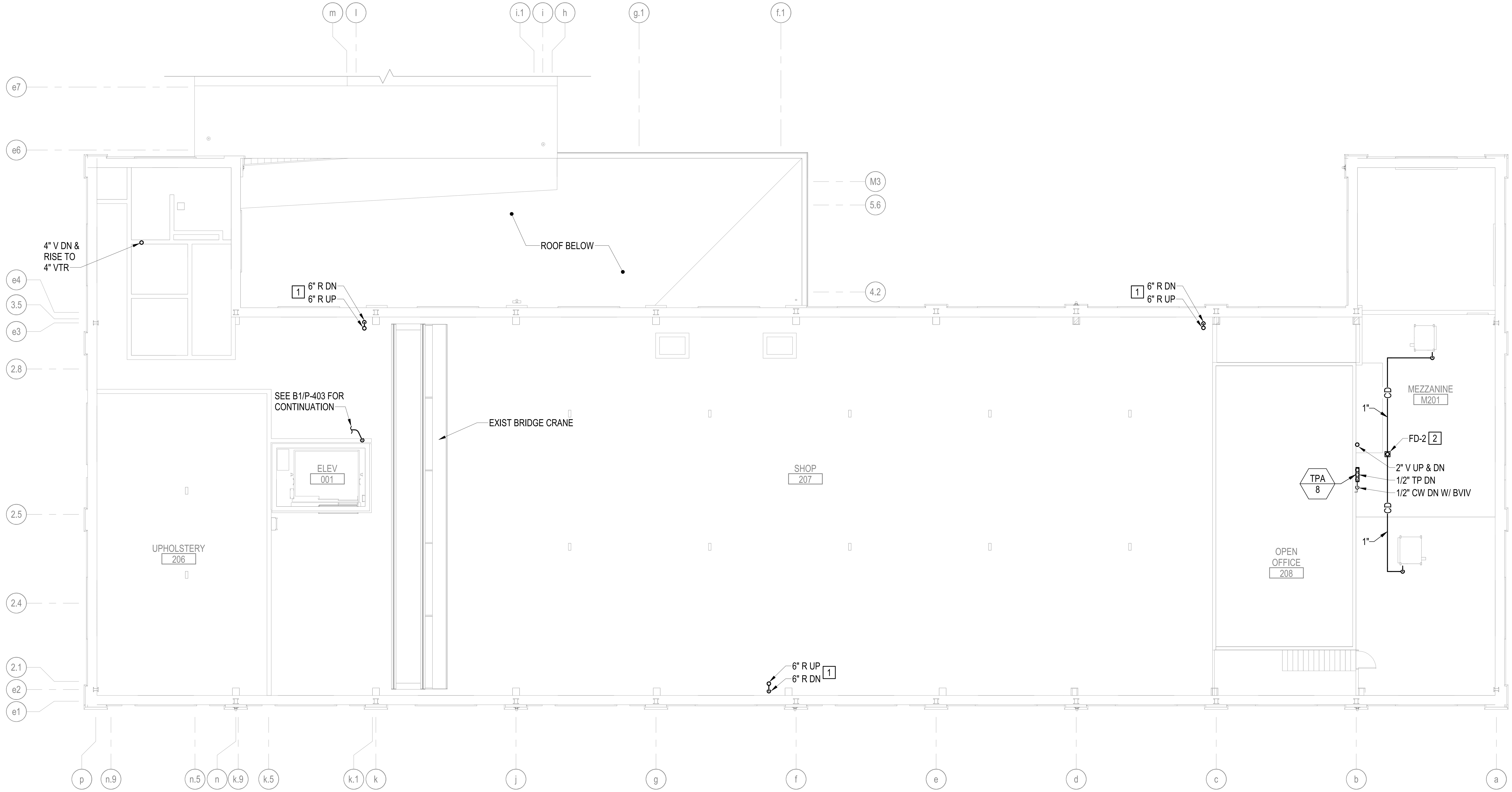


B1 SECOND FLOOR MEZZANINE PLUMBING PLAN - WEST

SCALE: 1/8" = 1'-0"

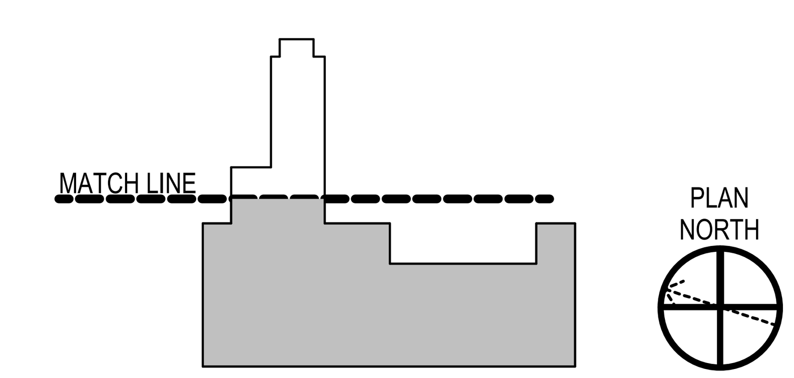
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1. PROVIDE PIPING RISER OFFSET USING 45 DEGREE ELS TO AVOID CONFLICT WITH HEATING WATER PIPING AND CRANE RAIL. SEE SHEET MP106 FOR HYDRONIC HEATING PIPING ARRANGEMENT.
2. PIPE 1" CONDENSATE DRAINS FROM BCU-2 AND BCU-3 TO FUNNEL ON FD-2.

KEY PLAN



GRAPHIC SCALE

CHECK GRAPHIC SCALE BEFORE USING

FINAL SUBMISSION 10/01/2025 RSH

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RICHARD CRESWELL No 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC

DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

SECOND FLOOR MEZZANINE PLUMBING PLAN

EPROJECT NO.: 1740579

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DRAWFORM REVISION: 14 SEP 2023

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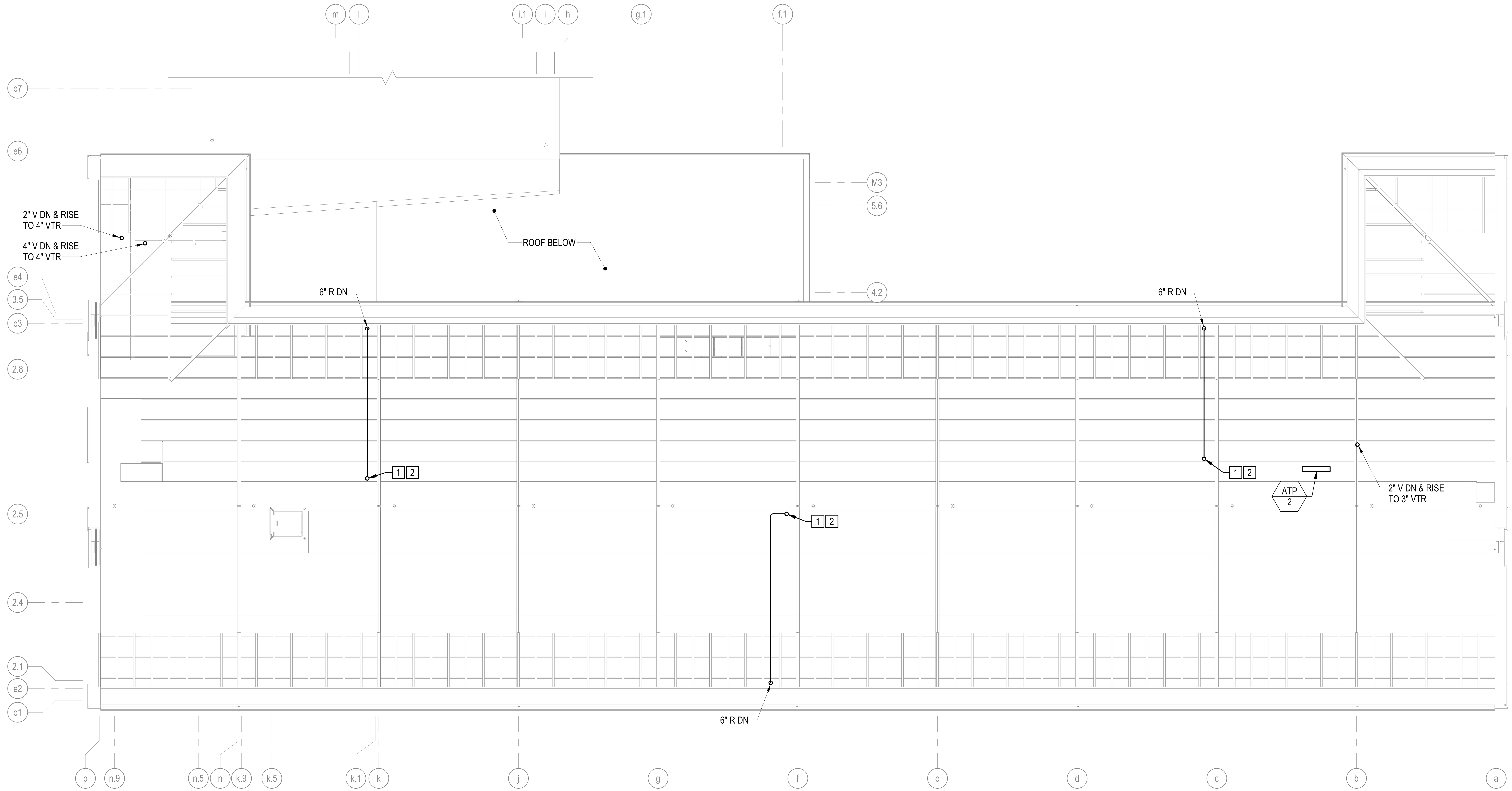


B3 ATTIC PLUMBING PLAN - WEST

SCALE: 1/8" = 1'-0"

GENERAL NOTES

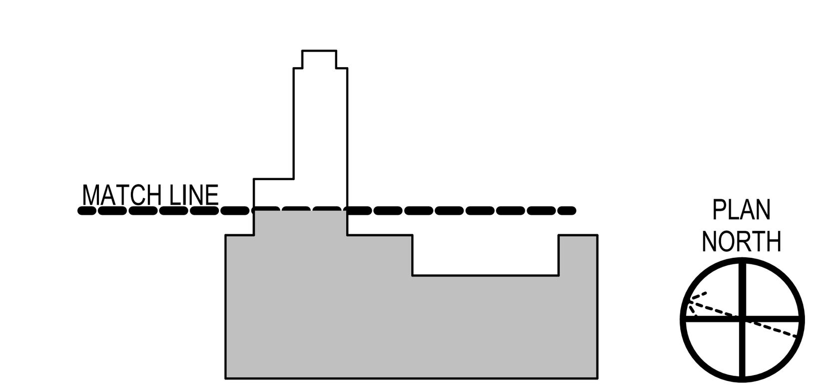
1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1 PROVIDE RADON VENT UP THROUGH ROOF, EXTENDING A MINIMUM THREE FEET ABOVE ROOF PENETRATION. SEE ARCHITECTURAL SHEET AE521 FOR ROOF PENETRATION, FLASHING, AND SEALING REQUIREMENTS. SEE ELECTRICAL SHEET EP107 FOR REQUIRED PRE-WIRING FOR INLINE EXHAUST FAN ALONG VENT RISER TO FACILITATE AN EXHAUST FAN INSTALLATION. IN THE EVENT RADON TESTING IDENTIFIES ELEVATED RADON LEVEL WITHIN THE BUILDING AND THE GOVERNMENT AWARDS OPTION 3.

2 OPTION 3: PROVIDE CERTIFIED PROFESSIONAL RADON MITIGATION SPECIALIST SELECTED INLINE FAN AT PRE-WIRED LOCATIONS (3) IN ATTIC. PROVIDE 4'-0" OF CLEARANCE AROUND FAN FOR MAINTENANCE.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

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STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD

B60 REPAIRS

ATTIC PLUMBING PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.

SHEET 177 OF 282

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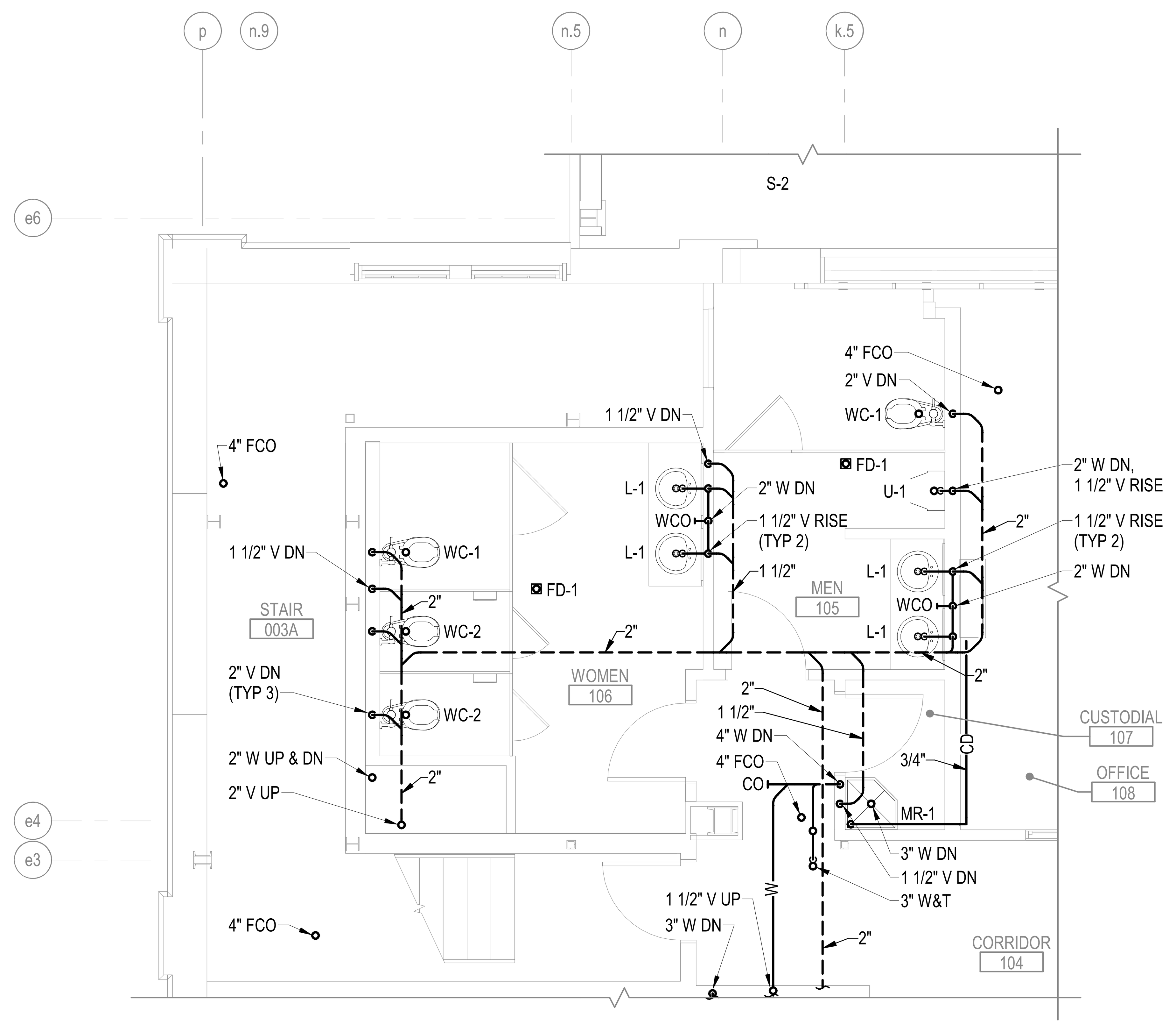
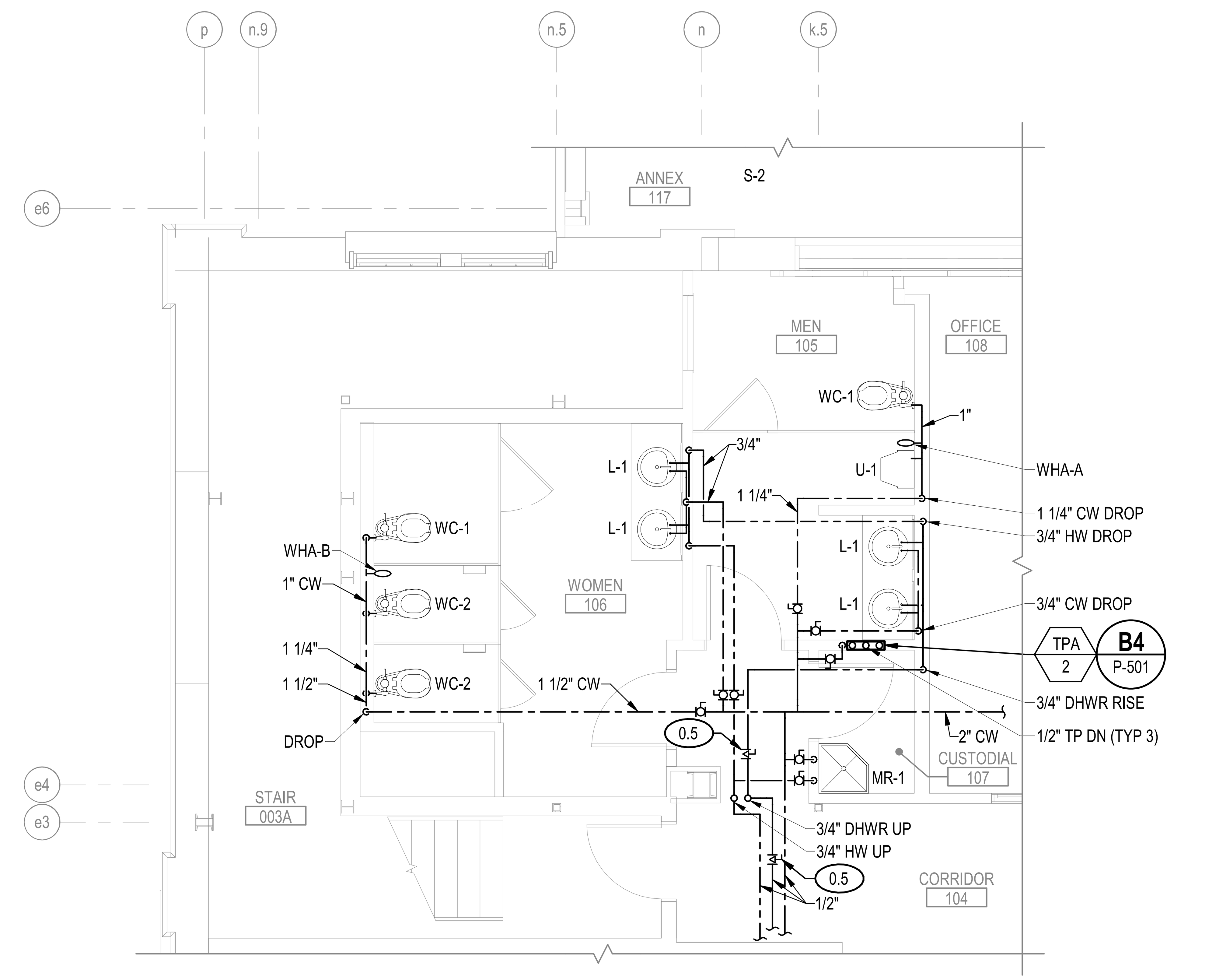


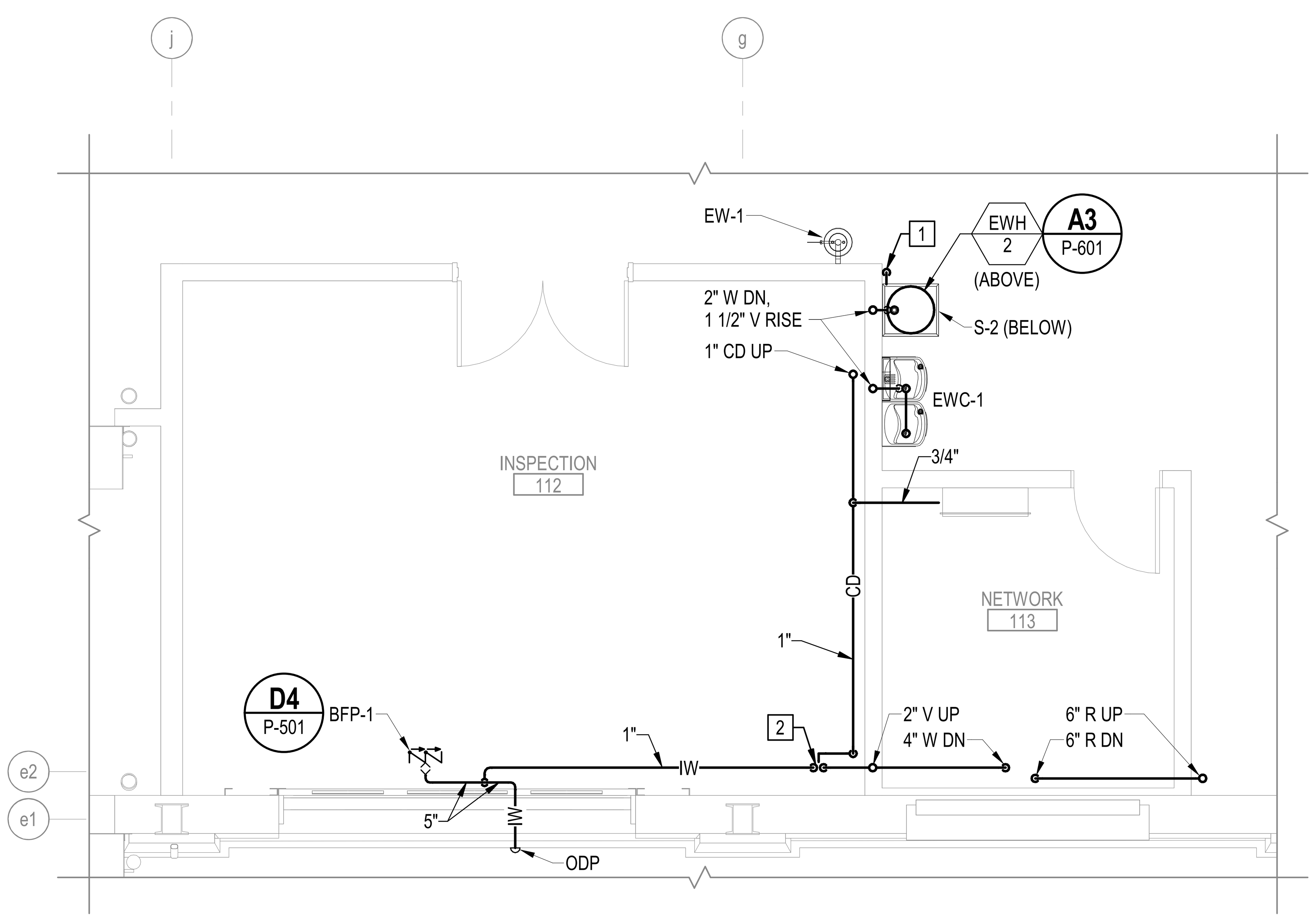
C1 ENLARGED FIRST FLOOR SANITARY WASTE AND VENT PART PLAN

SCALE: 1/4" = 1'-0" (P-104)



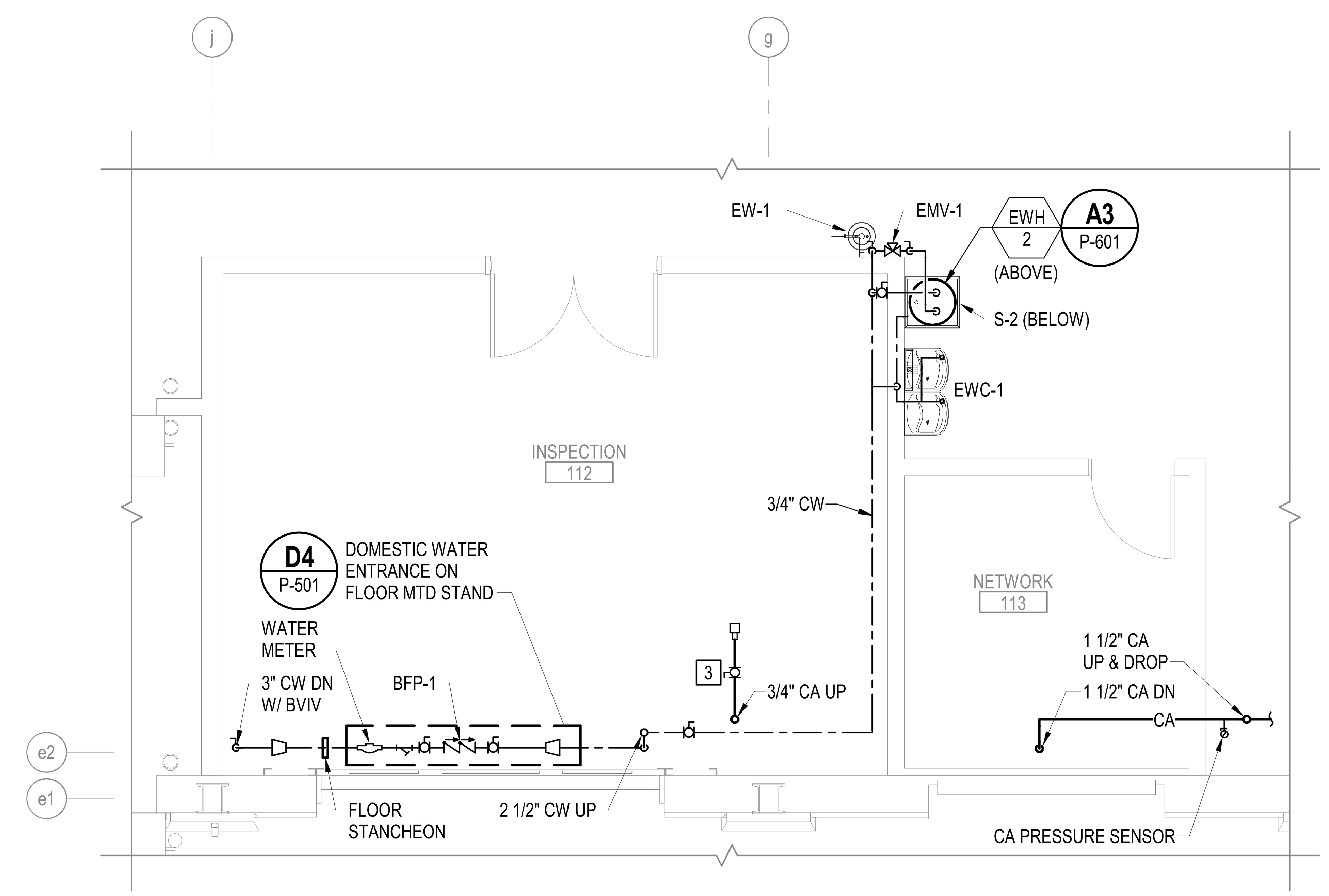
C3 ENLARGED FIRST FLOOR DOMESTIC WATER SUPPLY PART PLAN

SCALE: 1/4" = 1'-0" (P-104)



A1 ENLARGED FIRST FLOOR SANITARY WASTE AND VENT PART PLAN

SCALE: 1/4" = 1'-0" (P-104)



A3 ENLARGED FIRST FLOOR DOMESTIC WATER SUPPLY PART PLAN

SCALE: 1/4" = 1'-0" (P-104)

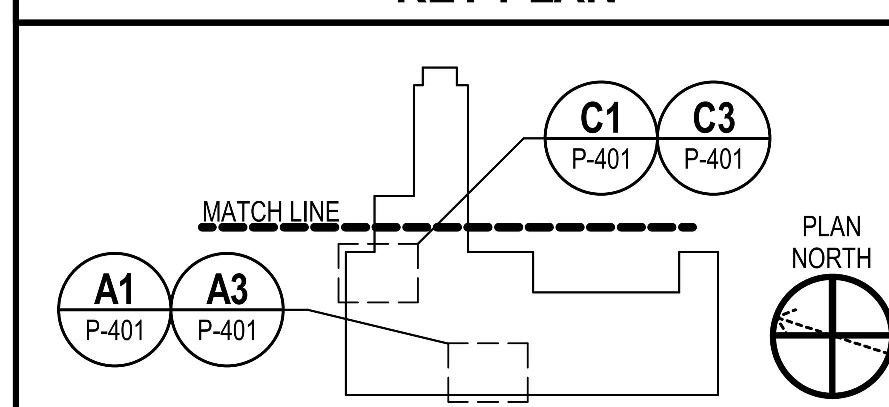
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1. PIPE 3/4" DRAIN DOWN TO SINK S-2. SECURE TO WALL.
2. 2" INDIRECT WASTE DRAIN TRAP WITH AIR GAP FITTING. PIPE DRAIN FROM BFP-1 AND CONDENSATE DRAINS FROM BCU-1 AND IU-2 TO AIR GAP FITTING.
3. PROVIDE 1/2" CA WITH ISOLATION BALL VALVE AND QUICK CONNECT FEMALE HOSE CONNECTION FITTING 80" AFF. COORDINATE QUICK CONNECT FITTING SELECTION WITH CONTRACTING OFFICER.

KEY PLAN



GRAPHIC SCALE

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CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
ENLARGED PLUMBING PART PLANS 1

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 178 OF 282
P-401

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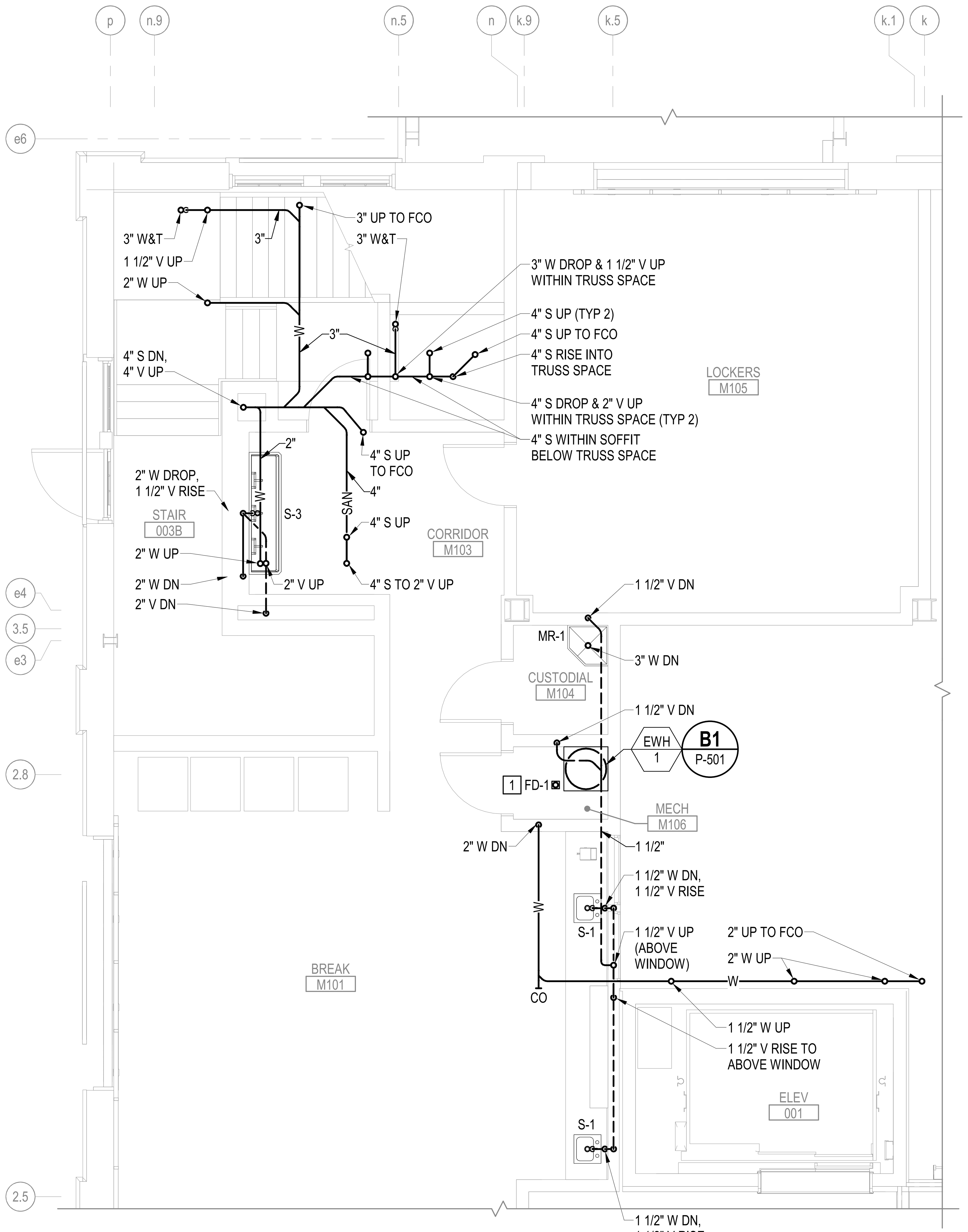
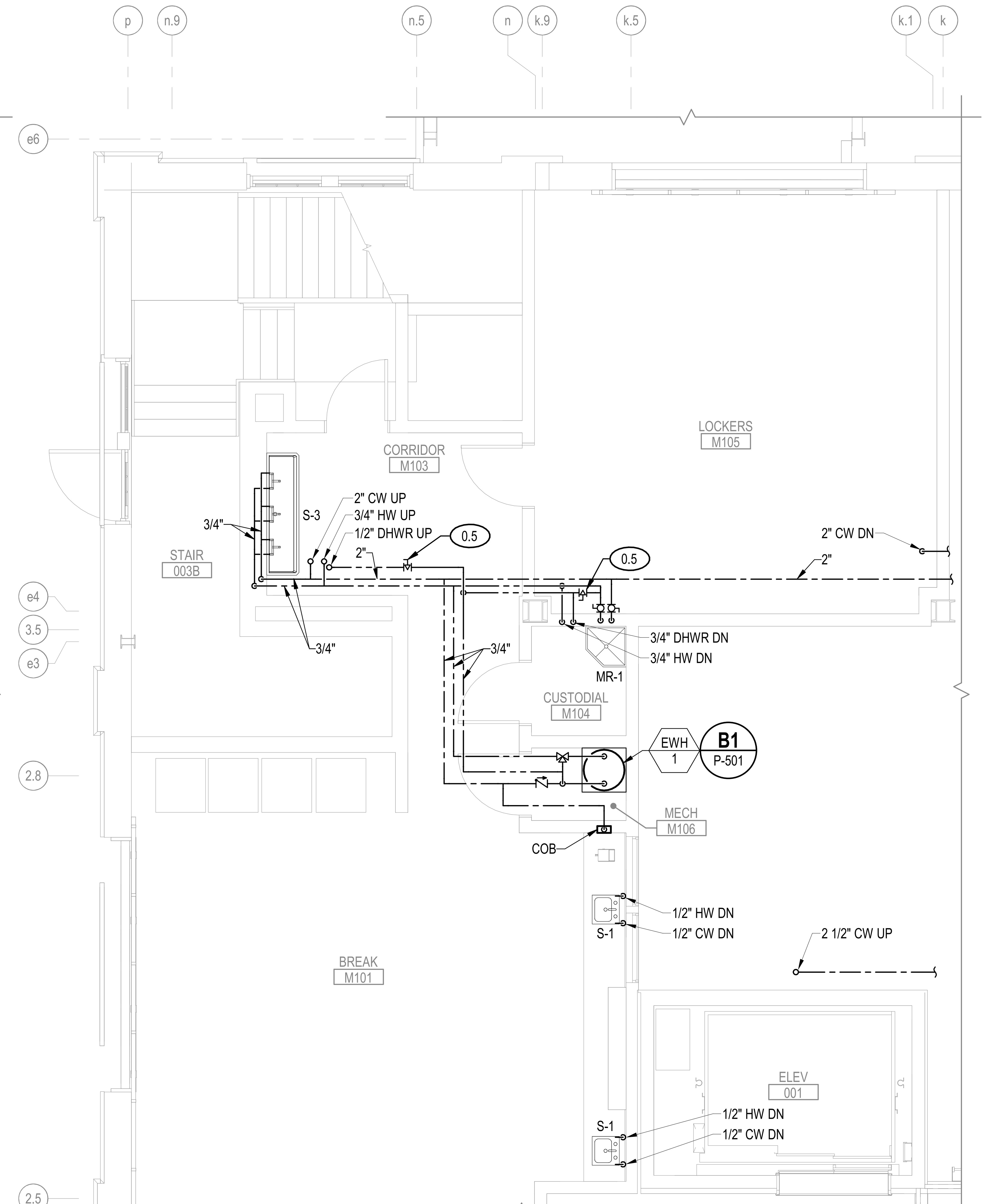


A1 ENLARGED FIRST FLOOR MEZZANINE SANITARY WASTE AND VENT PART PLAN

SCALE: 1/4" = 1'-0" (P-106)



A3 ENLARGED FIRST FLOOR MEZZANINE DOMESTIC WATER SUPPLY PART PLAN

SCALE: 1/4" = 1'-0" (P-106)

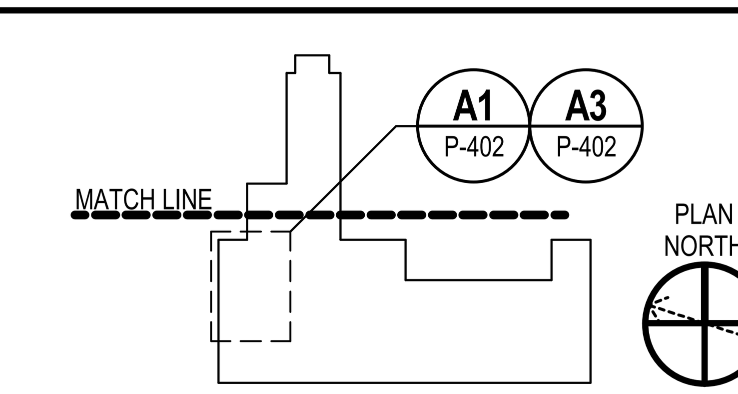
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1 RUN THE WATER HEATER AUXILIARY DRAIN PAN PIPING TO FD-1.

KEY PLAN



GRAPHIC SCALE

1/4" = 1' - 0" 0 1' 5' 10' 15'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

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STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

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CHECKED BY MSA
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PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

ENLARGED PLUMBING PART PLANS 2

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 179 OF 282
P-402

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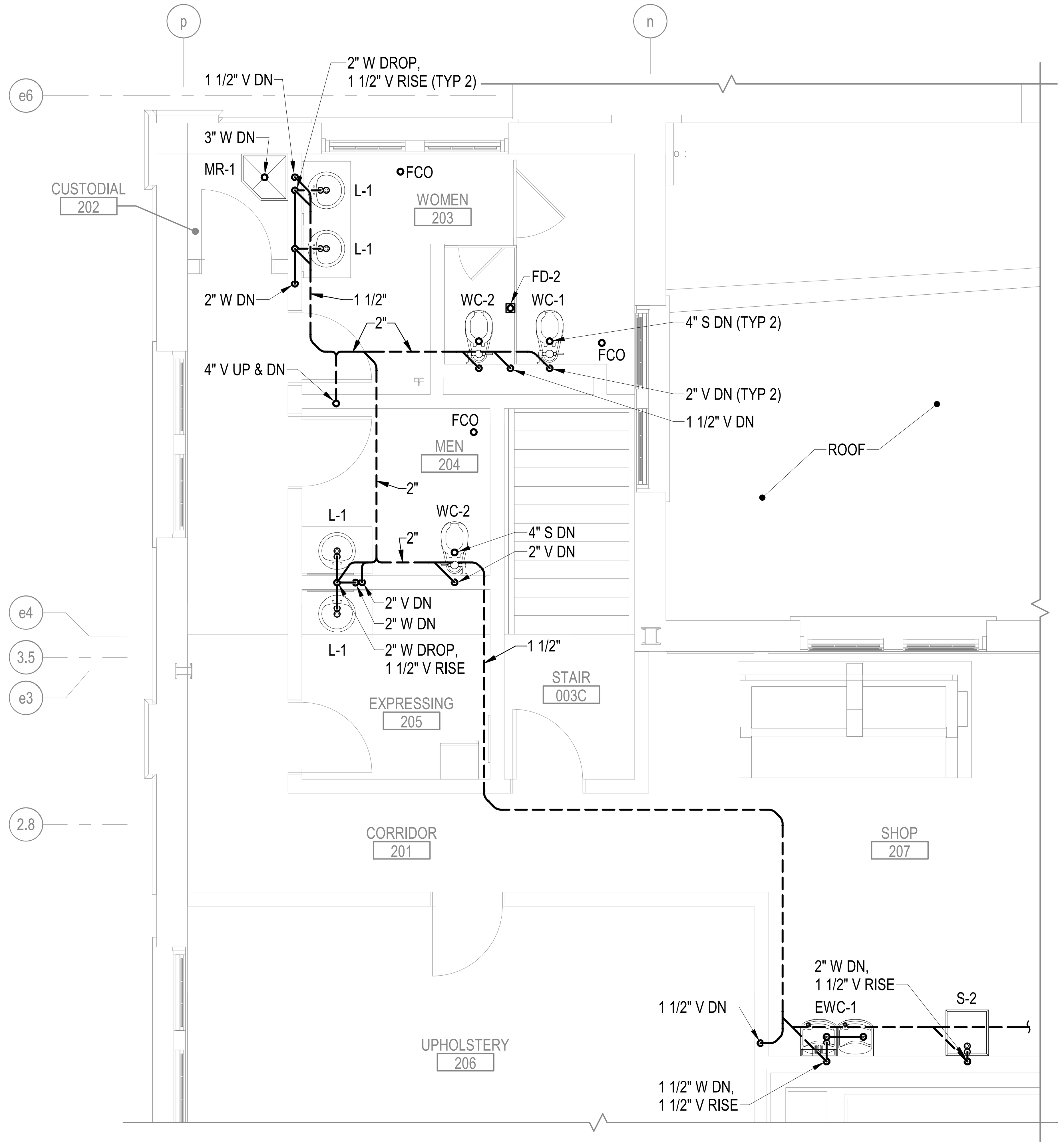
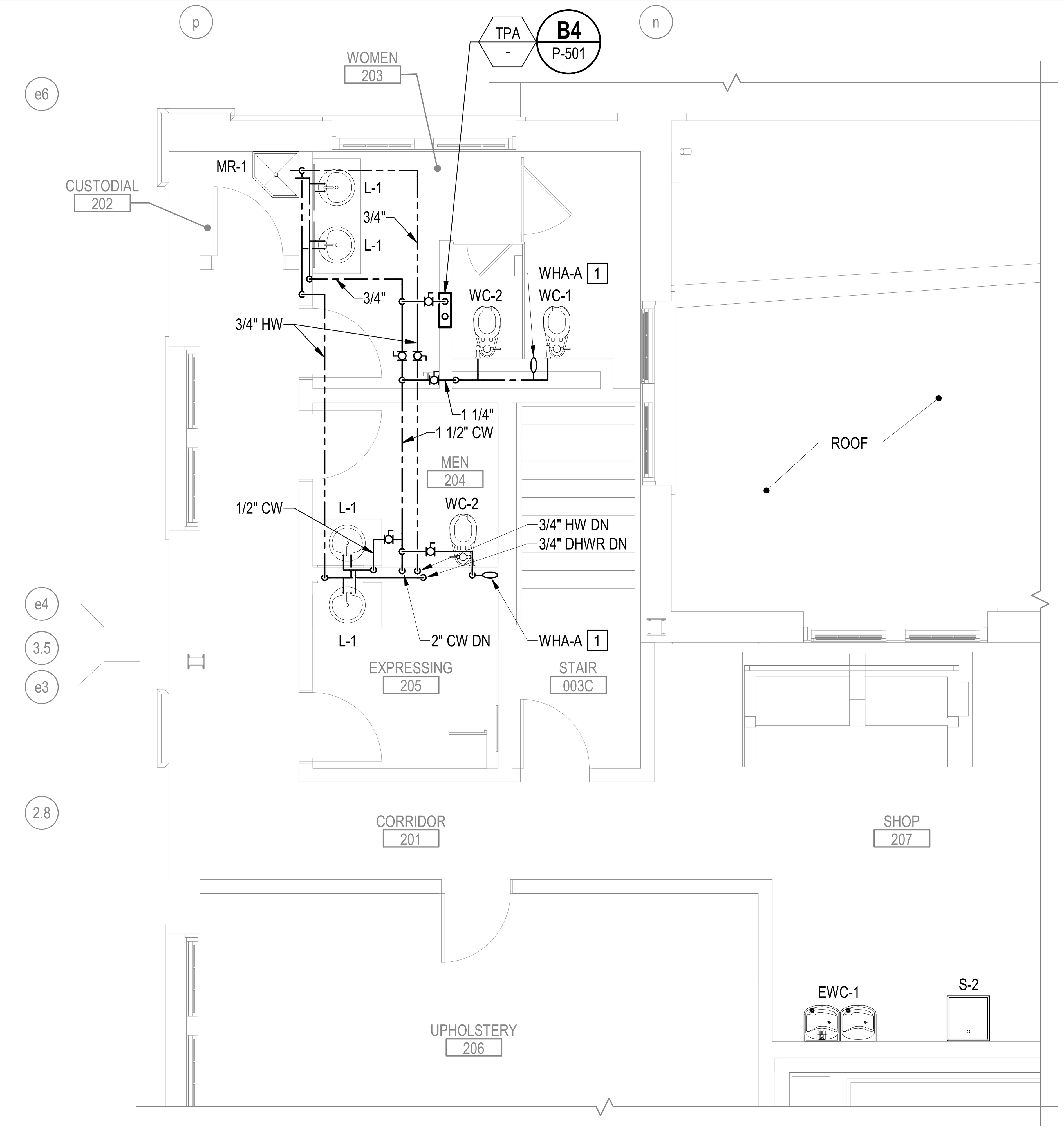


B1 ENLARGED SECOND FLOOR SANITARY WASTE AND VENT PART PLAN

SCALE: 1/4" = 1'-0" (P-107)



B3 ENLARGED SECOND FLOOR DOMESTIC WATER SUPPLY PART PLAN

SCALE: 1/4" = 1'-0" (P-107)

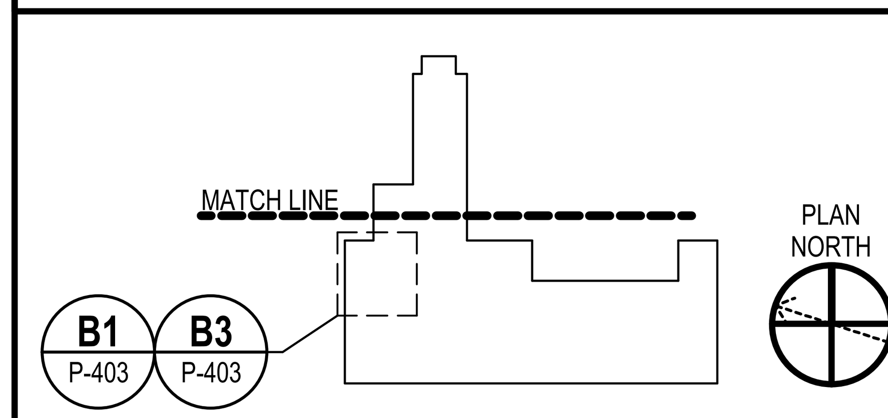
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. COORDINATE THE INSTALLATION OF PIPING AND EQUIPMENT TO MAINTAIN MANDATORY CLEARANCE ZONES AROUND CRANE RAILS AND BRIDGE CRANE OPERATIONAL TRAVEL PATH.

KEYNOTES (THIS SHEET ONLY)

1 PROVIDE 8"x8" WALL MOUNTED WATER HAMMER ARRESTOR SERVICE ACCESS DOOR.

KEY PLAN



GRAPHIC SCALE

1/4" = 1' - 0" 0 1' 5' 10' 15'

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	FINAL SUBMISSION	10/01/2025	RSH

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RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	SRW
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

ENLARGED PLUMBING PART PLANS 3

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 180 OF 282
P-403

DRAWFORM REVISION: 14 SEP 2023

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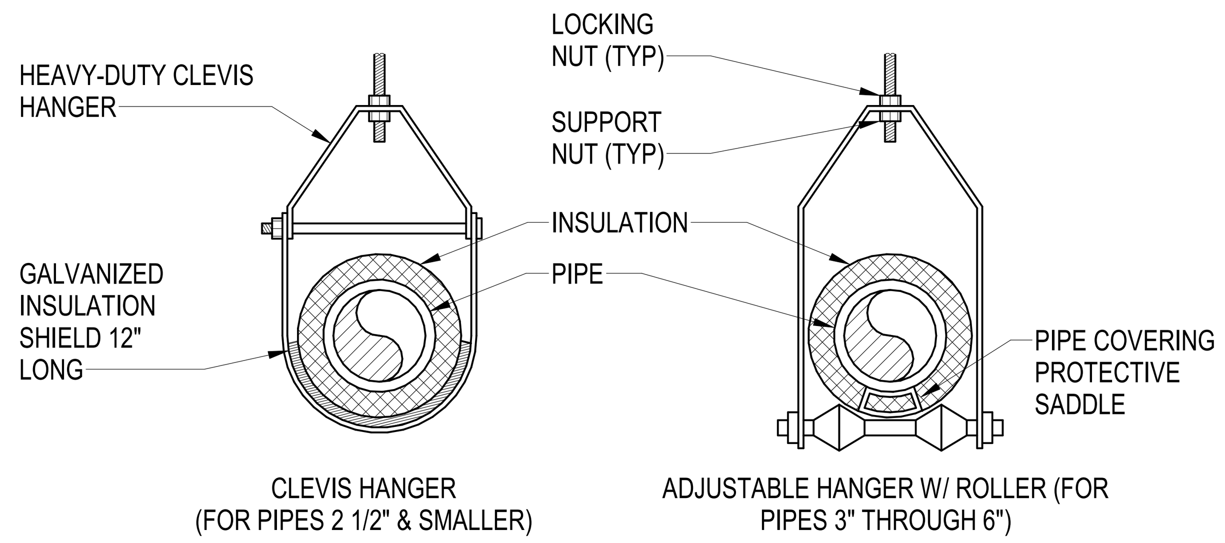
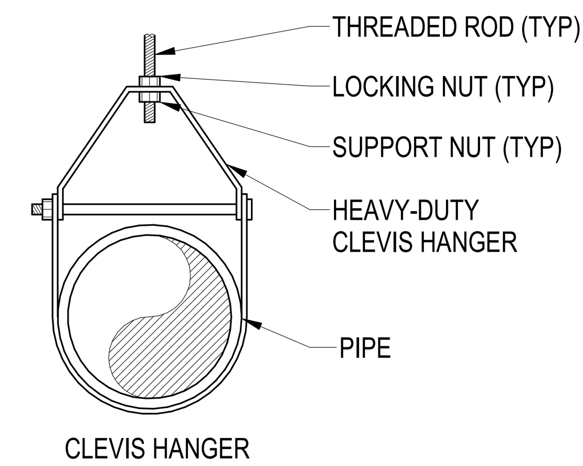
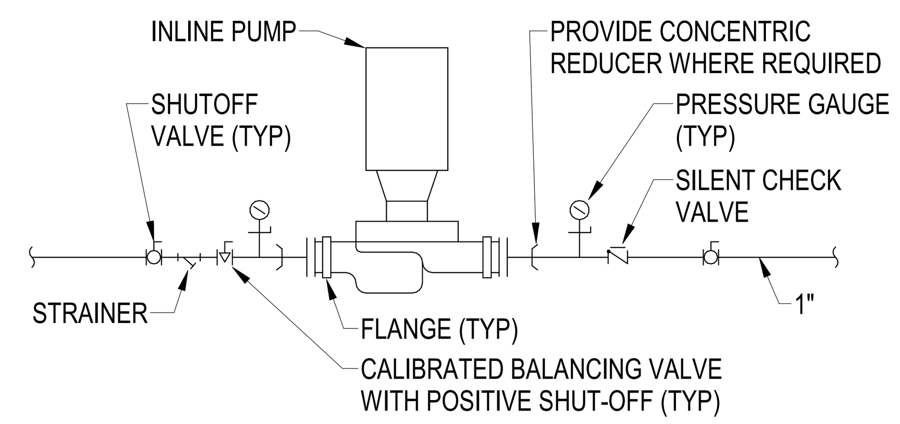


D1 PIPE HANGER ATTACHMENT DETAILS
SCALE: NOT TO SCALE

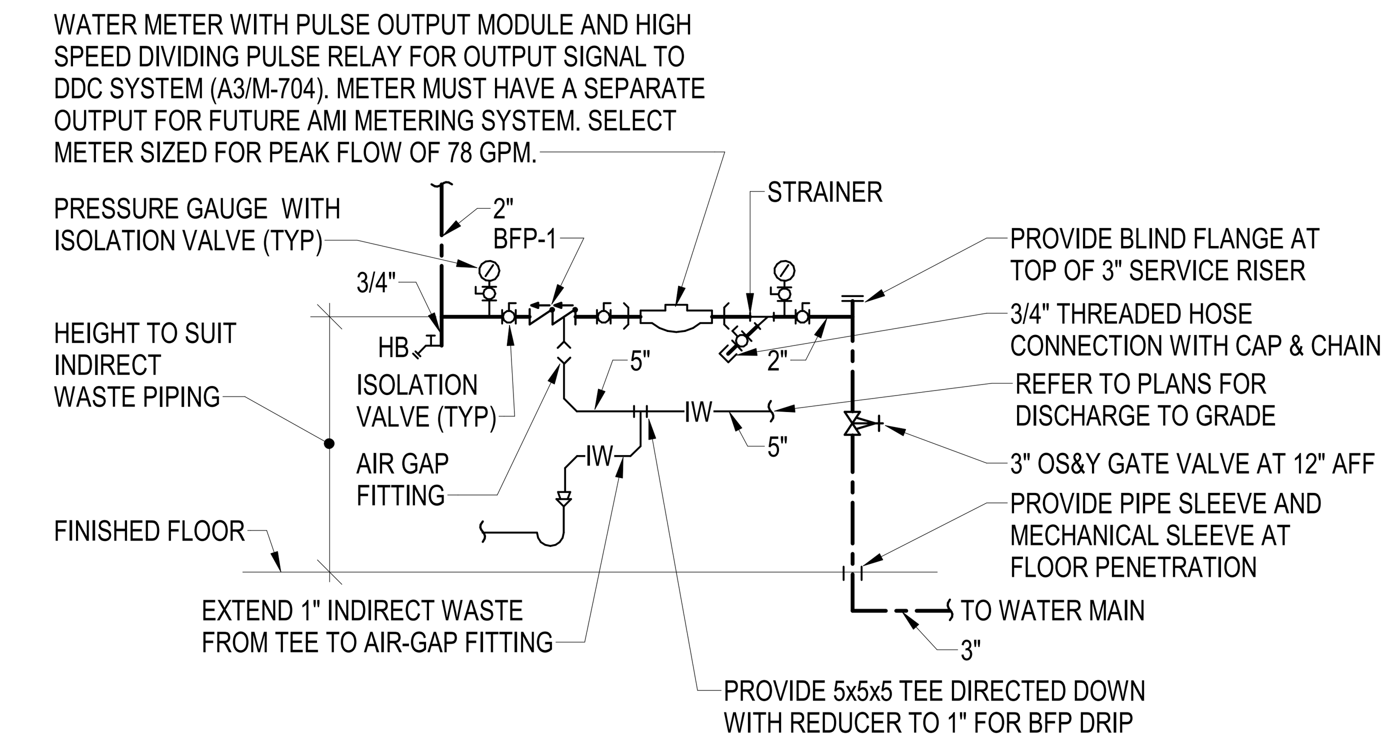


D2 PIPE HANGER ATTACHMENT DETAIL
SCALE: NOT TO SCALE

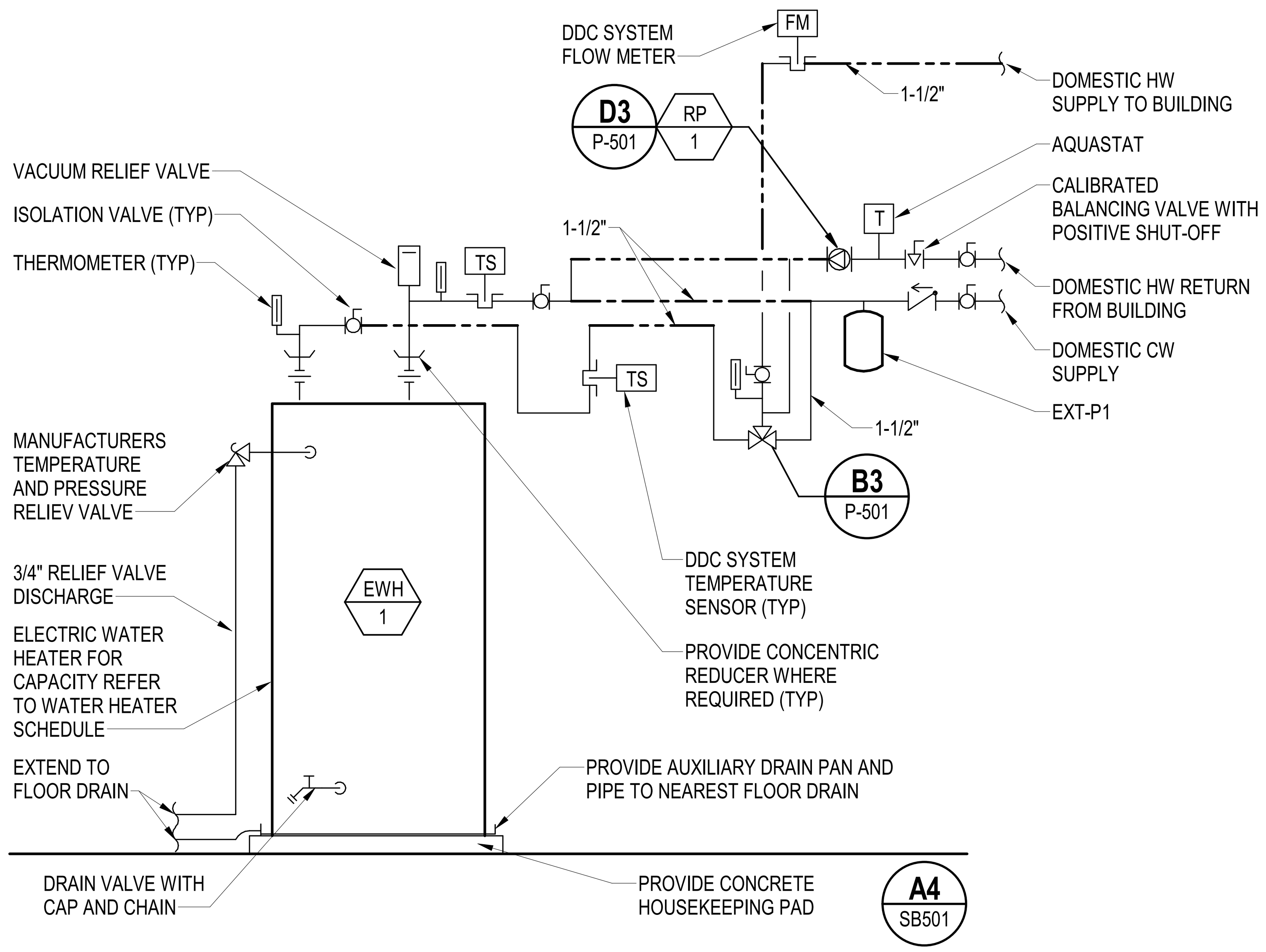


NOTE
1. VALVES SHALL BE SAME SIZE AS PIPE.

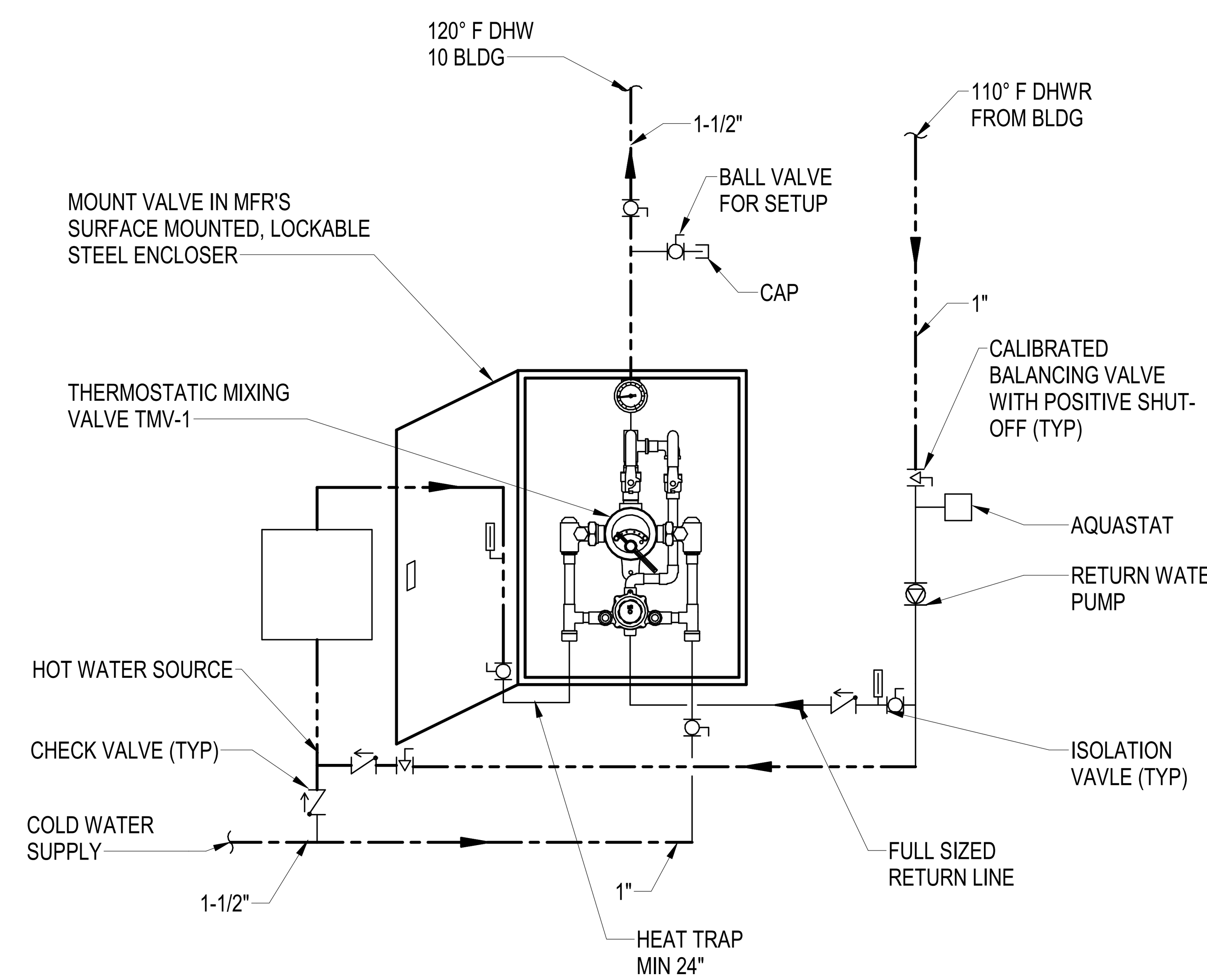
D3 IN-LINE PUMP PIPING DETAIL
SCALE: NOT TO SCALE



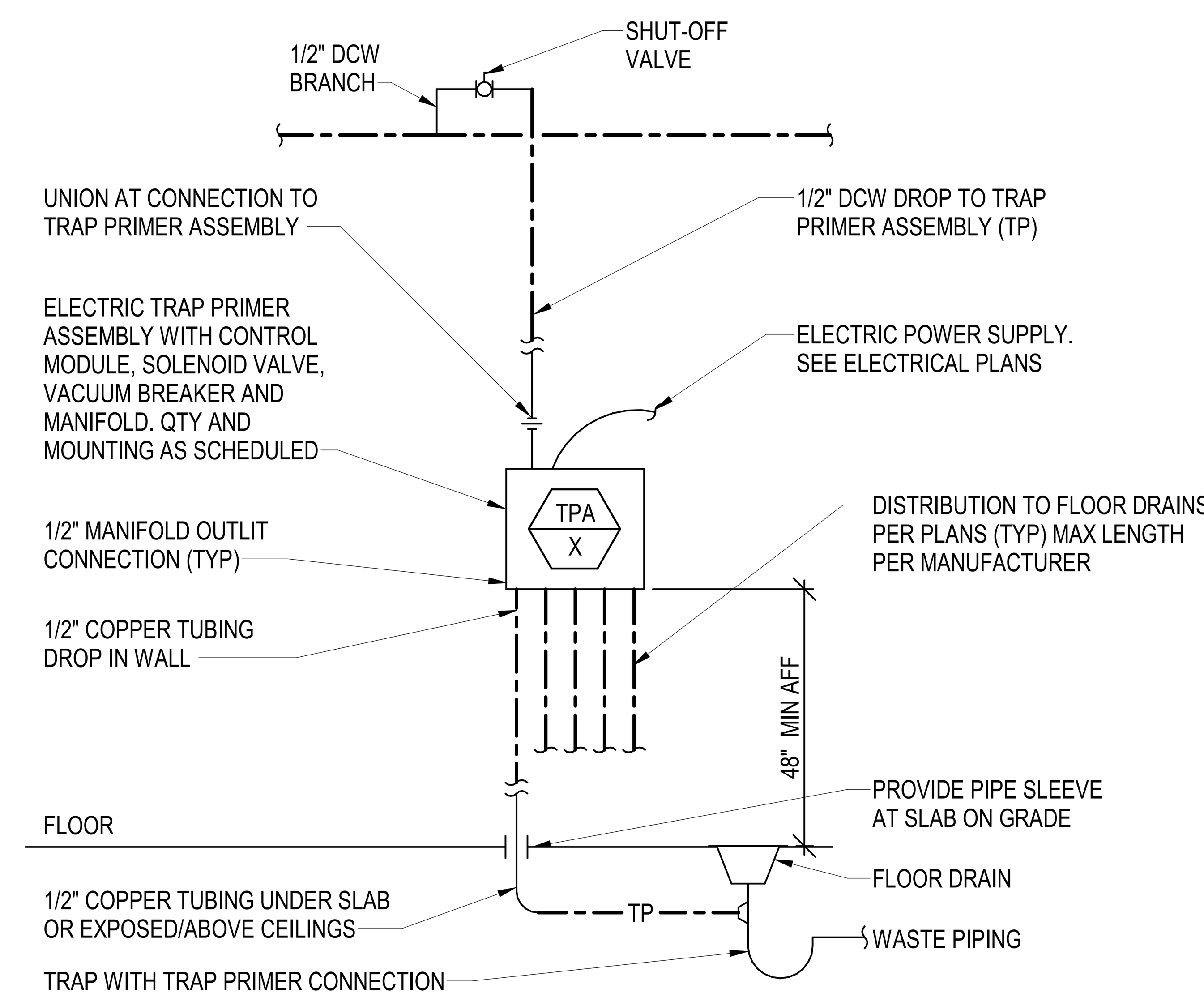
D4 WATER SUPPLY ENTRANCE DETAIL
SCALE: NOT TO SCALE



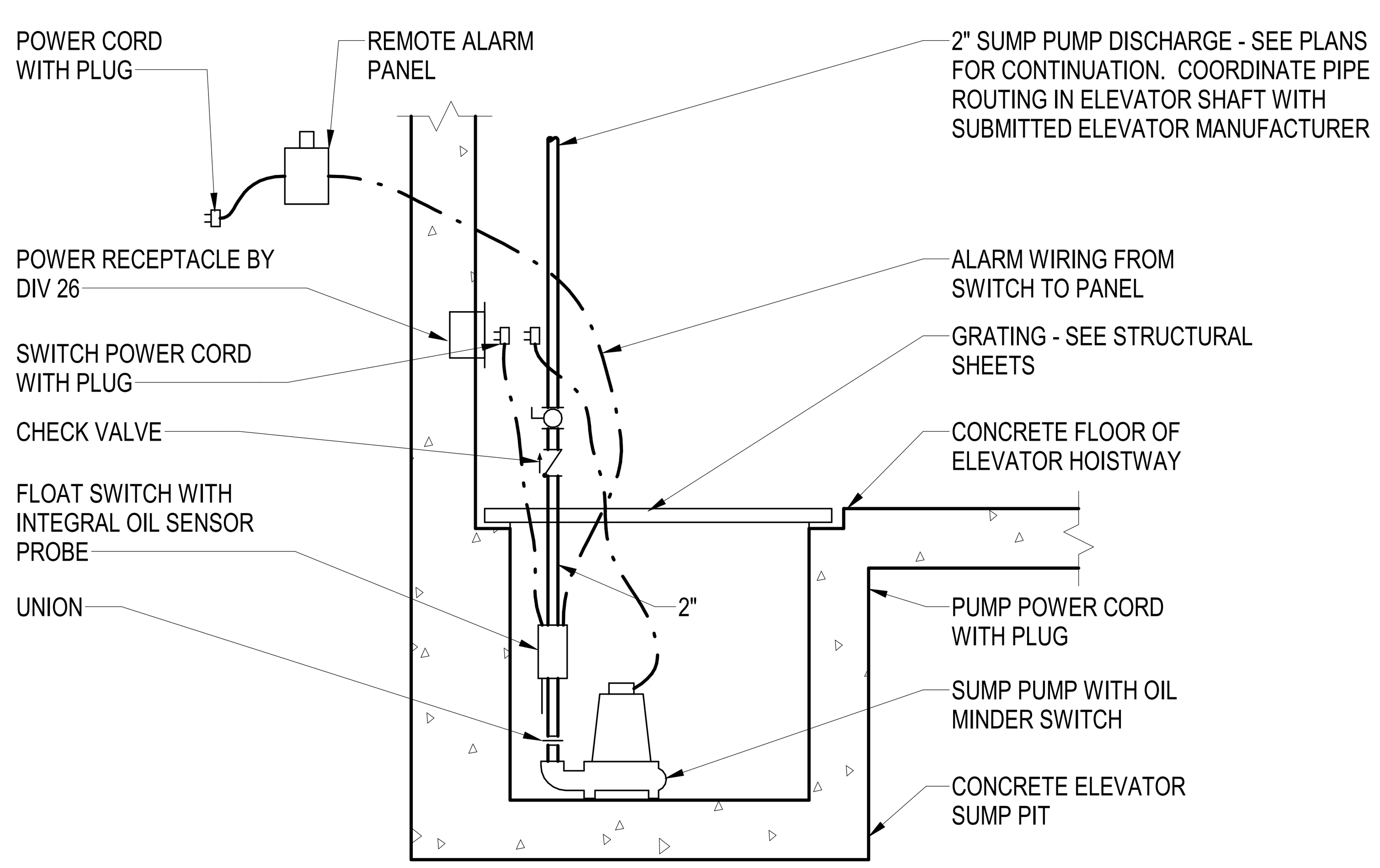
B1 ELECTRIC HOT WATER HEATER DETAIL
SCALE: NOT TO SCALE



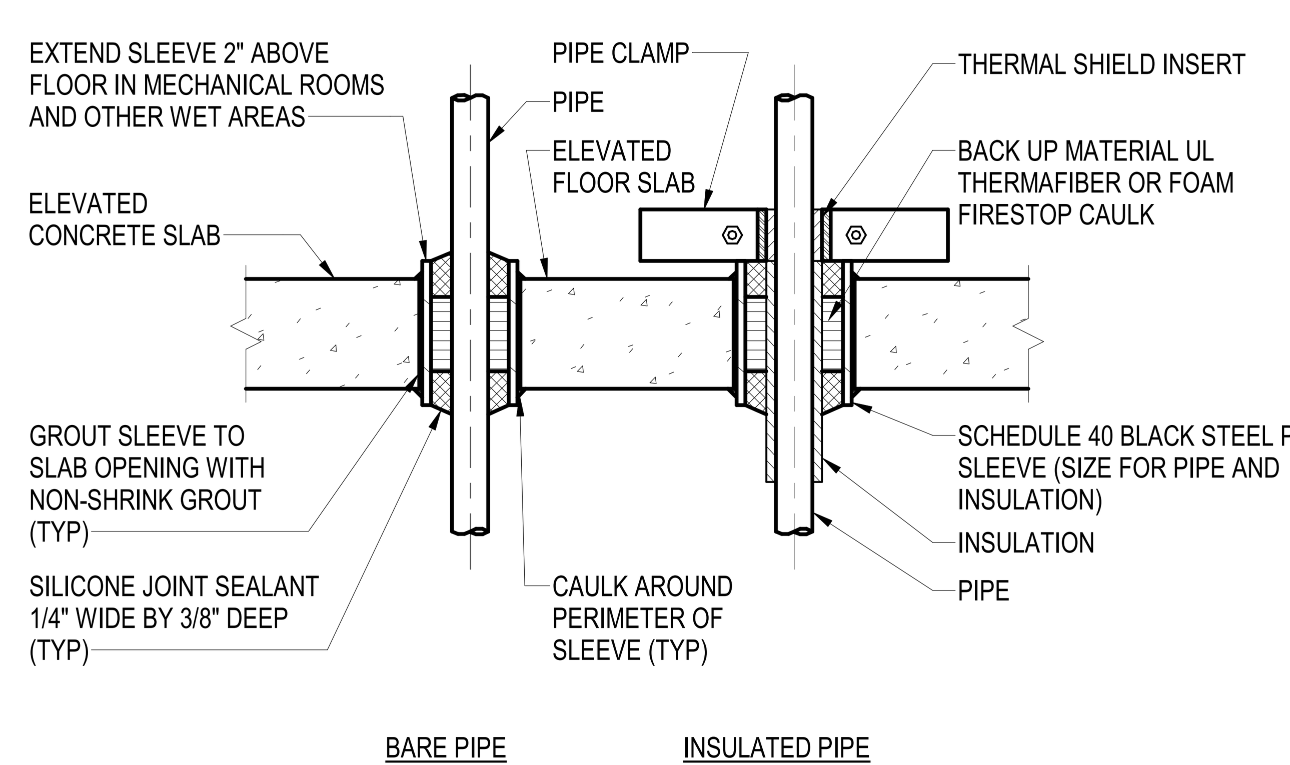
B3 TMV-1 HI-LO MIXING VALVE PIPING SCHEMATIC
SCALE: NOT TO SCALE



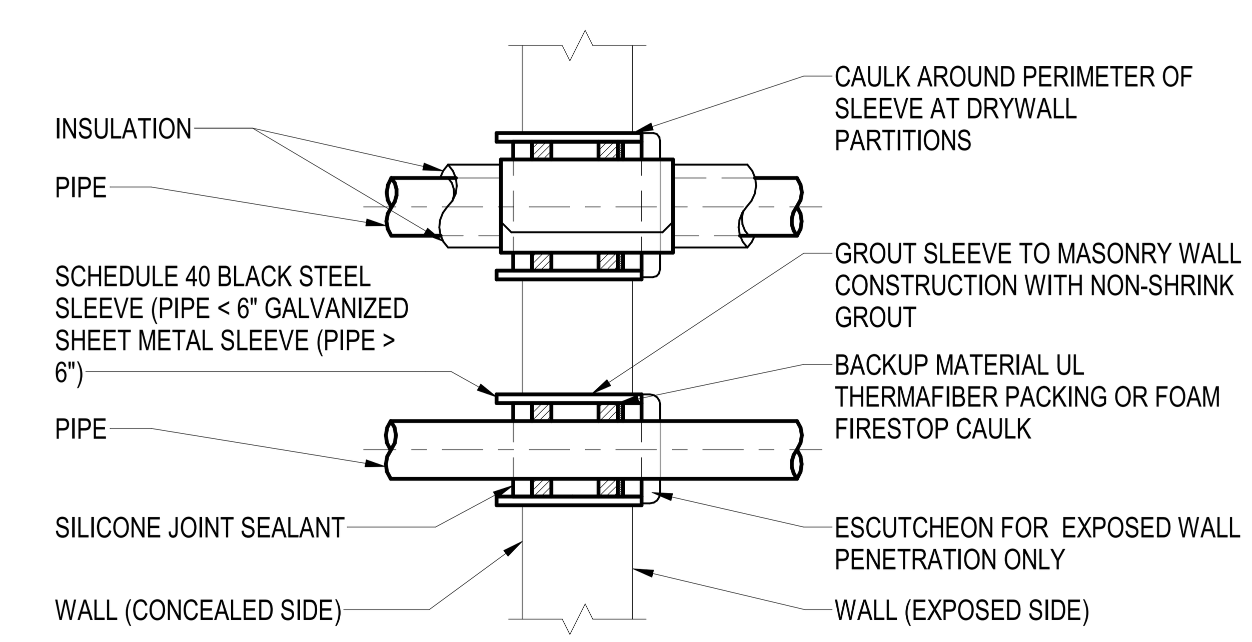
B4 TYPICAL ELECTRONIC TRAP PRIMER ASSEMBLY SCHEMATIC
SCALE: NOT TO SCALE



A1 ELEVATOR SUMP PUMP DETAIL
SCALE: NOT TO SCALE



A3 PIPE SLEEVE THRU ELEVATED FLOOR
SCALE: NOT TO SCALE



A4 PIPE SLEEVE THROUGH WALL
SCALE: NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
PLUMBING DETAILS

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 181 OF 282
P-501
DRAWFORM REVISION: 14 SEP 2023

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PLUMBING FIXTURE ROUGH-IN SCHEDULE

UNIT NO	DESCRIPTION	WASTE	VENT	HW	CW	REMARKS	BASIS OF DESIGN
WC-1	WATER CLOSET	4"	2"	-	1"	FLOOR MOUNTED, FLOOR OUTLET, MANUAL FLUSH VALVE, ADA HEIGHT	AMERICAN STANDARD MADERA 2854.016
WC-2	WATER CLOSET	4"	2"	-	1"	FLOOR MOUNTED, FLOOR OUTLET, MANUAL FLUSH VALVE	AMERICAN STANDARD MADERA 2858.016
U-1	URINAL	2"	1 1/2"	-	3/4"	WALL HUNG, MANUAL FLUSH VALVE, ADA HEIGHT	AMERICAN STANDARD WASHBROOK 6590.501
L-1	LAVATORY, COUNTER	1 1/2"	1 1/2"	1/2"	1/2"	OVAL, STAINLESS STEEL, COUNTERTOP, UNDERMOUNT, MANUAL FAUCET, ADA HEIGHT	ELKAY ASANA ELUH1511
L-2	LAVATORY, WALL HUNG	1 1/2"	1 1/2"	1/2"	1/2"	WALL HUNG, CONCEALED CARRIER, MANUAL FAUCET, ADA HEIGHT	SLOAN SS-3003
S-1	SINK, KITCHENETTE	2"	1 1/2"	1/2"	1/2"	SINGLE-BASIN, STAINLESS STEEL, COUNTERTOP, UNDERMOUNT, ADA HEIGHT	ELKAY ELUHAD 191655
S-3	3-STATION WASH SINK	2"	1 1/2"	3 @ 1/2"	3 @ 1/2"	WALL MOUNTED, STAINLESS STEEL, THREE MANUAL FAUCETS, ADA HEIGHT	ELKAY EWMA6020C
S-2	UTILITY SINK	2"	1 1/2"	1/2"	1/2"	PLASTIC DEEP BASIN W/ STEEL LEGS, MANUAL FAUCET	SWAN MODEL MF-F
MR-1	MOP RECEPTOR	3"	1 1/2"	1/2"	1/2"	FLOOR MOUNTED, 24x24 NEO-ANGLE DROP FRONT TERAZZO, MANUAL WALL FAUCET	FLORESTONE MODEL 96
EW-1	EMERGENCY EYEWASH			1/2"*		WALL MOUNTED W/ COVER (*TEMPERED WATER), WITH EMERGENCY ALARM SYSTEM	HAWS MODEL 7360 BTWC, HAWS 9001.5
EWC-1	ELECTRIC WATER COOLER	1 1/2"	1 1/2"	-	1/2"	WALL MOUNTED, BI-LEVEL, WITH BOTTLE FILLER, FIXTURE CARRIER, NO FILTER	ELKAY EZSTLG8WSSK
HB	HOSE BIBB	-	-	-	3/4"	SURFACE MOUNTED, CHROME FINISH, WHEEL HANDLE	
COB	COFFE MAKER OUTLET BOX	-	-	-	1/2"	RECESSED, WITH INTEGRAL BALL VALVE	
FD-1	FLOOR DRAIN	2"	1 1/2"	-	1/2" TP	SEE DRAIN SCHEDULE	
FD-2	FLOOR DRAIN	3"	1 1/2"	-	1/2" TP	SEE DRAIN SCHEDULE	
FD-3	FLOOR DRAIN	4"	2"	-	1/2" TP	SEE DRAIN SCHEDULE	
FS-1	FLOOR SINK	4"	2"	-	1/2" TP	SEE DRAIN SCHEDULE	

NOTES:
1. FOR FIXTURE, FAUCET, AND/OR FAUCET MOUNTING HEIGHTS, SEE AE002.
2. PROVIDE WITH ASSE 1070 TEMPERATURE LIMITING DEVICE, EITHER INTEGRAL WITH FAUCET OR SUPPLIED ON HOT WATER LINE TO FAUCET.

PUMP SCHEDULE

UNIT NO	SERVES	LOCATION	TYPE	GPM	TOTAL HEAD	MOTOR DATA HP	RPM	VOLTS/PHASE	SUCTION/ DISCHARGE (IN)	BASIS OF DESIGN	NOTES
RP-1	DHW RECIRC	MECH. M106	IN-LINE	3.0	25	1/8	3250	115/1	1/1	TACO GENIE 008	1
SP-1	ELEV PIT	ELEV SHAFT	SUMP	50.0	10	4/10	-	115/1	11.5	ZOELLER N152 OIL GUARD SYSTEM	2

NOTES:
1. PROVIDE STAINLESS STEEL PUMP FOR POTABLE WATER.
2. PROVIDE PACKAGED PUMP SYSTEM WITH OIL SWITCH, FLOAT SWITCH, AND NEMA 4x ALARM AND CONTROL PANEL WITH H-O-A SWITCH, AUDIO AND VISUAL ALARM DEVICES, AND DRY ALARM CONTACT FOR DDC SYSTEM INTERFACE.

BACKFLOW PREVENTER SCHEDULE

UNIT NO	LOCATION	SYSTEM SERVED	TYPE	SIZE	DESIGN FLOW (GPM)	MAX WPD @ DESIGN (PSI)	MAX RELIEF FLOW RATE @ 55 PSI (GPM)	BASIS OF DESIGN	NOTES
BFP-1	INSPECTION 112	BUILDING WATER SERVICE	RPZ	2"	88	14	150	WATTS LP 009 QTS	1
BFP-2	MECH 115	HEATING WATER MAKE-UP	RPZ	1"	15	12	50	WATTS LP 009 QTS	1

NOTES: 1. PROVIDE WITH BALL TYPE ISOLATION VALVES AND AIR GAP FITTING FOR DRAIN.

ELECTRIC WATER HEATER SCHEDULE

UNIT NO	LOCATION	STORAGE GALLONS	CW INLET TEMP °F	HW OUTLET TEMP °F	MIN GPH RECOVERY @95°ΔT	WATTAGE PER ELEMENT	NUMBER OF ELEMENTS	TOTAL INPUT KW	VOLTS / PHASE	BASIS OF DESIGN	NOTES
EWH-1	CUSTODIAL M106	119	45	140	95	6000	2	12.0	460 / 3	LOCHINVAR CHPA120PD	
EWH-2	SHOP 111	20	45	140	95	2000	1	2.0	120 / 1	RHEEM XE20	
EWH-3	OPEN OFF 2ND FLOOR	2.5	45	120		1440	1	1.44	120 / 1	RHEEM XE02	

NOTES: 1.

MIXING VALVE SCHEDULE

UNIT NO	LOCATION	SERVES	INLET	OUTLET	SUPPLY TEMP °F	DESIGN GPM	MIN GPM	PRESSURE DROP (PSIG)	BASIS OF DESIGN	NOTES
TMV-1	MECH M106	DOMESTIC HOT WATER	3/4"	1"	120°	20	1.0	15.0	LEONARD XL32A-LF	1,2
TMV-2	SHOP 111	DOMESTIC HOT WATER	1/2"	1/2"	120°	3	0.3	5.0	LEONARD 170-LF-STSTL EXP	1,2
EMV-1	SHOP 111	EYEWASH	1/2"	1/2"	78°	3		5.0	GUARDIAN G6020	

NOTES:
1. PROVIDE VALVE IN MANUFACTURER'S LOCKABLE, PAINTABLE STEEL CABINET.
2. PROVIDE HIGH-LOW, DUAL MIXING VALVE ASSEMBLY WITH INTEGRAL MANUFACTURER'S OUTLET THERMOMETER.

EXPANSION TANK SCHEDULE

UNIT NO	LOCATION	SERVES	MIN TANK VOLUME (GAL)	ACCEPTANCE VOLUME	DIMENSIONS (IN)	BASIS OF DESIGN	NOTES
EXT-P1	MECH M106	DOMESTIC WATER	6.4	3.2	12ø x 18	AMTROL ST-12C-DD	1

NOTES: 1. PROVIDE DIAPHRAGM TANK LISTED FOR USE WITH POTABLE WATER.

AUTOMATIC TRAP PRIMER ASSEMBLY SCHEDULE

UNIT NO	LOCATION	MOUNTING	NO OF MANIFOLD OPENINGS	NO OF DESIGN CONNECTIONS	CONN SIZE	VOLTS / PHASE	BASIS OF DESIGN	NOTES
TPA-1	CUSTODIAL 107	RECESSED	4	3	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2
TPA-2	INSPECTION 112	RECESSED	4	1	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2
TPA-3	MECH 115	RECESSED	4	1	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2
TPA-4	FIRE PUMP M107	RECESSED	4	1	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2
TPA-5	MECH M106	RECESSED	4	1	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2
TPA-6	MECH MEZZ M107	RECESSED	4	1	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2
TPA-7	WOMEN 203	RECESSED	4	1	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2
TPA-8	MEZZANINE M201	RECESSED	4	1	1/2"	120 / 1	PRECISION PLUMBING PRODUCTS PT-4	1, 2

NOTES:
1. CAP UNUSED OUTLETS.
2. PROVIDE WITH MANUFACTURER'S LOCKABLE, PAINTABLE STEEL PANEL ACCESS DOOR.

DRAIN SCHEDULE

UNIT NO	TYPE	DESCRIPTION	PIPE SIZE	STRAINER DIMENSIONS	BASIS OF DESIGN	NOTES
FD-1	FLOOR DRAIN	SQUARE DRAIN FOR FINISHED SPACES	2"	6" x 6"	ZURN EZ-PV2-S6	1,3
FD-2	FLOOR DRAIN	ROUND DRAIN FOR UNFINISHED SPACES W/ FUNNEL	3"	7ø	ZURN Z415E	1,2
FD-3	FLOOR DRAIN	ROUND DRAIN FOR UNFINISHED SPACES W/ FUNNEL	4"	7ø	ZURN Z415E	1,2
FS-1	FLOOR SINK	1/2" GRATE, 6" DEEP	4"	6" x 6"	ZURN	1,2

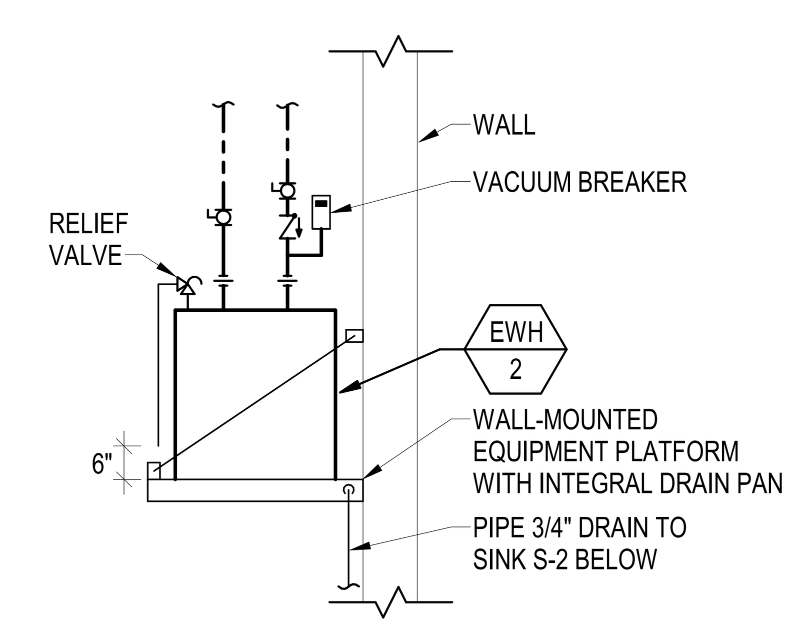
NOTES:
1. PROVIDE TRAP PRIMER CONNECTION TO FLOOR DRAIN OR FLOOR SINK.
2. PROVIDE WITH SEDIMENT BUCKET.
3. PROVIDE WITH CHROME PLATED FINISH.

RADON FAN SCHEDULE (OPTION 3)

UNIT NO	SERVES	CFM	ESP IN WC	DRIVE TYPE	FAN TYPE	FAN RPM	WATTS	VOLTS / PHASE	BASIS OF DESIGN	NOTES	ACCESSORIES
RF-1	RADON MITIGATION	200	2	DIRECT	IN-LINE	4099	175	120 / 1	FANTECH 4EC-4	1,2	A, B, C

NOTES:
1. PROVIDE UNDER OPTION 3 ONLY
2. TYPICAL FOR 3 FANS
3. FOR DDC SYSTEM FAN MONITORING AND ALARM CONTROL POINTS, SEE C2/M-705.

ACCESSORIES:
A. MFR INTEGRAL FAN SPEED CONTROLLER.
B. MFR ANTI-VIBRATION COUPLERS FOR 4" PVC PIPE.
C. WALL MOUNTING BRACKET FOR VERTICAL INSTALLATION.



A3 **ELECTRIC WATER HEATER EWH-2 PIPING DETAIL**

SCALE: NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

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FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
PLUMBING SCHEDULES AND DETAIL

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 182 OF 282
P-601
DRAWFORM REVISION: 14 SEP 2023

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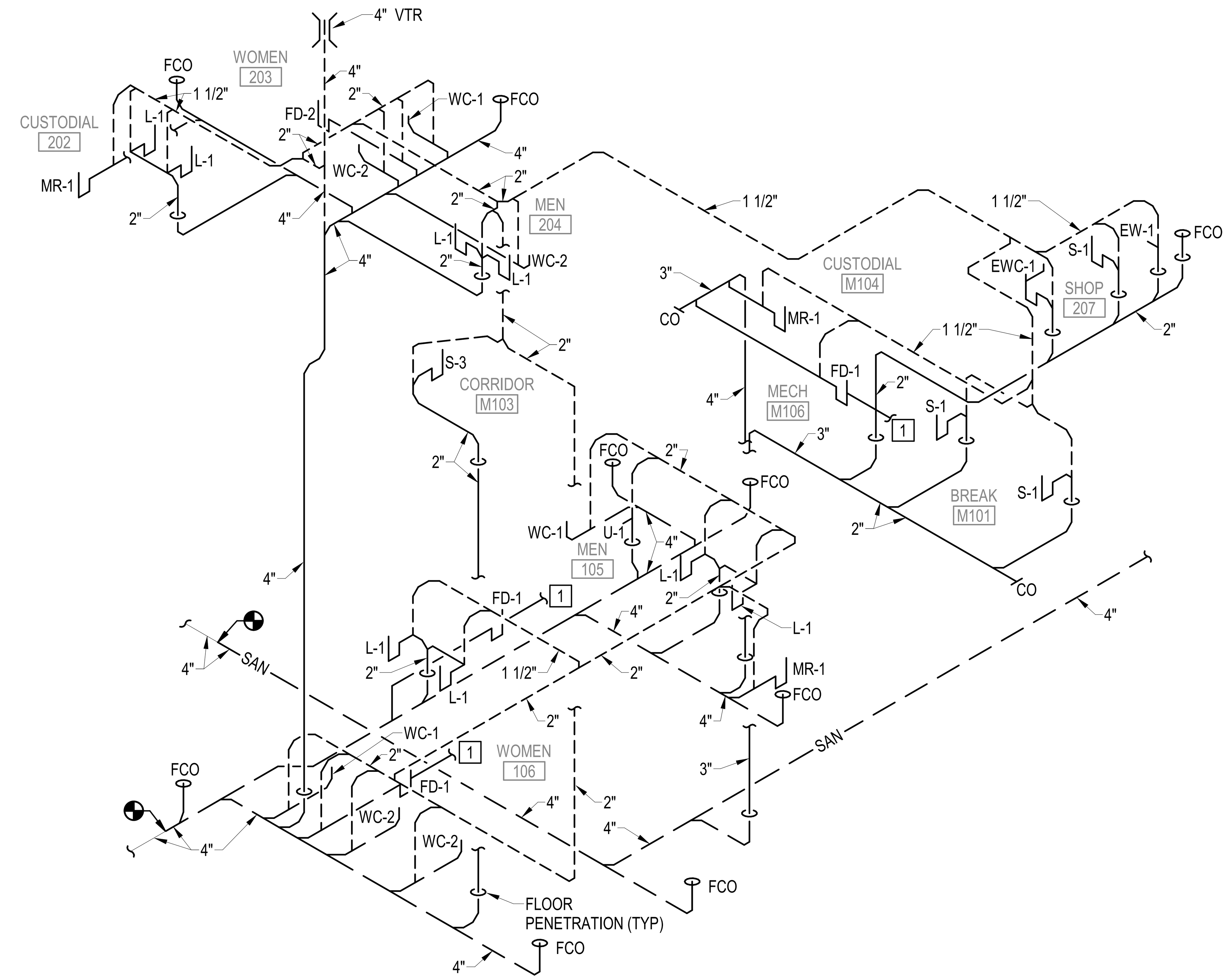

C1 SANITARY AND VENT PLUMBING RISER DIAGRAM

SCALE: NOT TO SCALE

GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.

KEYNOTES (THIS SHEET ONLY)

1 1/2" TRAP PRIMER CONNECTION. SEE PLANS FOR PIPE ROUTING TO TRAP PRIMER ASSEMBLY.

SYM	DESCRIPTION	DATE	APPR
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RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

SEAL

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ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME

PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

PLUMBING RISER DIAGRAM

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 183 OF 282

P-701

DRAWFORM REVISION: 14 SEP 2023

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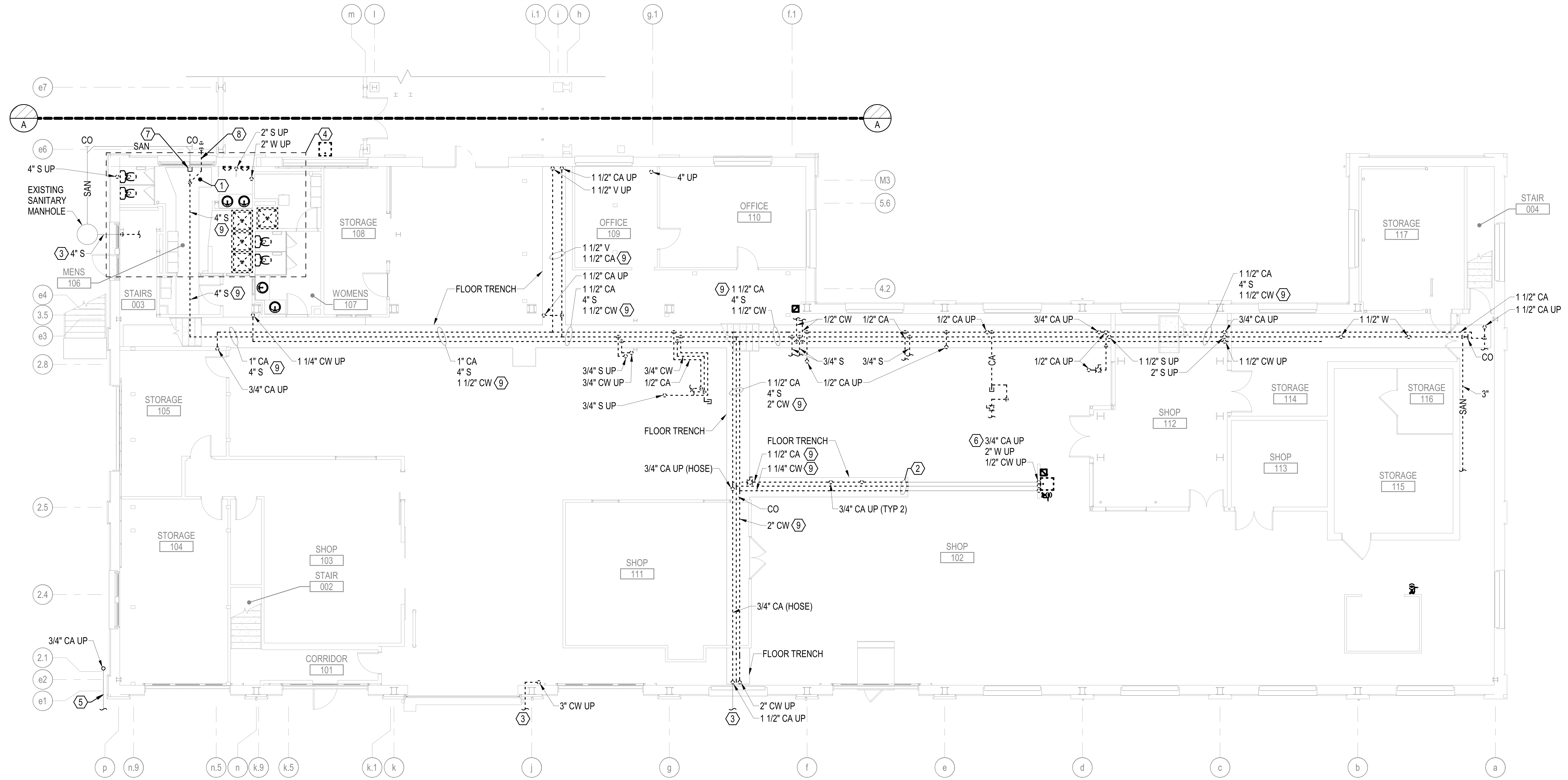


B1 UNDERSLAB AND FIRST FLOOR TRENCH PLUMBING REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

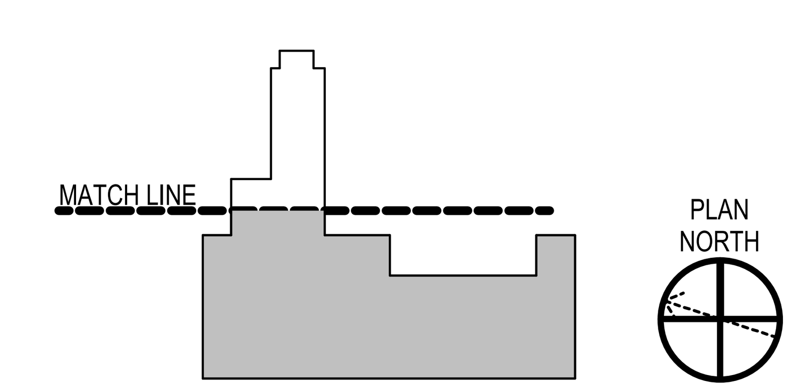
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. LIMITED INFORMATION WAS AVAILABLE DOCUMENTING CURRENT PIPING WITHIN FLOOR TRENCHES. THESE CONTAIN ACTIVE AS WELL AS ABANDONED PIPING. REFERENCES TO THE KINDS OF PIPING AND PIPE SIZES ARE INDICATED TO PROVIDE A REPRESENTATIVE LIST OF HVAC AND PROCESS PIPING BUT NOT NECESSARILY ALL PIPING. SANITARY, WASTE, VENT, DOMESTIC WATER AND COMPRESSED AIR ARE SHOWN ON PLUMBING REMOVALS SHEETS. REMOVE EQUIPMENT AND PIPING WITHIN THE FLOOR TRENCHES, UNLESS OTHERWISE NOTED. FOR BIDDING PURPOSES, ASSUME 1200 FEET OF 4" METAL STEAM AND HEATING WATER TYPE PIPE TO BE REMOVED FROM THE TRENCHES.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVALS WITH HAZARDOUS MATERIALS REMOVALS SHEETS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE FLOOR TRENCH SUMP PUMP ASSEMBLY, PIPING, AND ASSOCIATED CONTROLS AND WIRING.
2. CUT AND CAP PIPING THAT CONTINUES BEYOND TRENCH UNDERSLAB (SAN, CW, CA).
3. SEE CIVIL SHEET CX101 FOR CONTINUATION.
4. RECORD OF UNDERSLAB SANITARY, WASTE, AND VENT PIPING IS UNAVAILABLE. REMOVE UNDERSLAB PIPING THROUGHOUT EXISTING MENS AND WOMENS ROOMS. REMOVE SANITARY, WASTE, AND VENT PIPING IN FLOOR TRENCH OR PASSING ACROSS FLOOR TRENCH. COORDINATE REMOVALS WITH FLOOR SAW CUTTING CONSTRAINTS IN PLACE TO PRESERVE EXISTING HISTORIC PARTITION WALLS. COORDINATE WITH STRUCTURAL AND ARCHITECTURAL DRAWINGS.
5. EXISTING 3/4" COMPRESSED AIR FROM BUILDING 60 TO BUILDING 292.
6. CUT AND CAP PIPING BELOW SLAB SURFACE, COORDINATE PATCHING AND REPAIR OF FLOOR SLAB.
7. CUT AND TEMPORARILY CAP FOR REUSE.
8. REMOVE VENT. COORDINATE PATCHING AND REPAIR OF WALL PENETRATION.
9. REMOVE SANITARY WASTE, VENT, DOMESTIC WATER AND COMPRESSED AIR, PIPING THROUGHOUT FLOOR TRENCH. ELLIPSES SHOWN ALONG THE FLOOR, WITH LEADERS FROM PIPE TYPES, AND SIZES CHARACTERIZED THE TYPE AND SIZE BY PIPES INTENT ACCORDING TO BRACING RECORDS. SEE SHEET MD104 FOR IDENTIFICATION OF HVAC TYPE PIPING REMOVALS REQUIREMENTS.

KEY PLAN



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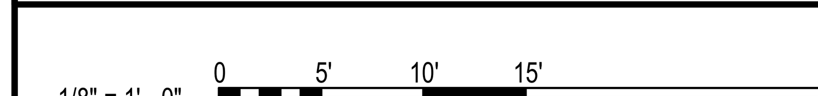

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SYM	DESCRIPTION	DATE	APPR
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NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC

DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FIRE PROTECTION S. RUCINSKI

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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
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ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

UNDERSLAB AND FIRST FLOOR TRENCH PLUMBING REMOVALS PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.

SHEET 163 OF 282

PD101

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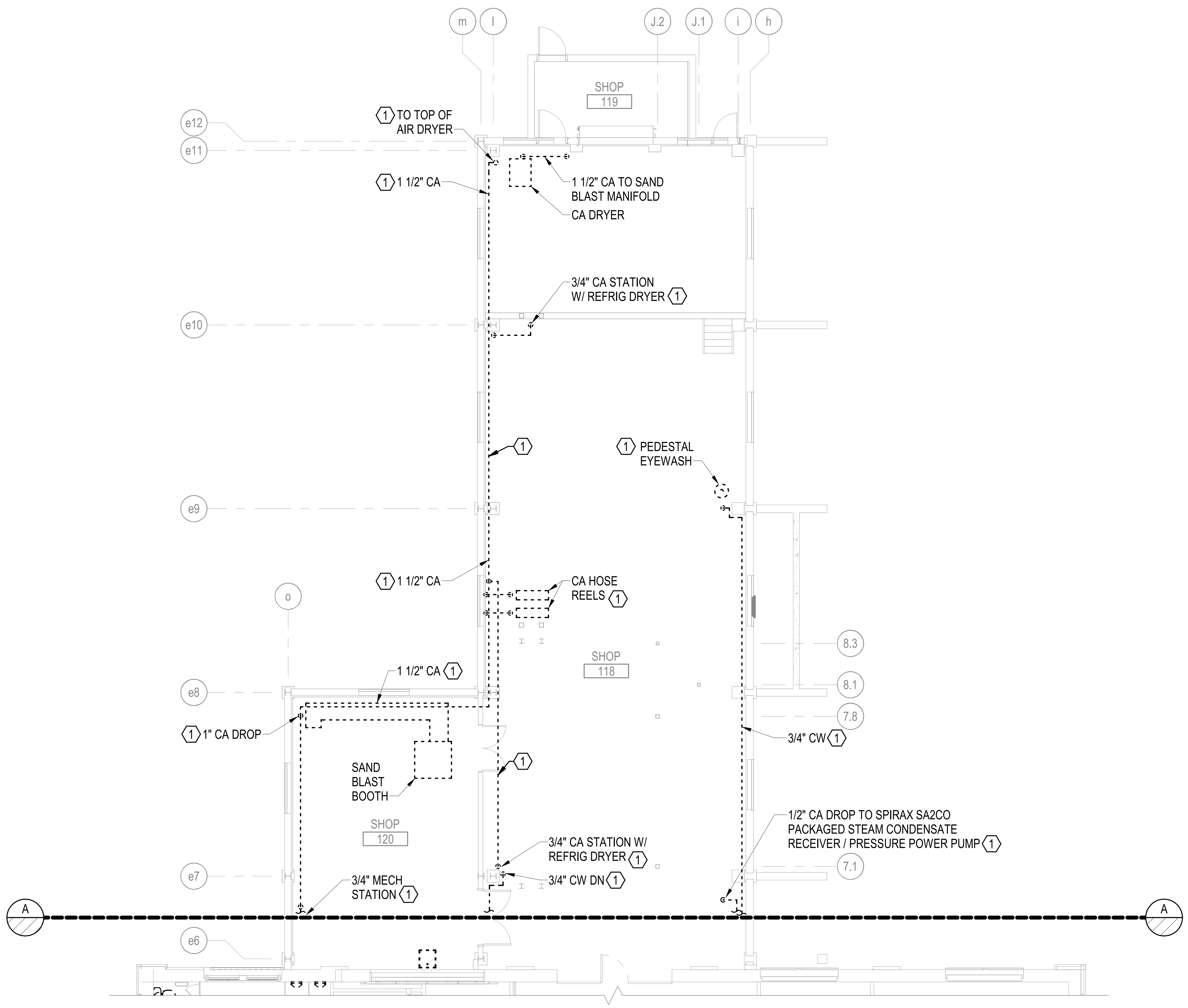


A1 FIRST FLOOR PLUMBING REMOVALS PLAN - EAST

SCALE: 1/8" = 1'-0"

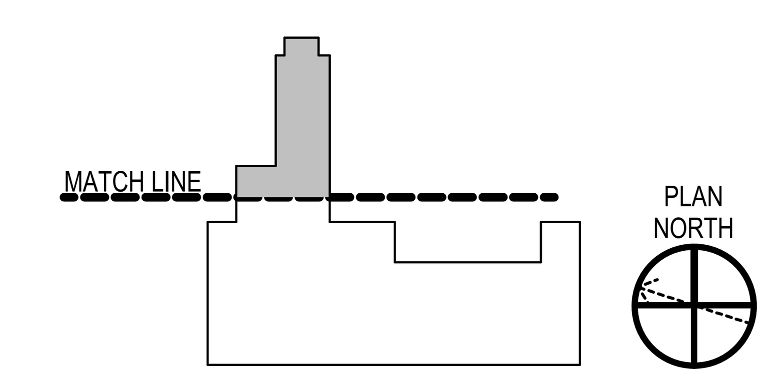
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. CUT AND CAP PIPING THAT CONTINUES BEYOND TRENCH UNDERSLAB (SAN, CW, CA).
3. SEE CIVIL SHEET CX101 FOR CONTINUATION.
4. RECORD OF UNDERSLAB SANITARY, WASTE, AND VENT PIPING IS UNAVAILABLE. REMOVE UNDERSLAB PIPING THROUGHOUT EXISTING MENS AND WOMENS ROOMS. REMOVE SANITARY, WASTE, AND VENT PIPING IN FLOOR TRENCH OR PASSING ACROSS FLOOR TRENCH. COORDINATE REMOVALS WITH FLOOR SAW CUTTING CONSTRAINTS IN PLACE TO PRESERVE EXISTING HISTORIC PARTITION WALLS.
5. EXISTING 3/4" COMPRESSED AIR FROM BUILDING 60 TO BUILDING 292.
6. CUT AND CAP PIPING BELOW SLAB SURFACE. COORDINATE MATCHING AND REPAIR OF FLOOR SLAB

REMOVALS KEYNOTES (THIS SHEET ONLY)

1 REMOVE FIXTURES AND PIPING.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

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SYM	DESCRIPTION	DATE	APPR
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NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

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FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR PLUMBING REMOVALS PLAN - EAST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 164 OF 282

PD102

DRAWFORM REVISION: 14 SEP 2023

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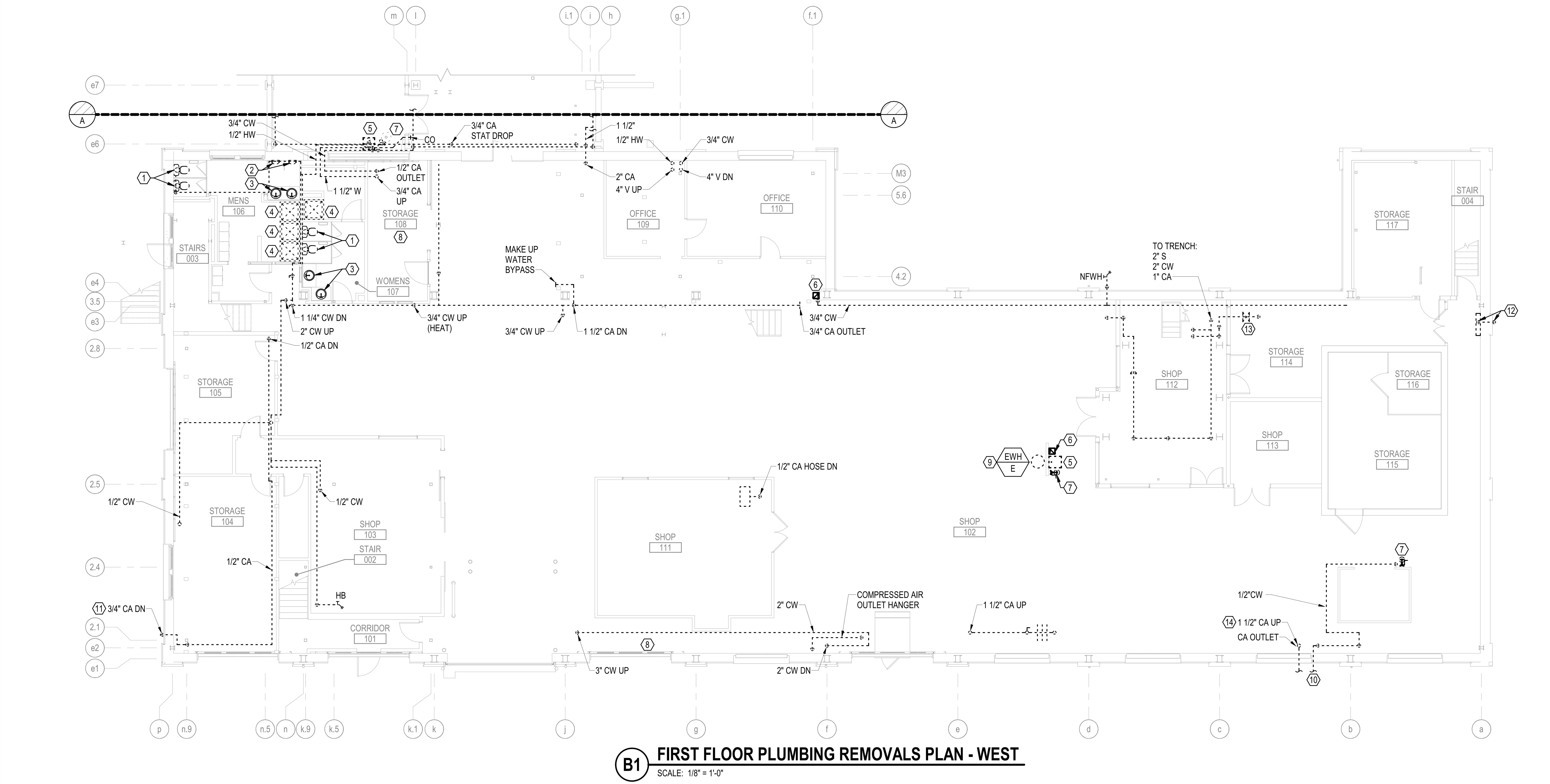


B1 FIRST FLOOR PLUMBING REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

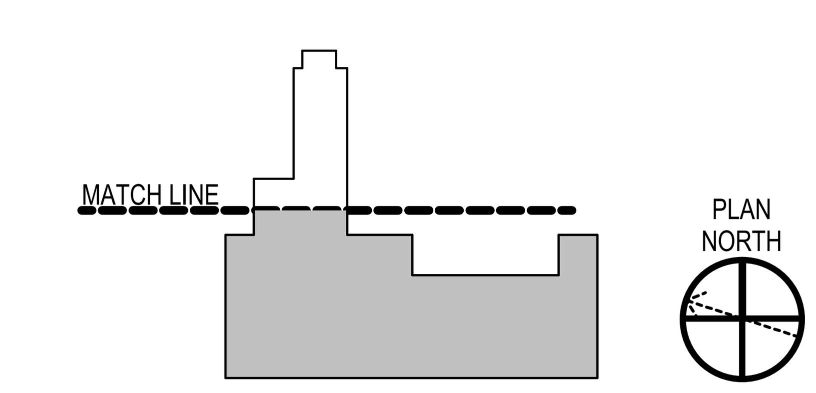
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. LIMITED INFORMATION WAS AVAILABLE DOCUMENTING CURRENT PIPING WITHIN FLOOR TRENCHES. THESE CONTAIN ACTIVE AS WELL AS ABANDONED PIPING. REFERENCES TO THE KINDS OF PIPING AND PIPE SIZES ARE INDICATED TO PROVIDE A REPRESENTATIVE LIST OF HVAC AND PROCESS PIPING BUT NOT NECESSARILY ALL PIPING. SANITARY, WASTE, VENT, DOMESTIC WATER AND COMPRESSED AIR ARE SHOWN ON PLUMBING REMOVALS SHEETS. REMOVE EQUIPMENT AND PIPING WITHIN THE FLOOR TRENCHES, UNLESS OTHERWISE NOTED. FOR BIDDING PURPOSES, ASSUME 1200 FEET OF 4" METAL STEAM AND HEATING WATER TYPE PIPE TO BE REMOVED FROM THE TRENCHES.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVALS WITH HAZARDOUS MATERIALS REMOVALS SHEETS.
7. REMOVE PLUMBING VENTS AND DOMESTIC WATER PIPING MAINS SERVING MENS/WOMENS RESTROOMS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE WATER CLOSET AND ASSOCIATED SANITARY, VENT, AND DOMESTIC WATER PIPING.
2. REMOVE URINAL AND ASSOCIATED SANITARY, VENT AND DOMESTIC WATER PIPING.
3. REMOVE LAVATORY AND ACCLIMATED WASTE, VENT AND DOMESTIC WATER PIPING.
4. REMOVE SHOWER AND ASSOCIATED WASTE, WENT, AND DOMESTIC WATER PIPING.
5. REMOVE SINK AND ASSOCIATED WASTE, VENT, AND DOMESTIC WASTER PIPING.
6. REMOVE ELECTRIC WATER COOLER AND ASSOCIATED WASTE WENT, AND DOMESTIC WATER PIPING.
7. REMOVE EMERGENCY EYEWASH AND ASSOCIATED WASTE VENT AND DOMESTIC WATER PIPING.
8. REMOVE ELECTRIC WATER HEATER AND ASSOCIATED VENT AND DOMESTIC WATER PIPING.
9. REMOVE POINT OF USE ELECTRIC WATER HEATER AND ASSOCIATED VENT AND DOMESTIC WATER PIPING.
10. REMOVE DOMESTIC WATER ENTRANCE PIPING. CAP AT WALL. PROVIDE METAL EMBOSSED TAG SECURED TO CAPPED PIPE.
11. REMOVE COMPRESSED PIPING TO WITH IN 6" OF GROUND LEVELS AND CAP. PROVIDE ALUMINUM TAG EMBOSSED WITH NOTE (ABND C.A. MONTH/YEAR).
12. REMOVE ABANDONED CA SERVICE ENTRANCE MOUNTED ALONG WALL TO INCLUDE BUT NOT LIMITED TO BREACH REGULATORY SAFETY VALVE CA SPEC AND PIPING. COORDINATE PATCHING OF WALL AND TRENCH.
13. REMOVE MULTIPLE CW AND HW RISERS SERVING 1ST FLOOR MEZZ AND 2ND FLOOR PLUMBING FIXTURES.
14. REMOVE CW PIPING AND RADON VALVE BACK TO WALL PENETRATION CAP. PROVIDE EMBOSSED ALUMINUM TAG WITH THE PLUMBING STATEMENT.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER, STATE OF MAINE 10-01-25

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC

DESIGNER	RNC
DRAWN BY	SRW
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
FIRST FLOOR PLUMBING REMOVALS PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 165 OF 282
PD103

DRAWFORM REVISION: 14 SEP 2023

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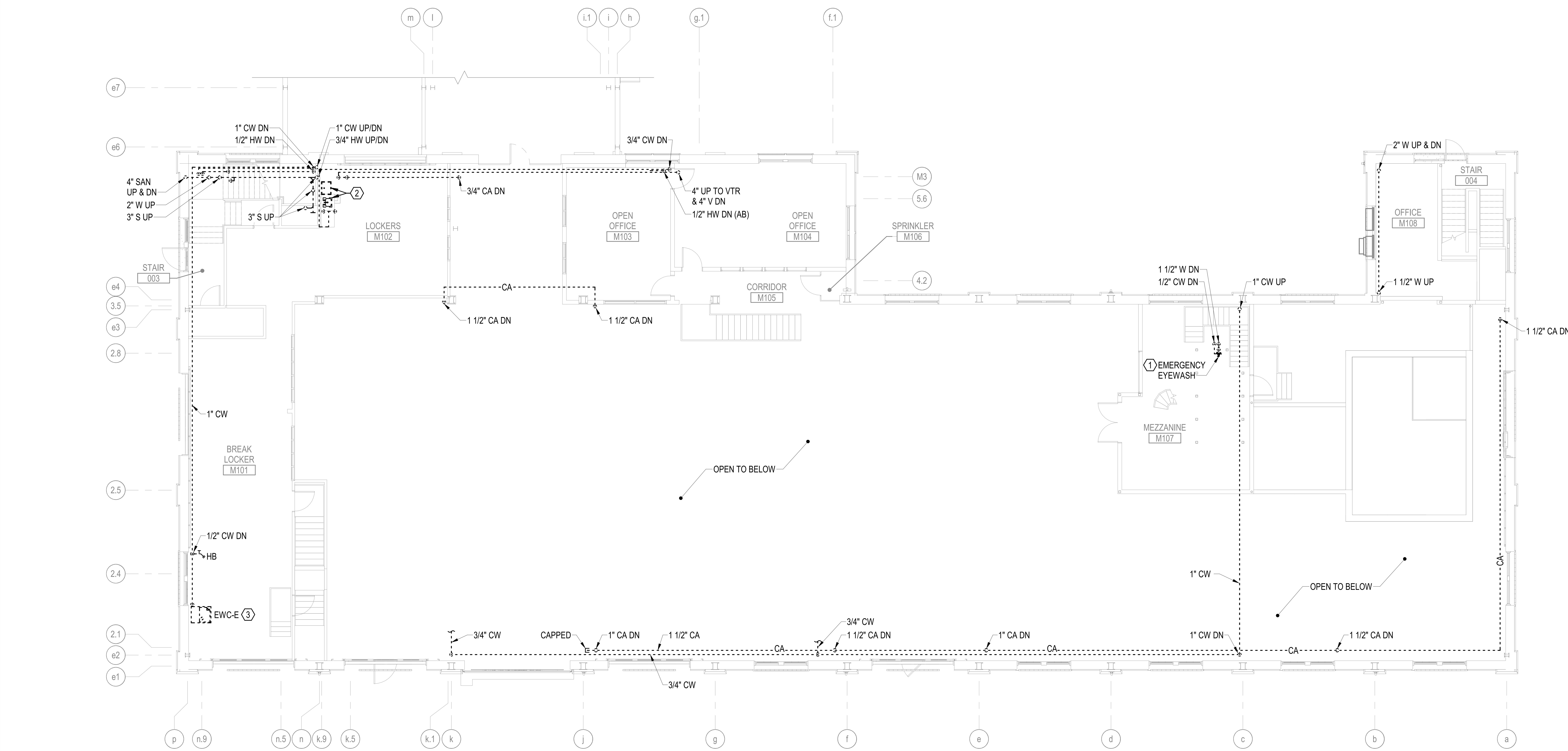


B1 FIRST FLOOR MEZZANINE PLUMBING REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

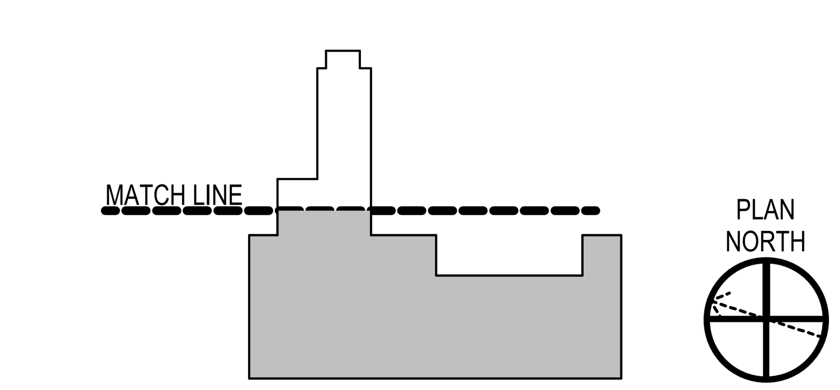
GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. LIMITED INFORMATION WAS AVAILABLE DOCUMENTING CURRENT PIPING WITHIN FLOOR TRENCHES. THESE CONTAIN ACTIVE AS WELL AS ABANDONED PIPING. REFERENCES TO THE KINDS OF PIPING AND PIPE SIZES ARE INDICATED TO PROVIDE A REPRESENTATIVE LIST OF HVAC AND PROCESS PIPING BUT NOT NECESSARILY ALL PIPING. SANITARY, WASTE, VENT, DOMESTIC WATER AND COMPRESSED AIR ARE SHOWN ON PLUMBING REMOVALS SHEETS. REMOVE EQUIPMENT AND PIPING WITHIN THE FLOOR TRENCHES, UNLESS OTHERWISE NOTED. FOR BIDDING PURPOSES, ASSUME 1200 FEET OF 4" METAL STEAM AND HEATING WATER TYPE PIPE TO BE REMOVED FROM THE TRENCHES.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE EMERGENCY EYEWASH AND ASSOCIATED WASTE, VENT, AND DOMESTIC WATER PIPING ASSEMBLIES.
2. REMOVE SINK AND ASSOCIATED WASTE, VENT, AND DOMESTIC WATER PIPING ASSEMBLIES.
3. REMOVE ELECTRIC WATER COOLER AND ASSOCIATED WASTE, VENT, AND DOMESTIC WATER PIPING ASSEMBLIES. THE CONTRACTOR MUST PROVIDE THE MIDLANT ODS FORMS TO DOCUMENT REFRIGERANT RECOVERY, DISPOSAL, AND CHARGE ACTIVITIES. SEE SPECIFICATION SECTION 02 41 00 "DEMOLITION AND DECONSTRUCTION".

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

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SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	SRW
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MEZZANINE PLUMBING REMOVALS PLAN - WEST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 166 OF 282

PD104

DRAWFORM REVISION: 14 SEP 2023

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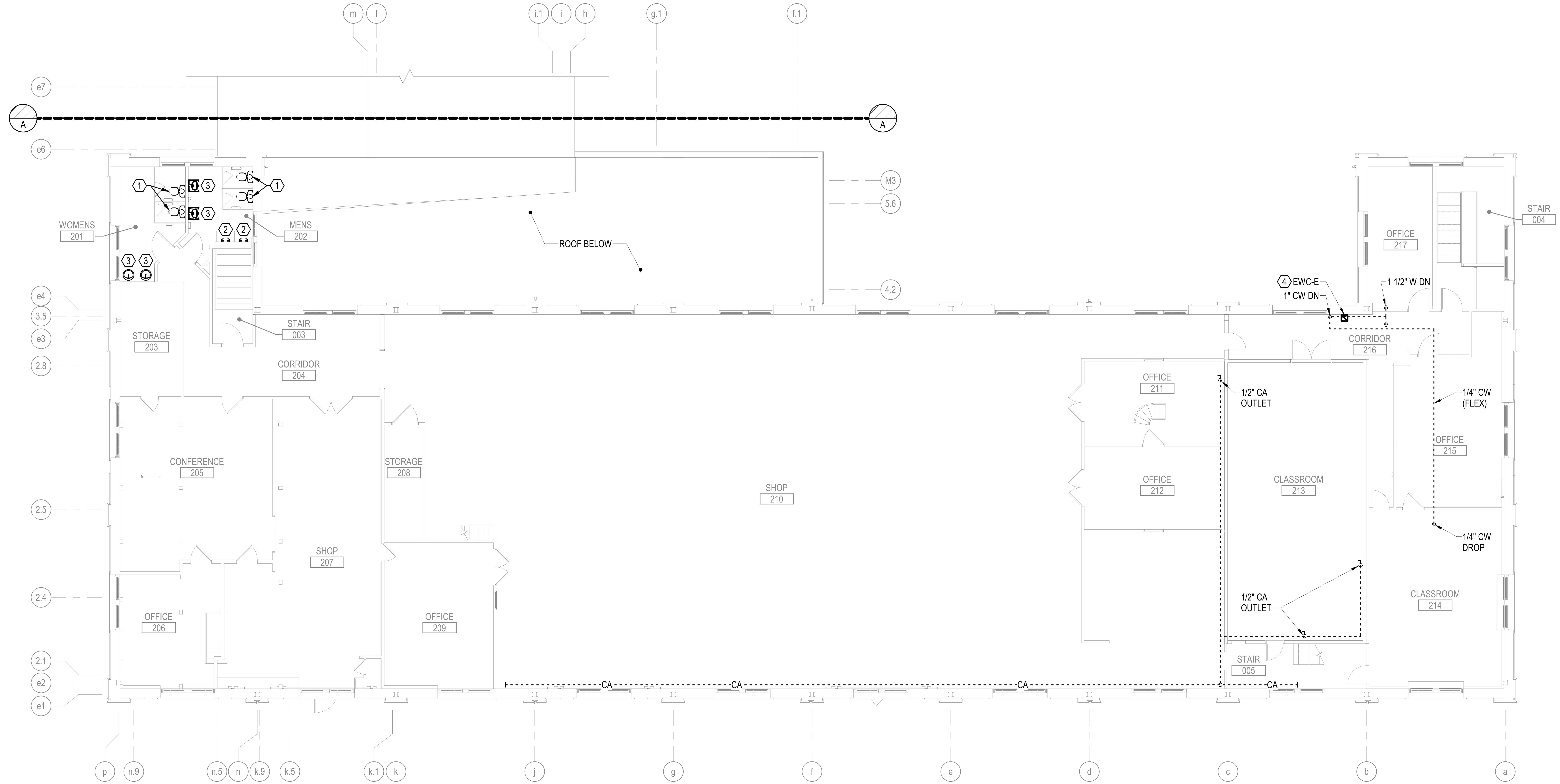

B1 SECOND FLOOR PLUMBING REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
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3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE WATER CLOSET AND ASSOCIATED SANITARY, VENT, AND DOMESTIC WATER PIPING.
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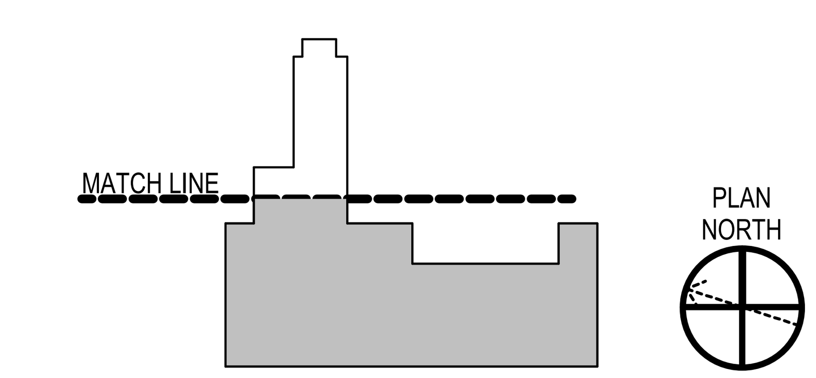
3. REMOVE LAVATORY AND ASSOCIATED WASTE, VENT, AND DOMESTIC WATER PIPING.
4. REMOVE ELECTRIC WATER COOLER AND ASSOCIATED WASTE, VENT, AND DOMESTIC WATER PIPING ASSEMBLIES. THE CONTRACTOR MUST PROVIDE THE MIDLANT ODS FORMS TO DOCUMENT REFRIGERANT RECOVERY, DISPOSAL, AND CHARGE ACTIVITIES. SEE SPECIFICATION SECTION 02 41 00 "DEMOLITION AND DECONSTRUCTION".

KEY PLAN



GRAPHIC SCALE

1/8" = 1'-0" 0 5' 10' 15' 30'

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SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

SECOND FLOOR PLUMBING REMOVALS PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 167 OF 282
PD105

DRAWFORM REVISION: 14 SEP 2023

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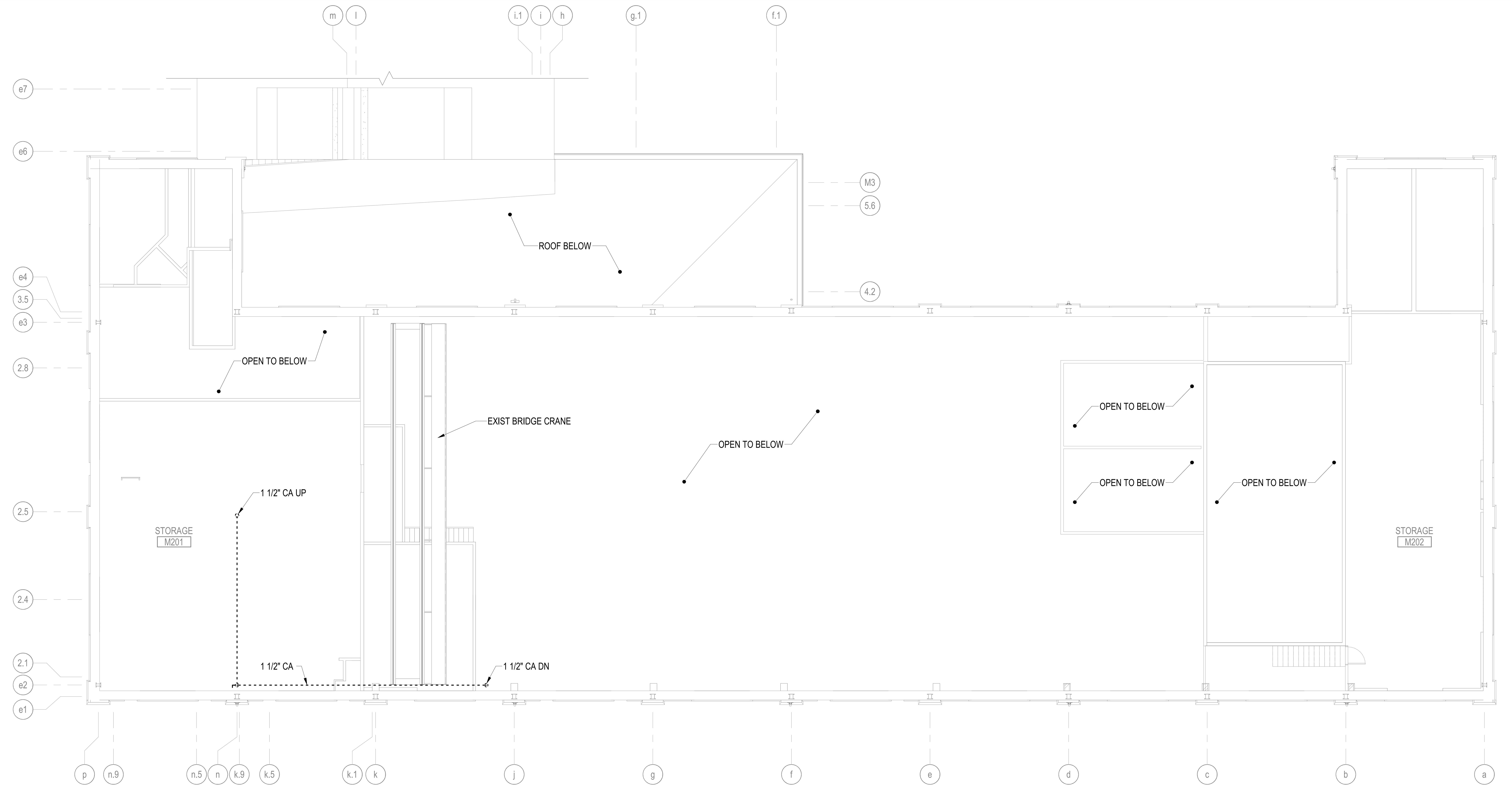

B1 SECOND FLOOR MEZZANINE PLUMBING REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE P-001 FOR PLUMBING LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
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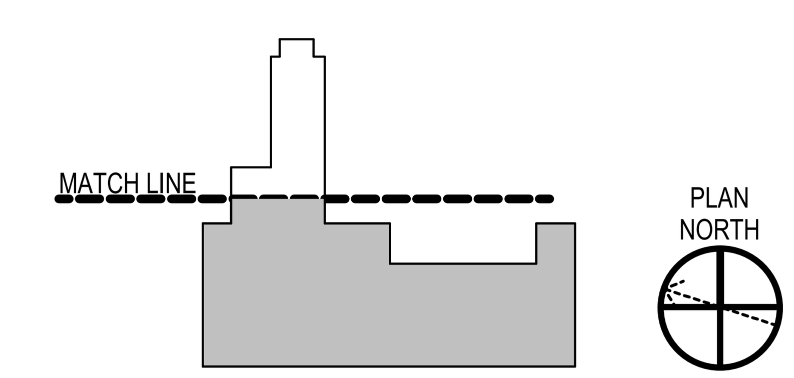
6. COORDINATE REMOVELS WITH HAZARDOUS MATERIALS REMOVALS SHEETS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

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SYM / DESCRIPTION / DATE / APPR	FINAL SUBMISSION — 10/01/2025 — RSH
	NAVFAC
SEAL	STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25
	OAK POINT ASSOCIATES
APPROVED	FOR COMMANDER NAVFAC
ACTIVITY	
SATISFACTORY TO DATE	
DESIGNER	RNC
DRAWN BY	SRW
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI
	DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
	B60 REPAIRS
	SECOND FLOOR MEZZANINE PLUMBING REMOVALS PLAN
EPROJECT NO.:	1740579
NAVFAC DRAWING NO.	
SHEET	168 OF 282
	PD106

DRAWFORM REVISION: 14 SEP 2023

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MECHANICAL SYMBOLS LEGEND

ANNOTATIONS

- ERV 1 — EQUIPMENT ABBREVIATION (SEE MECHANICAL ABBREVIATIONS LIST)
- EQUIPMENT SEQUENCE NUMBER (SEE SCHEDULES)
- S-1 100 — AIR INLET OR OUTLET WITH CFM
- RH-1 2.2 — FINTUBE DESIGNATION (SEE SCHEDULES)
- MBH
- 1.2 — GPM SETTING FOR BALANCING VALVE
- 1 — KEYNOTE
- 1 — REMOVALS KEYNOTE
- CONNECT TO EXISTING
- DIRECTION OF AIRFLOW (EXHAUST OR RETURN)
- DIRECTION OF AIRFLOW (SUPPLY)

DUCTWORK

- RETURN GRILLE/REGISTER
- SUPPLY DIFFUSER/GRILLE/REGISTER
- EXHAUST GRILLE/REGISTER
- TRANSFER GRILLE
- SIDEWALL REGISTER/GRILLE
- AD — ACCESS DOOR
- DUCT
- DUCT WITH FLEXIBLE CONNECTION
- UP DN — DUCT RISE / DROP
- RECTANGULAR / ROUND RETURN DUCT UP
- RECTANGULAR / ROUND RETURN DUCT DOWN
- RECTANGULAR / ROUND SUPPLY DUCT UP
- RECTANGULAR / ROUND SUPPLY DUCT DOWN
- RECTANGULAR / ROUND EXHAUST DUCT UP
- RECTANGULAR / ROUND EXHAUST DUCT DOWN
- RECTANGULAR ELBOW WITH TURNING VANES
- FLEXIBLE DUCT
- MANUAL DAMPER
- FD — FIRE DAMPER
- M — MOTORIZED DAMPER, PARALLEL BLADE
- M — MOTORIZED DAMPER, OPPOSED BLADE
- H C — HEATING COIL
- DX — DIRECT EXPANSION COOLING COIL

PIPING AND VALVES

- PIPE ELBOW DOWN
- PIPE ELBOW UP, PIPE UP AND DOWN
- PIPE TEE DOWN
- PIPE TEE UP, PIPE UP AND DOWN
- PIPE CAP
- DIRECTION OF FLOW
- STRAINER
- BALL VALVE
- BUTTERFLY VALVE
- UNION
- GATE VALVE
- PRESSURE GAUGE
- CHECK VALVE
- CALIBRATED BALANCING VALVE WITH POSITIVE SHUTOFF
- AUTOMATIC FLOW CONTROL VALVE
- P — PIPE PITCH DOWN
- PRESSURE RELIEF VALVE
- 2-WAY AUTOMATIC CONTROL VALVE
- 3-WAY AUTOMATIC CONTROL VALVE
- PRESSURE REDUCING VALVE
- OS&Y VALVE
- THERMOMETER
- AIR VENT, AUTOMATIC
- AIR VENT, MANUAL
- CONCENTRIC REDUCER
- ECCENTRIC REDUCER
- VACUUM BREAKER
- FLEXIBLE CONNECTOR
- IN-LINE PUMP
- AQUASTAT
- AIR SEPARATOR

CONTROLS AND METERING

- OS&Y VALVE IN VERTICAL
- BALL VALVE IN VERTICAL
- TS — TEMPERATURE SENSOR
- LT — LOW TEMPERATURE FREEZESTAT
- OC — OCCUPANCY SENSOR
- PS — PRESSURE SWITCH
- FS — FLOW SWITCH
- FM — FLOW METER
- CS CS — CURRENT SENSOR
- DP — DIFFERENTIAL PRESSURE SWITCH
- DPS — DIFFERENTIAL PRESSURE SENSOR
- VFD — VARIABLE FREQUENCY DRIVE
- $ — MANUAL SWITCH
- EAS — EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH
- ES — END SWITCH
- UN-OCC — UN-OCCUPIED OVERRIDE BUTTON
- AFMS — OUTSIDE AIRFLOW MEASURING STATION
- SD — DUCT MOUNTED SMOKE DETECTOR
- S S — START/STOP CONTROLLER
- T — AVERAGING DUCT MOUNTED TEMPERATURE SENSOR
- T — PROBE TYPE DUCT MOUNTED TEMPERATURE SENSOR
- T — FLUID TEMPERATURE SENSOR WITH WELL
- T — SPACE TEMPERATURE SENSOR WITH USER ADJUSTABLE SETPOINTS AND OCCUPANCY OVERRIDE
- T OA — WALL MOUNTED OUTSIDE AIR TEMPERATURE SENSOR
- T — WALL MOUNTED THERMOSTAT
- PSD — PUMP SUCTION DIFFUSER
- RS — RELATIVE HUMIDITY SENSOR
- F T — STEAM TRAP, FLOAT & THERMOSTATIC
- HOSE BIBB
- ROOF VENTILATOR, EXHAUST/RELIEF
- PUMP
- MOTOR
- F T — STEAM TRAP, THERMODYNAMIC

MECHANICAL LINE TYPE LEGEND

Line	Description
- - - - - -	REMOVE ITEMS
(gray line)	EXISTING ITEMS TO REMAIN
———	PROVIDE ITEMS
RS	REFRIGERANT SUCTION
RL	REFRIGERANT LIQUID
HWS	HOT WATER SUPPLY
HWR	HOT WATER RETURN
RS/RL	REFRIGERANT SUCTION / REFRIGERANT LIQUID
D	DRAIN LINE
HPS	HIGH PRESSURE STEAM
HPC	HIGH PRESSURE CONDENSATE
LPS	LOW PRESSURE STEAM
LPC	LOW PRESSURE CONDENSATE
CD	CONDENSATE DRAIN
PCR	PUMPED CONDENSATE RETURN
CA	COMPRESSED AIR
— · — · —	CONTROL WIRING

MECHANICAL ABBREVIATIONS

A	AMPERE, AIR
AC	AIR CONDITIONING, AIR CONDITIONER
AD	ACCESS DOOR
ADA	AMERICANS WITH DISABILITIES ACT
AFF	ABOVE FINISHED FLOOR
AFMS	AIRFLOW MEASURING STATION
AHU	AIR HANDLING UNIT
AIC	AMPERE INTERRUPTING CAPACITY
AMB	AMBIENT
AMS	AIRFLOW MEASURING STATION
APD	AIR PRESSURE DROP
APPROX	APPROXIMATELY
AS	AIR SEPARATOR
ASME	AMERICAN SOCIETY OF MECHANICAL ENGINEERS
ASS'Y	ASSEMBLY
ASTM	AMERICAN SOCIETY FOR TESTING AND MATERIALS ASSEMBLY
ATFP	ANTI-TERRORISM FORCE PROTECTION
BDD	BACKDRAFT DAMPER
BHP	BRAKE HORSEPOWER
BLDG	BUILDING
BTU	BRITISH THERMAL UNIT
BTUH	BTU PER HOUR
CAP	CAPACITY
CC	COOLING COIL
CD	CONDENSATE DRAIN
CF	CHEMICAL FEED
CFM	CUBIC FEET PER MINUTE
C	CENTERLINE
CLG	CEILING
CO	CLEANOUT
CONN	CONNECTION
CONC	CONCRETE
COND	CONDENSATE, CONDENSING, CONDITIONS
COP	COEFFICIENT OF PERFORMANCE
CR	CONDENSATE RECEIVER
CS	CURRENT SENSOR
CU	CONDENSING UNIT
CUH	CABINET UNIT HEATER
CW	COLD WATER
CWS	COLD WATER SUPPLY
D	DEPTH, DAMPER
DB	DRY BULB
dB	DECIBELS
DDC	DIRECT DIGITAL CONTROLS
DEG	DEGREES
ø, DIA	DIAMETER
DIFF	DIFFERENTIAL
DN	DOWN
DOAS	DEDICATED OUTDOOR AIR SYSTEM
DP	DIFFERENTIAL PRESSURE
DPS	DIFFERENTIAL PRESSURE SWITCH
DWG	DRAWING
E	EXISTING, EXHAUST
EA	EXHAUST AIR, EACH
EAS	EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH
EAT	ENTERING AIR TEMPERATURE
EDB	ENTERING DRY BULB TEMPERATURE
EER	ENERGY EFFICIENCY RATIO
EF	EXHAUST FAN
EFF	EFFICIENCY
EFT	ENTERING FLUID TEMPERATURE
ELEC	ELECTRIC/ELECTRICAL
ELEV	ELEVATION, ELEVATOR
EQUIP	EQUIPMENT
ERV	ENERGY RECOVERY VENTILATOR
ESP	EXTERNAL STATIC PRESSURE
EWB	ENTERING WET BULB TEMPERATURE
EWT	ENTERING WATER TEMPERATURE
EXIST	EXISTING
EXP	EXPANSION
EXT	EXPANSION TANK

°F	DEGREES FAHRENHEIT
FACP	FIRE ALARM CONTROL PANEL
FC	FLEX CONNECTOR
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN, FIRE DAMPER
FF	FINISHED FLOOR
FLA	FULL LOAD AMPS
FLR	FLOOR
FM	FLOW METER
FPM	FEET PER MINUTE
FR	FINTUBE RADIATION
FS	FLOW SWITCH
FT	FOOT, FEET
G	GAS
GA	GAUGE
GAL	GALLONS
GALV	GALVANIZED
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
GUI	GRAPHICAL USER INTERFACE
GWB, GYP	GYPSUM WALLBOARD
H	HEIGHT
H2O	WATER
HB	HOSE BIBB
HC	HEATING COIL
HGT	HEIGHT
HORIZ	HORIZONTAL
HP	HORSEPOWER, HEAT PUMP
HR	HOUR
HT	HEIGHT
HTG	HEATING
HUMID	HUMIDIFIER
HVAC	HEATING, VENTILATION, AND AIR CONDITIONING (UNIT)
HW	HOT WATER
HWR	HOT WATER RETURN
HWS	HOT WATER SUPPLY
HX	HEAT EXCHANGER
HZ	HERTZ
ID	INSIDE DIAMETER
IN	INCH, INCHES
IW	INDIRECT WASTE
KW	KILOWATT
L	LOUVER, LENGTH
LAT	LEAVING AIR TEMPERATURE
LBS	POUNDS
LCU	LOCAL CONTROL UNIT
LDB	LEAVING DRY BULB
LF	LINEAR FEET
LFT	LEAVING FLUID TEMPERATURE
LOC	LOCATION, LOCATED
LRA	LOCKED ROTOR AMPS
LT	LOW TEMPERATURE
LWB	LEAVING WET BULB
LWT	LEAVING WATER TEMPERATURE
M	MOTOR
MAU	MAKE-UP AIR UNIT
MAX	MAXIMUM
MAX PD	MAXIMUM PRESSURE DROP
MBH	1000 BTU PER HOUR
MBU	1000 BTU
MCA	MINIMUM CIRCUIT AMPERES
MECH	MECHANICAL
MERV	MINIMUM EFFICIENCY REPORTING VALUE
MFR	MANUFACTURER
MIN	MINIMUM
MOCP	MAIN OVERCURRENT PROTECTION
MTG	MOUNTING
N/A	NOT APPLICABLE
NC	NOISE CRITERIA, NORMALLY CLOSED
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NO	NUMBER, NORMALLY OPEN
NPT	NATIONAL PIPE THREAD
NTS	NOT TO SCALE

OA	OUTSIDE AIR
OAT	OUTSIDE AIR TEMPERATURE
OBD	OPPOSED BLADE DAMPER
OC	ON CENTER, OCCUPANCY
OD	OUTSIDE DIAMETER
OED	OPEN ENDED DUCT
OEP	OPEN ENDED PIPE
OS&Y	OUTSIDE STEM & YOKE
P	PUMP, PITCH, PRESSURE
PCR	PUMPED CONDENSATE RETURN
PD	PRESSURE DIFFERENCE
PH	PHASE
PPM	PARTS PER MILLION
PRESS	PRESSURE
PRV	PRESSURE REDUCING VALVE
PSI	POUNDS PER SQUARE INCH
PSIG	POUNDS PER SQUARE INCH GAUGE
QTY	QUANTITY
R	RADIUS, RETURN
RA	RETURN AIR
RAT	RETURN AIR TEMPERATURE
REFRIG	REFRIGERANT
REQ'D	REQUIRED
RH	RELATIVE HUMIDITY
RL	REFRIGERANT LIQUID
RLA	RUNNING LOAD AMPERES
RM	ROOM
RO	RETURN/TRANSFER OPENING
RPM	REVOLUTIONS PER MINUTE
RPZ	REDUCED PRESSURE ZONE
RS	REFRIGERANT SUCTION
S	SUPPLY
SA	SUPPLY AIR, SOUND ATTENUATOR
SAT	SUPPLY AIR TEMPERATURE, SUSPENDED ACOUSTICAL TILE
SC	SENSIBLE COOLING
SCCR	SHORT CIRCUIT CURRENT RATING
SEER	SEASONAL ENERGY EFFICIENCY RATIO
SF	SQUARE FOOT/FEET, SUPPLY FAN
SIM	SIMILAR
SMACNA	SHEET METAL AND AIR CONDITIONING CONTRACTOR'S NATIONAL ASSOCIATION
SP	STATIC PRESSURE
SQ	SQUARE
SS	STAINLESS STEEL
STM	STEAM
T	THERMOSTAT, TRANSFER
TA	TRANSFER AIR
TC	TOTAL COOLING
TEMP	TEMPERATURE
TS	TEMPERATURE SENSOR
TSP	TOTAL STATIC PRESSURE
TYP	TYPICAL
UH	UNIT HEATER
UL	UNDERWRITERS LABORATORY
UN-OCC	UN-OCCUPIED OVERRIDE BUTTON
V	VENT, VALVE, VOLT(S)
VAC	VOLTS, ALTERNATING CURRENT
VD	VOLUME DAMPER
VDC	VOLTS, DIRECT CURRENT
VEL	VELOCITY
VFD	VARIABLE FREQUENCY DRIVE
VIF	VERIFY IN FIELD
VTR	VENT THRU ROOF
W	WIDTH
W/	WITH
WB	WET BULB
WC	WATER COLUMN
WG	WATER GAUGE
WH	WATER HEATER
WPD	WATER PRESSURE DROP

GENERAL MECHANICAL SYSTEM NOTES

1. PIPING AND DUCTWORK ARE SHOWN DIAGRAMMATICALLY, EXACT LOCATION SHALL BE DETERMINED IN THE FIELD. COORDINATE IN FIELD WITH STRUCTURAL ELEMENTS AND THE FIRE SUPPRESSION, FIRE ALARM, PLUMBING AND ELECTRICAL TRADES.
2. PIPING AND DUCTWORK SHALL BE SUPPORTED FROM BUILDING STRUCTURE. NO STRUCTURAL MEMBERS SHALL BE CUT OR DRILLED..
3. COMPLY WITH THE INTERNATIONAL MECHANICAL CODE (2015), INTERNATIONAL ENERGY CONSERVATION CODE (2015), ASHRAE 90.1 (2016), ASHRAE 62.1 (2016) AND NFPA 90A.
4. COORDINATE MECHANICAL WORK WITH OUTAGES SCHEDULE.
5. REFER TO SHEET G-007 FOR CONSTRUCTION SEQUENCE STRATEGY AND SHEET G-008 FOR OUTAGE SCHEDULE.
6. OVERALL PROJECT COORDINATION DRAWINGS ARE REQUIRED . SUBMIT UTILITY LAYOUTS, INCLUDING PIPE DIMENSIONS AND ELEVATIONS WITH ALLOWANCE FOR INSULATION WHERE REQUIRED, AND PIPE HANGERS AND SUPPORTS. SUBMIT APPROVED EQUIPMENT CUT SHEETS, CLEARLY INDICATING REQUIRED SERVICE CLEARANCES TO GC FOR COMPILATION OF REQUIRED DRAWINGS. REFER TO SPECIFICATION SECTION 01 30 10.00 22 FOR COORDINATION PROCEDURES AND COORDINATION DRAWING REQUIREMENTS.
7. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

FINAL SUBMISSION 10/01/2025 RSH
SYM DESCRIPTION DATE APPR

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 10-01-25 LICENSED PROFESSIONAL ENGINEER
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 184 OF 282
M-001

DRAWFORM REVISION: 14 SEP 2023

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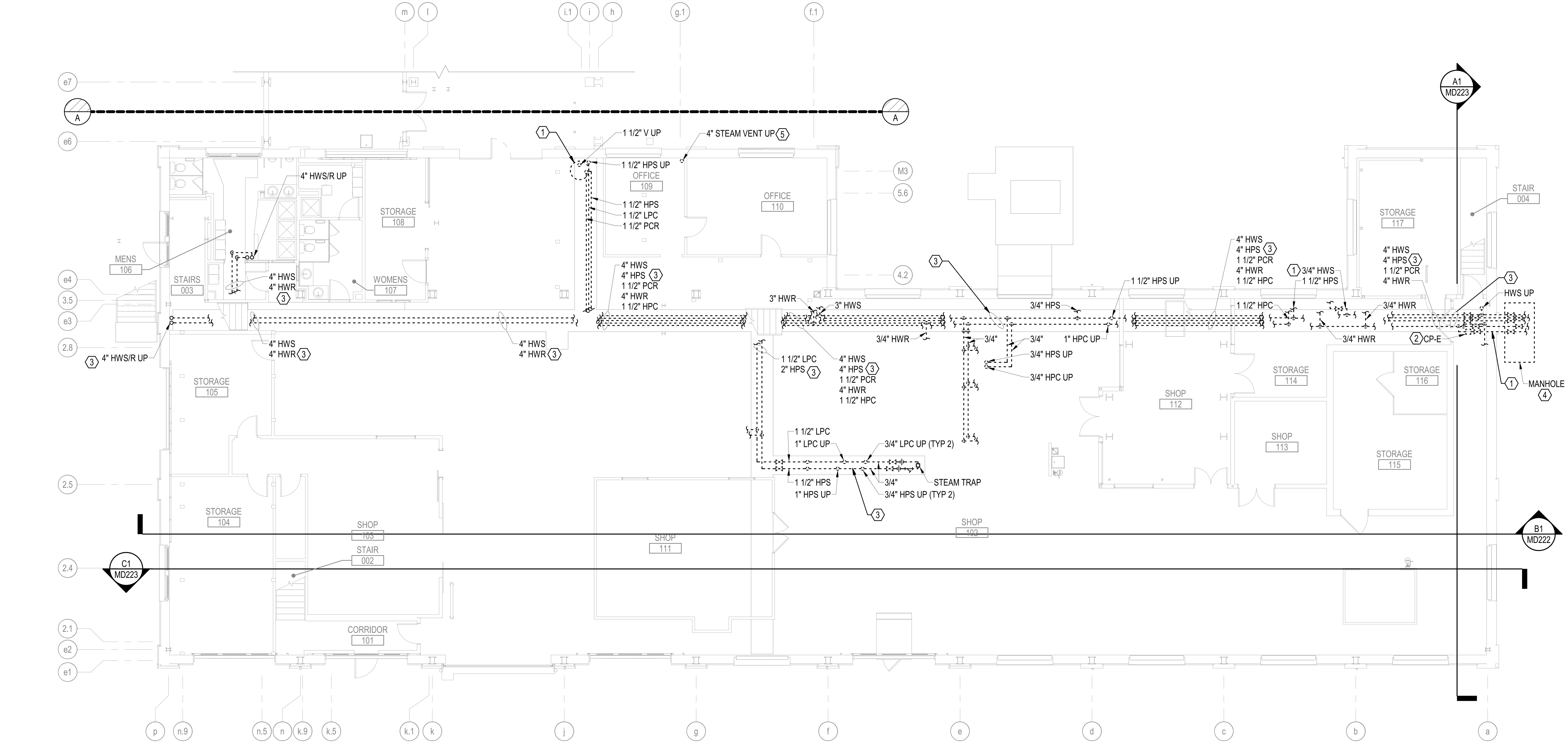


B1 FIRST FLOOR TRENCH MECHANICAL PIPING REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

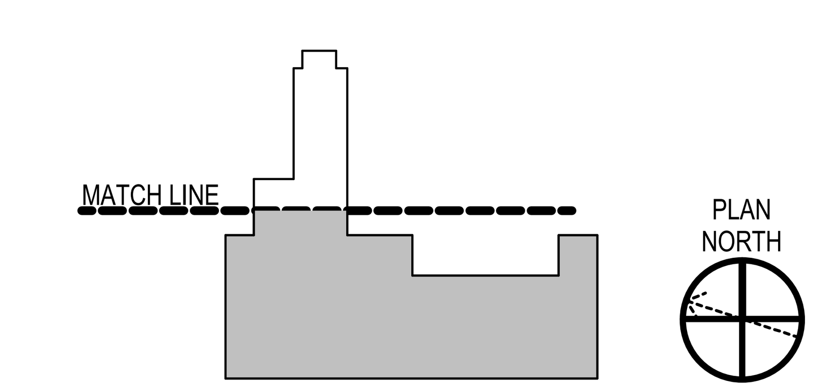
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. LIMITED INFORMATION WAS AVAILABLE DOCUMENTING CURRENT PIPING WITH-IN FLOOR TRENCHES. THESE CONTAIN ACTIVE AS WELL AS ABANDONED PIPING. REFERENCES TO THE KINDS OF PIPING AND PIPE SIZES ARE INDICATED TO PROVIDE A REPRESENTATIVE LIST OF HVAC AND PROCESS PIPING BUT NOT NECESSARILY ALL PIPING, SANITARY, WASTE, VENT, DOM WATER AND COMPRESSED AIR ARE SHOWN ON PLUMBING REMOVALS SHEETS. REMOVE EQUIPMENT AND PIPING WITHIN THE FLOOR TRENCHES, UNLESS OTHERWISE NOTED. FOR BIDDING PURPOSES, ASSUME 1200 FEET OF 4" METAL STEAM AND HEATING WATER TYPE PIPE TO BE REMOVED FROM THE TRENCHES.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE STEAM CONDENSATE RECEIVER, DUPLEX PUMP SET, AND ASSOCIATED CONTROLS, WIRING, AND PIPING.
2. PIPE AND ASSOCIATED VALVE(S) LOCATED IN TRENCH. OFFSET FOR CLARITY.
3. REMOVE PIPING THROUGHOUT TRENCH.
4. REMOVE HPS AND PCR PIPING BACK TO MAINS IN MANHOLE, LEAVING NO BRANCH ISOLATION VALVE, CAP BRANCH TEE AT MAIN. SCHEDULE UTILITY SHUTDOWN DURING NON HEATING SEASON USING MUB UTILITY SHUTDOWN APPLICATION.
5. REMOVE VENT PIPE AND CAP FLUSH WITH FLOOR SURFACE.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 STATE OF MAINE

SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR TRENCH MECHANICAL PIPING REMOVALS PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 185 OF 282
MD101

DRAWFORM REVISION: 14 SEP 2023

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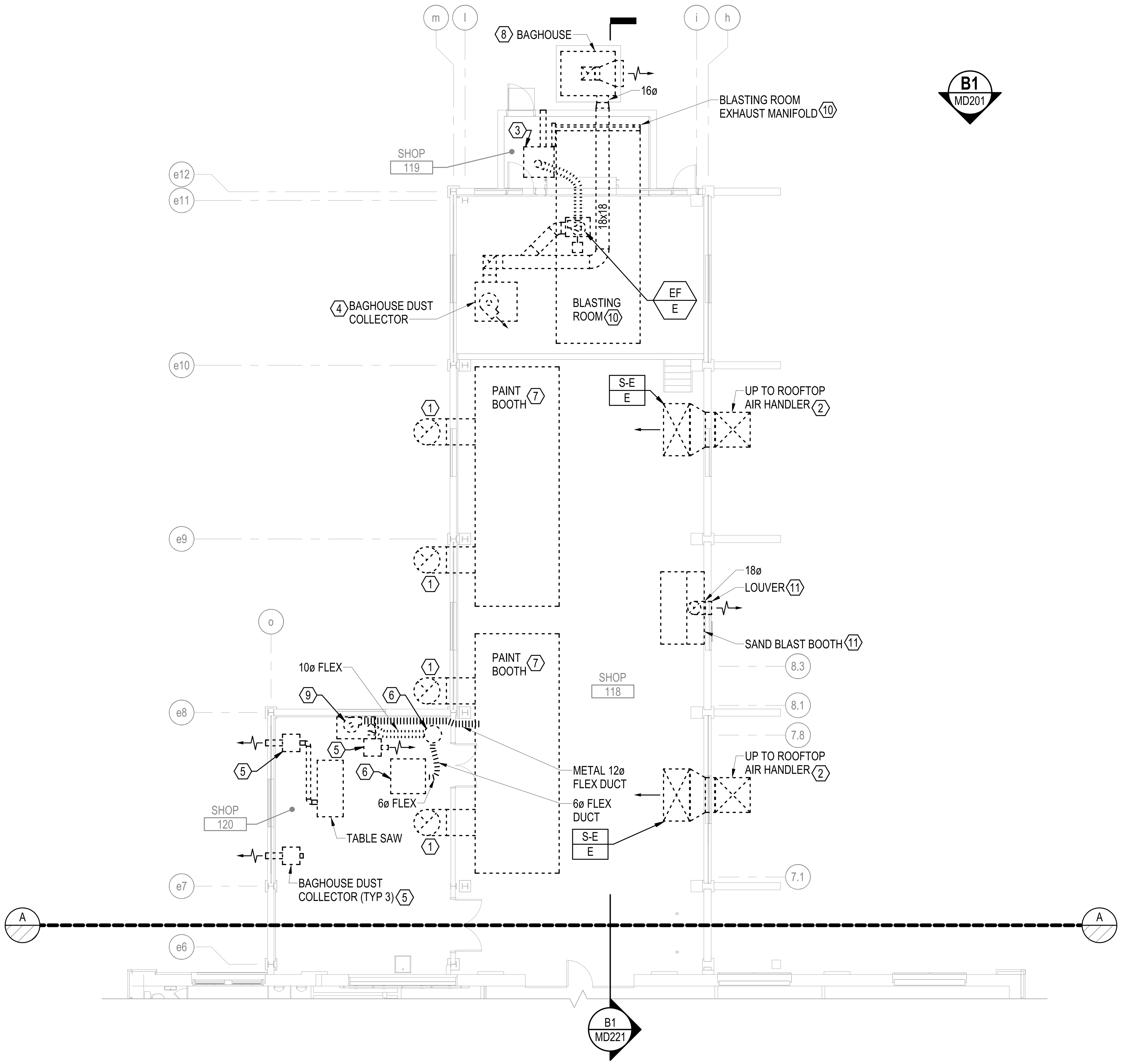


A1 FIRST FLOOR MECHANICAL DUCTWORK REMOVALS PLAN - EAST

SCALE: 1/8" = 1'-0"

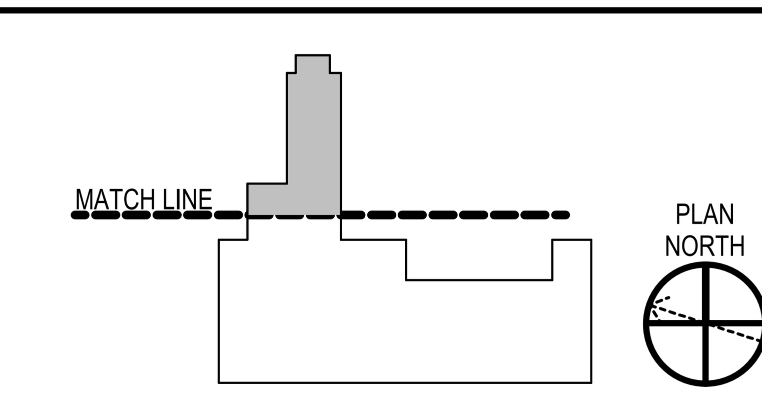
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL HVAC AND INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND PIPING THROUGHOUT THE SPACES SHOWN ON THIS SHEET, UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. [CLIN 002] REMOVE EXHAUST DUCTWORK FROM PAINT BOOTH ON OUT TO THE WALL OR CEILING. PROVIDE TEMPARARY COVERS FOR WALL OPENINGS.
2. [CLIN 002] REMOVE MAKE-UP AIR DUCTWORK IN ENTIRETY FROM SUPPLY AIR GRILLE TO ROOFTOP AIR HANDLER.
3. [CLIN 002] REMOVE SHOT BLAST COLLECTION SYSTEM, EXHAUST FAN, AND ASSOCIATED CONTROLS, WIRING, AND DUCTWORK.
4. [CLIN 002] REMOVE BLASTING ROOM EXHAUST AND DUST COLLECTION SYSTEM AND ASSOCIATED CONTROLS, WIRING, AND DUCTWORK.
5. [CLIN 002] REMOVE DUCTED EXHAUST SYSTEM WITH BAGHOUSE AIR FILTRATION, INCLUDING FAN, CONTROLS, WIRING, AND DUCTWORK.
6. [CLIN 002] REMOVE BLASTING BOOTH AND SHOT BLAST COLLECTION SYSTEM, INCLUDING ASSOCIATED CONTROLS, WIRING, AND DUCTWORK.
7. [CLIN 002] REMOVE PAINT BOOTH, AND ASSOCIATED CONTROLS, WIRING, AND FILTERS.
8. [CLIN 002] REMOVE BAGHOUSE ASSEMBLY, INCLUDING CONTROLS, WIRING, AND DUCTWORK.
9. [CLIN 002] REMOVE EXHAUST FAN ON ELEVATED WALL MOUNTED PLATFORM AND ASSOCIATED DUCTWORK PLATFORM TO REMAIN IN PLACE.
10. [CLIN 002] REMOVE BLASTING ROOM AND ASSOCIATED EQUIPMENT AND DUCTWORK.
11. [CLIN 002] REMOVE SAND BLAST BOOTH AND ASSOCIATED DUCTWORK AND LOUVER.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
FIRST FLOOR MECHANICAL DUCTWORK REMOVALS PLAN - EAST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 186 OF 282
MD102

DRAWFORM REVISION: 14 SEP 2023

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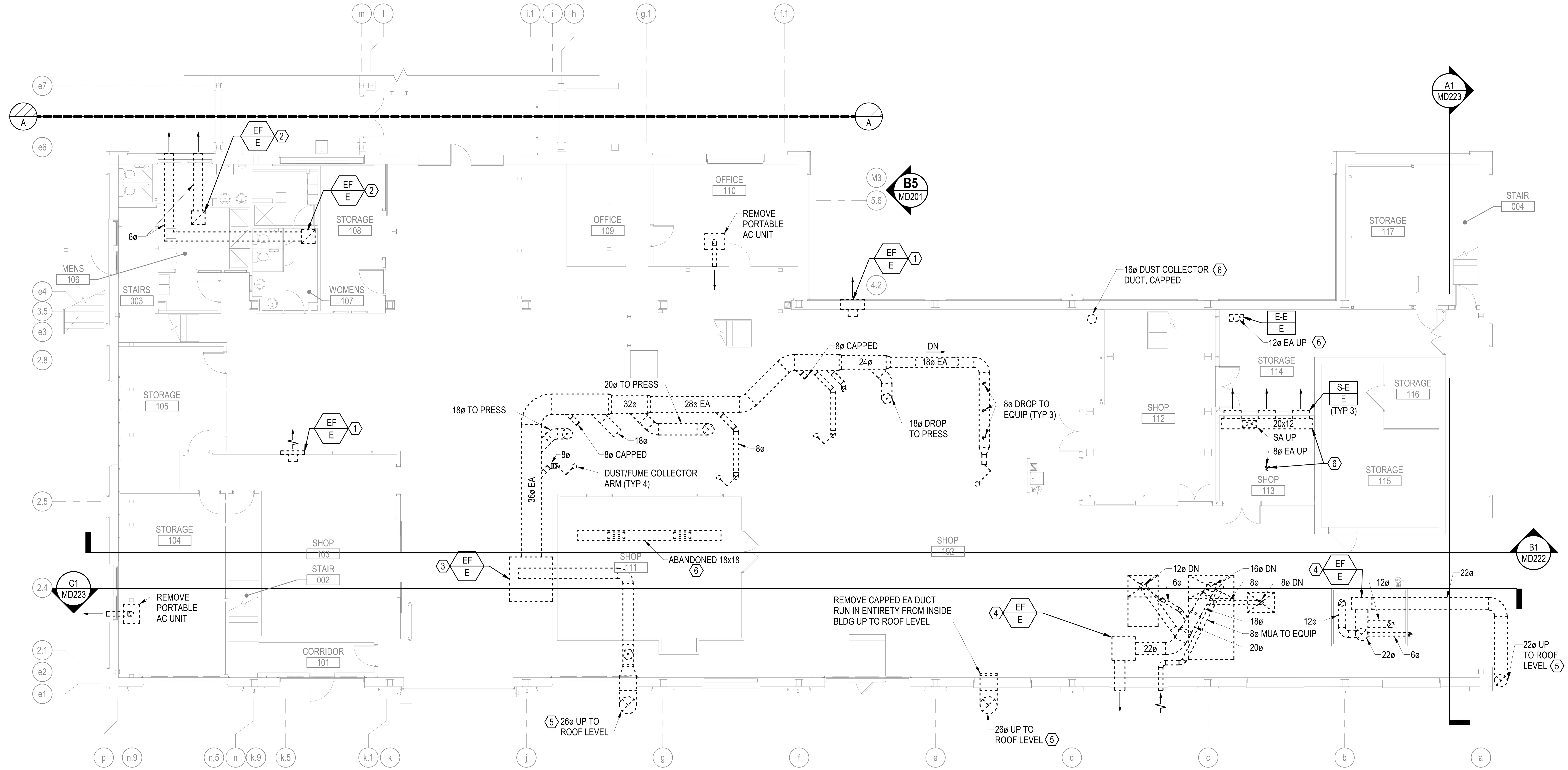


B1 FIRST FLOOR MECHANICAL DUCTWORK REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

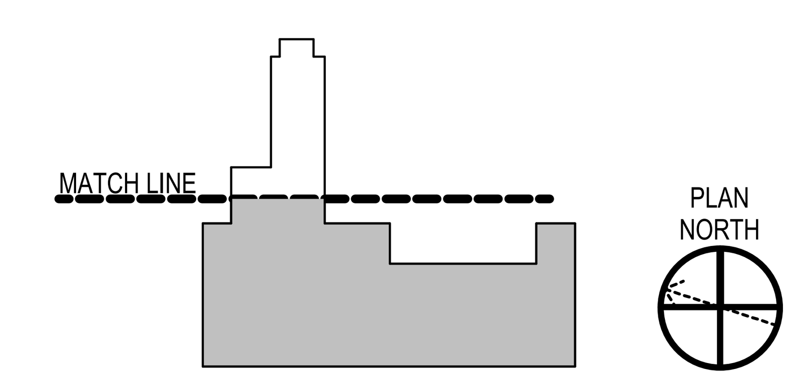
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL HVAC AND INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND DUCTWORK THROUGHOUT THE SPACES SHOWN ON THIS SHEET UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE THROUGH-WALL EXHAUST FAN, AND ASSOCIATED CONTROLS AND WIRING.
2. REMOVE EXHAUST FAN AND ASSOCIATED DUCTWORK, LOUVERS AND CONTROLS.
3. REMOVE EXHAUST FAN AND ASSOCIATED DUCTWORK, HOODS, VFD, CONTROLS, AND WIRING.
4. REMOVE EXHAUST FAN AND ASSOCIATED DUCTWORK, ENCLOSURES, HOODS, CONTROLS, AND WIRING.
5. REMOVE ENTIRE EXHAUST DUCT RISER AND ASSOCIATED SUPPORTS.
6. REMOVE DUCTWORK ASSEMBLY.

KEY PLAN



GRAPHIC SCALE

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NAVFAC

OAK POINT ASSOCIATES

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	CBM
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

UNCLASSIFIED

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MECHANICAL DUCTWORK REMOVALS PLAN - WEST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 187 OF 282

MD103

DRAWFORM REVISION: 14 SEP 2023

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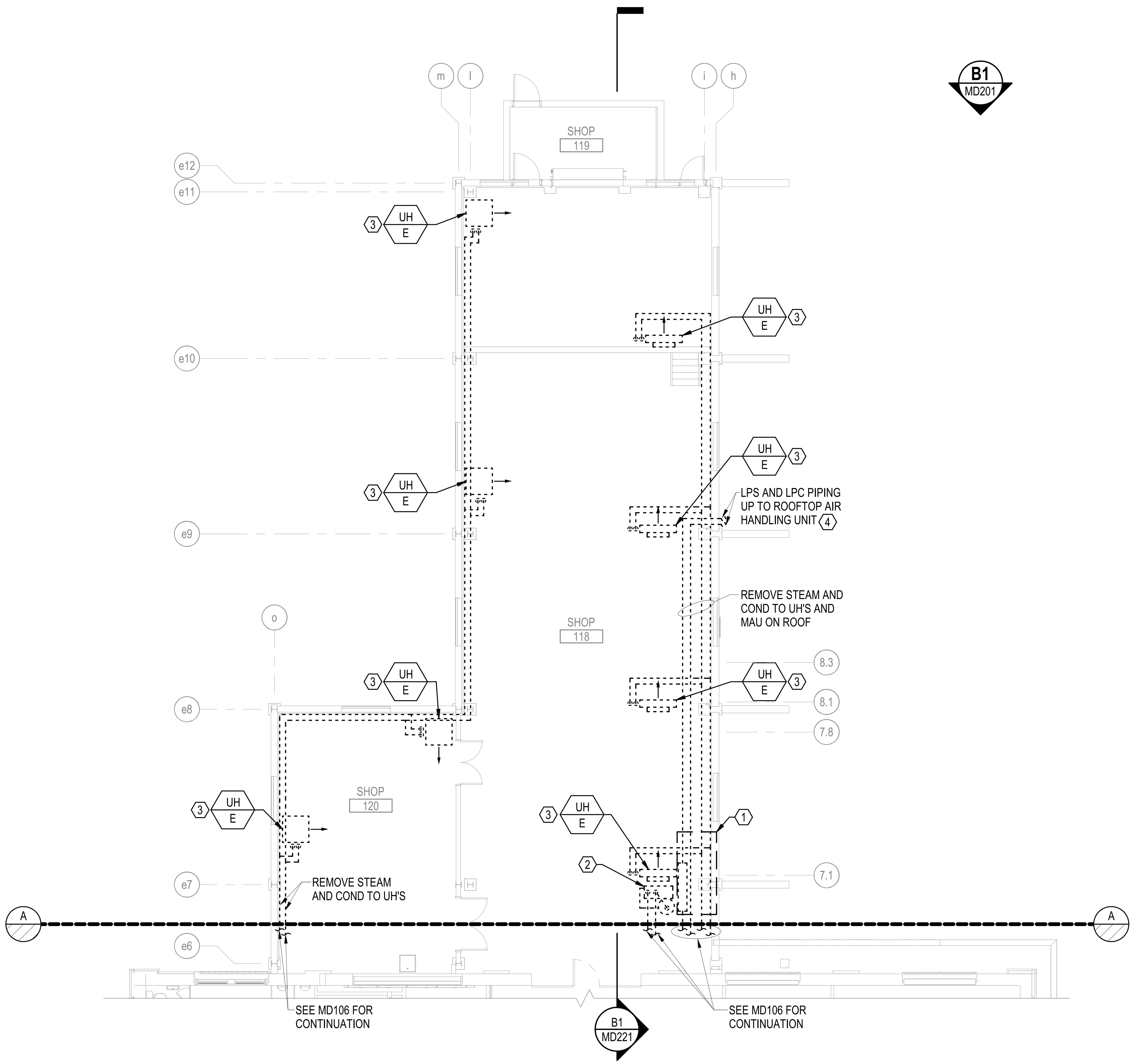


A1 FIRST FLOOR MECHANICAL PIPING REMOVALS PLAN - EAST

SCALE: 1/8" = 1'-0"

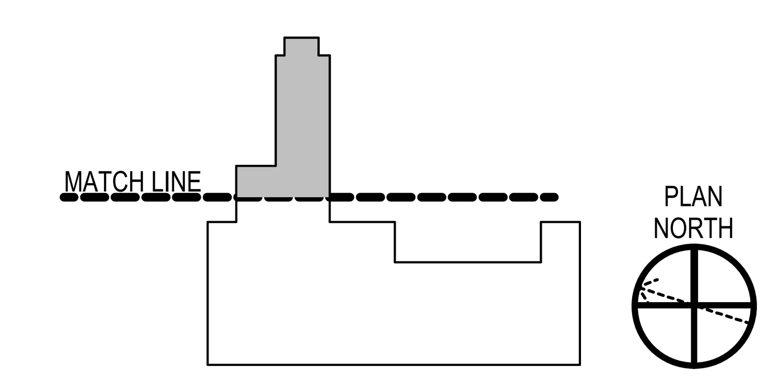
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL HVAC AND INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND PIPING THROUGHOUT THE SPACES SHOWN ON THIS SHEET, UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE FOUR PRESSURE REDUCING VALVE ASSEMBLIES AND ASSOCIATED BYPASS LINES, SAFETY VALVES, CONTROLS, AND PIPING.
2. REMOVE STEAM CONDENSATE RECEIVER, PRESSURE POWERED PUMP, ASSOCIATED CONTROLS AND PIPING. REMOVE COMPRESSED AIR SERVICE BRANCH PIPING AND REGULATOR SERVING PRESSURE POWERED PUMP.
3. REMOVE UNIT HEATER AND ASSOCIATED CONTROLS, WIRING, STEAM CONDENSATE TRAP ASSEMBLY AND BRANCH PIPING.
4. SEE SHEET MD120 FOR EXTERIOR PIPING REMOVAL.

KEY PLAN



GRAPHIC SCALE

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SYM	DESCRIPTION	DATE	APPR
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NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC

DRAWN BY CBM

CHECKED BY MSA

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MECHANICAL PIPING REMOVALS PLAN - EAST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 188 OF 282

MD104

DRAWFORM REVISION: 14 SEP 2023

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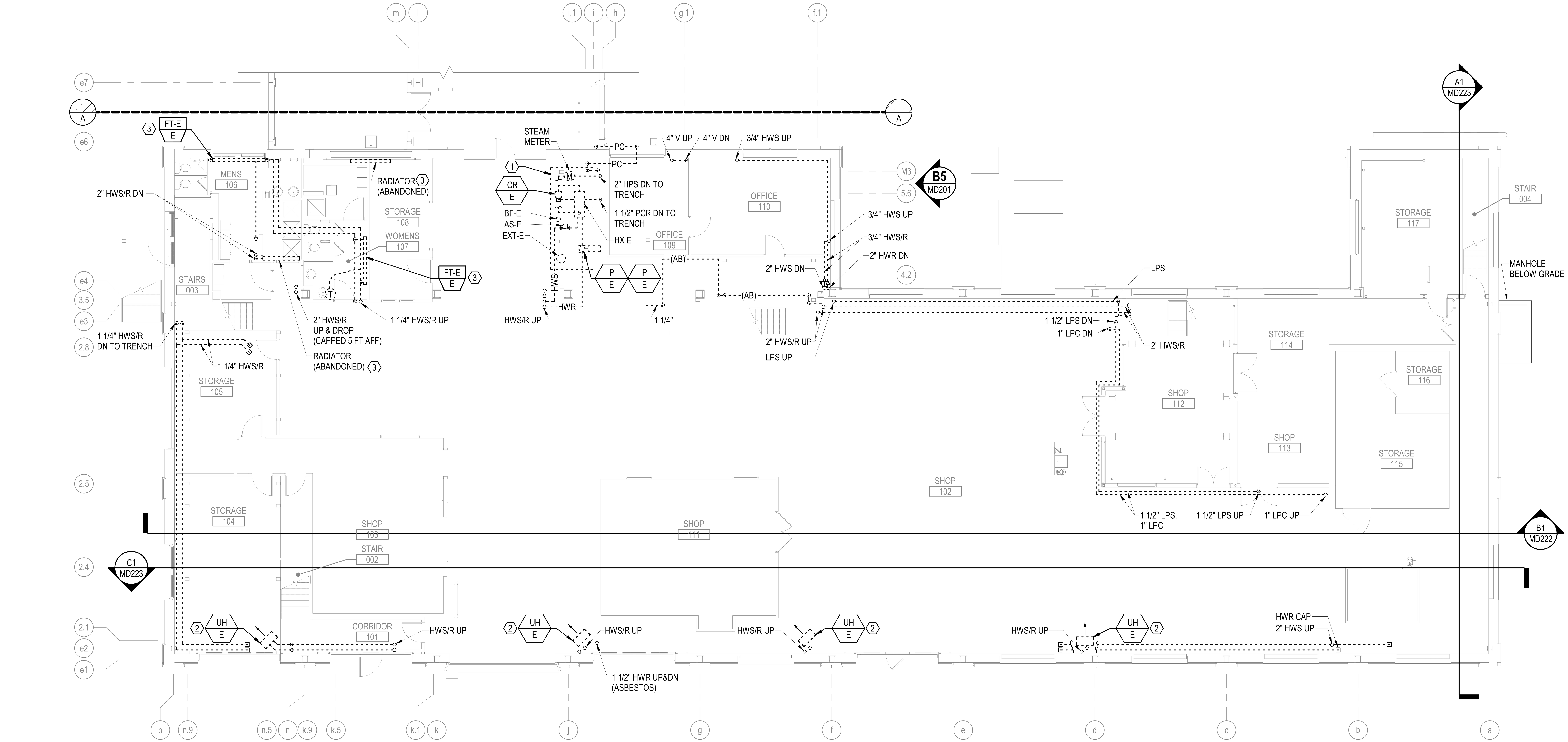


B1 FIRST FLOOR MECHANICAL PIPING REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND PIPING THROUGHOUT THE SPACES SHOWN ON THIS SHEET, UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

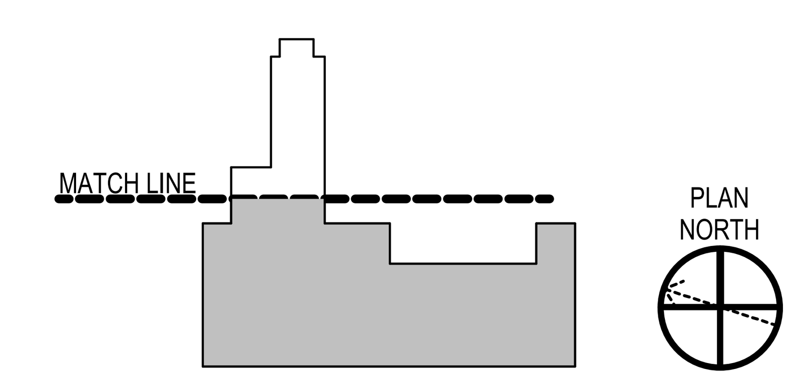
REMOVALS KEYNOTES (THIS SHEET ONLY)

1 REMOVE EXISTING STEAM TO HOT WATER HEATING SYSTEM, INCLUDING STEAM METER ASSEMBLY, STEAM PRESSURE REDUCING STATION, STEAM CONTROL VALVES ASSEMBLY, FLASH TANK ASSEMBLY, SHELL FINTUBE HEAT EXCHANGER ASSEMBLY, INLINE HYDRONIC CIRCULATORS WITH VFDS, AIR SEPERATOR, EXPANSION THINK, HYDRONIC SPECIALTIES, STEAM TRAP ASSEMBLES, STEAM CONDENSATE RECEIVER WITH DUPLEX PUMP SET, AND ASSOCIATED CONTROLS, WIRING, AND PIPING.

2 REMOVE UNIT HEATER AND ASSOCIATED BRANCH PIPING, CONTROLS AND WIRING.

3 REMOVE TERMINAL HEATING EQUIPMENT AND ASSOCIATED BRANCH PIPING, CONTROLS AND WIRING.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MECHANICAL PIPING REMOVALS PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 189 OF 282
MD105

DRAWFORM REVISION: 14 SEP 2023

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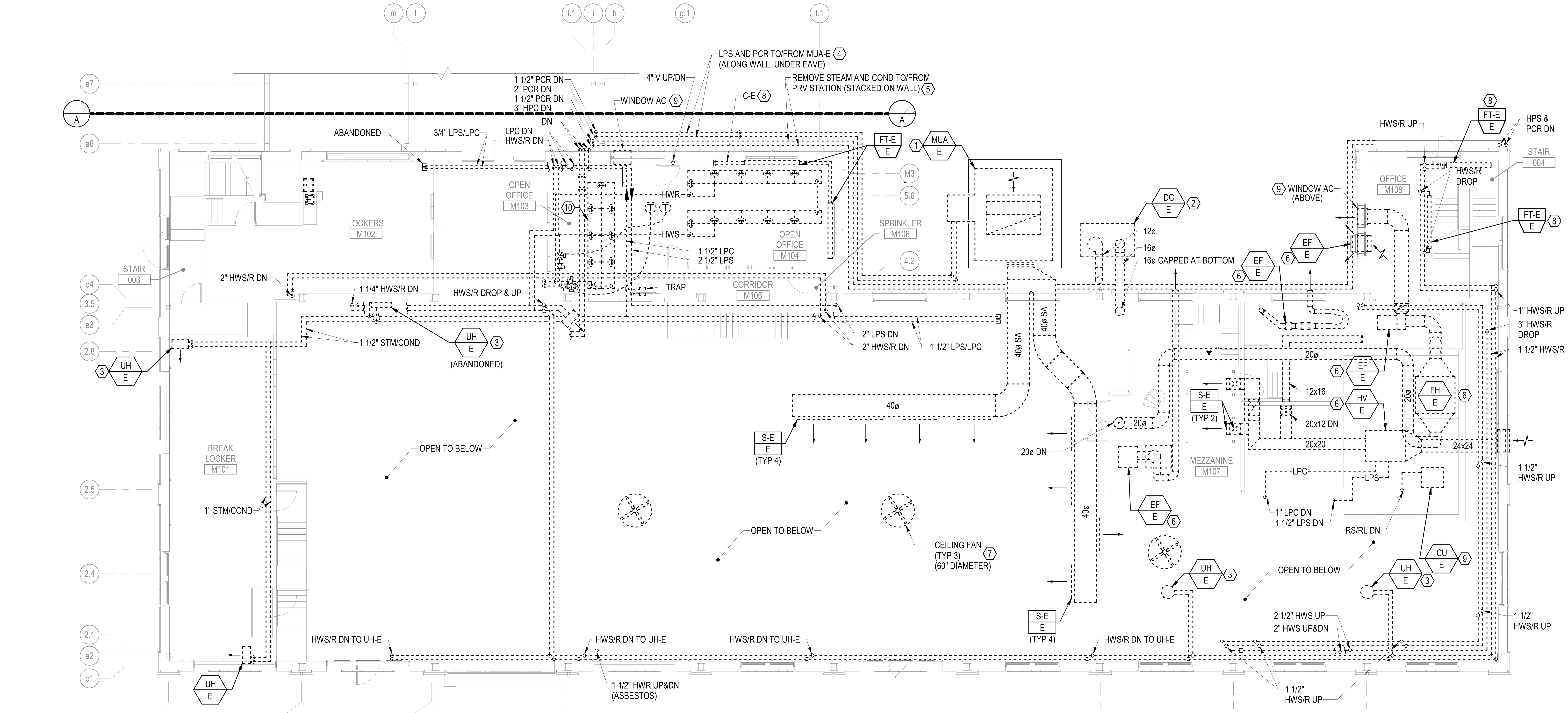


B1 **FIRST FLOOR MEZZANINE MECHANICAL REMOVALS PLAN - WEST**

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND PIPING THROUGHOUT THE SPACES SHOWN ON THIS SHEET, UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

(1) [CLIN 002] REMOVE MAKE UP AIR UNIT MUA-E AND ASSOCIATED PRESSURE POWERED CONDENSATE PUMP, CONTROLS, WIRING, DUCTWORK AND PIPING. THE EXISTING CONCRETE SERVICE PAD MUST REMAIN IN PLACE FOR REUSE.

(2) [CLIN 002] REMOVE DUST COLLECTION SYSTEM, INCLUDING EXHAUST FAN, BAGHOUSE, BAGHOUSE SUPPORT STAND, CONTROLS, WIRING, AND DUCTWORK.

(3) REMOVE UNIT HEATER AND ASSOCIATED BRANCH PIPING, CONTROLS AND WIRING.

(4) [CLIN 002] REMOVE LPS AND LPC PIPING FROM MAKE-UP AIR UNIT OUTSIDE OF THE BUILDING AT GROUND LEVEL TO THE WALL PENETRATIONS AT THE ANNEX WING OF THE BUILDING. REMOVE PIPING WALL SUPPORTS, AND COORDINATE PATCHING / REPAIR OF BRICK WORK. SEE SHEET MD201 FOR ELEVATION VIEW OF PIPING.

(5) REMOVE HPS AND PCR PIPING FROM ANNEX WING WALL PENETRATIONS TO CORNER OF EXTERIOR WALL OUTSIDE OF OPEN OFFICE AS SHOWN. REMOVE PIPING WALL SUPPORTS, AND COORDINATE PATCHING / REPAIR OF BRICK WORK.

(6) [CLIN 002] REMOVE INDUSTRIAL PROCESS EXHAUST AND MAKE UP AIR SYSTEMS, INCLUDING BUT NOT LIMITED TO AIR HANDLING UNITS, FANS, CONTROLS, MIRING, DUCTWORK, HOODS, AND PIPING.

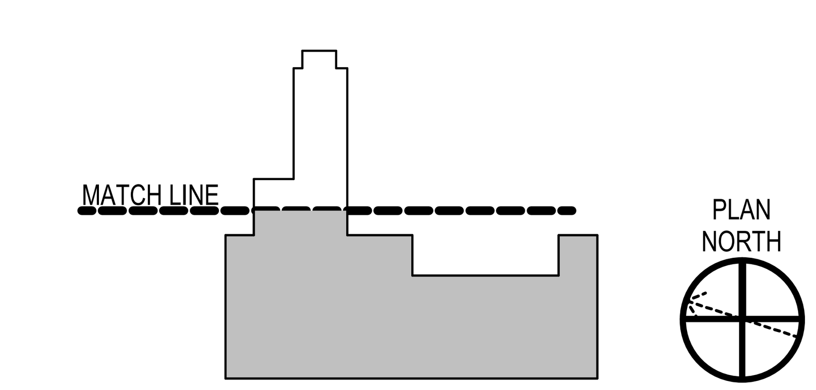
(7) REMOVE PADDLE TYPE CEILING FANS AND ASSOCIATED CONTROLS AND WIRING.

(8) REMOVE TERMINAL HEATING UNIT AND ASSCOIATED BRANCH PIPING, CONTROLS, AND WIRING.

(9) REMOVE AIR CONDITIONER UNIT: DISPOSAL OF UNITS CONTAINING REFRIGERANT SHALL BE DONE IN ACCORDANCE WITH CLEAN AIR ACT, LOCAL, STATE, AND FEDERAL REGULATIONS. ALL REFRIGERANT MUST BE REMOVED PRIOR TO EQUIPMENT REMOVAL. THE USED REFRIGERANT SHALL BE RECLAIMED BY EPA APPROVED, SPECIALIZED EQUIPMENT AND DISPOSED OF BY CONTRACTOR. THE CONTRACTOR MUST PROVIDE THE MIDLANT ODS FORMS TO DOCUMENT REFRIGERANT RECOVERY, DISPOSAL, AND CHARGE ACTIVITIES. SEE SPECIFICATION SECTION 02 41 00 "DEMOLITION AND DECONSTRUCTION".

(10) REMOVE HYDRONIC RADIANT CEILING PANELS AND ASSOCIATED HYDRONIC BRANCH PIPING, CONTROLS, AND WIRING.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

FIRST FLOOR MEZZANINE MECHANICAL REMOVALS PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 190 OF 282
MD106

DRAWFORM REVISION: 14 SEP 2023

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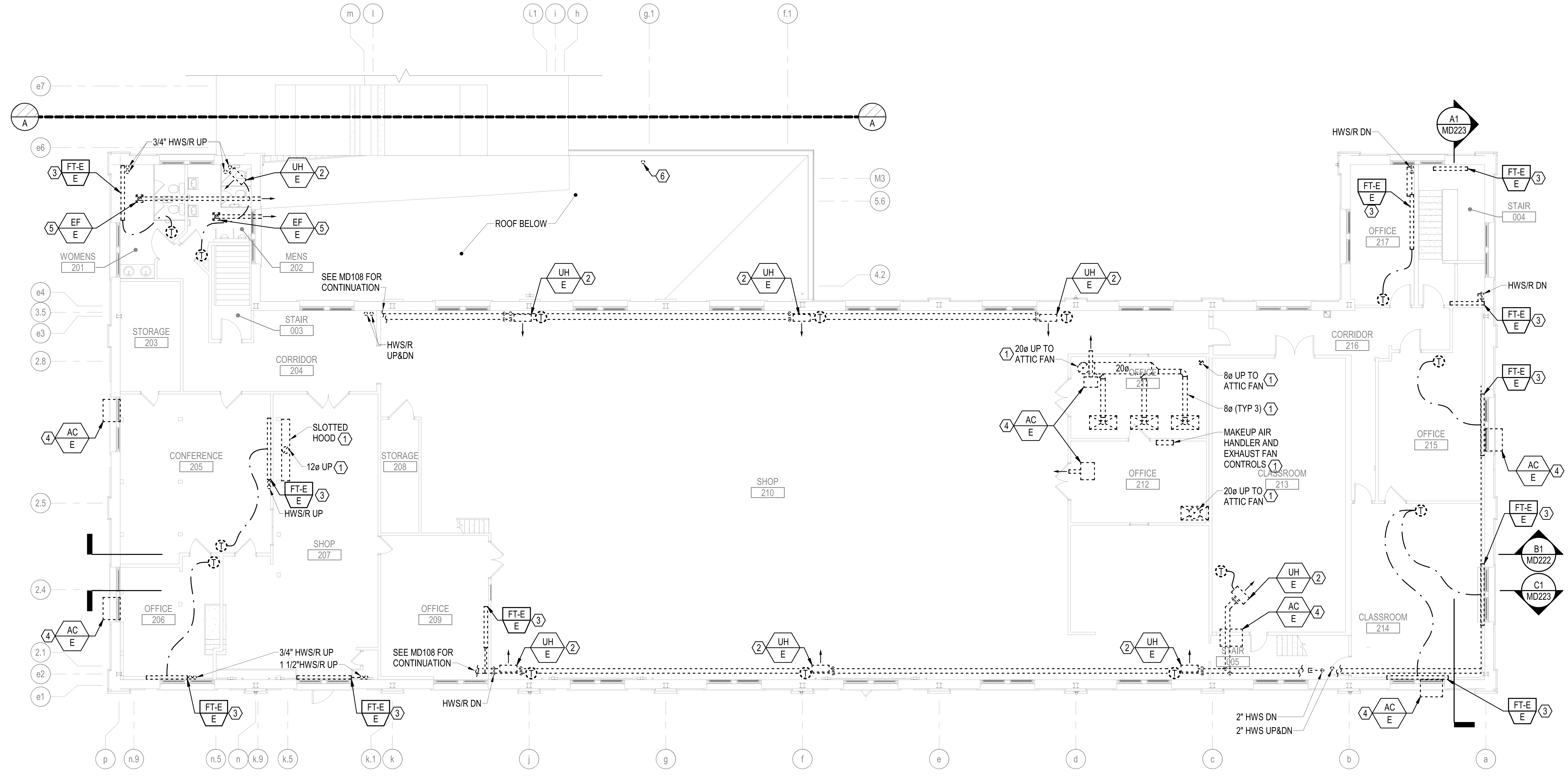


B1 SECOND FLOOR MECHANICAL REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

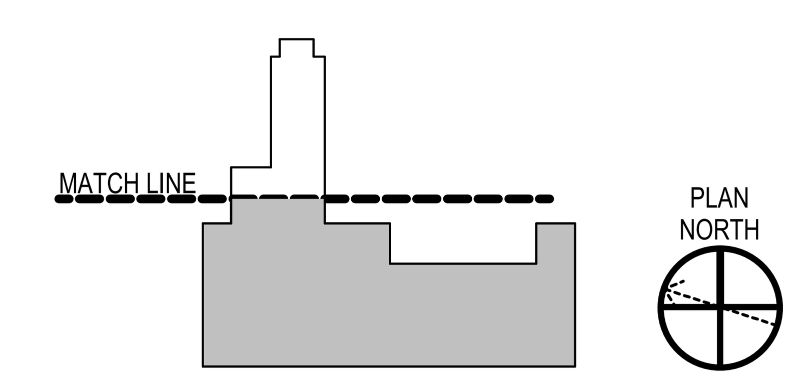
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. LIMITED INFORMATION WAS AVAILABLE DOCUMENTING CURRENT PIPING WITH-IN FLOOR TRENCHES. THEY CONTAIN ACTIVE AS WELL AS ABANDONED PIPING. REFERENCES TO THE KINDS OF PIPING AND PIPE SIZES ARE INDICATED TO PROVIDE A REPRESENTATIVE LIST OF HVAC AND PROCESS PIPING BUT NOT NECESSARILY ALL PIPING, SANITARY, WASTE, VENT, DOM WATER AND COMPRESSED AIR ARE SHOWN ON PLUMBING REMOVALS SHEETS. REMOVE EQUIPMENT AND PIPING WITHIN THE FLOOR TRENCHES, UNLESS OTHERWISE NOTED. FOR BIDDING PURPOSES, ASSUME 1200 FEET OF 4" METAL STEAM AND HEATING WATER TYPE PIPE TO BE REMOVED FROM THE TRENCHES.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. [CLIN 002] REMOVE INDUSTRIAL PROCESS EXHAUST AND MAKE UP AIR SYSTEMS, INCLUDING BUT NOT LIMITED TO AIR HANDLING UNITS, FANS, CONTROLS, MIRING, DUCTWORK, HOODS, AND PIPING.
2. REMOVE UNIT HEATER AND ASSOCIATED BRANCH PIPING, CONTROLS AND WIRING.
3. REMOVE TERMINAL HEATING UNIT AND ASSCOIATED BRANCH PIPING, CONTROLS, AND WIRING.
4. REMOVE AIR CONDITIONER UNIT: DISPOSAL OF UNITS CONTAINING REFRIGERANT SHALL BE DONE IN ACCORDANCE WITH CLEAN AIR ACT, LOCAL, STATE, AND FEDERAL REGULATIONS. ALL REFRIGERANT MUST BE REMOVED PRIOR TO EQUIPMENT REMOVAL. THE USED REFRIGERANT SHALL BE RECLAIMED BY EPA APPROVED, SPECIALIZED EQUIPMENT AND DISPOSED OF BY CONTRACTOR. THE CONTRACTOR MUST PROVIDE THE MIDLANT ODS FORMS TO DOCUMENT REFRIGERANT RECOVERY, DISPOSAL, AND CHARGE ACTIVITIES. SEE SPECIFICATION SECTION 02 41 00 "DEMOLITION AND DECONSTRUCTION".
5. REMOVE EXHAUST FANS AND ASSOCIATED DUCTWORK, HOODS, CONTROLS, AND WIRING.
6. REMOVE STEAM VENT. COORDINATE PATCHING AND REPAIR OF ROOF OPENING.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

OAK POINT ASSOCIATES

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
SECOND FLOOR MECHANICAL REMOVALS PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 191 OF 282
MD107

DRAWFORM REVISION: 14 SEP 2023

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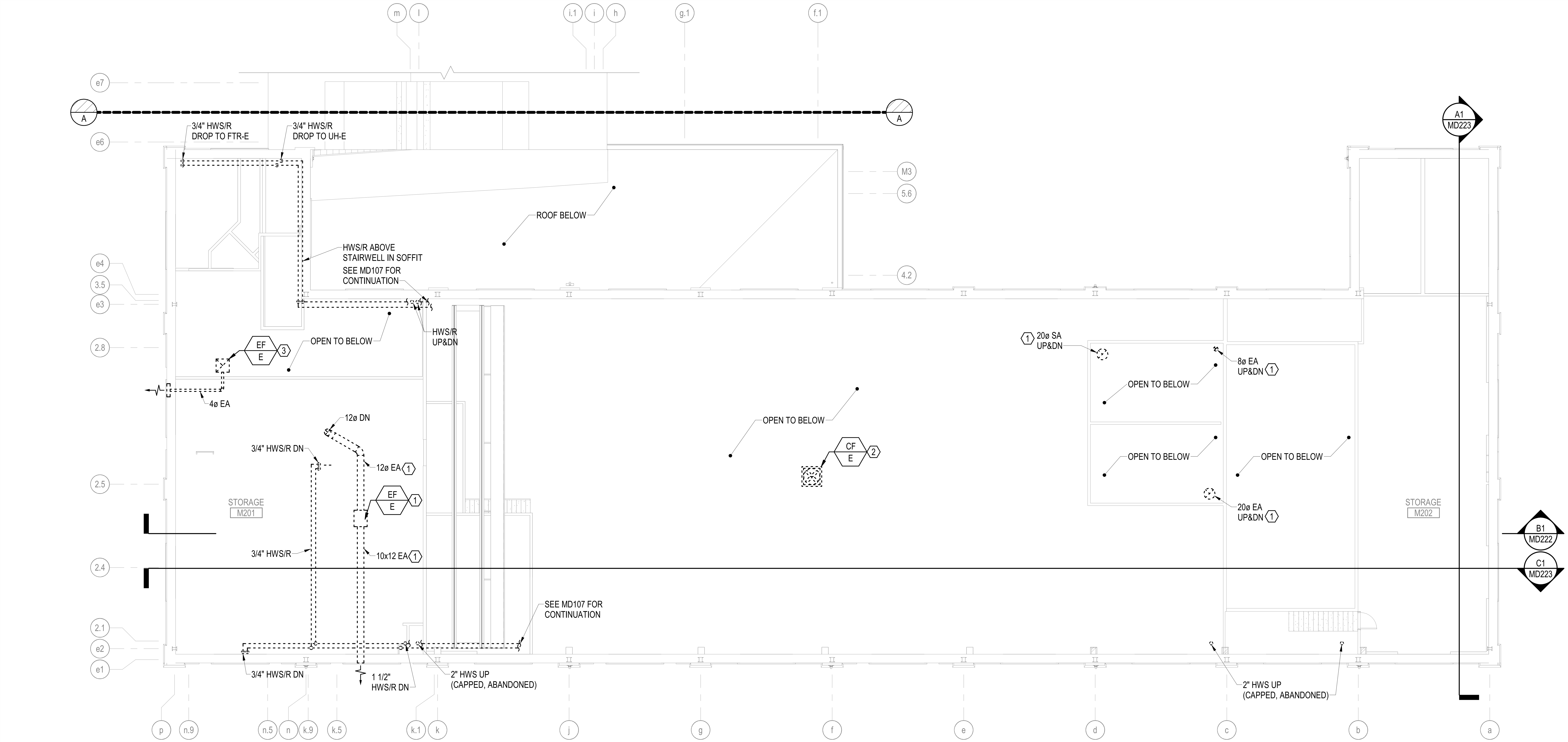


B1 SECOND FLOOR MEZZANINE MECHANICAL REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

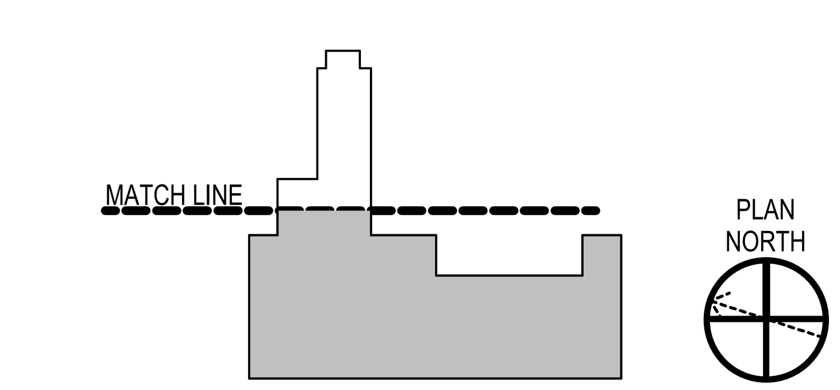
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL HVAC AND INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND DUCTWORK THROUGHOUT THE SPACES SHOWN ON THIS SHEET UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. [CLIN 002] REMOVE INDUSTRIAL PROCESS EXHAUST AND MAKE UP AIR SYSTEMS, INCLUDING BUT NOT LIMITED TO AIR HANDLING UNITS, FANS, CONTROLS, MIRING, DUCTWORK, HOODS, AND PIPING.
2. REMOVE CEILING FAN, ASSOCIATED CONTROLS AND WIRING.
3. REMOVE EXHAUST FAN AND ASSOCIATED DUCTWORK HOODS, CONTROLS AND WIRING.

KEY PLAN



GRAPHIC SCALE

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NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
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PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

SECOND FLOOR MEZZANINE MECHANICAL REMOVALS PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 192 OF 282
MD108

DRAWFORM REVISION: 14 SEP 2023

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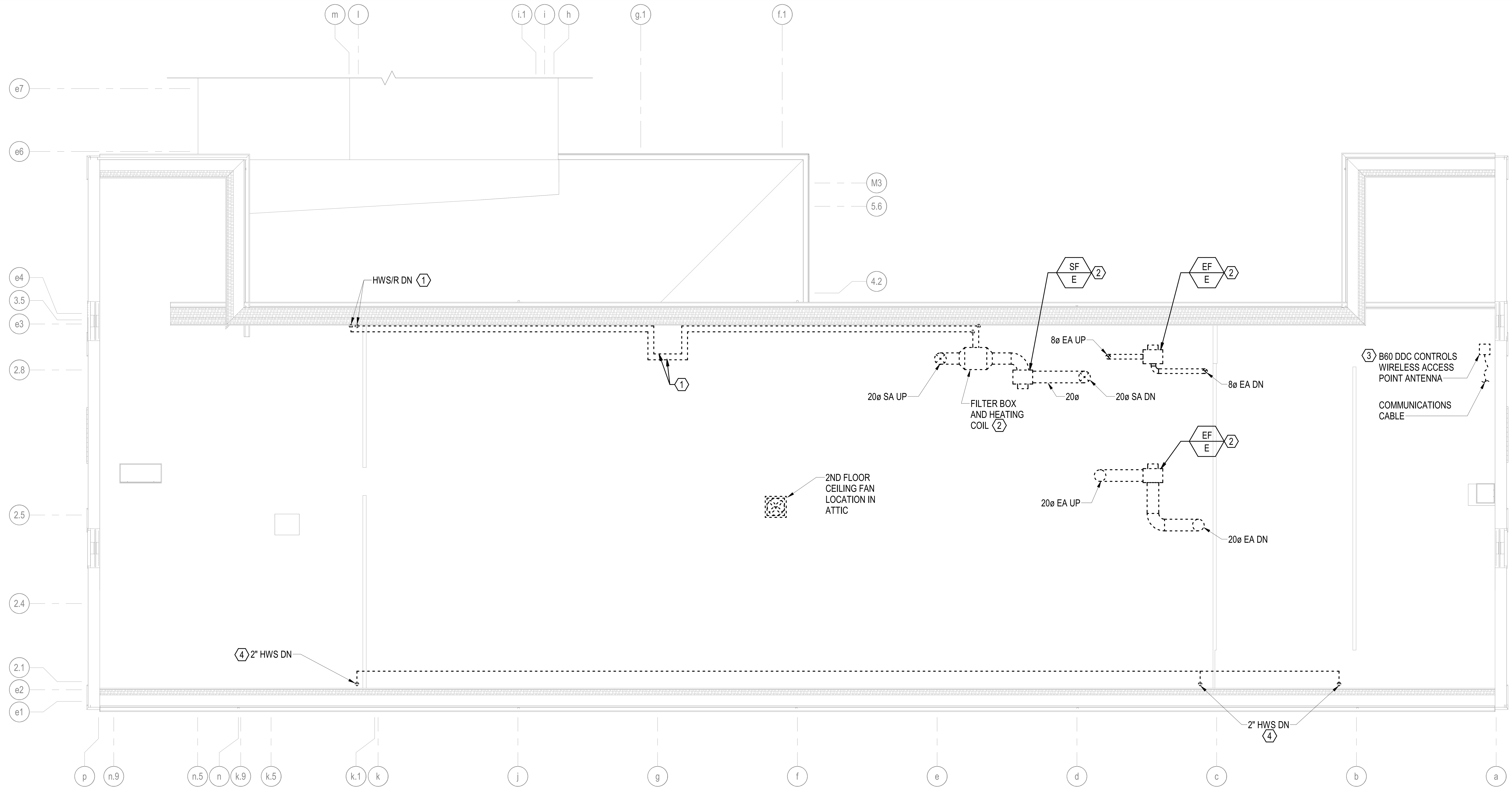


B1 ATTIC MECHANICAL REMOVALS PLAN - WEST

SCALE: 1/8" = 1'-0"

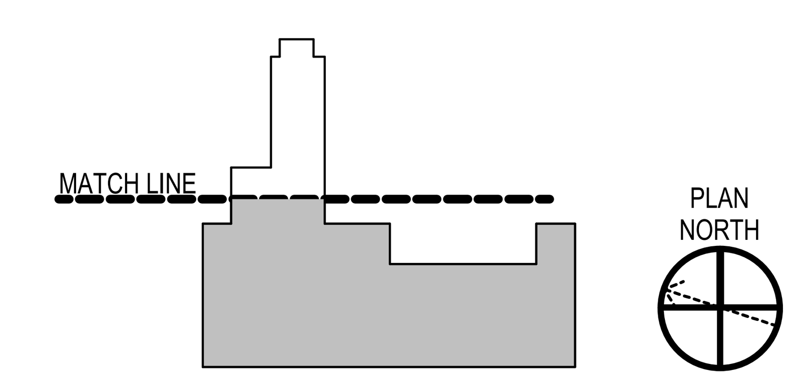
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND PIPING THROUGHOUT THE SPACES SHOWN ON THIS SHEET, UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE HWS/R PIPING.
2. [CLIN 002] REMOVE INDUSTRIAL PROCESS EXHAUST AND MAKE UP AIR SYSTEMS, INCLUDING BUT NOT LIMITED TO AIR HANDLING UNITS, FANS, CONTROLS, MIRING, DUCTWORK, HOODS, AND PIPING.
3. REMOVE DDC CONTROLS WIRELESS ACCESS POINT ANTENNA AND ASSOCIATED CONTROLS, WIRING, AND CABLES.
4. REMOVE ABANDONED HWS/R PIPING.

KEY PLAN



GRAPHIC SCALE

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NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
ATTIC MECHANICAL REMOVALS PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 193 OF 282
MD109

DRAWFORM REVISION: 14 SEP 2023

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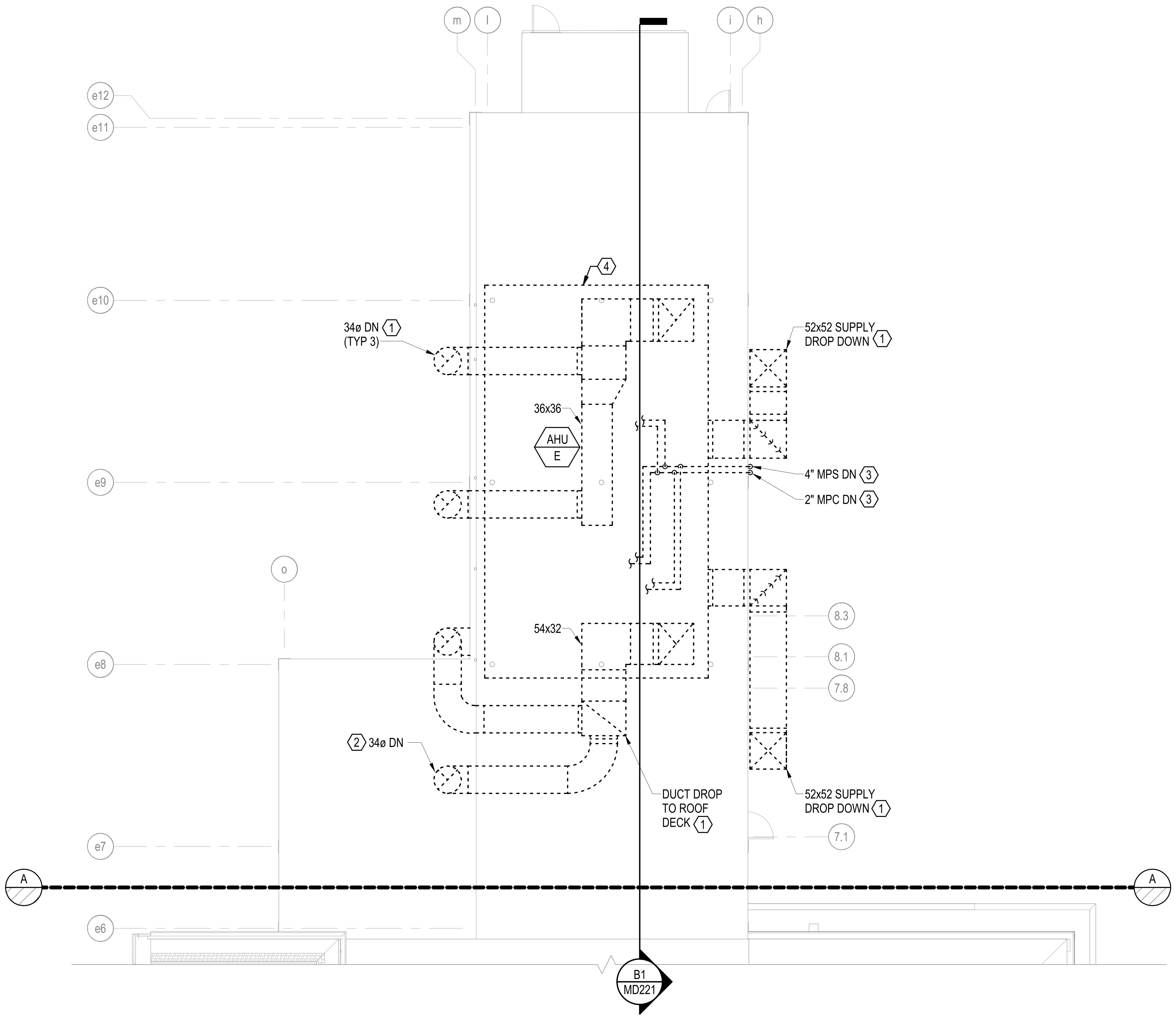


A1 **ROOF MECHANICAL REMOVALS PLAN - EAST**

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL EQUIPMENT, CONTROLS AND ASSOCIATED DUCTWORK SHOWN WITHIN ANNEX.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF PENETRATIONS REMOVED.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

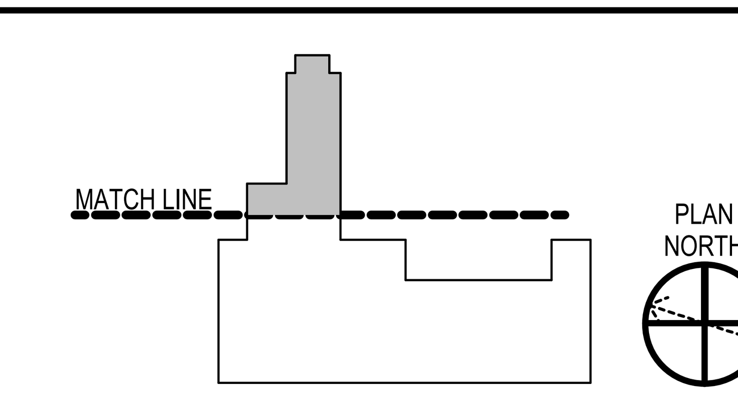
1 [CLIN 002] REMOVE DUCTWORK FROM MAKE UP AIR HANDLING UNIT TO EXTERIOR WALL PENETRATION.

2 [CLIN 002] REMOVE DUCTWORK FROM MAKE UP AIR HANDLING UNIT TO EXTERIOR ROOF PENETRATION.

3 [CLIN 002] REMOVE MPS AND MPC PIPING FROM CONNECTIONS TO MAKE UP AIR HANDLER DOWN TO THE ANNEX WALL PENETRATION.

4 [CLIN 002] REMOVE MAKE UP AIR HANDLER IN ITS ENTIRETY

KEY PLAN



GRAPHIC SCALE

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CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
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NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC

DRAWN BY CBM

CHECKED BY MSA

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM/ME

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

ROOF MECHANICAL REMOVALS PLAN - EAST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 194 OF 282

MD120

DRAWFORM REVISION: 14 SEP 2023

10/15/2025 12:13:25 PM MD120
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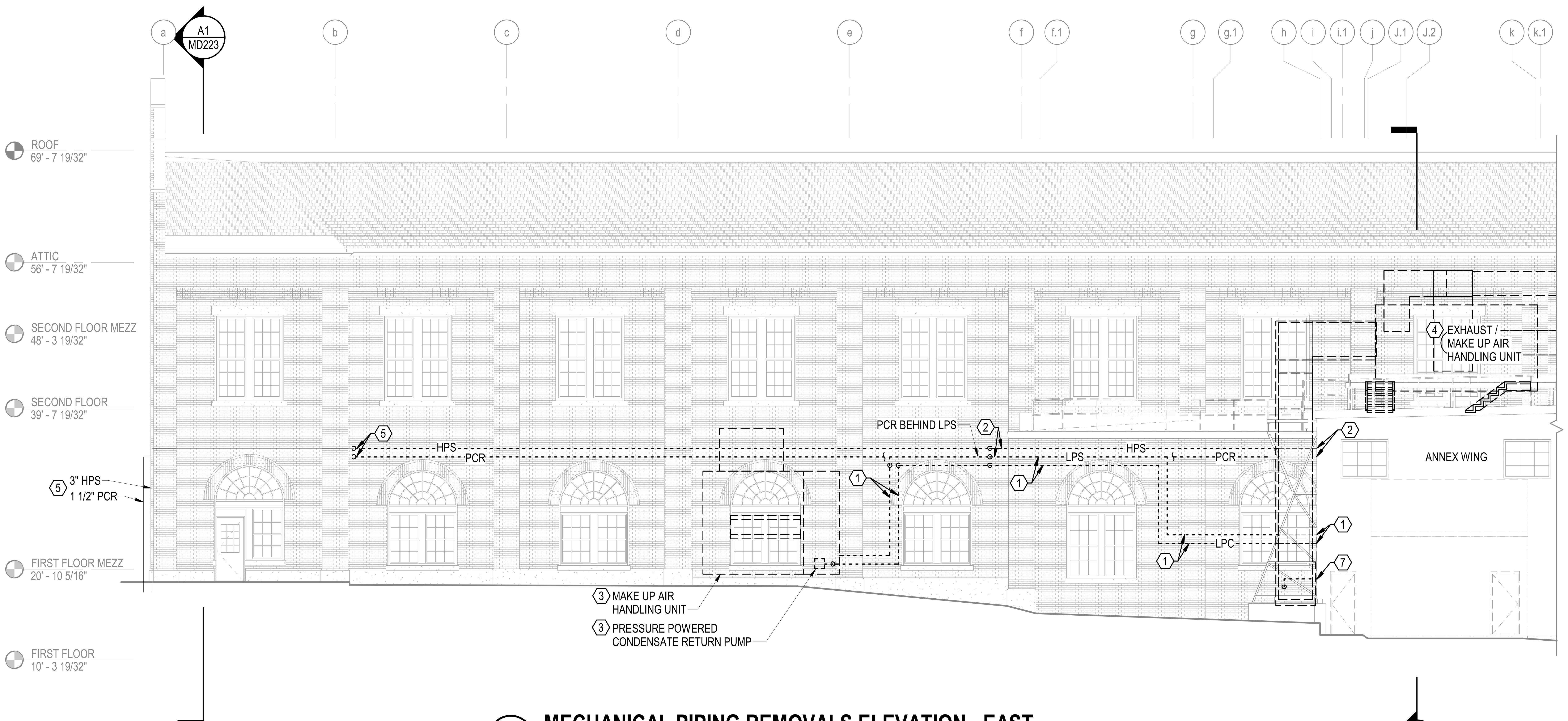
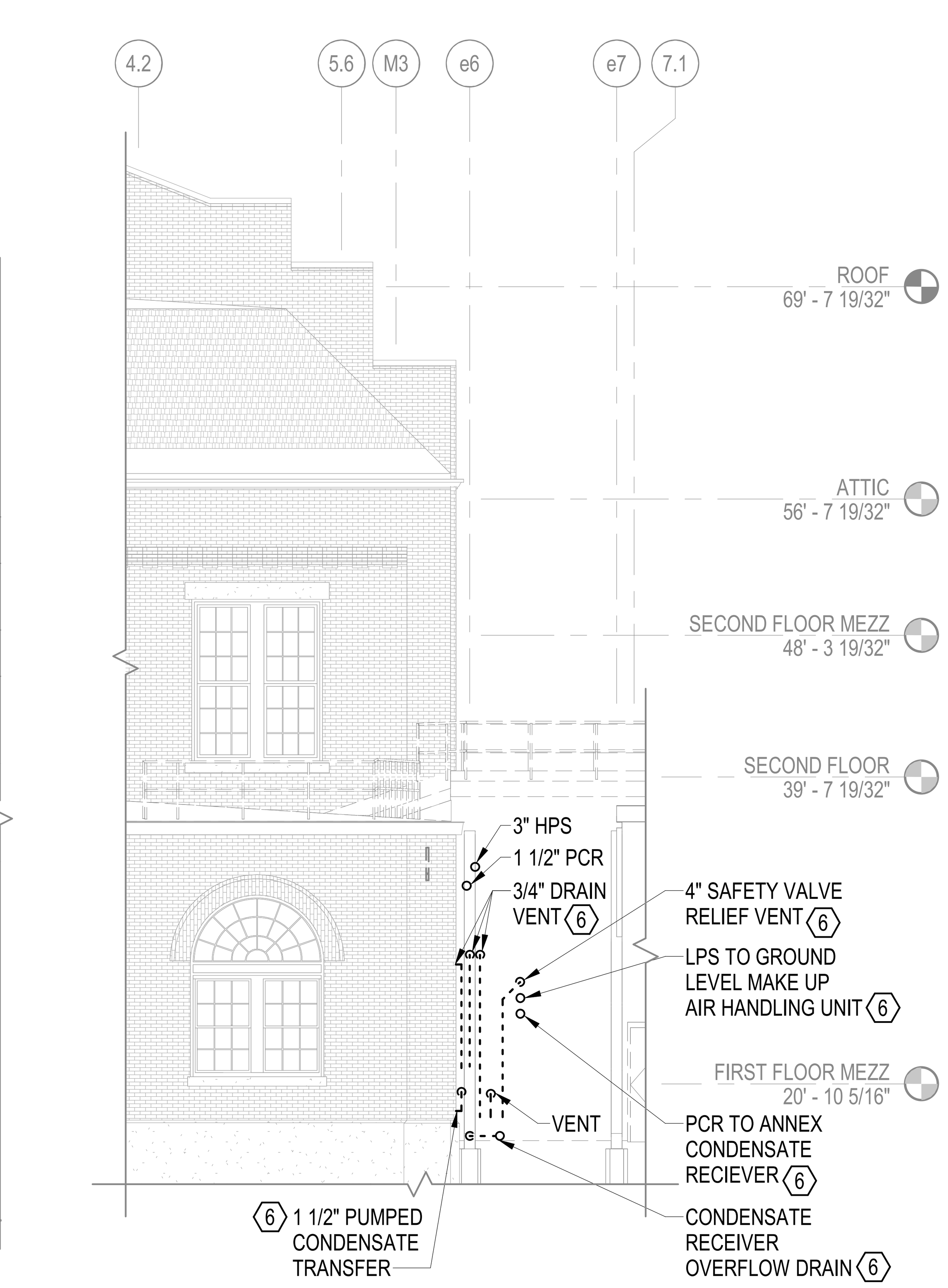


B1 MECHANICAL PIPING REMOVALS ELEVATION - EAST

SCALE: 1/8" = 1'-0" (MD102)



B5 PARTIAL SOUTH ELEVATION MECHANICAL PIPING REMOVALS

SCALE: 1/8" = 1'-0" (MD103)

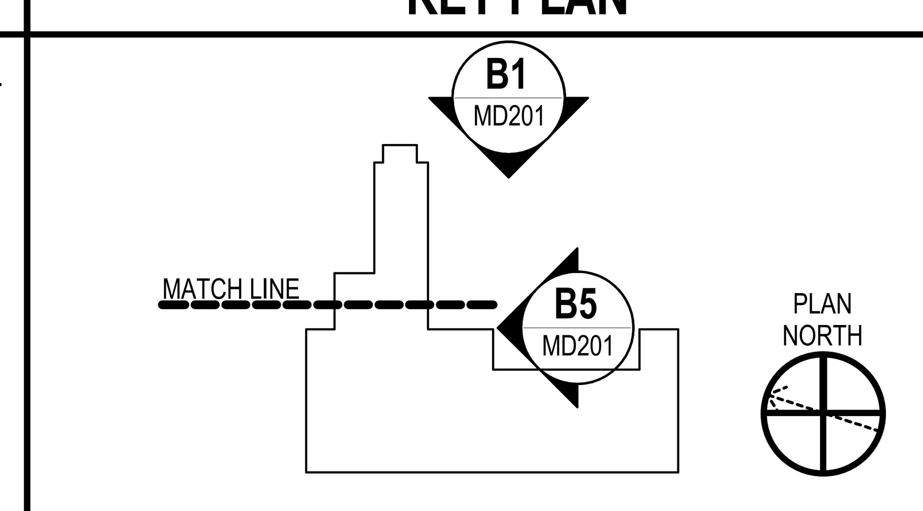
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND PIPING THROUGHOUT THE SPACES SHOWN ON THIS SHEET, UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL, AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT, DUCTWORK, AND PIPING THROUGHOUT THE BUILDING TO BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STEAM CONDENSATE SYSTEM RETURN LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING 60.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. [CLIN 002] REMOVE LPS AND PCR PIPING FROM GROUND LEVEL MAKE UP AIR HANDLING UNIT AND PRESSURE POWERED PUMP TO ANNEX WALL PENETRATIONS.
2. REMOVE HPS AND PCR PIPING FROM THE ANNEX WING WALL PENETRATIONS TO THE BUILDING OUTSIDE OF THE FIRST FLOOR MEZZANINE OPEN OFFICE. SEE MD106 FOR PLAN VIEW PIPING REMOVALS.
3. SEE SHEET MD106 FOR MAKE UP AIR HANDLING UNIT AND ASSOCIATED PRESSURE POWERED CONDENSATE PUMP REMOVAL REQUIREMENTS.
4. SEE SHEET MD120 FOR EXHAUST/MAKE UP AIR HANDLING UNIT SYSTEM REMOVALS.
5. HPS AND PCR PIPING TO REMAIN IN PLACE. REFER TO SHEET M-401 FOR PLAN SHOWING HPS AND PCR CONNECTIONS TO THE EXISTING PIPING TO IDENTIFY HOW MUCH OF THE EXISTING PIPING ALONG THE NORTH WALL SHOULD REMAIN. PIPE SUPPORTS FOR THESE PIPES MUST REMAIN IN PLACE.
6. REMOVE PIPING.
7. REMOVE STEAM CONDENSATE TRANSFER PIPING.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL EXTERIOR ELEVATION REMOVALS

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 195 OF 282
MD201

DRAWFORM REVISION: 14 SEP 2023

UNCLASSIFIED

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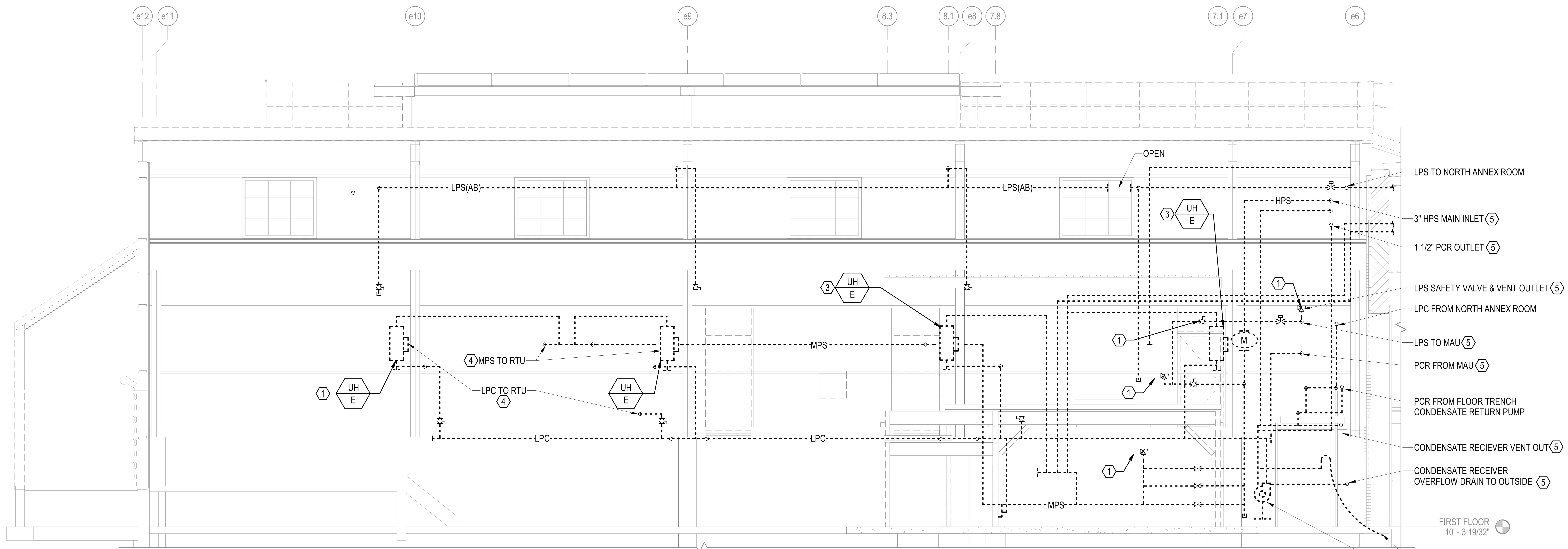


B1 FIRST FLOOR MECHANICAL PIPING REMOVALS BUILDING SECTION

SCALE: 1/4" = 1'-0" (MD102)

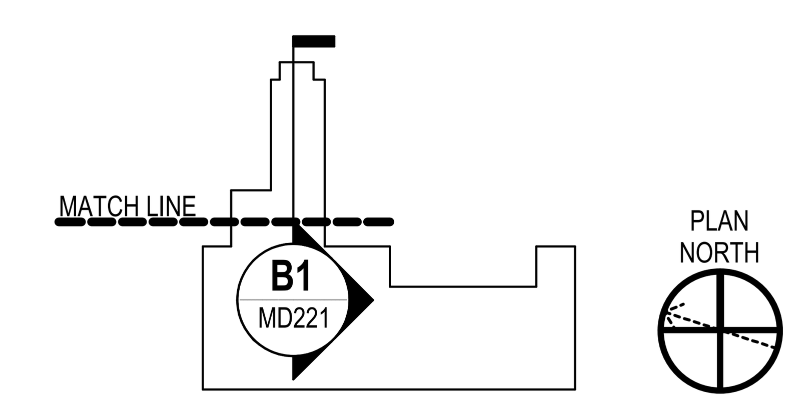
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL HVAC AND INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND DUCTWORK THROUGHOUT THE BUILDING UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT DUCTWORK, AND PIPING THROUGHOUT THE BUILDING MUST BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STREAM CONDENSATE SYSTEM LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING.
9. THE REMOVAL ELEVATION SHEETS ARE INCLUDED TO SHOW SPACIAL PERSPECTIVE. NOT ALL EQUIPMENT, DUCTWORK, OR PIPING IS SHOWN. REFER TO REMOVAL PLANS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1. REMOVE THROUGH-WALL EXHAUST FAN, AND ASSOCIATED CONTROLS AND WIRING.
2. REMOVE EXHAUST FAN AND ASSOCIATED DUCTWORK, LOUVERS AND CONTROLS.
3. REMOVE EXHAUST FAN AND ASSOCIATED DUCTWORK, HOODS, VFD, CONTROLS, AND WIRING.
4. REMOVE EXHAUST FAN AND ASSOCIATED DUCTWORK, ENCLOSURES, HOODS, CONTROLS, AND WIRING.
5. SEE SHEET MD201 FOR EXTERIOR PIPE REMOVAL.

KEY PLAN



GRAPHIC SCALE

1/4" = 1' - 0" 0 1' 5' 10' 15'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

MECHANICAL BUILDING SECTION REMOVALS 1

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.

SHEET 196 OF 282

MD221

DRAWFORM REVISION: 14 SEP 2023

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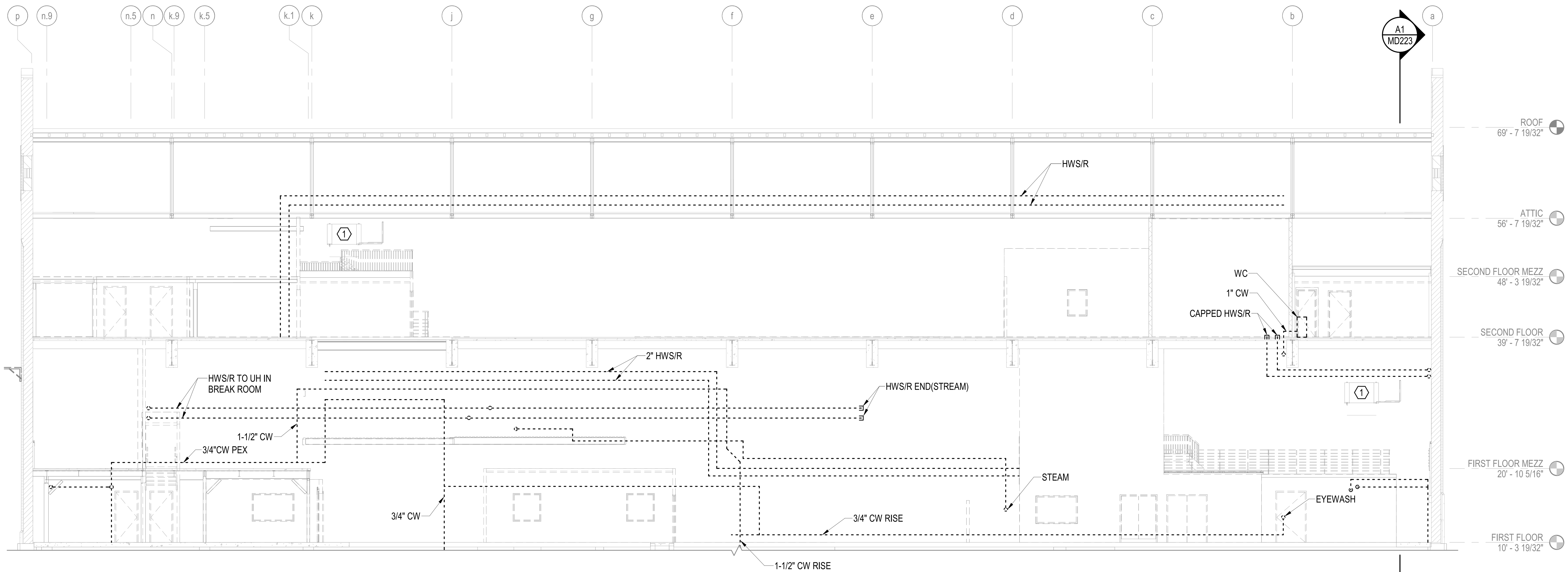


B1 EAST WALL MECHANICAL PIPING REMOVALS BUILDING SECTION

SCALE: 1/8" = 1'-0" (MD101)

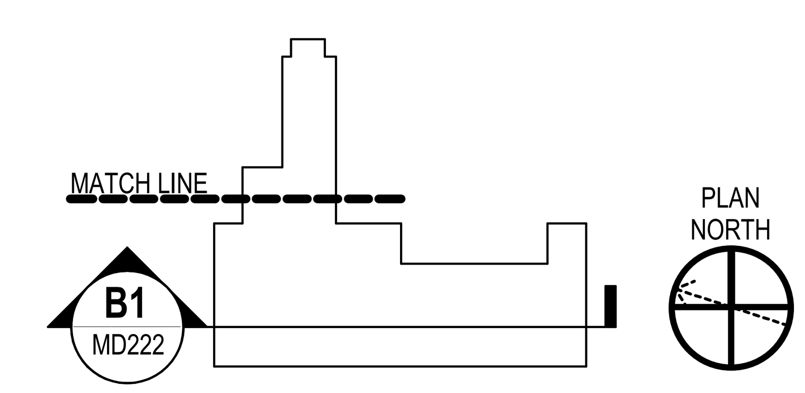
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL HVAC AND INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND DUCTWORK THROUGHOUT THE BUILDING UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT DUCTWORK, AND PIPING THROUGHOUT THE BUILDING MUST BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STREAM CONDENSATE SYSTEM LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING.
9. THE REMOVAL ELEVATION SHEETS ARE INCLUDED TO SHOW SPACIAL PERSPECTIVE. NOT ALL EQUIPMENT, DUCTWORK, OR PIPING IS SHOWN. REFER TO REMOVAL PLANS.

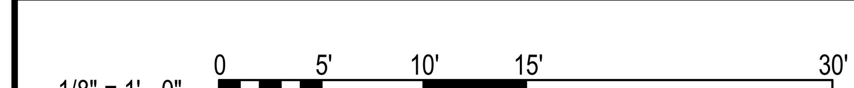
REMOVALS KEYNOTES (THIS SHEET ONLY)

1 OVERHEAD CRANE TO REMAIN IN PLACE.

KEY PLAN



GRAPHIC SCALE



CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO	DATE
DESIGNER	RNC
DRAWN BY	CBM
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM/E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

MECHANICAL BUILDING SECTION REMOVALS 2

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 197 OF 282
MD222

DRAWFORM REVISION: 14 SEP 2023

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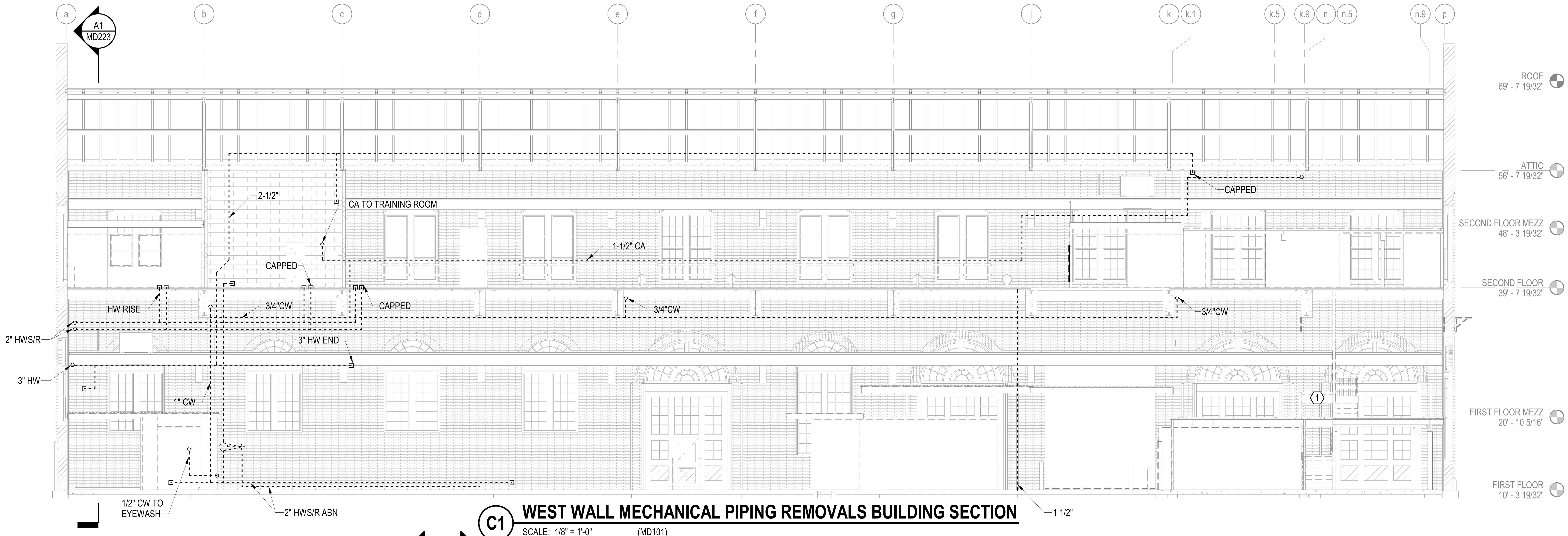
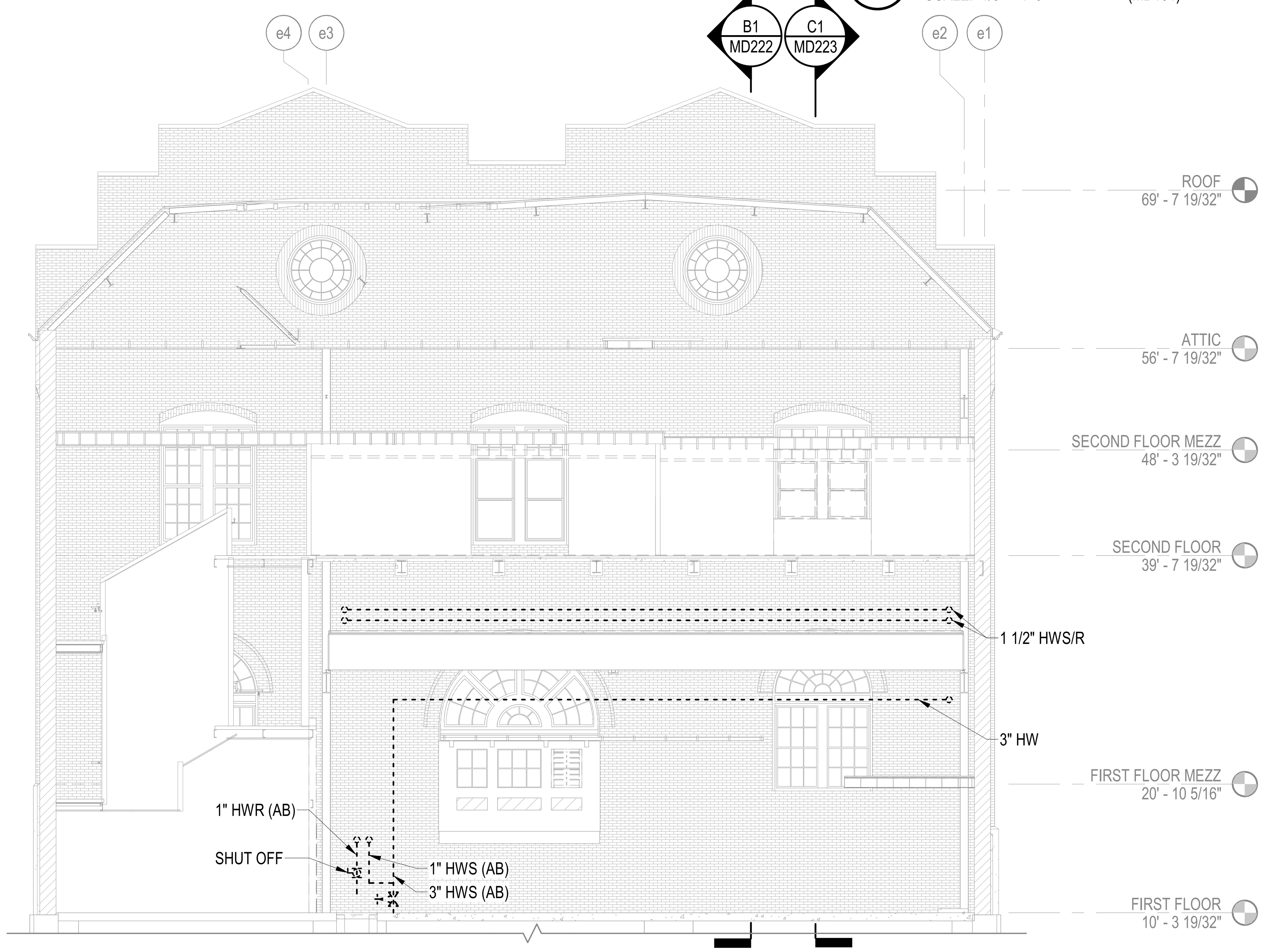


C1 WEST WALL MECHANICAL PIPING REMOVALS BUILDING SECTION

SCALE: 1/8" = 1'-0" (MD101)



A1 SOUTH WALL MECHANICAL PIPING REMOVALS BUILDING SECTION

SCALE: 1/8" = 1'-0" (MD101)

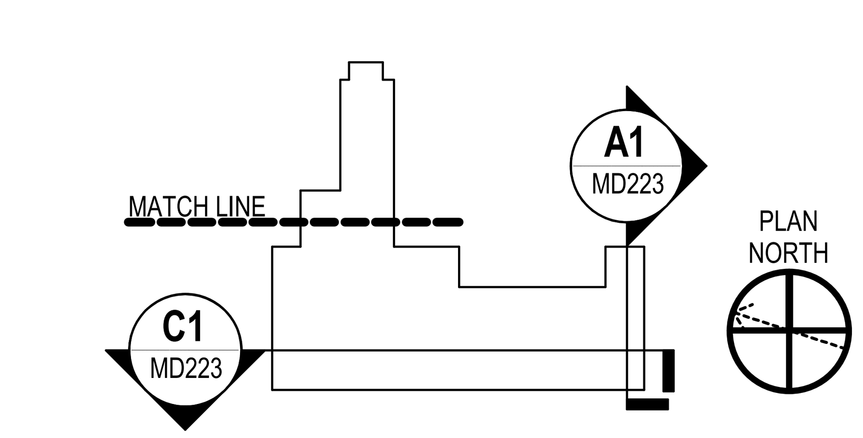
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. REMOVE ALL HVAC AND INDUSTRIAL PROCESS EQUIPMENT AND ASSOCIATED CONTROLS, WIRING AND DUCTWORK THROUGHOUT THE BUILDING UNLESS OTHERWISE NOTED.
3. COORDINATE WITH ARCHITECTURAL DRAWINGS FOR PATCHING OF OPEN FLOOR, WALL AND CEILING PENETRATIONS ASSOCIATED WITH REMOVALS.
4. MECHANICAL SUBCONTRACTOR SHALL IDENTIFY, MAKE SAFE, CUT LOOSE AND CAP ALL ITEMS SPECIFIED TO BE DEMOLISHED. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION AND REMOVAL OFF SITE TO AN AUTHORIZED SALVAGE FACILITY.
5. REFER TO CLIN 002 FOR LIST OF INDUSTRIAL EQUIPMENT REMOVAL REQUIREMENTS.
6. COORDINATE REMOVAL WITH HAZMAT DRAWINGS AND REQUIREMENTS.
7. EXISTING MECHANICAL EQUIPMENT DUCTWORK, AND PIPING THROUGHOUT THE BUILDING MUST BE REMOVED, UNLESS OTHERWISE NOTED.
8. CONDUCT TESTING OF EXISTING PUMPED STREAM CONDENSATE SYSTEM LINE BACK PRESSURE AND REPORT RESULTS TO CONTRACTING OFFICER PRIOR TO COMMENCING WITH REMOVALS IN BUILDING.
9. THE REMOVAL ELEVATION SHEETS ARE INCLUDED TO SHOW SPACIAL PERSPECTIVE. NOT ALL EQUIPMENT, DUCTWORK, OR PIPING IS SHOWN. REFER TO REMOVAL PLANS.

REMOVALS KEYNOTES (THIS SHEET ONLY)

1 OVERHEAD CRANE TO REMAIN IN PLACE.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	CBM
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

MECHANICAL BUILDING SECTION REMOVALS 3

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 198 OF 282

MD223

DRAWFORM REVISION: 14 SEP 2023

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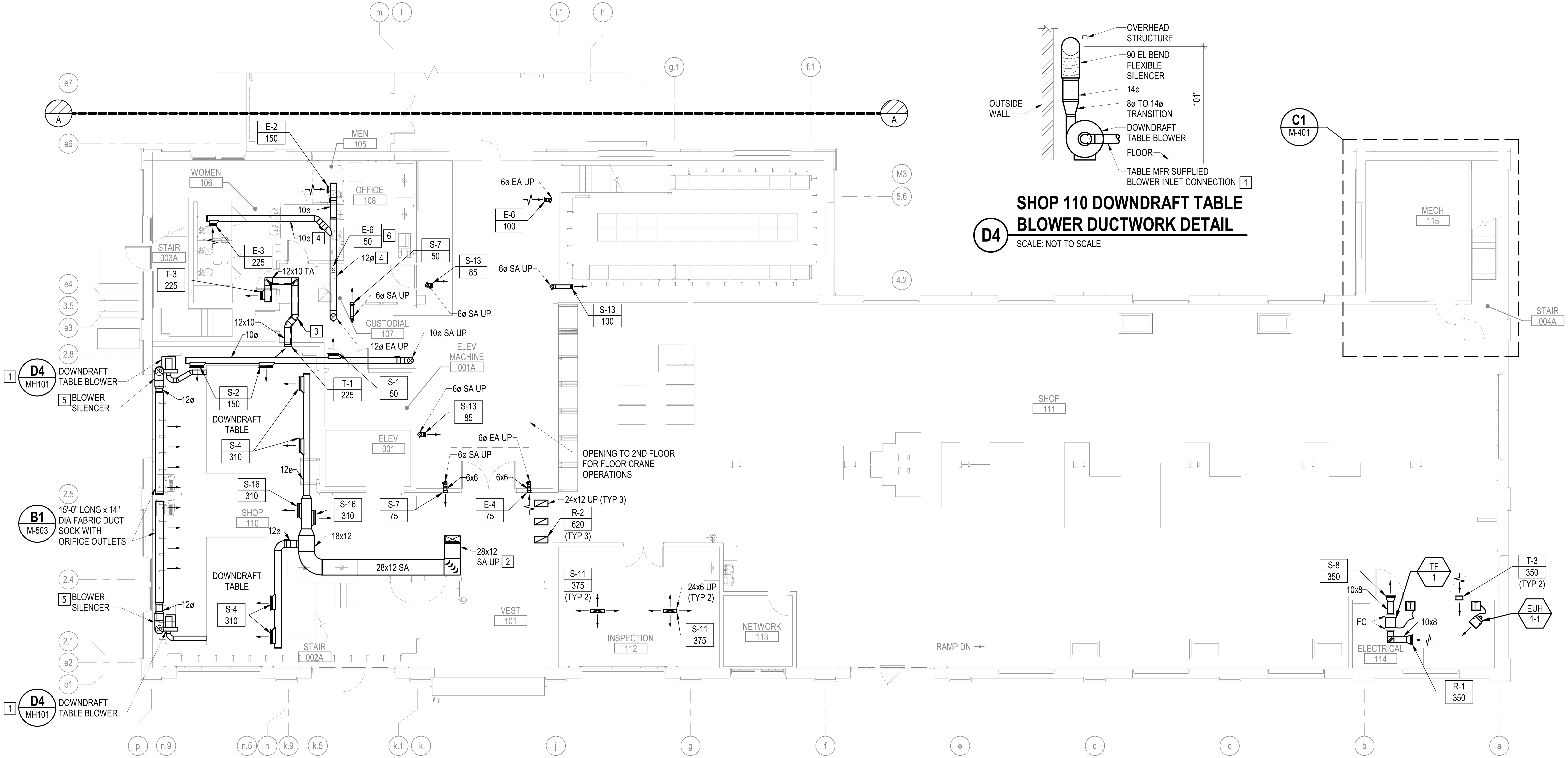


D4 SHOP 110 DOWNDRAFT TABLE BLOWER DUCTWORK DETAIL

SCALE: NOT TO SCALE

B1 FIRST FLOOR MECHANICAL DUCTWORK PLAN - WEST

SCALE: 1/8" = 1'-0"

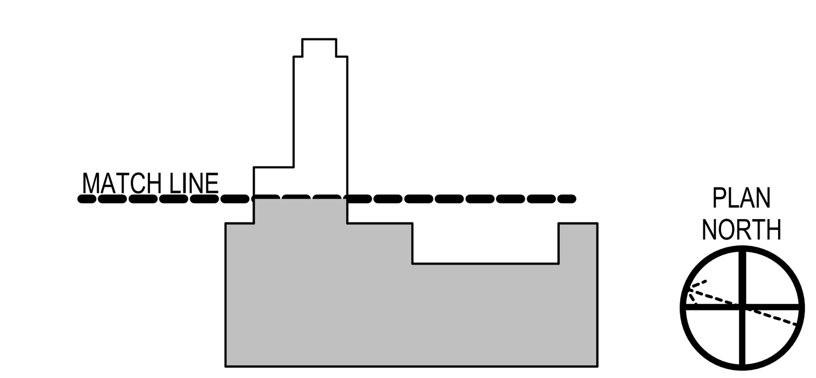
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.

KEYNOTES (THIS SHEET ONLY)

1. CONNECTIONS FROM DOWNDRAFT TABLE TO BLOWER MUST BE PROVIDED BY TABLE MANUFACTURER CERTIFIED INSTALLERS, AS PART OF THE EQUIPMENT PACKAGE.
2. PROVIDE TURNING VANES.
3. PROVIDE 1/2" THICK CLOSED CELL ELASTOMERIC RUBBER INSULATION DUCT LINER FOR TRANSFER DUCT.
4. RUN DUCTWORK EXPOSED NEAR CEILING.
5. PROVIDE TRANSITION DUCTWORK, FLEXIBLE SILENCER AND CONNECTING HARDWARE FROM DOWNDRAFT TABLE BLOWER OUTLET THROUGH DUCT SOCK DIFFUSER ASSEMBLY. COORDINATE INSTALLATION OF BLOWER WITH TABLE SYSTEM INSTALLERS AND VERIFY IN FIELD STRUCTURAL CONSTRAINTS ON PLACEMENT OF SILENCER AND DUCT SOCK ASSEMBLY, REVISE DUCT LENGTHS AS NEEDED TO FIT WHERE INTENDED.
SILENCER B00: SOUNDOWN DUCT SILENCER; INLET DIAMETER-14", OUTSIDE DIAMETER-16".
DUCTSOCK B00: DUCT SOX, FABRIC-VERONA NON-POROUS, INTERNAL HOOP SYSTEM SUSPENSION, ORIFICE OUTLETS.
6. PROVIDE DAMPER IN BRANCH DUCT.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY KLG
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MECHANICAL DUCTWORK PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 199 OF 282
MH101

DRAWFORM REVISION: 14 SEP 2023

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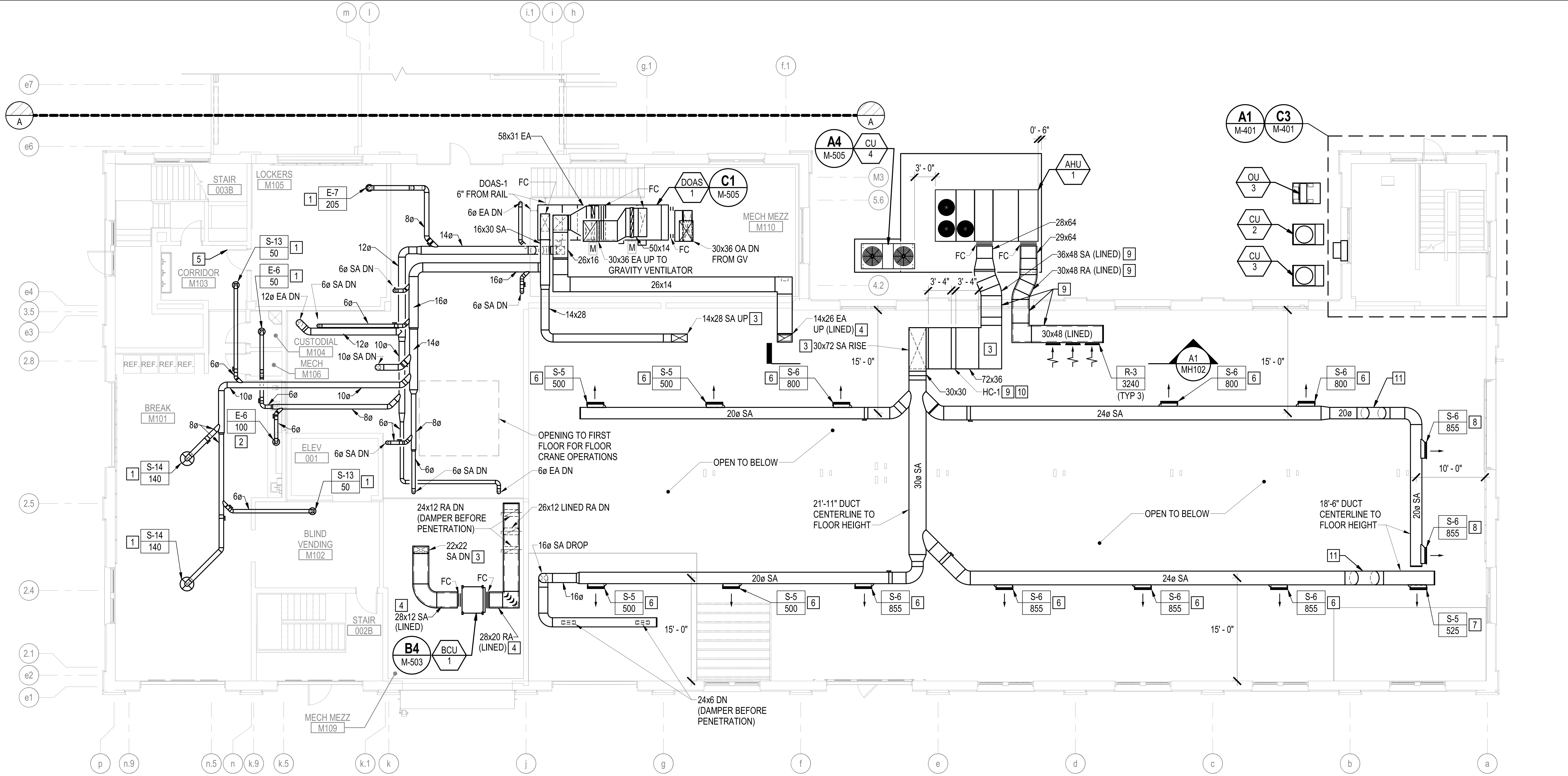
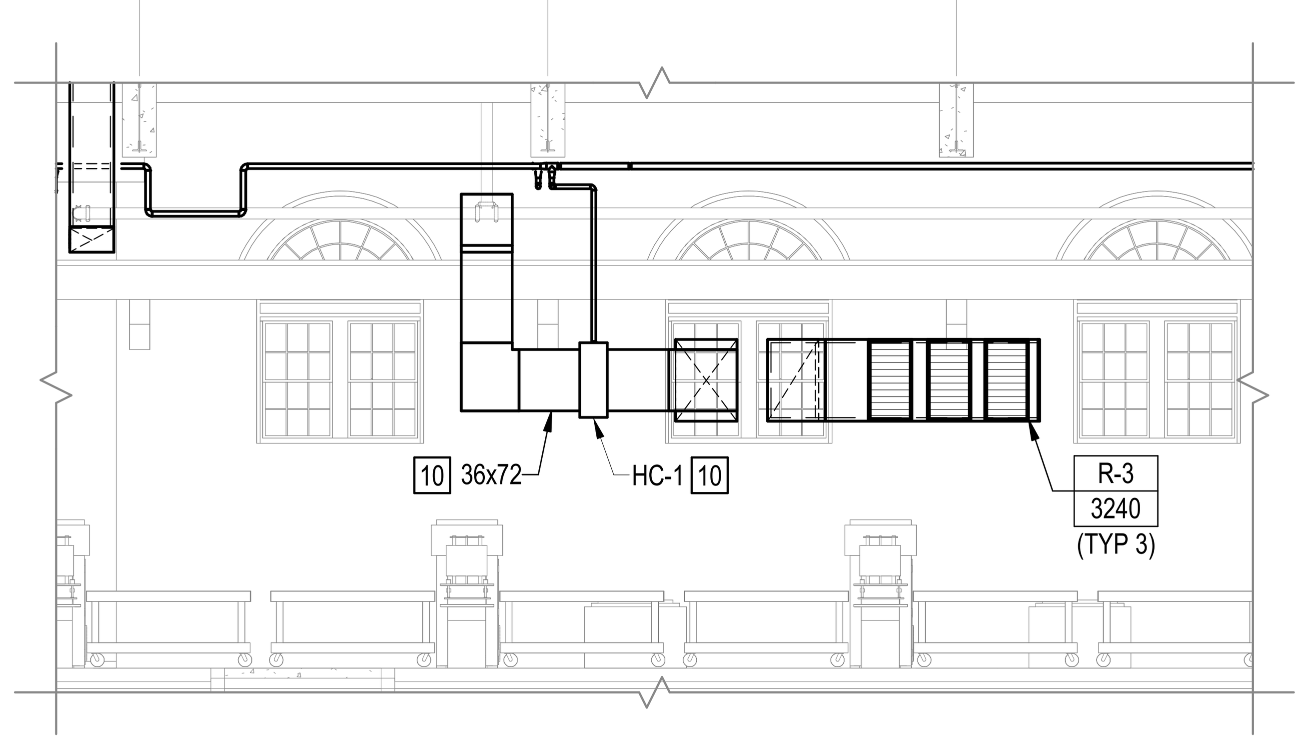


B1 FIRST FLOOR MEZZANINE MECHANICAL DUCTWORK PLAN - WEST

SCALE: 1/8" = 1'-0"



A1 AHU-1 DUCT HEATING COIL (HC-1) SECTION

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. PROVIDE 1/2" THICK CLOSED CELL ELASTOMERIC INSULATION FOR DUCT LINER.

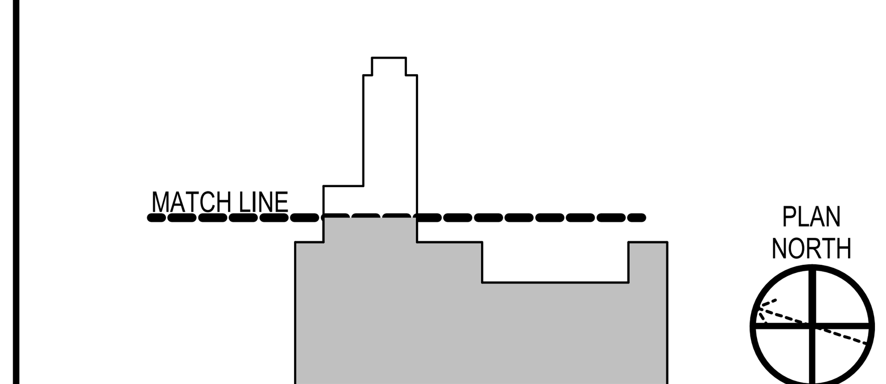
KEYNOTES (THIS SHEET ONLY)

1. LOCATE DIFFUSER/GRILLE 10 FEET ABOVE FINISHED FLOOR.
2. LOCATE EXHAUST GRILLE 84" ABOVE FINISHED FLOOR.
3. PROVIDE TURNING VANES IN DUCT ELBOWS/TRANSITIONS.
4. PROVIDE 1/2" THICK CLOSED CELL ELASTOMERIC RUBBER INSULATION DUCT LINER.
5. LOUVERED DOOR.

KEYNOTES (CONTINUED) (THIS SHEET ONLY)

6. ORIENT THE HORIZONTAL CENTERLINE OF THE GRILLE 25 DEGREES BELOW HORIZONTAL.
7. ORIENT THE HORIZONTAL CENTERLINE OF THE GRILLE 20 DEGREES BELOW HORIZONTAL.
8. ORIENT THE HORIZONTAL CENTERLINE OF THE GRILLE 0 DEGREES BELOW HORIZONTAL.
9. PROVIDE DUCT INSULATION ON AHU-1 RETURN DUCTWORK. PROVIDE DUCT INSULATION ON AHU-1 SUPPLY DUCTWORK FROM UNIT UP TO AND INCLUDING HEATING COIL HC-1.
10. SIZE THE SECTIONS OF DUCTWORK TO EACH SIDE OF THE DUCT HEATING COIL TO CONNECT WITHOUT TRANSITION.
11. DUCT OFFSET DOWNWARD TO RUN UNDER OVERHEAD CRANE ASSEMBLY.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	CBM
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
FIRST FLOOR MEZZANINE MECHANICAL DUCTWORK PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 200 OF 282
MH102

DRAWFORM REVISION: 14 SEP 2023

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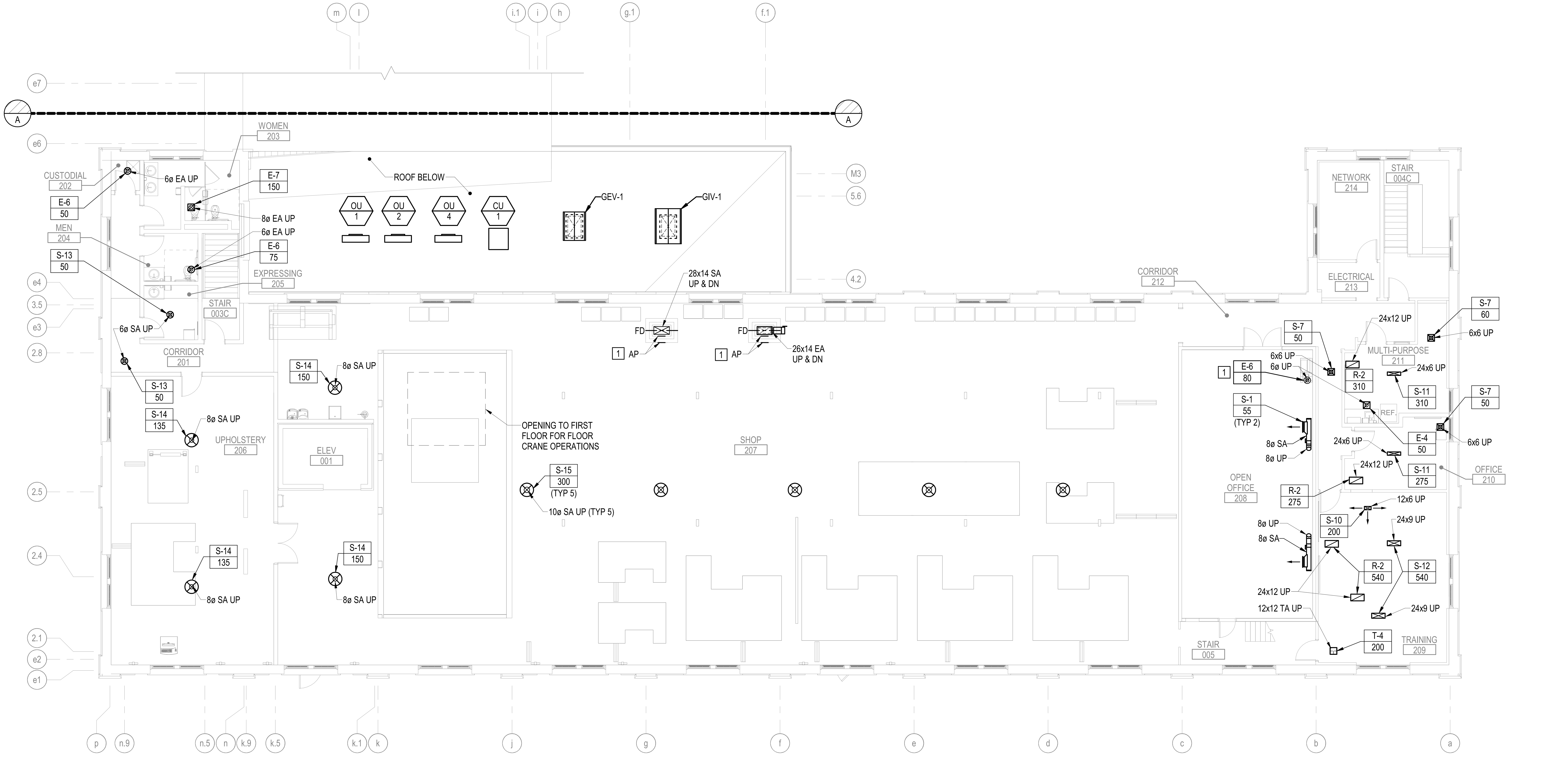


B1 SECOND FLOOR MECHANICAL DUCTWORK PLAN - WEST

SCALE: 1/8" = 1'-0"

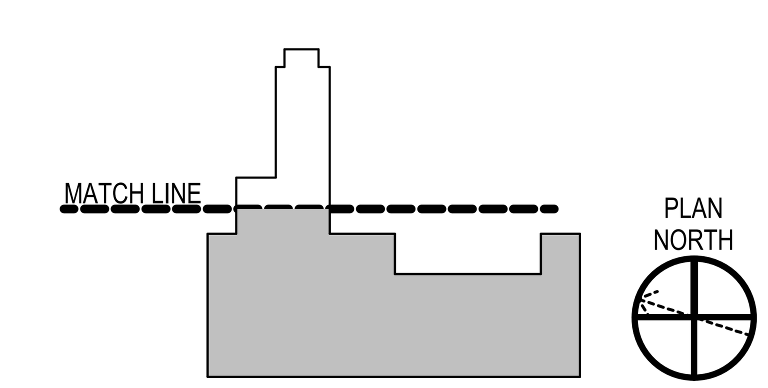
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.

KEYNOTES (THIS SHEET ONLY)

1 LOCATE ACCESS PANELS TO PERMIT ACCESS TO FIRE DAMPERS FUSEABLE LINE INSIDE DUCT.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" — 0, 5', 10', 15', 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

SECOND FLOOR MECHANICAL DUCTWORK PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 201 OF 282
MH103

DRAWFORM REVISION: 14 SEP 2023

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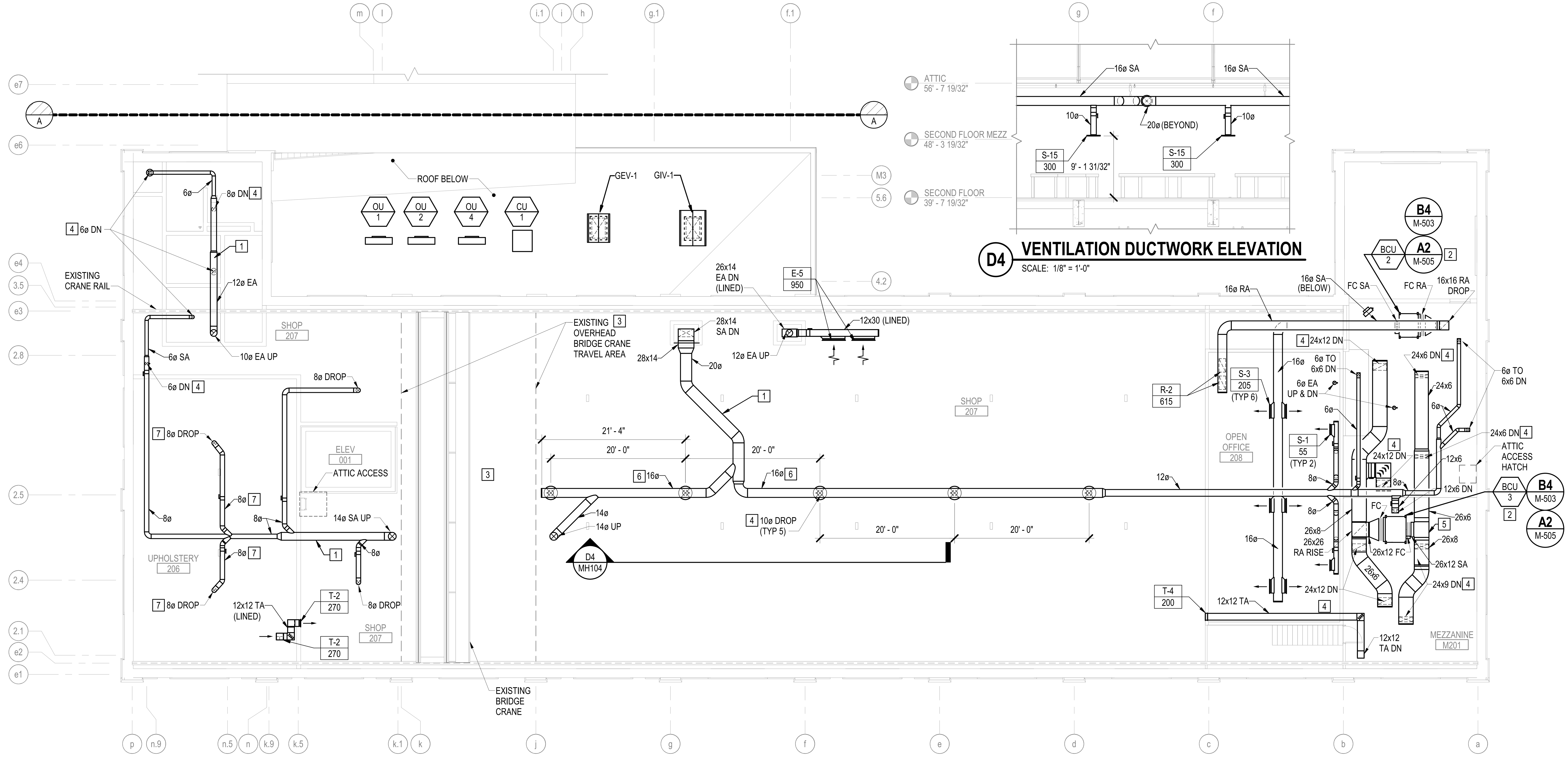


D4 VENTILATION DUCTWORK ELEVATION

SCALE: 1/8" = 1'-0"

B1 SECOND FLOOR MEZZANINE MECHANICAL DUCTWORK PLAN - WEST

SCALE: 1/8" = 1'-0"

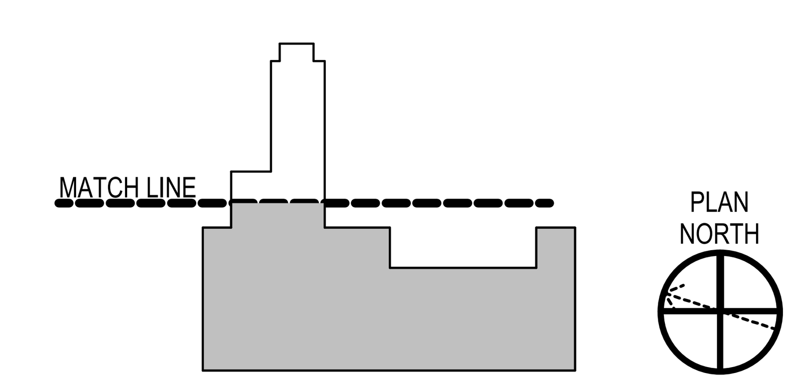
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. FOR DUCTWORK LINER WITH 1/2" THICK CLOSED COIL ELASTOMERIC RUBBER INSULATION.

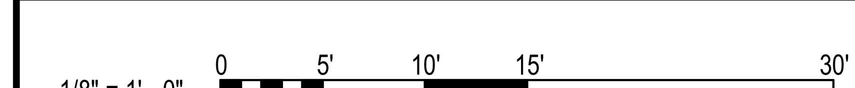
KEYNOTES (THIS SHEET ONLY)

1. LOCATE CENTERLINE OF DUCT 22" BELOW CEILING SURFACE.
2. SUPPORT BLOWER COIL UNIT ON A CHANNEL STRUT SUPPORT STAND WITH ELASTOMERIC VIBRATION ISOLATORS. BASE OF BLOWER COIL UNIT ENCLOSURE TO BE 36 INCHES ABOVE FINISHED FLOOR.
3. DUCTWORK NOT PERMITTED BETWEEN COLUMN LINES K AND J DUE TO OVERHEAD BRIDGE CRANE TRAVEL AREA.
4. PROVIDE BALANCING DAMPER ON DUCT DROP.
5. PROVIDE TURNING VANES.
6. LOCATE CENTERLINE OF DUCT OUTLET 24" BELOW CEILING.
7. COORDINATE DUCT LOCATIONS WITH SHOP CEILING LIGHTS.

KEY PLAN



GRAPHIC SCALE



CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY KLG
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

SECOND FLOOR MEZZANINE MECHANICAL DUCTWORK PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 202 OF 282
MH104

DRAWFORM REVISION: 14 SEP 2023

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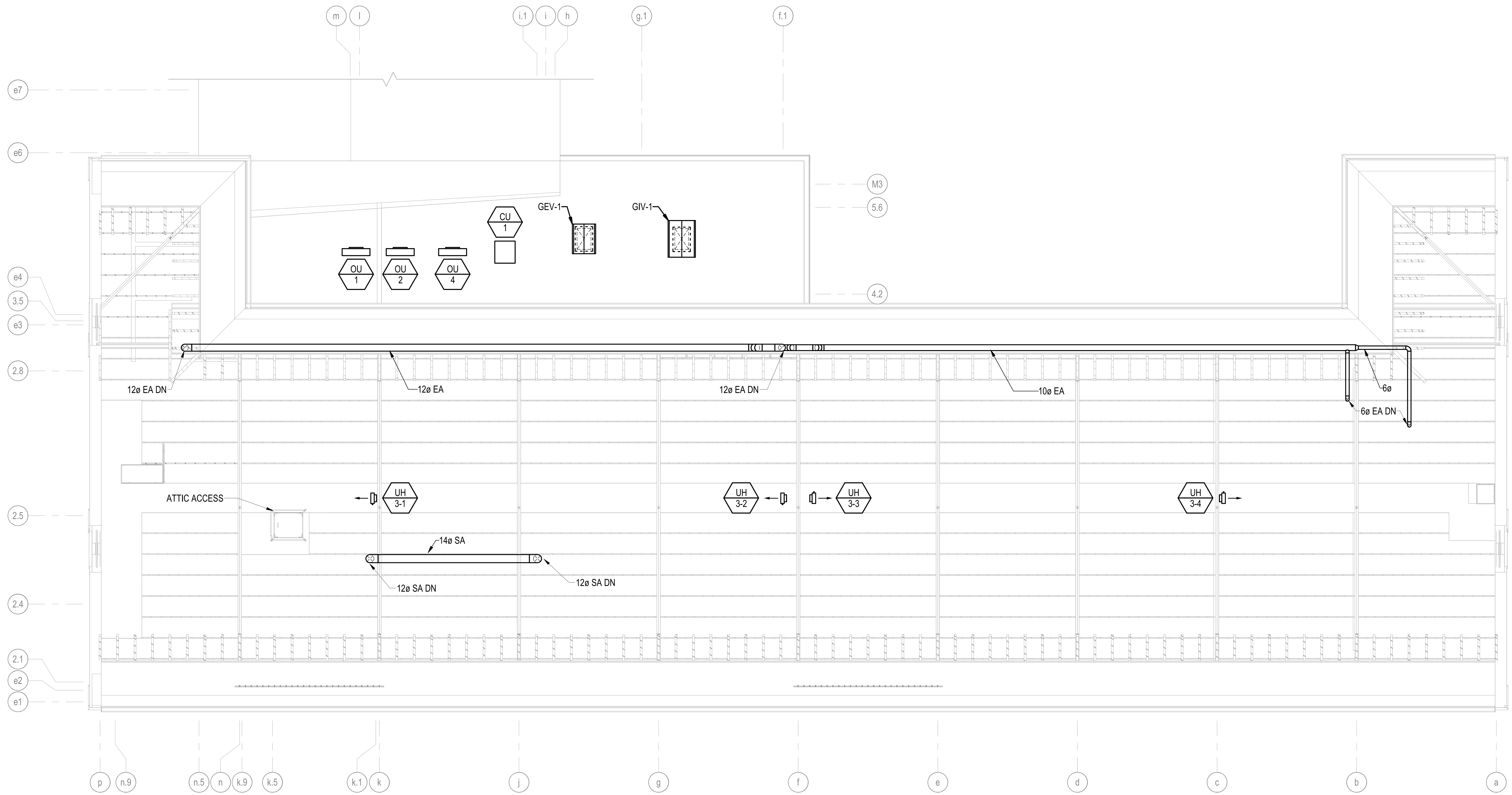

B1 ATTIC MECHANICAL DUCTWORK PLAN - WEST

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.

KEYNOTES (THIS SHEET ONLY)

1 -

KEY PLAN

MATCH LINE

PLAN NORTH

GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC

DRAWN BY SRW

CHECKED BY MSA

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

ATTIC MECHANICAL DUCTWORK PLAN

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 203 OF 282

MH105

DRAWFORM REVISION: 14 SEP 2023

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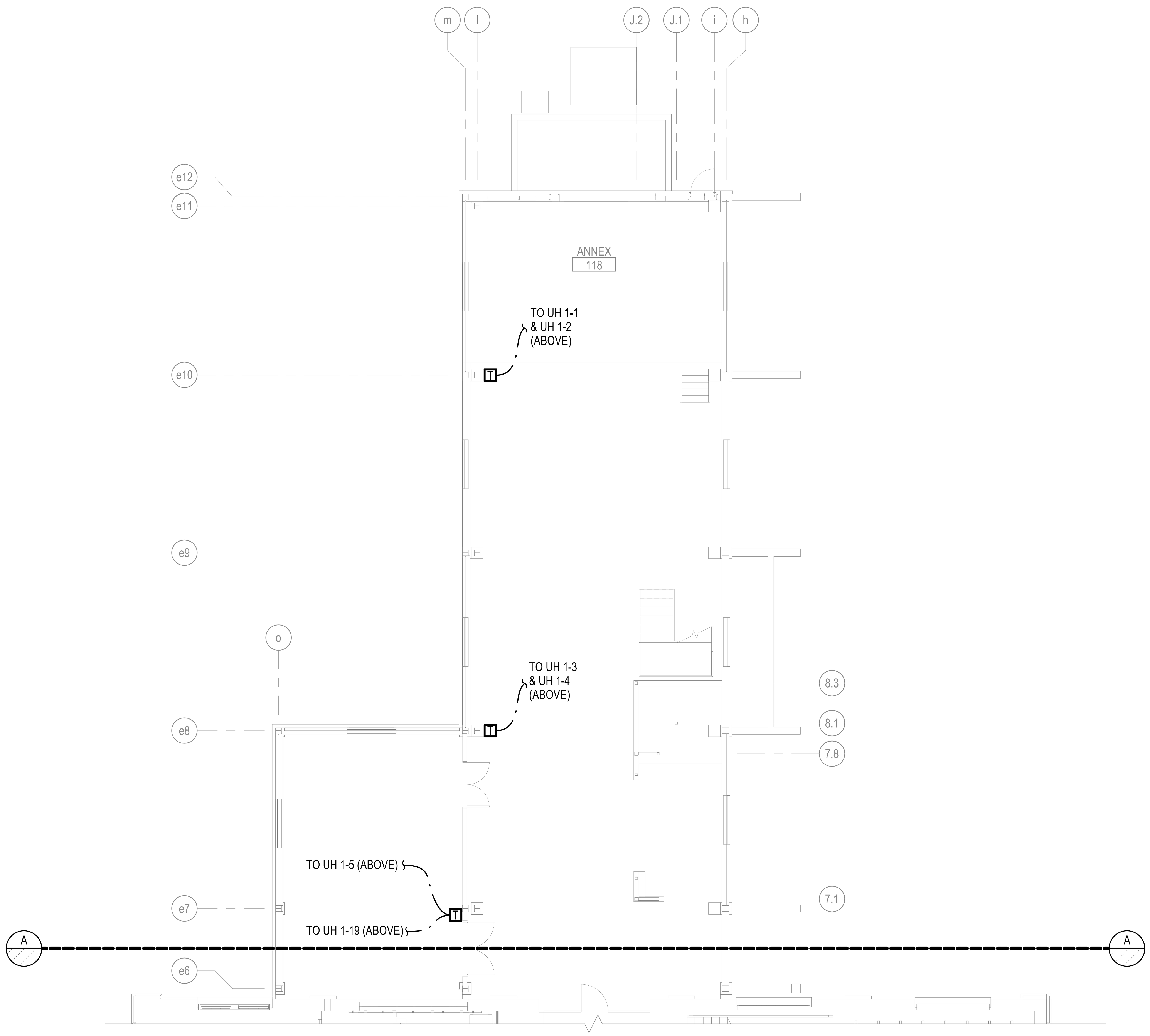


A1 **FIRST FLOOR MECHANICAL PIPING PLAN - EAST**

SCALE: 1/8" = 1'-0"

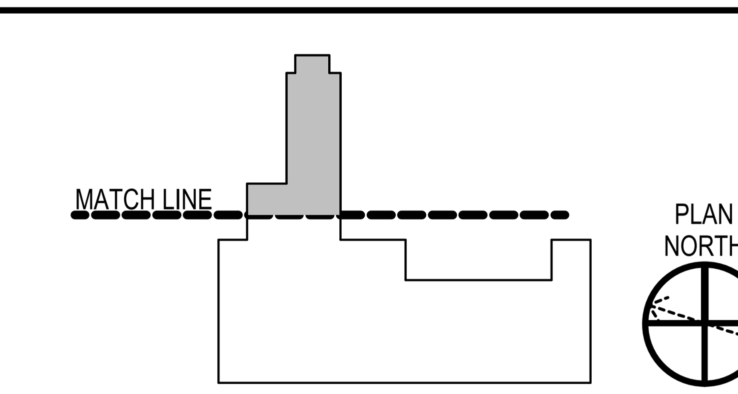
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.

KEYNOTES (THIS SHEET ONLY)

1 -

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC

DRAWN BY SRW

CHECKED BY MSA

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME

PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MECHANICAL PIPING PLAN - EAST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 204 OF 282

MP101

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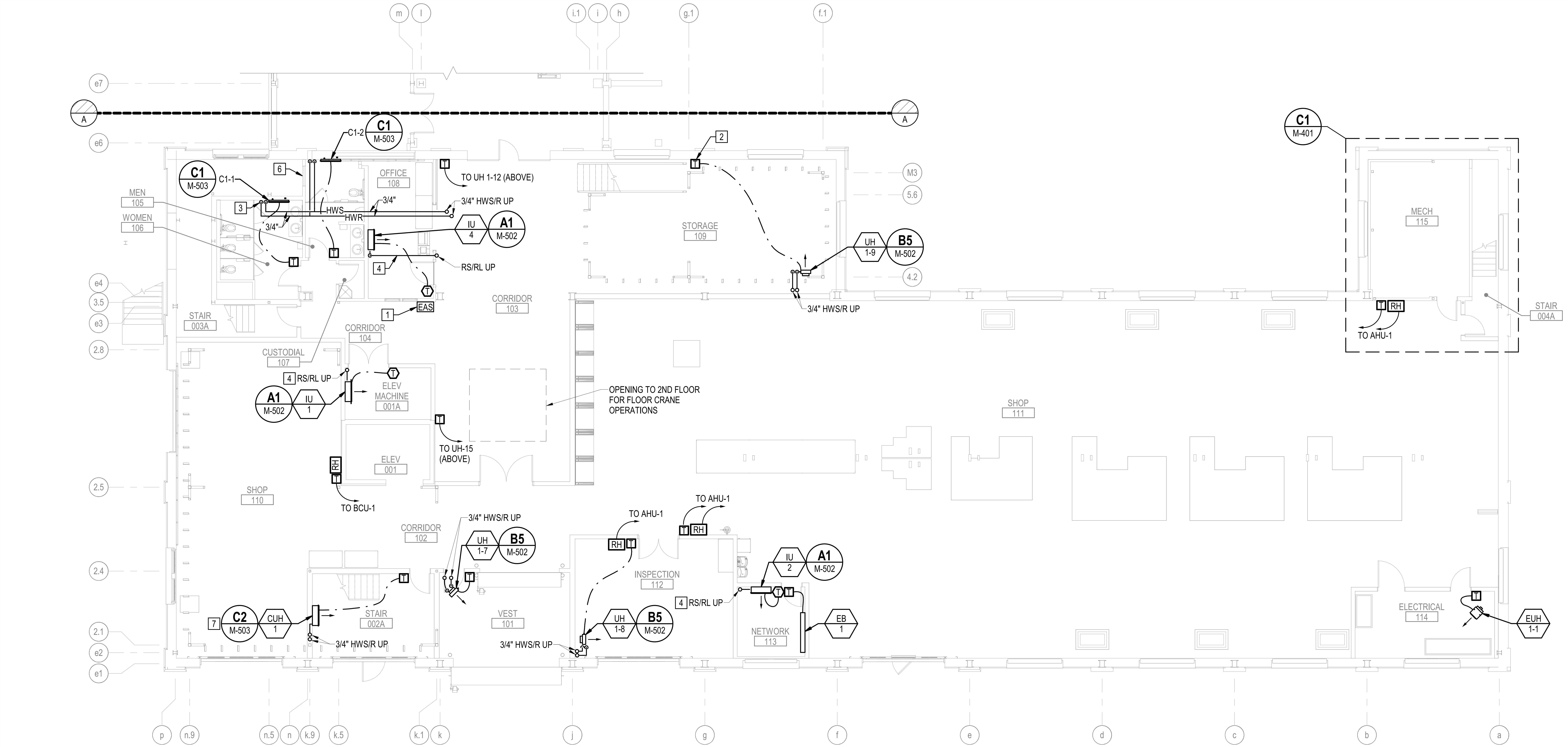


B1 FIRST FLOOR MECHANICAL PIPING PLAN - WEST

SCALE: 1/8" = 1'-0"

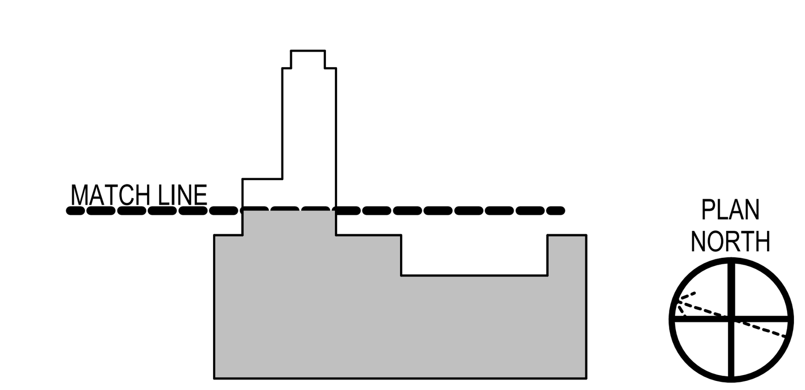
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. HWS/R TERMINAL HEATER BRANCH PIPING IS 3/4" UNLESS OTHERSWISE INDICATED.

KEYNOTES (THIS SHEET ONLY)

1. EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH. SEE A1/M-701 FOR MORE INFORMATION.
2. PROVIDE INSULATED BACKER PLATE FOR T-STAT MOUNTED TO AN OUTSIDE WALL.
3. PROVIDE PVC JACKET ON EXPOSED PIPE INSULATION UP TO 8'-0" AFF.
4. PROVIDE PRE-MANUFACTURED REFRIGERANT LINE ENCLOSURES WITH REMOVABLE COVERS FOR EXPOSED LINES WITHIN THE ROOM SERVED BY IU.
5. BELOW STAIRWAY ACCESS PANEL, MAINTAIN ACCESS TO IT.
6. LOCATE CENTERLINE OF CABINET UNIT HEATER CABINET DIRECTLY UNDER THE CENTERLINE OF THE STAIR TREADS DIRECTLY OVERHEAD.
7.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 STATE OF MAINE LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MECHANICAL PIPING PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 205 OF 282
MP102

DRAWFORM REVISION: 14 SEP 2023

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GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. TERMINAL UNIT HEATER PIPING IS 3/4", UNLESS OTHERWISE NOTED.

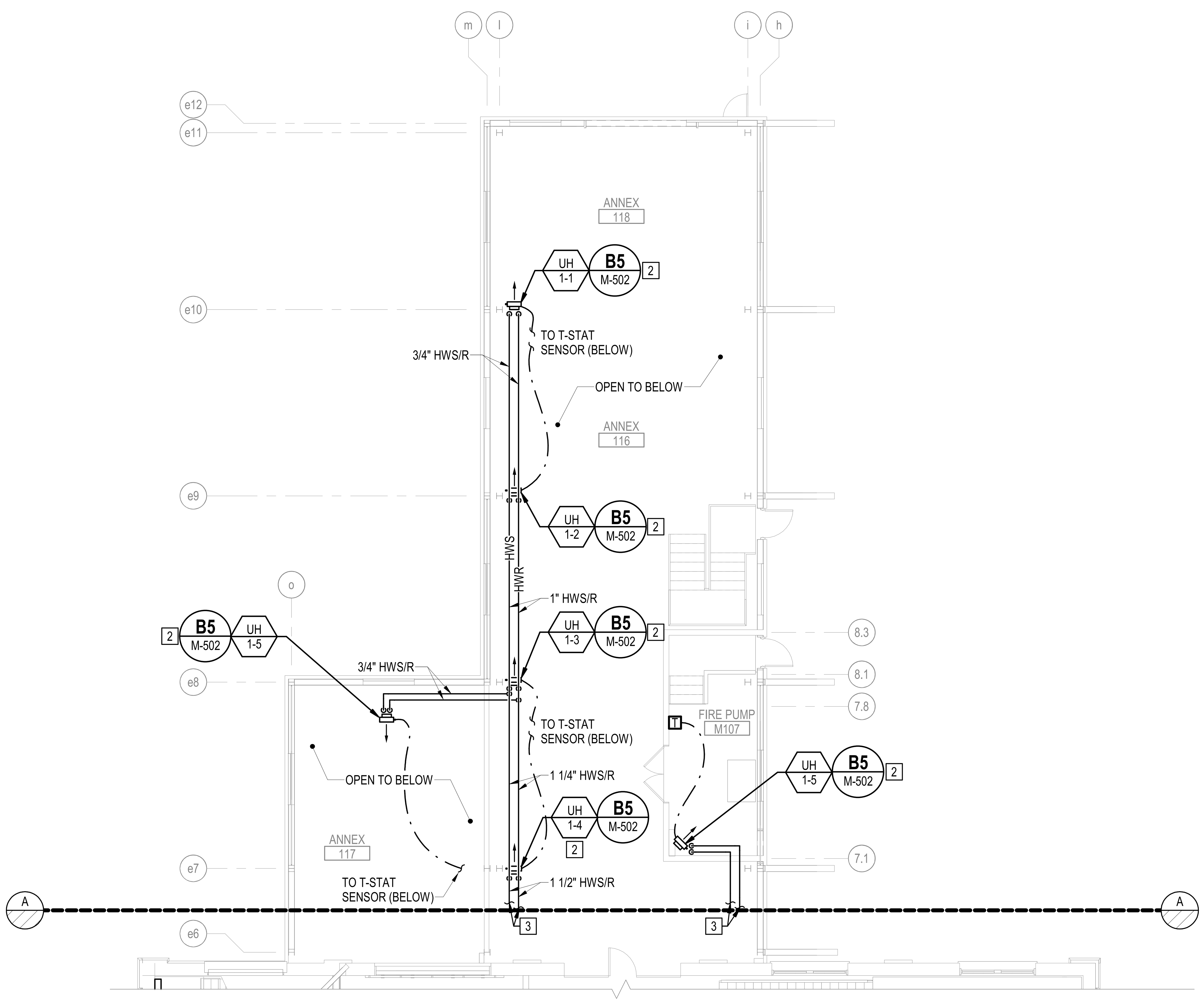
SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE
RICHARD CRESWELL
No. 13178
10-01-25
LICENSED
PROFESSIONAL ENGINEER
SEAL

KEYNOTES (THIS SHEET ONLY)

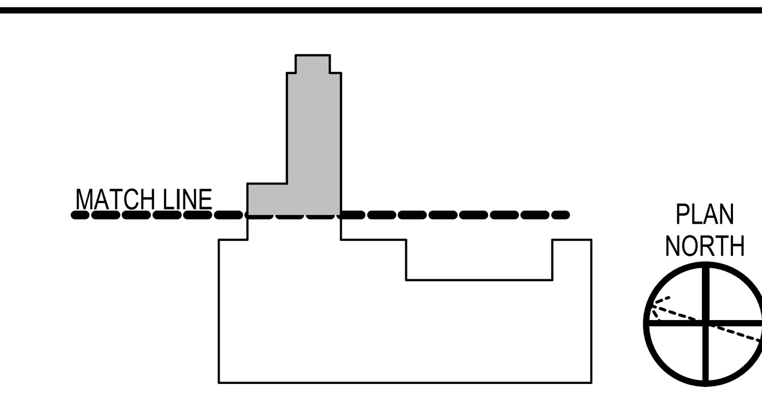
1. SUSPEND UNIT HEATER BELOW EXISTING ELEVATED EQUIPMENT STAND. PROVIDE ADDITIONAL CHANNEL STRUT AS NEEDED TO COMPLETE A PROPER SUPPORT OF THE UNIT HEATER.
2. PROVIDE UNIT HEATER SUPPORTS FOR BOTTOM OF UNIT HEATER ENCLOSURE TO BE POSITIONED 13'-0" AFF.
3. SEE SHEET MP104 FOR CONTINUATION.



A1 FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - EAST

SCALE: 1/8" = 1'-0"

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0"   0   5'   10'   15'   30'

CHECK GRAPHIC SCALE BEFORE USING

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER	RNC
DRAWN BY	SRW
CHECKED BY	MSA
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - EAST

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 206 OF 282

MP103

DRAWFORM REVISION: 14 SEP 2023

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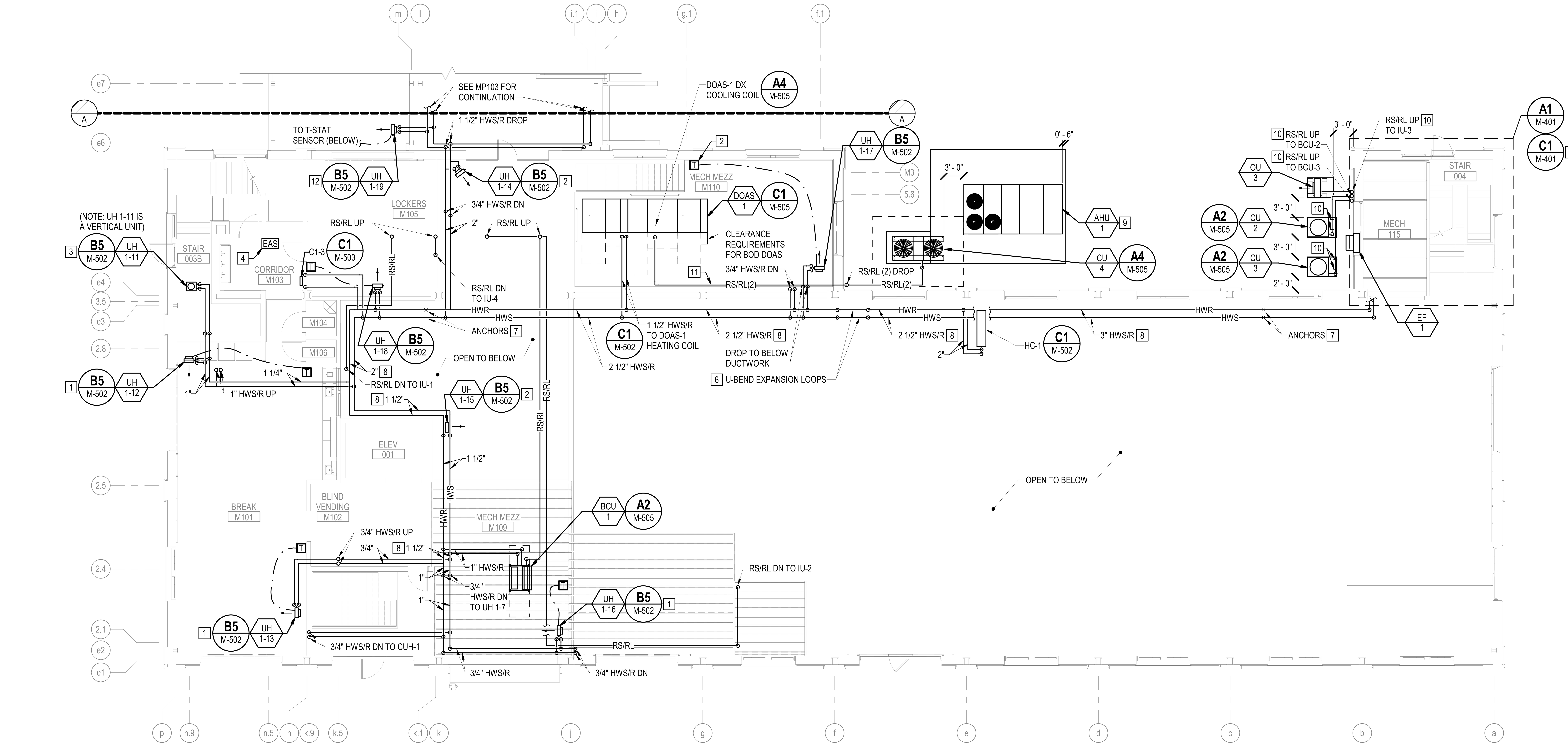


B1 FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - WEST

SCALE: 1/8" = 1'-0"

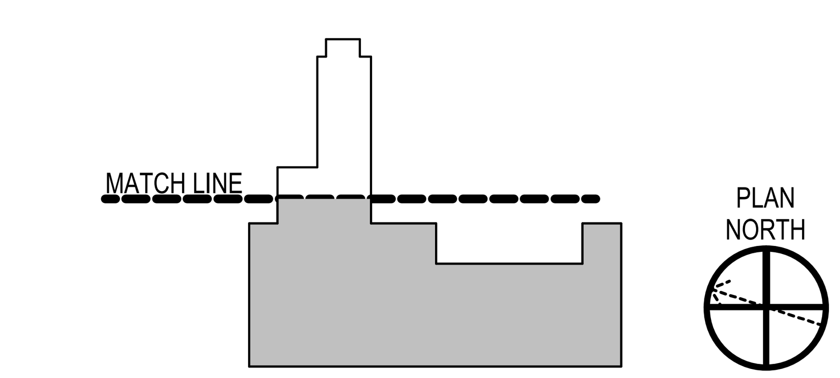
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. HWS/R TERMINAL HEATER BRANCH PIPING IS 3/4" UNLESS OTHERSWISE INDICATED.

KEYNOTES (THIS SHEET ONLY)

1. SUSPEND UNIT FROM WALL MOUNTED SUPPORT BRACKETS WITH BOTTOM OF ENCLOSURE AT 9'-0" AFF.
2. SUSPEND UNIT FROM WALL MOUNTED SUPPORT BRACKETS WITH BOTTOM OF ENCLOSURE AT 10'-0" AFF.
3. SUSPEND UNIT FROM WALL MOUNTED SUPPORT BRACKETS WITH BOTTOM OF ENCLOSURE AT 12'-0" ABOVE STAIR FLOOR DIRECTLY BELOW.
4. EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH. SEE A1/M-701 FOR MORE INFORMATION.
5. SEE SHEET M-401 FOR STAIR 004 HEATING REQUIREMENTS AND FOR REFRIGERANT LINE WALL PENETRATION TRANSITION INTO MECH 115.
6. PROVIDE HWS/R U-BEND EXPANSION LOOPS WITH 64" SIDES AND 32" END.
7. PROVIDE DELEGATED DESIGN FOR EXPANSION ANCHOR POINTS FOR REVIEW AND APPROVAL.
8. SUSPEND PIPING FROM WALL MOUNTED SUPPORT BRACKETS 23'-0" TO 25'-0" ABOVE FIRST FLOOR.
9. PROVIDE GROUND MOUNTED STANCHEON: FOR AHU-1 REFRIGERANT PIPING FROM UNIT TO EAST MECH MEZZ M110. RISE ON WALL AND PENETRATE INTO OPEN CEILING IN STORAGE 109.
10. PROVIDE PREMANUFACTURED LINESET ENCLOSURES WITH REMOVABLE COVERS FROM OUTDOOR UNIT TO PIPING PENETRATIONS INTO BUILDING SUPPORT HORIZONTAL PIPE RUNS AT BACK OF UNITS, STANCHEONS SECURED TO EQUIPMENT PAD EXTENSIONS.
11. RUN REFRIGERANT PIPING ALONG WALL APPROXIMATELY 7'-0" ABOVE THE MEZZANINE FLOOR.
12. PROVIDE UNIT HEATER SUPPORTS FOR BOTTOM OF UNIT HEATER ENCLOSURE TO BE POSITIONED 13'-0" AFF.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0"   0   5'   10'   15'   30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

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RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
FIRST FLOOR MEZZANINE MECHANICAL PIPING PLAN - WEST

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 207 OF 282
MP104

DRAWFORM REVISION: 14 SEP 2023

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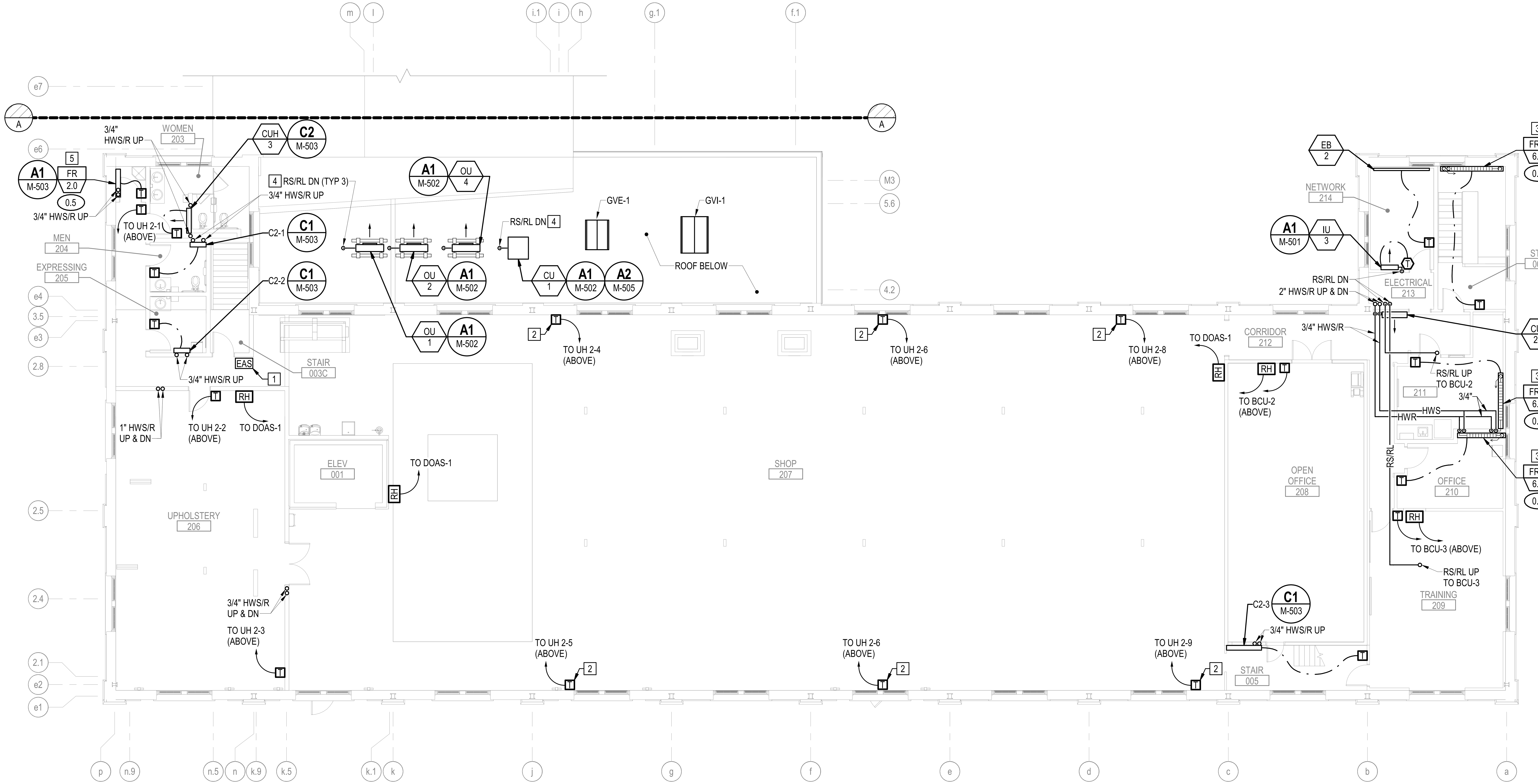


B1 SECOND FLOOR MECHANICAL PIPING PLAN - WEST

SCALE: 1/8" = 1'-0"

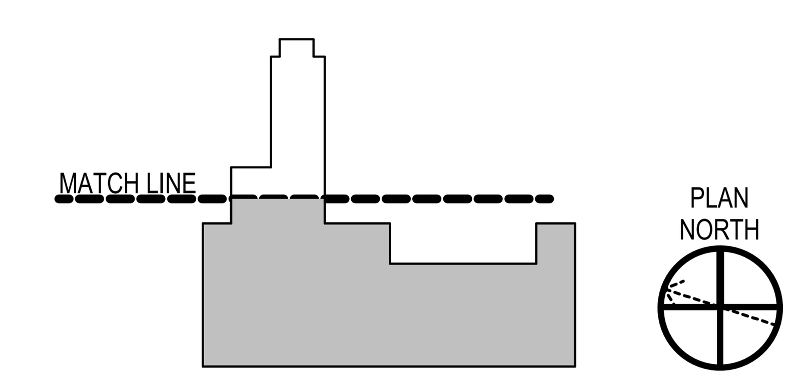
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. HWS/R TERMINAL HEATER BRANCH PIPING IS 3/4" UNLESS OTHERSWISE INDICATED.

KEYNOTES (THIS SHEET ONLY)

1. EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH. SEE A1/M-701 FOR MORE INFORMATION.
2. PROVIDE INSULATED BACKER PLATE FOR THERMOSTAT.
3. RUN HWR PIPING WITHIN FINTUBE RADIATOR ENCLOSURE, BACK TO RISER IN WALL.
4. RS/RL LINESET PIPING AND POWER CABLE TO PENETRATE ROOF THROUGH WATER TIGHT PITCH POCKET, COORDINATE PITCH POCKET LOCATION WITH CONDENSING UNIT PIPE CONNECTING TO MINIMIZED PIPE LENGTH. SEE CONDENSING UNIT SCHEDULE FOR LINSET PIPE SIZES.
5. LOCATE BASE OF FINTUBE RADIATOR ENCLOSURE 72" AFF.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0"   0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
SECOND FLOOR MECHANICAL PIPING PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 208 OF 282
MP105

DRAWFORM REVISION: 14 SEP 2023

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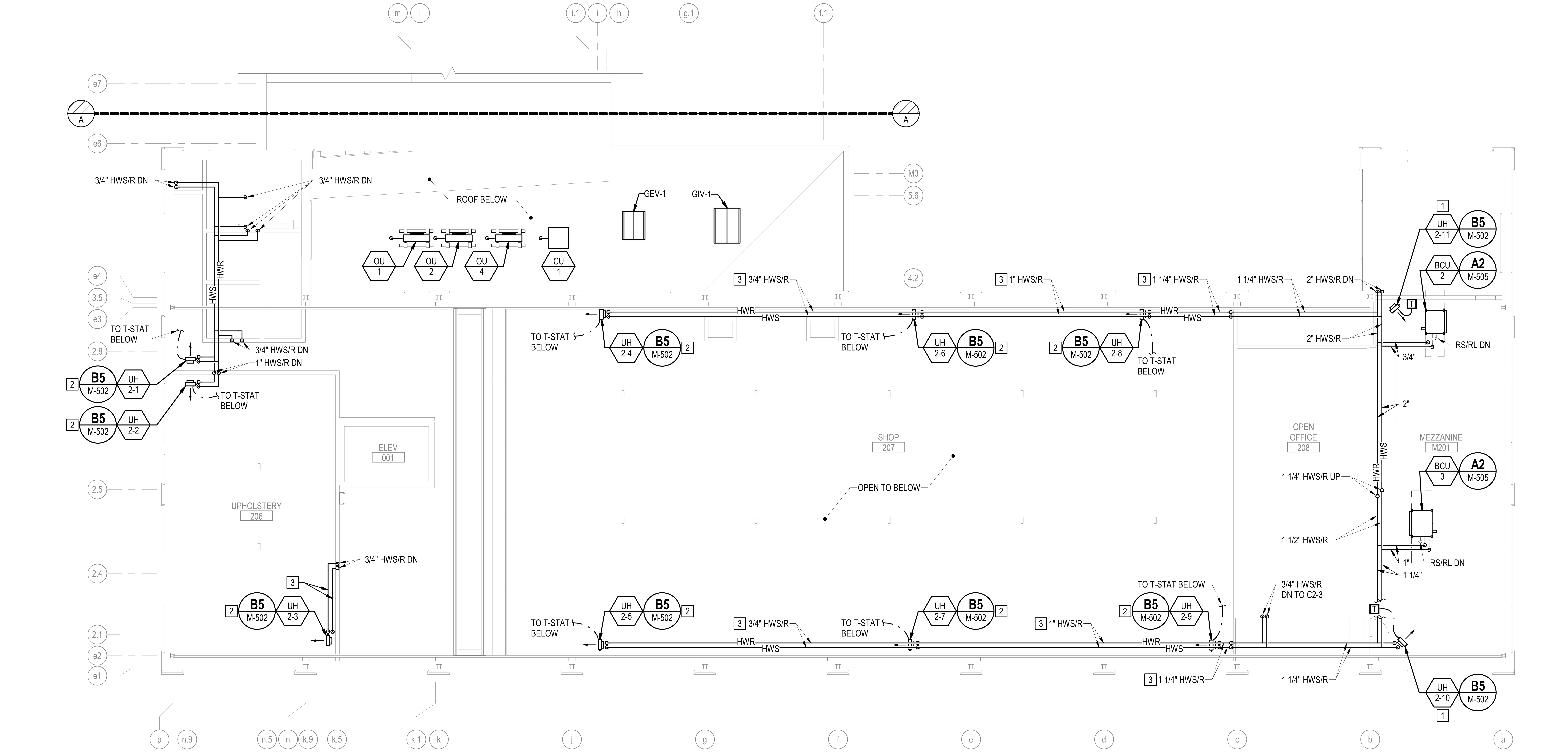


B1 SECOND FLOOR MECHANICAL MEZZANINE PLAN - WEST

SCALE: 1/8" = 1'-0"

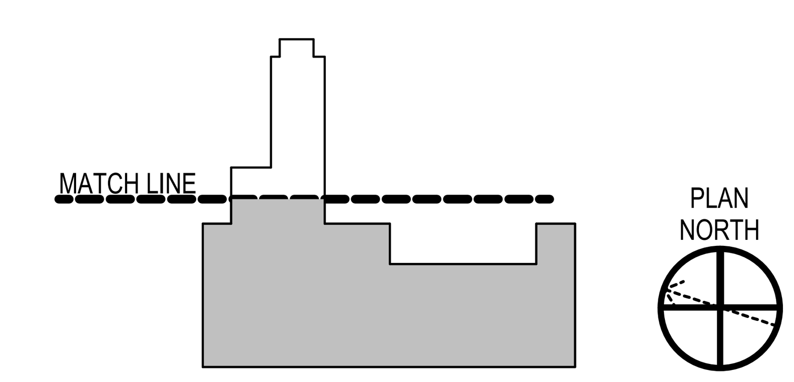
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. HWS/R TERMINAL HEATER BRANCH PIPING IS 3/4" UNLESS OTHERSWISE INDICATED.

KEYNOTES (THIS SHEET ONLY)

1. PLACE TOP OF UNIT HEATER APPROXIMATELY 12" BELOW CEILING
2. LOCATE BASE OF UNIT HEATER APPROXIMATELY 8'-0" AFF, BELOW HWS/R PIPING.
3. SUSPEND PIPING FROM WALL MOUNTED SUPPORT BRACKETS APPROXIMATELY 10'-0" AFF.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

SECOND FLOOR MEZZANINE MECHANICAL PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 209 OF 282
MP106

DRAWFORM REVISION: 14 SEP 2023

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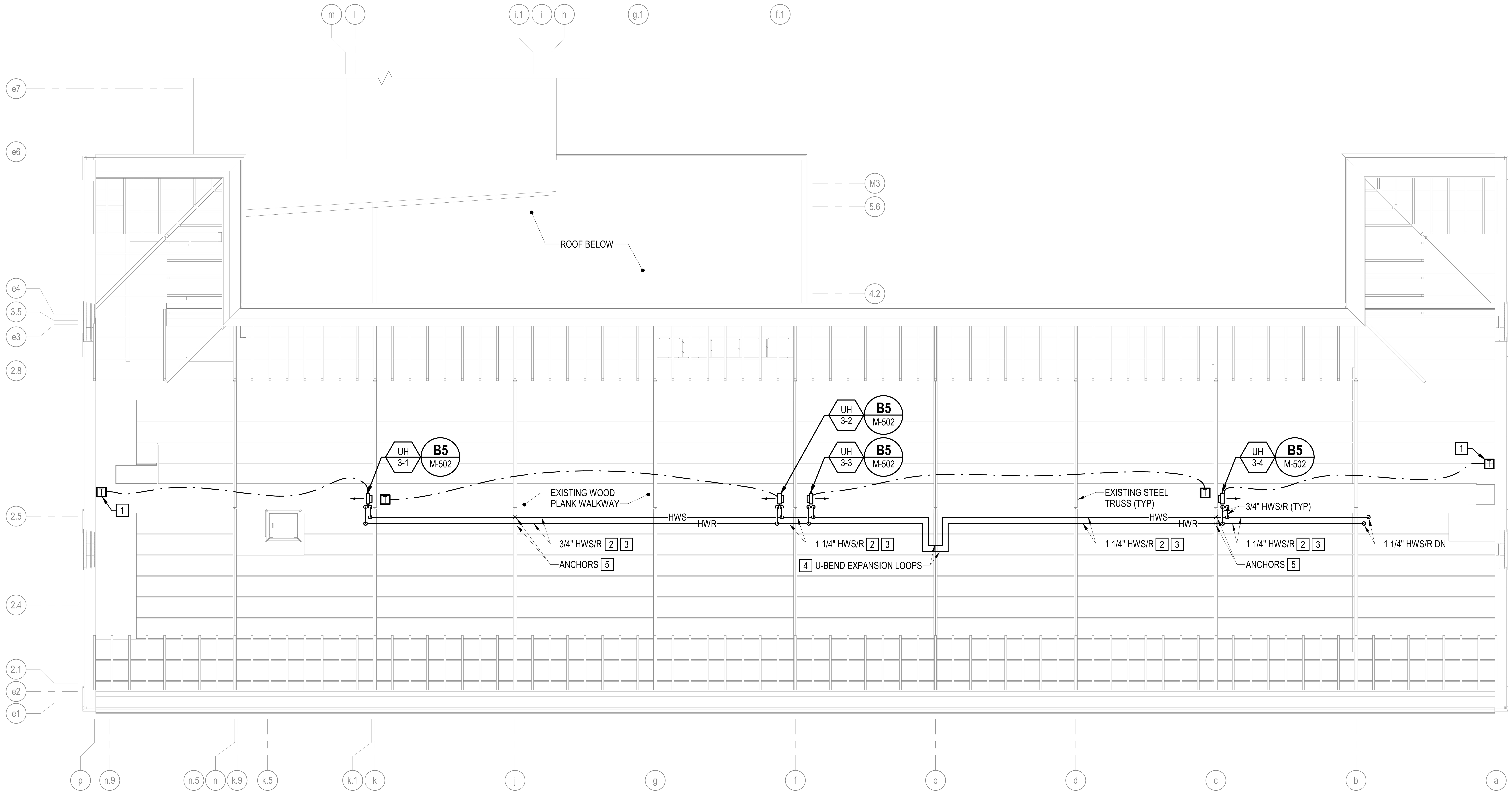


B1 ATTIC MECHANICAL PIPING PLAN - WEST

SCALE: 1/8" = 1'-0"

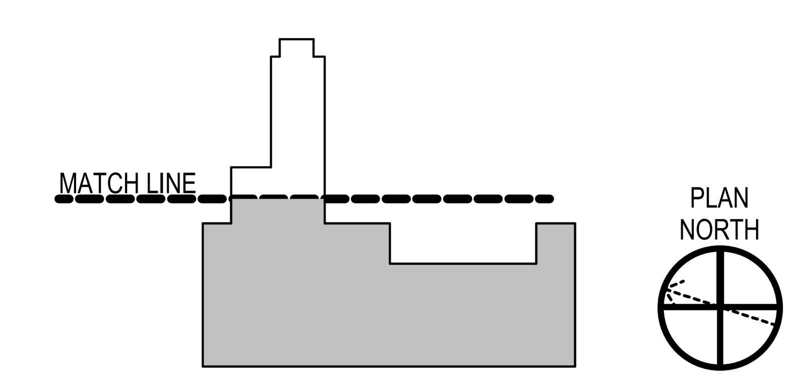
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. HWS/R TERMINAL HEATER BRANCH PIPING IS 3/4" UNLESS OTHERSWISE INDICATED.

KEYNOTES (THIS SHEET ONLY)

1. PROVIDE INSULATED BACKER FOR T-STAT SENSOR.
2. RUN HWS/R PIPING NEAR ATTIC FLOOR ON STANCHEONS SECURED TO CONTINUOUS STRUCTURAL CHANNEL RAILS RUNNING FROM HWS/R RISERS TO ATTIC UNIT HEATER 3-1.
3. PROVIDE PVC JACKET ON HWS/R PIPING UP TO 6'-0" ABOVE WALKWAY.
4. PROVIDE NESTED U-BEND EXPANSION LOOPS WITH HWS HAVING 48" SIDES AND A 24" END. HWR U-BEND TO HAVE 48" SIDES.
5. PROVIDE DELEGATED DESIGN FOR EXPANSION ANCHOR POINTS FOR REVIEW AND APPROVAL.

KEY PLAN



GRAPHIC SCALE

1/8" = 1' - 0" 0 5' 10' 15' 30'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
ATTIC MECHANICAL PIPING PLAN

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 210 OF 282
MP107
DRAWFORM REVISION: 14 SEP 2023

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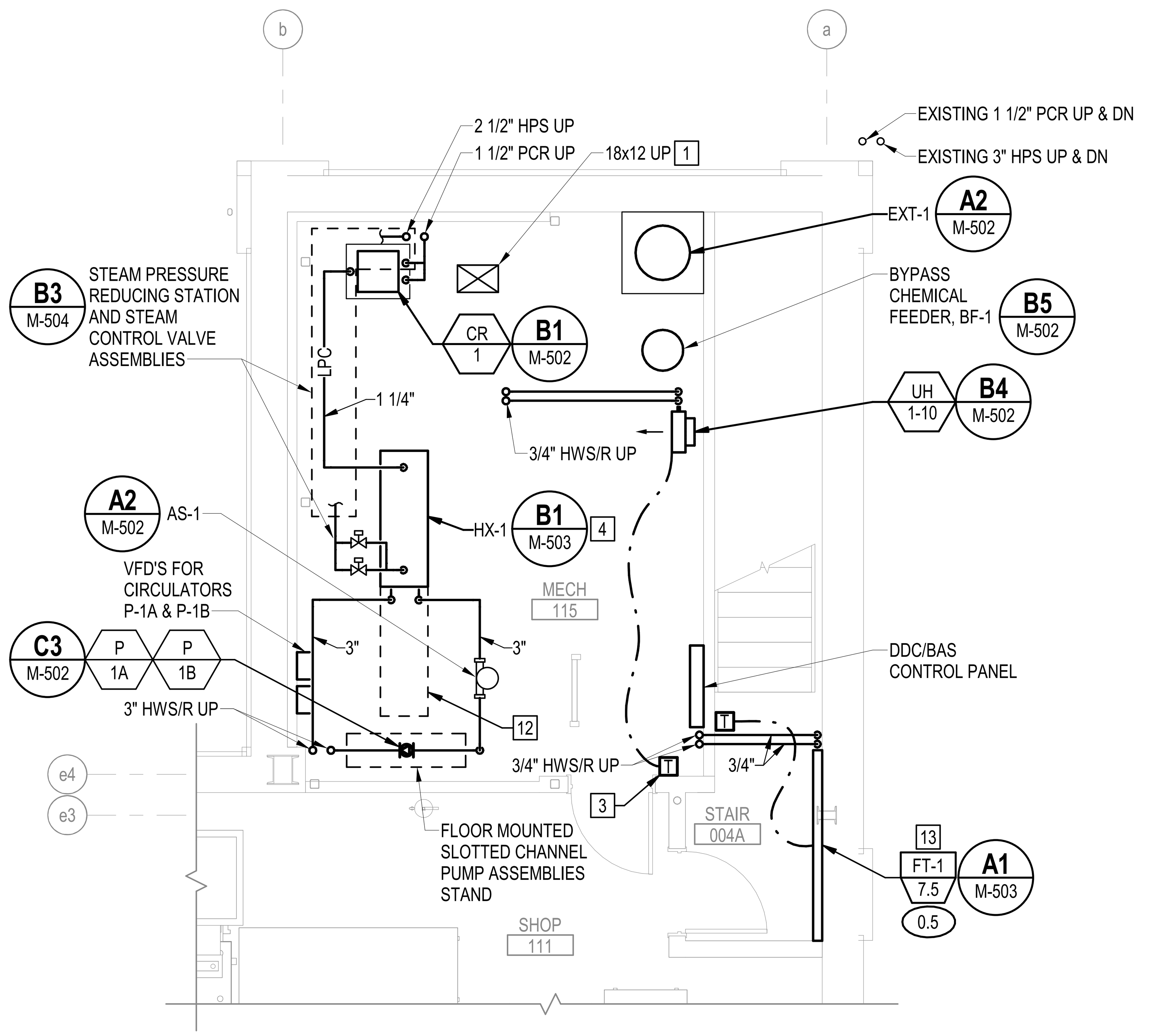
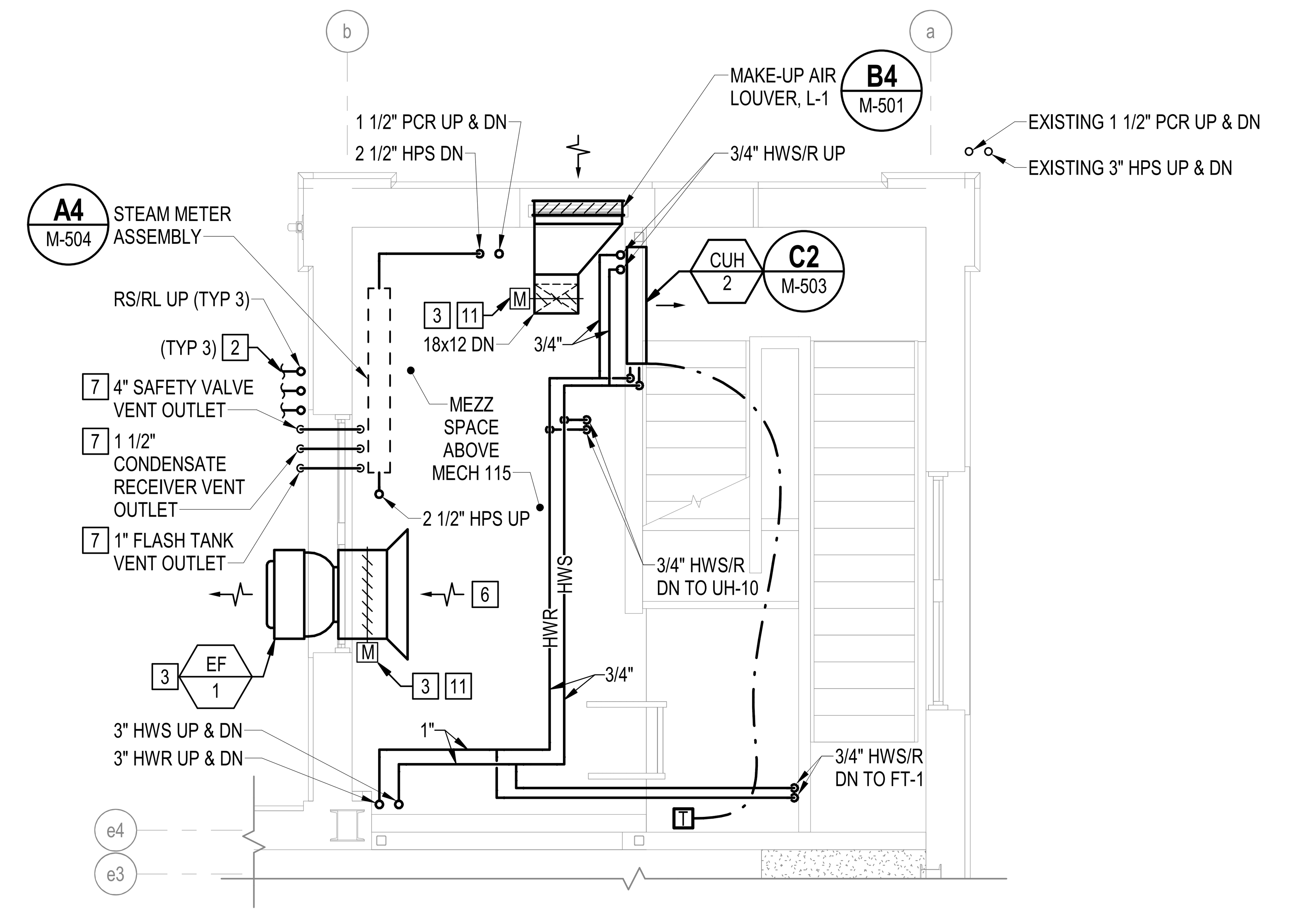


ENLARGED MECH 115 AND STAIR 004 FIRST FLOOR MECHANICAL PART PLAN

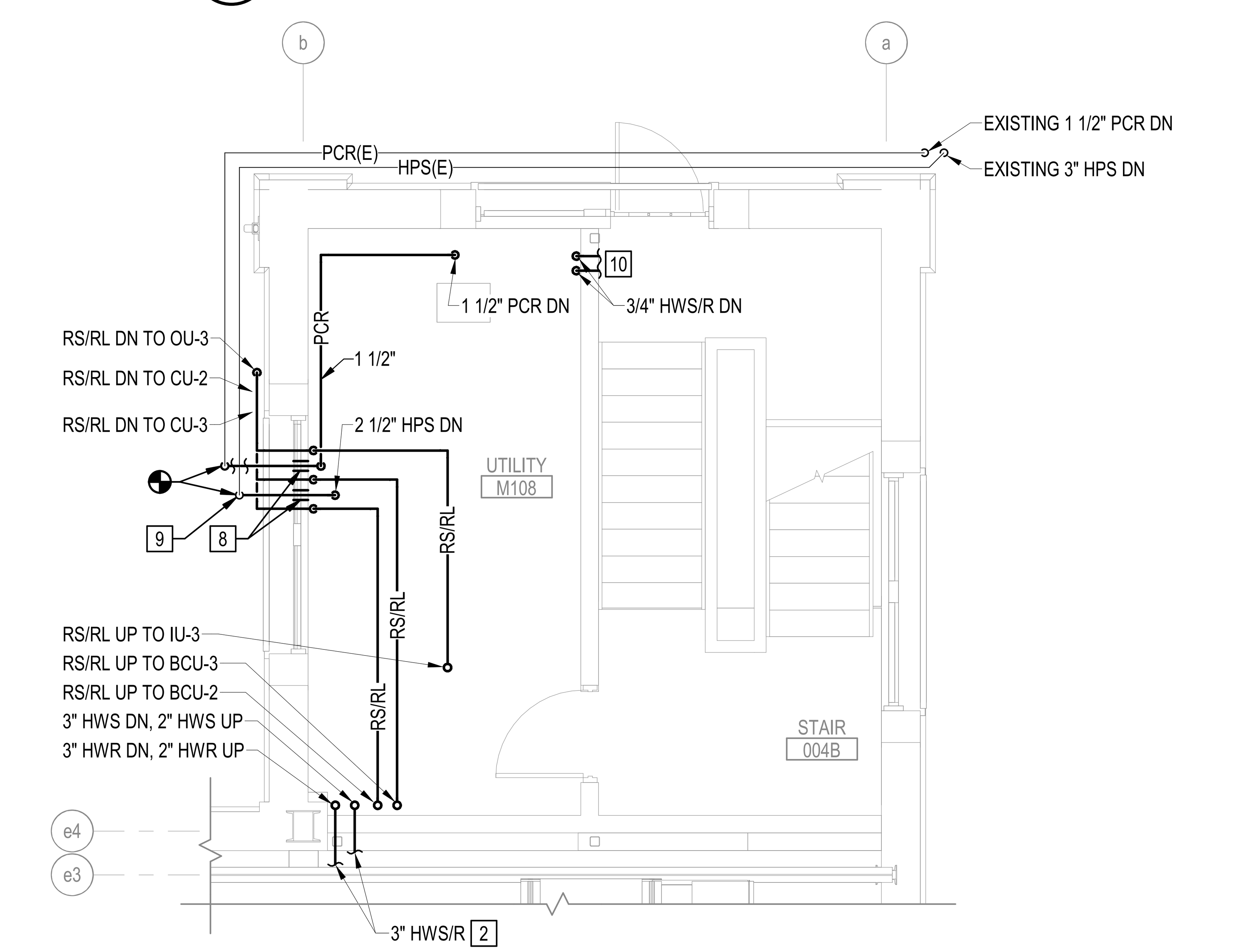
C1 SCALE: 1/4" = 1'-0" (MH101, MP102)



NOTE: CEILING LEVEL IS AT THE TOP OF THE DOUBLE HUNG WINDOWS BOTTOM SASH.

ENLARGED MECH 115 MEZZANINE AND STAIR 004 MECHANICAL PART PLAN

C3 SCALE: 1/4" = 1'-0" (MH102, MP104)



ENLARGED UTILITY M108 AND STAIR 004 INTERMEDIATE LANDING MECHANICAL PART PLAN

A1 SCALE: 1/4" = 1'-0" (MH102, MP104)

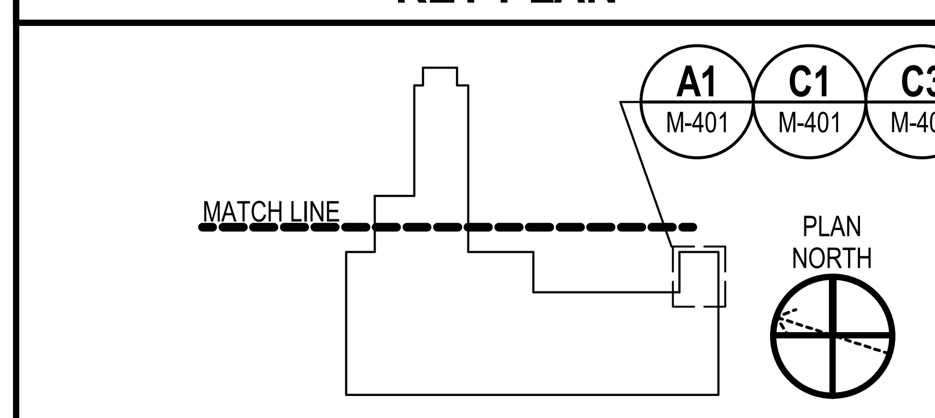
GENERAL NOTES

1. SEE M-001 FOR MECHANICAL LEGENDS, ABBREVIATIONS AND GENERAL NOTES.
2. HWS/R TERMINAL HEATER BRANCH PIPING IS 3/4" UNLESS OTHERSWISE INDICATED.

KEYNOTES (THIS SHEET ONLY)

1. PROVIDE OPEN END DUCT DROP TO WITHIN 18" AFF. PROVIDE BIRD SCREEN MESH AT DUCT OPENING. COORDINATE DUCT RISER OPENING LOCATION WITH PLATFORM OPENING.
2. SEE SHEET MP104 FOR CONTINUATION.
3. CONNECT THERMOSTATIC SENSOR WITH EXHAUST FAN, EF-1, AND MAKE-UP AIR LOUVER MOTOR OPERATED DAMPERS, AS INDICATED IN CONTROL DIAGRAM A4/M-705.
4. PLACE HEAT EXCHANGER IN A LOCATION THAT PROVIDES ADEQUATE CLEARANCE FOR THE REMOVAL OF TUBE BUNDLE.
5. EXTEND INTO ROOM AS NEEDED TO AVOID DUCT RISER CONFLICT WITH PLATFORM BEAMS.
6. PROVIDE TRANSITION OPENING WITH BIRD SCREEN MESH COVERED INLET.
7. DROP OUTLET TO 6" ABOVE GROUND LEVEL.
8. PASS CORROSION RESISTANT METAL JACKET AND INSULATED PIPING THROUGH INSULATED ARCHITECTURAL PANEL. SEAL AIR AND WATER TIGHT.
9. LOCATE 3" TO 2 1/2" HPS REDUCER IN PIPE DROP. OR PROVIDE AN ECCENTRIC REDUCER.
10. PROVIDE HWS/R PIPING TO FINTUBE RADIATOR LOCATED ON STAIR LANDING ABOVE THE ONE SHOWN ON THIS PLAN. SEE SECOND FLOOR MECHANICAL PIPING PLAN ON MP105 FOR IMAGE OF THE FINTUBE RADIATOR ON THE STAIR LANDING.
11. PROVIDE LOW LEAKAGE INSULATED MOTOR OPERATED DAMPER WITH BLADES THERMALLY BROKEN FROM FRAME.
12. LOCATE HEAT EXCHANGER WITH ADEQUATE CLEARANCE TO REMOVE THE TUBE BUNDLE FROM THE SHELL OF THE HEAT EXCHANGER.
13. LOCATE BASE OF FINTUBE RADIATOR ENCLOSURE 72" AFF.

KEY PLAN



GRAPHIC SCALE

1/4" = 1' - 0" 0 1' 5' 10' 15'

CHECK GRAPHIC SCALE BEFORE USING

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
ENLARGED MECHANICAL PART PLANS

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 211 OF 282
M-401

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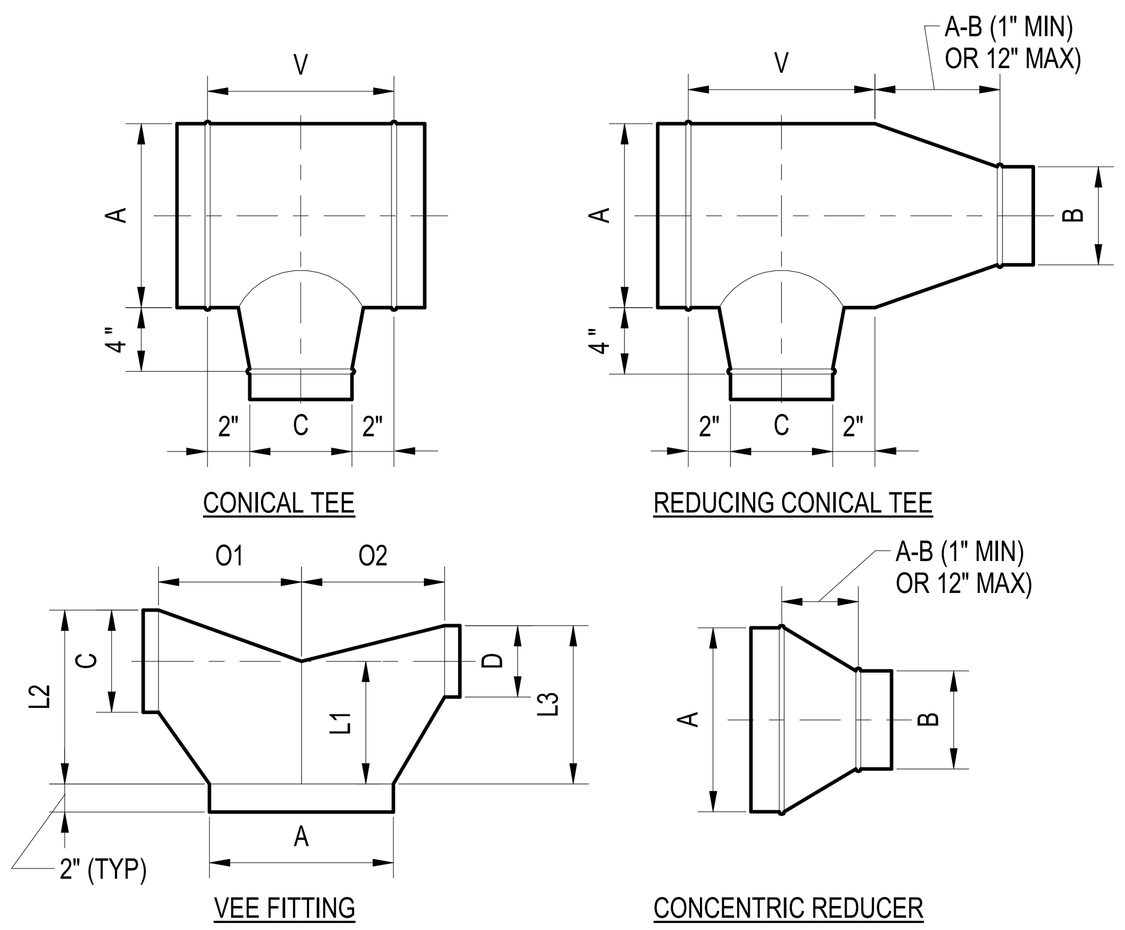
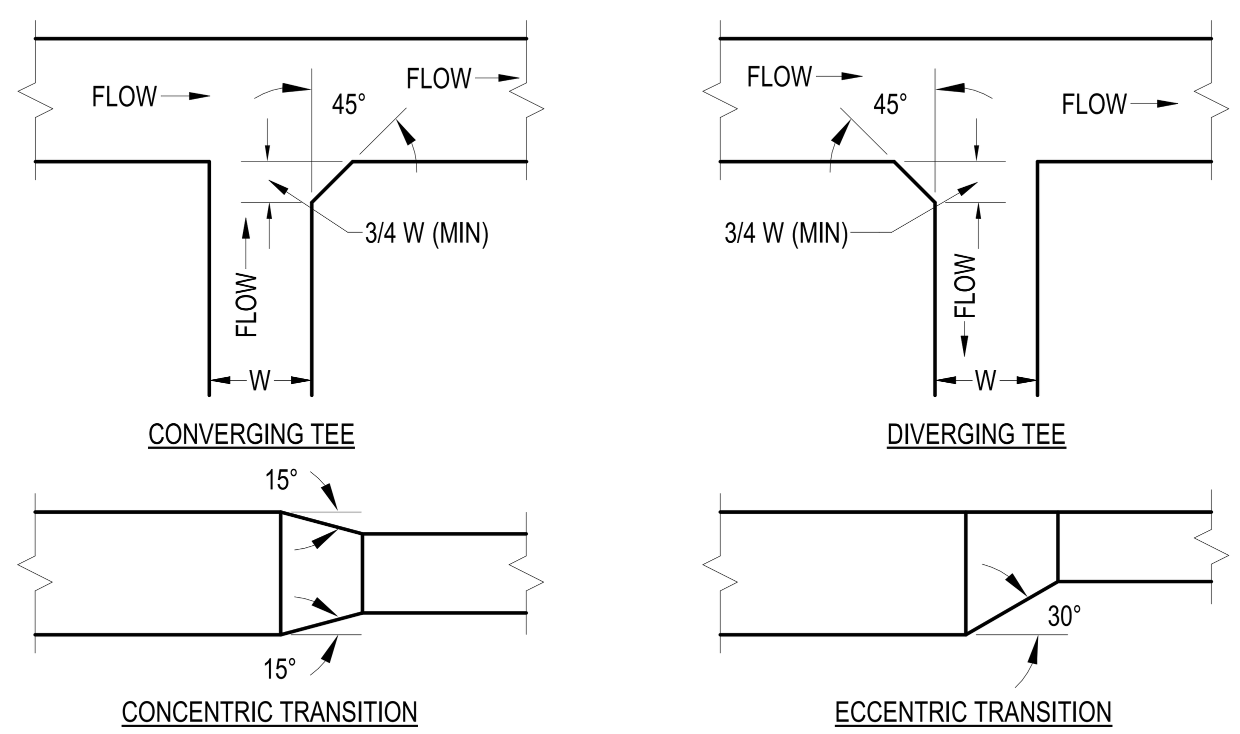
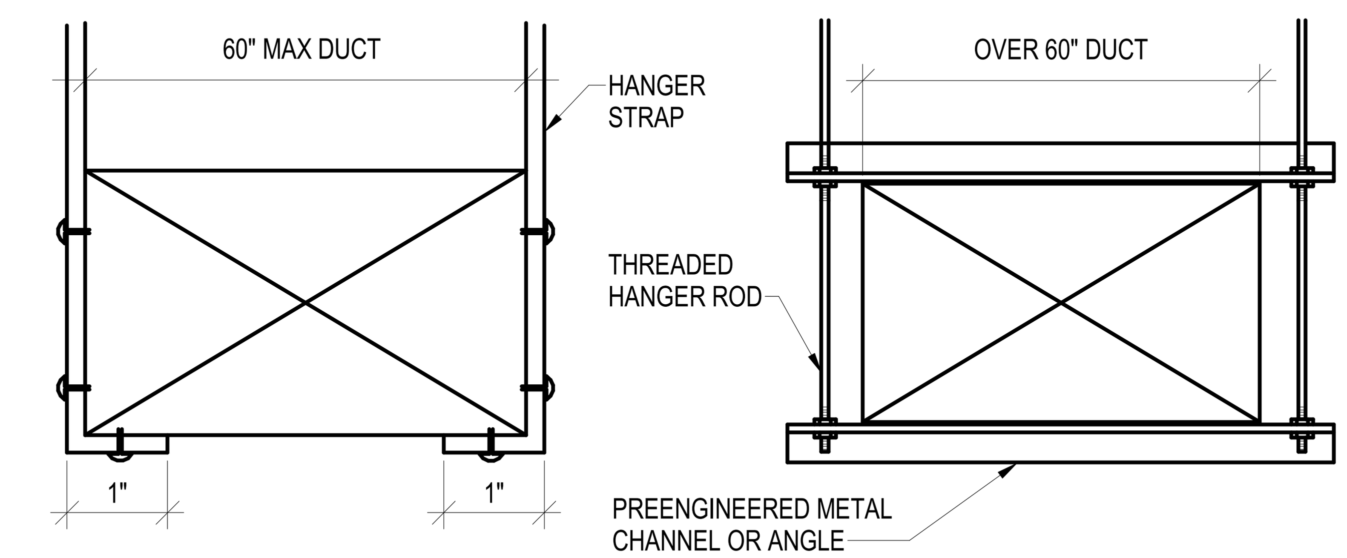


C1 TYPICAL ROUND DUCT FITTINGS DETAIL
SCALE: NOT TO SCALE

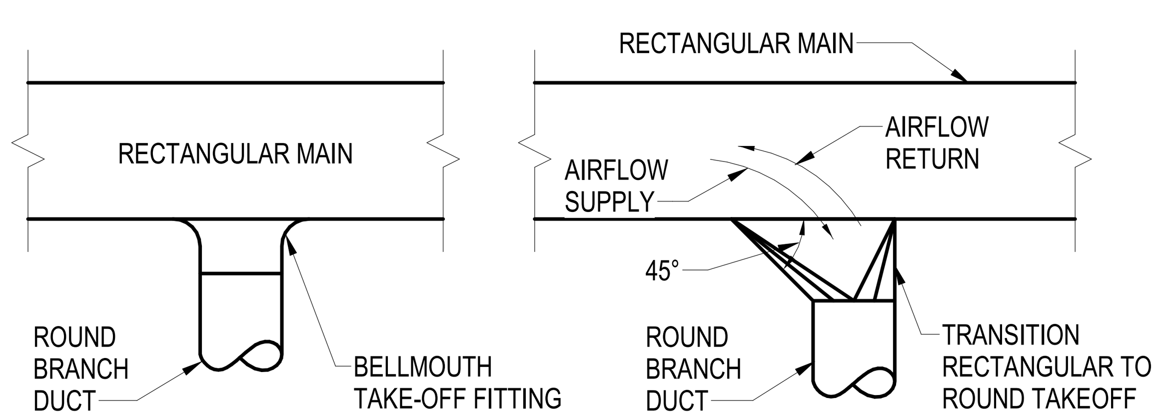


C2 RECTANGULAR DUCT TRANSITIONS DETAIL
SCALE: NOT TO SCALE



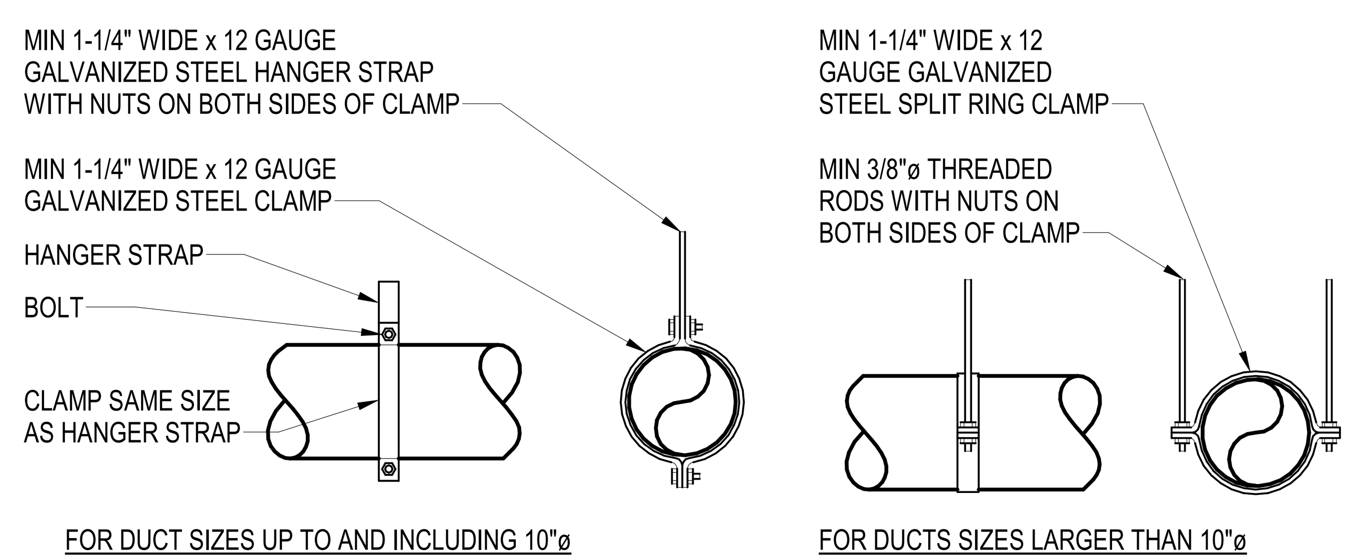
NOTE
SIZE AND RATING OF HANGERS AND SUPPORTS SHALL BE PER LATEST SMACNA STANDARDS.

C4 DUCT SUPPORT ATTACHMENTS DETAIL
SCALE: NOT TO SCALE

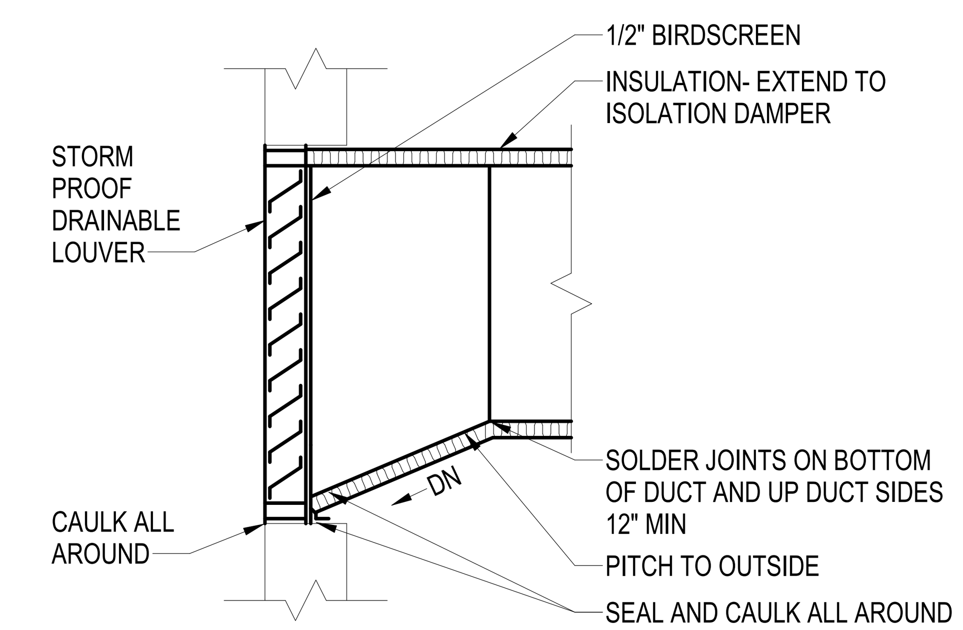


NOTE
SPIN-IN TYPE STRAIGHT TAKE-OFF FITTINGS SHALL NOT BE PERMITTED.

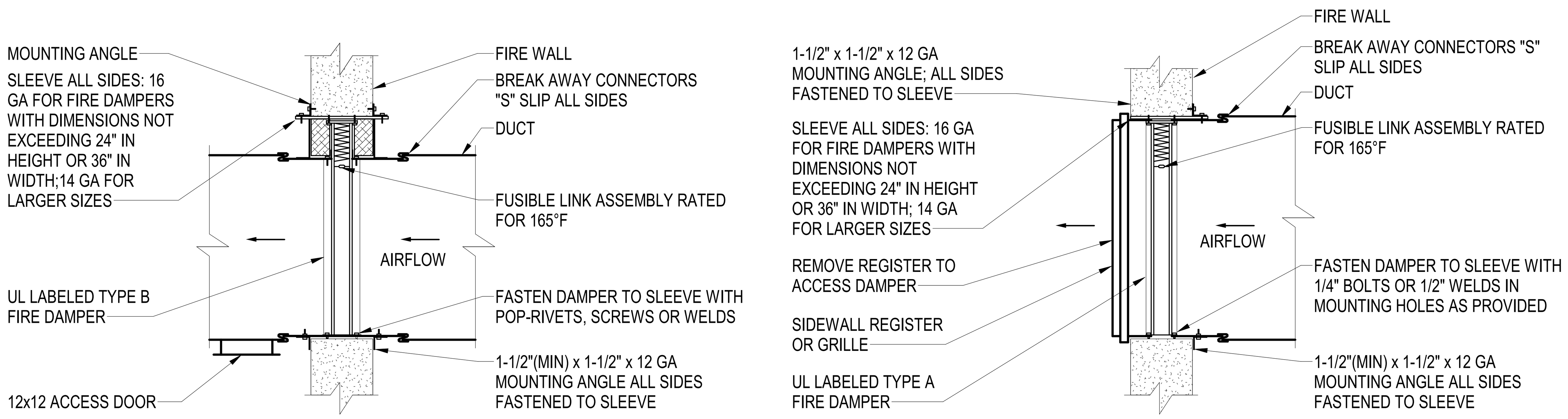
B1 TYPICAL ROUND BRANCH DUCT CONNECTIONS TO RECTANGULAR MAIN
SCALE: NOT TO SCALE



B2 ROUND DUCT HANGER DETAIL
SCALE: NOT TO SCALE

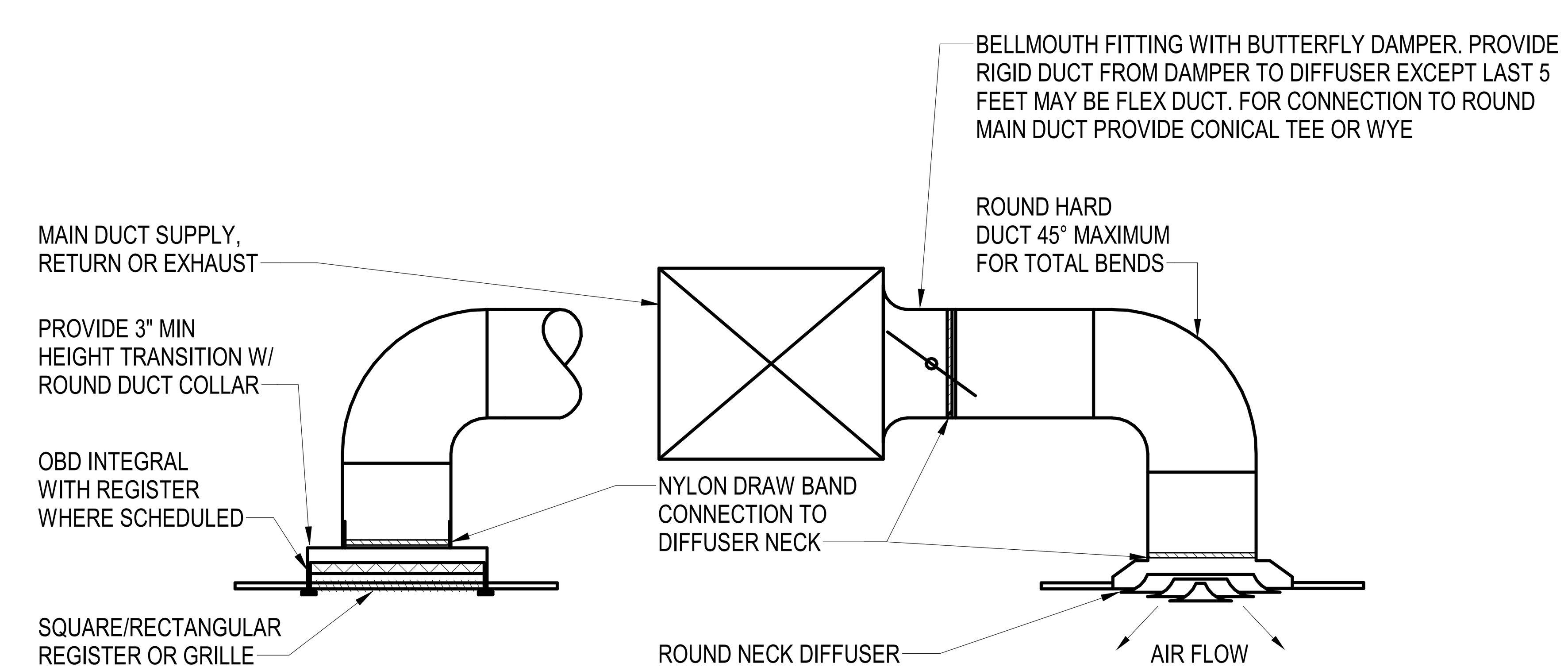


B4 LOUVER CONNECTION DETAIL
SCALE: NOT TO SCALE



NOTES
1. FOR HORIZONTAL FLOOR INSTALLATION PROVIDE FIRE DAMPER WITH SPRING LOADED DAMPER BLADES.
2. FASTEN ANGLES TO SLEEVE WITH POP-RIVETS, SCREWS OR WELDS AT 8-INCH ON CENTER, MINIMUM 2 PER SIDE, 4 SIDES OF OPENING ON BOTH SIDES OF DAMPER.
3. PROVIDE CLEARANCE BETWEEN DAMPER SLEEVE OR DAMPER AND THE FIRE-RATED WALL, PARTITION OR FLOOR IN ACCORDANCE WITH THE FIRE DAMPER MANUFACTURER. IF REQUIRED BY THE DAMPER MANUFACTURER, PROVIDE FIRESTOPPING IN THIS SPACE ON ALL SIDES TO MAINTAIN FIRE RATING.
4. USE FOR GALVANIZED SHEET METAL DUCT ONLY.

A1 FIRE DAMPER DETAIL
SCALE: NOT TO SCALE



A4 DIFFUSER AND REGISTER RUNOUT DETAIL
SCALE: NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR
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NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL DETAILS 1

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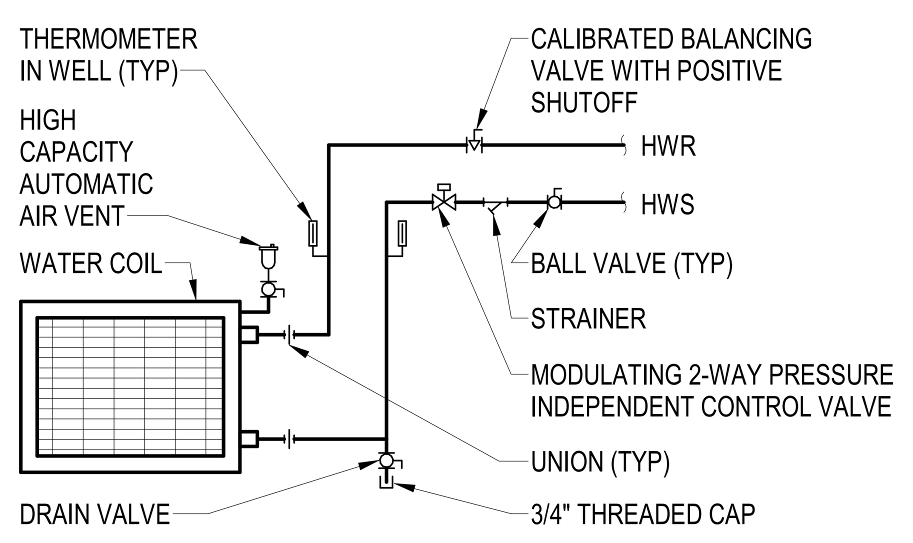
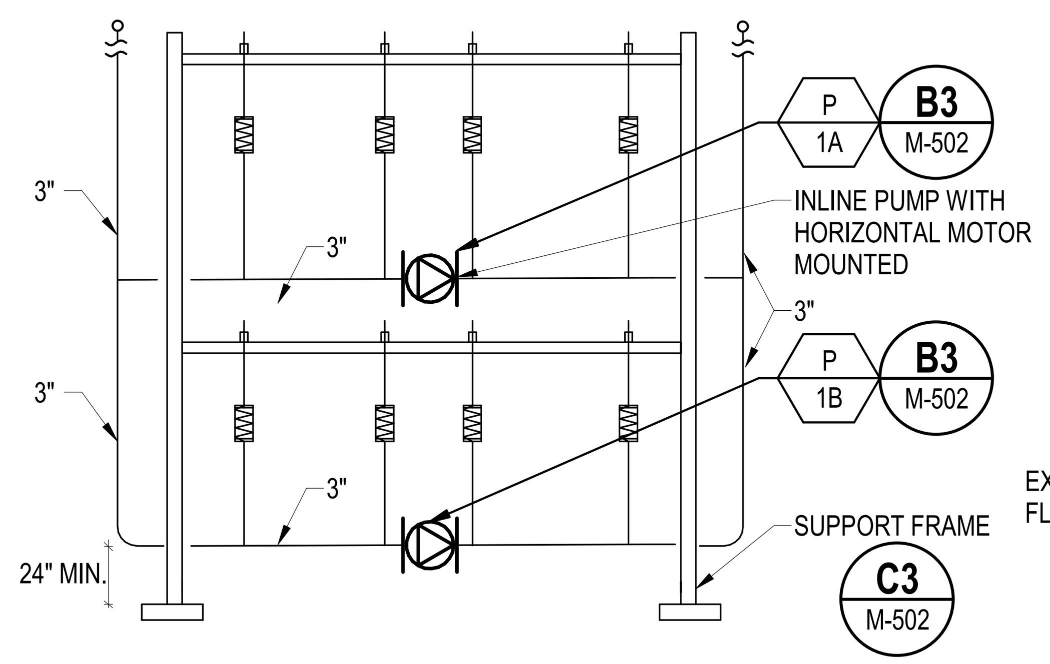
NAVFAC DRAWING NO.
SHEET 212 OF 282
M-501



NOTES
1. VALVES SHALL BE SAME SIZE AS PIPE.
2. SHUT-OFF VALVES MAY BE BALL OR BUTTERFLY. SEE SPECIFICATIONS FOR VALVE TYPE AND SIZE RESTRICTIONS.
3. PROVIDE UNIONS AND/OR FLANGES AS REQUIRED FOR FIT-UP AND REMOVAL OF COIL.

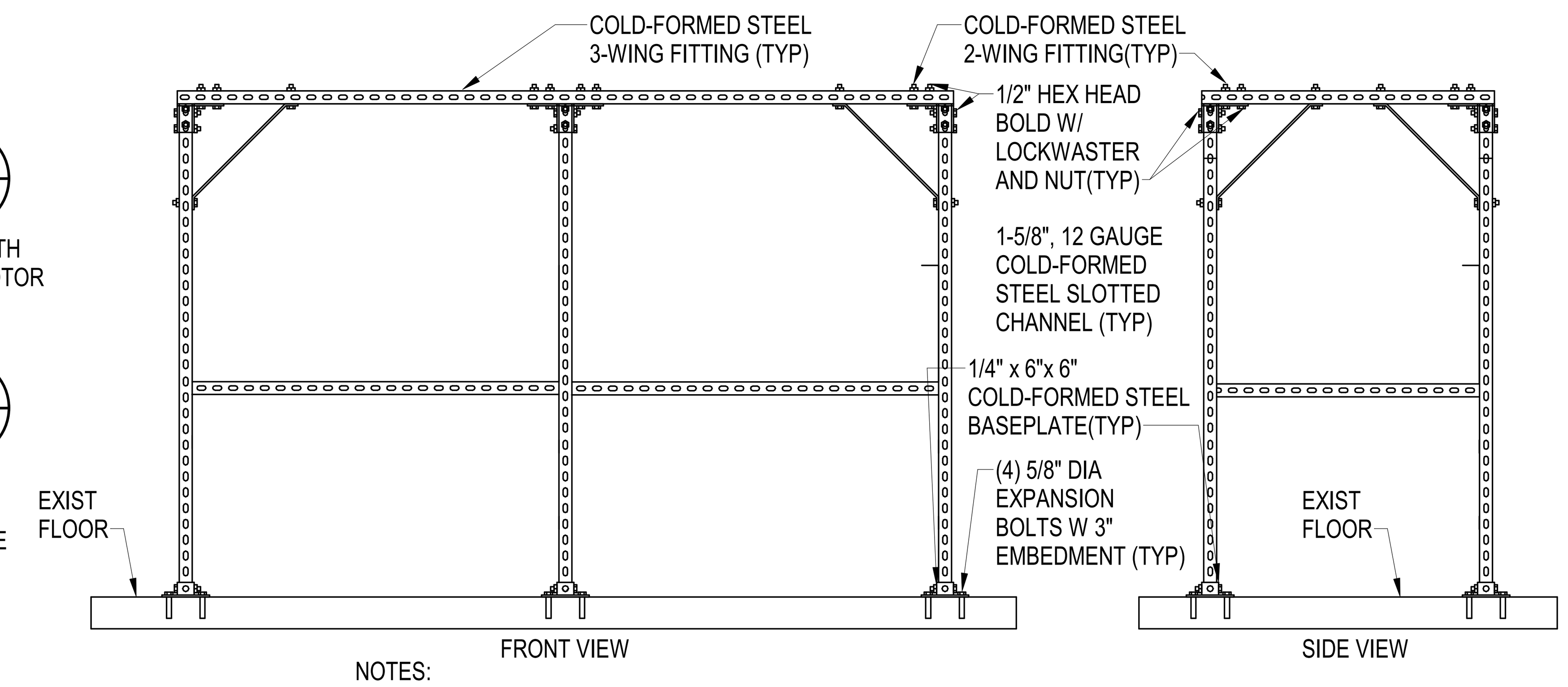
C1 DOAS-1 HOT WATER COIL PIPING SCHEMATIC

SCALE: NOT TO SCALE (MP104)



C2 HEATING WATER CIRCULATOR ASSEMBLY ON FLOOR MOUNTED FRAME DETAIL

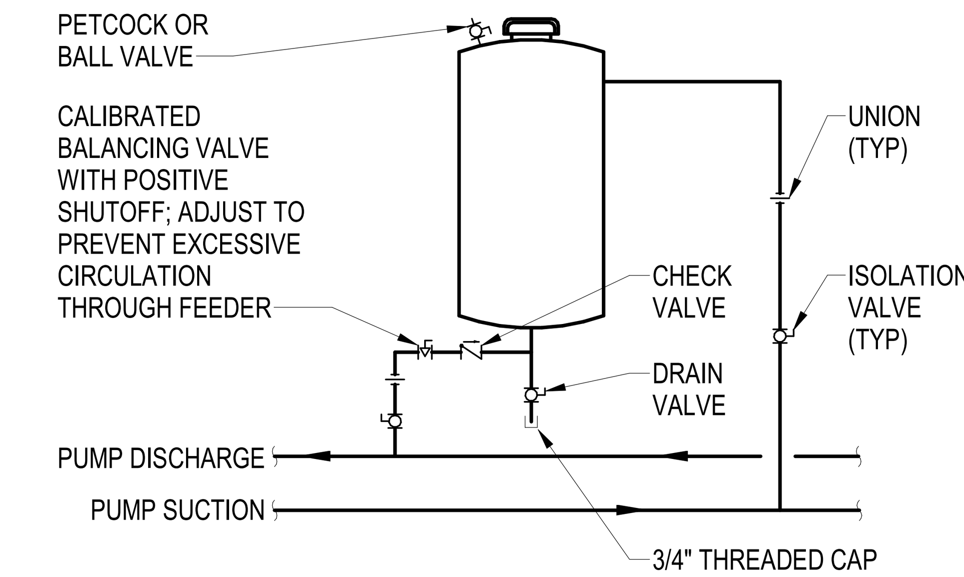
SCALE: NOT TO SCALE (M-401)



NOTES:
1. SECURE POST BASE TO EXSISTING CONCRETE PAD WITH MECHANICAL EXPANSION ANCHORS.
2. FASTENERS AND FITTINGS SHALL BE INSTALLED IN ACCORDANCE WITH THE MFG PRINTED INSTRUCTIONS.

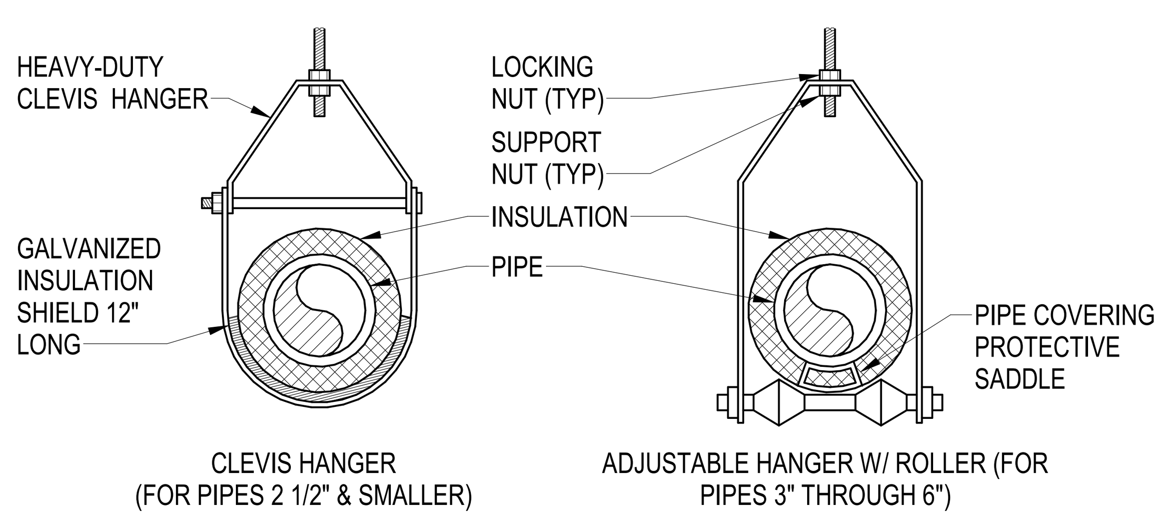
C3 GALVANIZED SLOTTED STEEL CHANNEL FRAME

SCALE: NOT TO SCALE (M-502)



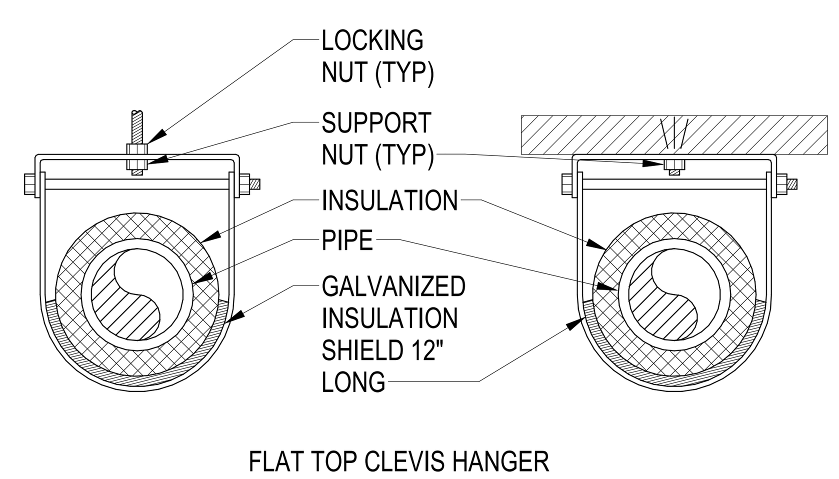
C5 BYPASS CHEMICAL FEEDER PIPING SCHEMATIC

SCALE: NOT TO SCALE (M-401)



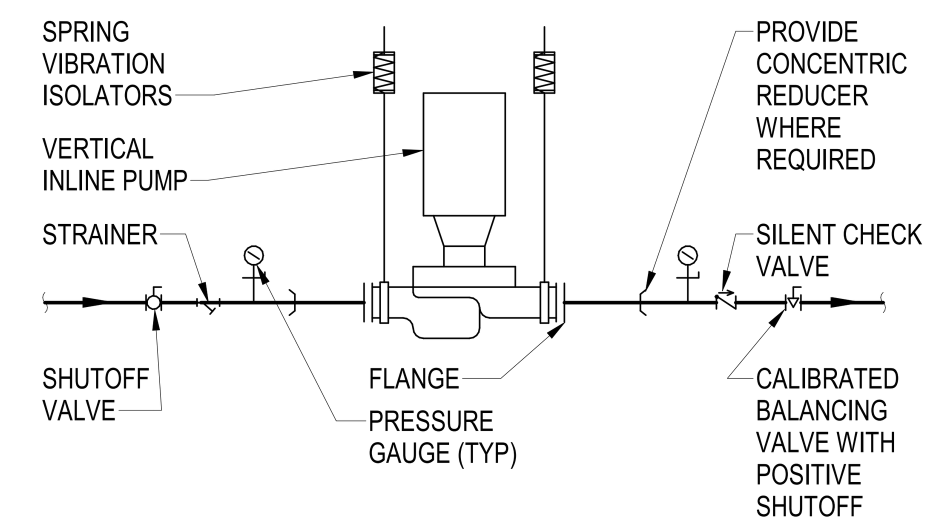
B1 PIPE HANGER ATTACHMENT DETAILS

SCALE: NOT TO SCALE



B2 LOW CEILING PIPE HANGER ATTACHMENT DETAIL

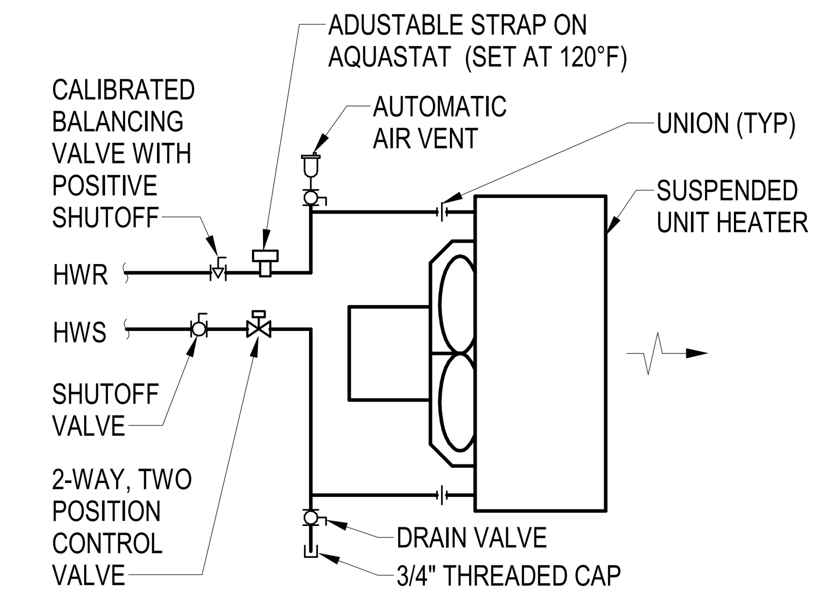
SCALE: NOT TO SCALE



NOTES
1. VALVES SHALL BE SAME SIZE AS PIPE.
2. PROVIDE SPRING VIBRATION ISOLATORS FOR PUMP.

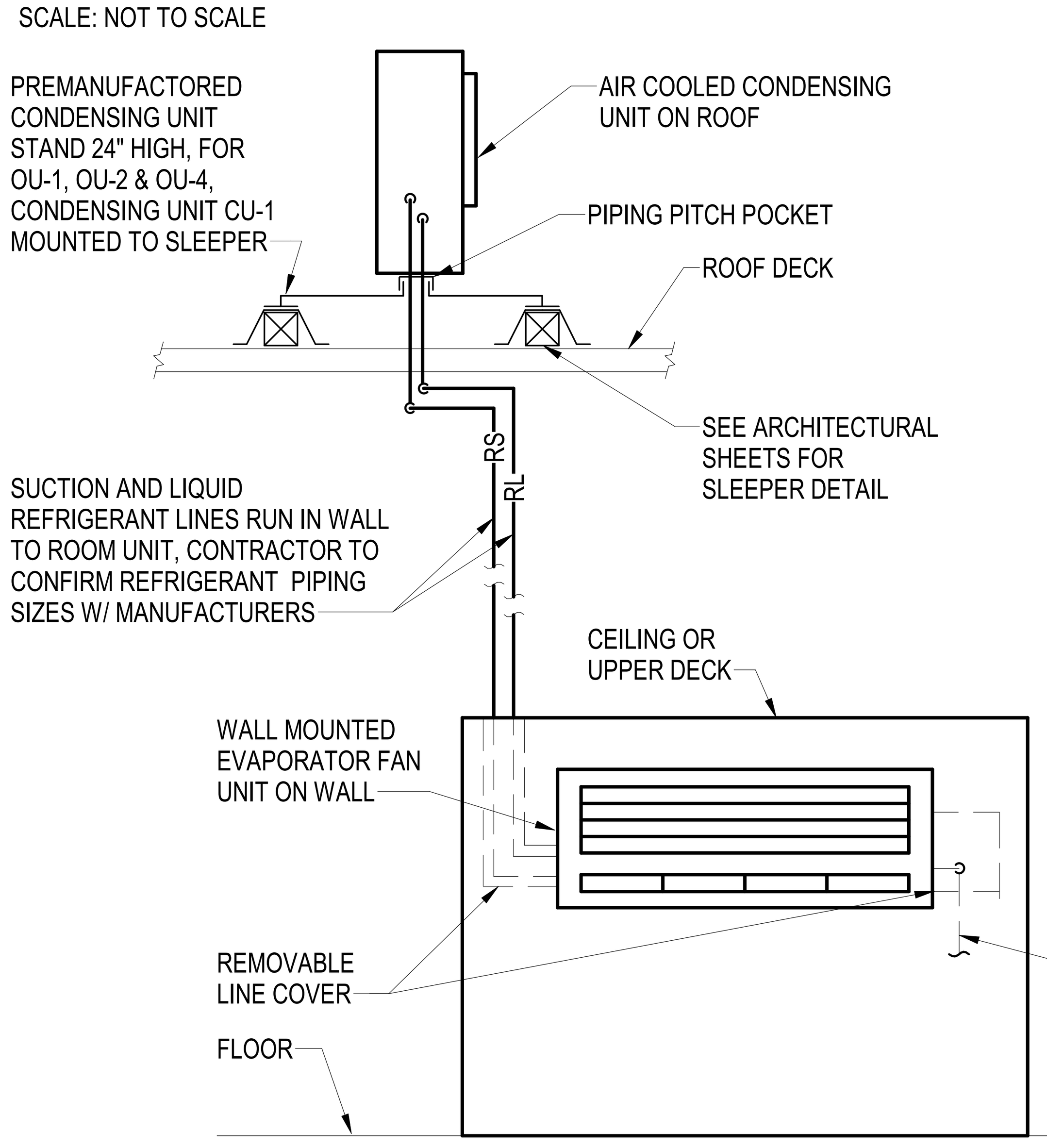
B3 INLINE PUMP PIPING DETAIL

SCALE: NOT TO SCALE (M-502)



B5 TYPICAL UNIT HEATER WATER COIL PIPING SCHEMATIC

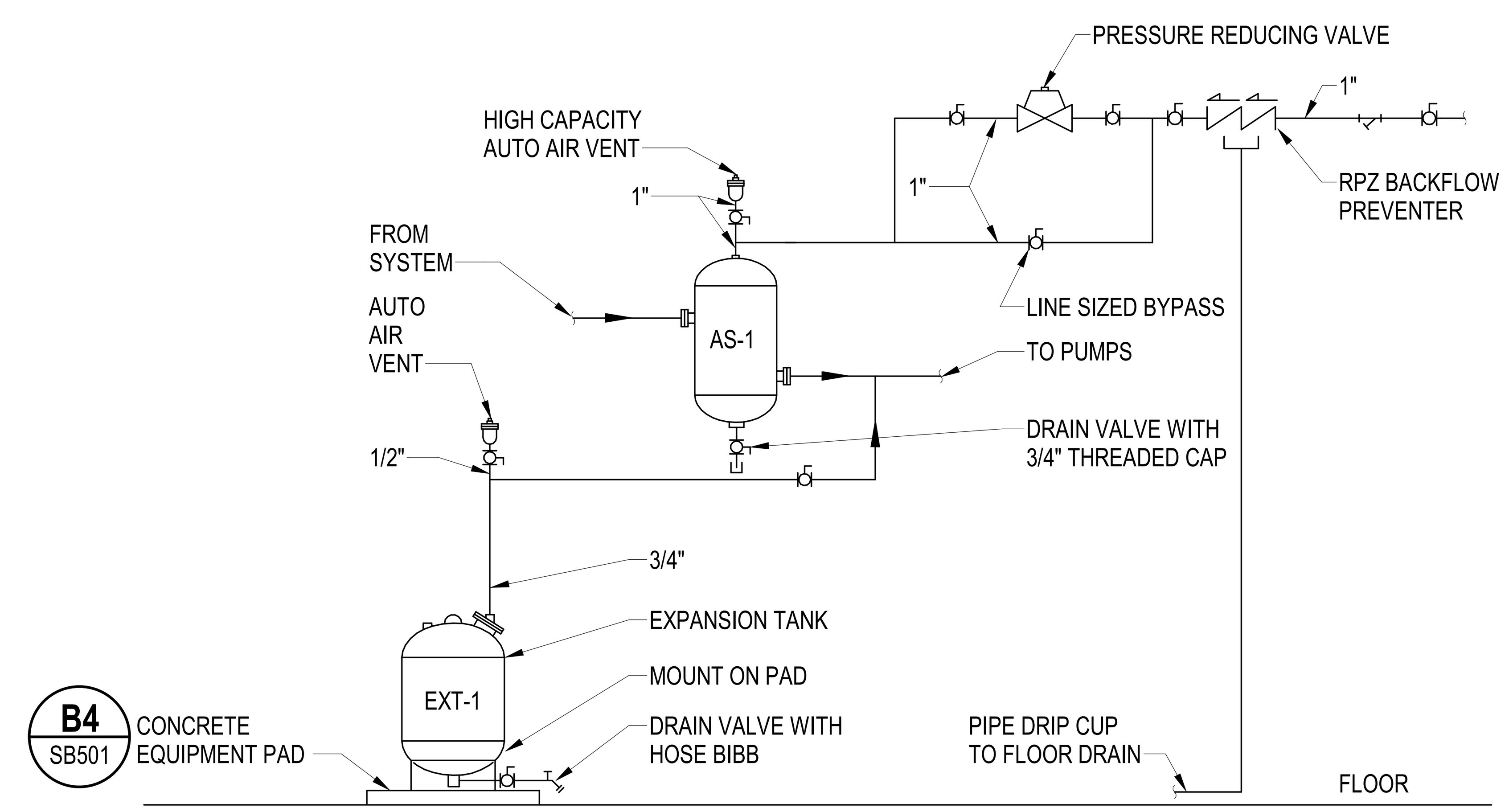
SCALE: NOT TO SCALE (MP102, MP103, MP104, MP106, MP107)



NOTE: CU-1 SERVES BLOWER COIL UNIT BCU-1, MOUNTED ON A FLOOR STAND.

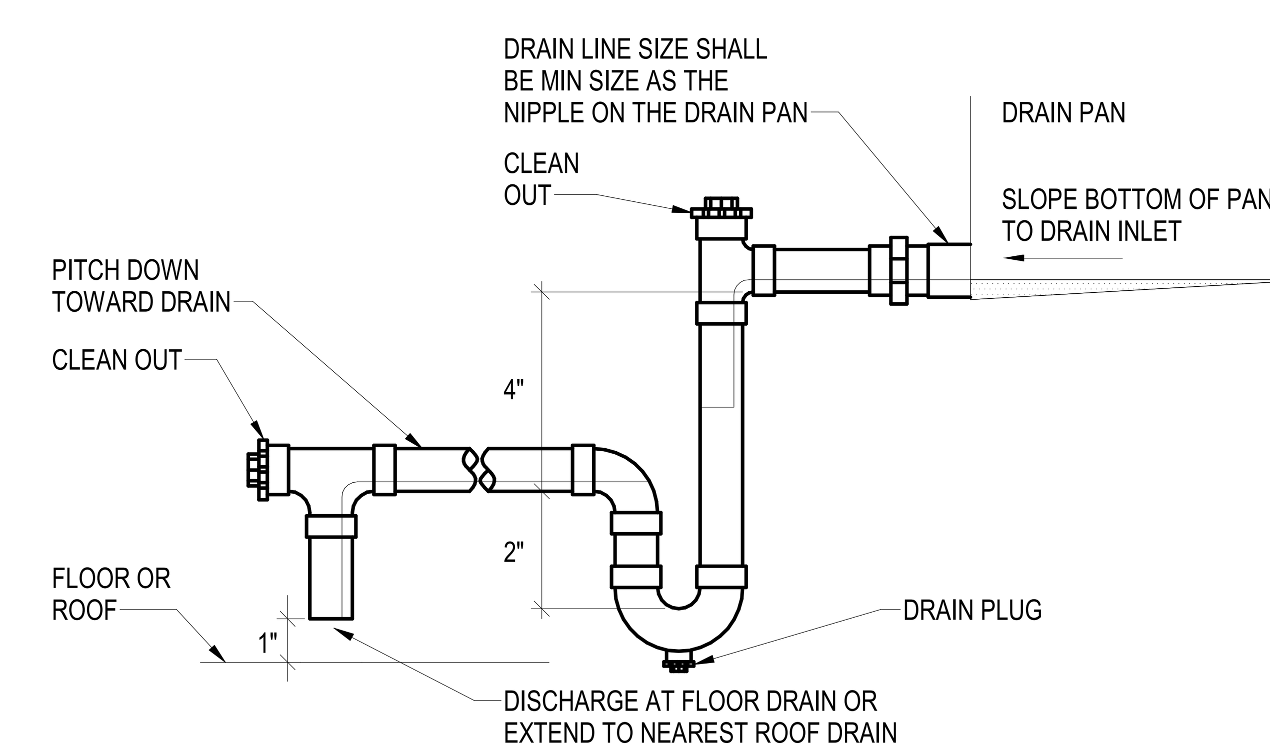
A1 TYPICAL ROOF MOUNTED CONDENSATE UNIT SPLIT SYSTEM ELEVATION

SCALE: NOT TO SCALE (MP102, MP105)



A2 AIR CONTROL PIPING DIAGRAM - HYDRONIC HEATING SYSTEM

SCALE: NOT TO SCALE (M-401)



NOTE
THIS DETAIL FOR NEGATIVE PRESSURE TRAPS ONLY.

A4 CONDENSATE TRAP DETAIL

SCALE: NOT TO SCALE

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RICHARD CRESWELL No. 13178 STATE OF MAINE LICENSED PROFESSIONAL ENGINEER 10-01-25 SEAL

OAK POINT ASSOCIATES

APPROVED
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ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
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DESIGN MANAGER T. CARLETON
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DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL DETAILS 2

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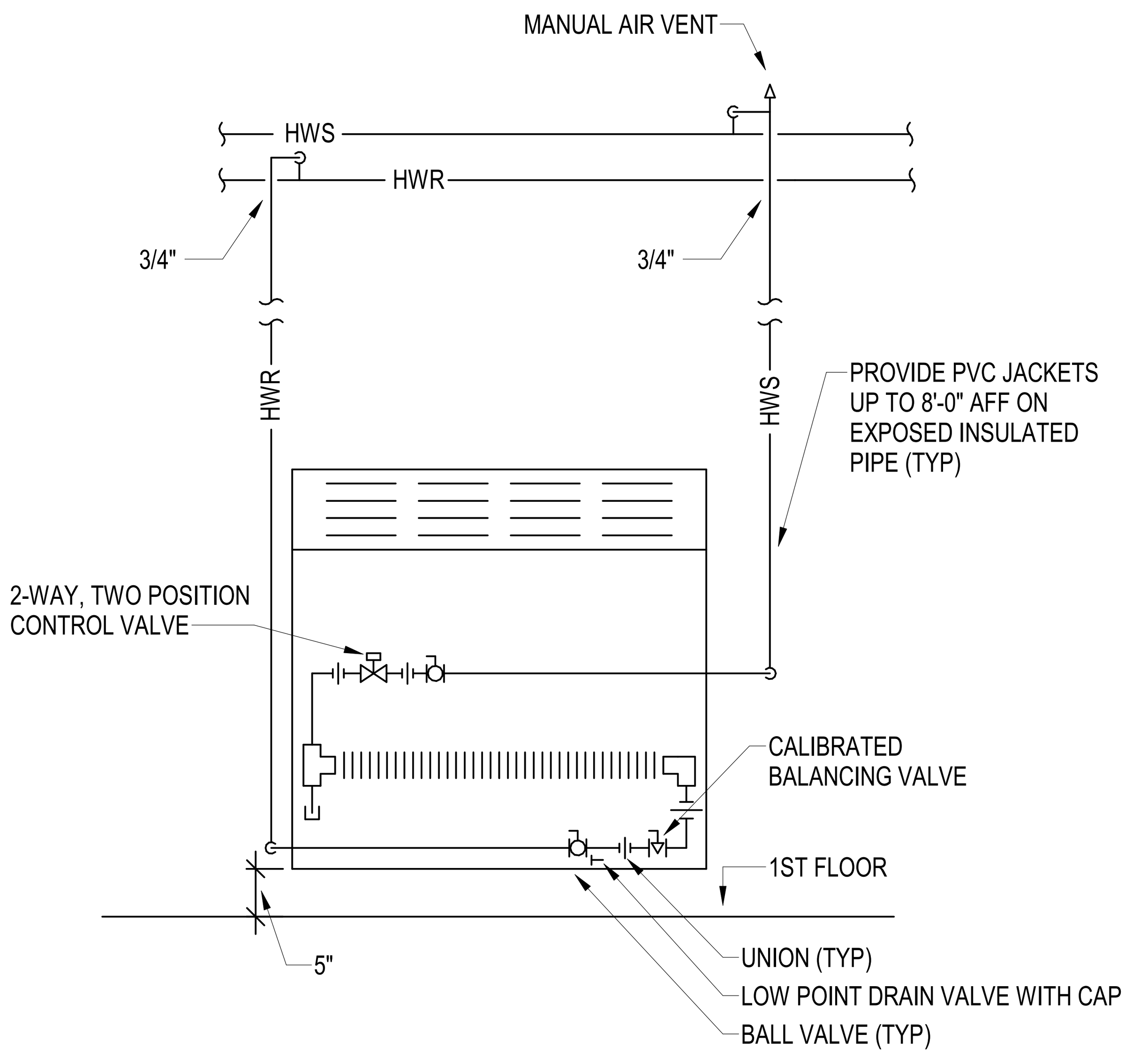
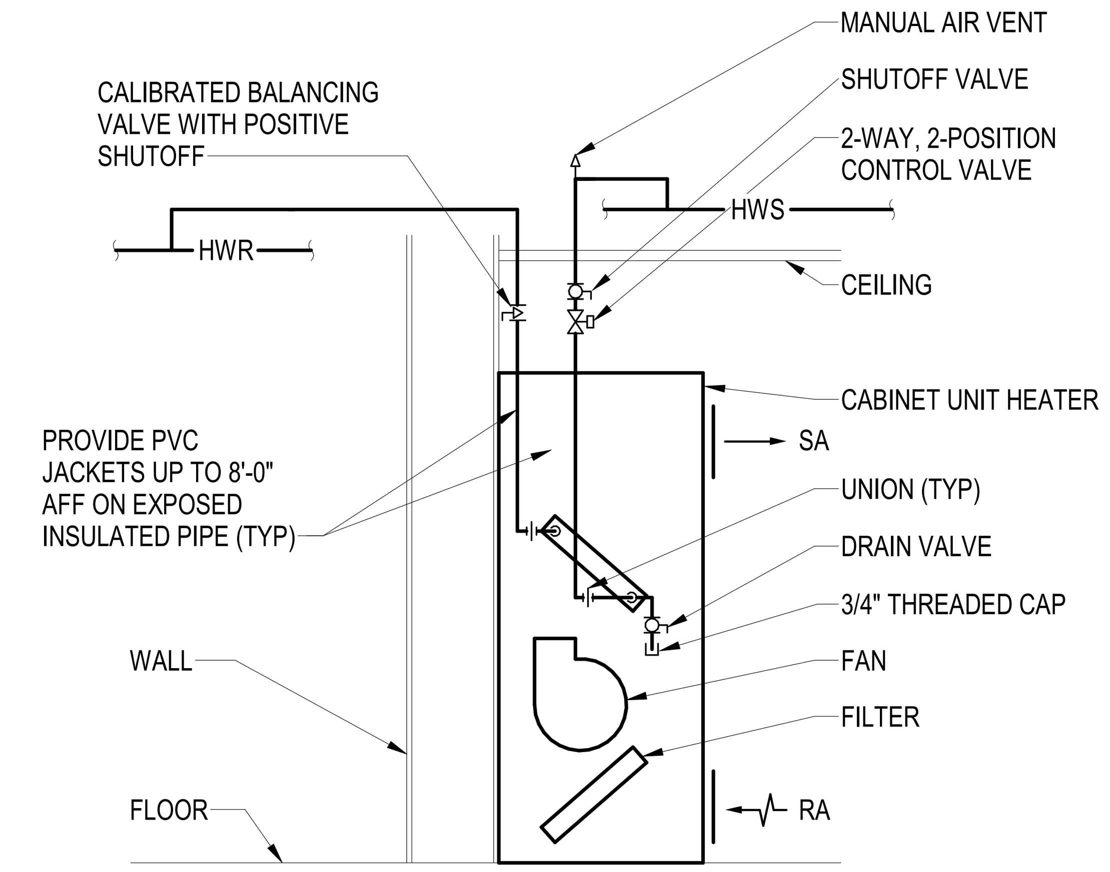


C1 TYPICAL WALL MOUNTED CONVECTOR PIPING SCHEMATIC

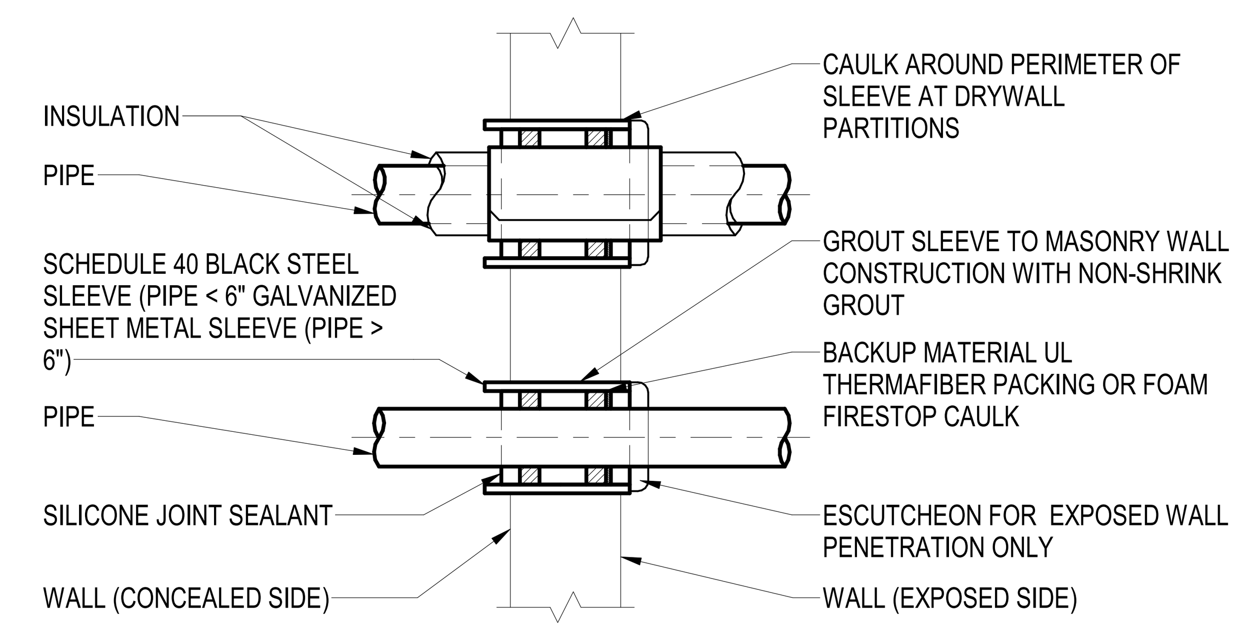
SCALE: NOT TO SCALE (MP102, MP105)



NOTE: CUH-2 SIMILAR, BUT RECESSED IN WALL.

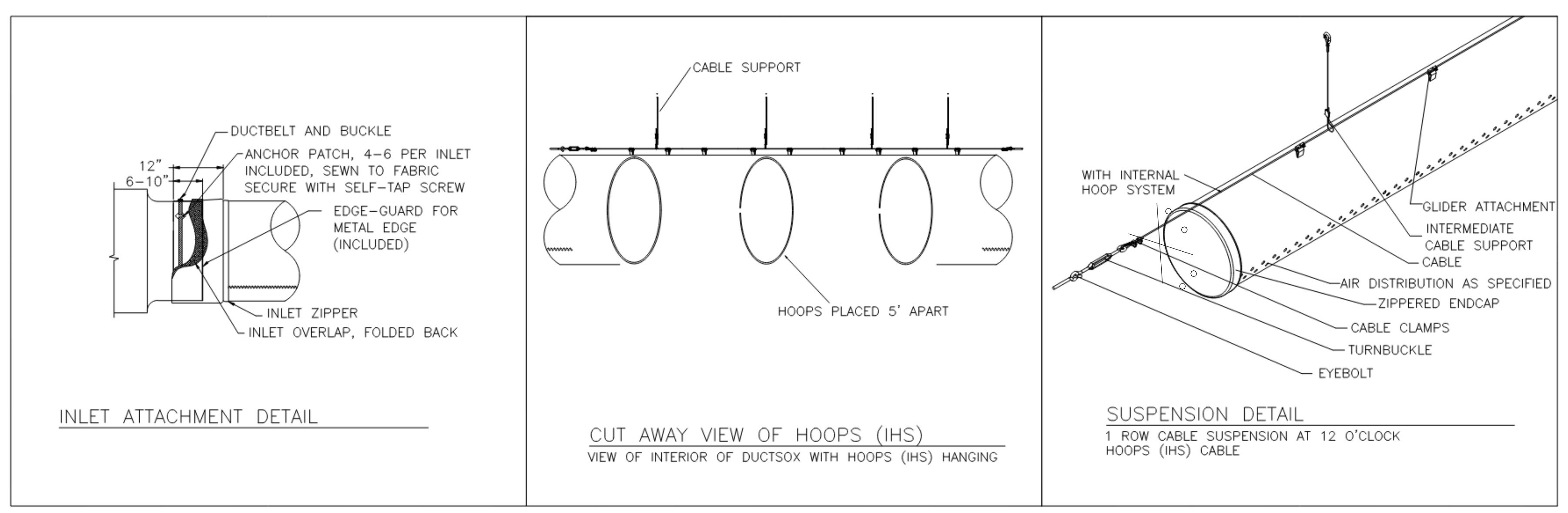
C2 TYPICAL CABINET HEATER PIPING SCHEMATIC

SCALE: NOT TO SCALE (MP102, MP105, M-401)



C4 PIPE SLEEVE THRU WALL

SCALE: NOT TO SCALE

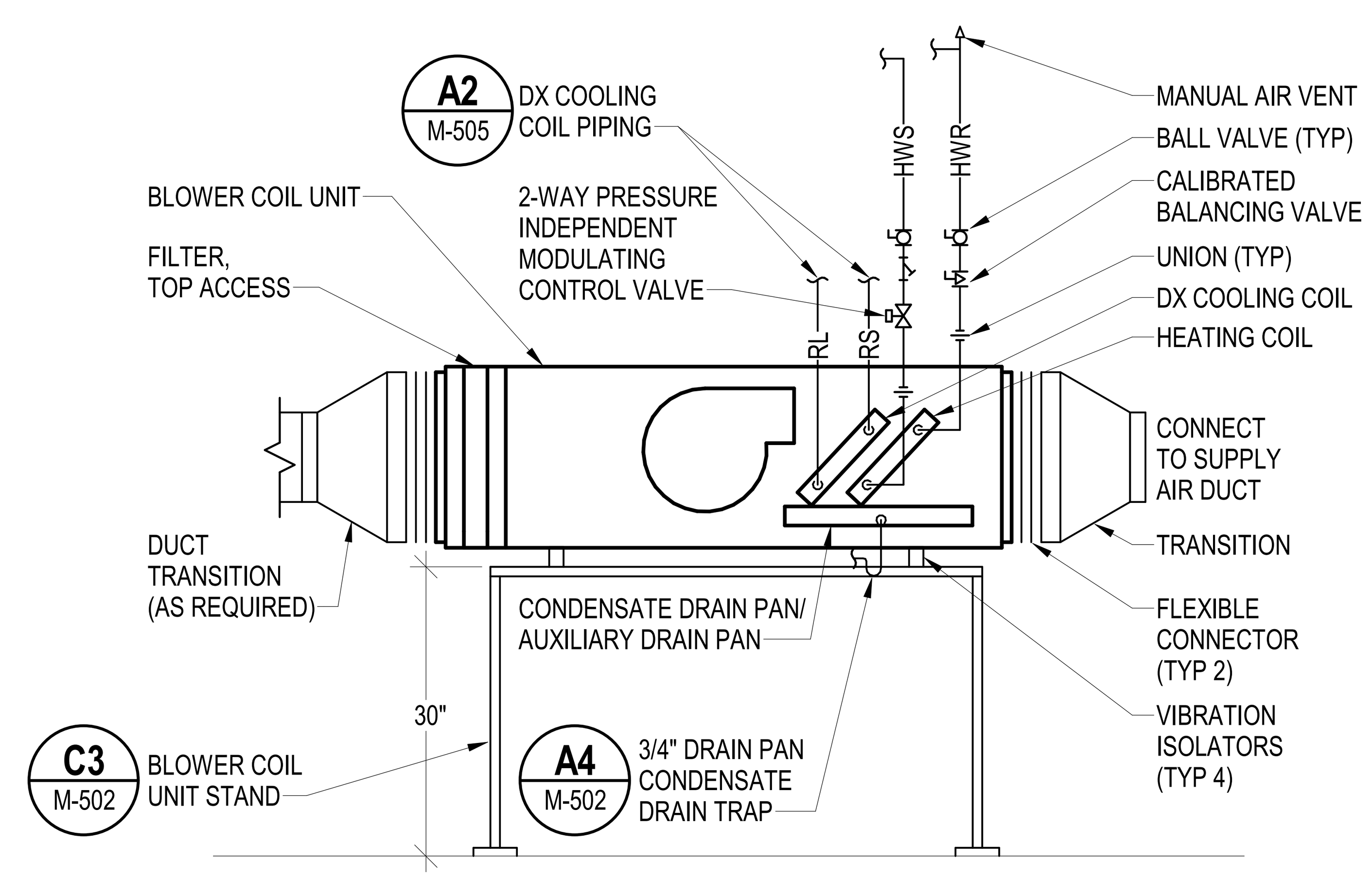


DISPERSION INFORMATION (VELOCITY OF 150, 100, 50 FPM)											
TAG #	CFM DISPERSED	DISPERSION TYPE	DISPERCION SET 1	DISPERSION SET 2	DISPERSION SET 3	DISPERSION SET 4	FABRIC	COLOR	SUSPENSION	HARDWARE	HANG HEIGHT
DS-1	1500	ORIFICE	SIZE 0.75 AT 2:00- 6' 9' 12'	SIZE 0.75 AT 3:00- 6' 9' 12'	SIZE 0.75 AT 4:00- 6' 9' 12'	SIZE 0.75 AT 5:00- 6' 9' 12'	VERONA NP	WHITE	IHS	GALV CABLE	NOT SPEC.
DS-2	1500	ORIFICE	SIZE 0.75 AT 7:00- 6' 9' 12'	SIZE 0.75 AT 8:00- 6' 9' 12'	SIZE 0.75 AT 9:00- 6' 9' 12'	SIZE 0.75 AT 10:00- 6' 9' 12'	VERONA NP	WHITE	IHS	HARDWARE	NOT SPEC.

BASIS OF DESIGN: DUCTSOX

B1 DUCTSOCKS SCHEMATICS & DISPERSION INFORMATION

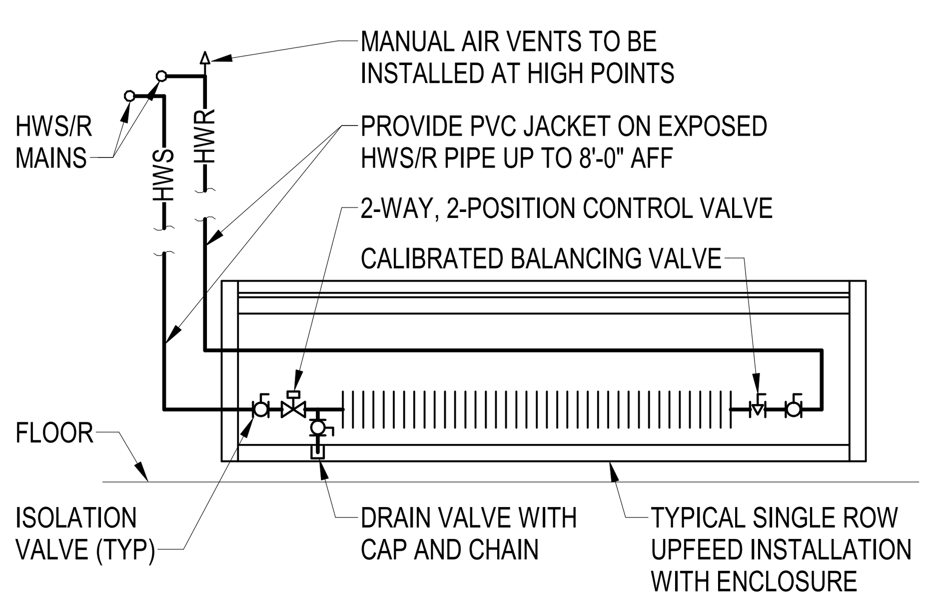
SCALE: NOT TO SCALE (MH101)



NOTE: VALVES AND SPECIALTIES ASSOCIATED WITH COOLING COIL MUST BE INSTALLED ABOVE DRAIN PAN.

B4 TYPICAL BLOWER COIL UNIT CONNECTION DETAIL

SCALE: NOT TO SCALE (MH102, MH104, MP104, MP106)



A1 TYPICAL FINTUBE RADIATION DETAIL

SCALE: NOT TO SCALE (MP105, M-401)

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SYM	DESCRIPTION	DATE	APPR
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NAVFAC

STATE OF MAINE
RICHARD CRESWELL
No. 13178
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PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
MECHANICAL DETAILS 3

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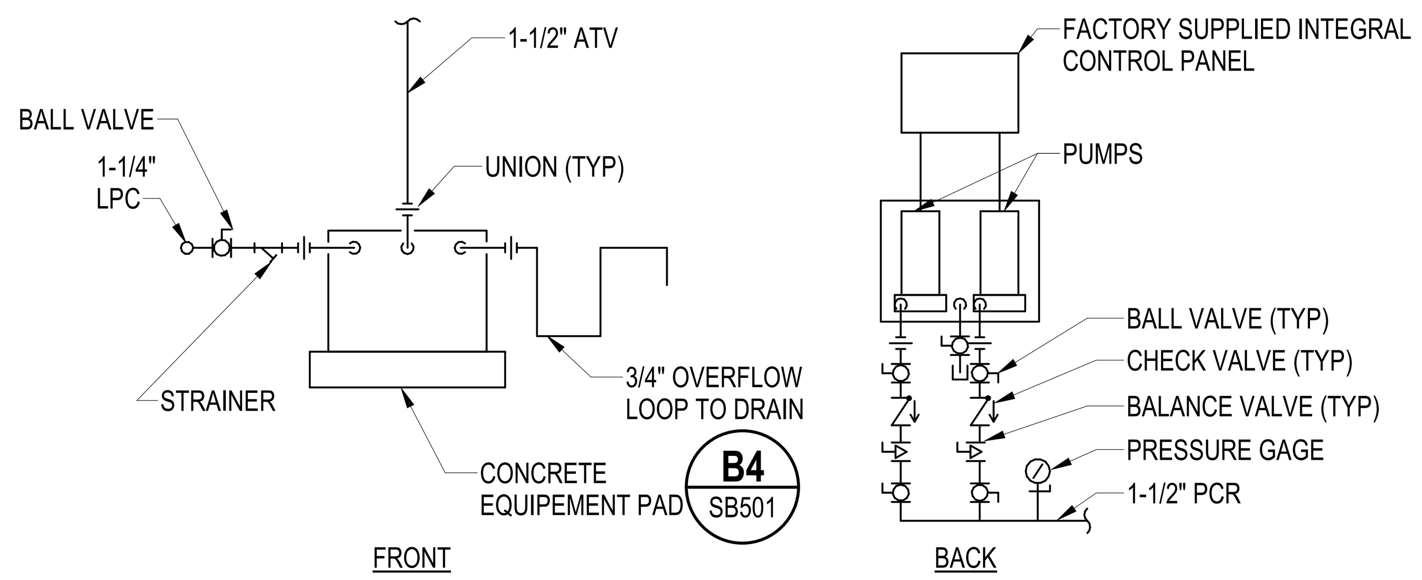
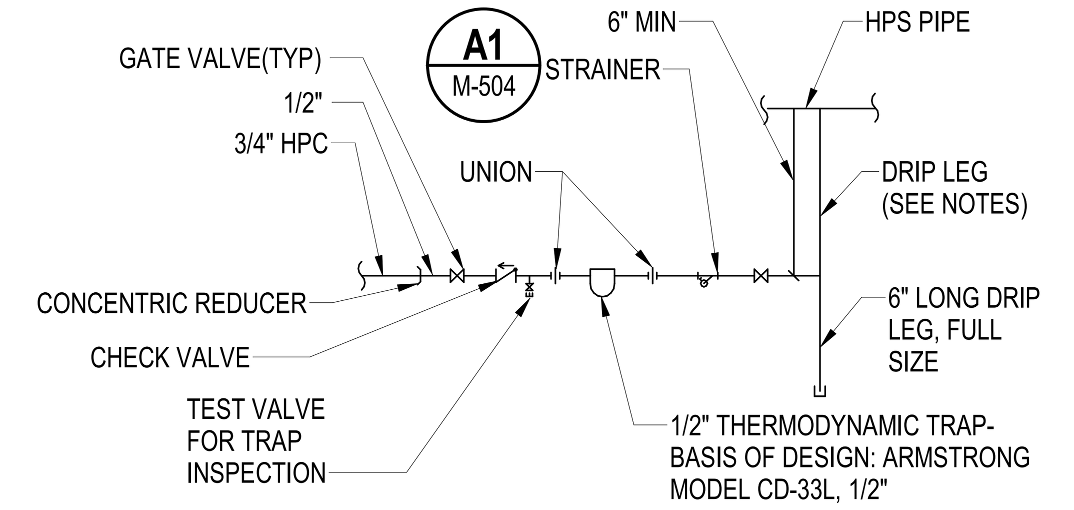


D1 CONDENSATE RECEIVER AND DUPLEX PUMP PIPING SCHEMATIC

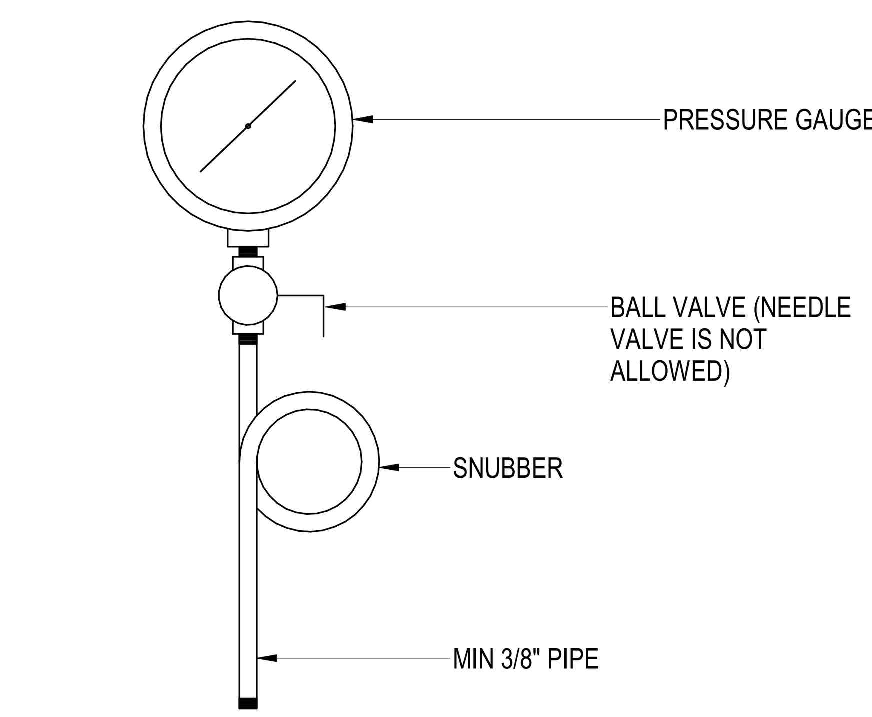
SCALE: NOT TO SCALE (M-401)



NOTES
1. DRIP LEG OFF STEAM MAIN CHALL BE SCHEDULED 80 STEEL.
2. PROVIDE TRAP SIZED WITH A SAFETY FACTOR OF 2 TIMES DESIGN LOAD

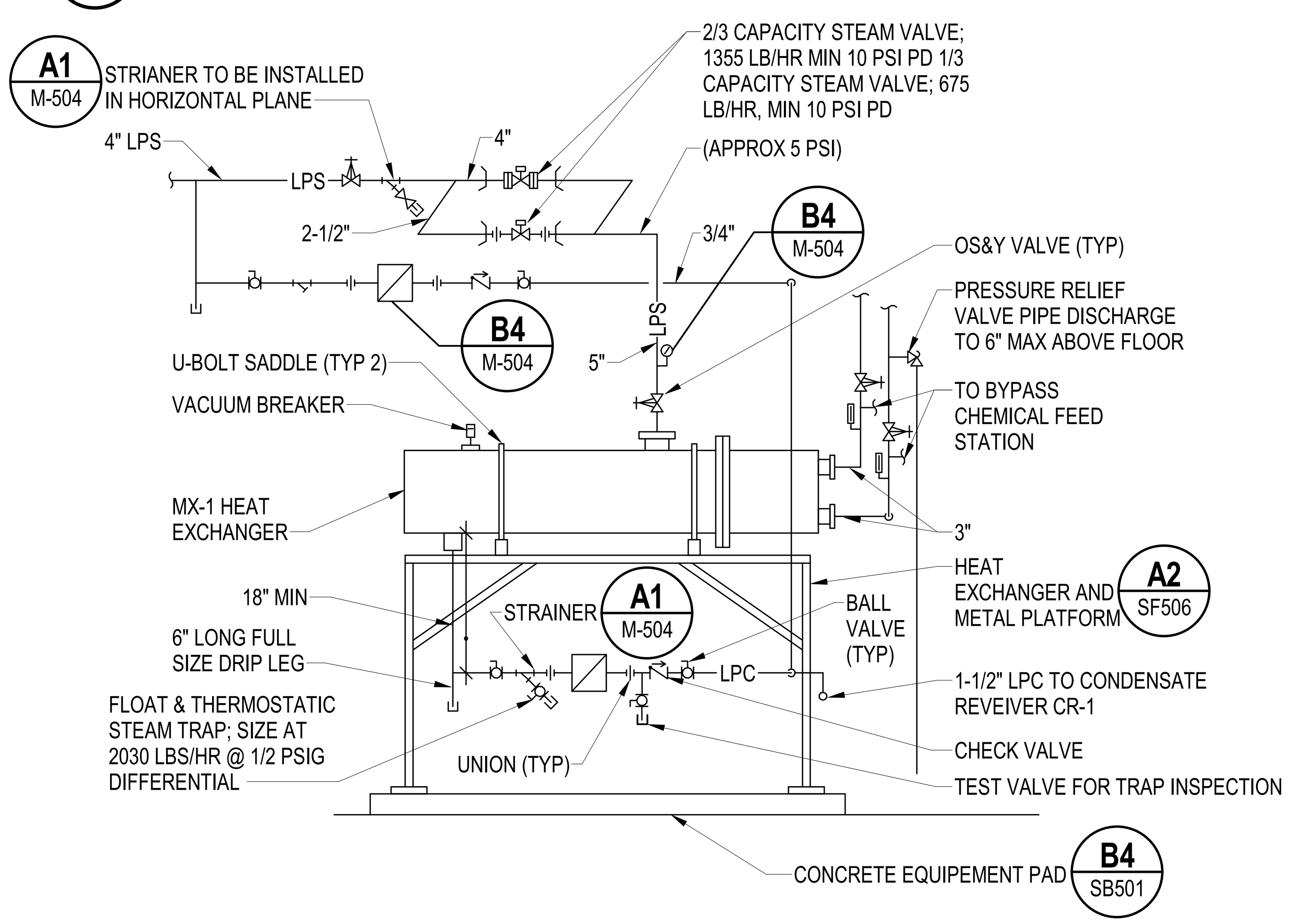
D3 TYPICAL THERMODYNAMIC TRAP HIGH PRESSURE DRIP LEG PIPING DETAIL

SCALE: NOT TO SCALE (M-401, M-504)



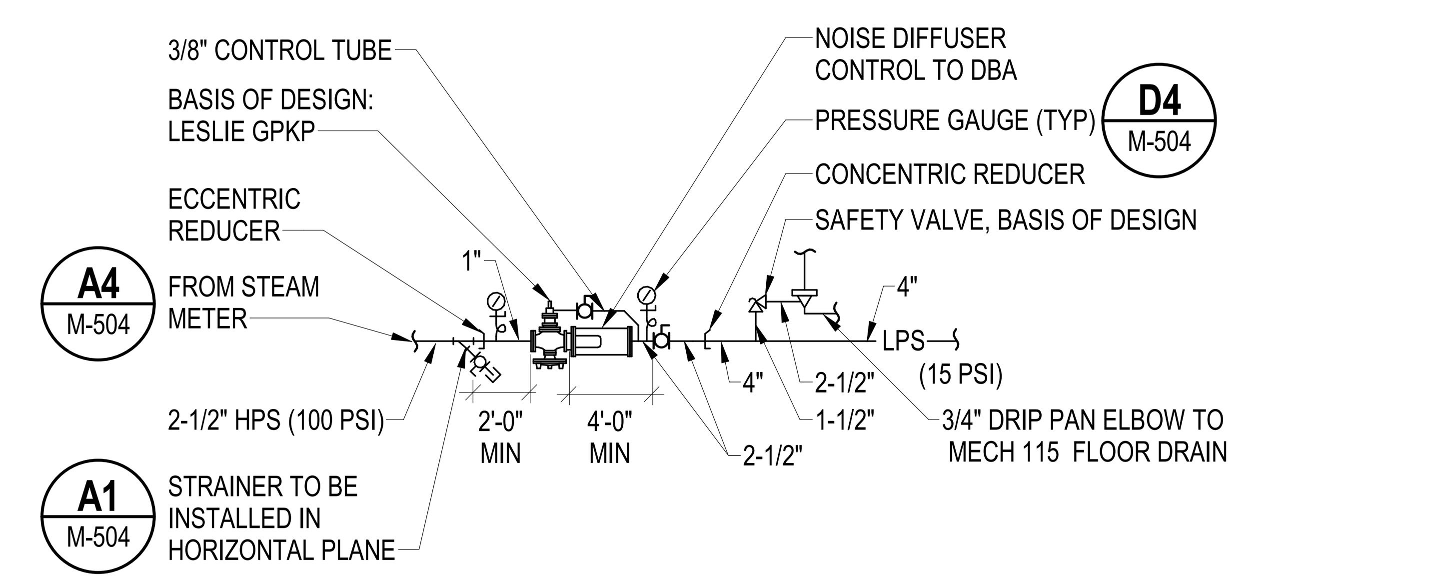
D4 STEAM PRESSURE GAUGE ASSEMBLY

SCALE: NOT TO SCALE (M-504)



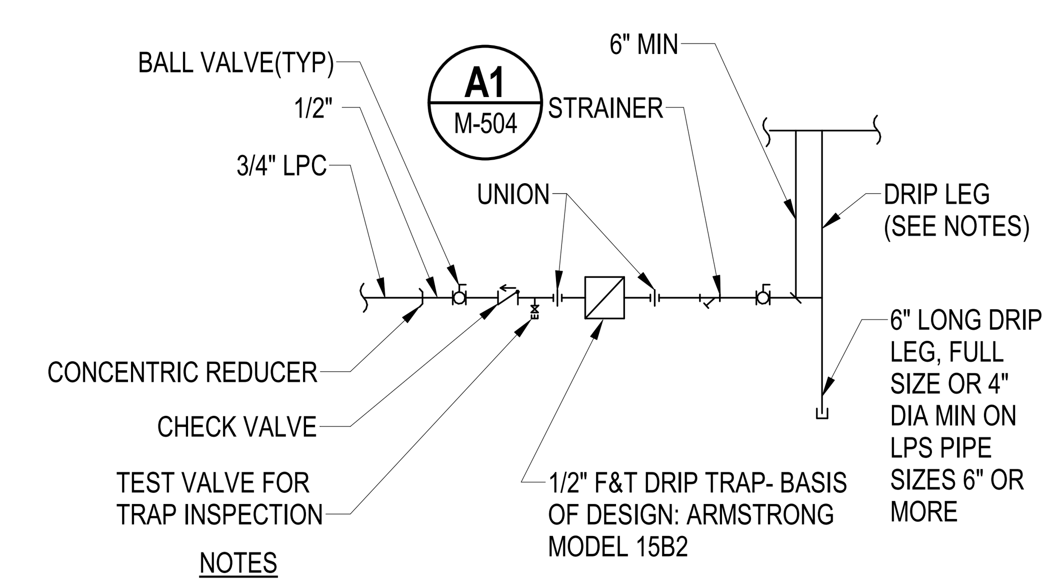
B1 STEAM TO HOT WATER CONVERTOR HX-1 PIPING SCHEMATIC

SCALE: NOT TO SCALE (M-401, M-504)



B3 STEAM PRESSURE REDUCING VALVE ASSEMBLY SCHEMATIC

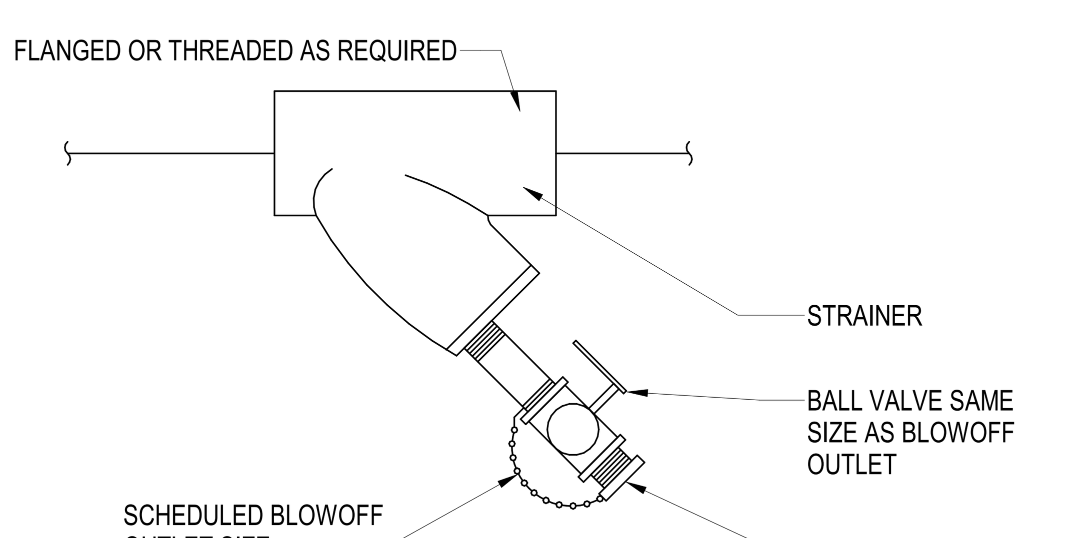
SCALE: NOT TO SCALE (M-401)



NOTES
1. DRIP LEG OFF STEAM MAIN CHALL BE SCHEDULED 80 STEEL.
2. PROVIDE TRAP SIZED WITH A SAFETY FACTOR OF 2 TIMES DESIGN LOAD

B4 TYPICAL F&T TRAP LOW PRESSURE DRIP LEG PIPING DETAIL

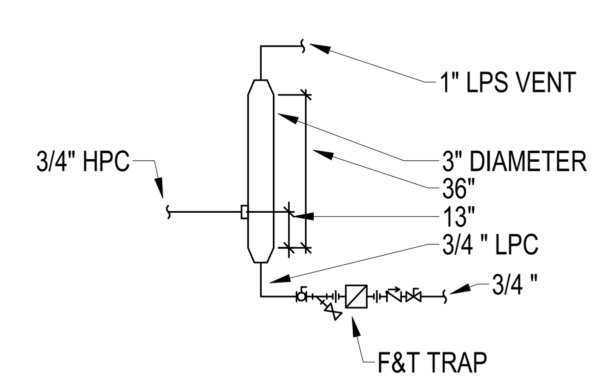
SCALE: NOT TO SCALE (M-504)



STRAINER SIZE (IN)	SCHEDULED BLOWOFF OUTLET SIZE (IN)
3/4	1/2
1 TO 2-1/2	3/4
3 & 4	1

A1 STRAINER ASSEMBLY DETAIL (TYP)

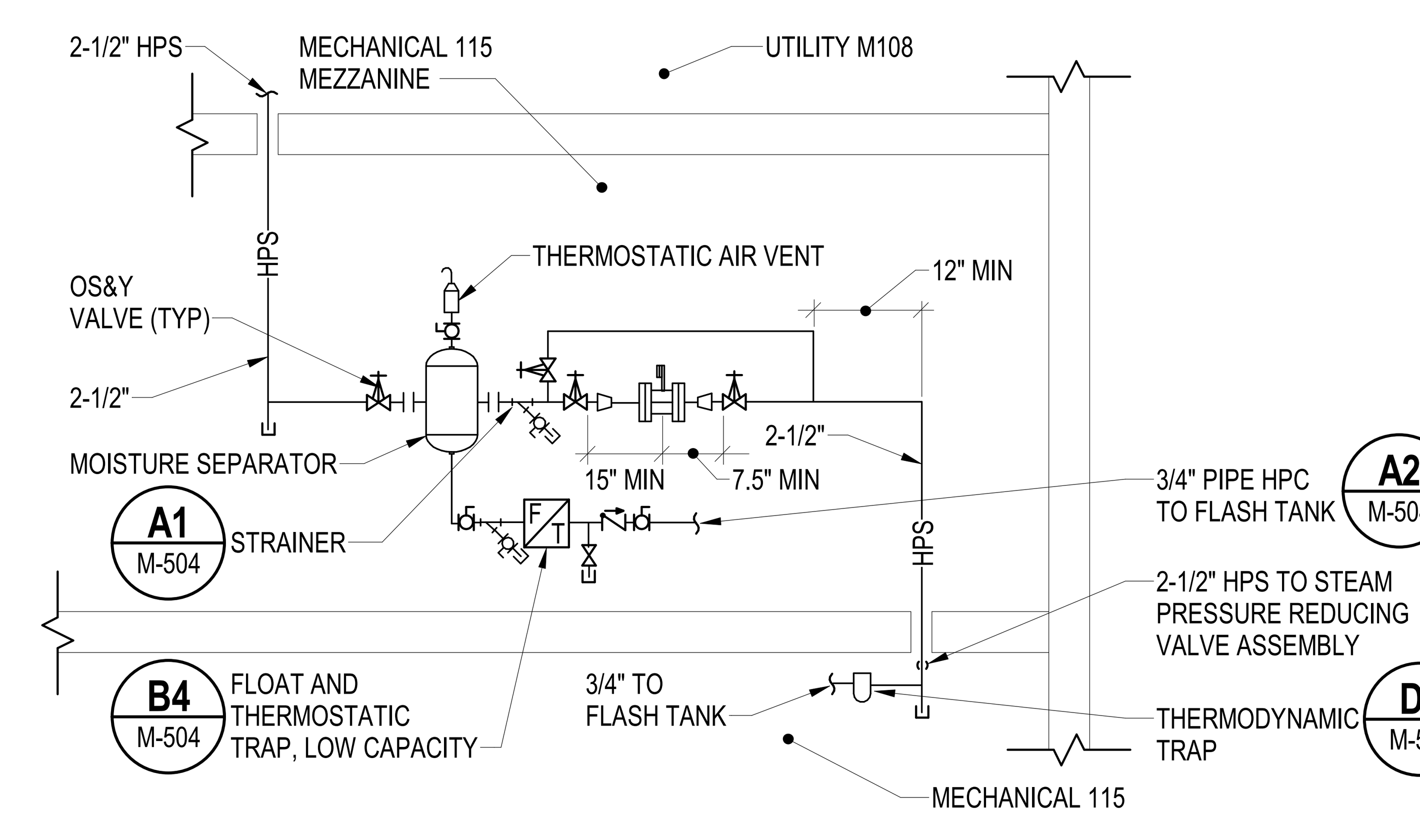
SCALE: NOT TO SCALE (M-504)



A2 FLASH TANK DETAIL

SCALE: NOT TO SCALE (M-401, M-504)

NOTES
1. BASIS OF DESIGN: EMCO SPIRAX SARCO MODEL VLM20 SIZE 2", METER.
2. INSTALL IN ACCORDANCE WITH FLOW METER MANUFACTURERS WRITTEN RECOMMENDED INSTALLATION INSTRUCTIONS.
3. REFER TO DIAGRAM A1/M-704 FOR STEAM METER OUTPUT CONNECTION TO BAS FOR BUILDING ENERGY USE METERING. METER MUST ALSO HAVE AN ADDITIONAL OUTPUT CONNECTION FOR POSSIBLE FUTURE CONNECT TO ADVANCED METERING INFRASTRUCTURE SYSTEMS.



A4 STEAM METER SCHEMATIC

SCALE: NOT TO SCALE (M-401)

FINAL SUBMISSION | 10/01/2025 | RSH
SYM | DESCRIPTION | DATE | APPR

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RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL DETAILS 4

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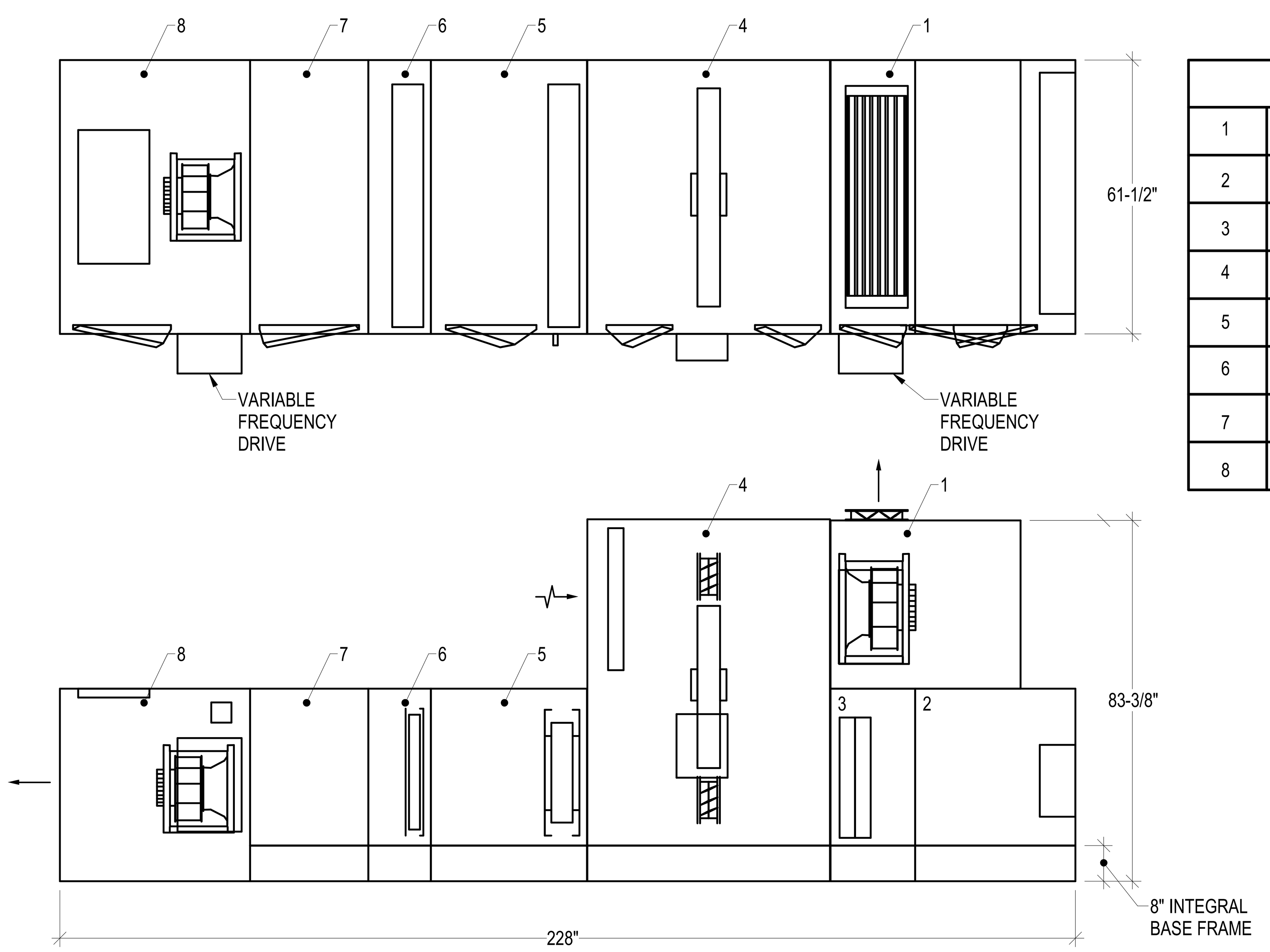
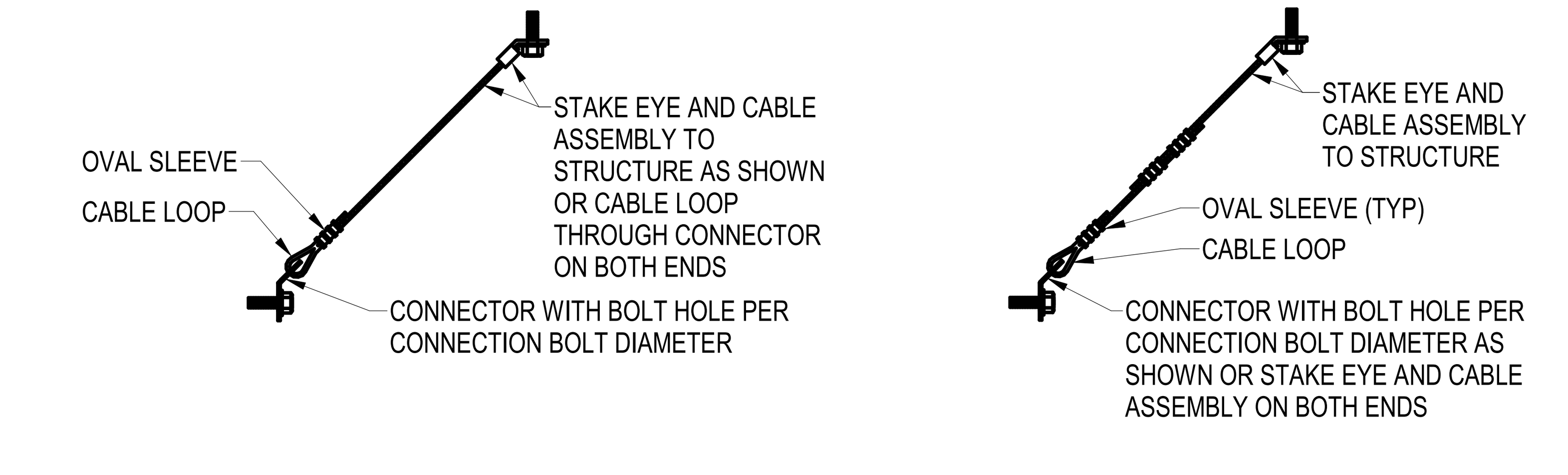
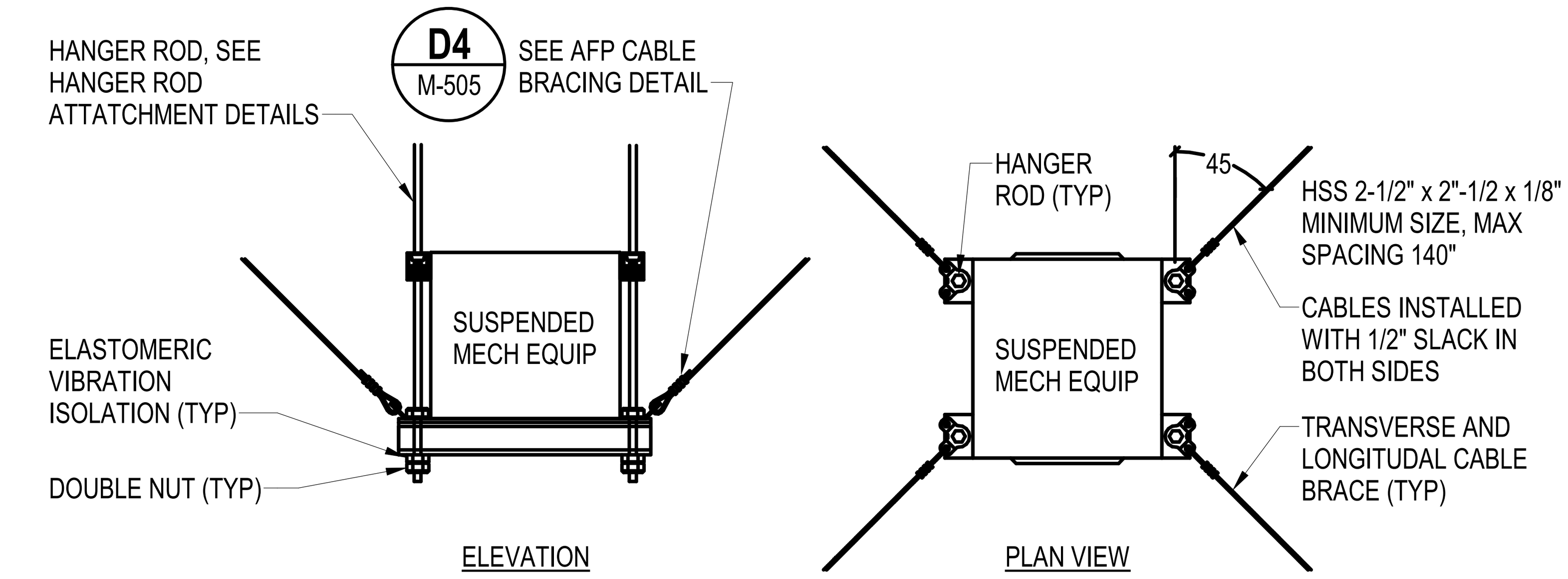


DOAS-1 COMPONENT SCHEDULE	
1	PLENUM EXHAUST FAN SECTION
2	AIR MIXING SECTION
3	FILTER SECTION
4	ENERGY WHEEL W/ BYPASS DAMPER
5	DX COOLING COIL SECTION
6	WATER HEATING COIL SECTION
7	ACCESS SECTION
8	PLENUM SUPPLY FAN SECTION

C1 **DOAS-1 AIR HANDLING UNIT DIAGRAM (BOD)**
SCALE: NOT TO SCALE (MH102, MP104)

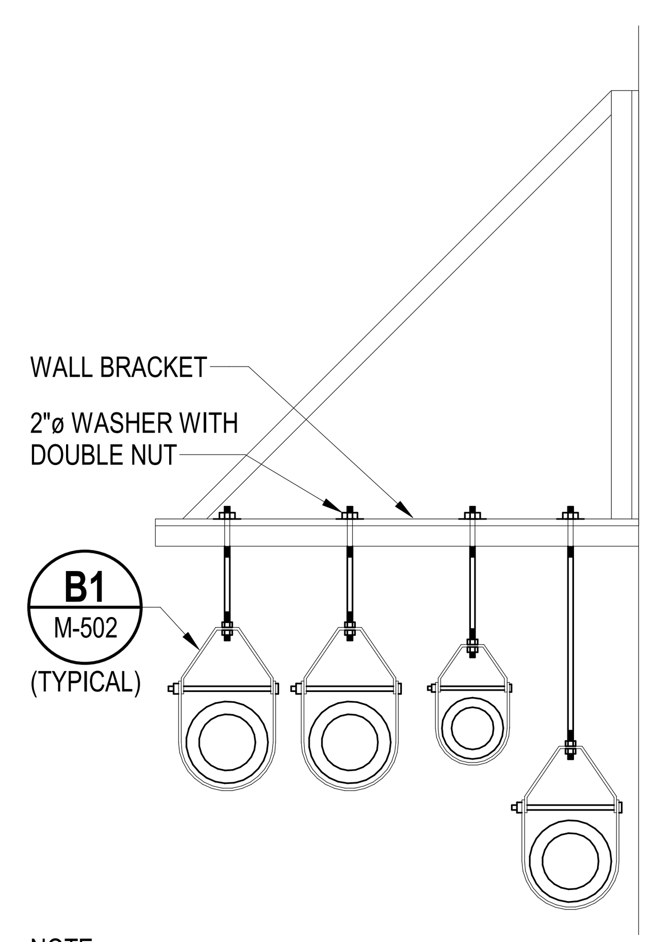


D4 **ATFP CABLE BRACING ASSEMBLY DETAIL**
SCALE: NOT TO SCALE



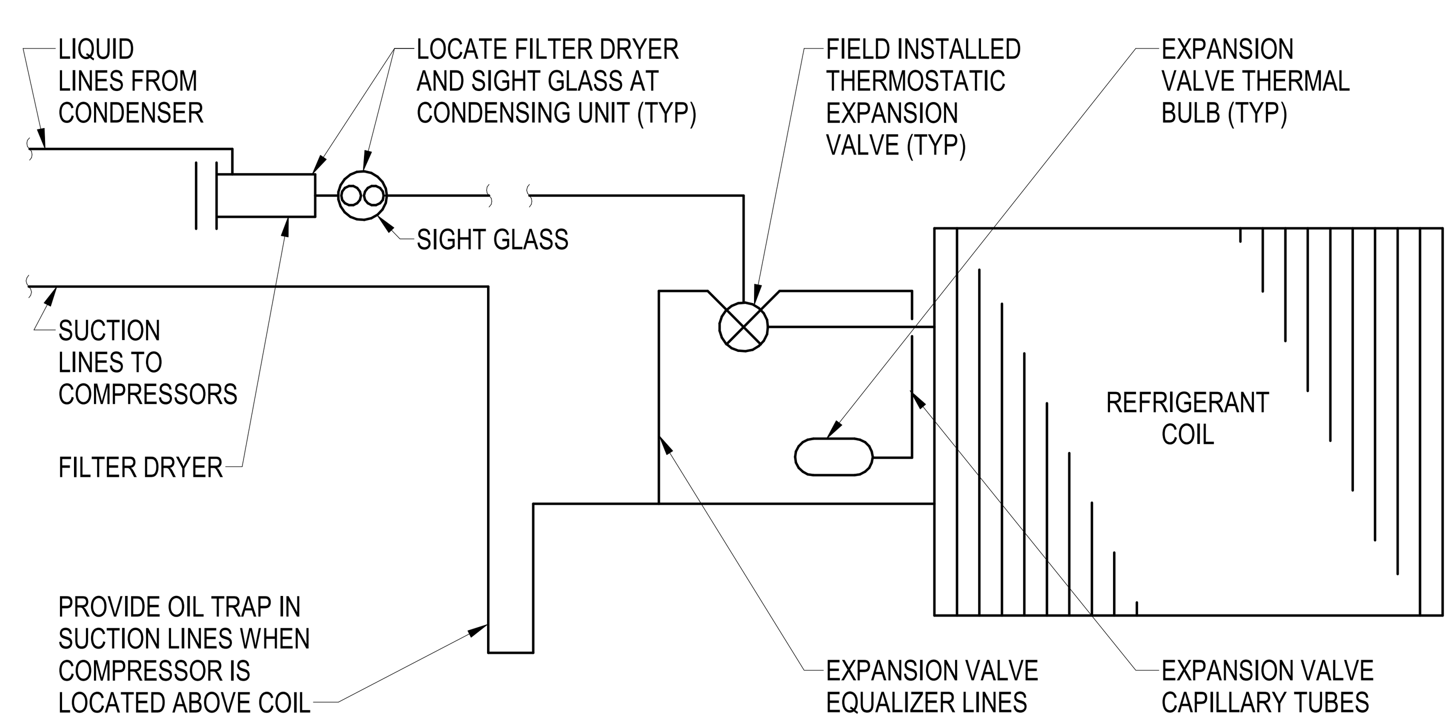
NOTES:
1. PROVIDE BRACING ON ALL OVERHEAD MOUNTED EQUIPMENT WEIGHING OVER 31 LBS PER UFC 4-010-01 DOD MINIMUM ANTI-TERRORISM STANDARDS FOR BUILDINGS, STANDARD 19
2. BRACING DESIGN SHALL BE PERFORMED BY A REGISTERED DESIGN PROFESSIONAL (REGISTERED IN THE PROJECT'S JURISDICTION) AND SUBMITTED TO THE CONTRACTING OFFICER FOR REVEIW AND APPROVAL
3. BRACING CHALL BE DESIGNED TO RESIST FORCES OF 0.5 TIMES THE EQUIPMENT WEIGHT IN ANY HORIZONTAL DIRECTION AND 1.5 TIMES THE EQUIPMENT WEIGHT IN THE DOWNWARD DIRECTION
4. DETAIL PROVIDED AS GUIDENCE ONLY FOR THE SUPPORT OF SUSPENDED EQUIPMENT AND DOES NOT RELIEVE THE CONTRACTOR FROM THE DELEGATED DESIGN RESPONSIBILTY IN MEETING THE ATFP STANDARD REQUIREMENTS

B4 **ATFP BRACING DETAIL FOR SUSPENDED MECHANICAL EQUIPMENT**
SCALE: NOT TO SCALE



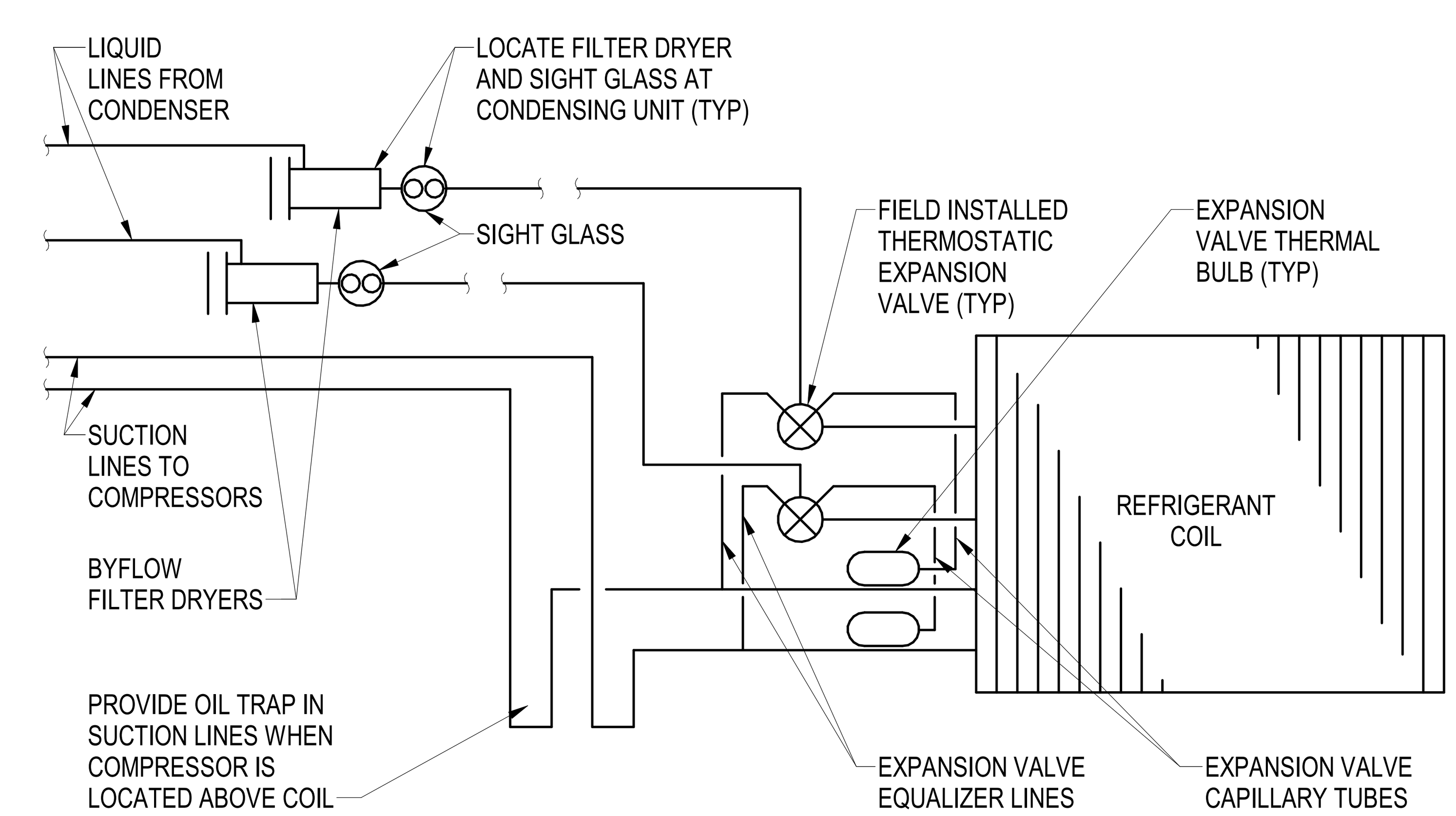
NOTE
PROVIDE SUPPORTS EVERY 10 LINEAR FEET MINIMUM

A1 **WALL MOUNTED PIPE SUPPORT SCHEMATIC**
SCALE: NOT TO SCALE



NOTES:
1. PROVIDE MANUFACTURER RECOMMENDED SIZE ISOLATION VALVES (3) AND MOUNTING MATERIALS FOR THE CIRCUIT.
2. PROVIDE EQUIPMENT MANUFACTURER'S RECOMMENDED REFRIGERANT LINE SIZES. BOD REFRIGERANT LINE SIZES ARE LISTED ON SCHEDULE.

A2 **DUCTED SPLIT SYSTEMS BCU-1/CU-1, BCU-2/CU-2, AND BCU-3/CU-3 REFRIGERANT COIL PIPING SCHEMATIC - SINGLE CIRCUIT**
SCALE: NOT TO SCALE (MP104, MP105, MP106)



NOTES:
1. PROVIDE MANUFACTURER RECOMMENDED SIZE ISOLATION VALVES (3) AND MOUNTING MATERIALS FOR EACH CIRCUIT.
2. PROVIDE RAWAL SERVICES, INC APR CONTROL, SIZED AND INSTALLED IN ACCORDANCE WITH MANUFACTURER'S WRITTEN INSTRUCTIONS.
3. PROVIDE EQUIPMENT MANUFACTURER'S RECOMMENDED REFRIGERANT LINE SIZES. ADD LINE SIZE ESTIMATE; RS: 1-1/8", RL: 0.5"

A4 **DOAS-1 / C-4 REFRIGERANT COIL PIPING SCHEMATIC-2 CIRCUIT**
SCALE: NOT TO SCALE (MP104)

SYM	DESCRIPTION	DATE	APPR
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PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
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DEDICATED OUTSIDE AIR UNIT WITH ENERGY RECOVERY SCHEDULE

UNIT NO	SERVES	SUPPLY			EXHAUST			SUMMER RECOVERY				WINTER RECOVERY				DX COOLING COIL				HOT WATER COIL							ELECTRICAL DATA		MAX UNIT WEIGHT (LBS)	BASIS OF DESIGN	NOTES
								SUPPLY		EXHAUST		SUPPLY		EXHAUST																	
		CFM MAX/MIN	ESP IN WC	FAN HP	CFM MAX/MIN	ESP IN WC	FAN HP	EAT DB/WB °F	LAT DB/WB °F	EAT DB/RH	LAT DB/WB	EAT °F	LAT °F	EAT@ RH °F	LAT °F	ENTERING AIR DB/WB °F	LEAVING AIR DB/WB °F	TC/SC MBH	REFRIG TYPE	EAT °F	LAT °F	GPM	WPD FT WC	EWT °F	LWT °F	MBH	VOLTS/ PHASE	MCA			
DOAS-1	1ST & 2ND FLRS	3770	1.25	3.0	3260	1.25	3.0	86.8/73.9	78/67	75.0/62.5	83.5/70.8	-5.3	49	70.0	15.3	79.7/66.8	52.2/52.4	178/116	R454B	0.0	70.0	18.7	1.2	180	150	280.9	460/3	18.1	3100	TRANE CSRR010	1,2,3,4,5,6,7,8,9,10,11,12

NOTES:
1. PROVIDE LOW LEAKAGE, MOTORIZED DAMPERS FOR OUTSIDE AIR AND EXHAUST.
2. PROVIDE THE FOLLOWING ADDITIONAL OPTIONS AND ACCESSORIES: UL\cUL 1995, FROST CONTROL: SPRING ISOLATION, NETWORK PROTOCOL: BACNetIP, ROTATION SENSOR, DUCT FLANGE
3. MERV 13 OUTSIDE AIR FILTER, MERV 8 RETURN AIR FILTER.
4. SEE DIAGRAM C1/M-505 FOR BASIS OF DESIGN UNIT LAYOUT AND LIST OF MODULES. UNIT TO BE SHIPPED AND BROUGHT INTO BUILDING IN SHIPPING SPLITS.
5. PROVIDE PLENUM FANS WITH VFD CONTROLS.
6. PROVIDE A SINGLE POINT OF CONNECTION FOR FANS AND WHEEL.
7. PROVIDE SERVICE LIGHTS IN THE FAN AND WHEEL SECTIONS WITH EXTERIOR MOUNTED SWITCHES.
8. PROVIDE SINGLE POINT OF CONNECTION FOR LIGHTS.
9. PROVIDE DOUBLE WALL CONSTRUCTION.
10. PROVIDE COASTAL CORROSION PROTECTION ON THE COILS.
11. PROVIDE MINIMUM SCCR OF 10,000 AIC.
12. PROVIDE WITH REFRIGERANT LEAK DETECTION SYSTEM.

SPLIT SYSTEM AIR COOLED CONDENSING (OUTDOOR) UNIT SCHEDULE

UNIT NO	SERVES	COOLING LOAD MBH	REFRIG TYPE	SEER	DESIGN TEMP °F	CONDENSER FAN(S)	COMPRESSOR(S)	RS/RL LINESET TUBING SIZES (OUSIDE Ø)	ELECTRICAL			BASIS OF DESIGN	NOTES
					DB/WB	QTY	QTY		VOLTS/ PHASE	MCA	MOCP		
OU-1	IU-1	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUY-AK12NL	1,2,4,5,6,7,8,10
OU-2	IU-2	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUY-AK12NL	1,2,4,5,6,7,8,10
OU-3	IU-3	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUY-AK12NL	1,2,4,5,6,7,8,10
OU-4	IU-4	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUZ-AK12NL	1,3,4,5,6,7,8,10
CU-1	BCU-1	48	R-454B	14.0	90.0/72.0	1	1	7/8" / 5/16"	460/3	9.4	15	TRANE 5TWA4048A4	3,4,5,6,7,8,9,10
CU-2	BCU-2	42	R-454B	14.0	90.0/72.0	1	1	7/8" / 5/16"	460/3	8.6	15	TRANE 5TWA4042A4	3,4,5,6,7,8,9,10
CU-3	BCU-3	48	R-454B	14.0	90.0/72.0	1	1	7/8" / 5/16"	460/3	9.4	15	TRANE 5TWA4048A4	3,4,5,6,7,8,9,10

NOTES:
1. PROVIDE MANUFACTURER'S RECOMMENDED 24" HIGH STAND.
2. COOLING ONLY - PROVIDE LOW AMBIENT PACKAGE TO COOL DOWN TO -20°F AMBIENT.
3. PROVIDE LOW AMBIENT PACKAGE TO COOL DOWN TO 30°F AMBIENT.
4. SPLIT SYS. MUST HAVE AT LEAST A MAX. 150 FT. REFRIGERANT PIPING LENGTH & 50 FEET PIPING HEIGHT DIFFERENCE CAPACITY.
5. CORROSION RESISTANCE RATING OF 2000 HRS. OF DURABILITY UNDER ASTM 117 TESTING STANDARDS.
6. PROVIDE MANUFACTURERS RECOMMENDED PRE-INSULATED REFRIGERANT LINESET SIZES
7. PROVIDE MANUFACTURED ENCLOSURE FOR EXPOSED INSULATED REFRIGERANT LINESET AND POWER CABLE FOR SPLIT SYSTEM
8. PROVIDE FIELD INSTALLED CRANK CASE HEATER, INSTALL IN ACCORDANCE WITH MANUFACTURER'S WRITTEN INSTRUCTIONS
9. HEAT PUMP
10. PROVIDE WITH EQUIPMENT MANUFACTURER'S VIBRATION ISOLATORS.

SPLIT SYSTEM INDOOR UNIT (IU) SCHEDULE

UNIT NO	SERVES	HEATING LOAD (MSH)	REFRIG TYPE	NOMINAL COOLING TONS	HEATING CAPACITY @4°F (MBH)	CABINET TYPE	NOMINAL CFM	SEER	ELECTRICAL	WEIGHT LBS	BASIS OF DESIGN	NOTES
									VOLTS/PH			
IU-1	ELEV MACHINE RM	N/A	R-454B	1.0	N/A	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	1,3,4
IU-2	NMCI 110	N/A	R-454B	1.0	N/A	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	1,3,4
IU-3	NMCI 209B	N/A	R-454B	1.0	N/A	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	1,3,4
IU-4	OFFICE 108	12	R-454B	1.0	7.0	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	2,3,4

NOTES:
1. COOLING ONLY UNIT.
2. HEAT PUMP UNIT.
3. PROVIDE WITH MANUFACTURER'S DELUXE WIRED, WALL MOUNTED CONTROLLER (BOD: TRANE-MITSUBISHI TAR-42MAAUS).
4. PROVIDE PRE-MANUFACTURERED ENCLOSURE FOR EXPOSED PIPING WITHIN SERVED ROOM.

STEAM CONDENSATE RECIEVER AND RETURN PUMP SCHEDULE

UNIT NO	LOCATION	CAPACITY		DISCHARGE PRESSURE (PSIG)	RECEIVER CAPACITY (GAL)	MOTOR DATA		RPM	BASIS OF DESIGN	NOTES
		LB/HR	GPM			HP	VOLTS/ PHASE			
CR-1	MECHANICAL 115	2100	9.0	75	23	3.0	460/3	3500	BELL & GOSSETT MODEL 23 CB9 - 75	1,2,3,4,5,6,7

NOTES:
1. PROVIDE DUPLEX PUMP SET.
2. PROVIDE FACTORY MOUNTED CONTROL PANEL, (NEMA 2 ENCLOSURE) FLOAT SWITCHES, AND ELECTRIC ALTERNATOR.
3. PROVIDE WATER LEVEL GAGE WITH SHUT OFF VALVE.
4. PROVIDE DISCHARGE PRESSURE GAUGE.
5. PROVIDE HIGH LEVEL ALARM.
6. PROVIDE DUPLEX DISCHARGE MANIFOLD.
7. SEE SPECIFICATIONS FOR FURTHER DETAILS.

EXHAUST FAN SCHEDULE

UNIT NO	SERVES	CFM	ESP IN WC	DRIVE TYPE	FAN TYPE	WEIGHT (LB)	FAN RPM	SONES	MAX BHP	HP	VOLTS/P HASE	BASIS OF DESIGN	ACCESSORIES
EF-1	MECH 115	650	0.35	DIRECT	SIDE WALL	56	1595	7.8	0.09	1/10	120/1	GREENHECK CUE - 090 - VG	A,B,C
TF-1	ELEC 114	350	0.25	DIRECT	INLINE	34	1234	5.0	0.03	1/6	120/1	GREENHECK SQ - 90 - VG	A,C

ACCESSORIES:
A. ELECTRONICALLY COMMUTATED MOTOR ECM.
B. PROVIDE WITH LOW LEAKAGE, INSULATED MOTOR OPERATED DAMPER, WITH THERMALLY BROKEN INSULATED FRAME.
C. MFR FAN MOUNTED DISCONNECT SWITCH.

FINTUBE RADIATION SCHEDULE

UNIT NO	ELEMENT				ENCLOSURE		CAPACITY BTUH/LF	AVERAGE WATER TEMP °F	BASIS OF DESIGN	NOTES
	TUBE SIZE	ROWS	FINS PER FOOT	SIZE (IN)	MOUNTING HEIGHT (IN)	ENCLOSURE HEIGHT (IN)				
FR-1	3/4	1	40	3-5/8 x 4-1/4	18	14	1040	170	VULCAN LV4-S14	1
FR-2	3/4	1	50	3-5/8 x 4-1/4	79 1/2	19 1/2	970	170	VULCAN LV4-DS19	2

NOTES:
1. SLOPE TOP STYLE
2. DOUBLE SLOPE TOP STYLE

PUMP SCHEDULE

UNIT NO	SERVES	LOCATION	TYPE	GPM	TOTAL HEAD	PUMP EFF (%)	MOTOR DATA				SUCTION/ DISCHARGE (IN)	BASIS OF DESIGN	NOTES
							BHP	HP	RPM	VOLTS/PHASE			
P-1A	HW CIRCULATION	MECH 115	INLINE	133.6	69.5	65.7	3.5	7.5	1800	460/3	2.5/2.5	BELL & GOSSETT E-80	1
P-1B	HW CIRCULATION	MECH 115	INLINE	133.6	69.5	65.7	3.5	7.5	1800	460/3	2.5/2.5	BELL & GOSSETT E-80	1

NOTES: 1. PROVIDE WITH VARIABLE FREQUENCY DRIVE.

LOUVER SCHEDULE

UNIT NO	SERVES	TYPE	CFM	DIMENSIONS (INCHES)			MIN FREE AREA SQUARE FEET	BASIS OF DESIGN	NOTES
				LENGTH	HEIGHT	DEPTH			
L-1	MECH 115 MAKE UP	FIXED	650	36	36	6	0.77	RUSKIN/EUF6375DX	1,2,3,4,5,6

NOTES:
1. PROVIDE WITH BIRDSCREEN.
2. LOUVER SHALL BE DRAINABLE.
3. PROVIDE WITH COASTAL CORROSION PROTECTION FINISH IN CUSTOM COLOR SELECTED BY CONTRACTING OFFICER.
4. PROVIDE LOUVER RATED FOR MINIMUM 35 PSF WIND LOAD.
5. PROVIDE WITH FLANGE OPTION.
6. LOUVER REPLACE LOWER WINDOW SASH. VERIFY IN FIELD SIZE OF LOUVER HEIGHT TO REPLACE LOWER WINDOW SASH.

STEAM-TO-WATER CONVERTER SCHEDULE

UNIT NO	SERVES	LOCATION	STEAM		HOT WATER				FOULING FACTOR	HEATING MBH	BASIS OF DESIGN	NOTES
			INLET PRESSURE	LB/HR	FLOW GPM	EWT °F	EWT °F	MAX WPD				
HX-1	BLDG 60 HYDRONIC HEAT SYST	MECH 115	5 PSI	2030	133.8	180	150	4.0 FT	0.0005	1963	BELL & GOSSETT SU-104-4	1

NOTES: 1. PROVIDE WITH OVERFLOW ALARM OUTPUTS.

STEAM PRESSURE REDUCING VALVE SCHEDULE

TAG NO	LOCATION	CAPACITY LB/HR	INLET PRESSURE (PSIG)	OUTLET PRESSURE (PSIG)	VALVE SIZE	SAFETY VALVE		BASIS OF DESIGN	NOTES
						TAG	CAPACITY W/30 PSIG SET PRESS LB/HR		
PRV-1	MECH 115	2030	100	15	1"	SV-1	4631	LESLIE GPKP-1	1,2

NOTES:
1. CONTROL NOISE LEVEL TO BELOW 80 dBA BY PROVIDING ATTENUATION EQUIPMENT AS REQUIRED.
2. PRESSURE REDUCING VALVE MUST HAVE A 100 TO 1 PRESSURE TURN DOWN RATIO CAPACITY, OR PROVIDE PARALLEL 1/3 - 2/3 PRESSURE REDUCING VALVE ASSEMBLIES (2) AND SAFETY VALVES (2) FOR PRVS WITH SMALLER TURNDOWN CAPABILITY.

AIR COOLED HEAT PUMP CONDENSING UNIT SCHEDULE FOR DOAS-1

UNIT NO	SERVES	NOMINAL COOLING CAPACITY FULL LOAD	LOW TEMP HEATING CAPACITY (MBH)	REFRIG TYPE	REFRIG Ø LINE SIZES (IN) RS/RL	EER FULL LOAD	DESIGN TEMP	CONDENSER FAN(S)				COMPRESSOR(S)			CAPACITY STEPS	ELECTRICAL				BASIS OF DESIGN	NOTES
							COND AMBIENT	QTY	FAN HP EA	FLA EA	LRA EA	QTY	FLA EA	LRA EA		VOLTS/PHASE	MCA	MDP	WEIGHT		
CU-4	DOAS-1	15.0	105	R-454B	1-18"/1/2"	10.8	90° F	2	1.0	2.2	10.1	2	11.9	2.2	2	460/3	33	40	800LB	TRANE ODYSEY TWA 180K	1,2,3,4,5

NOTES:
1. DUAL COMPRESSORS/ DUAL CIRCUIT
2. COASTAL CORROSION LEVEL - COATED COMMERCIAL COIL PROTECTION
3. MANUFACTURERS MICRO PROCESSED CONTROLS; BOD; TRANE54MB10 7000
4. HAIL GUARDS, SPRING VIBRATION, ISOLATORS, BACNET INTERFACE
5. REFRIGERANT LEAK DETECTION SYSTEM

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 10-01-25 STATE OF MAINE LICENSED PROFESSIONAL ENGINEER

SEAL

OAK POINT ASSOCIATES

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FIRE PROTECTION S. RUCINSKI

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ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL SCHEDULES 1

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OUTDOOR PACKAGED HVAC UNIT SCHEDULE

UNIT NO	AREA SERVED	REFIGERANT TYPE	NORMAL COOLING CAPACITY, TONS	MIN COOLING SENSIBLE COOLING MBH	MIN COOLING TOTAL COOLING MBH	ENTERING AIR DB/WB	CFM SUPPLY AIR	CFM MIN OUTSIDE AIR	EXTERNAL STATIC PRESSURE IN WC	ECONOMIZER	HEATING SOURCE	ELECTRICAL BHP	ELECTRICAL HP	ELECTRICAL VAULTS/ PHASE	WEIGHT (LB)	BASIS OF DESIGN	NOTES
AHU-1	SHOP 111	R454B	27.5	243	324	79/60	10,800	2,000	1.5	EXHAUST FANS (2)	HC-1	7.3	10.0	460/3	5200	TRANE VOYAGER TCH330D40	1 - 10

NOTES:
1. UNIT WILL BE INSTALLED AT GRADE.
2. MAXIMUM UNIT LENGTH MUST NOT EXCEED 180-INCHES, WITHOUT HOODS.
3. CABINET OUTDOOR UNIT, DOUBLE WALL CONSTRUCTION CONDENSER COIL HAIL GUARDS, HINGED SERVICE ACCESS, HORIZONTAL AIRFLOW CONFIGURATION, STAINLESS STEEL COIL DRAIN PANS, 18" HIGH CORROSION RESISTANT CURB.
4. CONTROLS: OUTSIDE AIRFLOW MEASUREMENT, BACNET COMMUNICATION INTERFACE.
5. ELECTRICAL: HIGH FAULT SCCR(MINIMUM 22,000 AIC), CONVIENENCE OUTLET, DISCONNECT SWITCH.
6. FILTRATION: 4" MERV 14 THROW AWAY FILTERS, CLOGGED FILTER SWITCH.
7. MECHANICAL: SINGLE ZONE VAV, SHAFT GROUND RING, 100% POWER EXHAUST WITH ULTRA LOW LEAK POWER, EXHAUST DAMPER, ECONOMIZER WITH DIFFERENTIAL ENTHALPY CONTROL, AND ULTRA LOW LEAK OUTSIDE AIR DAMPER.
8. REFRIGERATION: COASTAL CORROSION PROTECTED CONDENSER COIL, MODULATING HOT GAS REHEAT, SERVICE VALVES.
9. PROVIDE SINGLE POINT OF POWER CONNECTION.
10. PROVIDE UNIT WITH FACTORY FURNISHED REFRIGERANT LEAK DETECTION SYSTEM.

BLOWER COIL UNIT SCHEDULE

UNIT NO	SERVES	DESIGN CFM	ESP	DX COOLING REFRIGERANT	DX COOLING APD (FT HD)	DX COOLING TC MBH	DX COOLING SC MBH	DX COOLING EAT DB / WB	DX COOLING LAT DB / WB	DISTRIBUTEES	HEATING EWT °F	HEATING MBH	HEATING GPM	HEATING ROWS	CABINET STYLE / NOMINAL CABINET SIZE (CFM)	ELECTRICAL VOLTS/ PHASE	ELECTRICAL MCA	ELECTRICAL BHP / HP MOTOR	WEIGHT LBS	BASIS OF DESIGN	NOTES
BCU-1	SHOP 110	1850	1.0	R454B	0.53	50.2	46.3	76.6 / 69.9	68.8 / 68.5	1	180	62.8	4.2	1	HORIZONTAL	460/3	3.13	0.91/1.0	210	TRANE BCHE054	1,2,3,4
BCU-2	OPEN OFFICE 208	1225	1.0	R454B	0.53	43.3	31.3	76.2 / 64.8	59.2 / 58.7	1	180	46.2	3.1	1	HORIZONTAL	460/3	3.13	0.91/1.0	180	TRANE BCHE036	1,2,3,4
BCU-3	TRAINING 209, 210, 211	1670	1.0	R454B	0.53	47.8	43.0	76.6 / 69.0	63.2 / 62.7	1	180	52.6	3.5	1	HORIZONTAL	460/3	3.13	1.0/1.0	210	TRANE BCHE054	1,2,3,4

NOTES:
1. PROVIDE UNIT WITH ECM FAN MOTORS AND DRAIN PAN FLOAT SWITCHES.
2. PROVIDE UNIT WITH HEATING COIL IN REHEAT POSITISON, DOWNSTREAM OF COOLING COIL.
3. PROVIDE TOP ACCESS FILTER SECTION WITH 2" MERV 8 FILTERS, AUXILIARY DRIP PAN, AND VIBRATION ISOLATORS.
4. PROVIDE UNIT WITH FACTORY FURNISHED REFRIGERANT LEAK DETECTION SYSTEM.

HOT WATER UNIT HEATER SCHEDULE

UNIT NO	LOCATION	CFM	HEATING MBH	AIR FLOW DIRECTION	WATER TEMP °F ENTERING	WATER TEMP °F LEAVING	GPM	MAX PRESSURE DROP IN WC	ELECTRICAL DATA HP	ELECTRICAL DATA RPM	ELECTRICAL DATA VOLTS/ PHASE	WEIGHT LBS	BASIS OF DESIGN	NOTES
UH 1-1	ANNEX 118	1100	45.0	HORIZONTAL	180	150	3.0	0.5	1/12	1000	115/1	45	VULCAN HV-84	1
UH 1-2	ANNEX 116	1100	45.0	HORIZONTAL	180	150	3.0	0.5	1/12	1000	115/1	45	VULCAN HV-84	1
UH 1-3	ANNEX 115	1100	45.0	HORIZONTAL	180	150	3.0	0.5	1/12	1000	115/1	45	VULCAN HV-84	1
UH 1-4	ANNEX 116	1100	45.0	HORIZONTAL	180	150	3.0	0.5	1/12	1000	115/1	45	VULCAN HV-84	1
UH 1-5	ANNEX 117	900	30.0	HORIZONTAL	180	150	2.0	0.5	1/20	1000	115/1	40	VULCAN HV-60	1
UH 1-6	FIRE PUMP M107	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-7	VEST 101	1400	49.0	HORIZONTAL	180	150	3.3	0.5	1/20	1000	115/1	30	VULCAN HV-96	1
UH 1-8	INSPECTION 112	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-9	STORAGE 109	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-10	MECH 115	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-11	STAIR 003	1200	37.0	VERTICAL	180	150	2.5	0.5	1/8	850	115/1	60	VULCAN VV-104	1
UH 1-12	BREAK M101	900	30.0	HORIZONTAL	180	150	2.0	0.5	1/20	1000	115/1	45	VULCAN HV-60	1
UH 1-13	BREAK M101	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-14	CORRIDOR 103	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-15	CORRIDOR 103	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-16	MECH MEZZ M109	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-17	MECH MEZZ M107	900	30.0	HORIZONTAL	180	150	2.0	0.5	1/20	1000	115/1	40	VULCAN HV-60	1
UH 1-18	LOCKERS 105	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 1-19	ANNEX 117	900	30.0	HORIZONTAL	180	150	2.0	0.5	1/20	1000	115/1	40	VULCAN HV-60	1
UH 2-1	CORRIDOR 201	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 2-2	UPHOLSTERY 206	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 2-3	UPHOLSTERY 206	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 2-4	SHOP 207	1100	40.0	HORIZONTAL	180	150	2.8	0.5	1/20	1000	115/1	45	VULCAN HV-72	1
UH 2-5	SHOP 207	1100	40.0	HORIZONTAL	180	150	2.8	0.5	1/20	1000	115/1	45	VULCAN HV-72	1
UH 2-6	SHOP 207	1100	40.0	HORIZONTAL	180	150	2.8	0.5	1/20	1000	115/1	45	VULCAN HV-72	1
UH 2-7	SHOP 207	1100	40.0	HORIZONTAL	180	150	2.8	0.5	1/20	1000	115/1	45	VULCAN HV-72	1
UH 2-8	SHOP 207	1100	40.0	HORIZONTAL	180	150	2.8	0.5	1/20	1000	115/1	45	VULCAN HV-72	1
UH 2-9	SHOP 207	1100	40.0	HORIZONTAL	180	150	2.8	0.5	1/20	1000	115/1	45	VULCAN HV-72	1
UH 2-10	MEZZANINE M201	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 2-11	MEZZANINE M201	550	20.0	HORIZONTAL	180	150	1.4	0.5	25W	1550	115/1	30	VULCAN HV-36	1
UH 3-1	ATTIC	900	33.0	HORIZONTAL	180	150	2.2	0.5	1/20	1000	115/1	40	VULCAN HV-60	1
UH 3-2	ATTIC	900	33.0	HORIZONTAL	180	150	2.2	0.5	1/20	1000	115/1	40	VULCAN HV-60	1
UH 3-3	ATTIC	900	33.0	HORIZONTAL	180	150	2.2	0.5	1/20	1000	115/1	40	VULCAN HV-60	1
UH 3-4	ATTIC	900	33.0	HORIZONTAL	180	150	2.2	0.5	1/20	1000	115/1	40	VULCAN HV-60	1

NOTES:
1. PROVIDE WITH SPEED CONTROLLER AND DISCONNECT SWITCH.

GRAVITY VENTILATOR SCHEDULE

UNIT NO	SERVES	CFM	MAX THROAT VELOCITY (FPM)	MAX PRESSURE DROP IN WC	THROAT SIZE (IN)	CURB CAP SIZE (IN)	BASIS OF DESIGN	NOTES
GIV-1	DOAS-1 OA	3770	630	0.06	24W x 36L	30W x 42L	GREENHECK FGI	1,2,3
GEV-1	DOAS-1 EA	3260	550	0.05	24W x 36L	30W x 42L	GREENHECK FGR	1,2,3

NOTES:
1. PROVIDE WITH 0.5 IN GALVANIZED BIRD SCREEN.
2. PROVIDE ALUMINUM CONSTRUCTION.
3. PROVIDE WITH 18 IN HIGH INSULATED ROOF CURB.

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

STATE OF MAINE
RICHARD CRESWELL
No. 13178
10-01-25
LICENSED
PROFESSIONAL ENGINEER
SEAL

NAVFAC

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
MECHANICAL SCHEDULES 2

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 218 OF 282
M-602

DRAWFORM REVISION: 14 SEP 2023

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UNCLASSIFIED

DIFFUSER / GRILLE SCHEDULE

UNIT NO	FACE SIZE IN	NECK SIZE IN	MAX PRESSURE DROP IN WC	MAX NOISE CRITERIA	CFM RANGE	TYPE	BASIS OF DESIGN	NOTES
S-1	10x4	-	0.05	20	0-100	SPIRAL DUCT SUPPLY GRILLE	PRICE SDGE	5,6,9,10,11,14
S-2	14x5	-	0.05	20	101-190	SPIRAL DUCT SUPPLY GRILLE	PRICE SDGE	5,6,9,10,11,14
S-3	16x6	-	0.05	20	191-250	SPIRAL DUCT SUPPLY GRILLE	PRICE SDGE	5,6,9,10,11,14
S-4	18x8	-	0.05	20	251-450	SPIRAL DUCT SUPPLY GRILLE	PRICE SDGE	5,6,9,10,11,14
S-5	28x8	-	0.05	20	451-600	SPIRAL DUCT SUPPLY GRILLE	PRICE SDGE	5,6,9,10,11,14
S-6	28x12	-	0.05	20	601-900	SPIRAL DUCT SUPPLY GRILLE	PRICE SDGE	5,6,9,10,11,14
S-7	6x6	-	0.05	20	0-100	LOUVERED FACE SUPPLY GRILLE	PRICE 520	2,3,4,5,7,8,10,11
S-8	16x8	-	0.05	20	350	LOUVERED FACE SUPPLY GRILLE	PRICE 520	2,3,4,5,7,8,10,11
S-9	18x8	-	0.05	20	300-350	LOUVERED FACE SUPPLY GRILLE	PRICE 520	2,3,4,5,7,8,10,11
S-10	12x6	-	0.10	20	150-200	LOUVERED FACE SUPPLY GRILLE	PRICE 520	2,3,4,5,7,8,10,11
S-11	24x6	-	0.10	20	275-400	LOUVERED FACE DIRECTIONAL DIFFUSER	PRICE SMD	1,2,3,4,11,18
S-12	24x9	-	0.10	20	401-600	LOUVERED FACE DIRECTIONAL DIFFUSER	PRICE SMD	1,2,3,4,11,18
S-13	8ø	6ø	0.05	20	0-100	ROUND GRILLE	PRICE RSG	6,10,11,16
S-14	18	8ø	0.05	20	101-200	ROUND PLAQUE DIFFUSER	PRICE RPD	2,3,11
S-15	22.5	10ø	0.05	20	201-350	ROUND PLAQUE DIFFUSER	PRICE RPD	2,3,11
R-1	16x8	-	0.10	20	350	LOUVERED FACE RETURN GRILLE	PRICE 530	2,3,7,8,11
R-2	24x12	-	0.10	20	600-620	LOUVERED FACE RETURN GRILLE	PRICE 530	1,2,3,4,5,7,8,11,12,19
R-3	24x46	-	0.10	20	3000-3400	HEAVY DUTY RETURN GRILLE	PRICE 96	2,3,4,7
E-1	10x4	-	0.10	20	0-100	SPIRAL DUCT EXHAUST GRILLE	PRICE SDGER	5,6,11,14,15
E-2	14x6	-	0.10	20	101-200	SPIRAL DUCT EXHAUST GRILLE	PRICE SDGER	5,6,11,14,15
E-3	12x8	-	0.10	20	201-300	SPIRAL DUCT EXHAUST GRILLE	PRICE SDGER	5,6,11,14,15
E-4	6x6	-	0.10	20	0-90	LOUVERED FACE EXHAUST GRILLE	PRICE 530	2,3,4,5,7,11
E-5	20x28	-	0.05	20	980	LOUVERED FACE EXHAUST GRILLE	PRICE 530	2,3,4,5,7,11,13
E-6	6ø	6ø	0.05	20	0-100	ROUND EGGCRATE GRILLE	PRICE RECG	6,11,12,17
E-7	8ø	8ø	0.05	20	101-200	ROUND EGGCRATE GRILLE	PRICE RECG	6,11,12,17
T-1	12x10	-	0.05	20	225	LOUVERED FACE TRANSFER GRILLE	PRICE 530	2,3,4,7,8,11
T-2	12x12	-	0.05	20	200-270	LOUVERED FACE TRANSFER GRILLE	PRICE 530	2,3,4,7,11
T-3	16x8	-	0.05	20	225	LOUVERED FACE TRANSFER GRILLE	PRICE 530	2,3,4,7,8,11

NOTES:
1. FOR INSTALLATION IN T-BAR CEILING.
2. STEEL CONSTRUCTION.
3. PROVIDE TRANSITION FROM DUCTWORK TO DIFFUSER / GRILLE.
4. SURFACE MOUNTED.
5. PROVIDE WITH INTEGRAL OPPOSED BLADE DAMPER.
6. ALUMINUM CONSTRUCTION.
7. 3/4" BLADE SPACING, 45-DEG DEFLECTION.
8. BLADES PARALLEL TO LONG DIMENSION.
9. 3/4" BLADE SPACING, ADJUSTABLE DEFLECTION.
10. DOUBLE DEFLECTION.
11. PROVIDE COLOR AND FINISH AS SELECTED BY CONTRACTING OFFICER.
12. DUCT MOUNTED.
13. BLADE PARALLEL TO SHORT DIMENSION.
14. DIRECT DUCT MOUNT AND FRAMES MATCH DUCT CURVATURE.
15. PERFORATED SCREEN FACE.
16. 1" BLADE SPACING, ADJUSTABLE BLADES.
17. 1/2" x 1/2" EGG CRATE.
18. FOUR-DIRECTION DISTRIBUTION.

HOT WATER CABINET UNIT HEATER SCHEDULE

UNIT NO	LOCATION	NOMINAL LENGTH	3-SPEED CFM	HEATING MBH	WATER TEMP °F		GPM	MAX PRESSURE DROP FT WC	CFM	FAN DATA				WEIGHT (LBS)	BASIS OF DESIGN	NOTES
					ENTERING	LEAVING				HP	RPM	A	VOLTS/ PHASE			
CUH 1	STAIR 002	47"	280/375/430	23	180	150	1.5	0.5	430	1/15	1050	0.65	115/1	130	VULCAN FI-1050-04	1,3,4
CUH 2	STAIR 004	47"	280/375/430	35	180	150	2.3	0.5	430	1/15	1050	0.65	115/1	130	VULCAN RWI 1130-04	2,3,5
CUH 3	WOMEN 203	43"	150/240/335	15	180	150	1.0	0.5	335	1/15	1050	0.45	115/1	115	VULCAN RWI 1130-03	2,3,4
CUH 4	CORRIDOR 212	43"	150/240/335	15	180	150	1.0	0.5	335	1/15	1050	0.45	115/1	110	VULCAN FI 1050-03	1,3,4

NOTES:
1. FLOOR MOUNTED INVERTED FLOW CABINET WITH UPPER-FRONT GRILLE INLET AND LOWER-FRONT GRILLE DISCHARGE.
2. RECESSED WALL INVERTED FLOW CABINET WITH UPPER-FRONT GRILLE INLET AND LOWER-FRONT GRILLE DISCHARGE.
3. PROVIDE WITH ECM MOTOR, CONTROL BOARD, AND SERVICE DISCONNECT SWITCH.
4. 1-ROW COIL.
5. 2-ROW COIL.

HOT WATER CONVECTOR SCHEDULE

UNIT NO	LOCATION	ENCLOSURE				CAPACITY BTUH	GPM	PD FT WC	AVERAGE WATER TEMP (°F)	BASIS OF DESIGN	NOTES
		NOMINAL LENGTH	NOMINAL HEIGHT	DEPTH	MOUNTING HEIGHT						
C 1-1	WOMEN 106	28"	32"	4"	5"	3.0	0.5	0.15	170	VULCAN W-A	1,4
C 1-2	MEN 105	28"	32"	4"	5"	3.0	0.5	0.15	170	VULCAN W-A	1,4
C 1-3	CORRIDOR M103	36"	32"	6"	5"	4.0	0.5	0.15	170	VULCAN PWG-A	2,4
C 2-1	MEN 204	28"	32"	6"	5"	3.0	0.5	0.15	170	VULCAN PWG-A	2,4
C 2-2	EXPRESS 205	28"	32"	4"	5"	3.0	0.5	0.15	170	VULCAN PWG-A	2,4
C 2-3	STAIR 005	60"	32"	8"	5"	11.0	0.8	0.15	165	VULCAN FS-A	3,4

NOTES:
1. WALL MOUNTED EXPOSED.
2. PARTIALLY RECESSED UNIT WITH INTEGRAL AIR OUTLET AND INLET GRILLES.
3. FREE STANDING CABINET ENCLOSURE WITH INTEGRAL AIR OUTLET GRILLE AND ARCHED INLET.
4. PROVIDE WITH END POCKET.

AIR SEPARATOR SCHEDULE

UNIT NO	LOCATION	SERVES	TYPE	CAPACITY (GPM)	PRESSURE DROP (FT)	PIPE SIZE (IN)	STRAINER FREE AREA (SQ IN)	WEIGHT (LBS)	BASIS OF DESIGN	NOTES
AS-1	MECHANICAL 115	HEATING WATER	TANGENTIAL	190	2.9	3"	51	188	BELL & GOSSETT MODEL R-3F	1

NOTES: 1. PROVIDE WITH BRACKET SUPPORTS AND INTEGRAL STRAINER.

EXPANSION TANK SCHEDULE

UNIT NO	LOCATION	SERVES	MINIMUM ACCEPTANCE VOLUME (GAL)	MINIMUM TANK VOLUME (GAL)	BASIS OF DESIGN	NOTES
EXT-1	MECHAINCAL 115	HEATING WATER SYSTEM	12.7	24	BELL & GOSSETT B10D	1

NOTES:
1. PROVIDE ASME 125 PSIG RATED BLADDER TYPE TANK.

ELECTRIC UNIT HEATER SCHEDULE

UNIT NO	SERVES	CFM	HEATING MBH	INPUT KW	VOLTS/ PHASE	MOTOR HP (W)	SCCR	THROW (FT)	MAX HEIGHT (FT)	WEIGHT (LB)	BASIS OF DESIGN	NOTES
EUH 1-1	ELECT 113	380	10.2	3.0	208/1	16	5000	12	8	34	MODINE HER	1

NOTES: 1. PROVIDE WITH DISCONNECT SWITCH.

ELECTRICAL BASEBOARD HEATER SCHEDULE

UNIT NO	SERVES	LENGTH	HEATING MBH	INPUT KW	VOLTS/ PHASE	BASIS OF DESIGN	NOTES
EB 2-1	NETWORK 113	72"	5.1	1.5	208/1	KING - MODEL 6K2015	1,2,3
EB 2-2	NETWORK 214	96"	6.8	2.0	208/1	KING - MODEL 6K2020	1,2,3

NOTES:
1. PROVIDE EQUIPMENT MANUFACTURER SUPPLIED END CAPS FOR EACH HEATER ENCLOSURE.
2. PROVIDE DISCONNECT SWITCH.
3. PROVIDE FINISH COLOR SELECTED BY CONTRACTING OFFICER FROM MANUFACTURER'S STANDARD AND CUSTOM COLORS.

DESIGN CONDITIONS SCHEDULE

LOCATION	INDOOR COOLING DESIGN TEMP °F	OUTDOOR COOLING DESIGN TEMP °F	INDOOR HEATING DESIGN TEMP °F	OUTDOOR HEATING DESIGN TEMP °F	NOTES
OFFICES	76	89	68	0	-
OPEN OFFICES	76	89	68	0	-
TRAINING	76	89	68	0	-
LOCKER ROOMS	76	89	68	0	-
MECH	N/A	89	68	0	-
RESTROOMS	N/A	89	68	0	-
SHOPS	76	89	68	0	-

NOTES: 1. -

DUCT MOUNTED HOT WATER HEATING COIL SCHEDULE

UNIT NO	SERVES	AIR SIDE				DUCT SIZE		WATER SIDE					NOTES
		CFM	MAX APD IN WC	EAT °F	LAT °F	WIDTH IN	HEIGHT IN	GPM	MAX WPD FT HD	EWT °F	LWT °F	INSTALLED WT (LB.)	
HC-1	AHU-1	10,800	0.2	55	87	72	36	25.0	2.0	180	150	112	1,2,3,4

NOTES:
1. FOULING FACTOR: 0,00025 HR - SQ FT - DEG F/BTU
2. PROVIDE COASTAL CORROSION TYPE EPOXY E-COAT COIL COATING
3. GALVANIZED CASING, NO TURBULATORS
4. BASIS OF DESIGN: TRANE 1-ROW, HOT WATER 5/8" SHIPPING COIL, MODEL D5WB36072 GOAAIIIEABA

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

RICHARD CRESWELL No. 13178 10-01-25 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE
SEAL

NAVFAC

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL SCHEDULES 3

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 219 OF 282
M-603

DRAWFORM REVISION: 14 SEP 2023

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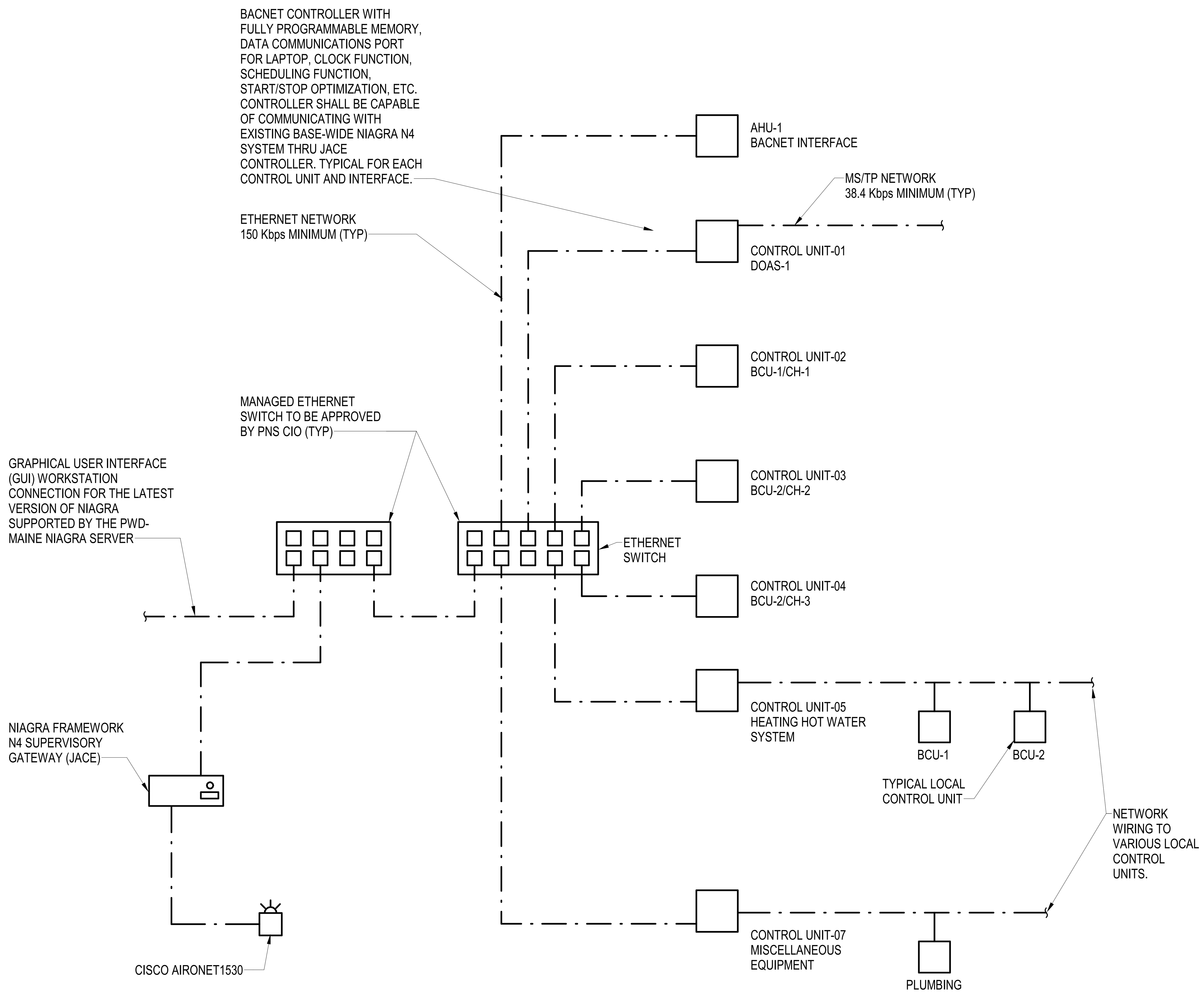
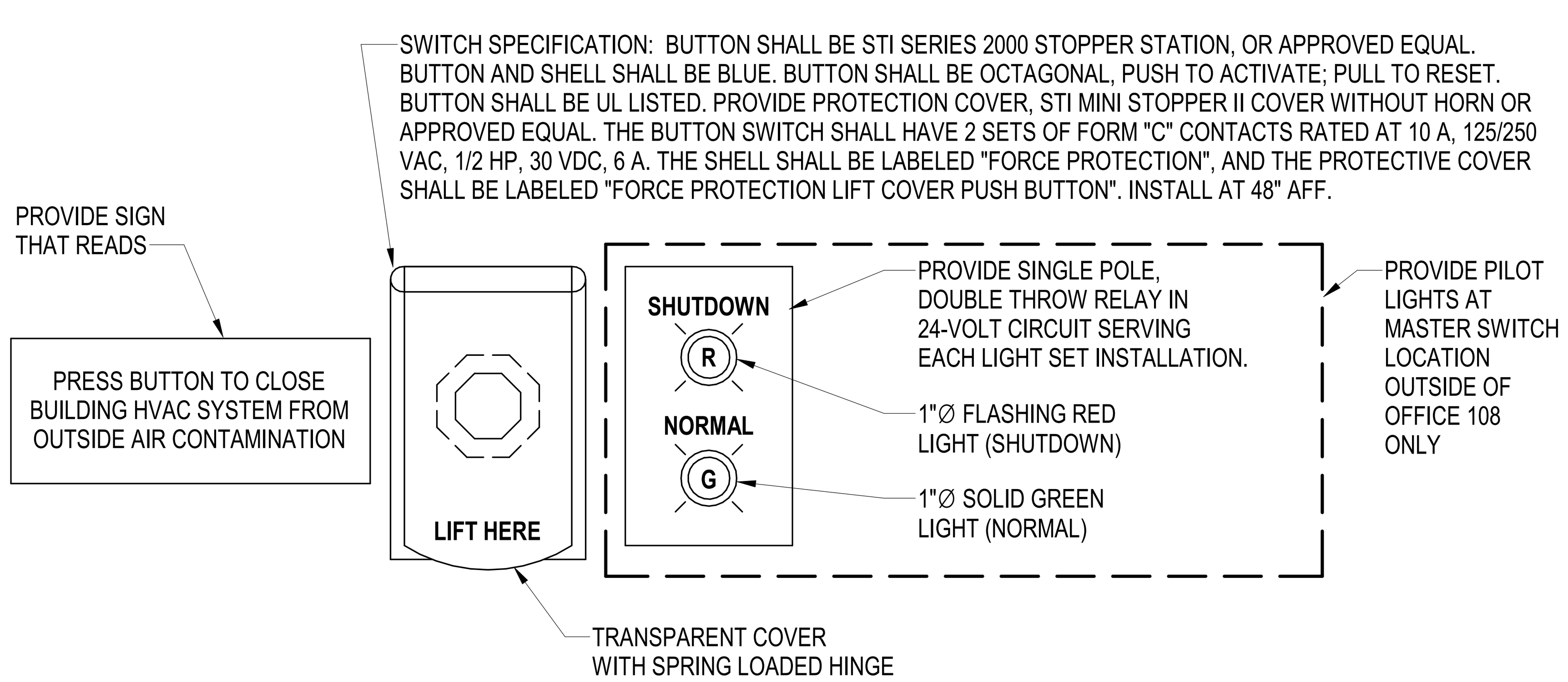
CONTROL SYSTEM GENERAL NOTES

1. PROVIDE A COMPLETE OPERATIONAL DIRECT DIGITAL CONTROL (DDC) SYSTEM IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS AND DRAWINGS. THE SYSTEM SHALL CONSIST OF NATIVE BACNET DEVICES AT ALL LEVELS COMMUNICATING VIA ASHRAE 135 BACNET PROTOCOL OVER APPROPRIATE BACNET LISTED LANS. WHEREVER POSSIBLE, LIMIT THE USE OF SENSORS AND DEVICES WHICH COMMUNICATE IN IP PROTOCOL OVER IP LANS. WHEN THIS IS NOT POSSIBLE, PLACE ALL IP LAN CABLE IN METAL CONDUIT WITH METALLIC LOCKABLE ENCLOSURES AT BOTH ENDS. THE DDC SYSTEM SHALL CONNECT TO A NIAGARA SUPERVISORY GATEWAY (COMMONLY REFERRED TO AS A JACE). PROVIDE A BUILDING WORKSTATION LAPTOP COMPUTER WITH NIAGRA FRAMEWORK SOFTWARE VERSION 4.2 OR LATER THE BUILDING WORKSTATION SHALL INCLUDE NIAGARA WORKBENCH OR EQUIVALENT NIAGARA FRAMEWORK ENGINEERING TOOL SOFTWARE. THE CONTRACTOR SHALL CREATE SYSTEM GRAPHICS, DATA TRENDING, DATA LOGGING, ALARMING AND TROUBLESHOOTING FOR THE DDC SYSTEM. THE CONTRACTOR SHALL DISCOVER THE DDC SYSTEM GRAPHICS VIA THE JOIN PROCESS IN THE BUILDING 374 BASE-WIDE EMCS. THE FACILITY POINT OF CONNECTION SHALL BE A REPLACEMENT CISCO AIRONET 1530 WIRELESS ACCESS POINT. PROVIDE LICENSE FOR NIAGRA JACE TO SUPPORT THE NUMBER OF NETWORK DEVICES REQUIRED FOR THIS PROJECT. LICENCE FOR JACE AND LAPTOP IS TO INCLUDE 5 YEARS OF SOFTWARE, UPDATES AND WARRANTY.
2. ALARMS SHALL BE ANNUNCIATED ON THE GRAPHICAL USER INTERFACE (GUI).
3. SETTINGS, MODES AND SET POINTS, THAT ARE INDICATED AS BEING ADJUSTABLE, SHALL BE ADJUSTABLE BY THE SYSTEM OPERATORS THROUGH POINT-AND-CLICK ICONS ON THE GUI WITHOUT THE NEED TO CHANGE OR EDIT PROGRAMMING.
4. ANALOG DATA SHALL BE TRENDED AT REGULAR INTERVALS, DETERMINED BY THE EXPECTED RATE OF CHANGE OF THE DATA, AND SHALL BE ARCHIVED AND STORED ON THE GUI COMPUTER. TREND DATA SHALL BE CONFIGURED AND STORED TO SATISFY NAVY REQUIREMENTS FOR FUTURE MEASUREMENT AND VERIFICATION.
5. BINARY DATA SHALL BE TRENDED ON A CHANGE OF STATE BASIS AND SHALL BE ARCHIVED AND STORED ON THE GUI COMPUTER. TREND DATA SHALL BE CONFIGURED AND STORED TO SATISFY NAVY REQUIREMENTS FOR FUTURE MEASUREMENT AND VERIFICATION.
6. PROVIDE LOCAL CONTROL UNITS AS REQUIRED TO CONTROL BLOWER COIL UNITS OR OTHER TERMINAL EQUIPMENT .
7. CONTROL CONTRACTOR SHALL PROVIDE DDC/UMCS SYSTEM WITH TWO UNIQUE SETS OF ACCESS PASSWORDS. ONE SET OF PASSWORDS, PRIMARY USER, SHALL BE FOR NAVFAC PWD-ME SHOP PERSONNEL WITH COMPLETE ACCESS TO ALL POINTS, GRAPHICS AND PROGRAMMING. THE SECOND SET OF PASSWORDS, SECONDARY USER, SHALL BE FOR LIMITED ACCESS BY CODE 980 FOR SUPPORT AND MAINTENANCE OF SPECIFIC SYSTEMS AND/OR EQUIPMENT. LIST OF SECONDARY USER POINTS AND ACCESS SHALL BE SUBMITTED TO CONTRACTING OFFICER FOR APPROVAL.
8. COMPLY WITH CYBERSECURITY REQUIREMENTS AND PERFORM REQUIRED PROGRAMMING AND TESTING PER SPECIFICATIONS.
9. ALL NIAGRA INTEGRATION SHALL FOLLOW THE LATEST VERSION OF THE NAVFAC PWD-ME ICS STANDARDS.
10. BAS CONTRACTOR SHALL PROVIDE A COMPLETE CONTROL POINT MATRIX INDICATING THE FUNCTION OF THE INDIVIDUAL COMPONENTS OF THE BUILDING AUTOMATON. THE BAS CONTRACTOR SHALL SUBMIT A DETAILED NARRATIVE INDICATING THE SPECIFIC FUNCTION OF EACH COMPONENT AND DEVICE. THE NARRATIVE SHALL NOT BE A REPLICATION OR DUPLICATION OF THE ENGINEER'S SEQUENCE OF OPERATION.
11. THE BAS SUBCONTRACTOR SHALL PROVIDE UTILITY MONITORING OR ELECTRICAL USAGE AND DEMAND FOR THE CHILLED WATER SYSTEM (IE:CHILLERS, PUMPS, ETC.).
12. THE BAS SUBCONTRACTOR SHALL ASSIST THE BALANCING CONTRACTOR WITH THE BALANCING AS WELL THE COMMISSIONING AGENT IN SYSTEM VERIFICATION.
13. PRIOR TO FINAL COMPLETION, THE BAS CONTRACTOR SHALL DEMONSTRATE TO THE COORDINATE OFFICE AND ENGINEER THAT ALL SYSTEMS ARE FUNCTIONING AS DESIGNED AND MEETS THE INTENT OF THE DRAWINGS AND SPECIFICATIONS.



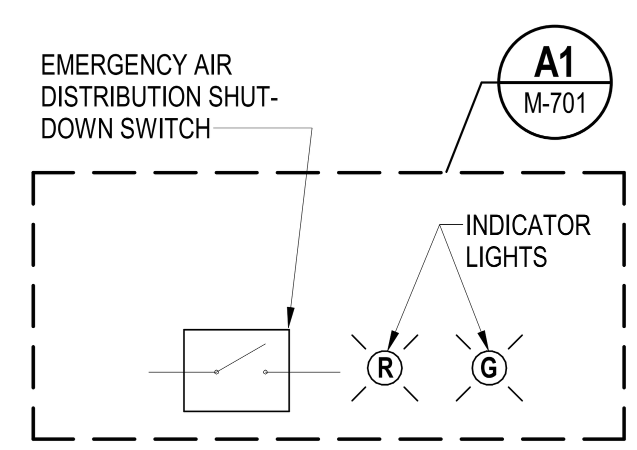
B1 DIRECT DIGITAL CONTROL SYSTEM ARCHITECTURE

SCALE: NOT TO SCALE



A1 EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH (EAS) DETAIL

SCALE: NOT TO SCALE



SEQUENCE OF OPERATION

IN THE EVENT THAT THE EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH IS ACTIVATED, THE FANS IN AIR HANDLING EQUIPMENT SHALL STOP AND THE OUTSIDE AND EXHAUST AIR DAMPERS ASSOCIATED WITH THESE UNITS SHALL CLOSE. THE FANS SHALL REMAIN STOPPED AND THE DAMPERS SHALL REMAIN CLOSED UNTIL THE SWITCH IS RELEASED. REFER TO EQUIPMENT SPECIFIC CONTROL DIAGRAMS FOR FORCE PROTECTION SHUTDOWN PROTOCOLS.

EADS POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
EM. AIR DISTRIBUTION SHUTDOWN SWITCH	X			X		X			X		1,2,4
SHUTDOWN LIGHT (RED)	X				X				X		3
NORMAL LIGHT (GREEN)	X				X				X		3

NOTES:
1. GENERATE AN ALARM IF THE EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH IS CLOSED.
2. TYPICAL OF 3. REFER TO SHEETS MP102, MP104, AND MP105 FOR LOCATIONS.
3. PROVIDE AT SINGLE LOCATION ONLY. REFER TO SHEET MP102.
4. PROVIDE "VIRTUAL" EADS SWITCH ON GUI FOR REMOTE OPERATION AND TESTING THROUGH DDC SYSTEM.

A3 EMERGENCY AIR DISTRIBUTION CONTROL DIAGRAM

SCALE: NOT TO SCALE

FINAL SUBMISSION 10/01/2025 RSH
SYM DESCRIPTION DATE APPR

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL CONTROL DIAGRAMS 1

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SEQUENCE OF OPERATION

OCCUPIED / UNOCCUPIED MODES:
THE OCCUPIED AND UNOCCUPIED MODES SHALL BE DETERMINED BY A USER- ADJUSTABLE 7 DAY / 24-HOUR SCHEDULE WHICH SHALL BE ADJUSTABLE BY THE BUILDING OPERATOR THROUGH THE GUI. SEPARATE SCHEDULES SHALL BE PROVIDED FOR HOLIDAYS, EVENTS AND EVERYDAY USE.

DEHUMIDIFICATION MODE:
DEHUMIDIFICATION MODE SHALL BE ENABLED MAJORITY OF THE SPACE RH SENSORS INDICATE RH LEVELS ABOVE THE RH MAXIMUM SET POINT, 65%RH (ADJUSTABLE). DEHUMIDIFICATION MODE SHALL BE DISABLED WHEN ALL THE SPACE RH LEVELS ARE 5%RH (ADJUSTABLE) BELOW THE MAXIMUM RH SET POINT.

OCCUPIED:
DURING OCCUPIED MODE THE SUPPLY AIR FAN SHALL RUN CONTINUOUSLY AND THE OUTSIDE AIRFLOW SHALL NOT FALL BELOW THE MINIMUM OUTSIDE AIRFLOW SET POINT.

THE SPACE COOLING SET POINT SHALL BE 75°F (ADJUSTABLE) AND THE SPACE HEATING SET POINT SHALL BE 68°F (ADJUSTABLE).

FOR COOLING THE AVERAGE SPACE TEMPERATURE SHALL BE DETERMINED BY THE AVERAGE OF THREE WALL-MOUNTED SPACE TEMPERATURE SENSORS – TWO IN SHOP 111 AND ONE IN INSPECTION ROOM 112. FOR HEATING THE AVERAGE SPACE TEMPERATURE SHALL BE DETERMINED BY THE AVERAGE OF THE TWO SENSORS IN SHOP 111 (THE INSPECTION ROOM HAS ITS OWN HEAT).

OCCUPIED HEATING:
HEATING MODE SHALL BE ENABLED WHEN THE HEATING AVERAGE SPACE TEMPERATURE FALLS BELOW THE HEATING SET POINT. THE HEATING MODE SHALL REMAIN ENABLED UNTIL THE COOLING MODE IS ENABLED. DURING HEATING MODE THE OUTSIDE AIRFLOW SHALL REMAIN AT THE SCHEDULED MINIMUM, AND THE DX COOLING SHALL REMAIN OFF. THE SUPPLY AIR SET POINT SHALL BE RESET ACCORDING TO THE FOLLOWING USER ADJUSTABLE RESET SCHEDULE:

SPACE TEMPERATURE	SUPPLY AIR SET POINT
1°F < HEATING SET POINT	95°F
1°F > HEATING SET POINT	SPACE HEATING SET POINT

THE SUPPLY FAN SHALL RUN AT 50% (ADJUSTABLE) SPEED UNTIL THE SUPPLY AIR SETPOINT REACHES MAXIMUM. IF THE SUPPLY AIR SET POINT IS AT MAXIMUM FOR MORE THAN 15 MINUTES (ADJUSTABLE) AND THE AVERAGE SPACE HEATING TEMPERATURE IS STILL BELOW SET POINT THEN THE FAN SHALL SLOWLY SPEED UP AND MODULATE TO MAINTAIN THE HEATING SET POINT. THE FAN SHALL SLOWLY REDUCE SPEED AS THE TEMPERATURE APPROACHES SET POINT AND SHALL RETURN TO 50% SPEED WHEN THE SET POINT IS SATISFIED FOR 15-MINUTES.

OCCUPIED COOLING:
COOLING MODE SHALL BE ENABLED WHEN THE COOLING AVERAGE SPACE TEMPERATURE RISES ABOVE THE COOLING SET POINT. THE COOLING MODE SHALL REMAIN ENABLED UNTIL THE HEATING MODE IS ENABLED. DURING COOLING MODE THE HEATING COIL CONTROL VALVE SHALL REMAIN CLOSED.
ECONOMIZER COOLING SHALL BE ENABLED WHENEVER THE OUTSIDE AIR ENTHALPY IS AT LEAST 10% LOWER THAN THE RETURN AIR ENTHALPY AND THE OUTSIDE AIR TEMPERATURE IS BELOW 55°F (ADJUSTABLE). DURING ECONOMIZER COOLING THE DX COOLING SHALL REMAIN OFF, THE MIXING DAMPERS (OUTSIDE, EXHAUST, & RETURN) SHALL OPERATE IN UNISON AND SHALL MODULATE TO MAINTAIN THE SUPPLY AIR SET POINT, AND THE SUPPLY AIR TEMPERATURE SET POINT SHALL BE REST ACCORDING TO THE FOLLOWING USER ADJUSTABLE REST SCHEDULE:

SPACE TEMPERATURE	SUPPLY AIR SET POINT
2°F > COOLING SET POINT	55°F
1°F < COOLING SET POINT	65°F

THE SUPPLY FAN SHALL RUN AT 50% UNTIL THE SUPPLY AIR TEMPERATURE REACHES MINIMUM SET POINT. IF THE SUPPLY AIR IS AT MINIMUM SET POINT FOR 15-MINUTES (ADJUSTABLE) AND THE SPACE IS STILL GREATER THAN 2°F ABOVE SET POINT THEN THE FAN SHALL SLOWLY SPEED UP AND MODULATE TO MAINTAIN THE COOLING SET POINT. THE FAN SHALL SLOWLY REDUCE SPEED AS THE TEMPERATURE APPROACHES SET POINT AND SHALL RETURN TO 50% SPEED WHEN THE SET POINT IS SATISFIED FOR 15-MINUTES.

WHENEVER ECONOMIZER COOLING IS DISABLED THE MINIMUM OUTSIDE AIRFLOW SHALL BE MAINTAINED, THE HEATING COIL SHALL REMAIN OFF, AND THE DX COOLING SHALL CYCLE TO MAINTAIN THE SPACE COOLING SET POINT. DX COOLING SHALL BE ENABLED WHEN THE AVERAGE SPACE TEMPERATURE IS 1°F (ADJUSTABLE) ABOVE THE COOLING SET POINT AND DISABLED 1°F (ADJUSTABLE) BELOW THE COOLING SET POINT. THE SUPPLY FAN SHALL RUN AT 50% SPEED UNLESS SPACE CONDITIONS CANNOT BE MAINTAINED. IF DX COOLING HAS BEEN ENABLED FOR 20 MINUTES (ADJUSTABLE) AND THE SPACE IS STILL ABOVE SET POINT OR THE SPACE TEMPERATURE IS 2°F (ADJUSTABLE) ABOVE SET POINT THE FAN SHALL SLOWLY SPEED UP AND THEN MODULATE T MAINTAIN THE SPACE SET POINT. WHEN THE SPACE TEMPERATURE IS SATISFIED FOR 10-MINUTES THE FAN SHALL RETURN TO 50% SPEED.

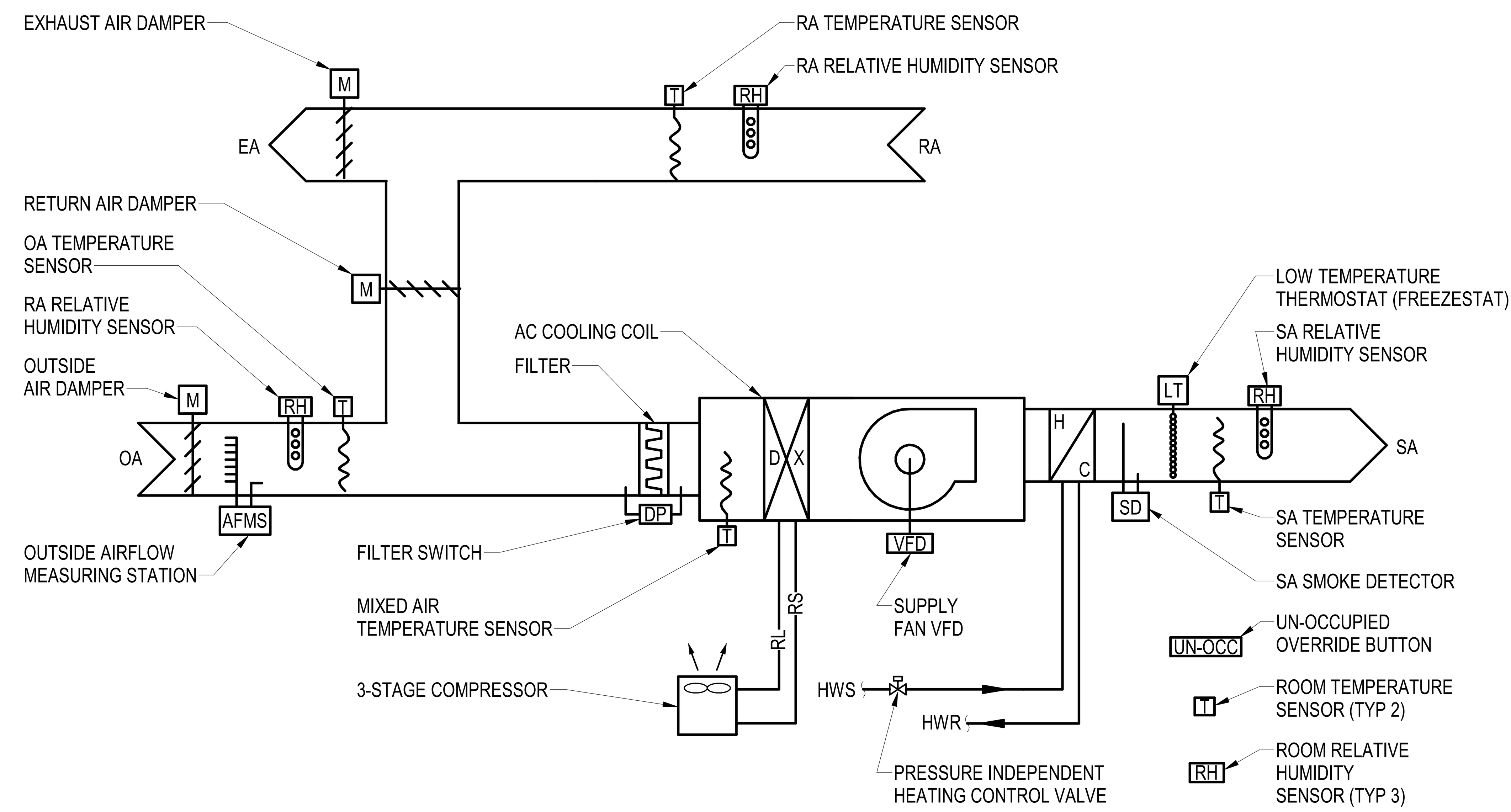
OCCUPIED DEHUMIDIFICATION:
IF DEHUMIDIFICATION IS NEEDED DURING OCCUPIED MODE THEN THE MINIMUM OUTSIDE AIRFLOW WILL BE MAINTAINED, THE FAN SHALL RUN AT FULL SPEED (ADJUSTABLE) THE DX COOLING SHALL BE ENABLED AND THE HEATING COIL CONTROL VALVE SHALL MODULATE TO MAINTAIN THE CURRENT SUPPLY AIR SET POINT.

UNOCCUPIED:
DURING UNOCCUPIED MODE OUTSIDE AND EXHAUST DAMPERS SHALL REMAIN CLOSED, THE RETURN AIR DAMPER SHALL REMAIN OPEN, AND THE SUPPLY FAN SHALL REMAIN OFF UNLESS THERE IS A CALL FOR NIGHT SETBACK HEATING. THE NIGHT SETBACK HEATING SET POINT SHALL BE 65°F (ADJUSTABLE). IF THE AVERAGE HEATING TEMPERATURE FALLS 1°F (ADJUSTABLE) BELOW THE NIGHT SETBACK SET POINT THEN THE SUPPLY FAN SHALL RUN AT FULL SPEED (ADJUSTABLE) AND THE HEATING COIL CONTROL VALVE SHALL REMAIN 100% OPEN UNTIL THE NIGHT SET BACK SET POINT IS SATISFIED.

IF DEHUMIDIFICATION IS NEEDED DURING UNOCCUPIED TIMES THE SUPPLY FAN SHALL RUN AT FULL SPEED (ADJUSTABLE), THE DX COOLING SHALL BE ENABLED, AND THE HEATING CONTROL VALVE SHALL MODULATE TO MAINTAIN THE NIGHT SET BACK SET POINT. OTHERWISE, THE FAN SHALL REMAIN OFF, THE DX COOLING SHALL REMAIN OFF, AND THE HEATING COIL VALVE SHALL REMAIN OFF.

SAFETIES:
IF THE FREEZESTAT INDICATES A LOW SUPPLY TEMPERATURE CONDITION THE SUPPLY FAN SHALL STOP, THE OUTSIDE AND EXHAUST DAMPERS SHALL CLOSE, THE DX COOLING SHALL STOP, AND THE HEATING COIL CONTROL VALVE SHALL BE FULLY OPEN.

IF THE SMOKE DETECTOR INDICATES SMOKE IN THE DUCT THE SUPPLY FAN SHALL STOP AN THE OUTSIDE AND EXHAUST AIR DAMPERS SHALL CLOSE.



AHU-1 POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
SUPPLY AIR TEMPERATURE	X	X				X	X		X		1
SUPPLY AIR RELATIVE HUMIDITY	X	X							X		
RETURN AIR TEMPERATURE	X	X							X		
RETURN AIR RELATIVE HUMIDITY	X	X							X		
RETURN AIR CARBON DIOXIDE	X	X				X			X		7
SUPPLY AIR SMOKE DETECTOR	X			X		X			X		3
FILTER SWITCH	X			X		X			X		5
MIXED AIR TEMPERATURE	X	X							X		
COOLING CONTROL SIGNAL	X		X		X				X		10
HEATING COIL CONTROL VALVE	X		X						X		
SUPPLY FAN VFD ENABLE	X				X				X		10
SUPPLY FAN VFD SIGNAL	X		X						X		10
SUPPLY FAN VFD ALARM	X			X		X			X		6,10
OUTSIDE AFMS	X	X							X		10
OUTSIDE AIRFLOW DAMPER	X		X						X		
ROOM TEMPERATURE	X	X							X		8
ROOM RELATIVE HUMIDITY SENSOR	X	X							X		9
SUPPLY DUCT STATIC PRESSURE	X	X							X		
FREEZESTAT	X			X					X		
OUTSIDE AIRFLOW	X	X							X		
UN-OCCUPIED OVERRIDE BUTTON	X			X					X		

NOTES:
1. GENERATE ALARM ON GUI IF TEMPERATURE IS NOT ±5°F OF SET POINT.
2. GENERATE ALARM ON GUI IF TEMPERATURE IS BELOW 15°F.
3. GENERATE ALARM ON GUI IF SMOKE DETECTOR INDICATES AN ALARM CONDITION.
4. GENERATE ALARM ON GUI IF FREEZESTAT INDICATES A LOW TEMPERATURE CONDITION.
5. GENERATE MAINTENANCE ALARM ON GUI IF FILTER PRESSURE DROP EXCEEDS 0.70 IN H2O.
6. GENERATE ALARM ON GUI IF VFD INDICATES AN ALARM CONDITION.
7. GENERATE ALARM ON GUI IF CARBON DIOXIDE LEVEL EXCEEDS 900 PPM.
8. TYPICAL OF TWO.
9. TYPICAL OF THREE.
10. CONNECTION VIA BACNET.

A1 **AHU-1 CONTROL DIAGRAM**
SCALE: NOT TO SCALE

FINAL SUBMISSION 10/01/2025 RSH
SYM DESCRIPTION DATE APPR

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY SRW
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
MECHANICAL CONTROL DIAGRAMS 2

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 221 OF 282
M-702
DRAWFORM REVISION: 14 SEP 2023

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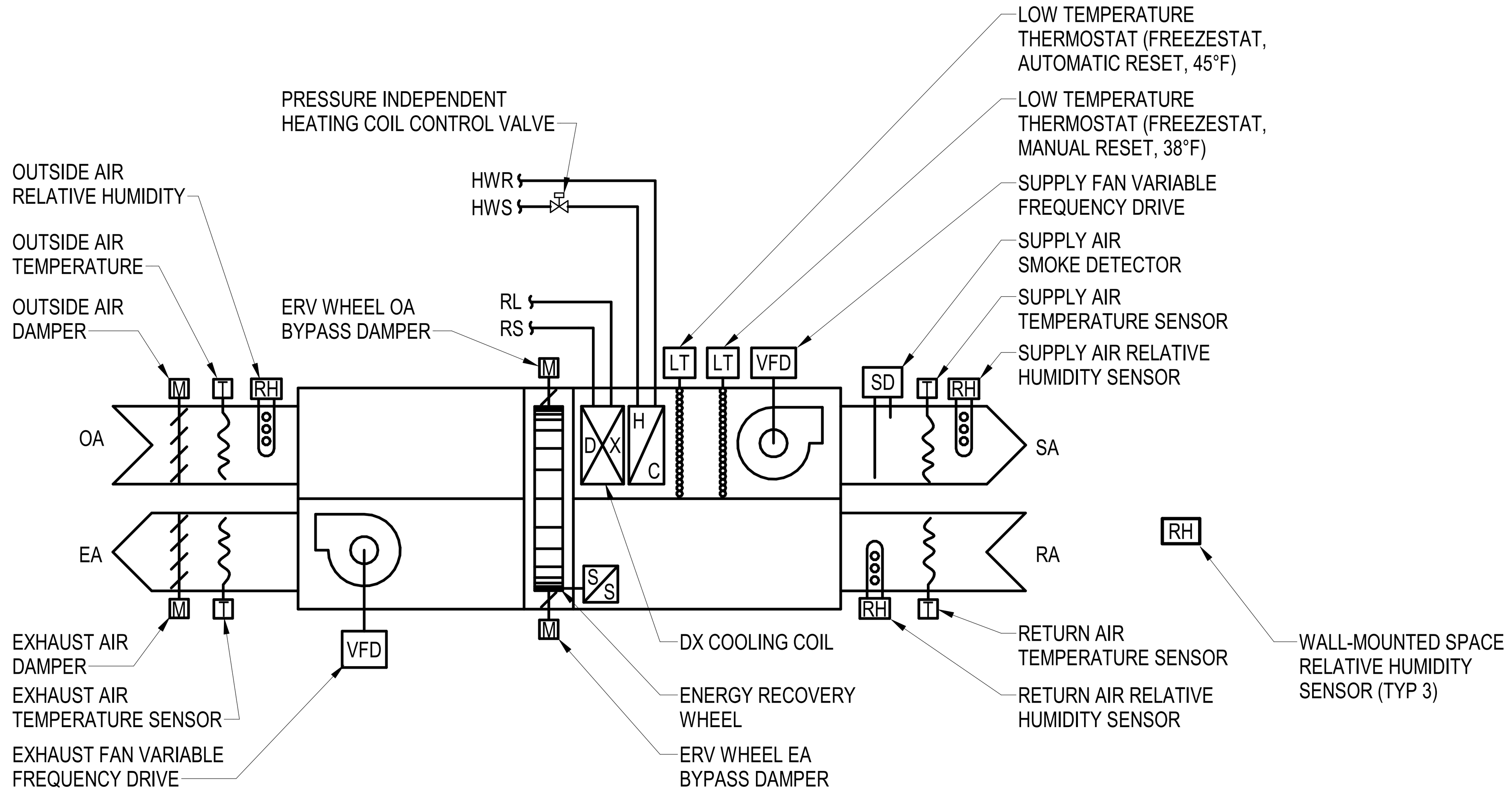


SEQUENCE OF OPERATION - ENERGY RECOVERY

THE DEDICATED OUTDOOR AIR SYSTEM (DOAS) SHALL OPERATE IN THE OCCUPIED MODE WHENEVER AN ASSOCIATED ZONE IS OPERATING IN THE OCCUPIED MODE OR THE UN-OCCUPIED OVERRIDE MODE; OTHERWISE THE DOAS SHALL BE IN THE UN-OCCUPIED MODE.

OCCUPIED MODE:
DURING THE OCCUPIED MODE THE EXHAUST FAN SHALL RUN CONTINUOUSLY AND THE OUTSIDE AND EXHAUST AIR DAMPERS SHALL BE OPEN. WHENEVER THE OUTSIDE AIR TEMPERATURE IS BELOW 35°F (ADJUSTABLE) THE SUPPLY FAN SHALL STOP FOR 5 MINUTES (ADJUSTABLE) ONCE EVERY HOUR, TO ALLOW THE DOAS TO DEFROST, OTHERWISE IT SHALL RUN CONTINUOUSLY. THE ENERGY RECOVERY WHEEL SHALL RUN CONTINUOUSLY.

THE HEATING COIL, ERV WHEEL BYPASS DAMPERS, AND VRF COOLING SHALL MODULATE, WITHOUT OVERLAP, TO MAINTAIN THE SUPPLY AIR TEMPERATURE SET POINT (65°F ADJUSTABLE).

OUTSIDE AIR BYPASS DAMPER AND EXHAUST AIR BYPASS DAMPER:
THE DAMPERS SHALL MODULATE IN UNISON TO MAINTAIN THE MIXED AIR TEMPERATURE. IN HEATING MODE, IF THE DAMPERS ARE MORE THAN 50% (ADJUSTABLE) OPEN THE WHEEL SHALL STOP.

DEHUMIDIFICATION MODE:
IF THE SPACE RELATIVE HUMIDITY SHALL BE DETERMINED BY THE AVERAGE OF THREE WALL-MOUNTED RELATIVE HUMIDITY SENSORS IN THE SPACE. WHEN THE AVERAGE RELATIVE HUMIDITY LEVEL RISES ABOVE THE RH MAXIMUM SET POINT, 60%RH ADJUSTABLE), DOAS-1 SHALL ENTER DEHUMIDIFICATION MODE (DURING OCCUPIED AND UNOCCUPIED TIMES). DURING DEHUMIDIFICATION MODE THE DX COIL SHALL PROVIDE FULL COOLING AND THE HEATING COIL SHALL MODULATE TO MAINTAIN THE SUP[PLY AIR SET POINT. DOAS-1 DEHUMIDIFICATION MODE SHALL END WHEN THE AVERAGE RH LEVEL FALLS 5%RH BELOW THE SET POINT.

UNOCCUPIED MODE:
DURING THE UN-OCCUPIED MODE THE EXHAUST AND SUPPLY FAN SHALL REMAIN OFF AND THE OUTSIDE AND EXHAUST AIR DAMPERS SHALL REMAIN CLOSED. THE HEATING COIL CONTROL VALVE SHALL MODULATE TO MAINTAIN THE SUPPLY AIR TEMPERATURE SENSOR AT 50° F (ADJUSTABLE).

IF THE UN-OCCUPIED OVERRIDE BUTTON ON A ROOM SENSOR IS PRESSED THE DOAS SHALL OPERATE AS LONG AS ANY ZONE IS OCCUPIED.

SAFETY - FREEZE STAT:
DURING A FREEZING CONDITION THE FANS SHALL REMAIN OFF, THE OUTSIDE AND THE EXHAUST AIR DAMPERS SHALL REMAIN CLOSED, THE HEATING COIL CONTROL VALVE SHALL BE OPEN AND AN ALARM SHALL BE GENERATED ON THE GRAPHICAL USER INTERFACE.

SAFETY - SMOKE DETECTOR/FIRE ALARM:
UPON ACTIVATION OF A SMOKE DETECTOR OR THE FIRE ALARM THE DOAS SHALL ENTER UNOCCUPIED MODE AND SHALL REMAIN IN UNOCCUPIED MODE UNTIL THE ALARM IS CLEARED. THE SMOKE DETECTOR SHALL BE HARD WIRED TO SHUT DOWN THE SUPPLY FAN AND CLOSE THE OUTSIDE AND EXHAUST AIR DAMPERS.

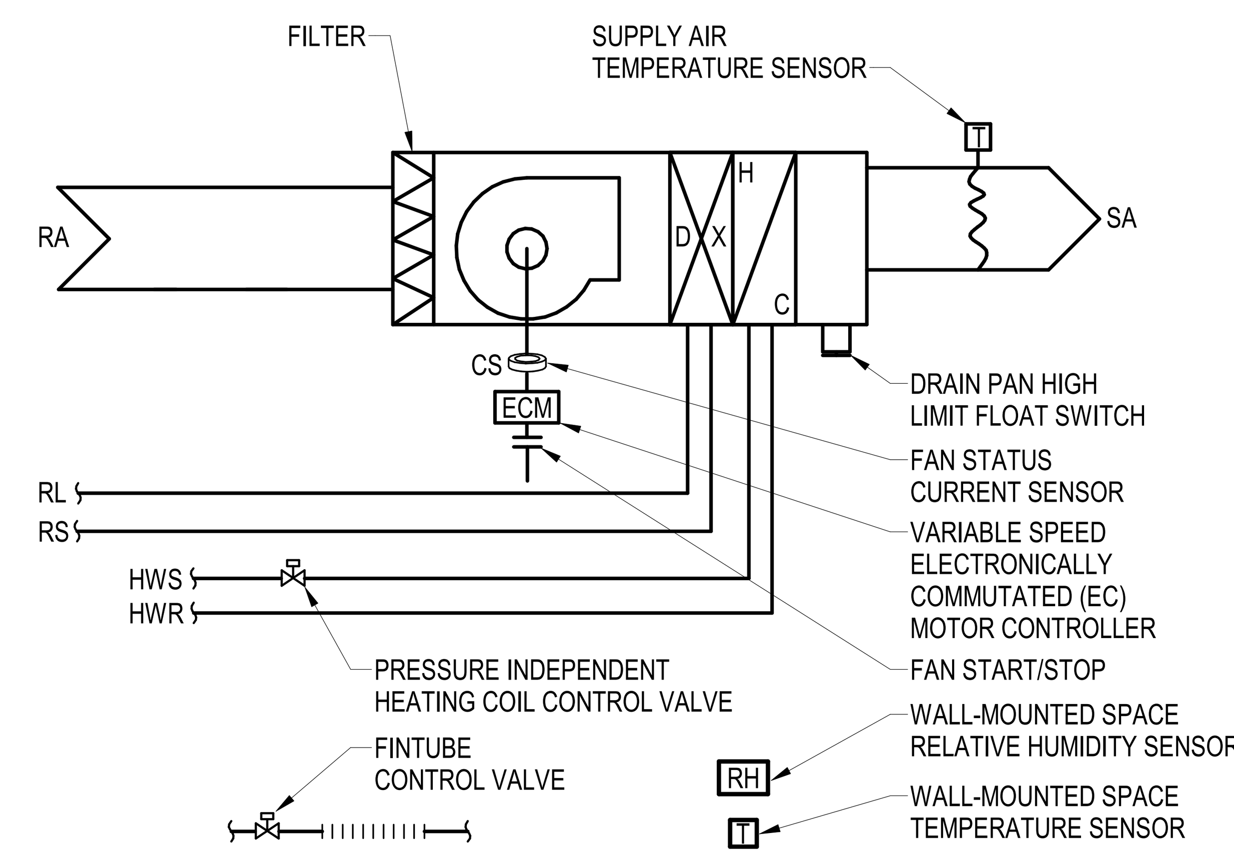
SAFETY - EAS SHUTDOWN:
UPON ACTIVATION OF THE EAS SHUTDOWN THE DOAS FANS SHALL STOP AND THE OA & EA DAMPERS SHALL REMAIN CLOSED.

DOAS-1 POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
SUPPLY FAN VFD START/STOP	X				X				X		
SUPPLY FAN SIGNAL	X		X						X		1
SUPPLY FAN VFD ALARM	X					X			X		1
EXHAUST FAN VFD START/STOP	X				X				X		
EXHAUST FAN SIGNAL	X		X						X		1
EXHAUST FAN VFD ALARM	X						X		X		1
SUPPLY AIR TEMPERATURE	X	X				X	X		X		2
RETURN AIR TEMPERATURE	X	X							X		
EXHAUST AIR TEMPERATURE	X	X				X	X		X		3
OUTSIDE AIR TEMPERATURE	X	X							X		
OUTSIDE AIR RELATIVE HUMIDITY	X	X							X		
OUTSIDE AIR DAMPER	X				X				X		
EXHAUST AIR DAMPER	X				X				X		
OUTSIDE AIR BYPASS DAMPER	X		X						X		
EXHAUST AIR BYPASS DAMPER	X		X						X		
HEATING COIL VALVE	X		X						X		
VRF CONTROL SIGNAL	X		X						X		
FREEZESTAT MANUAL RESET	X			X		X			X		4
FREEZESTAT AUTOMATIC RESET	X			X		X			X		4
ENERGY RECOVERY WHEEL START/STOP	X				X				X		
SUPPLY AIR SMOKE DETECTOR	X			X		X			X		5
ERV WHEEL OA BYPASS DAMPER	X		X						X		
ERV WHEEL EA BYPASS DAMPER	X		X						X		
SPACE RELATIVE HUMIDITY SENSOR	X	X							X		6
SUPPLY AIR RELATIVE HUMIDITY SENSOR	X	X							X		
RETURN AIR RELATIVE HUMIDITY SENSOR	X	X							X		

NOTES:
1. GENERATE ALARM IF FAN FAILS TO SHOW PROOF OF AIRFLOW.
2. GENERATE ALARM IF SUPPLY AIR TEMPERATURE FALLS BELOW 32°F (ADJUSTABLE).
3. GENERATE ALARM IF EXHAUST AIR TEMPERATURE FALLS BELOW 15°F (ADJUSTABLE).
4. GENERATE ALARM IF FREEZESTAT INDICATES A LOW TEMPERATURE CONDITION.
5. GENERATE ALARM IF SMOKE DETECTOR INDICATES AN ALARM CONDITION.
6. TYPICAL OF 3.

A1 **DOAS-1 ENERGY RECOVERY VENTILATOR CONTROL DIAGRAM**
SCALE: NOT TO SCALE



TYPICAL BCU POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
SPACE TEMPERATURE SENSOR	X	X		X		X	X	X	X		2,4,5,6
SUPPLY AIR TEMPERATURE	X	X							X		
HEATING COIL CONTROL VALVE	X		X						X		
SUPPLY AIR FAN STATUS (CURRENT SENSOR)	X	X				X			X		1
FAN START/STOP	X				X				X		
FAN SPEED SIGNAL	X		X						X		
DRAIN PAN HIGH LIMIT SWITCH	X			X		X			X		3
NIGHT SETBACK TEMPERATURE	X						X		X		
SPACE RELATIVE HUMIDITY SENSOR	X	X							X		
FINTUBE CONTROL VALVE	X				X				X		7

NOTES:
1. GENERATE ALARM AT GUI IF FAN FAILS TO SHOW PROOF OF OPERATION.
2. GENERATE ALARM AT GUI IF THE ROOM TEMP. IS NOT WITHIN +/- 5°F OF SET POINT FOR MORE THAN 10 MINUTES.
3. GENERATE ALARM AT GUI IF THE DRAIN PAN FLOAT SWITCH INDICATES THE PRESENCE OF WATER.
4. REFER TO MECHANICAL PIPING PLANS FOR SENSOR LOCATIONS.
5. ROOM TEMPERATE SENSORS SHALL INCLUDE NIGHT-SETBACK OVERRIDE BUTTON.
6. TYPICAL OF 3 FOR BCU-3, AND 1 EACH FOR BCU-1 AND BCU-2.
7. TYPICAL OF 3 FOR BCU-3 ONLY.

A3 **TYPICAL BCU CONTROL DIAGRAM**
SCALE: NOT TO SCALE

SEQUENCE OF OPERATIONS - BLOWER COIL UNITS

ZONE OCCUPANCY MODE:
A ZONE SHALL ENTER THE OCCUPIED MODE WHEN THE DOAS-1 SYSTEM IS IN THE OCCUPIED MODE.

OCCUPIED MODE:
DURING THE OCCUPIED MODE THE SUPPLY FAN SHALL RUN CONTINUOUSLY. THE OCCUPANT ADJUSTABLE SET POINT RANGE SHALL BE LIMITED BETWEEN 68°F AND 72°F (ADJUSTABLE). THE HEATING SET POINT SHALL BE EQUAL TO THE OCCUPANT SET POINT AND THE COOLING SET POINT SHALL BE 2°F ABOVE THE HEATING SET POINT.

THE SPACE COOLING TEMPERATURE FOR BCU-3 SHALL BE DETERMINED BY THE AVERAGE OF THE THREE ROOM TEMPERATURE SENSORS ASSOCIATED WITH BCU-3. THE COOLING SET POINT ASSOCIATED WITH THE TRAINING ROOM SHALL BE USED TO ENABLE AND OPERATE BCU-3 COOLING. THE SPACE HEATING TEMPERATURE FOR BCU-3 SHALL BE DETERMINED BY THE SPACE SET PONT IN THE TRAINING ROOM.

THE FAN SHALL RUN AT LOW SPEED WHENEVER THE ROOM TEMPERATURE IS LESS THAN 1°F BELOW THE HEATING SET POINT OR LESS THAN 1°F ABOVE THE COOLING SET POINT.

THE FAN SHALL RUN AT MEDIUM SPEED WHENEVER THE ROOM TEMPERATURE IS BETWEEN 1°F AND 2° F BELOW THE HEATING SET POINT OR BETWEEN 1°F AND 2°F ABOVE THE COOLING SET POINT.

THE FAN SHALL RUN AT HIGH SPEED WHENEVER THE ROOM TEMPERATURE IS MORE THAN 2°F BELOW THE HEATING SET POINT OR MORE THAN 2°F ABOVE THE COOLING SET POINT.

THE HEATING COIL VALVE SHALL MODULATE TO MAINTAIN THE ROOM HEATING SET POINT. THE HEATING COIL VALVE SHALL BE FULLY OPEN WHENEVER THE ROOM TEMPERATURE IS MORE THAN 3°F BELOW THE HEATING SET POINT, AND SHALL BE CLOSED WHENEVER ROOM TEMPERATURE IS ABOVE THE HEATING SET POINT.

THE DX COOLING COIL SHALL CYCLE TO MAINTAIN THE ROOM COOLING SETPOINT. THE HEATING CONTROL VALVE AND FINTUBE VALVES SHALL REMAIN CLOSED.

DEHUMIDIFICATION MODE:
IF THE SPACE RELATIVE HUMIDITY LEVEL RISES ABOVE THE RH MAXIMUM SET POINT, 60%RH ADJUSTABLE), THE BCU SHALL ENTER DEHUMIDIFICATION MODE (DURING OCCUPIED AND UNOCCUPIED TIMES). DURING DEHUMIDIFICATION MODE THE DX COOLING SHALL BE ENABLED AND THE HEATING COIL SHALL MODULATE TO MAINTAIN THE SUPPLY AIR TEMPERATURE ACCORDING TO THE FOLLOWING USER ADJUSTABLE REST SCHEDULE:

ROOM TEMPERATURE	SUPPLY AIR SET POINT
2°F > ROOM SET POINT	55°F
2°F < ROOM SET POINT	95°F

UNOCCUPIED MODE:
DURING THE UN-OCCUPIED MODE THE FCU AND/OR BCU SHALL OPERATE IN NIGHT SET-BACK MODE. DURING NIGHT SET-BACK MODE THE SUPPLY FAN SHALL BE OFF, THE HEATING VALVE SHALL BE CLOSED UNLESS THE ROOM TEMPERATURE FALLS BELOW THE NIGHT SET-BACK SET POINT (65°F, ADJUSTABLE). IF THE ROOM TEMPERATURE FALLS BELOW THE NIGHT SET-BACK SET POINT THE FAN SHALL RUN AT HIGH SPEED AND THE HEATING COIL VALVE SHALL REMAIN FULLY OPEN UNTIL NIGHT SET-BACK IS SATISFIED.

UNOCCUPIED OVERRIDE MODE:
WHEN THE OCCUPANT OVERRIDE BUTTON IS PRESSED DURING THE UNOCCUPIED MODE THE FC/BC UNIT SHALL OPERATE AND THE ERV SHALL OPERATE IN THE OCCUPIED MODE FOR 1 HOUR (ADJUSTABLE) AND THEN SYSTEMS SHALL RETURN TO THE UNOCCUPIED MODE. WHEN THE DOAS-3 IS CHANGED TO THE OCCUPIED MODE BY A LOCAL OVERRIDE SWITCH, ONLY THE OCCUPIED LOCAL ZONE SHALL ENTER THE OCCUPIED MODE; THE REMAINING LOCAL ZONES WILL REMAIN IN THE UNOCCUPIED MODE.

SAFETY - DRAIN PAN HIGH LIMIT:
THE CONDENSATE PAN FLOAT SWITCH SHALL STOP FAN, CLOSE COOLING COIL CONTROL VALVE, AND SIGNAL ALARM WHEN A HIGH WATER LEVEL IS PRESENT IN THE PAN.

STAGGERED START:
WHEN SWITCHING FROM UNOCCUPIED TO OCCUPIED MODE, EACH FAN SHALL START AFTER A RANDOMLY GENERATED NUMBER OF SECONDS TO AVOID EXCESSIVE SYSTEM CURRENT DRAW.

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC
STATE OF MAINE RICHARD CRESWELL No. 13178 10-01-25 LICENSED PROFESSIONAL ENGINEER
SEAL
OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER SRW
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
MECHANICAL CONTROL DIAGRAMS 3

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M-703
DRAWFORM REVISION: 14 SEP 2023

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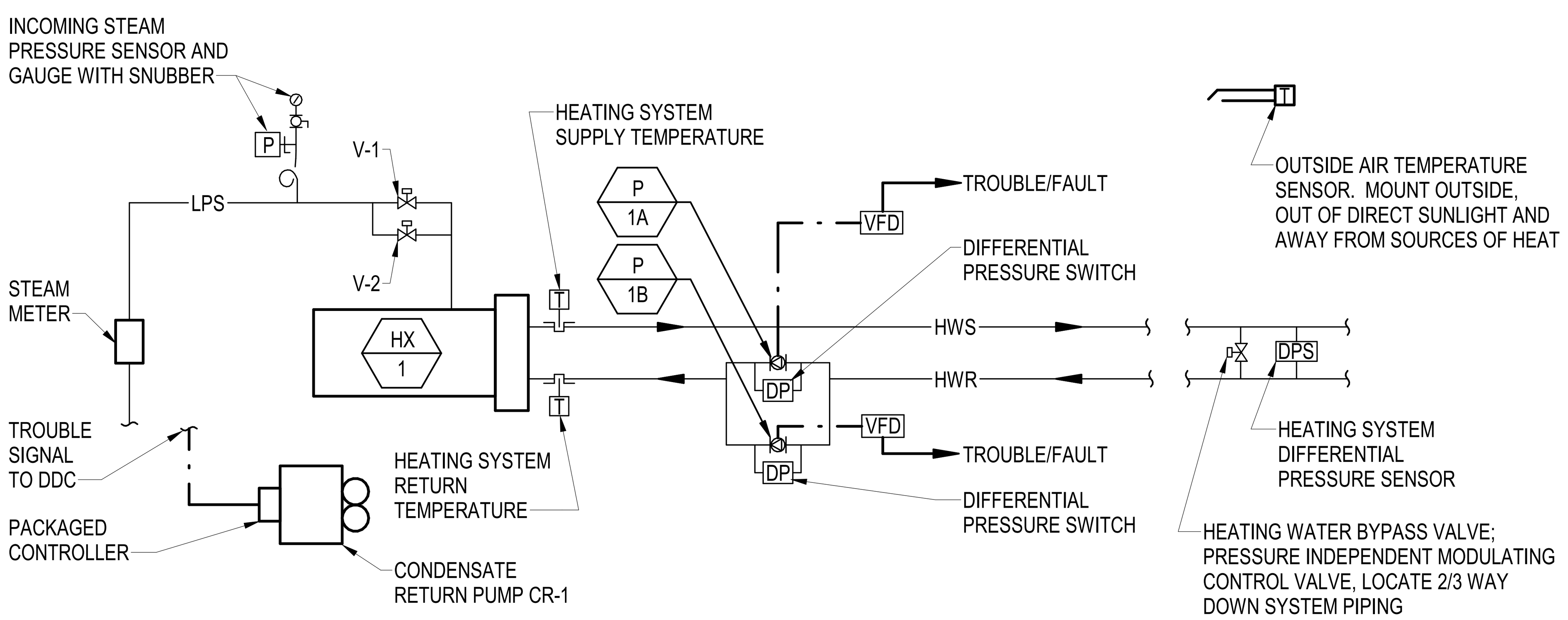


SEQUENCE OF OPERATION

HEATING SYSTEM SHALL BE ENABLED IF OA TEMPERATURE IS BELOW 60°F (ADJUSTABLE). HWS TEMPERATURE SHALL BE MAINTAINED ACCORDING TO THE USER ADJUSTABLE VALVE POSITION METHOD BASED ON THE FOLLOWING SCHEDULE:

VALVE POSITION	HWS SET POINT
BELOW 50%	140
BETWEEN 50% AND 90%	160
ABOVE 90%	180

IF ANY THERMOSTAT IS 5 DEGREES OR MORE BELOW SETPOINT FOR LONGER THAN 20 MINUTES, RAISE HWS SETPOINT TO 180.

THE 1/3 AND 2/3 STEAM CONTROL VALVES SHALL MODULATE IN SEQUENCE (1/3 VALVE FIRST) TO MAINTAIN THE HWS SET POINT.

THE HEATING WATER PUMPS P-1A AND P-1B SHALL OPERATE IN A LEAD/LAG ARRANGEMENT. THE LEAD PUMP SHALL SWITCH MONTHLY. IF THE LEAD PUMP FAILS TO SHOW PROOF OF FLOW OR THE LEAD PUMP VARIABLE FREQUENCY DRIVE (VFD) INDICATES AN ALARM CONDITION THE LEAD PUMP SHALL STOP AND THE LAG PUMP SHALL AUTOMATICALLY START.

THE LEAD PUMP SHALL RUN CONTINUOUSLY WHENEVER THE HEATING SYSTEM IS ENABLED AND IT'S SPEED SHALL MODULATE TO MAINTAIN THE HEATING WATER SYSTEM DIFFERENTIAL PRESSURE SETPOINT OF 30 PSI (ADJUSTABLE). THE PUMPS SHALL NOT OPERATE BELOW MINIMUM SPEED (30% OF FULL SPEED, ADJUSTABLE).

MEASUREMENT AND VERIFICATION:

THE STEAM METER SHALL BE USED TO MEASURE AND RECORD THE INCOMING STEAM CONSUMPTION IN BTU'S. THE FOLLOWING ENERGY USAGE IN BTU SHALL BE AVAILABLE AT GUI:

STEAM ENERGY USED IN THE PAST HOUR.
STEAM ENERGY USED IN THE PAST DAY.
STEAM ENERGY USED IN THE PAST WEEK.
STEAM ENERGY USED IN THE PAST MONTH.
STEAM ENERGY USED IN THE PAST YEAR.

THE VFDS SHALL BE CONFIGURED TO RECORD THE ENERGY USAGE OF THE PUMPS, AND THIS DATA SHALL BE COLLECTED BY THE DDC SYSTEM. THE FOLLOWING PUMP DATA SHALL BE ACCESSIBLE THROUGH THE GUI:

ELECTRICAL USAGE (KWH) FOR THE PAST HOUR.
ELECTRICAL USAGE (KWH) FOR THE PAST DAY.
ELECTRICAL USAGE (KWH) FOR THE PAST WEEK.
ELECTRICAL USAGE (KWH) FOR THE PAST MONTH.
ELECTRICAL USAGE (KWH) FOR THE PAST YEAR.

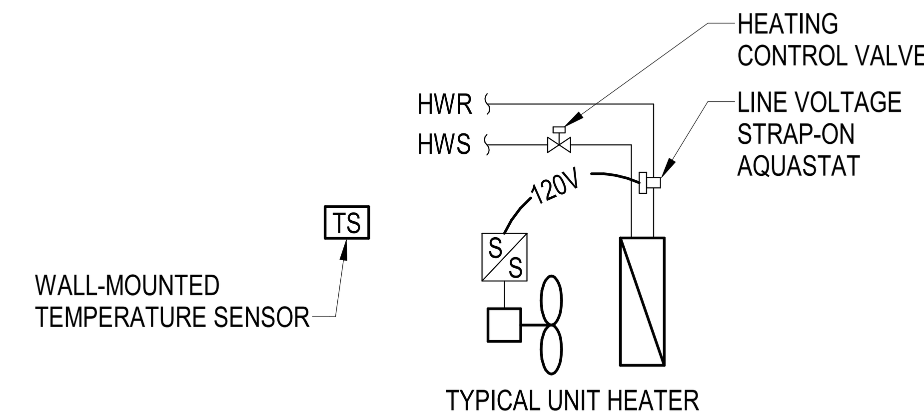
HEATING SYSTEM POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
HEATING SYSTEM SUPPLY TEMPERATURE	X	X				X	X		X		3
HEATING SYSTEM RETURN TEMPERATURE	X	X							X		
SUPPLY TEMP RESET UPPER CONDITION	X						X				
SUPPLY TEMP RESET LOWER CONDITION	X						X				
INCOMING STEAM PRESSURE	X	X							X		
V-1 STEAM VALVE (1/3-STEAM VALVE)	X		X						X		
V-2 STEAM VALVE (2/3-STEAM VALVE)	X		X						X		
P-1A VFD ENABLE	X				X				X		
P-1A VFD SIGNAL	X		X						X		
P-1A VFD TROUBLE ALARM	X			X		X			X		1
P-1A DIFFERENTIAL PRESSURE SWITCH	X			X		X			X		2
P-1B VFD ENABLE	X				X				X		
P-1B VFD SIGNAL	X		X						X		
P-1B VFD TROUBLE ALARM	X			X		X			X		1
P-1B DIFFERENTIAL PRESSURE SWITCH	X			X		X			X		2
HEATING SYSTEM DIFFERENTIAL PRESSURE	X						X		X		
CONDENSATE RECIEVER TROUBLE SIGNAL	X			X		X			X		
OUTSIDE AIR TEMPERATURE	X	X							X		
STEAM METER	X	X							X		
P-1A VFD ENERGY USAGE	X	X							X		
P-1B VFD ENERGY USAGE	X	X							X		

NOTES:
1. GENERATE ALARM IF VFD INDICATES A FAULT.
2. GENERATE ALARM IF PUMP FAILS TO SHOW PROOF OF FLOW.
3. GENERATE ALARM IF HEATING SYSTEM SUPPLY TEMPERATURE IS MORE THAN ± 10 DEG F (ADJUSTABLE) FROM SET POINT.

A1 HEATING SYSTEM CONTROL DIAGRAM

SCALE: NOT TO SCALE



SEQUENCE OF OPERATION

CONNECT AQUASTAT TO FAN POWER CIRCUIT TO DISABLE FAN WHEN WATER TEMPERATURE IS 120°F OR LOWER.

IF THE ROOM TEMPERATURE FALLS 2°F BELOW THE ROOM SET POINT (65°F, ADJUSTABLE) THE CONTROL VALVE SHALL OPEN AND THE FAN SHALL START. WHEN THE ROOM SET POINT IS SATISFIED THE VALVE SHALL CLOSE AND THE FAN SHALL STOP.

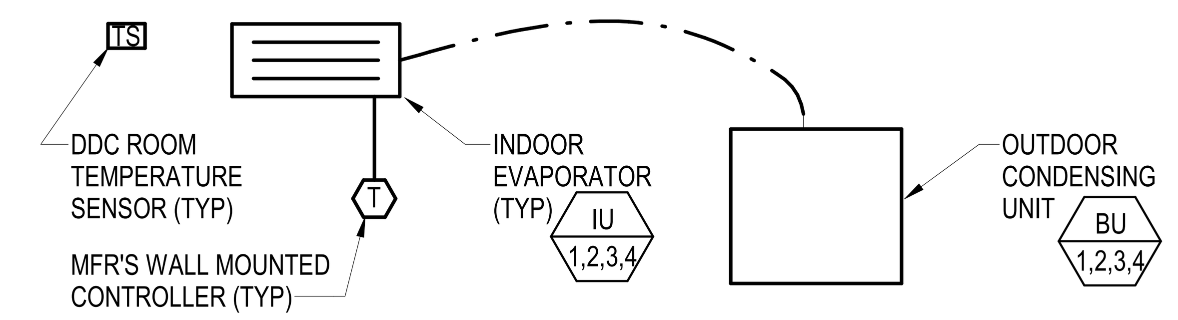
TYPICAL UNIT HEATER POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
ROOM TEMPERATURE	X	X				X			X		1
FAN START/STOP	X				X				X		
HEATING CONTROL VALVE	X				X				X		
HEATING SETPOINT	X						X		X		2

NOTES: 1. GENERATE ALARM IF THE ROOM TEMPERATURE FALLS 10°F (ADJUSTABLE) BELOW SET POINT FOR MORE THAN 30 MINUTES.
2. VARIABLE ADJUSTMENT THROUGH GUI ONLY.

C3 TYPICAL UNIT HEATER CONTROL DIAGRAM

SCALE: NOT TO SCALE



SEQUENCE OF OPERATION

DUCTLESS AIR CONDITIONING UNIT SHALL HAVE MANUFACTURER'S STAND ALONE, WALL MOUNTED CONTROLLER. EACH CONTROLLER SHALL HAVE A 24 HOUR, 7 DAY PROGRAMMABLE THERMOSTAT AND A COOLING ON/OFF SWITCH.

DUCTLESS SPLIT SYSTEM NOTE:

THE CONTROLS CONTRACTOR SHALL PROVIDE INTERCONNECTING WIRING BETWEEN SYSTEM DEVICES AS REQUIRED BY THE EQUIPMENT MANUFACTURER'S WRITTEN INSTRUCTIONS.

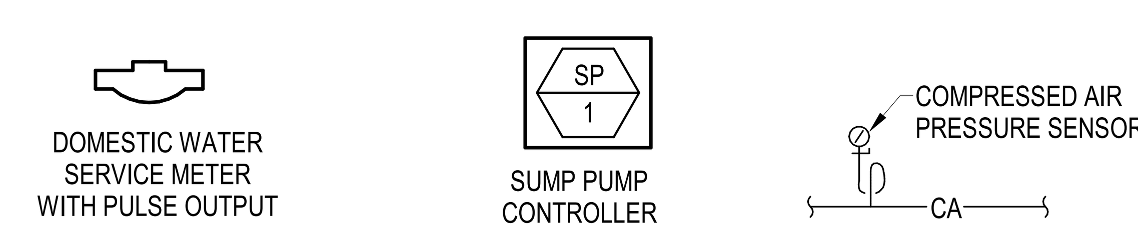
TYPICAL ELEVATOR ROOM AND NETWORK ROOM DUCTLESS AIR CONDITIONING UNIT POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
ROOM TEMPERATURE	X	X				X	X		X		1

NOTES:
1. GENERATE ALARM IF TEMPERATURE IN NOT ±5°F (ADJUSTABLE) OF SET POINT FOR MORE THAN 10 MINUTES.

C4 TYPICAL ELEVATOR MACHINE ROOM, NETWORK ROOM, AND OFFICE 108 DUCTLESS AIR CONDITIONING UNIT CONTROL DIAGRAM

SCALE: NOT TO SCALE



SEQUENCE OF OPERATION

THE DOMESTIC WATER METER SHALL MONITOR AND RECORD PEAK LOADS ON DAILY, WEEKLY AND MONTHLY INTERVALS, AND SHALL CALCULATE TOTAL USAGE FOR THE PAST HOUR, DAY, MONTH AND YEAR. PROVIDE INFORMATION ON AN ENERGY/UTILITY DASH BOARD ON THE GUI. FOR DOMESTIC HOT WATER ENERGY METER REFER TO A5/M-704.

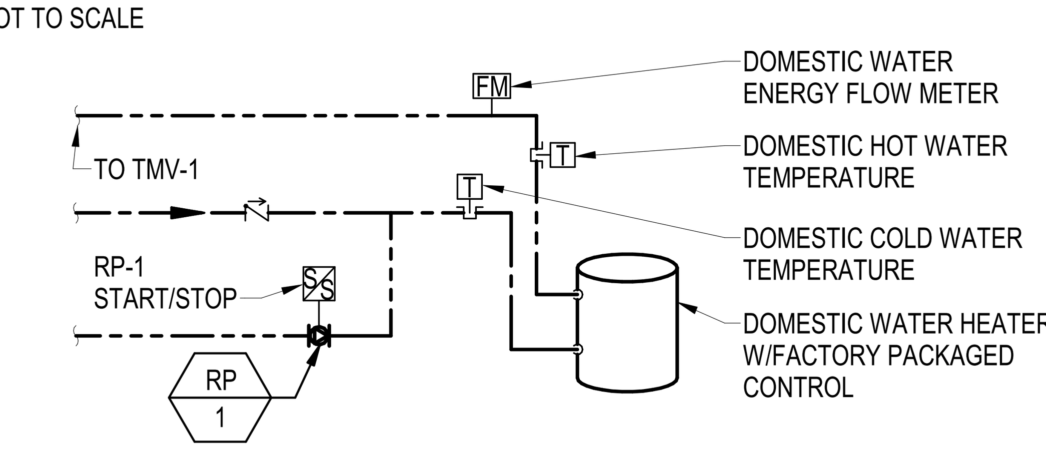
GLOBAL BUILDING POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
DOMESTIC WATER METER INPUT	X			X					X		1
PUMP SP-1 STATUS	X			X		X			X		2
PUMP SP-1 ALARM	X			X		X			X		2
COMPRESSED AIR PRESSURE SENSOR	X			X		X			X		3

NOTES:
1. COORDINATE REQUIREMENTS WITH EQUIPMENT SUPPLIER TO ACHIEVE COMPATIBILITY BETWEEN METER PULSE OUTPUT AND DDC SYSTEM.
2. GENERATE AN ALARM AT GUI IF PUMP CONTROLLER INDICATES AN ALARM CONDITION.
3. GENERATE AN ALARM AT GUI IF SENSOR DETECTS A SUPPLY PRESSURE BELOW 100 PSI (ADJUSTABLE) FOR MORE THAN 5 MINUTES (ADJUSTABLE).

A3 BUILDING GLOBAL POINTS CONTROL DIAGRAM

SCALE: NOT TO SCALE



SEQUENCE OF OPERATION

THE WATER HEATER PACKAGED CONTROLS SHALL CYCLE THE HEATER TO MAINTAIN THE DOMESTIC HOT WATER AT 140°F (ADJUSTABLE), CONTINUOUSLY. RP-1 SHALL RUN CONTINUOUSLY WHENEVER ANY OF THE BUILDING DOAS UNITS ARE RUNNING IN OCCUPIED MODE.

THE DOMESTIC HOT WATER METER (FLOW AND CHANGE IN TEMPERATURE) SHALL MONITOR, CALCULATE AND RECORD PEAK LOADS ON DAILY, WEEKLY AND MONTHLY INTERVALS, AND SHALL ALSO CALCULATE TOTAL USAGE FOR THE PAST HOUR, DAY, MONTH AND YEAR. PROVIDE INFORMATION ON AN ENERGY/UTILITY DASH BOARD ON THE GUI.

DOMESTIC WATER HEATER POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
DOMESTIC COLD WATER TEMPERATURE	X	X							X		
DOMESTIC HOT WATER TEMPERATURE	X	X				X	X		X		1
RP-1 START/STOP	X				X				X		
DOMESTIC WATER ENERGY FLOW METER	X	X							X		

NOTES:
1. GENERATE ALARM IF THE DHW TEMPERATURE FALLS BELOW 120°F FOR MORE THAN 10 MINUTES

A4 DOMESTIC WATER HEATER CONTROL DIAGRAM

SCALE: NOT TO SCALE

FINAL SUBMISSION — 10/01/2025 — RSH
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RICHARD CRESWELL No. 13178 10-01-25 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE
SEAL

OAK POINT ASSOCIATES

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER RNC
DRAWN BY CBM
CHECKED BY MSA
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
MECHANICAL CONTROL DIAGRAMS 4

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DRAWFORM REVISION: 14 SEP 2023

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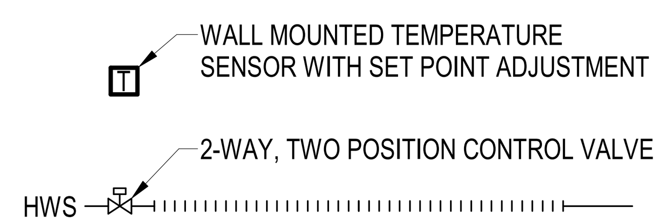


SEQUENCE OF OPERATION

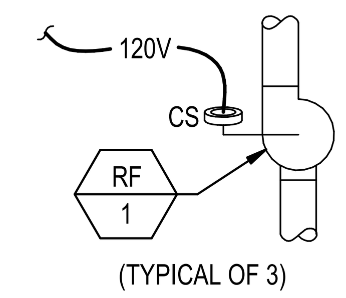
IF THE ROOM TEMPERATURE FALLS 2°F (ADJUSTABLE) BELOW THE ROOM SET POINT, 68°F (ADJUSTABLE), THE CONTROL VALVE SHALL OPEN. THE VALVE SHALL CLOSE WHEN THE SET POINT IS SATISFIED.

TYPICAL FINTUBE OR CONVECTOR POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
ROOM TEMPERATURE	X	X				X	X		X		1
CONTROL VALVE	X				X				X		

NOTES:
1. GENERATE ALARM IF THE ROOM TEMPERATURE FALLS 10°F (ADJUSTABLE) BELOW SET POINT.

C1 TYPICAL FINTUBE OR CONVECTOR CONTROL DIAGRAM
SCALE: NOT TO SCALE

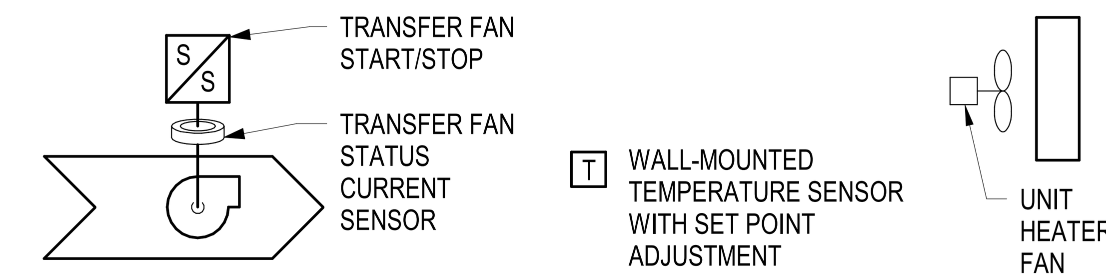


TYPICAL RADON EXHAUST SYSTEM POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
EXHAUST FAN 3 STATUS	X				X				X		1,2,3

NOTES:
1. TYPICAL OF 3.
2. GENERATE ALARM IF FAN FAILS TO SHOW PROOF OF FLOW.
3. PROVIDE AS PART OF OPTION 3.

C2 TYPICAL RADON EXHAUST FAN CONTROL DIAGRAM (OPTION 3)
SCALE: NOT TO SCALE



SEQUENCE OF OPERATION

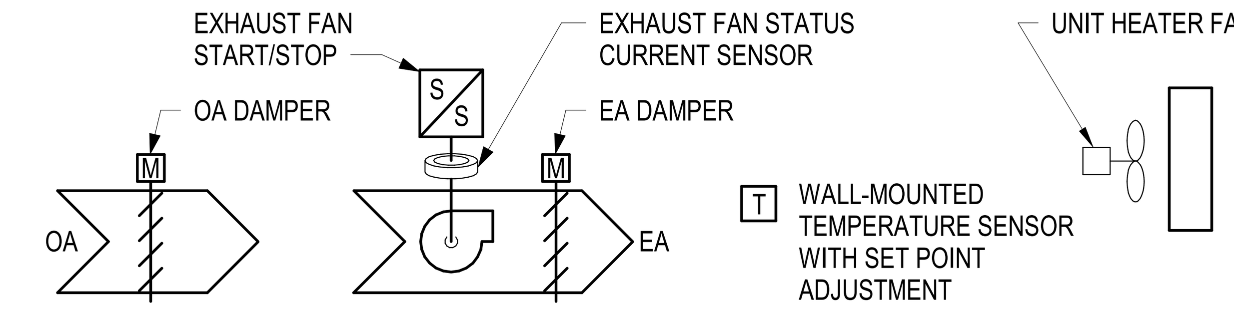
WHEN THE ROOM TEMPERATURE RISES 1°F (ADJUSTABLE) ABOVE THE ROOM SET POINT THEN THE TRANSFER FAN SHALL START AND RUN CONTINUOUSLY UNTIL THE ROOM IS SATISFIED. WHEN THE ROOM FALLS 1°F BELOW THE HEATING SETPOINT, ADJUSTABLE ON THE ROOM TEMPERATURE SENSOR BETWEEN 50°F AND 70°F, THE UNIT HEATER FAN SHALL START AND RUN CONTINUOUSLY UNTIL THE ROOM IS SATISFIED.

ELECTRICAL 114 POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
TRANSFER FAN START/STOP	x				x				x		
TRANSFER FAN STATUS CURRENT SENSOR	x	x				x			x		1
ROOM TEMPERATURE	x	x				x			x		2
ROOM SET POINT	x	x							x		
UNIT HEATER FAN	x				x				x		

NOTES:
1. GENERATE AN ALARM ON THE GUI IF THE FAN FAILS TO SHOW PROOF OF AIRFLOW.
2. GENERATE AN ALARM ON THE GUI IF THE ROOM TEMPERATURE FALLS BELOW 50°F (ADJUSTABLE) OR RISES ABOVE 85°F (ADJUSTABLE).

B4 ELECTRICAL 114 HEATING & COOLING CONTROL DIAGRAM
SCALE: NOT TO SCALE



SEQUENCE OF OPERATION

WHEN THE ROOM TEMPERATURE RISES 1°F ABOVE THE ROOM COOLING SETPOINT, 80°F (ADJUSTABLE), THE OUTSIDE AND EXHAUST AIR DAMPERS SHALL OPEN AND REMAIN OPEN UNTIL THE ROOM IS SATISFIED. IF THE ROOM RISES 3°F ABOVE THE COOLING SET POINT THEN THE EXHAUST FAN SHALL START AND RUN CONTINUOUSLY UNTIL THE ROOM IS SATISFIED. REFER TO CONTROL DIAGRAM C3/M-704, FOR UNIT HEATER SEQUENCE OF OPERATION AND CONTROL POINTS LIST.

MECHANICAL 115 POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
EXHAUST FAN START/STOP	x				x				x		
EXHAUST FAN STATUS CURRENT SENSOR	x	x				x			x		1
ROOM TEMPERATURE	x	x				x			x		2
ROOM SET POINT	x	x							x		
UNIT HEATER FAN	x				x				x		
OA DAMPER	x				x				x		
EA DAMPER	x				x				x		

NOTES:
1. GENERATE AN ALARM ON THE GUI IF THE FAN FAILS TO SHOW PROOF OF AIRFLOW.
2. GENERATE AN ALARM ON THE GUI IF THE ROOM TEMPERATURE FALLS BELOW 50°F (ADJUSTABLE) OR RISES ABOVE 85°F (ADJUSTABLE).

A4 MECHANICAL 115 HEATING & COOLING CONTROL DIAGRAM
SCALE: NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER STATE OF MAINE 10-01-25
SEAL

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DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
ROICC-ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
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GENERAL FIRE ALARM AND VOICE NOTIFICATION NOTES

1. PROVIDE A NEW, COMPLETE, BUILDING VOICE FIRE ALARM AND MASS NOTIFICATION SYSTEM FOR PORTSMOUTH NAVAL STATION, BUILDING 60. THE SYSTEM MUST INCLUDE ALL WIRING, RACEWAYS, PULL BOXES, TERMINAL CABINETS, OUTLET AND MOUNTING BOXES, CONTROL EQUIPMENT, ALARM AND SUPERVISORY SIGNALS, INITIATING DEVICES, ALARM NOTIFICATION APPLIANCES, NOTIFICATION APPLIANCE BOOSTER PANELS, AMPLIFIERS, INTERFACE EQUIPMENT, AND ALL OTHER ACCESSORIES AND MISCELLANEOUS ITEMS INCLUDING SOFTWARE, SYSTEM PROGRAMMING, TESTING AND CONSTRUCTION PERMITS REQUIRED FOR A COMPLETE OPERATING SYSTEM EVEN THOUGH EACH ITEM IS NOT SPECIFICALLY MENTIONED OR DESCRIBED. PROVIDE SYSTEM COMPLETE AND READY FOR OPERATION. THE SYSTEM MUST COMPLY WTH UFC 1-200-01, UFC 3-600-01, UFC 4-021-01, NFPA STANDARDS 70 (2023), 72 (2025), AND 101 (2024).

2. PROVIDE THE EQUIPMENT FOR THE FIRE ALARM SYSTEM AND VOICE CONTROL SYSTEM BY A SINGLE MANUFACTURER AND LISTED FOR THE INTENDED PURPOSE BY A NATIONALLY RECOGNIZED TESTING LABORATORY.

3. VERIFY CEILING HEIGHTS, WALL AND PARTITION LOCATIONS, AND DOOR LOCATIONS PRIOR TO INSTALLATION. FIELD DETERMINE ARRANGEMENTS OF EQUIPMENT, DUCTWORK, PIPING AND OTHER COMPONENTS WITH DUE CONSIDERATION OF STRUCTURAL, ELECTRICAL, AND ARCHITECTURAL SYSTEMS.

4. CONTRACT DRAWINGS AND SPECIFICATIONS ARE C0MPLIMENTARY. COORDINATE DRAWINGS AND SPECIFICATIONS PRIOR TO COMMENCING WORK. PROMPTLY REPORT QUESTIONS REGARDING DESIGN INTENT OR SYSTEM OPERATION TO THE CONTRACTING OFFICER. PROCEED WITH THE WORK ONLY AFTER THE DISCREPANCIES HAVE BEEN RESOLVED WITH THE CONTRACTING OFFICER.

5. SUBMIT A COMPLETE SET OF SHOP DRAWINGS THAT DELINEATES THE DEVICES BEING SUPPLIED FOR THIS PROJECT TO THE CONTRACTING OFFICER FOR DISTRIBUTION TO THE GOVERNMENT FIRE PROTECTION ENGINEER AND TO THE ENGINEER OF RECORD BEFORE INSTALLATION OR PROCUREMENT OF EQUIPMENT COMMENCES. PROVIDE SHOP DRAWINGS WITH MANUFACTURER'S NAMES AND CATALOG NUMBERS. PROVIDE DEVICE LOCATIONS AND LOCATIONS OF WIRING/CONDUIT BETWEEN DEVICES.

6. PROVIDE A NICET III TECHNICIAN (OR HIGHER) AS PART OF THE INSTALLATION TEAM INCLUDING THE PREPARATION OF SHOP DRAWINGS.

7. SECURE NECESSARY CONSTRUCTION PERMITS, INSPECTIONS AND FINAL SYSTEM ACCEPTANCE. COORDINATE WORK WITH THE CONTRACTING OFFICER, INCLUDING ACCESS TO THE BUILDING. SCHEDULE WORK TO MINIMIZE DISRUPTION TO OCCUPANTS OF THE BUILDING. NOTIFY THE CONTRACTING OFFICER 14 DAYS IN ADVANCE BEFORE THE EXISTING FIRE ALARM SYSTEM IS SHUTDOWN.

8. PROVIDE NEW FIRE ALARM SIGNAL TRANSMITTER, INCLUDING ANTENNAE CABLE, TRANSIENT VOLTAGE SURGE SUPPRESSOR, CONDUIT, AND WIRING REQUIRED. PROVIDE PROGRAMMING AND CONNECTIONS FOR A COMPLETE AND READY FOR OPERATION REPORTING SYSTEM CONNECTED TO THE BASE REPORTING SYSTEM. ENSURE FIRE ALARM NOTIFICATION SIGNALS ARE TRANSMITTED TO THE BASE FIRE DEPARTMENT VIA RADIO TRANSMITTER.

9. MAINTAIN EXISTING FIRE ALARM SYSTEMS FULLY 0PERATIONAL UNTIL THE NEW FIRE ALARM SYSTEM HAS BEEN INSTALLED, TESTED AND ACCEPTED BY THE NAVFAC FIRE PROTECTION ENGINEER AND ENGINEER OF RECORD. AS NEW FIRE ALARM EQUIPMENT AND APPLIANCES ARE INSTALLED, THEY SHALL BE LABELED "NOT IN SERVICE" UNTIL THE NEW FIRE ALARM SYSTEM IS ACCEPTED. AFTER THE NEW SYSTEM IS ACCEPTED REMOVE LABELS FROM NEW FIRE ALARM EQUIPMENT AND APPLIANCES, AND LABEL EXISTING FIRE ALARM EQUIPMENT "NOT IN SERVICE" UNTIL REMOVED.

10. UNLESS NOTED OTHERWISE REMOVE FIRE ALARM APPLIANCES LOCATED THROUGHOUT THE BUILDING INCLUDING HEAT DETECTORS, MANUAL PULL STATIONS, HORN/STROBE APPLIANCES, ADDRESSABLE MONITOR MODULES, ASSOCIATED CONDUIT/CIRCUITS AND FIRE ALARM SIGNAGE.

11. PROVIDE BUILDING FIRE ALARM/MASS NOTIFICATION (SPEAKERS AND STROBES), LED TEXT SIGNS, MANUAL PULL STATIONS AND ADDRESSABLE MONIT0R MODULES THROUGHOUT THE BUILDING. PROVIDE SUPPLEMENTARY BATTERIES, NOTIFICATION APPLIANCE CIRCUIT PANELS AND SPEAKER EXTENDER PANELS AS REQUIRED. LOCATE AS INDICATED OR IN ELECTRICAL ROOMS. PROVIDE WITH SPOT SMOKE DETECTION ABOVE PANELS.

12. PROVIDE FIRE ALARM CIRCUITS SUPERVISED AS FOLLOWS IN ACCORDANCE WITH NFPA 72:
    - 12.1. INITIATING DEVICE CIRCUITS (IDC) - CLASS A
    - 12.2. NOTIFICATION APPLIANCE CIRCUITS (NAC) - CLASS A, SYNCHR0NIZED
    - 12.3. SIGNALING LINE CIRCUITS (SLC) - CLASS A
    - 12.4. FIRE ALARM AND DIGITAL AUDIO LOOP NETWORKS - CLASS A

13. PROVIDE MANUAL PULL STATIONS 44 INCHES ABOVE FINISHED FLOOR TO OPERATING HANDLE. PROVIDE MANUAL PULL STATIONS WITHIN 5 FEET OF THE ENTRANCE TO EACH EXIT.

14. PROVIDE SMOKE DETECTORS A MINIMUM OF 3 FEET FROM AIR DIFFUSERS.

15. PROVIDE GENERAL EVACUATI0N FOR THE BUILDING. PROVIDE THE BUILDING AS A SINGLE FIRE ALARM NOTIFICATION ZONE. PROVIDE VISUAL DEVICES SYNCHRONIZED ACROSS THE FIRE ALARM NOTIFICATION ZONE.

16. PROVIDE STROBE NOTIFICATION APPLIANCE CIRCUIT POWER REQUIREMENTS AND CONDUCTOR SIZES NOT TO EXCEED A 15% VOLTAGE DROP. EACH VISUAL NOTIFICATI0N APPLIANCE CIRCUIT MUST NOT EXCEED 75% OF ITS RATED OUTPUT WHICH ASSUMES THE MAXIMUM AVAILABLE LOAD IS DISTRIBUTED EQUALLY OVER THE MAXIMUM AVAILABLE CIRCUITS OF THE PANEL.

17. PROVIDE MINIMUM BATTERY SIZING BASED ON FIRE ALARM/MASS NOTIFICATION SYSTEM - 48 HOURS OF SUPERVISORY POWER FOLLOWED BY 15 MINUTES OF ALARM, OR 60 MINUTES OF ALARM, WHICHEVER IS GREATER.

18. THE SYSTEM MUST HAVE AUDIBILITY LEVELS BETWEEN 70 DBA AND 110 DBA, AND INTELLIGIBILITY LEVELS OF A MINIMUM 0.7 CIS. THESE VALUES MUST BE MET WITH DOORS IN THE CLOSED POSITION. SPEAKER APPLIANCES ARE SHOWN ON CONTRACT DRAWINGS. DURING PRE-FUNCTIONAL TESTING PROVIDE ADDITIONAL SPEAKER APPLIANCES TO OBTAIN INTELLIGIBILITY THROUGHOUT THE BUILDING.

19. TAP SPEAKER APPLIANCES AS FOLLOWS: OFFICE AREAS AT 1/4 WATT, MECHANICAL SPACES AT 1 WATT, EXTERIOR AT 4 WATT. PROVIDE ADJUSTMENTS IN TAP SETTING DURING TESTING TO MEET AUDIBILITY AND INTELLIGIBILITY REQUIREMENTS. INDIVIDUAL SPEAKER AMPLIFIER/CIRCUITS MUST NOT BE LOADED MORE THAN 70% OF RATED AMPLIFIER POWER OUTPUT.

20. PROVIDE SPEAKERS/STROBES WITH A CLEAR STROBE LENS. PROVIDE HOUSING WHITE IN COLOR AND FACTORY MARKED WITH THE WORD "ALERT" IN RED.

21. PROVIDE EXTERIOR WEATHERPROOF SPEAKERS AT EACH FACILITY ENTRANCE, AND AT PRIMARY GATHERING LOCATIONS FOR BUILDING OCCUPANTS. DRILL THROUGH EXTERIOR WALLS AT EXTERIOR APPLIANCES. PROVIDE CONDUIT AT BACK OF APPLIANCE BACK BOX. INSTALL SURFACE MOUNTED EXTERIOR APPLIANCES WITH NO EXPOSED CONDUIT ON BUILDING EXTERIOR. MAKE OPENING WEATHERTIGHT. SEAL CONDUIT INTERIOR TO PREVENT PASSAGE OF MOISTURE.

22. LABEL ADDRESSABLE INPUT/OUTPUT FIRE ALARM COVER PLATES AND APPLIANCES (INCLUDING SPEAKERS, STROBES, SMOKE DETECTORS, MANUAL PULL STATIONS AND ADDRESSABLE MODULES) WITH THEIR ADDRESS UTILIZING A LABELING MACHINE. ADDRESS MUST INCLUDE NAC OR SLC NUMBER FOLLOWED BY APPLIANCE NUMBER. EACH SLC FIELD APPLIANCE ADDRESS MUST MATCH THE FIRE ALARM DISPLAY ADDRESS.

23. PROVIDE ALL CIRCUITS COMING INTO CONTROL CABINETS WITH LABEL TO INDICATE THE LOCATION, END OF LINE RESISTOR VALUE, AND FUNCTION OF THE CIRCUIT. LABEL CONDUCTORS INDIVIDUALLY WITH PERMANENT MARKINGS UTILIZING A LABELING MACHINE. PERMANENTLY AFFIX LABEL TO CONDUCTOR WITH SHRINK WRAP.

24. UNLESS NOTED OTHERWISE, THE PROJECT ELECTRICAL CONDUIT STANDARD IS ELECTRICAL METALLIC TUBING (EMT) AND PROVIDED THROUGHOUT THE PROJECT. REFERENCE TO CONDUIT IS EQUIVALENT TO STATING EMT. PROVIDE WIRING IN CONDUIT. FLEXIBLE METAL CONDUIT IS LIMITED TO 6 FEET AND SECURED PER THE NATIONAL ELECTRIC CODE. PROVIDE ONLY WIRE ASSOCIATED WITH THE FIRE ALARM SYSTEM IN THE CONDUIT. PROVIDE CONDUIT INSTALLED ABOVE CEILINGS WHERE DROP CEILINGS ARE PRESENT OR TO BE PROVIDED. PROVIDE COMPRESSION TYPE CONNECTORS BETWEEN EMT TRANSITIONS.

25. PROVIDE CONDUIT AS UNOBTRUSIVELY AS POSSIBLE, PARALLEL AND AS CLOSE AS POSSIBLE TO FLOOR/ CEILING SLAB, AND AT RIGHT ANGLES TO STRUCTURAL STEEL OR CONCRETE ELEMENTS.

26. PROVIDE CAST METAL BELL TYPE BOXES WITH THREADED HUBS ON EXPOSED WORK, EXCEPT WHERE FACTORY BOXES ARE PROVIDED. CONCEALED BOXES MAY BE STAMPED STEEL BOXES. PROVIDE TERMINAL CABINETS OF S0LID WALL C0NSTRUCTI0N WITH NO FACTORY CONDUIT KNOCK OUTS.

27. PROVIDE ELECTRICAL CONDUIT (SERVING FIRE ALARM EQUIPMENT), FIRE ALARM CONDUIT, BACK BOXES, COUPLING, JUNCTION BOXES AND COVERS FACTORY PAINTED RED IN UNFINISHED AREAS (ELECTRICAL ROOM, MECHANICAL ROOM, ETC.). WHEN EXPOSED IN FINISHED AREAS, PROVIDE CONDUIT AND JUNCTION BOXES PAINTED TO MATCH ROOM FINISH, THE INSIDE COVER OF THE JUNCTION BOX IDENTIFIED AS "FIRE ALARM" AND THE CONDUIT PAINTED WITH RED BANDS 2 INCH WIDE AT 10 FOOT CENTERS AND AT EACH SIDE OF A FLOOR, WALL OR CEILING PENETRATION.

28. DO NOT DRILL OR CUT STRUCTURAL MEMBERS FOR THE INSTALLATION OF EQUIPMENT, DEVICES, APPLIANCES, CONDUIT, OR WIRE. STRUCTURAL MEMBERS WITH UNAPPROVED HOLES OR MODIFICATIONS MUST BE REPLACED AT NO COST TO THE GOVERNMENT.

29. PROVIDE PENETRATIONS IN FIRE RESISTIVE WALLS AND FLOOR/CEILING AND ROOF SLABS CLEANED AND PROTECTED WITH A COMPLETE, TESTED THROUGH-PENETRATION ASSEMBLY SYSTEM OF EQUAL OR GREATER FIRE RESISTANCE TO THE ASSEMBLY BEING PENETRATED. PROVIDE EACH SYSTEM LABELED TO IDENTIFY FIRESTOP UL SYSTEM NUMBER, FIRESTOP RATING, DATE OF INSTALLATION AND CONTRACTOR'S NAME. RESTORE PENETRATIONS THROUGH SLAB OR CUTTING OF FIRE RATED ASSEMBLIES TO THEIR ORIGINAL FIRE RATING. THIS REQUIREMENT ALSO APPLIES TO ANY PENETRATION RESULTING FROM THE REMOVAL OF FIRE ALARM CONDUIT EXCEPT FOR THOSE PENETRATI0NS FILLED WITH THE SAME MATERIAL TYPE AS THE EXISTING C0NSTRUCTI0N.

30. UNLESS NOTED OTHERWISE PROVIDE CONDUIT 3/4 INCH MINIMUM TRADE SIZE. PROVIDE VERTICAL RISERS ONE INCH DIAMETER MINIMUM. PROVIDE WIRING CONDUCTORS TYPE FPLP. PROVIDE WIRING UL LISTED FOR ITS INTENDED USE. PROVIDE WIRING TEST FREE OF OPENS, SHORTS AND GROUNDS. PROVIDE SOLID OR STRANDED COPPER WIRING CONDUCTORS IN CONDUIT FOR THE FOLLOWING CIRCUITS (AT MINIMUM):
    - 30.1. INITIATING DEVICE CIRCUIT (IDC) - 14/2 AWG,
    - 30.2. SIGNALING LINE CIRCUIT (SLC) - 16/2 AWG LISTED PAIR,
    - 30.3. VISUAL NOTIFICATION APPLIANCES (NAC) - 14/2 AWG,
    - 30.4. AUDIO NOTIFICATION APPLIANCES (ANC) - 16/2 AWG LISTED SHIELDED PAIR,
    - 30.5. TEXT SIGNS - 14/2AWG

31. PROVIDE WIRING FROM APPLIANCE TERMINAL TO APPLIANCE TERMINAL WITH SCREW-TYPE TERMINAL BLOCKS. WIRE NUTS AND SIMILAR DEVICES ARE PROHIBITED.

32. "PROVIDE" MEANS CONTRACTOR MUST FURNISH NEW AND INSTALL COMPLETE AND READY FOR INTENDED USE.

33. "REMOVE" MEANS CONTRACTOR MUST DISCONNECT, REMOVE FROM SITE, AND LEGALLY DISPOSE.

34. PROVIDE SHOP DRAWINGS TO THE GENERAL CONTRACTOR FOR THEIR USE IN PREPARING COORDINATION DRAWINGS.

35. SEE ARCHITECTURAL FLOOR PLANS AND WALL TYPES, ARCHITECTURAL REFLECTED CEILING PLANS, ELECTRICAL POWER AND LIGHTING PLANS, SPRINKLER PLANS, AND MECHANICAL DUCTWORK AND PIPING PLANS FOR CEILING FEATURES, SOFFIT LOCATIONS, AND UTILITIES. COORDINATE WORK TO PREVENT INTERFERENCES AND MAINTAIN COMPLIANT COVERAGE. LOCATE CEILING APPLIANCES AND DEVICES IN A CONSISTENT PATTERN, CENTERED IN CEILING TILES WHERE TILES ARE PROVIDED. DO NOT ROUTE PIPING ABOVE THE DEDICATED WORKING CLEARANCES OF THE FMCU, LOC'S, OR NAC'S.

36. COORDINATE WITH MECHANICAL DRAWINGS, SPECIFICATIONS, AND SUBCONTRACTOR FOR HVAC CONTROLS AND INTERFACE.

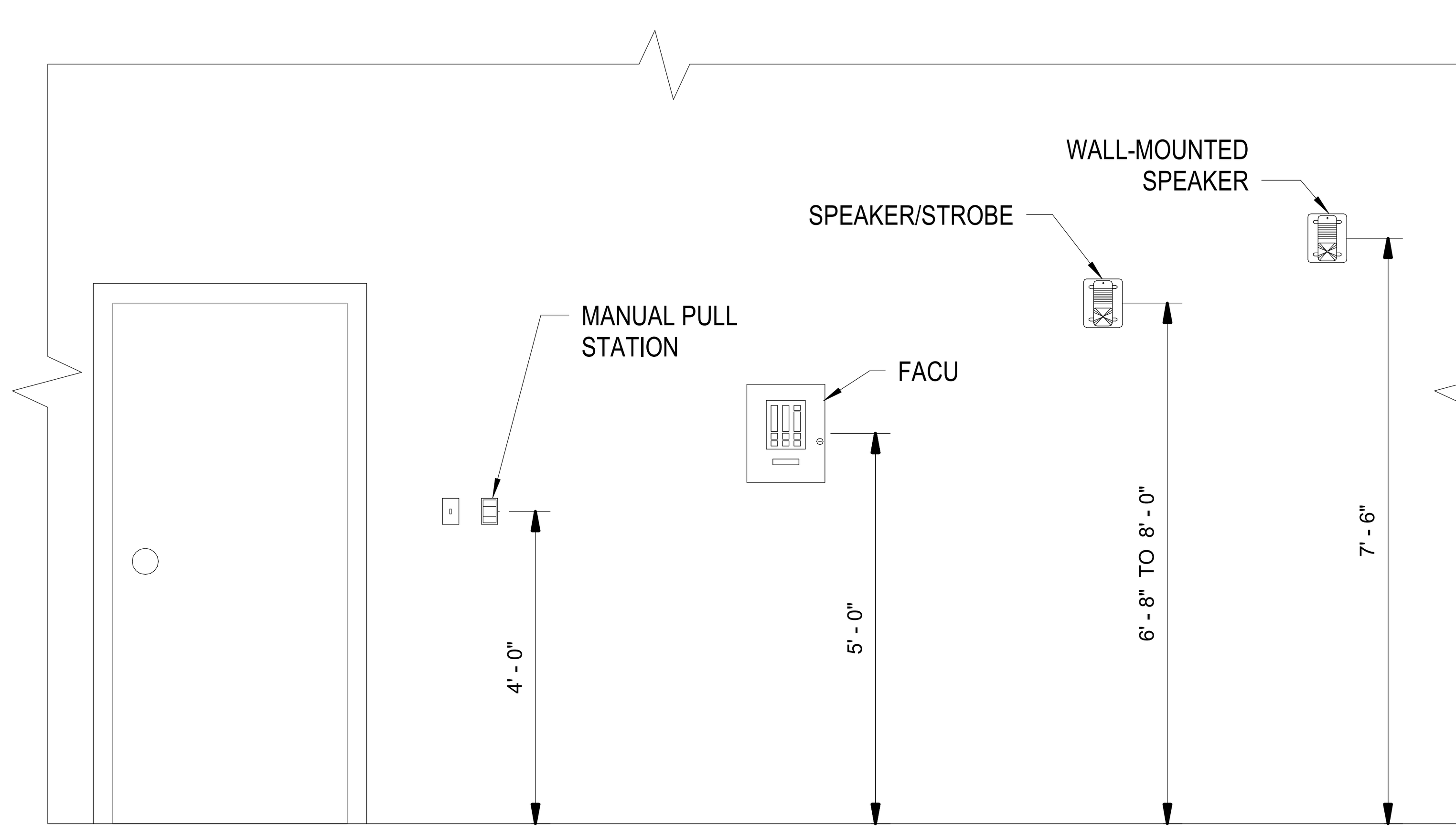
37. DEVICE AND APPLIANCE LOCATIONS AND QUANTITIES ARE SHOWN FOR REFERENCE ONLY. DEVICES AND APPLIANCES IN ADDITION TO THOSE SHOWN MUST BE PROVIDED AT NO ADDITIONAL COST TO THE GOVERNMENT TO MEET REQUIREMENTS OF THE UFC'S AND NFPA CODES.

38. LOCATIONS OF ADDRESSABLE INPUT/OUTPUT MODULES FOR LIGHTING CONTROLS RELAYS ARE NOT INDICATED. REFER TO ELECTRICAL LIGHTING DRAWINGS FOR QUANTITIES AND LOCATIONS OF LIGHTING CONTROLS REQUIRING FIRE ALARM SYSTEM CONTROL.

39. PROVIDE NOTIFICATION APPLIANCES LOCATED TO AVOID OBSTRUCTIONS AND OTHER WALL OR CEILING MOUNTED ITEMS. PROVIDE CANDELA RATING TO ACCOUNT FOR FINAL APPLIANCE LOCATION WITHIN SPACES.

FIRE ALARM LEGEND

Symbol	Description
P	MANUAL PULL STATION
XX	FIRE ALARM SPEAKER/STROBE WITH CLEAR LENS XX = CANDELA RATING
XX	FIRE ALARM SPEAKER/STROBE WITH CLEAR LENS (CEILING-MOUNTED)
	FIRE ALARM AND MASS NOTIFICATION SPEAKER
	PHOTOELECTRIC SMOKE DETECTOR
	DUCT SMOKE DETECTOR
FAMN	FIRE ALARM AND MASS NOTIFICATION CONTROL UNIT
NAC	NOTIFICATION APPLIANCE CIRCUIT PANEL
XMIT	TRANSMITTER
MM	MONITOR MODULE
LOC	LOCAL OPERATOR CONSOLE (RECESSED) WITH REMOTE MICROPHONE AND HVAC SHUTDOWN SWITCH
FSA	FIRE ALARM REMOTE ANNUNCIATION PANEL (RECESSED)
EOL	END-OF-LINE RESISTOR
R	RELAY MODULE
	TAMPER SWITCH
	WATERFLOW SWITCH
WP	INDICATES WEATHERPROOF



A4 **FIRE ALARM DEVICE ELEVATION**

SCALE: NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

JEFFREY J. DEMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER COMMONWEALTH OF MASSACHUSETTS 10-1-25

SEAL

Bowman FIRE & LIFE SAFETY

APPROVED	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DESIGNER	J. DEMAINE
DRAWN BY	C. MANTSCH
CHECKED BY	J. MCLEAN
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

FIRE ALARM NOTES

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 225 OF 282

FA001

DRAWFORM REVISION: 14 SEP 2023

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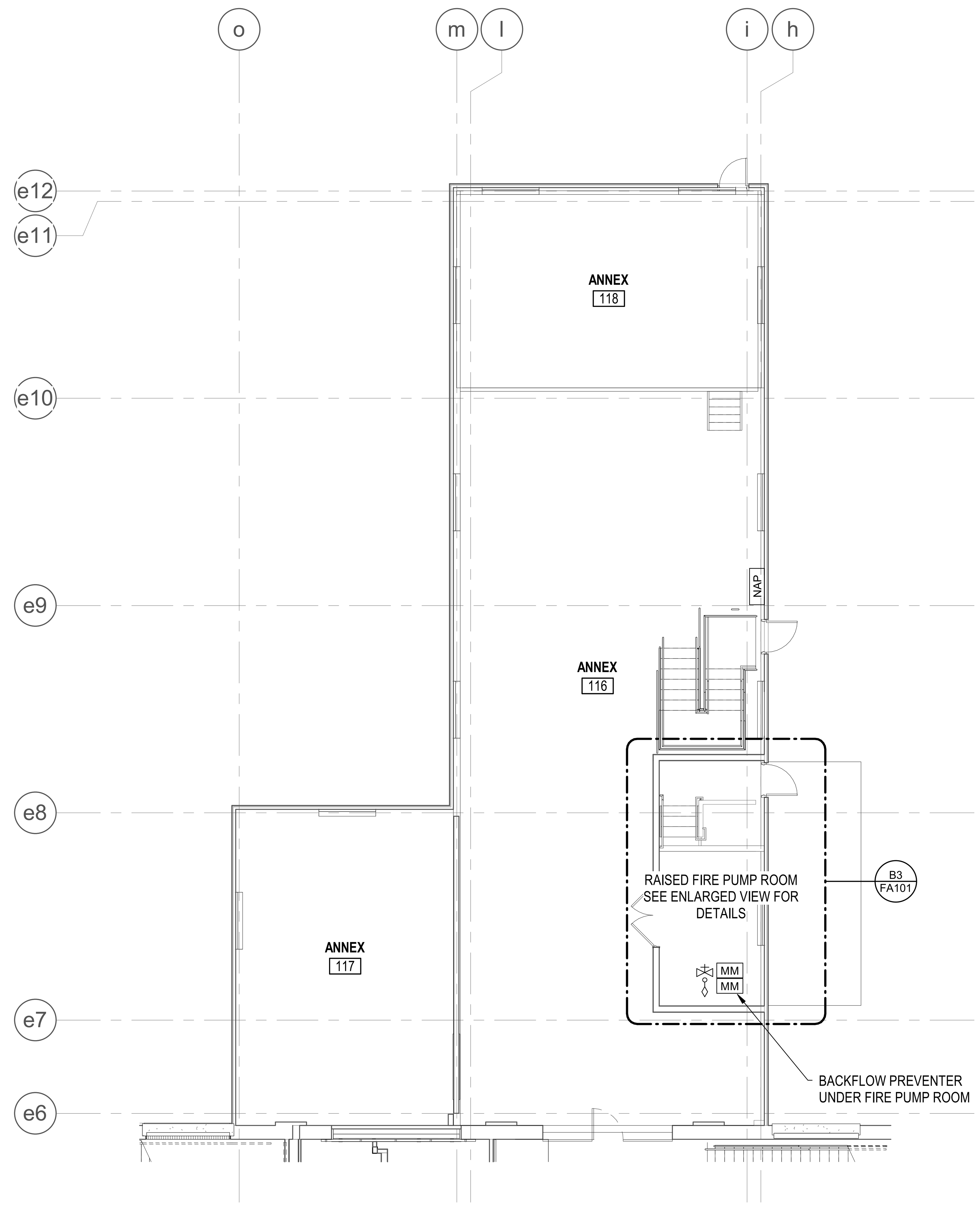
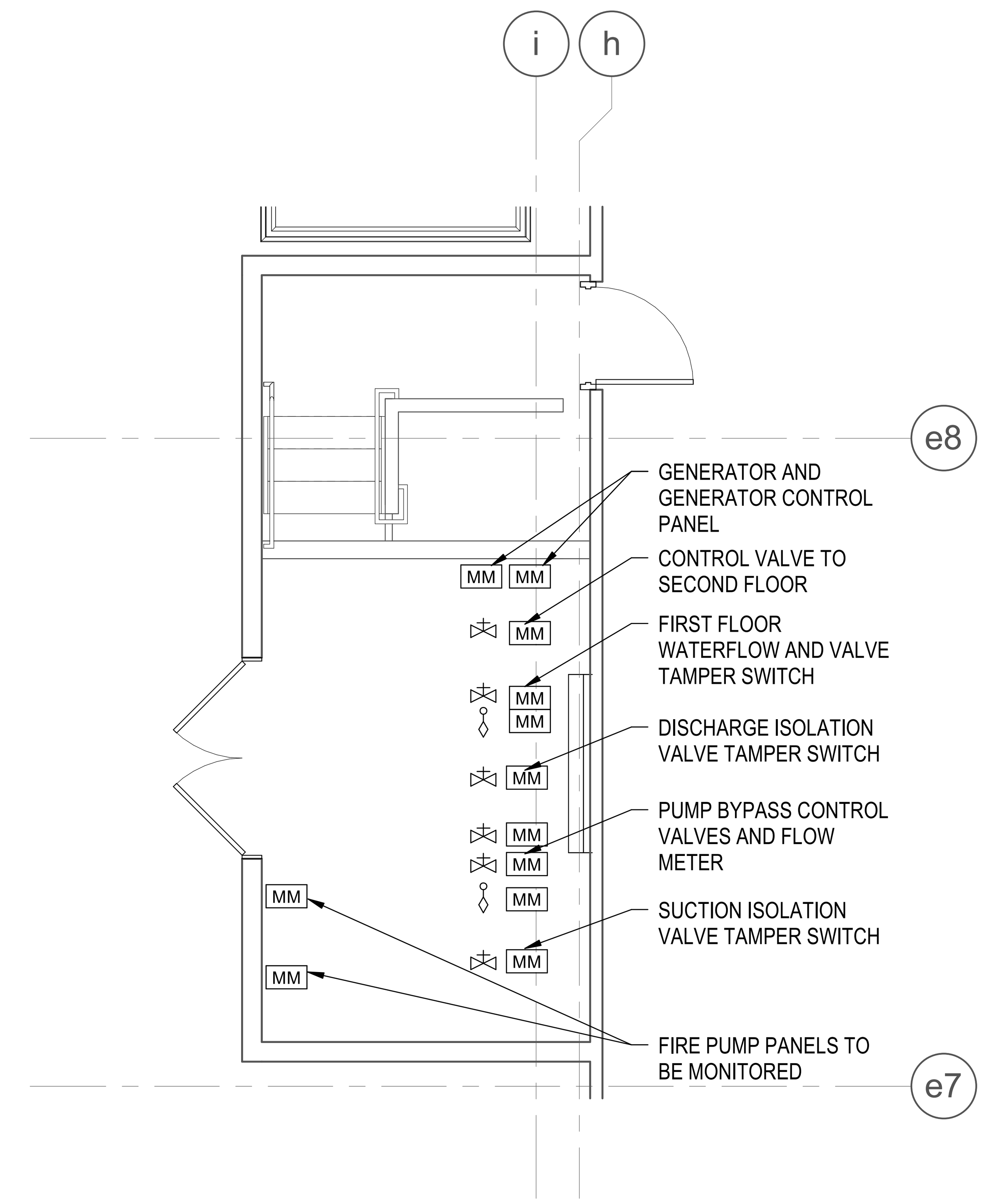


A1 FIRST FLOOR PART PLAN - EAST - FIRE ALARM

SCALE: 1/8" = 1'-0"



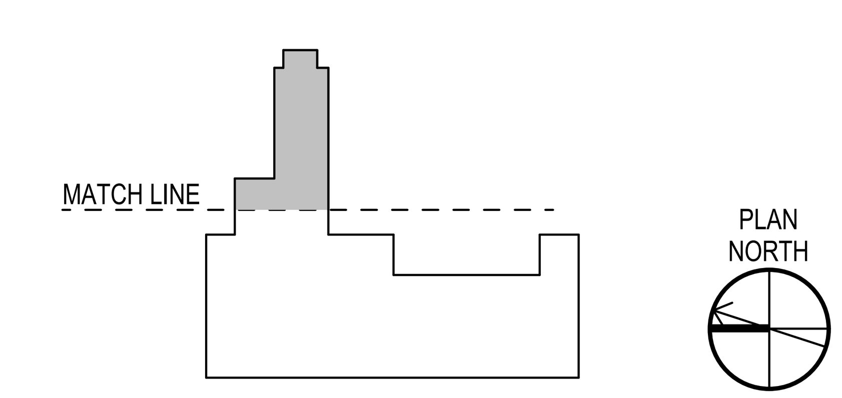
B3 FIRE PUMP ROOM - FIRE ALARM

SCALE: 1/4" = 1'-0"

FIRE ALARM NOTES

1. SEE SHEET FA001 AND FA501 FOR FIRE ALARM AND MASS NOTIFICATION SYSTEM SYMBOLS, NOTES AND DETAILS.
2. PROVIDE CONTROL MODULES ADJACENT TO LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROL DEVICES REFERENCE THE ELECTRICAL DRAWINGS FOR LOCATIONS OF LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROLS.

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

JEFFREY J. DeMAINE, FIRE PROTECTION, No. 41002, REGISTERED PROFESSIONAL ENGINEER, COMMONWEALTH OF MASSACHUSETTS, 10-1-25

SEAL

Bowman
FIRE & LIFE SAFETY

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER J. DEMAINE
DRAWN BY C. MANTSCH
CHECKED BY J. MCLEAN
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR PART PLAN - EAST - FIRE ALARM

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 226 OF 282
FA101

DRAWFORM REVISION: 14 SEP 2023

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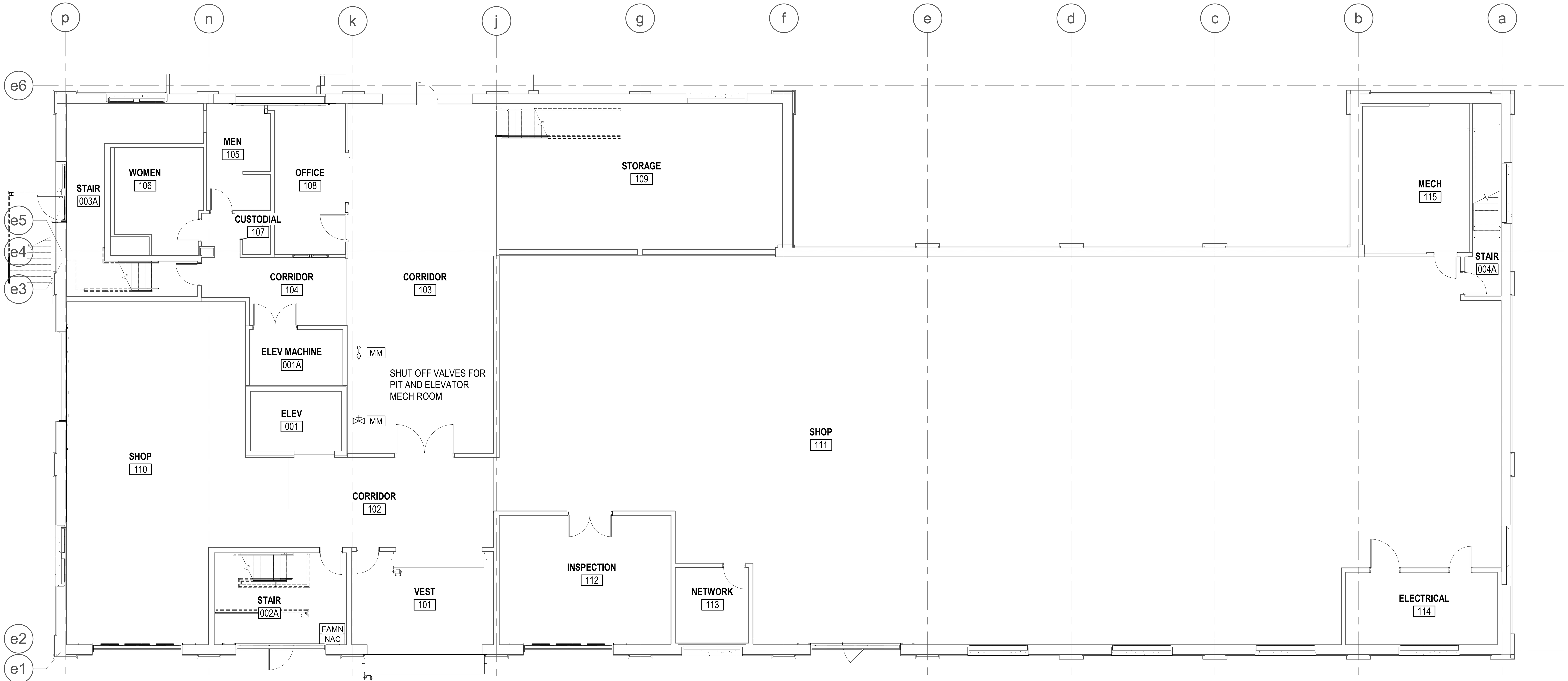

B1 FIRST FLOOR PART PLAN - WEST - FIRE ALARM

SCALE: 1/8" = 1'-0"

FIRE ALARM NOTES

1. SEE SHEET FA001 AND FA501 FOR FIRE ALARM AND MASS NOTIFICATION SYSTEM SYMBOLS, NOTES AND DETAILS.
2. PROVIDE CONTROL MODULES ADJACENT TO LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROL DEVICES REFERENCE THE ELECTRICAL DRAWINGS FOR LOCATIONS OF LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROLS.

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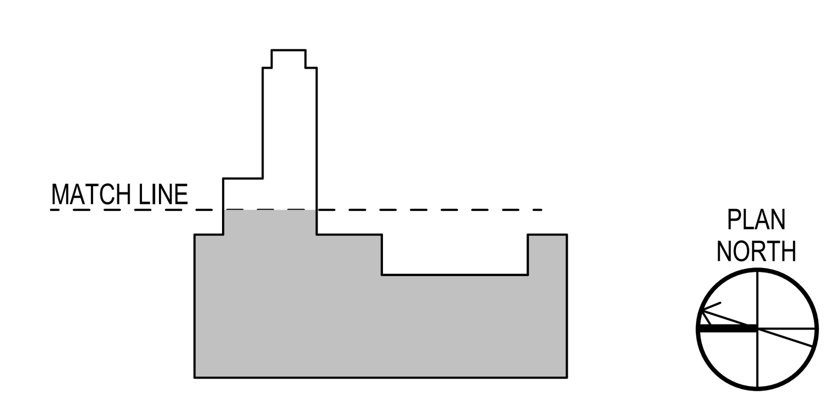

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COMMONWEALTH OF MASSACHUSETTS JEFFREY J. DeMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER 10-1-25

SEAL

Bowman FIRE & LIFE SAFETY

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FOR COMMANDER NAVFAC
ACTIVITY
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DEPARTMENT OF THE NAVY
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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
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PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR PART PLAN - WEST - FIRE ALARM

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 227 OF 282
FA102

DRAWFORM REVISION: 14 SEP 2023

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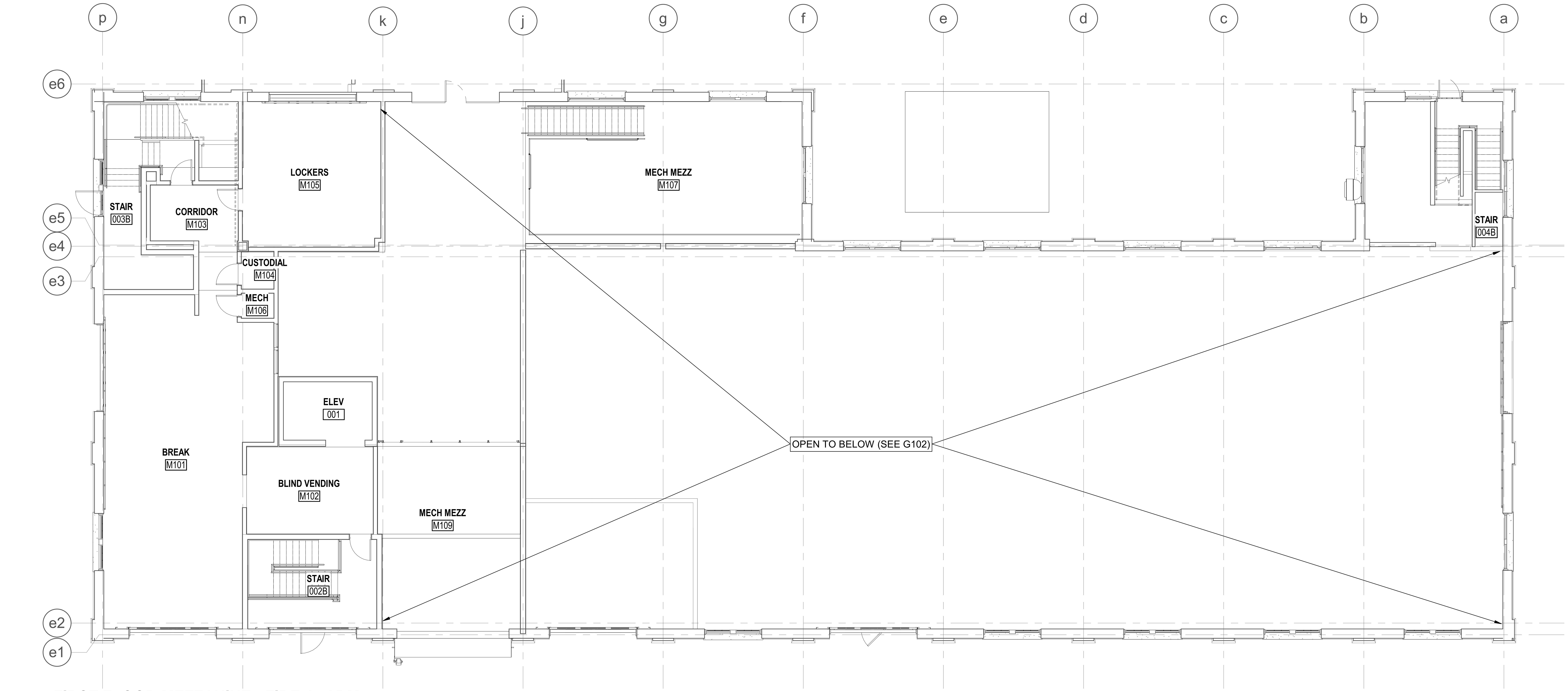


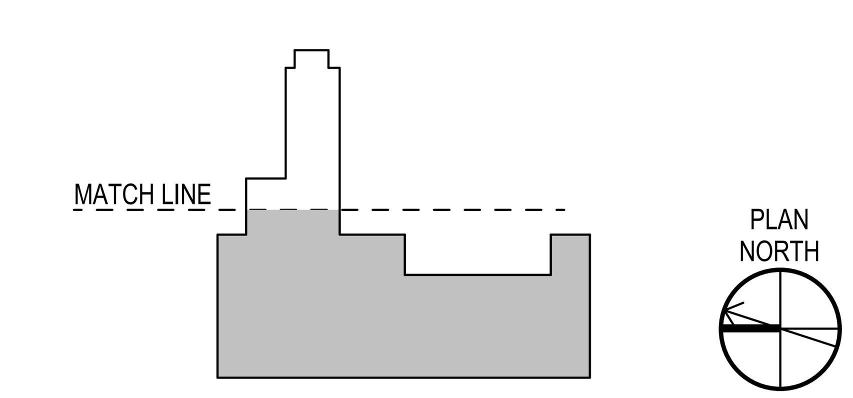
B1 FIRST FLOOR MEZZANINE - FIRE ALARM

SCALE: 1/8" = 1'-0"

FIRE ALARM NOTES

1. SEE SHEET FA001 AND FA501 FOR FIRE ALARM AND MASS NOTIFICATION SYSTEM SYMBOLS, NOTES AND DETAILS.
2. PROVIDE CONTROL MODULES ADJACENT TO LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROL DEVICES REFERENCE THE ELECTRICAL DRAWINGS FOR LOCATIONS OF LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROLS.

KEY PLAN



GRAPHIC SCALE(S)

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COMMONWEALTH OF MASSACHUSETTS JEFFREY J. DeMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER 10-1-25

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FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER J. DEMAINE

DRAWN BY C. MANTSCH

CHECKED BY J. MCLEAN

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

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DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR MEZZANINE - FIRE ALARM

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 228 OF 282

FA103

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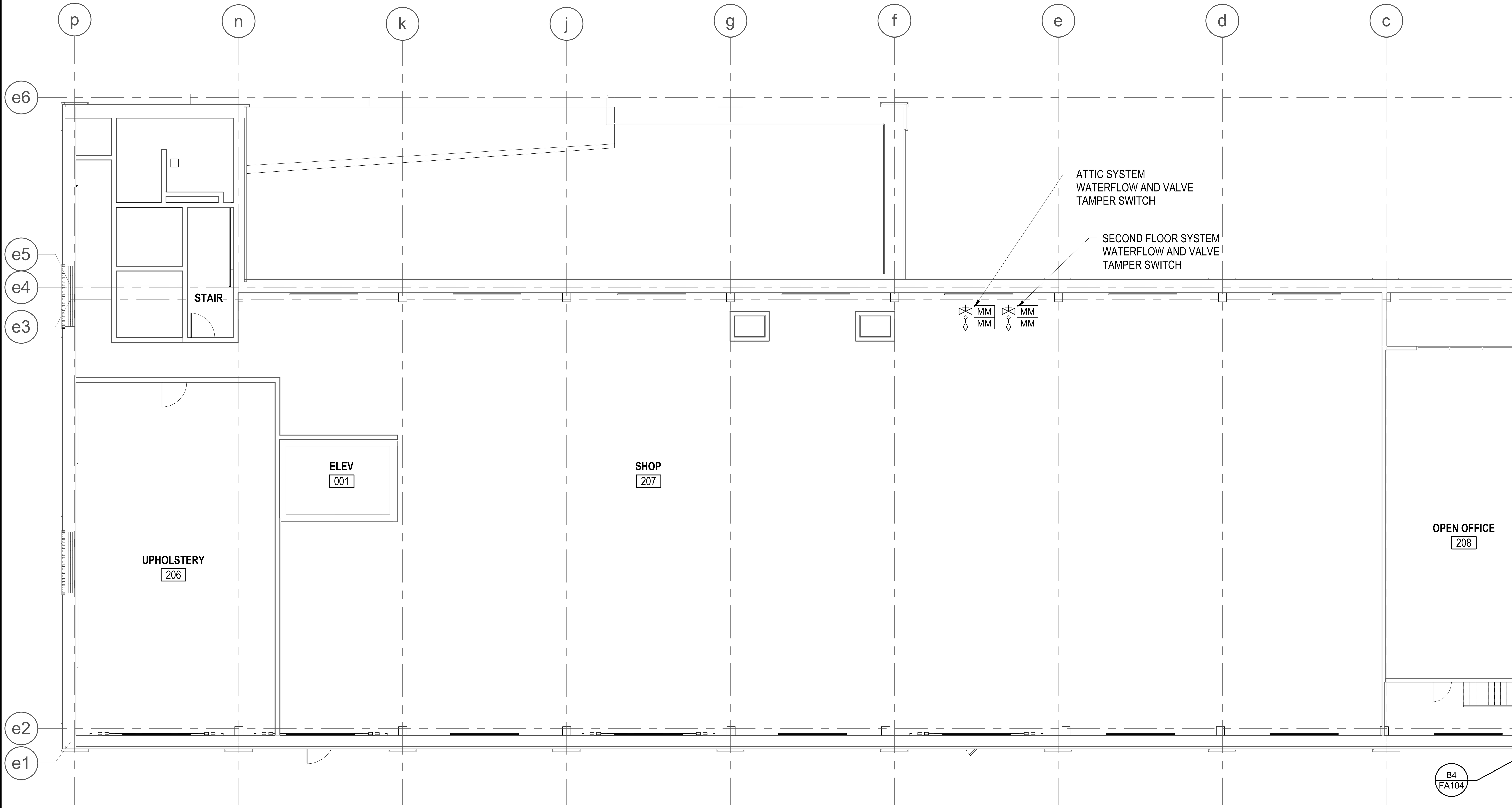
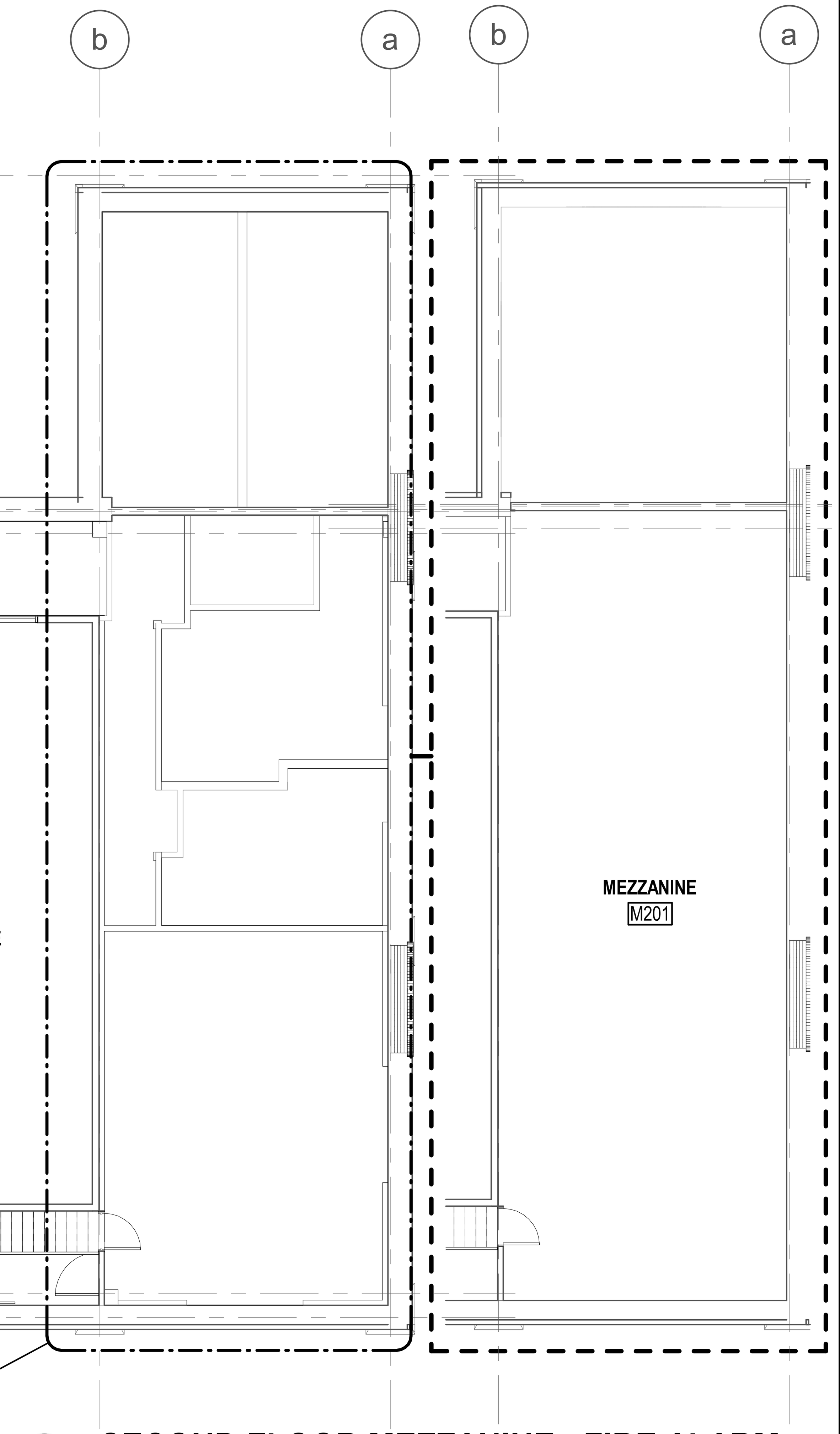


B1 SECOND FLOOR - FIRE ALARM

SCALE: 1/8" = 1'-0"



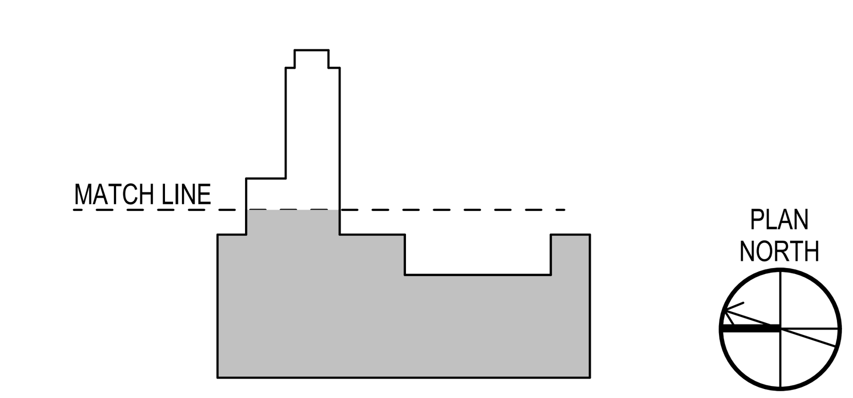
B4 SECOND FLOOR MEZZANINE - FIRE ALARM

SCALE: 1/8" = 1'-0"

FIRE ALARM NOTES

1. SEE SHEET FA001 AND FA501 FOR FIRE ALARM AND MASS NOTIFICATION SYSTEM SYMBOLS, NOTES AND DETAILS.
2. PROVIDE CONTROL MODULES ADJACENT TO LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROL DEVICES REFERENCE THE ELECTRICAL DRAWINGS FOR LOCATIONS OF LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROLS.

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
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NAVFAC

JEFFREY J. DeMAINE, FIRE PROTECTION, No. 41002, REGISTERED PROFESSIONAL ENGINEER, COMMONWEALTH OF MASSACHUSETTS, 10-1-25

SEAL

Bowman FIRE & LIFE SAFETY

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER J. DEMAINE

DRAWN BY C. MANTSCH

CHECKED BY J. MCLEAN

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM/IE

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

SECOND FLOOR - FIRE ALARM

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 229 OF 282

FA104

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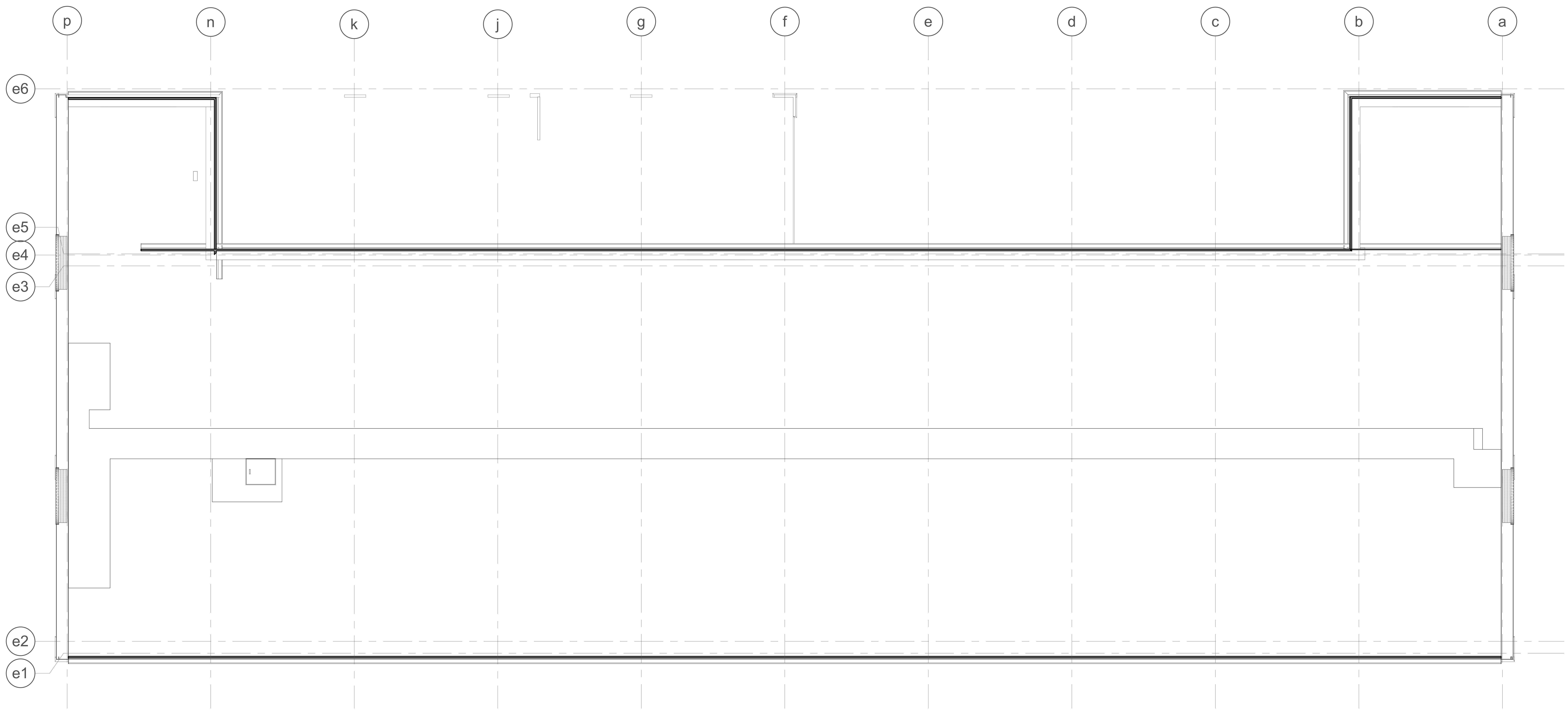

B1 ATTIC - FIRE ALARM

SCALE: 1/8" = 1'-0"

FIRE ALARM NOTES

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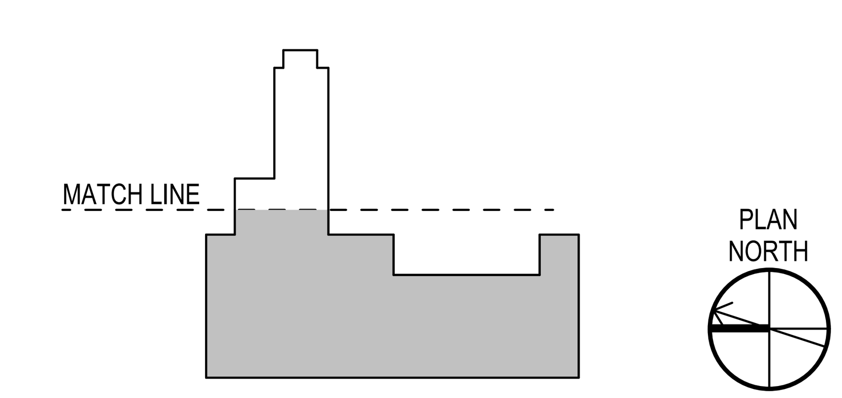
2. PROVIDE CONTROL MODULES ADJACENT TO LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROL DEVICES REFERENCE THE ELECTRICAL DRAWINGS FOR LOCATIONS OF LIGHTING CONTROL PANELS AND LOCAL OCCUPANCY CONTROLS.

KEY PLAN



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COMMONWEALTH OF MASSACHUSETTS JEFFREY J. DeMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER 10-1-25

SEAL

Bowman FIRE & LIFE SAFETY

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ACTIVITY

SATISFACTORY TO DATE

DESIGNER	J. DEMAINE
DRAWN BY	C. MANTSCH
CHECKED BY	J. MCLEAN
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

ATTIC - FIRE ALARM

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 230 OF 282

FA105

DRAWFORM REVISION: 14 SEP 2023

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FIRE ALARM SYSTEM SEQUENCE OF OPERATION MATRIX

	ALARM AT FAMN & FSA	SUPERVISORY AT FAMN	TROUBLE AT FAMN & FSA	ALARM AT RECEIVING STATION	SUPERVISORY AT RECEIVING STATION	TROUBLE AT RECEIVING STATION	ACTIVATE SPEAKERS & STROBES	SILENCE SPEAKERS & STROBES REMAIN ACTIVE	SHUTDOWN AHU / ERV	SHUNT ELEVATOR POWER	RECALL ELEVATOR TO FIRST FLOOR	RECALL ELEVATOR TO SECOND FLOOR	ACTIVATE FIREFIGHTER HAT	OVERRIDE FA MESSAGE	OVERRIDE PA SYSTEM	OVERRIDE OCCUPANCY SENSORS TO INTERIOR LIGHTING
MANUAL PULL STATION	●			●			●									●
SMOKE DETECTOR	●			●			●									●
WATERFLOW SWITCH	●			●			●									●
INPUT RECEIVED FROM MASS NOTIFICATION SYSTEM (MNS)		●			●									●	●	
TAMPER SWITCH		●			●											
FIRE ALARM TROUBLE CONDITION (OPENS, SHORTS OR GROUNDS)			●			●										
ALARM SILENCE AT FAMN OR FSA			●			●		●								
DUCT SMOKE DETECTOR		●			●				●							
SMOKE DETECTOR 2ND FLOOR AND MEZZANINE ELEVATOR LOBBY	●			●			●				●					●
SMOKE DETECTOR 1ST FLOOR ELEVATOR LOBBY	●			●			●					●				●
SMOKE DETECTOR ELEVATOR MACHINE ROOM	●			●			●					●	●			●
WATERFLOW SWITCH FOR ELEVATOR MACHINE ROOM	●			●			●			●						●
FIRE PUMP RUNNING		●			●											
FIRE PUMP PHASE REVERSAL		●			●											
COMMON PUMP ROOM ALARM		●			●											
COMMON MOTOR TROUBLE		●			●											
DIESEL GENERATOR		●			●											

B1 SEQUENCE OF OPERATION MATRIX
SCALE: NOT TO SCALE

MESSAGE #1: ATTENTION

"MAY I HAVE YOUR ATTENTION PLEASE. MAY I HAVE YOUR ATTENTION PLEASE. [INSERT INSTALLATION SPECIFIC MESSAGE HERE]"

MESSAGE #2: FIRE

"MAY I HAVE YOUR ATTENTION PLEASE. MAY I HAVE YOUR ATTENTION PLEASE. A FIRE EMERGENCY HAS BEEN REPORTED IN THE BUILDING. PLEASE LEAVE THE BUILDING BY THE NEAREST EXIT OR EXIT STAIRWAY. DO NOT USE THE ELEVATORS."

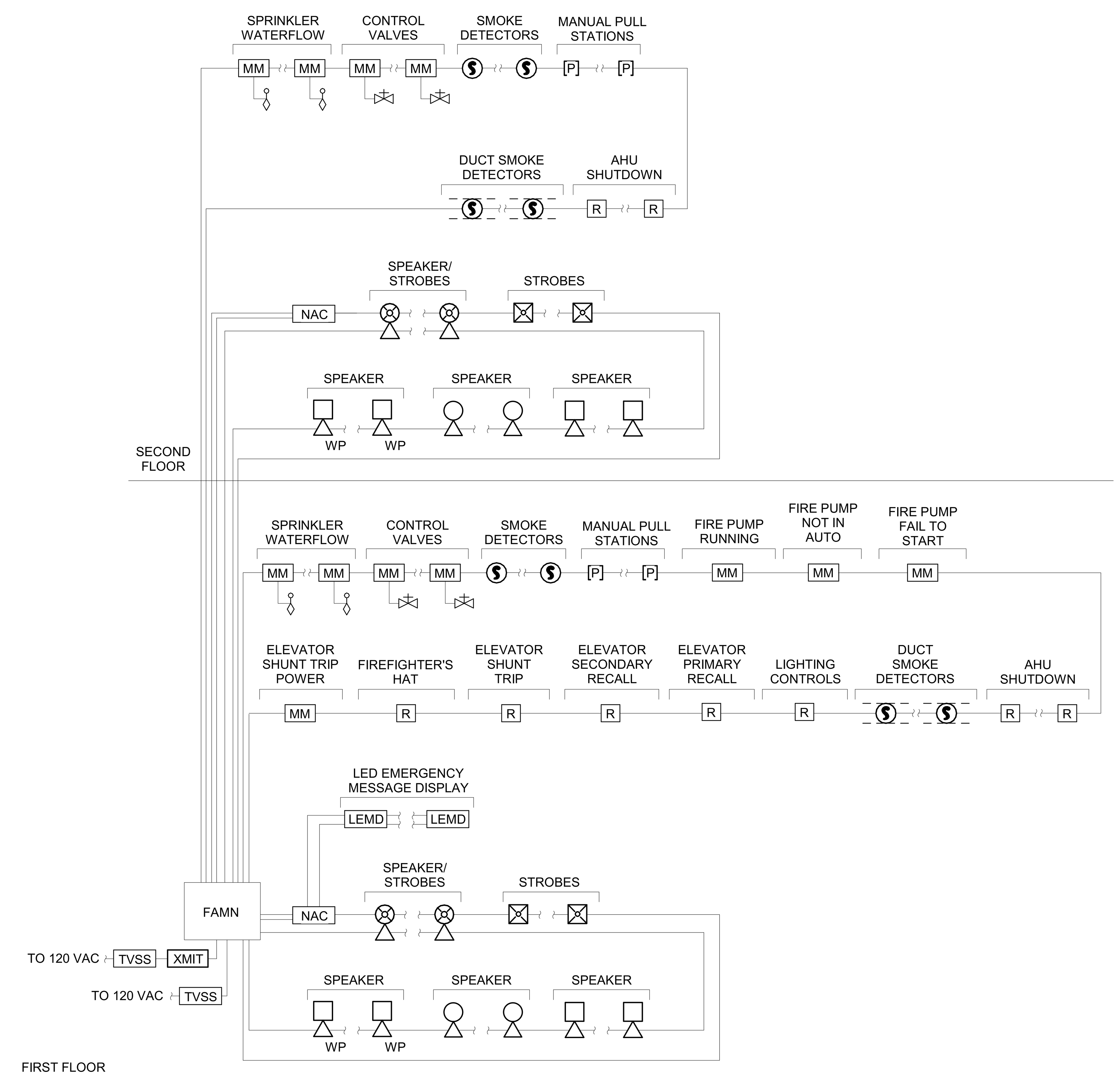
MESSAGE #3: TEST

"MAY I HAVE YOUR ATTENTION PLEASE. MAY I HAVE YOUR ATTENTION PLEASE. THIS IS A TEST OF THE BUILDING MASS NOTIFICATION SYSTEM. PLEASE CONTINUE YOUR NORMAL DUTIES. THIS IS ONLY A TEST."

MESSAGE #4: ALL CLEAR

"MAY I HAVE YOUR ATTENTION PLEASE. MAY I HAVE YOUR ATTENTION PLEASE. AN ALL CLEAR HAS BEEN ISSUED, RESUME NORMAL ACTIVITIES."

A1 PROGRAMMED MASS NOTIFICATION MESSAGES
SCALE: NOT TO SCALE



A4 RISER DIAGRAM
SCALE: NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
JEFFREY L. DeMAINE
FIRE PROTECTION
No. 41002
REGISTERED PROFESSIONAL ENGINEER
10-1-25
SEAL

Bowman
FIRE & LIFE SAFETY

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER J. DEMAINE
DRAWN BY C. MANTSCH
CHECKED BY J. MCLEAN
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
DETAILS - FIRE ALARM

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 231 OF 282
FA501

DRAWFORM REVISION: 14 SEP 2023

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FIRE SPRINKLER NOTES

1. FIRE SPRINKLER GENERAL NOTES, SYMBOLS LISTS & DETAILS APPLY TO SPRINKLER DRAWINGS FOR THIS PROJECT.
2. EXISTING PIPING AND EQUIPMENT NOT SHOWN ON DRAWINGS MUST REMAIN WITH NO CHANGE UNLESS NOTED OTHERWISE.
3. REPLACE THE EXISTING FIRE SPRINKLER SYSTEM THROUGHOUT THE BUILDING. WORK MUST COMPLY WITH UFC 3-600-01, NFPA 13, THE DRAWINGS, THE SPECIFICATION, THE MANUFACTURER'S RECOMMENDATIONS, AND STANDARD INDUSTRY PRACTICES.
4. THE DRAWINGS ARE DIAGRAMMATIC IN NATURE AND EXACT LOCATIONS AND ARRANGEMENTS OF EXISTING/NEW EQUIPMENT, DUCTWORK, PIPING AND OTHER COMPONENTS MUST BE DETERMINED IN THE FIELD WITH DUE CONSIDERATION OF STRUCTURAL, ELECTRICAL, AND ARCHITECTURAL SYSTEMS. PIPING AND SPRINKLER LOCATIONS SHALL BE SELECTED TO AVOID RUNNING IN FRONT OF CHARACTER-DEFINING FEATURES, LIKE WINDOWS, TRIM, DOORS, ETC. TO MINIMIZE IMPACTS ON HISTORIC FABRIC.
5. VERIFY PIPING AND SPRINKLER LOCATIONS IN THE FIELD PRIOR TO REMOVAL, REUSE, AND/OR INSTALLATION. TIE-IN POINT LOCATIONS ARE APPROXIMATE. CONTRACTOR MUST DETERMINE EXACT LOCATIONS IN THE FIELD BASED ON EXISTING CONDITIONS. THE CONTRACTOR MUST SIZE ALL PIPING ACCORDING TO THE AVAILABLE WATER SUPPLY. THE CONTRACTOR MUST VERIFY ELEVATIONS AND LOCATIONS OF NEW SPRINKLER PIPING AND EQUIPMENT PRIOR TO INSTALLATION.
6. INSTALLATION MUST PERMIT ACCESSIBILITY FOR SERVICE AND/OR REPLACEMENT OF EQUIPMENT PROVIDED.
7. IF INTERRUPTION OF SERVICE IS REQUIRED, COORDINATE SHUTDOWN WITH OWNER AND PROVIDE PLAN TO MITIGATE THE IMPAIRMENT. NOTIFY THE OWNER A MINIMUM OF 96 HOURS PRIOR TO SHUT-DOWN.
8. ALL NEW SPRINKLER PIPING MUST BE BLACK CARBON STEEL. SCHEDULE OF PIPE MUST BE PER SPECIFICATION.
9. DO NOT PAINT SPRINKLERS. REMOVE SPRINKLERS IF THEY ARE PAINTED AND PROVIDE NEW SPRINKLERS. ALL EXPOSED PIPE MUST BE PAINTED RED AND LABELED IN ACCORDANCE WITH THE SPECIFICATION.
10. ALL PIPING IN AREAS WITH SUSPENDED CEILINGS MUST BE CONCEALED ABOVE CEILINGS. ALL EXPOSED PIPE MUST BE PAINTED AND/OR LABELED IN ACCORDANCE WITH THE SPECIFICATION. ALL EXISTING PIPING THAT IS EXPOSED AND CURRENTLY NOT PAINTED, MUST BE PAINTED. ANY PIPING OBSERVED TO BE CORRODED THAT IS NOT IDENTIFIED FOR REPLACEMENT MUST BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER.
11. CONTRACTOR MUST SUBMIT SHOP DRAWINGS FOR APPROVAL BEFORE COMMENCING WORK.
12. SPRINKLER CONTRACTOR MUST COORDINATE ALL WORK WITH OTHER BUILDING TRADES. SPRINKLERS ARE TO PROVIDE ADEQUATE COVERAGE PER DESIGN SPECIFICATIONS.
13. PROVIDE ADEQUATE STRUCTURAL SUPPORTS AND SECURE MOUNTING METHODS WITH PROVISIONS FOR VIBRATION ISOLATION AND EXPANSION WHERE REQUIRED. CONTRACTOR MUST PROVIDE ALL NECESSARY PIPE HANGERS AND SUPPORTS AS REQUIRED BY NFPA 13. PROVIDE CLAMPS, OFFSETS, EXPANSION JOINTS, ANCHORS & GUIDES AS NECESSARY TO PREVENT UNDUE STRAIN ON PIPING. COORDINATE WITH ARCHITECTURAL TRADE.
14. CONTRACTOR MUST TAKE ALL NECESSARY MEASURES TO KEEP THE PREMISES DRY AT ALL TIMES AND TO PREVENT WATER DAMAGE. CONTRACTOR MUST REPAIR WATER DAMAGE RESULTING FROM THE WORK, WHETHER INTENTIONAL OR NOT, AT NO COST TO, AND TO THE SATISFACTION OF THE OWNER.
15. NEITHER THE OWNER NOR THE DESIGNER OF RECORD WILL BE RESPONSIBLE FOR PROVIDING A SAFE WORKING PLACE FOR THE CONTRACTOR, SUBCONTRACTORS, OR THEIR EMPLOYEES, OR ANY OTHER INDIVIDUAL RESPONSIBLE TO THEM FOR THE WORK. THIS RESPONSIBILITY RESTS SOLELY ON THE CONTRACTOR.
16. AT THE END OF EACH WORKING DAY, THE CONSTRUCTION SITE MUST BE LEFT IN A CLEAN AND NEAT CONDITION.
17. ALL PIPE PENETRATIONS THRU FIRE RATED WALL, SHAFT OR FLOOR ASSEMBLIES MUST BE PROVIDED WITH A UL LISTED PENETRATION ASSEMBLY THAT MEETS OR EXCEEDS THE RATING OF THE WALL, SHAFT OR FLOOR.
18. HYDRANT FLOW DATA FOR EXISTING UNDERGROUND PIPING AS FOLLOWS:
    DATE OF TEST: 22 JULY 2024
    TIME OF TEST: 0900
    STATIC PRESSURE: 58 PSI
    RESIDUAL PRESSURE: 57 PSI
    FLOW: 1,296 GPM
19. FOR HYDRANT LOCATIONS, SEE SHEET G001 - CODE SUMMARY.
20. BRANCHLINE OUTLETS WITHIN THE ANNEX AREA MUST BE 1 INCH DIAMETER THREADS TO ALLOW FOR FUTURE HIGH-RACK STORAGE.

FIRE PROTECTION LEGEND

LIGHT HAZARD
0.10 GPM/SF OVER HYDRAULICALLY MOST REMOTE 1,500 SF
K-5.6 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

ORDINARY HAZARD
0.20 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-8.0 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

EXTRA HAZARD
0.30 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-11.2 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 500 GPM

24 VDC ELECTRIC BELL

TEST HEADER

FIRE DEPARTMENT CONNECTION

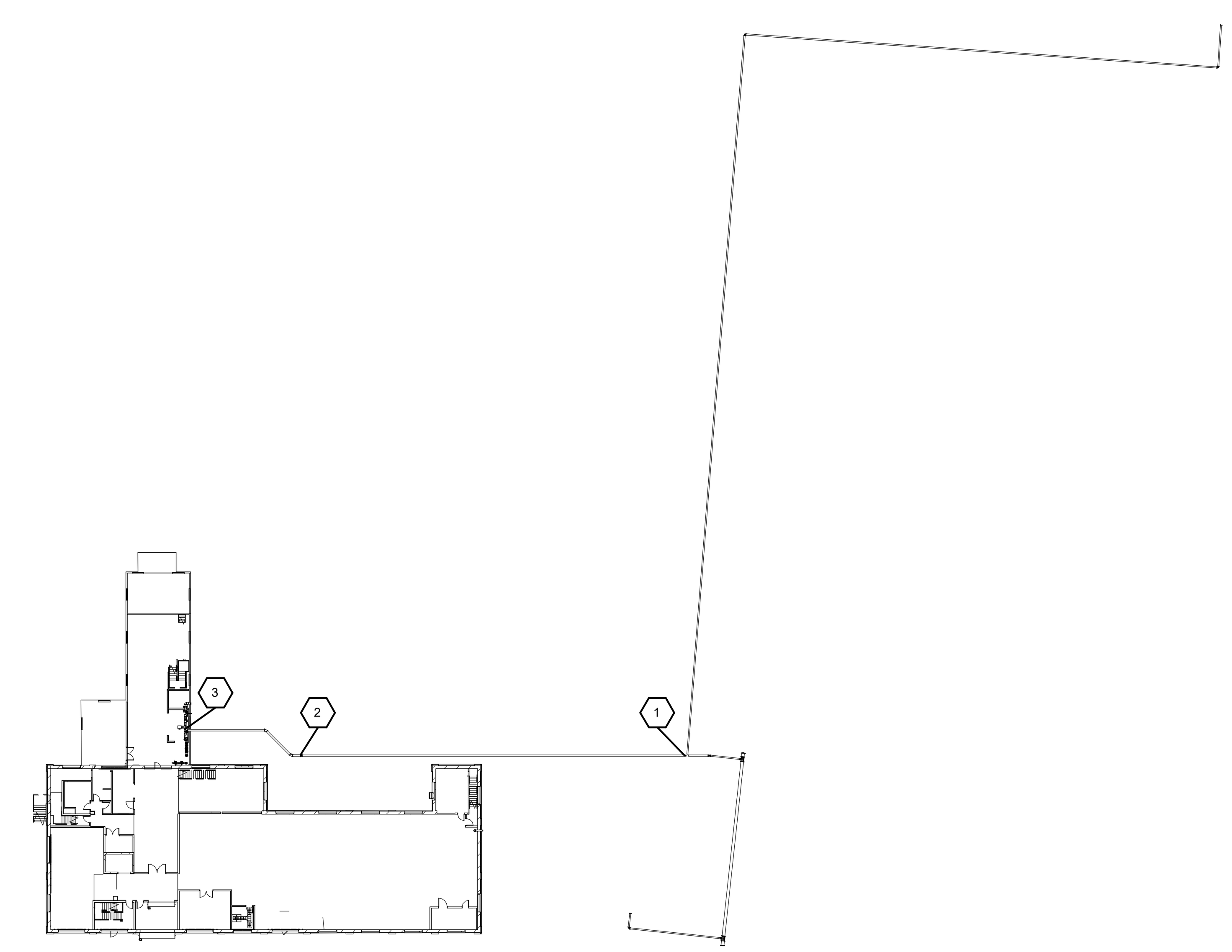
PENDENT SPRINKLER

RECESSED PENDENT SPRINKLER

UPRIGHT SPRINKLER

SCOPE OF WORK DESCRIPTION

PROVIDE FIRE PROTECTION DESIGN AND ENGINEERING SERVICES IN ACCORDANCE WITH REFERENCE (A)/BASIC IQ CONTRACT SAES, UFC 3-600-01 (FIRE PROTECTION ENGINEERING FOR FACILITIES), AND UFC 3-601-02 (OPERATION AND MAINTENANCE: INSPECTION, TESTING, AND MAINTENANCE OF FIRE PROTECTION SYSTEMS), AS WELL AS ALL OTHER APPLICABLE NATIONAL FIRE PROTECTION ASSOCIATION CODES/STANDARDS NOT COVERED BY THESE REFERENCES. FIRE AND LIFE SAFETY REQUIREMENTS MUST BE CONSIDERED AND ADDRESSED FOR EVERY FACILITY DESIGN.



PLAN NORTH

A2 **SITE PIPE**
SCALE: 1/64" = 1'-0"

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
JEFFREY J. DEMAINE
FIRE PROTECTION
No. 41002
REGISTERED PROFESSIONAL ENGINEER
10-1-25
SEAL

Bowman
FIRE & LIFE SAFETY

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER J. DEMAINE
DRAWN BY C. MANTSCH
CHECKED BY J. MCLEAN
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

FIRE PROTECTION NOTES

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 232 OF 282
FX001

DRAWFORM REVISION: 14 SEP 2023

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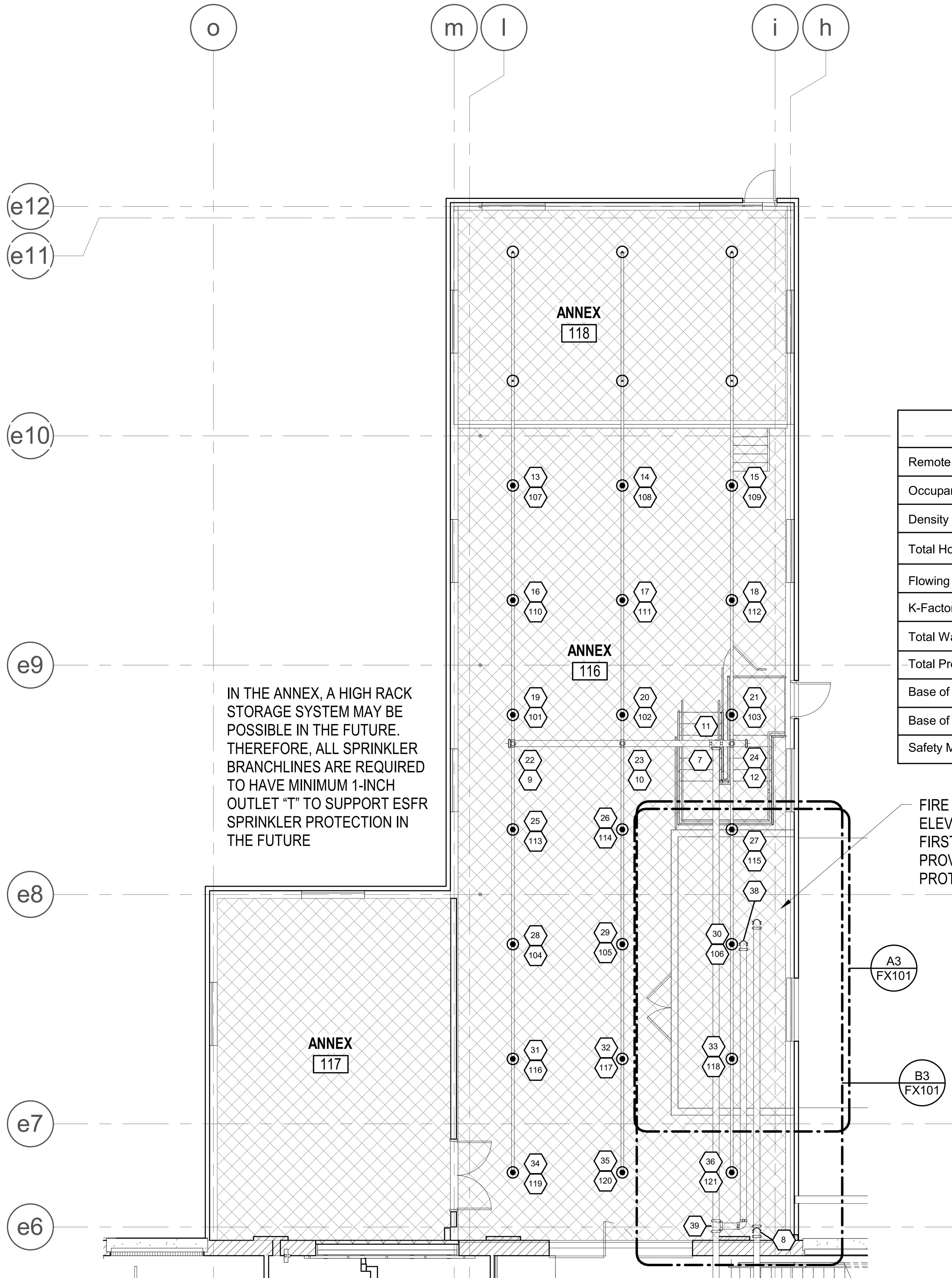
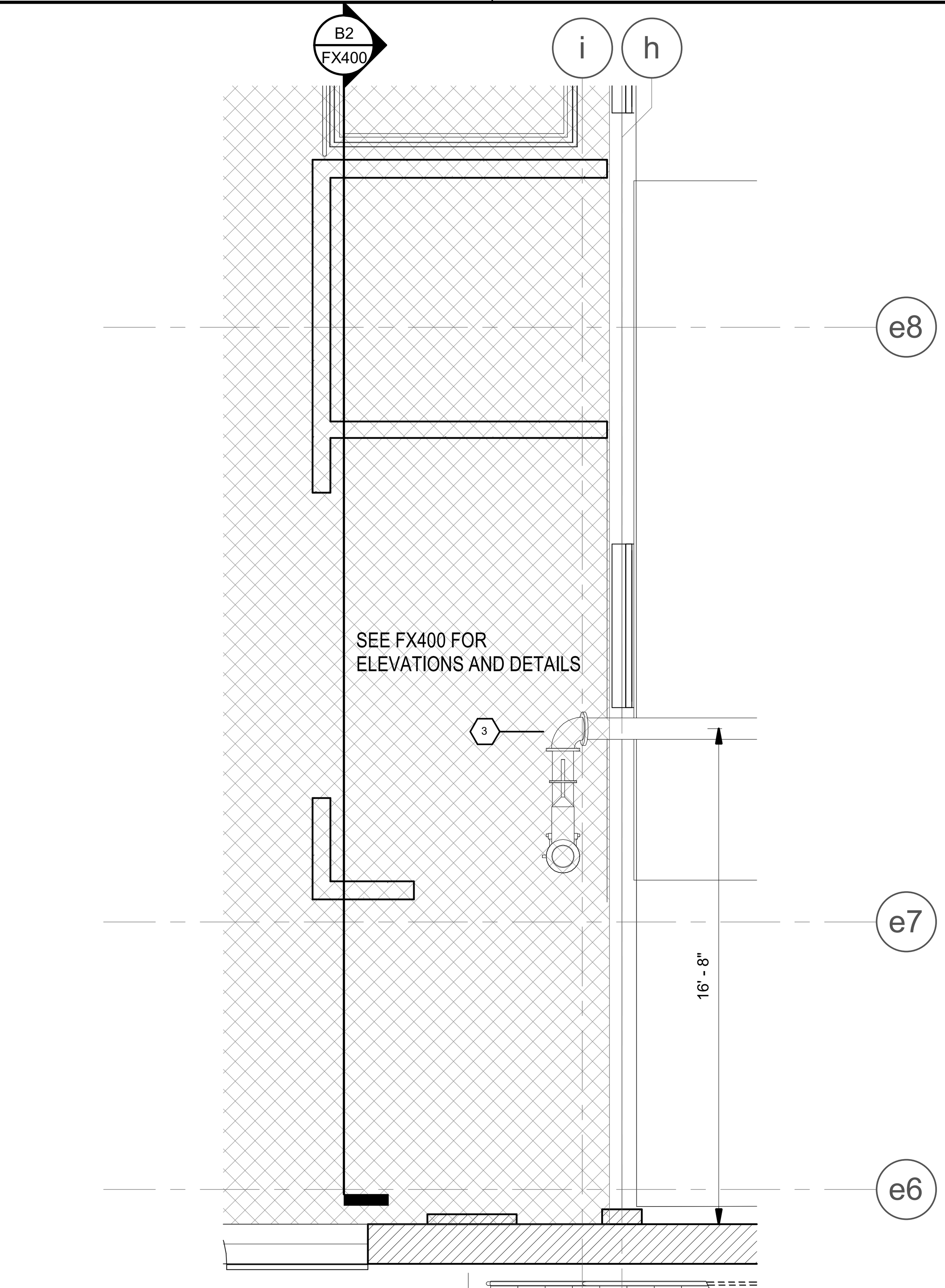


Hydraulic Information	
Remote Area Name	ANNEX
Occupancy Classification	Extra Hazard
Density	0.30 GPM
Total Hose Streams	500.00 GPM
Flowing Heads	21 @ 30.00 GPM
K-Factor	11.2
Total Water Required	1150.14 GPM
Total Pressure Required	-4.44 psi
Base of Riser	1150.14 GPM
Base of Riser	-4.44 psi
Safety Margin	+61.63 (107.76%)

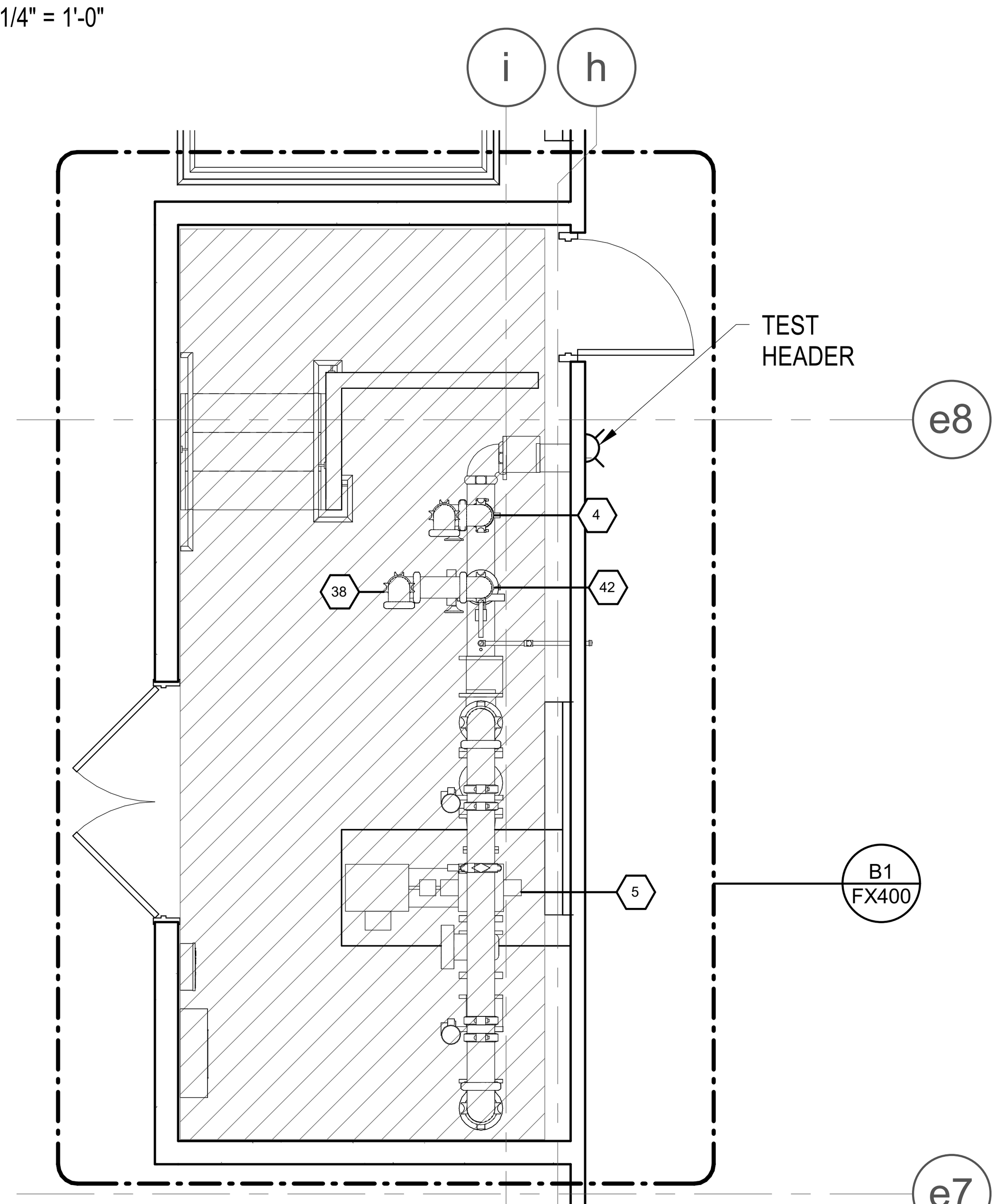
A1 FIRST FLOOR PART PLAN - EAST - FIRE PROTECTION

SCALE: 1/8" = 1'-0"



B3 UNDER FIRE PUMP ROOM - FIRE PROTECTION

SCALE: 1/4" = 1'-0"



A3 FIRE PUMP ROOM - FIRE PROTECTION

SCALE: 1/4" = 1'-0"

FIRE PROTECTION NOTES

1. SEE SHEET FX001, FX400, AND FX501 FOR FIRE PROTECTION NOTES AND DETAILS.

FIRE PROTECTION LEGEND

LIGHT HAZARD
0.10 GPM/SF OVER HYDRAULICALLY MOST REMOTE 1,500 SF
K-5.6 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

ORDINARY HAZARD
0.20 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-8.0 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

EXTRA HAZARD
0.30 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-11.2 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 500 GPM

24 VDC ELECTRIC BELL

TEST HEADER

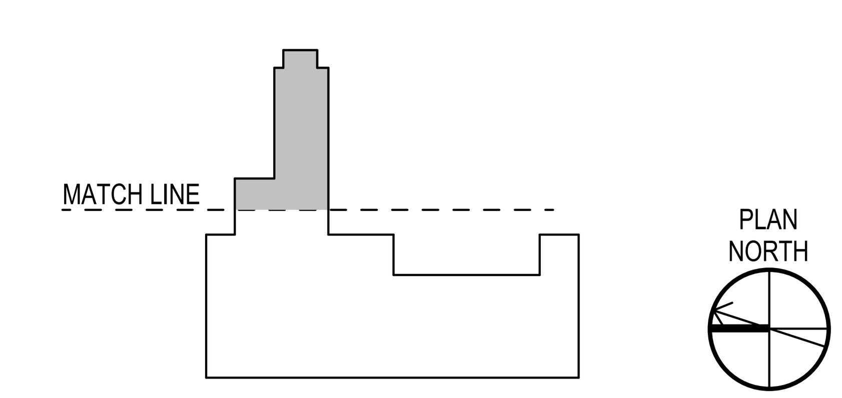
FIRE DEPARTMENT CONNECTION

PENDENT SPRINKLER

RECESSED PENDENT SPRINKLER

UPRIGHT SPRINKLER

KEY PLAN



GRAPHIC SCALE(S)

1/4" = 1' - 0" 0 1' 5' 10' 15'

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

COMMONWEALTH OF MASSACHUSETTS JEFFREY J. DeMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER 10-1-25

SEAL

Bowman FIRE & LIFE SAFETY

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
DESIGNER J. DEMAINE
DRAWN BY C. MANTSCH
CHECKED BY J. MCLEAN
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME
B60 REPAIRS
FIRST FLOOR PART PLAN - EAST - FIRE PROTECTION

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 233 OF 282
FX101

DRAWFORM REVISION: 14 SEP 2023

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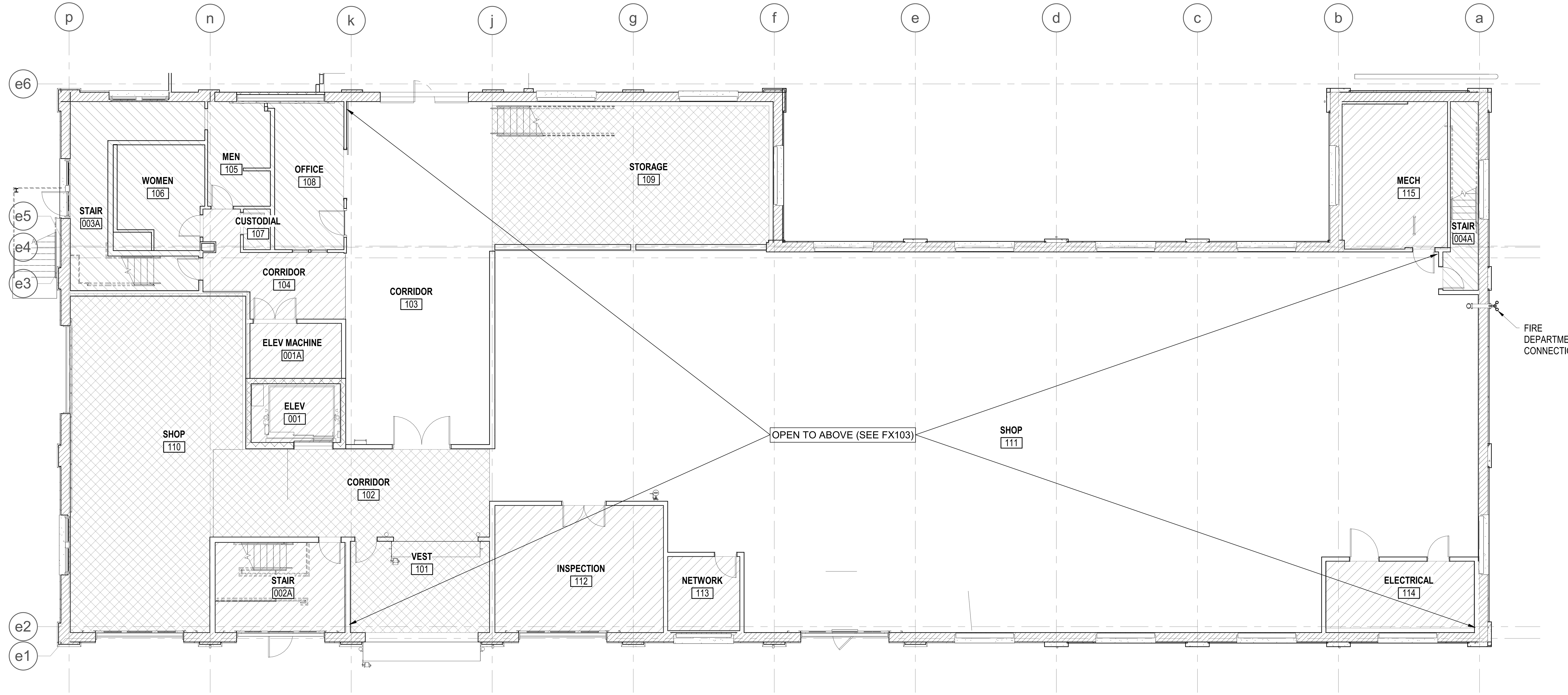


B1 FIRST FLOOR PART PLAN - WEST - FIRE PROTECTION

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE SHEET FX001, FX400, AND FX501 FOR FIRE PROTECTION NOTES AND DETAILS.

FIRE PROTECTION LEGEND

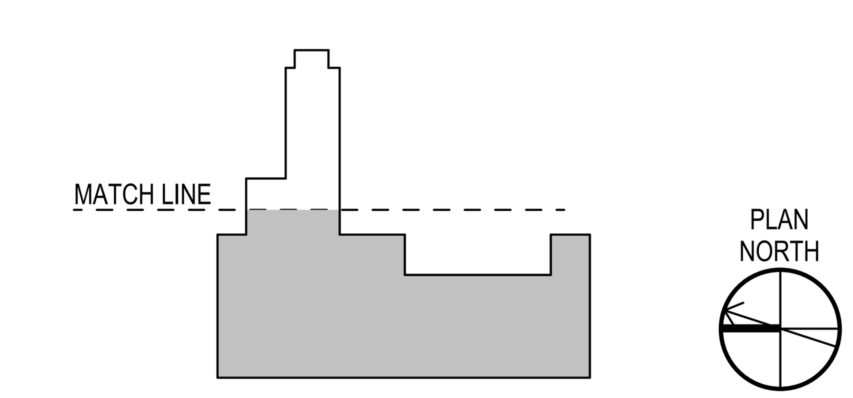
LIGHT HAZARD
0.10 GPM/SF OVER HYDRAULICALLY
MOST REMOTE 1,500 SF
K-5.6 SPRINKLERS WITH OUTSIDE
HOSE STREAM OF 250 GPM

ORDINARY HAZARD
0.20 GPM/SF OVER HYDRAULICALLY
MOST REMOTE 2,500 SF
K-8.0 SPRINKLERS WITH OUTSIDE HOSE
STREAM OF 250 GPM

EXTRA HAZARD
0.30 GPM/SF OVER HYDRAULICALLY
MOST REMOTE 2,500 SF
K-11.2 SPRINKLERS WITH OUTSIDE HOSE STREAM
OF 500 GPM

24 VDC ELECTRIC BELL
TEST HEADER
FIRE DEPARTMENT CONNECTION
PENDENT SPRINKLER
RECESSED PENDENT SPRINKLER
UPRIGHT SPRINKLER

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0"   0 5' 10' 15' 30'

FINAL SUBMISSION 10/01/2025 RSH

NAVFAC

JEFFREY T. DeMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER 10-1-25

Bowman FIRE & LIFE SAFETY

APPROVED FOR COMMANDER NAVFAC

DESIGNER J. DEMAINE
DRAWN BY C. MANTSCH
CHECKED BY J. MCLEAN
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

FIRST FLOOR PART PLAN - WEST - FIRE PROTECTION

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 234 OF 282
FX102

DRAWFORM REVISION: 14 SEP 2023

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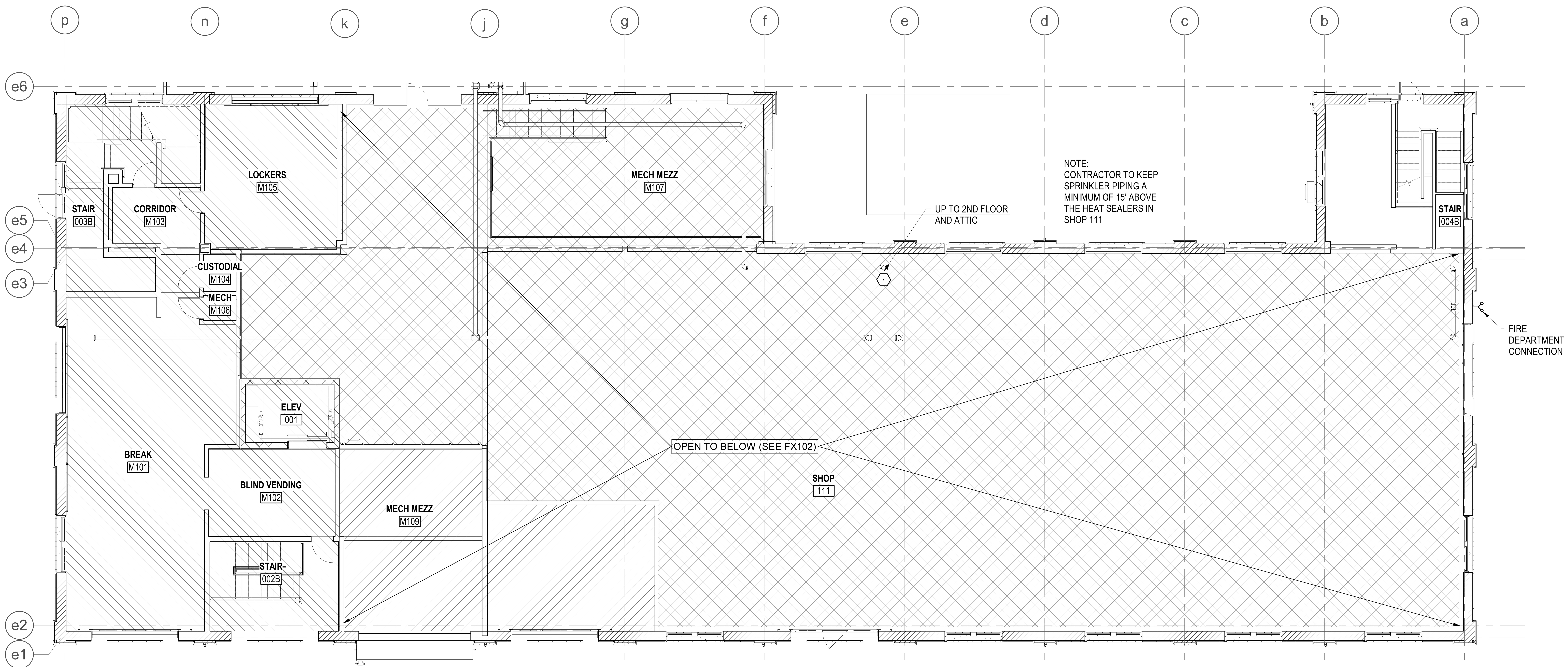


B1 FIRST FLOOR MEZZANINE - FIRE PROTECTION

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE SHEET FX001, FX400, AND FX501 FOR FIRE PROTECTION NOTES AND DETAILS.

FIRE PROTECTION LEGEND

LIGHT HAZARD
0.10 GPM/SF OVER HYDRAULICALLY
MOST REMOTE 1,500 SF
K-5.6 SPRINKLERS WITH OUTSIDE
HOSE STREAM OF 250 GPM

ORDINARY HAZARD
0.20 GPM/SF OVER HYDRAULICALLY
MOST REMOTE 2,500 SF
K-8.0 SPRINKLERS WITH OUTSIDE HOSE
STREAM OF 250 GPM

EXTRA HAZARD
0.30 GPM/SF OVER HYDRAULICALLY
MOST REMOTE 2,500 SF
K-11.2 SPRINKLERS WITH OUTSIDE HOSE STREAM
OF 500 GPM

24 VDC ELECTRIC BELL

TEST HEADER

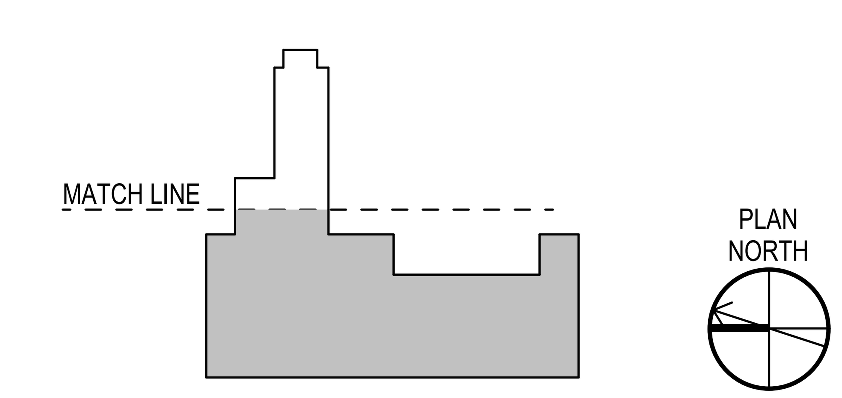
FIRE DEPARTMENT CONNECTION

PENDENT SPRINKLER

RECESSED PENDENT SPRINKLER

UPRIGHT SPRINKLER

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

JEFFREY J. DeMAINE, FIRE PROTECTION, No. 41002, COMMONWEALTH OF MASSACHUSETTS, REGISTERED PROFESSIONAL ENGINEER, 10-1-25

SEAL

Bowman
FIRE & LIFE SAFETY

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE

DESIGNER	J. DEMAINE
DRAWN BY	C. MANTSCH
CHECKED BY	J. MCLEAN
DESIGN MANAGER	T. CARLETON
PROJECT MANAGER	L. LYONS
FEAD/PM&E	
FIRE PROTECTION	S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

FIRST FLOOR MEZZANINE - FIRE PROTECTION

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
SHEET 235 OF 282
FX103

DRAWFORM REVISION: 14 SEP 2023

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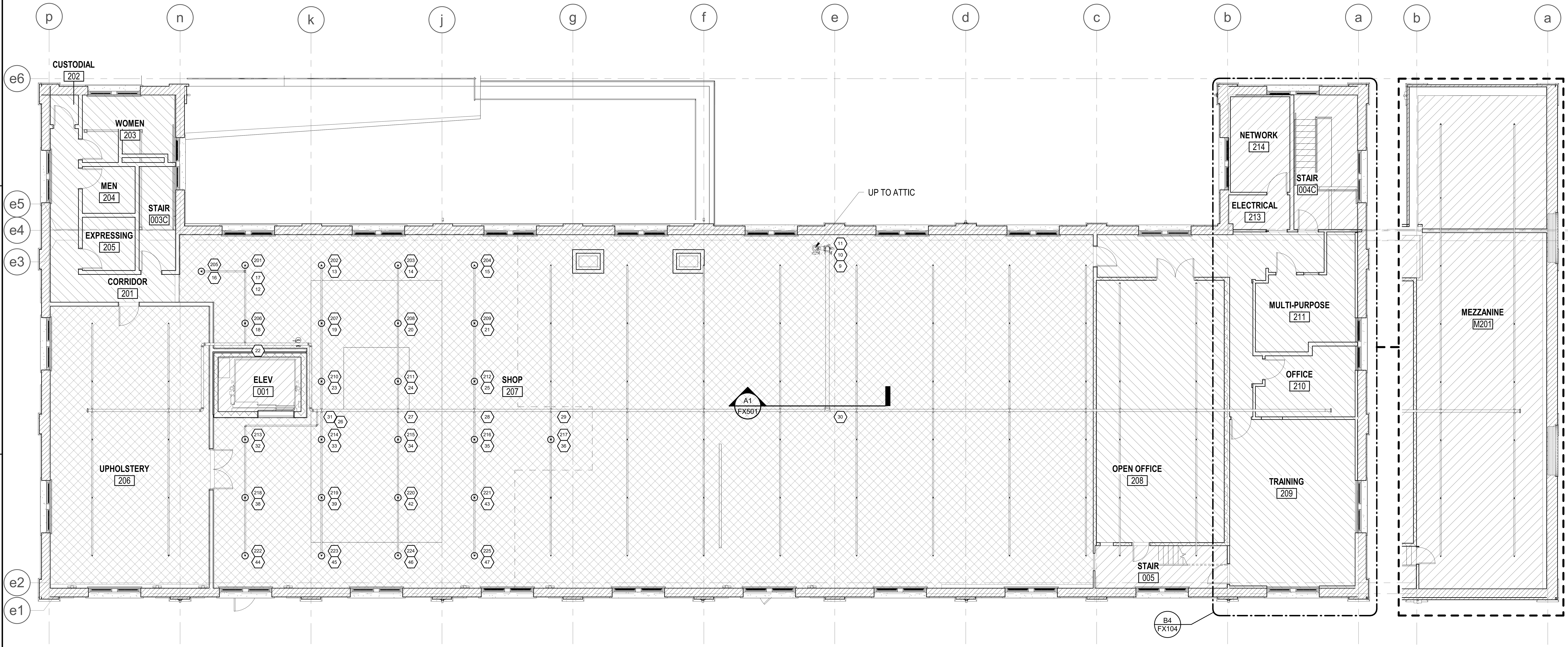


B1 SECOND FLOOR - FIRE PROTECTION

SCALE: 1/8" = 1'-0"

B4 SECOND FLOOR MEZZANINE - FIRE PROTECTION

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE SHEET FX001, FX400, AND FX501 FOR FIRE PROTECTION NOTES AND DETAILS.

FIRE PROTECTION LEGEND

LIGHT HAZARD
0.10 GPM/SF OVER HYDRAULICALLY MOST REMOTE 1,500 SF
K-5.6 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

ORDINARY HAZARD
0.20 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-8.0 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

EXTRA HAZARD
0.30 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-11.2 SPRINKLERS WITH AN OUTSIDE HOSE STREAM OF 500 GPM

24 VDC ELECTRIC BELL

TEST HEADER

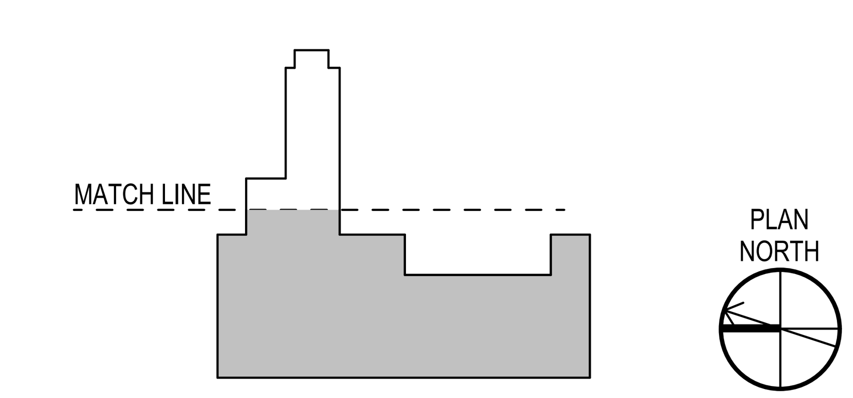
FIRE DEPARTMENT CONNECTION

RECESSED PENDENT SPRINKLER

UPRIGHT SPRINKLER

Hydraulic Information	
Remote Area Name	SECOND FLOOR
Occupancy Classification	Extra Hazard
Density	0.30 GPM
Total Hose Streams	500.00 GPM
Flowing Heads	25 @ 30.00 GPM
K-Factor	11.2
Total Water Required	1343.35 GPM
Total Pressure Required	32.73 psi
Base of Riser	1343.35 GPM
Base of Riser	32.73 psi
Safety Margin	+24.2 (42.5%)

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

COMMONWEALTH OF MASSACHUSETTS JEFFREY L. DeMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER 10-1-25

SEAL

Bowman FIRE & LIFE SAFETY

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE

DESIGNER J. DEMAINE
DRAWN BY C. MANTSCH
CHECKED BY J. MCLEAN
DESIGN MANAGER T. CARLETON
PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MIDLANT DC CORE
PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

SECOND FLOOR - FIRE PROTECTION

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.

SHEET 236 OF 282

FX104

DRAWFORM REVISION: 14 SEP 2023

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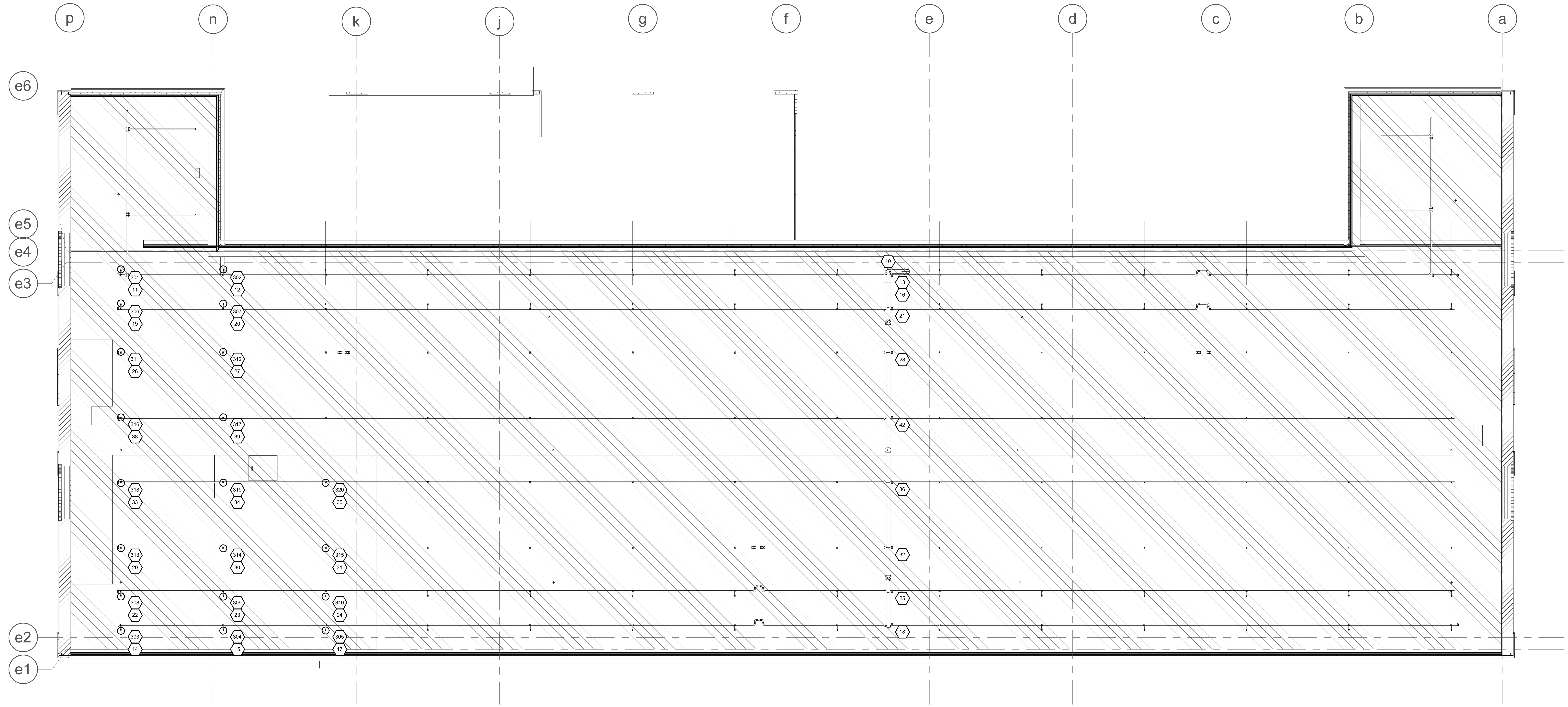


B1 ATTIC - FIRE PROTECTION

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. SEE SHEET FX001, FX400, AND FX501 FOR FIRE PROTECTION NOTES AND DETAILS.

FIRE PROTECTION LEGEND

Hydraulic Information	
Remote Area Name	ATTIC
Occupancy Classification	Light Hazard
Density	0.10 GPM
Total Hose Streams	250.00 GPM
Flowing Heads	20 @ 22.50 GPM
K-Factor	5.6
Total Water Required	723.25 GPM
Total Pressure Required	8.07 psi
Base of Riser	723.25 GPM
Base of Riser	8.07 psi
Safety Margin	+49.59 (86.01%)

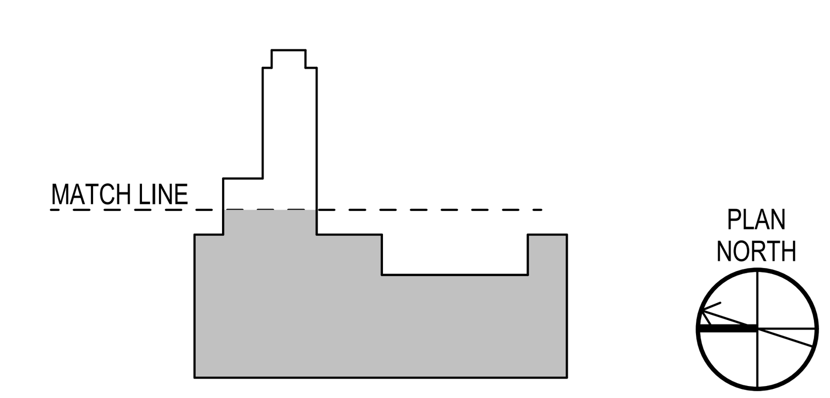
LIGHT HAZARD
0.10 GPM/SF OVER HYDRAULICALLY MOST REMOTE 1,500 SF
K-5.6 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

ORDINARY HAZARD
0.20 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-8.0 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 250 GPM

EXTRA HAZARD
0.30 GPM/SF OVER HYDRAULICALLY MOST REMOTE 2,500 SF
K-11.2 SPRINKLERS WITH OUTSIDE HOSE STREAM OF 500 GPM

24 VDC ELECTRIC BELL
TEST HEADER
FIRE DEPARTMENT CONNECTION
PENDENT SPRINKLER
RECESSED PENDENT SPRINKLER
UPRIGHT SPRINKLER

KEY PLAN



GRAPHIC SCALE(S)

1/8" = 1' - 0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR
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COMMONWEALTH OF MASSACHUSETTS
JEFFREY L. DeMAINE
FIRE PROTECTION
No. 41002
REGISTERED PROFESSIONAL ENGINEER
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FIRE & LIFE SAFETY

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PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME
B60 REPAIRS
ATTIC - FIRE PROTECTION

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NAVFAC DRAWING NO.
SHEET 237 OF 282
FX105

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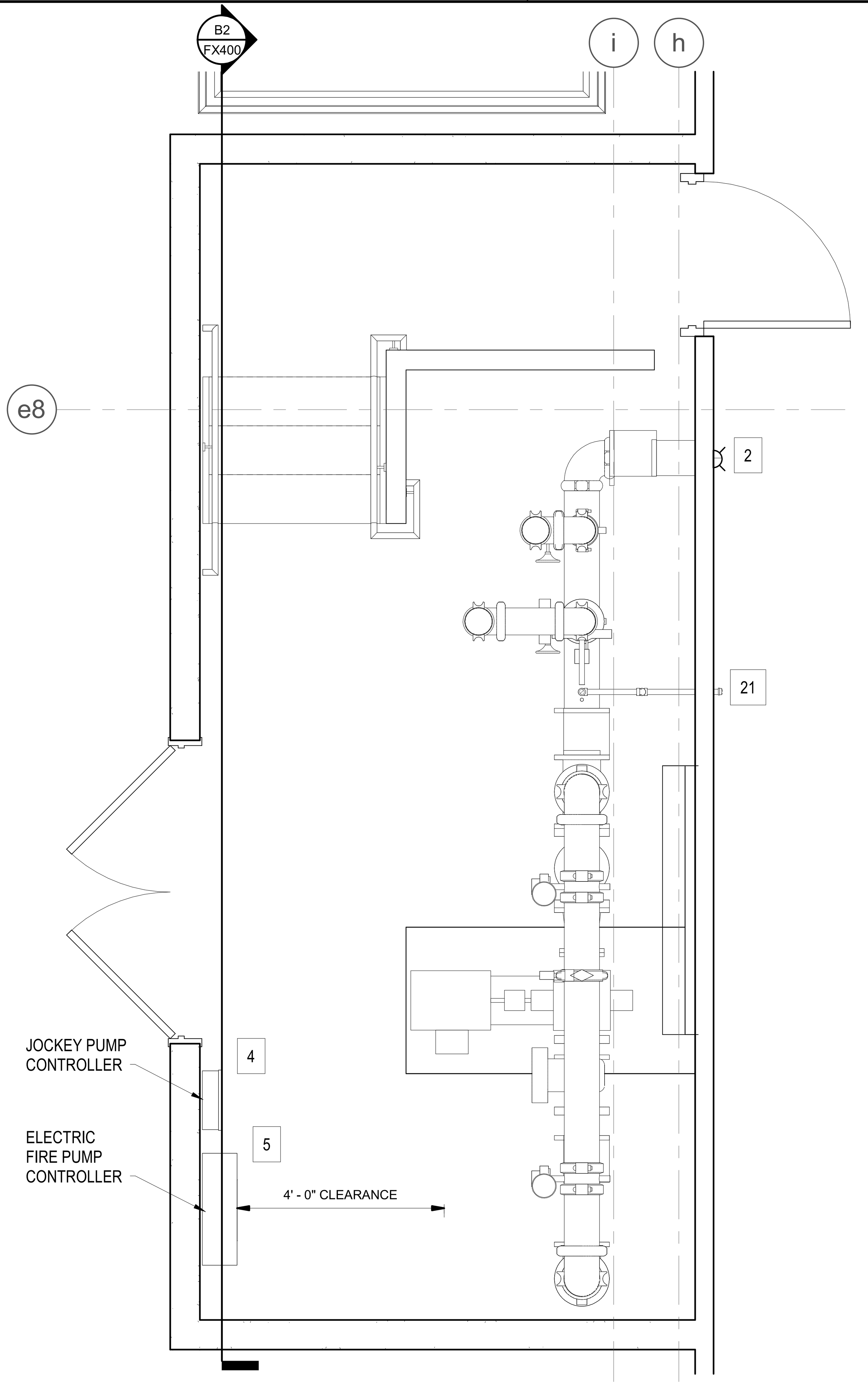
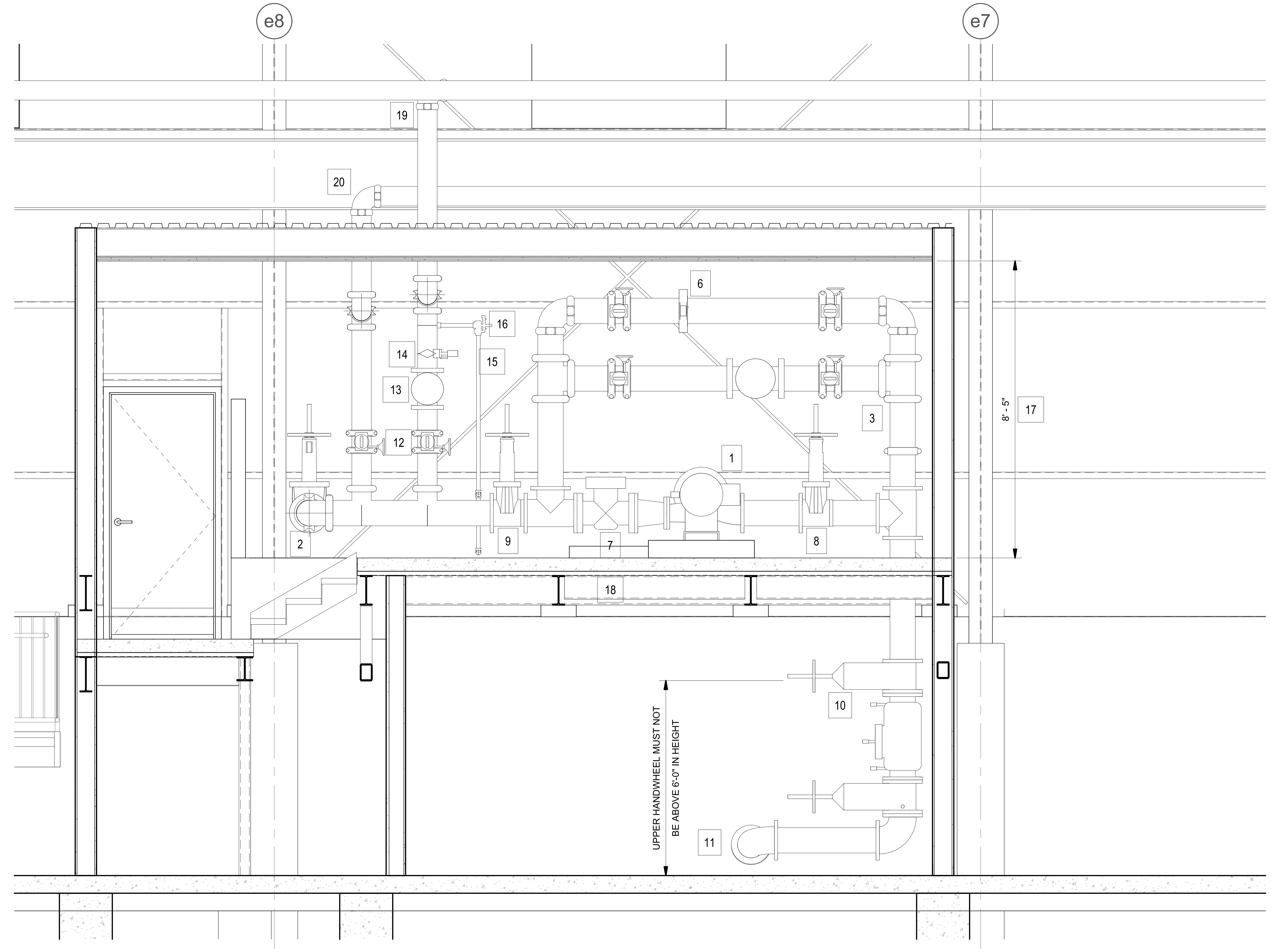


B1 ENLARGED FIRE PUMP ROOM

SCALE: 1/2" = 1'-0"



B2 FIRE PUMP ROOM SECTION

SCALE: 1/2" = 1'-0"

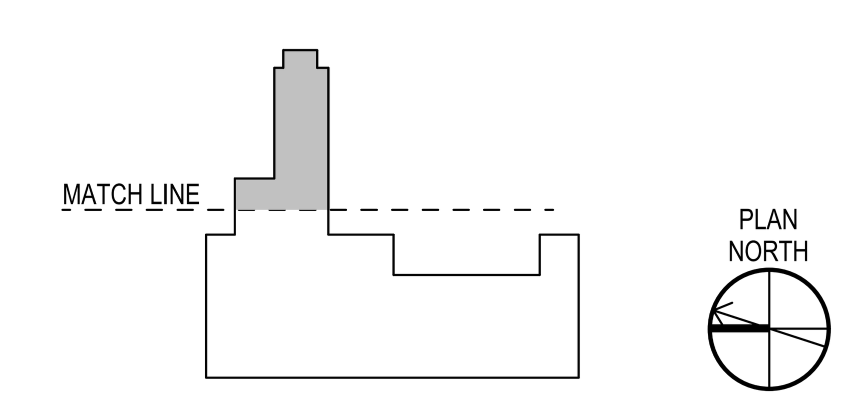
GENERAL NOTES

1. SEE SHEET FX001, FX400, AND FX501 FOR FIRE PROTECTION NOTES AND DETAILS.

FIRE PUMP KEYNOTES

1	NEW ELECTRIC FIRE PUMP, 1,500 GPM WITH 50-PSI BOOST	8	8" SUCTION ISOLATION VALVE	15	FIRST FLOOR SPRINKLER RISER DRAIN TO EXTERIOR
2	6" FEED TO TEST HEADER (SEE FX501/A4)	9	6" DISCHARGE ISOLATION VALVE	16	FIRST FLOOR SPRINKLER RISER INSPECTORS TEST AND DRAIN
3	8" FIRE PUMP BYPASS	10	8" BACKFLOW PREVENTER	17	CEILING HEIGHT
4	JOCKEY PUMP CONTROL PANEL	11	8" INCOMING FROM UNDERGROUND SUPPLY	18	NEW 1-HOUR FIRE PUMP ROOM ENCLOSURE
5	FIRE PUMP CONTROL PANEL	12	FIRST FLOOR SPRINKLER RISER OS&Y VALVE W/TAMPER SWITCH	19	6" TO FIRST FLOOR OVERHEAD SPRINKLER SYSTEM
6	FLOW METER	13	FIRST FLOOR SPRINKLER RISER CHECK VALVE	20	6" TO SECOND FLOOR SPRINKLER CONTROL VALVE ASSEMBLY (SEE FX501)
7	NEW CONCRETE PAD	14	FIRST FLOOR SPRINKLER RISER WATERFLOW SWITCH	21	LOCATE CONCRETE SPLASH BLOCK FOR EXTERIOR DRAIN

KEY PLAN



GRAPHIC SCALE(S)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

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JEFFREY L. DeMAINE FIRE PROTECTION No. 41002 REGISTERED PROFESSIONAL ENGINEER COMMONWEALTH OF MASSACHUSETTS 10-1-25

SEAL

Bowman FIRE & LIFE SAFETY

APPROVED
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO DATE
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CHECKED BY J. MCLEAN
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PROJECT MANAGER L. LYONS
FEAD/PM&E
FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
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MIDLANT DC CORE
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PORTSMOUTH NAVAL SHIPYARD
KITTERY, ME

B60 REPAIRS

ENLARGED FIRE PUMP ROOM

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FX400

DRAWFORM REVISION: 14 SEP 2023

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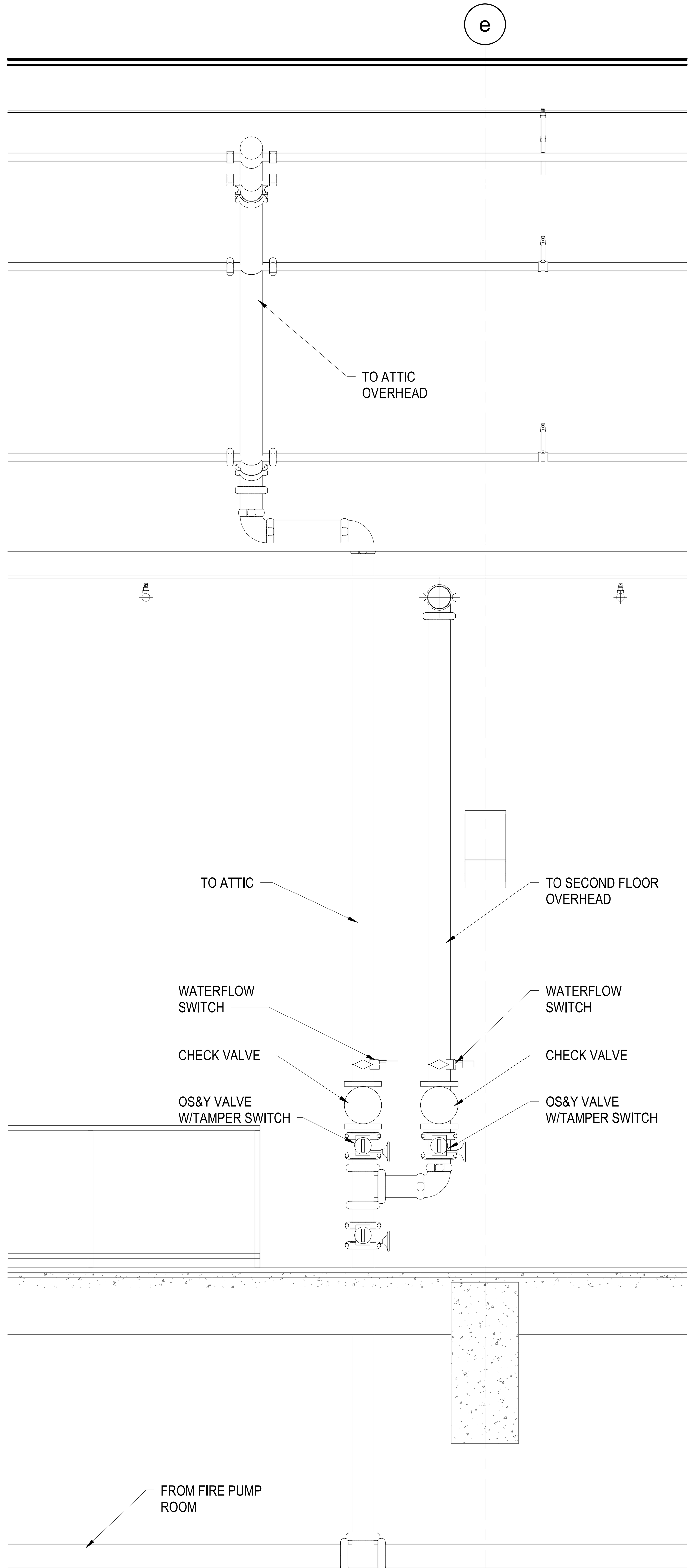
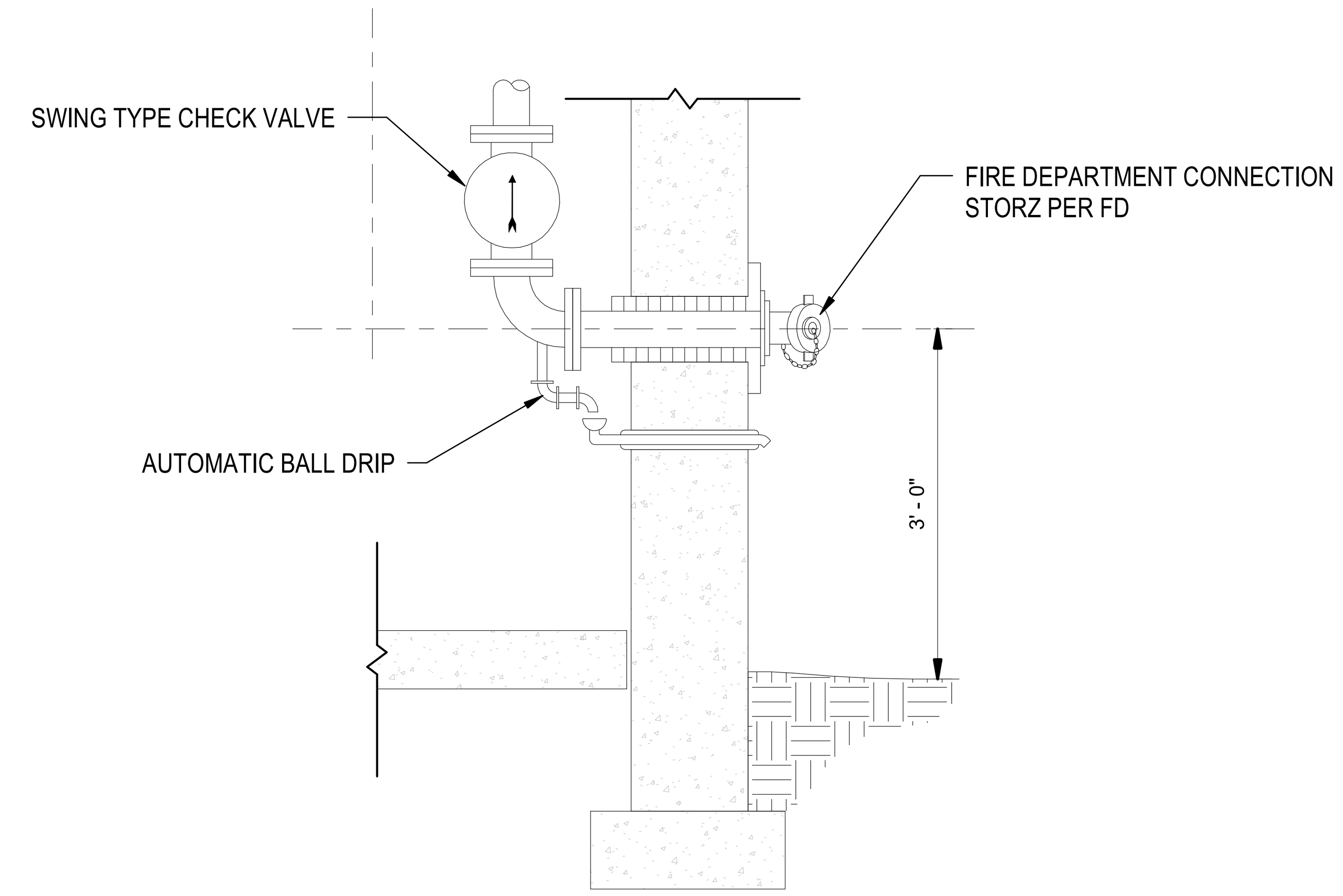
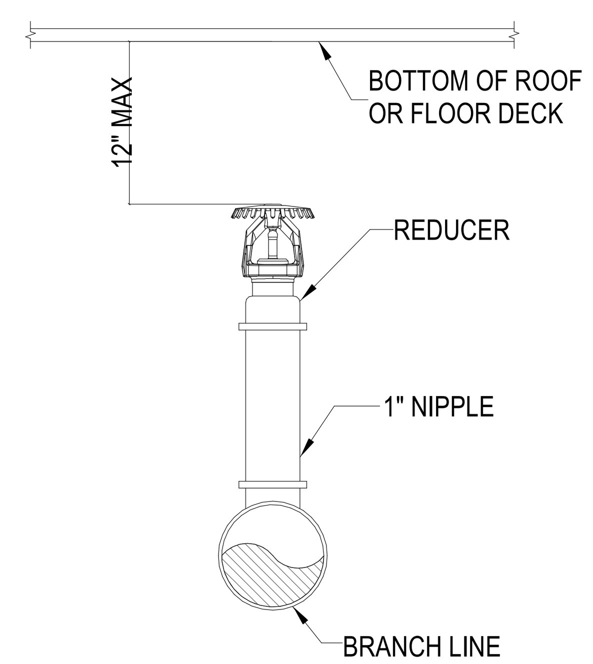


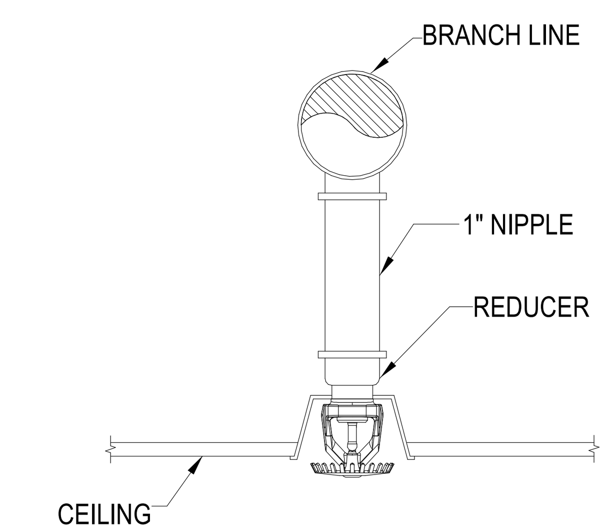
A1 **SECOND FLOOR/ATTIC VALVE ASSEMBLY**
SCALE: 1/2" = 1'-0"



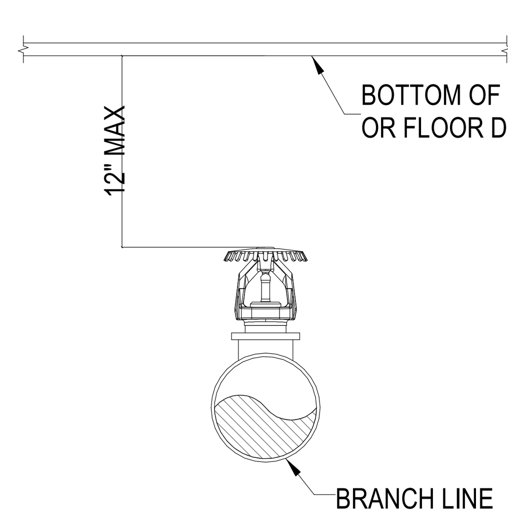
C3 **FIRE DEPARTMENT CONNECTION**
SCALE: 3/4" = 1'-0"



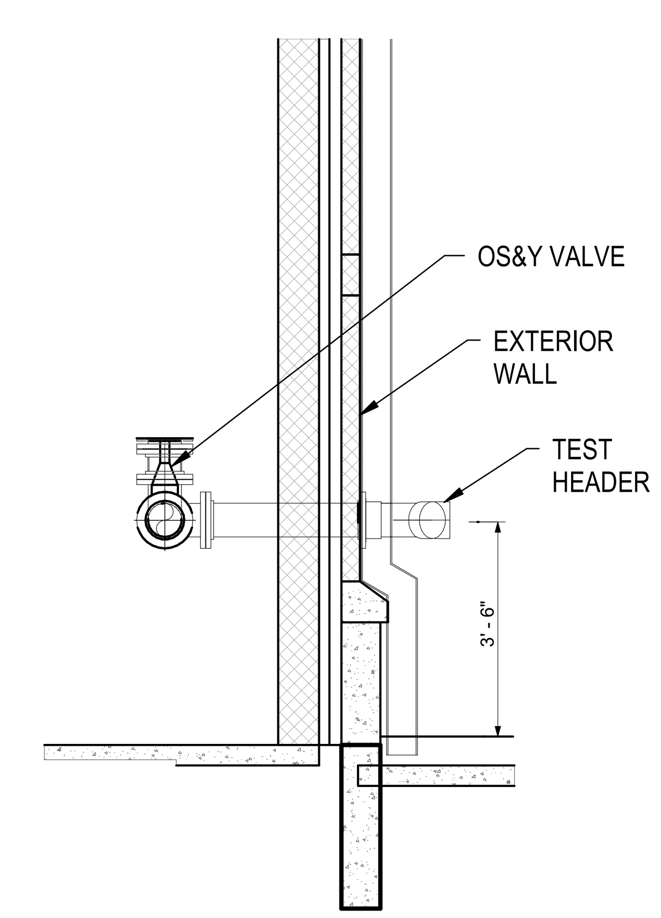
B3 **UPRIGHT SPRINKLER WITH SPRIG-UP**
SCALE: NOT TO SCALE



B4 **RECESSED PENDENT SPRINKLER DROP**
SCALE: NOT TO SCALE



A3 **UPRIGHT SPRINKLER ON BRANCHLINE**
SCALE: NOT TO SCALE



A4 **TEST HEADER DETAIL**
SCALE: NOT TO SCALE

GRAPHIC SCALE(S)

3/4" = 1' - 0" — 0 6" 1' 2' 3' 4' 5'

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COMMONWEALTH OF MASSACHUSETTS
JEFFREY T. DeMAINE
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FIRE PROTECTION	S. RUCINSKI

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KITTERY, ME
B60 REPAIRS
DETAILS - FIRE PROTECTION

EPROJECT NO.: 1740579
NAVFAC DRAWING NO.
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FX501

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